



**GREEN
CLIMATE
FUND**

Meeting of the Board
23 – 26 October 2023
Tbilisi, Georgia
Provisional agenda item 12

GCF/B.37/02/Add.08

2 October 2023

Consideration of funding proposals – Addendum VIII

Funding proposal package for FP217

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Funding Proposal

Project title:	Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration
Country(ies):	Rwanda
Accredited Entity:	Ministry of Environment
Date of first submission:	2022/02/25
Date of current submission	2023/09/11
Version number	V.6



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(a) PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Project	A.2. Public or private sector	Public	
A.3. Request for Proposals (RFP)	<u>Not applicable</u>			
A.4. Result area(s)			GCF contribution	Co-financiers' contribution
	Mitigation total		8 %	0 %
	☐ Energy generation and access		0 %	0 %
	☐ Low-emission transport		0 %	0 %
	☐ Buildings, cities, industries and appliances		0 %	0 %
	☒ Forestry and land use		8 %	0 %
	Adaptation total		92 %	100 %
	☒ Most vulnerable people and communities		13 %	49 %
	☒ Health and well-being, and food and water security		2 %	0 %
	☐ Infrastructure and built environment		0 %	0 %
☒ Ecosystems and ecosystem services		77 %	51 %	
A.5. Expected mitigation outcome	1,084,291tCO ₂ eq	A.6. Expected adaptation outcome	Direct Beneficiaries: 1,254,242 (654,404 females)	Indirect Beneficiaries: 9,260,745 (4,769,284 females)
			30.9 % of the CND population	69.9% of Rwanda's population
A.7. Total financing (GCF + co-finance)	50,000,000 USD	A.9. Project size	Small (Up to USD 50 million)	
A.8. Total GCF funding requested	39,056,421 USD			
A.10. Financial instrument(s) requested for the GCF funding	☒ Grant 39,056,421 USD ☐ Loan - ☐ Guarantee -		☐ Equity : ☐ Results-based payment :	
A.11. Implementation period	5 years	A.12. Total lifespan	20 years	
A.13. Expected date of AE internal approval	<u>TBD</u>	A.14. ESS category	B	
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐ M	
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐	
A.19. Complementarity and coherence	Yes ☒ No ☐			

A.20. Executing Entity information	Rwanda Forestry Agency (RFA) Country of Registration: Rwanda Ownership Type: Government agency.
A.21. Executive summary	
Rwanda's Congo-Nile Divide (CND), an area of 444,600 hectares, is a high-elevation landscape of remnant natural forest reserves embedded in a mosaic of intensively managed smallholder farms and commercial plantations. Extending along a north-south ridge with an altitudinal gradient from 1,900m to 4,507m, the CND separates the drainage basins of the Congo and Nile Rivers. It is part of the Albertine Rift, which is one of the most biologically rich regions in Africa and "one of the most important regions for conservation" on the continent¹, containing more than half of Africa's birds and 40% of its mammals. At its highest elevations, the region is crowned by the Volcanoes, Gishwati-Mukura, and Nyungwe National Parks, containing Rwanda's only remaining montane forests, one of the most biodiverse ecosystems on the African continent, with high levels of endemism (species found nowhere else on Earth) and dozens of threatened species, including populations of mountain gorillas and chimpanzees. Nyungwe has been recognized as one of the six highest priority protected areas in the Albertine Rift due to its high endemism and globally threatened species (Plumptre et al 2016). Though covering barely 5% of the country, this archipelago of protected forests is of critical importance for: • Its globally recognized high biodiversity values^{2,3, 4, 5}; • Its role in Rwanda's acclaimed ecotourism offerings serving as the primary source of Rwanda's foreign revenue and as a major economic development driver^{6, 7}; • The ecosystem services provided for surrounding subsistence farmlands, as well as for commercial tea and coffee plantations offering employment opportunities for women while providing the 2nd and 3rd highest sources of foreign revenue^{8,9}; • The regulation of water capture and release to growing urban centers on the CND periphery¹⁰, the nation's extensive river network, and downstream hydroelectric operations¹¹; and • The sequestration and storage of nearly 60% of Rwanda's national carbon stocks¹². These values and services are already compromised by a combination of anthropogenic and climatic factors; they will be much more severely threatened by the potential negative feedback loops associated with projected climate change impacts. The Project will promote an ecosystem-based approach to forest protection and restoration and will enhance the extent and resilience of forest ecosystems and species, including functionally linking currently isolated forest fragments. Through these approaches, the Project will protect, restore and rehabilitate forest ecosystems, enhancing biodiversity and reducing key threats, while	

1 Plumptre, A.J., et al. 2007. The biodiversity of the Albertine Rift. *Biological Conservation* 134:178-194.

2 Plumptre, et al. 2016. Conservation Action Plan for the Albertine Rift. Wildlife Conservation Society

3 UN/CBD. 2020. Rwanda 6th National Report to the Convention on Biological Diversity

4 2012. Eastern Afromontane Biodiversity Hotspot. Critical Ecosystems Partnership Fund.

5 2023. Nyungwe National Park. BirdLife International. www.birdlife.org.

6 RDB. 2021. Rwanda Development Board Annual Report

7 2010. *In the Kingdom of Gorillas*. Weber & Vedder.

8 Andrew, G., Masozera, M. 2010. *Payment for Ecosystem Services and Poverty Reduction in Rwanda*. Journal of Sustainable Development in Africa (V12, No.3)

9 RDB. 2021. Annual Report.

10 Karamage, F., et al., 2017. *Modeling Rainfall-Runoff Response to Land Use and Land Cover Change in Rwanda (1990–2016)*. *Water* 2017, 9(2), 14.

11 Rwanda Environment Management Authority (REMA), 2009. Rwanda State of Environment and Outlook Report, Chapter 8: Energy Resources.

12 Mugabowindekwe, M., Brandt, M., Chave, J. et al. Nation-wide mapping of tree-level aboveground carbon stocks in Rwanda. *Nat. Clim. Chang.* 13, 91–97 (2023). <https://doi.org/10.1038/s41558-022-01544-w>

increasing the provision of critical ecosystem services to strengthen adaptation of vulnerable communities.¹³
14

The CND landscape supports nearly one-third of Rwanda's population on barely one-fifth of its land base. Average population density is 736 people/km², with some districts now exceeding 1,000 people/km²¹⁵. The highest rates of rural poverty in the country are reported from the CND, including its most vulnerable subsistence farmers. While high rainfall has historically permitted generally sufficient food production, the region's rugged topography requires farming on extremely steep slopes. With more frequent extreme rainfall events in recent years, this agricultural landscape is recognized as the region of Rwanda most vulnerable to floods and landslides¹⁶. Between 2013-2016 heavy rains generated landslides that killed at least 174 people and destroyed the homes of more than 5,000. Many of these landslides caused flooding of highly productive raised-bed cultivation systems in the CND's many valleys and surrounding bottomlands and many hectares of crops being washed away annually¹⁷. An estimated 1.5 million tons of fertile soil are lost annually to erosion¹⁸.

CND forests are essential to the region's climate resilience. Since independence, the total natural forest cover has declined 40%, from 219,000 ha to 130,700 ha, mostly from episodic planned excisions as well as uncontrolled incursions. The loss around the Gishwati and Mukura core forest area has been most extreme. Of the original 32,098 ha in the 1980s, now only 3,460 ha remains as natural forest: almost a 90% loss (see Annex 2). In comparison, the Volcanoes and Nyungwe boundaries have been stable for decades. The recent creation of the Gishwati-Mukura National Park is an important step toward halting and partly restoring this forest, particularly for its land stabilization and other ecosystem services within the CND's most vulnerable lands and dependent communities.

IPCC models predict rising temperatures and increasingly intense precipitation events for the CND through the rest of this century. The rainfall implications are especially threatening for smallholders farming on steep, heavily deforested slopes, where they are most exposed to landslides, leading to loss of crops, homes, and lives. Yet their adaptive capacity is low because crop yields are vulnerable to rainfall variability and because high population density severely limits their options for relocation in the event of disaster. This is particularly true for women who, until very recently, had fewer legal land rights than men. The expected 2°C increase in temperature would further result in a more than 470m rise in vegetation zone elevations, impacting traditional crop selection and management practices for future farmers. An increase of this scale would also dramatically alter the optimal ranges for tea and coffee production, introducing potential competition between these important export crops and alternative land uses.

This Project is designed to sustain and expand critical forest ecosystem functions in the CND, and ultimately enhance the resilience of the vulnerable communities who depend on them. This will be done by strengthening government capacity for spatial planning, resulting in institutional and regulatory policies to promote and integrate climate adaptation strategies, based on the latest science, through coordinated planning at the national and district levels. It will also protect, restore and expand natural forest in and around

13 2003. Interlinkages Between Biological Diversity and Climate Change. CBD Technical Series Number 10.

14 2022. Ecosystems and ecosystem services Sectoral Guide. Green Climate Fund.

15 National Institute of Statistics of Rwanda (NISR); The Fifth Rwanda Population and Housing Census, Main Indicators Report, February 2023

16 https://www.gfdr.org/sites/default/files/publication/National_Risk_Atlas_of_Rwanda_electronic_version_0.pdf

17 Nsengiyumva JB, et al., 2018 Landslide Susceptibility Assessment Using Spatial Multi-Criteria Evaluation Model in Rwanda. International Journal of Environmental Research and Public Health. 15(2):243.
<https://doi.org/10.3390/ijerph15020243>

18 Bizimana, H.; Sönmez, O. 2015 *Landslide Occurrences in the Hilly Areas of Rwanda, Their Causes and Protection Measures*. Disaster Sci. Eng. 2015, 1, 1–7

the core national parks; consolidate and expand some smaller but critical "stepping stones"¹⁹ of remaining natural habitat between these national parks (mainly small forest patches and wetlands/ riparian areas); improve the landscape linkage function of key high altitude landscape linkages between the protected areas, while at the same time securing ecosystem services through afforestation and improved native species mix for plantation forests on steep slopes; silvopastoral interventions on cultivated pasture; and agroforestry interventions in regenerative agricultural areas. Complementary activities will address key drivers of deforestation while creating alternative livelihood pathways for youth and women through ecotourism and income-generating, climate-resilient crops, as well as strengthening supportive financial services.

Project interventions will include 10,000 ha of land (including both forest and silvopastoral sites) under improved, climate-resilient management; and improve agroforestry practices on 3,346 ha of on-farm plantations, directly benefiting 1,254,242 people (654,404 women) in CND with more climate resilient livelihoods from reduced exposure to landslides, floods and soil erosion and more knowledge on climate risks, value of forests and forests ecosystems, climate adaptation options, access to indigenous and agroforestry quality materials and improved ecosystem services. The project will also indirectly benefit 9,260,745 people nationally mainly from the education and outreach program on climate change and gender issues, climate risks, value of forests and forest ecosystems, climate adaptation options, and will create approximately 24,212 job opportunities in CND forest-dependent communities. The total mitigation impact from reduced deforestation, forest restoration introduction and adoption of improved cookstoves is estimated to be 1,084,291t CO₂-eq by 2043.

(b) PROJECT/PROGRAMME INFORMATION

B.1 (a). Climate context

This section draws from the Climate Change Vulnerability Assessment presented in Annex 2.1, where more comprehensive elaboration can be found. Discussion on climate predictions is based primarily upon multi-model means from the Coupled Model Intercomparison Project's sixth phase (CMIP6)²⁰, used in the Sixth IPCC Assessment Report. Results are shown for paired Shared Socioeconomic Pathways (SSP) and Representative Concentration Pathway (RCP) emission scenarios; for some variables emphasis is placed on a "pessimistic" high SSP-high RCP scenario (SSP5-8.5), for greater distinction between present day and future conditions as a consequence of anthropogenic climate change. Specialized modeling performed at ultra-high resolution for the East African Great Lake Environments project (EAGLE)²¹, and findings from scientific literature are utilized as well. Key findings are outlined in Table 1, and each climate component theme is discussed in the section on projected climate change impacts below.

¹⁹ Bagchi, R., et al. 2018. Forecasting potential routes for movement of endemic birds among important sites for biodiversity in the Albertine Rift under projected climate change. *Ecography*. 41: 401-413.

²⁰ Coupled Model Intercomparison Project Phase 6. [Link](#)

²¹ Grim, J. A., Pinto, J. A., Jensen, A. A., & Seimon, A., 2020: The East African Great Lake Environments (EAGLE) Climate Downscaling Dataset. [Link](#)

Table 1. Principal Climate Hazards of the Congo Nile Divide through Mid-century

Climate component	Character of change	Level of confidence	Near-term concern	Mid-century concern	Project Response
Temperature	Upward trend from global greenhouse gas emissions and land-use change	Very high for +2.6 °C net increase 1970-2040	Elevational range of biota including pests and pathogens increasingly out of balance	Major uphill displacements of biota including cultivars due to ~470 m rise in thermal conditions	Promotion of agroforestry practices: contribute to the conservation of soil moisture and its recharge through the infiltration of rainfall and runoff water, the creation of sheltered microclimatic conditions, and the inclusion into the soil of organic matter that contributes to moisture retention. Promotion of silvopastoral practices: Improve soil properties due to greater uptake of nutrients from deeper soil layers, enhanced availability of nutrients from leaf-litter and increased nitrogen input by N₂-fixing trees.⁸⁵ Moreover, silvopastoral systems enhance the resilience of the soil to degradation, nutrient loss, and climate change, while enhancing water holding and infiltration capacity of the soil which contributes to the regulation of the hydrological cycle by reducing runoff intensity^{22, 23}. Protective restoration measures at erosion-prone areas: Reduce exposure to soil erosion and floods of communities and their assets living in the road and river side. Restoration of degraded natural forests in protected areas Improve protected areas management to reduce potential human threats (forest encroachment, mining, fire). The project will support activities aimed at (1) promote alternative livelihoods that are resilient to climate change (2) decrease of demand and (3) increase of wood supply capacity. Dissemination of Improved Cook Stoves The project will support the development of an integrated
Annual precipitation	Increasing totals	Moderate, but considerable model variation	Low concern, with natural interannual variability still dominant	Moderate concern, with majority of models showing upward trend	
Precipitation intensity	Increasing short-period rainfall and storm totals	Almost certain to occur	Severe landslide hazard already present and increasing, building flash flood potential	Extreme hazard: widespread and frequent landslides and flash floods	
Cloud base height	Rising cloud base and levels of fog immersion in forests	Likely ongoing, will continue with temperature increase	Already significant in deforested highlands with some loss of moisture provision	Of major concern to forest ecology in protected landscapes	
Drought/dry spells	Increasing intensity	Moderate	Some increase in vegetation desiccation potential, possibly enhancing fire risk	Sustained risk of increase in desiccation and fire risk, possibly offset by rainfall increases	Improve protected areas management to reduce potential human threats (forest encroachment, mining, fire). The project will support activities aimed at (1) promote alternative livelihoods that are resilient to climate change (2) decrease of demand and (3) increase of wood supply capacity. Dissemination of Improved Cook Stoves The project will support the development of an integrated
Wildfire	Increasing occurrence and extent in forested areas	Low	Relatively random occurrences, as in the recent past, that could be managed by improved vigilance	If rainfall increases do not occur, increased vegetation desiccation would intensify risk for widespread fire	
Climatic seasonality	Disappearance of mid-year dry season	Low, but explicitly shown in some models	No concern	Increasing convective storm occurrences in June-Aug may eliminate dry season	

(b) PROJECT/PROGRAMME INFORMATION

					land use plan at landscape level that identifies and integrates the trade-offs necessary to improve delivery of ecosystem services, conserve natural resources and sustain economic growth.
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• Climate baseline and current impacts

The climate of western Rwanda and its CND landscape features conditions highly supportive of tropical montane forests, rich biodiversity and human activities alike. Despite being located deep in the tropics just south of the equator, the elevated terrain of the CND landscape, mostly above 2,000 meters, ensures relatively cool temperatures through the year (**Figure 1**).

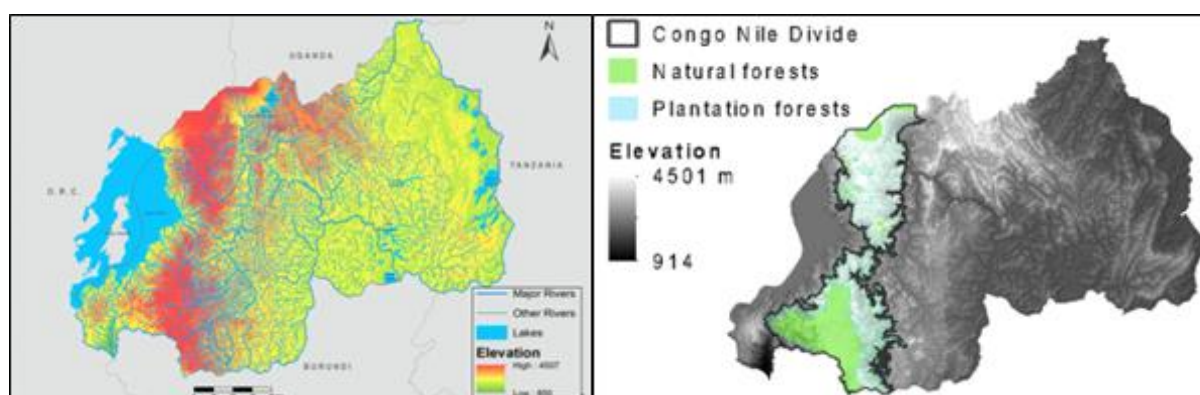


Figure 1 (left) Topographic map of Rwanda with riverine drainage channels overlaid. *Source: Rwanda (2019)²⁴*; **(right)** Congo Nile Divide, identified by the 1,900 m elevation contour and showing the presence of natural and plantation forests.

Rainfall is abundant for nine out of 12 months, and a reliable mid-year dry season adds dynamic character to phenological cycles, pollinator activity and agricultural practices (**Figure 2**). Prior to landscape conversion for agricultural purposes, continuous dense forest cover yielded augmentation of precipitation through fog interception in tree canopies, and near-daily cloud immersion sustained cloud forest ecosystems and species assemblages. The forests also maintained slope stability in steep terrain and yielded percolation and groundwater recharge rather than direct runoff of heavy precipitation. Such conditions are now constrained to within the CND's protected forests within national parks, covering just 30% of the CND landscape and just 5% of Rwanda's total land surface. These are Rwanda's only remaining montane forests yet represent

22 Ibrahim M, Guerra L, Casasola F, Neely N, 2010. Importance of silvopastoral systems for mitigation of climate change and harnessing of environmental benefits. In: Abberton M, Conant R, Batello C (Eds) Grassland carbon sequestration: management, policy and economics. Proceedings of the workshop on the role of grassland carbon sequestration in the mitigation of climate change. Integrated Crop Management, Vol. 11. FAO, Rome, Italy. <http://www.fao.org/docrep/013/i1880e/i1880e09.pdf>.

23 Jose S., 2009. Agroforestry for ecosystem services and environmental benefits: an overview. Agroforest Syst 76 (1):1–10.

24 Vital Signs, 2019: Rwanda Atlas. Environment, Agriculture and Livelihood Options. Vital Signs and WCS Rwanda. [Link](#).

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58.7% of national tree carbon stocks²⁵ and are imperiled by intense population pressures, with 30.7% of the country's population residing within the CND landscape districts.

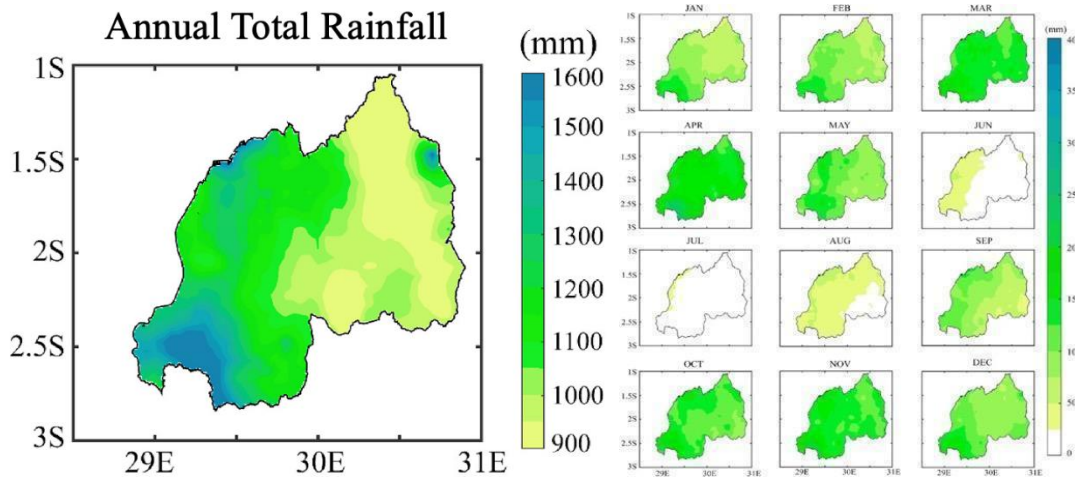


Figure 2. Annual (left) and monthly (right) rainfall (mm) climatology over Rwanda (period of record: 1981–2017), based on algorithms that calibrate satellite-based rainfall estimates with surface observations. Source: Jonah et al., 2021.²⁶

A reconstruction of temperatures for the past 140 years for Rwanda, based on quality controlled instrumental records and centered on Kigali, shows that global climate change already has strong local expression. Rapid temperature increase commenced in the 1970s, with a net gain of approximately +1.1 °C since 1970²⁷ (Figure 3).

²⁵ Mugabowindekwe, M., Brandt, M., Chave, J. et al. Nation-wide mapping of tree-level aboveground carbon stocks in Rwanda. *Nat. Clim. Chang.* 13, 91–97 (2023). <https://doi.org/10.1038/s41558-022-01544-w>

²⁶ Jonah, K., Wen, W., Shahid, S., Ali, M.A., Bilal, M., Habtemicheal, B.A., Iyakaremye, V., Qiu, Z., Almazroui, M., Wang, Y. and Joseph, S.N., 2021. Spatiotemporal variability of rainfall trends and influencing factors in Rwanda. *Journal of Atmospheric and Solar-Terrestrial Physics*, 219, p.105631.

²⁷ Berkeley Earth (website): Rwanda Country Profile. www.berkeleyearth.org

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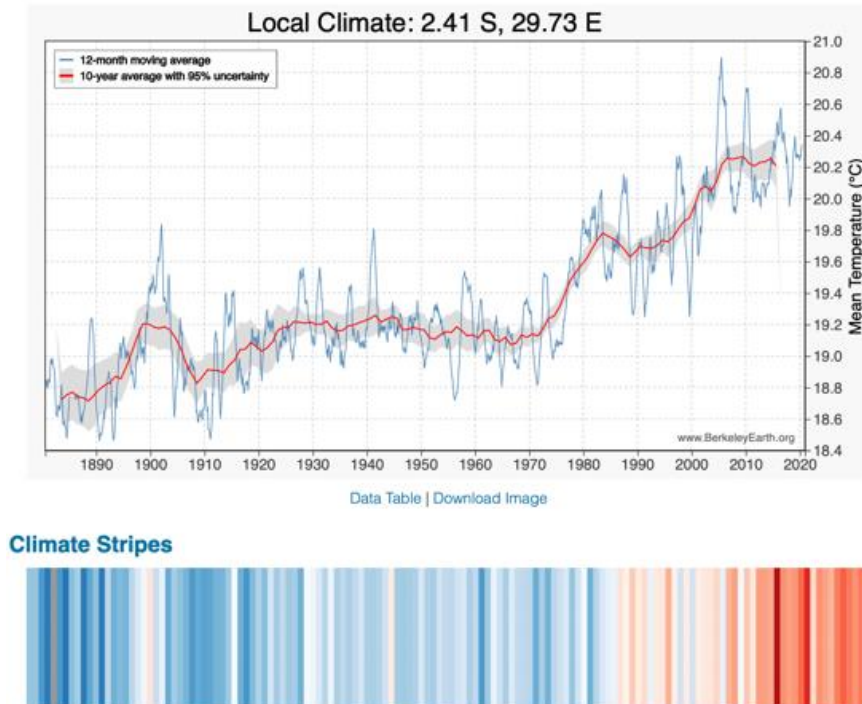


Figure 3. Berkeley Earth temperature reconstruction based on instrumental measurements for Kigali in central Rwanda. The “Climate Stripes” offers a visual display that effectively communicates both individual year temperature anomalies and the overall multi-decadal trend behavior. Source: Berkeley Earth Rwanda Country Profile. www.berkeleyearth.org

The regional warming of the climate over the past half-century coincided with widespread forest conversion to farmland along the CND. As a consequence, at fine spatial scales climatic conditions across the CND at present mimic the landscape mosaic, whereby only the protected forests will retain much of the past climatic character while converted landscapes are likely to be significantly warmer by day, have reduced cloud cover and higher cloud bases, and experience slightly reduced rainfall as well. Due to the increased moisture carrying capacity of warmer air (7% increase per °C of warming)²⁸, ongoing climatic warming promotes increasingly intense short-period rainfall, which acts synergistically with landscape conversion on steep slopes to cause quickflow runoff, yielding large increases in landslide occurrence and soil erosion more generally (**Figure 4**). With its high topographic relief, the CND is highly susceptible to soil erosion. **Throughout Rwanda, soil erosion has increased by 54% since 1990 with the bulk of the increase experienced and highest erosion levels over western Rwanda.**²⁹ Between January and June in 2018, landslides caused more than 200 mortalities, and in May 2020³⁰, 65 deaths from landslides and flash flooding occurred during a single night of storms focused on the eastern slopes of the CND.³¹

²⁸ Trenberth, K.E., 2011. Changes in precipitation with climate change. *Climate Research*, 47, pp.123-138.

²⁹ National Institute of Statistics of Rwanda, 2019: Rwanda Natural Capital Accounts-Ecosystems. p. 23

³⁰ Reuters 2018. <https://www.reuters.com/article/us-rwanda-floods-idUSKBN1I81I>

³¹ Associated Press 2020. <https://apnews.com/article/b7658b76a403fd9bc8cba249f9d484fd>

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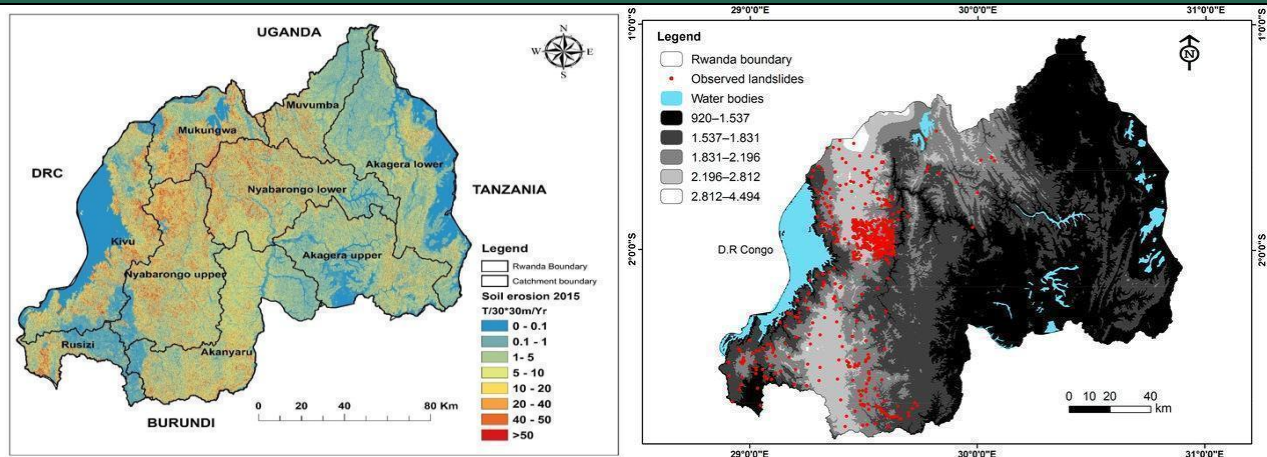


Figure 4. (left) Intensity of soil erosion in Rwanda assessed for the year 2015. Source: NISR: National Institute of Statistics of Rwanda, 2019. **(right)** Observed landslide occurrences in Rwanda against a topographic background, showing intense clusters along both the east and west sides of the CND landscape. Source: Nsengiyumva et al., 2018.³²

Wildfire occurrence in forested lands exacerbated by climatic warming is a rapidly mounting concern globally, and montane forests of the CND carry increasing vulnerability to loss from burning. Absence of fire-adapted flora in CND highland forests, however, suggests that fire has not been a major selective force within the forest ecosystem. The 1997 visitation of weeks-long uncontrolled wildfires in the Nyungwe forest had an ecological imprint that remains evident 25 years later, as burned areas revegetated with dense ferns rather than replacement forests,³³ representing a net loss of 12% of Nyungwe's forest cover and corresponding loss of its carbon endowment. Despite increased propensity for drying of vegetation as the climate warms, forests remain sufficiently moist such that fire causing forest losses are infrequent, and when they occur are usually tied to anomalous dry spells paired with human ignition, as was the case in 1997.³⁴

Cloud base altitude, which is highly influential in CND montane forest ecology and species composition, is likely rising above recent past conditions due to both climatic warming and conversion of forests to agricultural lands. Fog interception by vegetation augments montane forest precipitation in central African highland forests by an estimated 10%³⁵, so highland forest loss by human action therefore represents a local reduction to the ecosystem service of rainwater provision).

Projected climate change impacts

Temperature increases. Localized projections for the CND landscape of climatic changes for coming decades, generated for the most recent IPCC assessment, present a strong consensus on temperature increase for the next two decades. The magnitude of net temperature change by late-century, and the tendency of warming to accelerate, hold steady, decelerate or even begin an overall temperature decline is unknowable at present, but a large number of climate model simulations show common patterns, whereby divergence in possible temperature outcomes will only become apparent around 2040 (**Figure 5**). These

³² Nsengiyumva, J.B., Luo, G., Nahayo, L., Huang, X. and Cai, P., 2018. Landslide susceptibility assessment using spatial multi-criteria evaluation model in Rwanda. *International journal of environmental research and public health*, 15(2), p.243.

³³ Masozera, A.B. and Mulindahabi, F., 2012. Post-fire regeneration in Nyungwe National Park-Rwanda. WCS White paper report.

³⁴ Seimon, A., 2012: Climatology and Potential Climate Change Impacts of the Nyungwe Forest National Park, Rwanda. WCS White paper report, Wildlife Conservation Society, New York, USA, 44 pp.

³⁵ Bruijnzeel, L.A., Mulligan, M. and Scatena, F.N., 2011. Hydrometeorology of tropical montane cloud forests: emerging patterns. *Hydrological Processes*, 25(3), pp.465-498.

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results are also consistent with outputs from the previous CMIP5 model suite for western Rwanda (e.g., IFAD report 2020)³⁶. As such, using the consensus values of such models is a sensible choice for environmental planning, with a +2.0 °C increase over the IPCC-referenced 1995-2014 baseline conditions offering a representative value for warming by mid-century. However, the observed warming of +0.6 °C registered between 1970-95 prior to the IPCC baseline period must be recognized as well, suggesting **net warming of +2.6 °C from a more representative 20th century baseline**.

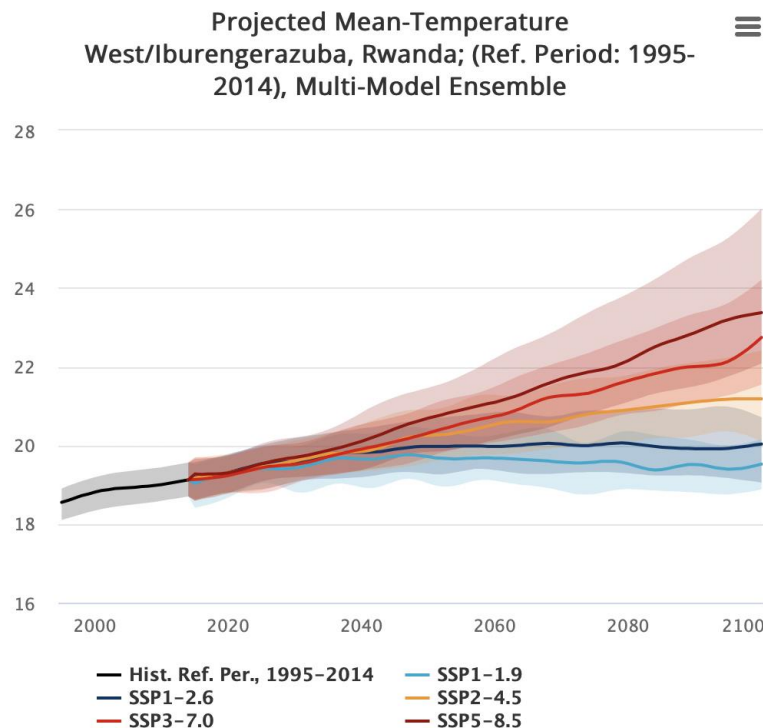


Figure 5. CMIP6 multi-model ensemble predictions for mean annual temperature over western Rwanda across the 21st century under a variety of SSP-RCP combinations. Source: World Bank³⁷

Annual mean precipitation changes. The CMIP6 results for precipitation changes over coming decades are less consistent than the temperature projections. Recent studies focused in East Africa have identified limitations to global model utilization for regionally specific predictions, such as failure to represent observed precipitation magnitude and distribution in the present, and reliance on approximations (i.e., parameterizations) that inadequately represent the role of convective storms in precipitation delivery³⁸. The CMIP6 ensemble means for annual precipitation over western Rwanda have a baseline value of approximately 2,000 mm per year, which is considerably higher than actual values (**Figure 6**; compare values with Figure 2 above). The higher emissions scenarios show increasing totals over time, which is plausible given the increasing water vapor carrying capacity of the warming atmosphere. Strangely, the most optimistic scenario (SSP1-1.9) shows strongly varying trends at decadal scales, increasing far more rapidly

³⁶ Hunter, R., Crespo, O., Coldrey, K., Cronin, K., New, M. 2020. Research Highlights – Climate Change and Future Crop Suitability in Rwanda. International Fund for Agricultural Development (IFAD), Rome.

³⁷ World Bank Climate Change Knowledge Portal. [Link](#)

³⁸ Wainwright, C.M., Marsham, J.H., Rowell, D.P., Finney, D.L. and Black, E., 2021. Future changes in seasonality in East Africa from regional simulations with explicit and parameterized convection. *Journal of Climate*, 34, pp.1367-1385.

(b) PROJECT/PROGRAMME INFORMATION

in the near-term than other scenarios. Such inconsistencies reduce confidence in the overall product suite for precipitation predictions.

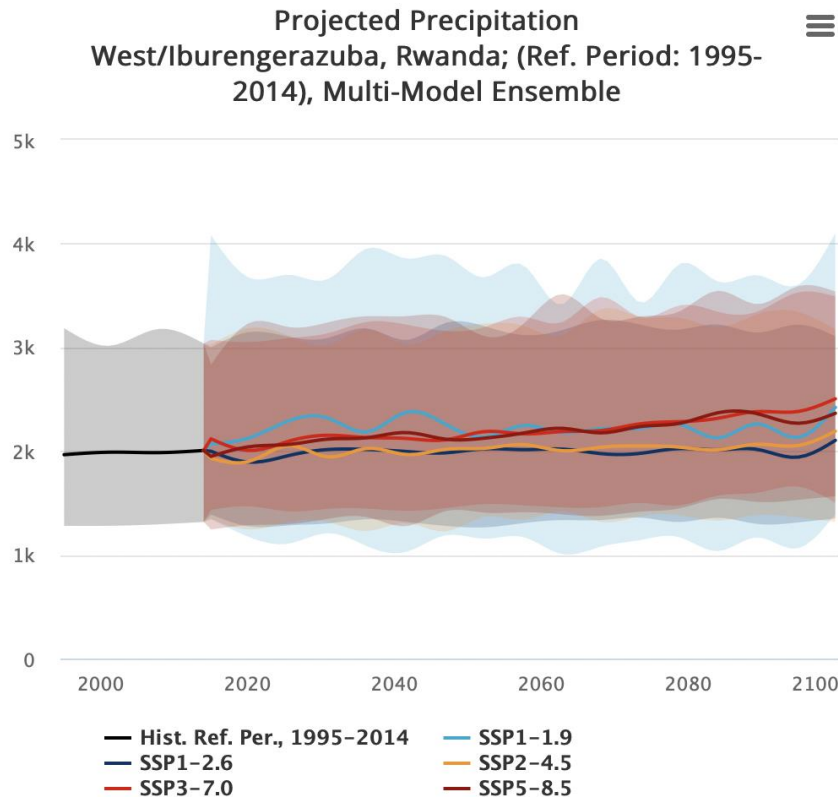


Figure 6. CMIP6 multi-model ensemble prediction for mean annual precipitation over western Rwanda across the 21st century under a variety of SSP-RCP combinations. Left-axis scale is thousands of mm per year. [Source:](#) World Bank³⁹

There is also major inconsistency between CMIP model suites. As assessment for western Rwanda from CMIP5 under RCP 8.5 comparing historical and mid-century outputs suggested rainfall decreases of 5-10% in most months, with net annual decrease of 86 mm⁴⁰; the baseline time periods and geographic domains of these two CMIP ensemble compilations for western Rwanda differ, but the reversal from drier to wetter gives low confidence in the utilization of precipitation predictions specific to the CND if derived from global models.

To generate more refined results, Grim et al., (2020)⁴¹ performed regional simulations of East African climate at ultra-fine, convection-allowing scale (WRF model at 3.3 km model resolution); these were based off the global CESM2 model (part of the CMIP6 suite) boundary conditions under the high emissions RCP8.5. The regional model was configured to optimize parity with regional climate station data. The resulting high spatial resolution output suggests a strongly amplified rainfall signal over the 21st century over western Rwanda relative to the CMIP6 ensemble mean for RCP 8.5 (**Figure 7**). Also notable is that the rest of Rwanda exhibits

³⁹ World Bank Climate Change Knowledge Portal. [Link](#)

⁴⁰ Hunter, R., Crespo, O., Coldrey, K., Cronin, K., New, M. 2020. Research Highlights – Climate Change and Future Crop Suitability in Rwanda. International Fund for Agricultural Development (IFAD), Rome.

⁴¹ Grim, J. A., Pinto, J. A., Jensen, A. A., & Seimon, A. (2020). The East African Great Lake Environments (EAGLE) Climate Downscaling Dataset (NCAR Technical Note TN-563+STR). [Link](#)

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far smaller increases, suggesting that the CND's hydrological significance as a water tower for the nation will be reinforced over time.

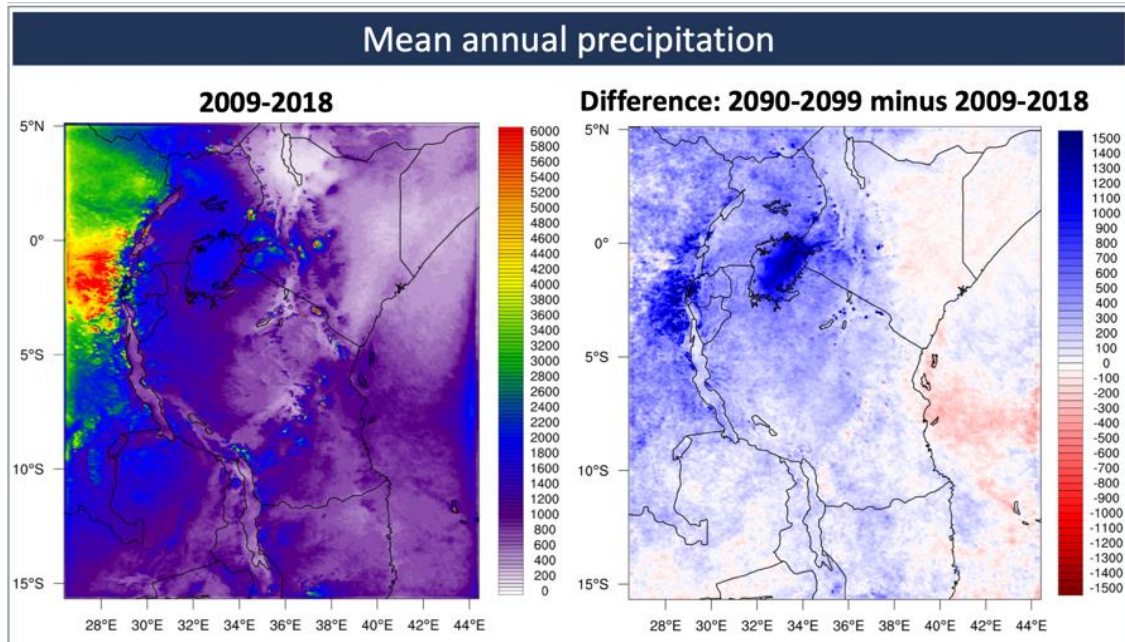


Figure 7. EAGLE projects high-resolution WRF simulation of annual mean precipitation (**left**) and difference between end of century and recent past (**right**) across East Africa, where values are in mm per year. The CND landscape and Lake Kivu show comparable increases to eastern Lake Victoria, and represent the largest rainfall increases across the modeled domain. Source: Grim et al. (2020)⁴²

Precipitation intensity changes. Due to well-understood physical properties of the atmosphere, as warming progresses short-period rainfall rates and individual storm totals, especially for the fraction of higher-end events that can cause flash flooding and promote landslides, will increase and should amplify very significantly over time⁴³. The CMIP6 ensemble means for monthly counts of days with heavy precipitation suggest an upward trend is now underway (**Figure 8**). Forthcoming increases are most pronounced for November through January. Such increases in heavy precipitation events are to be expected under a warming climate and are of major social and ecological concern, being a driver of landslide occurrences and major contributors to soil erosion intensity.

⁴² Grim, J. A., Pinto, J. A., Jensen, A. A., & Seimon, A. (2020). The East African Great Lake Environments (EAGLE) Climate Downscaling Dataset (NCAR Technical Note TN-563+STR). [Link](#)

⁴³ Fowler, H.J., Ali, H., Allan, R.P., Ban, N., Barbero, R., Berg, P., Blenkinsop, S., Cabi, N.S., Chan, S., Dale, M. and Dunn, R.J., 2021. Towards advancing scientific knowledge of climate change impacts on short-duration rainfall extremes. *Philosophical Transactions of the Royal Society A*, 379(2195)

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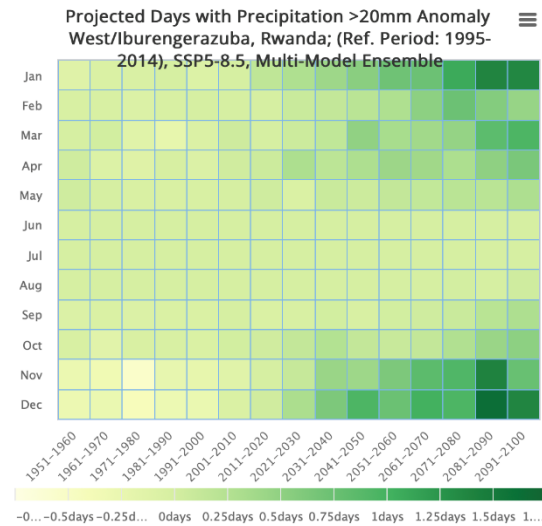


Figure 8. CMIP6 multi-model ensemble prediction for monthly changes in the number of days with rainfall >20 mm over western Rwanda across the 21st century under RCP8.5. Source: World Bank⁴⁴

Drought/dry spells. Warming will also increase the propensity for vegetation to desiccate during dry spells, intensifying wilting of natural vegetation and cultivars. The capacity of air to extract moisture from landscapes through direct evaporation and evapotranspiration of vegetation will also increase exponentially as temperatures rise. This means that land surfaces and vegetation will dry out more quickly between rainfall episodes in the future than in the past. It also means that to maintain hydrological balances that natural and human systems depend upon, total rainfall must increase in proportion to intensifying evaporation rates. The CMIP6 ensemble results for the CND region suggest such a favorable balance may indeed be maintained through at least mid-century (**Figure 9**). Under the full range of emissions scenarios considered, there is almost no change in ensemble means through mid-century.

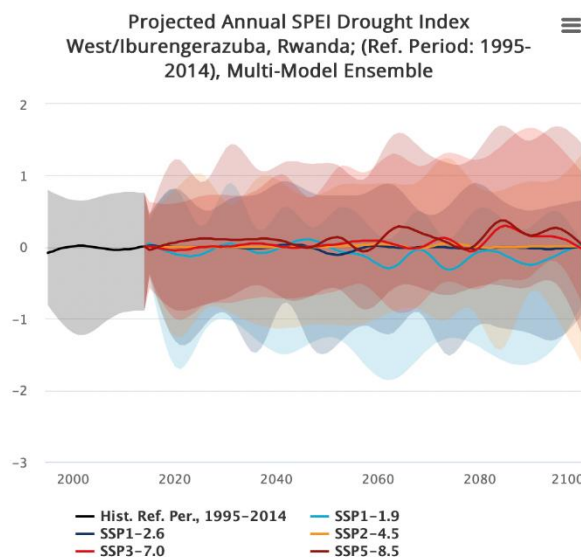


Figure 9. Western Rwanda CMIP6 model-model ensemble mean time series of the Standardized Precipitation Evapotranspiration index (SPEI) for a range of climate change scenarios across the 21st century. The SPEI is a measure of the integrated water deficit

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in a location, taking into account the contribution of temperature dependent evapotranspiration of a 12-month period. Source: World Bank⁴⁵

A more useful variable for assessing drought potential at daily-seasonal times scales is the Vapor Pressure Deficit (VPD). This is an absolute measure of the difference between the water vapor content of the air and its saturation value, and as such, represents a direct metric of the ability of the atmosphere to extract moisture from the land surface⁴⁶. High resolution VPD predictions for present day, mid-century and late-century under boundary conditions prescribed by the CESM model under the high emission RCP 8.5 scenario are shown in **Figure 10**. Annual cycles of VPD for model grid points high in the Nyungwe Forest and at lesser elevation in Kigali east of the CND both show remarkably little variation in VPD over the course of the century. Owing to the cool, persistently moist highland climate that characterize the CND landscape, the VPD values are consistently low in Nyungwe throughout the year for the three decadal time periods 2009-18, 2055-64 and 2090-99. This contrasts significantly with results from Kigali, where VPD registers four times higher than in Nyungwe during the mid-year dry season (20 vs. 5 hectopascal pressure deficit). This represents a favorable outlook over several decades to come for the CND in terms of limiting desiccating conditions that would elevate fire potential, stress crops and rapidly draw down hydrological resources. Importantly, however, not clearly evident in these model outputs is that rising cloud bases and associated warming temperatures increase the rate of drying of vegetation, increasing wildfire risk during rain-free periods, especially in lower reaches of forested terrain bordering settled areas where risks of anthropogenic ignition are highest.

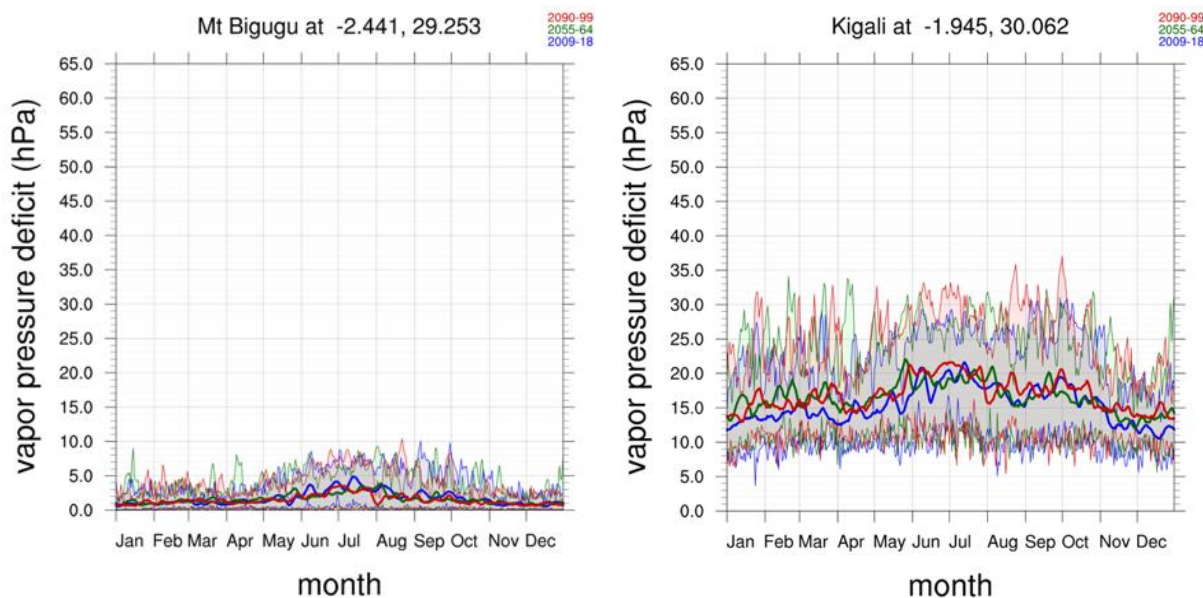


Figure 10. Modeled Vapor Pressure Deficit (VPD) on slope of Mt Bigugu in the Nyungwe Forest (**left**) and Kigali (**right**) for the recent past (2009-18, blue, mid-century (2055-64, green) and late-century (2090-99, red). Source: EAGLE project⁴⁷

Precipitation seasonality. The annual cycle of climate along the CND sees abundant rainfall for much of the year interrupted by a marked dry season in the June-August months when rainfall becomes light and sporadic, and during which hydrological deficits develop as evaporation far exceeds precipitation and resultant runoff. While highland orography largely explains the enhanced rainfall along the CND, the

⁴⁵ World Bank Climate Change Knowledge Portal. [Link](#)

⁴⁶ Seager, R., Hooks, A., Williams, A.P., Cook, B., Nakamura, J. and Henderson, N., 2015. Climatology, variability, and trends in the US vapor pressure deficit, an important fire-related meteorological quantity. *Journal of Applied Meteorology and Climatology*, 54(6), pp.1121-1141.

⁴⁷ East African Great Lake Environments (EAGLE) project. [Link](#)

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occurrence of the dry seasons occurs region wide irrespective of terrain elevation (**Figure 11**). This finds explanation in the seasonal north-south migration of a pan-equatorial band of enhanced cloudiness and rainfall known as the Inter Tropical Convergence Zone (ITCZ)⁴⁸. The slight reduction in rainfall experienced over several weeks along the CND in January and February reflects much more pronounced dryness to the north in Uganda; conversely, the more defined mid-year dry season along the CND spanning June through August is associated with enhanced southeasterly airflows that effectively push the zone of atmospheric convergence promoting rainfall across the equator into Uganda.

In this light, an unusual finding from the EAGLE project's high-resolution modeling studies is that the CND landscape has a heightened risk of major change in climatic seasonality. By the 2055-64 period: the mid-year dry season effectively disappears. Being the output of a single model run under a high-end greenhouse gas emissions scenario may make this seem implausible; It is highly dissimilar from mid-century predictions of the consensus of global models, which sustain the dry season. It finds corroboration, however, in a new study on East African rainfall, which found that the parameterization of convection in global models may be a source of uncertainty for projections of changes in seasonal timing, and that potentially impactful changes in seasonality are therefore quite plausible⁴⁹. Such an eventuality is exceptionally difficult to plan for, and since it would only come to fruition well after the project period, if at all, additional studies could be conducted to provide needed guidance.

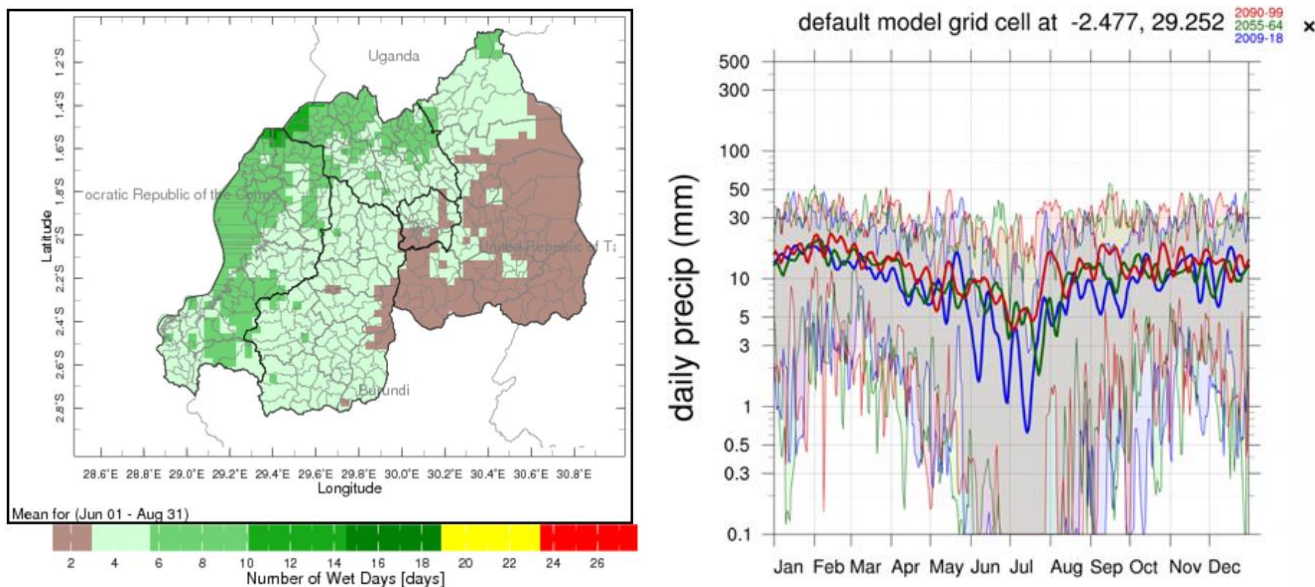


Figure 11. (left) The average number of “wet” days in Rwanda with rainfall exceeding 3 mm for the months of June-August. The wettest parts of the CND only receive significant rainfall on ~10% of days during the dry season. Source: Meteo Rwanda⁵⁰; **(right)** Annual cycle of rainfall rate (mm/day, note logarithmic scale) at Nyungwe Forest National Park showing daily means (thick lines) and extremes (thin lines) for 10-year time periods centered on 2014 (blue), 2060 (green) and 2095 (red). The dry season evident in the 2014 mean reaching below 1 mm/day, effectively disappears by the 2060 time period. Source: EAGLE project⁵¹

⁴⁸ Siebert, A., Dinku, T., Vuguziga, F., Twahirwa, A., Kagabo, D.M., delCorral, J. and Robertson, A.W., 2019. Evaluation of ENACTS-Rwanda: A new multi-decade, high-resolution rainfall and temperature data set—Climatology. *International Journal of Climatology*, 39(6), pp.3104-3120.

⁴⁹ Wainwright, C.M., Marsham, J.H., Rowell, D.P., Finney, D.L. and Black, E., 2021. Future changes in seasonality in East Africa from regional simulations with explicit and parameterized convection. *Journal of Climate*, 34, pp.1367-1385.

⁵⁰ ENACTS online tool at Meteo Rwanda. [Link](#)

⁵¹ EAGLE project: [Link](#)

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• Climate risks, vulnerability and impacts

Without strategic actions to confront mounting stresses borne by climate change interacting with unsustainable land-use practices, the CND would doubtless undergo a profound transformation over coming decades with significant loss of biodiversity and ecosystem services, reducing agricultural potential, severe hazards to humanity and loss of livelihoods. Temperature increase and loss of fog immersion will cause desiccation of the lower margins of protected forests promoting die-offs, arrival of pests, pathogens and invasive species, and increasing fire ignition potential. Current cultivation practices of both subsistence crops and valuable cash crops like tea and coffee will be drawn uphill, adding to pressure to convert protected forests to farmland. Whereas climatic changes will act as inexorably mounting stressors, unsustainable land surface conversion and land use practices will greatly exacerbate their impacts. The steep hillsides that characterize much of western Rwanda will experience intensifying erosion and risk of landslides, while built infrastructure such as road and bridges will be overwhelmed since they were engineered to withstand rainfall intensities expected under past climatic conditions that are sure to be exceeded, and with increasing frequency, as time progresses and temperatures steadily increase. Precipitation intensity will increase for short-period rainfall rates and individual storm totals, especially for the fraction of higher-end events that can cause flash flooding and promote landslides. This is certain to occur, is probably already discernible, and likely to amplify very significantly over time as warming progresses. This ties directly to landslides in particular, which already cause high mortality and significant destruction each year, and the CND's elevated vulnerability due to high topographic relief and deforested slopes makes this hazard of paramount concern to address (**Figure 12**).

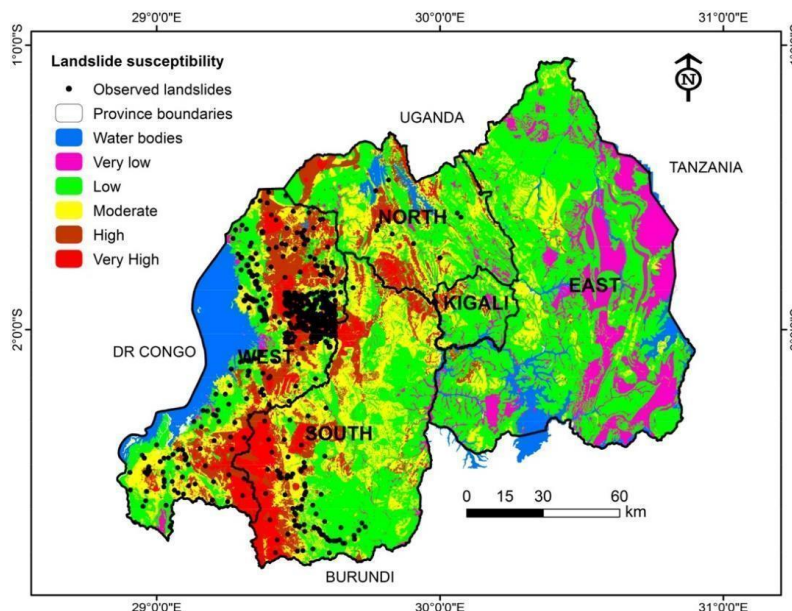


Figure 12. Rwanda landslide susceptibility map showing past observed landslides. Source: Nsengiyumva et al., 2018.⁵²

Significant temperature increases are almost certain to continue for decades to come, and will have mounting impacts on both natural and human systems along the CND, and some are already discernible. Over equatorial Africa observed environmental lapse rates feature temperature decreases by approximately 5.5 °C per kilometer of elevation increase⁵³, so 2 °C of warming expected by mid-century translates to a vertical rise on the order of 473 meters, promoting large uphill displacements of a wide range of organisms,

⁵² Nsengiyumva, J.B., Luo, G., Nahayo, L., Huang, X. and Cai, P., 2018. Landslide susceptibility assessment using spatial multi-criteria evaluation model in Rwanda. *International journal of environmental research and public health*, 15(2), p.243.

⁵³ Camberlin, P., 2018. Climate of Eastern Africa. In *Oxford Research Encyclopedia of Climate Science*. [Link](#)

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ecological processes, cultivars and human activities. While temperatures over the CND region are not expected to rise to levels for heat stress to elevate mortality⁵⁴ or reduce labor productivity, the indirect effect of rising temperatures through increased disease incidence is substantial⁵⁵.

Western Rwanda is the most productive region in the country for the cultivation of both tea and coffee, where conditions are optimized on the CND's sloping terrain. Both are cash crops with high export value, and are the target of active expansion through both governmental programs and private enterprise. Intensifying environmental stresses from climate change particularly due to thermal increases and the arrival of invasive pests from lowland regions, present challenges both for current plantings and in planning for expansion to new areas; the same concerns apply to other cultivars.

Cloud base altitude, which is highly influential in montane forest ecology and species composition, is likely already elevated above recent past conditions due to conversion of forests to agricultural lands, and certain to rise further in proportion to the degree of warming over coming decades. Ecosystems and species assemblages in the lower reaches of protected forests may already be out of balance with this important environmental variable, and this imbalance will only increase over time, promoting rapid species turnovers, enhancing potential for invasives, and act as a strong driver⁵⁶ of upward range extensions. Rising cloud bases in concert with warming temperatures also increase drying of vegetation, enhancing wildfire risk during rain-free periods. Fog interception by vegetation also augments montane forest precipitation by an estimated 10%, so rising cloud bases will act to remove this important hydrological input too. Highland forest loss therefore represents a local loss to the ecosystem service of rainwater provision that could be partially restored through reforestation.

Such challenges highlight how essential highland forests are to the climatic resilience of CND communities, for the ecosystem services and products they provide for both the region's vulnerable communities and the national economy. They also recharge aquifers; regulate water flow; control flooding; retain soil; provide wood fuel energy and timber; underpin the country's tourism, which provides the largest contribution to Rwanda's foreign exchange earnings⁵⁶; and provide wider benefits of atmospheric pollution control that sustain the country's economy and the wellbeing of its people⁵⁷.

Forest conversion to farmland in the CND highlands has until recently served as a release valve for lowland population pressure at the expense of drastically reduced national carbon stocks, diminished resilience to climate change, enhanced potential for destructive outcomes, reduced ecosystem service provision and disrupted biological connectivity. The resultant loss of ecosystem services and functioning and intensifying environmental stresses borne by climate change now require redirection towards forest restoration and other actions to ensure a sustainable future. Safeguarding the Congo-Nile Divide's remaining highland forests, and setting long-term goals of reforesting functional linkages connecting them, therefore serve Rwanda's national long-term interests on lessening the severity of impacts of climate change and sustaining critical ecosystem services both locally and downstream, while contributing to global efforts to draw down greenhouse gas concentrations.

B.1 (b). Summary of spatial biodiversity assessment

The majority of Rwanda's remaining montane forests are restricted to the national parks, which support a variety of threatened and endemic species. A few fragmented forest patches are situated outside of the National Park boundaries. Landcover change (especially for widespread smallholder agriculture), fuelwood

⁵⁴ Asefi-Najafabady, S., K. Vandecar, A. Seimon, P. Lawrence and D. Lawrence, 2018: Climate change, population and poverty: vulnerability and exposure to heat stress in East Africa. *Climatic Change*, 148, 561-573.

⁵⁵ World Bank, 2022: Rwanda Country Climate and Development Report. World Bank, Washington DC. [Link](#)

⁵⁶ Rwanda Development Board 2017 Annual Report [Link](#)

⁵⁷ Andrew, G. and Masozera, M., 2010. Payment for ecosystem services and poverty reduction in Rwanda. *Journal of sustainable development in Africa*, 12(3), pp.122-139.

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harvesting, and human-induced fires, coupled with climate change impacts, especially landslides, erosion and downstream flooding, have compromised the delivery of critical ecosystem services derived from these forests. A detailed spatial analysis of biodiversity in the Congo Nile Divide was conducted to delineate priority areas for the long-term conservation and restoration of forests, and the sustainable management of landscapes, in order to secure the ecosystem services needed to improve the resilience of vulnerable communities to climate change impacts.

The spatial biodiversity assessment for Rwanda's Congo-Nile Divide (CND) (see details in Annex 2 and Annex 2.2) is based on a rapid systematic conservation plan, using MARXAN decision-support software. The key analyses involved identifying and mapping the remaining areas of natural forest and other ecosystem types, modeling bioclimatic change and identifying climate change refugia, a Condatis landscape connectivity and bottlenecks analyses, and evaluating ecosystem threat and protection levels. The analyses thus build in landscape connectivity, climate change refugia, biodiversity values, ecosystem services and social costs.

The final outcome of the MARXAN and Condatis connectivity analyses split the CND domain into four major landscape categories with a set of priority implementation areas, each with their own place-bound project interventions. Interventions range from the restoration of natural forest, establishment and improvement of protective forest on steep slopes and along riparian areas; and to implement biodiversity-friendly agroforestry to reduce landslides, erosion and downstream flooding. Over and above these place-bound interventions are a variety of other mechanisms for promoting the sustainability of rural livelihoods and protecting montane forest in Rwanda's CND landscape.

The Systematic Conservation Plan analyses covered a range of biodiversity features, including:

- Terrestrial ecosystems, including their IUCN Red List threat status and protection level. The analysis focussed on identifying priority remaining intact areas, based on the development of a map of ecological condition.
- Protected Areas, Protected Forests and Protected Wetlands. All the identified ecosystems to be gazetted for protection are included.
- Climate change refugia based on projected changes in bioclimatic envelopes under a range of climate change scenarios.
- Identification of key landscape linkage areas and bottlenecks.
- Hydrological process areas – Rivers and Streams, including buffers.
- Hydrological process areas – Wetlands and Lakes, including buffers.
- Hydrological process areas – Areas with high rainfall.
- Landscape process areas - Steep slopes (over 55%) which are most important for minimizing erosion and landslide risk.

Individual species were not separately considered in the assessment as all key species are tightly linked to their underlying intact habitat (in particular intact forest patches as well as wetlands and riparian areas); and available distribution data for species was much broader than the ecosystem data. In selecting priority conservation areas, the SCP methodology always attempts to be spatially efficient by meeting conservation targets in as small an area as possible, while avoiding conflict with other land users, at the lowest possible cost for other sectors. The following was taken into account in the prioritization:

- Urban and dwellings, tea, roads, plantations, cultivated pasture, coffee, bamboo and agriculture.
- Areas of greatest population density.
- Areas with highly impacted land cover classes.

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Results of the MARXAN spatial analysis:

The MARXAN landscape prioritization (**Figure 13**), which builds in landscape connectivity, climate change refugia, biodiversity values, ecosystem services, and social costs (in terms of avoiding, where possible, areas with highest population density, agriculture etc), splits the landscape into four key planning categories (**Figure 14**):

- **Core Protected Area (PA) Nodes:** National Parks comprise the “Core PA Nodes” that need to be secured and well managed, which include Volcanoes, Gishwati-Mukura and Nyungwe National Parks. Priority activities include strengthening PA management and sustainability, rehabilitation and restoration of natural forests, other conservation-oriented land use activities that reduce stress on PAs and natural forests (e.g. improved wood stove efficiency to reduce pressure on natural forests) and supporting sustainable biodiversity compatible activities (e.g. improved beekeeping). These nodes also include buffer areas around the National Parks.
- **Stepping Stones:** These are priority nodes outside of the current National Parks that are critical for maintaining landscape connectivity, consisting of small, isolated patches of forest, at Dutake and Karehe-Gatuntu Protected Forests and the extensive Gishwati Pastures. These areas would be a sensible focus for some (patches of) forest restoration and protection, beekeeping and energy efficient stoves. The Gishwati Pastures are a focus for agroforestry on pastoral land to increase the coverage of native trees to secure reasonable landscape connectivity for forest species.
- **Landscape linkages:** These are key landscape linkages and knickpoints in the farming landscape that require afforestation on steep slopes and riparian areas to link the CND at a landscape scale. Compatible land use activities include agroforestry, increasing the use of native species, reforesting steep slopes, beekeeping and energy efficient stoves. **Note that the analysis is based on some level of improved connectivity via patches of protective forest and riparian strips, as well as overall improved species composition and tree coverage (within mixed agroforestry systems). It does not imply a continuous natural forest corridor; as this potential no longer exists, and its creation would not be possible in this highly used landscape without unacceptable impacts on livelihoods.**
- **Broader Farming Mosaic:** These are broader areas of moderate priority where conservation interventions can support broader sustainable landscapes and ecosystem service delivery but are likely to be beyond the scope of most project interventions except for those linked to land use planning.

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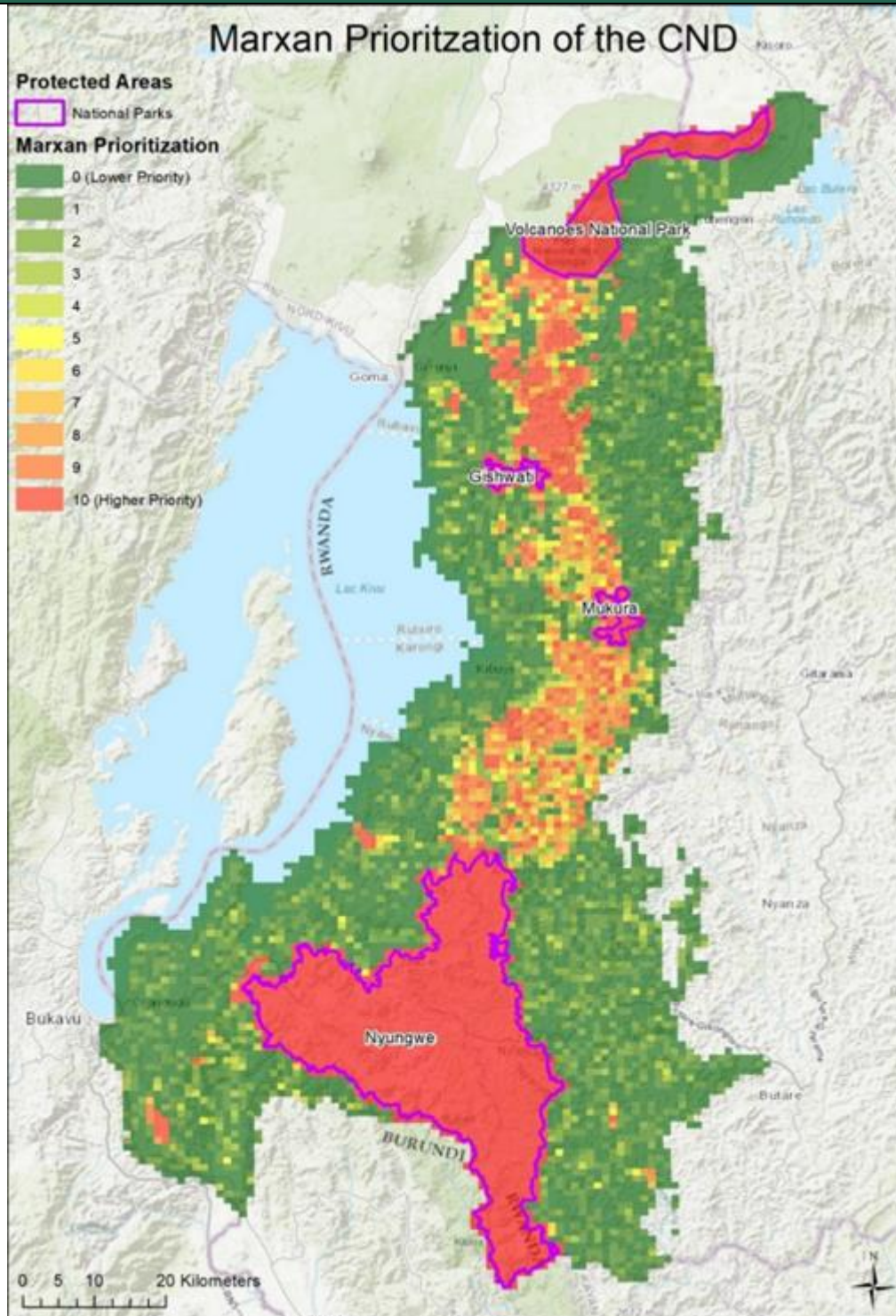


Figure 13. The MARXAN irreplaceability map for the Congo Nile Divide showing the landscape prioritization, ranging from high priority in red to low priority in green.

To aid prioritization and description of the landscape categories above, the four categories were further split into specific planning areas, referred to as “Landscape Implementation Sectors”, as shown **Figure 14** and in **Table 2** with a description of each planning category and associated ideas, required outcomes, core benefits and associated benefits in **Table 3**.

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Table 2. Summary table of landscape categories and associated implementation sectors for the CND.

Landscape Category	Landscape Implementation Sector	Area (ha)
Core PA Nodes	Volcanoes NP and "Buffer" ⁵⁸	19 487,0
	Nyungwe NP and Buffer	116 794,9
	Mukura NP and Buffer ⁵⁹	4 713,5
	Gishwati NP and Buffer ⁵⁹	4 013,7
Stepping Stones	Gishwati Pastures Stepping Stone	15 547,4
	Karehe-Gatuntu Stepping Stone	401,1
	Dutake Stepping Stone	903,0
Landscape Linkages	Nyungwe NP to Mukura NP Linkage	23 375,4
	Gishwati NP to Volcanoes NP Linkage	5 014,8
	Mukura N to Gishwati NP Linkage	7 823,7
Broader Farming Mosaic	Nyungwe to Mukura Broader Farming Mosaic	21 164,4
	Volcanoes Broader Farming Mosaic	11 694,4
	Mukura Broader Farming Mosaic	4 814,0
	Gishwati Broader Farming Mosaic	28 433,2
	Nyungwe Broader Farming Mosaic	11 437,0

Priority Areas for Implementation Activities:

Specific priority areas for a range of interventions were spatially identified, including:

- Natural forest rehabilitation and restoration with the National Parks (and their buffers) and in the Stepping Stones.
- Protective forests for steep slopes and riparian areas, particularly in the Landscape Linkages.
- Agroforestry interventions in key highland Landscape Linkages.
- Specific agroforestry priorities on pastoral land (silvo-pastoral practices).
- Indigenous shade trees for tea and coffee plantations.
- Beekeeping in National Park buffers and Stepping Stone buffers.

⁵⁸Note that buffers refer to both the specific legally designated buffers around Nyungwe and Gishwati-Makura NPs, as well as broader buffer areas adjacent to all these NPs, as well as Volcanoes NP.

⁵⁹In order to allow for specific landscape description, we separately refer to Gishwati NP and Makura NP where necessary, even though these areas are managed as Gishwati-Makura NP.

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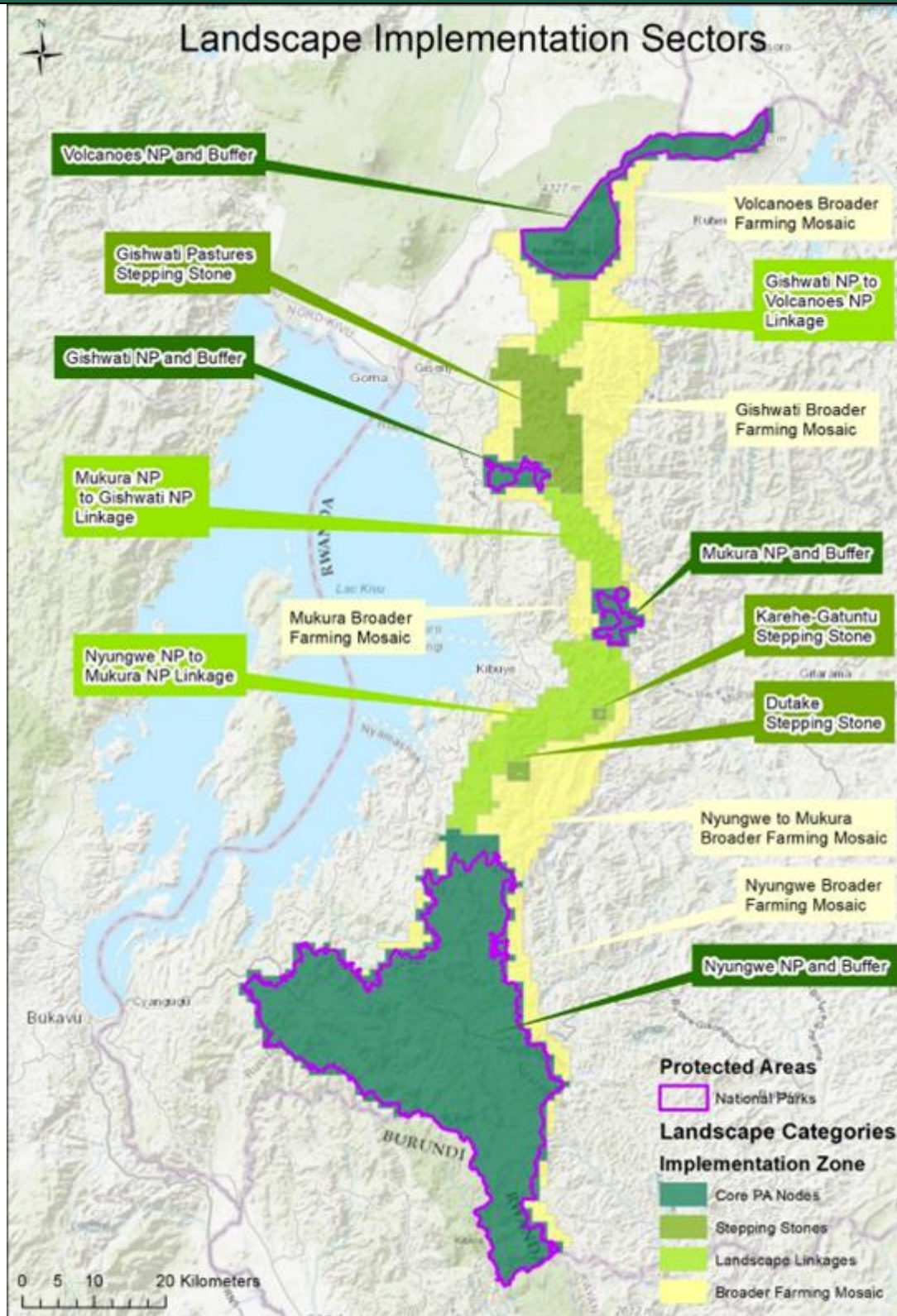


Figure 14. The four major landscape categories (Core PA Nodes, Stepping Stones, Landscape Linkages and the Broader Farming Mosaic) were split into specific areas to aid prioritization and description.

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Table 3. A description of the four key landscape planning categories and associated ideas, required outcomes, core benefits and associated benefits.

Landscape Component	Core PA Nodes The National Parks	Stepping Stones Priority nodes outside of current Pas	Landscape linkages Key landscape linkages and knickpoints in the farming landscape	Broader Farming Mosaic The broader landscape mosaic	Institutional issues
Description	The current core national parks and protected forests.	Critical pieces of biodiversity outside of the PAs required for landscape connectivity, maintenance of biodiversity and delivery of ecosystem services.	The parts of landscape within identified key corridors, where functional connectivity and ability to deliver ecosystem services needs to be urgently maintained or improved.	Remaining farmland areas of the CND.	The non-geographic specific elements of the CND system
Key ideas	Protect and manage for climate resilience.	Restore and protect to ensure landscape connectivity and ecosystem service delivery.	Functional linked farming landscapes delivering ecosystem services.	Diverse climate resilient farmland delivering ecosystem services.	Strong, well-capacitated and equitable environmental governance and land use planning.
Required Outcome	PAs effectively protect and manage natural forests improving resilience to climate change impacts and risks. Natural forests protected, connected, more resilient to climate change impacts and risks.	Critical landscape nodes / stepping-stones are secured and where necessary restored to a natural state.	Priority portions of the farming landscape are specifically managed to improve overall connectivity and ecosystem service delivery.	Sustainably managed farmland landscape is more biodiverse, supports delivery of ecosystem services and is resilient to climate risk.	Government and civil society are well capacitated to ensure robust landscape planning that supports climate resilience.
Core benefit	Maintain globally significant, species-rich natural forests. Core areas secure best possible source and/or refuge areas for species under climate change.	Biodiversity value of critical landscape nodes is maintained. Landscape connectivity supported through retention of key stepping-stones for species movement across the landscape.	Improved connectivity of the landscape ensures long term climate resilience. Value of core PAs and priority nodes is retained (i.e. the inevitable degradation of sites due to isolation is avoided).	Generally improved farmland management ensures rural sustainability and supports livelihoods.	Integrated land use plans, with community participation and spatial planning tools/ monitoring
Associated benefits	Improved delivery of ecosystem services (especially water) and support of rural economies and livelihoods.	Improved delivery of ecosystem services (especially water) and support of rural economies and livelihoods.	Improved delivery of ecosystem services (especially water) and support of rural economies and livelihoods.	Improved delivery of ecosystem services (especially water) and support of rural economies and livelihoods.	Cross sectoral planning and management.

This study was a rapid assessment to support project proposal development and does not replace a full conservation planning process. There are significant additional steps which are required to develop a product that is useful for land use planning during project implementation. These changes include a robust stakeholder engagement process, at a national, district and local scale; incorporation of issues relating to

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land use rights, both of landowners and farm tenants; incorporation of issues relating to social safeguards, especially for marginalized groups; inclusion of issues related to planning processes and strategies, at a national and local scale; finer scale planning (ideally at a 1:50 000 scale); and improved biodiversity data, including revised data on forest degradation, validation of the ecological condition map, and specific species data where possible.

B.2 (a). Theory of change narrative and diagram

The theory of change for this Project is founded on several key principles. Central to all, the CND represents a complex landscape – a geospatial mosaic of forest and farming patches, each dependent on the other. These coupled, biological and socio-economic systems are under great strain due to intense land pressure, which has resulted in a negative cycle of natural resource degradation that undermines the well-being of both natural and socio-economic systems. The advent of climate change is a forceful driver that intensifies this cycle of degradation, with the resultant loss of key ecosystem services that importantly link forests and people. This cycle must be disrupted in order to chart a path toward climate resilience for natural and human systems. Ecological connectivity must be established within this mosaic of land uses to promote landscape heterogeneity to help build resilience to climate change of forest ecosystems and people who depend on them. The approach to be taken by this Project is therefore based on principles of ecosystem-based adaptation: that healthy and resilient ecosystems form an essential foundation for adaptation to climate challenges⁶⁰. Climate-aware management of both natural and human-dominated ecosystems – interacting and integrated at a landscape level – can break the cycle of degradation and instead lead to a positive sequence of mutually reinforcing, climate-smart, sustainable, and resilient benefits..

In sum, the goal statement of the Project is:

IF Rwanda's CND landscape of native forests and neighboring farmlands is effectively managed for climate resilience

THEN the CND will comprise an interconnected and interdependent set of globally significant, species-rich natural forests in a matrix of sustainable farmlands, acting as buffers against damage from extreme weather events: assuring landscape integrity, stability and adaptability, providing vital ecosystem services and improving livelihoods for vulnerable people and the nation at large and contributing importantly to the conservation of biodiversity, the national economy, and mitigation of GHGs

BECAUSE building the capacity and management of integrated, climate-resilient forest and farming systems will ensure natural forest perpetuity and break and reverse the climate change-induced intensifying cycles of drought, flooding, and landslides that lead to degradation of livelihoods and resultant increased pressure on forest resources.

The status and future of the CND's forests and people are tightly intertwined: both heavily dependent on the natural resources and processes that provide a foundation for their ability to thrive. They represent ecological and socio-economic systems that are importantly coupled in a complex landscape, linked in particular by the ecosystem services that flow throughout these systems (hydrological and microclimate regulation, soil conservation, and nature-based tourism). At present these services, and the consequent relationships between forests and people – especially those who are most vulnerable in Rwandan society – are deeply

⁶⁰ Scarano, F.R. et al, 2017. *Ecosystem-based adaptation to climate change: concept, sustainability and a role for conservation science*, Perspectives in Ecology & Conservation 15(2), 65-73.

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strained⁶¹. Forests have been reduced in extent, fragmented, and degraded by fire⁶², overuse of resources⁶³, and introduction of exotic species⁶⁴. The sharply sloped rural lands of the CND have been farmed more and more extensively and intensively on steeper and steeper slopes^{65,66} and throughout riverine bottomlands, further reducing tree cover outside forest blocks: thereby further compromising the ability of the land to retain soils⁶⁷, avoid flooding⁶⁸, provide consistent and healthy water⁴⁴, ameliorate microclimate⁴⁹, store carbon⁶⁹, and remain productive. The coupling of these stressed systems has led to a negative feedback cycle of insecurity and constrained the well-being of farmers. Having little access to alternative sources of livelihoods or capital to enable change⁷⁰, smallholder farmers have consequently increased pressures on natural forests for land, fuel, and other forest resources⁴⁶. Such pressures in turn cause further forest loss and degradation and decline in biodiversity – reducing their ability to provide essential ecosystem services to people living in the landscape and across the nation, including tourism revenues and employment that are of national import. Thus, these systems are caught in a negative cycle that has proven difficult to break, despite significant sector-specific efforts. One such pressure – demand for fuelwood – is quite substantial given that a recent Government of Rwanda (GoR) census found that 77% of Rwandan households use wood as their primary fuel for cooking⁷¹.

With the advent of climate change, these already strained systems are showing clear signs of breakdown. Increased variability in the timing and intensity of rains along with rising temperatures are causing increased landslides⁷² and soil erosion⁵⁰, increased desiccation and drought⁵⁰, and unpredictable timing for agricultural plantings and harvest⁷³. This is reducing farm productivity and security, driving even greater pressures on the remaining forested estate, thereby compromising further the essential ecosystem services that are necessary to such well-being. At a larger scale, climate-induced forest degradation and decline of biodiversity in the CND compromise national-level benefits of water provision and quality¹, nature-based tourism⁷⁴, the supply of hydroelectricity,^{7,75} and carbon storage¹⁰ [see diagram A below].

61 Bagstad, K.J., 2019. *Towards ecosystem accounts for Rwanda: Tracking 25 years of change in flows and potential supply of ecosystem services*. British Ecological Society.

62 Masozera, A.B., Mulindahabi, F., 2007. *Post-Fire Regeneration in Nyungwe National Park*, Rwanda. Wildlife Conservation Society.

63 Republic of Rwanda, 2011. *Green Growth and Climate Resilience: National Strategy for Climate Change and Low Carbon Development*.

64 USAID, 2019. *Rwanda Tropical Forests and Biodiversity Analysis*.

65 Camberlin, P., 2018. *Climate of Eastern Africa*. In Oxford Research Encyclopedia of Climate Science.

66 Seimon, A., 2022. *An Overview of Climate Change and its Impacts along the Congo-Nile Divide in Rwanda*.

67 World Bank, 2019. *Rwanda Systematic Country Diagnostic*.

68 Karamage, F., et al., 2017. *Modeling Rainfall-Runoff Response to Land Use and Land Cover Change in Rwanda (1990–2016)*. *Water* 2017, 9(2), 14.

69 Mugabowindekwe, M., Brandt, M., Chave, J. et al. Nation-wide mapping of tree-level aboveground carbon stocks in Rwanda. *Nat. Clim. Chang.* 13, 91–97 (2023). <https://doi.org/10.1038/s41558-022-01544-w>

70 Clay N., King B., 2019. *Smallholders' uneven capacities to adapt to climate change amid Africa's 'green revolution': Case study of Rwanda's crop intensification program*. World Dev.

71 Government of Rwanda, 2021. *Rwanda Household Survey 2019/2020*. National Institute of Statistics of Rwanda.

72 Uwihirwe, J., et al., 2020. *Landslide precipitation thresholds in Rwanda*. *Landslides*.

73 World Bank; CIAT. 2015. *Climate-Smart Agriculture in Rwanda*. CSA Country Profiles for Africa, Asia, and Latin America and the Caribbean Series. Washington D.C.: The World Bank Group.

74 Rwanda Environment Management Authority (REMA), 2015. *Rwanda State of Environment and Outlook Report*.

75 Rwanda Environment Management Authority (REMA), 2009. *Rwanda State of Environment and Outlook Report*, Chapter 8: Energy Resources.

(b) PROJECT/PROGRAMME INFORMATION

Diagram A. Climate impacts & risks intensify negative feedback loop for people & forests

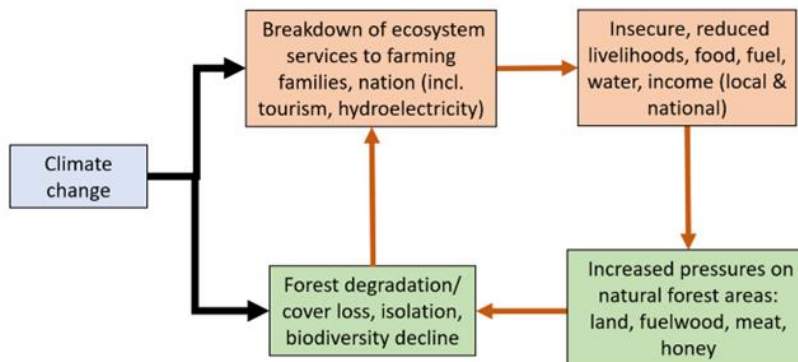
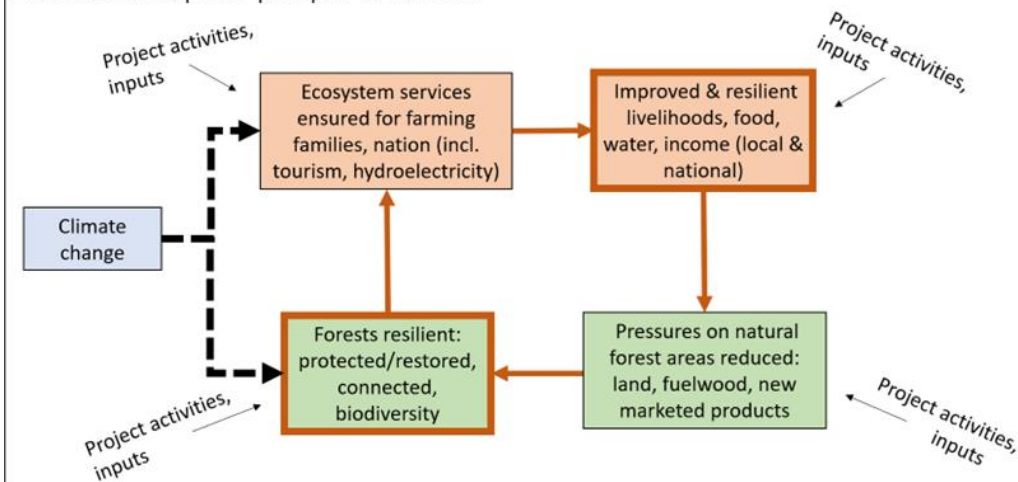


Diagram B. Adaptation to climate impacts & risks promotes positive feedback loop for people & forests



This context calls for new mechanisms of land-use planning and management that are based on consideration of the CND as a landscape, consisting of an inter-dependent mosaic of forests and farms, the management of each determining the health of the whole. Focusing on this larger scale, with coordination across geographic and sectoral boundaries, while mainstreaming climate risks into management planning will be key – and the foundation for a paradigm shift. Use of this framework to design and apply practical, precedent-setting Project actions in-forest and on-farm is expected to shift mindsets and management systems. This, in combination with establishing durable sources of financing, is thereby intended to shift the natural and human ecosystems themselves toward sustainable climate resilience. [see diagram B above, and the ToC (Theory of Change) diagram below.

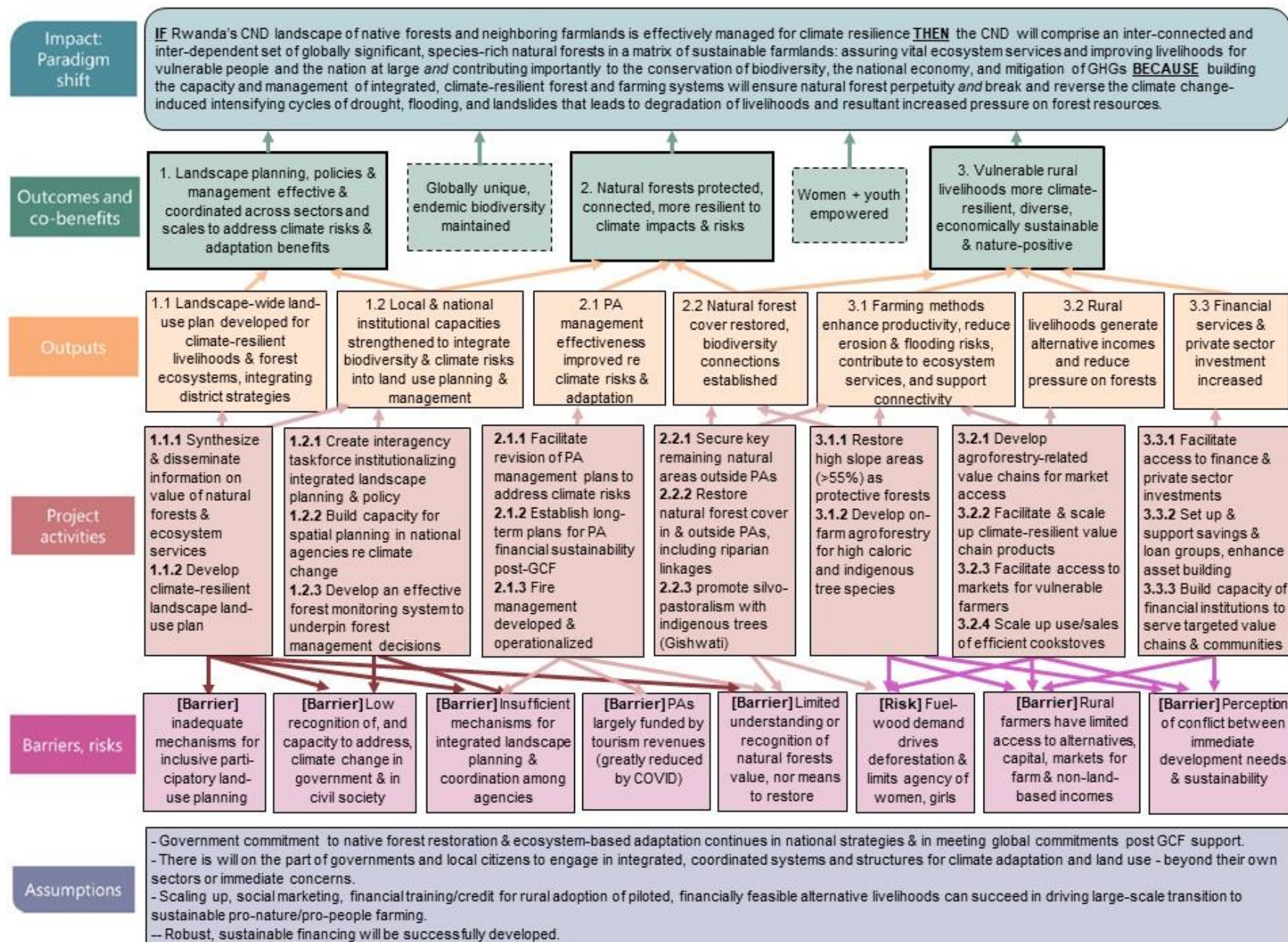


Figure SEQ Figure * ARABIC 15. Theory of change

Outcome 1. Landscape planning, policies and management are effective and coordinated across sectors and scales to address climate risks and adaptation benefits. Major barriers currently stand in the way of moving in this direction, as outlined in the ToC diagram. ***Awareness, understanding, attention, and action regarding climate risks to natural resources are currently insufficient.*** Planning and adaptation to climate impacts are constrained by a lack of information, site-focused modeling, spatial analysis, and real-time monitoring. Where some capacity exists, it ***operates within sector initiatives rather than across them.*** It also proceeds largely within governmental programs, with ***mechanisms for inclusive land-use planning inadequate***, not fully engaging rural peoples – particularly women and youth. Because of these barriers, coordination of landscape planning and management is weak across governmental and non-governmental sectors (agriculture, forestry, national parks, economic development), scales (local, district, national), and actors (women, youth, smallholder farmers, commercial businesses, as well as governmental agencies). Activities outlined in **Component 1**, aimed at integrated landscape land-use planning and the capacity for effective management, are intended to help ensure ecosystem services persist, and thereby align the ties between forests and people as they are faced with climate challenges. Activities under this component will provide the government and public with up-to-date information on climate adaptation (1.1.1), build agency capacities for incorporation into land-use planning – particularly spatial analysis and forest monitoring (1.2.2, 1.2.3), establish mechanisms for integrated landscape planning – including public participation and operations of an inter-agency task force (1.1.2, 1.2.1), and facilitate the creation of a climate-smart, integrated CND land-use plan (1.1.2).

Outcome 2. Natural forests are protected, connected, more resilient to climate impacts and risks. Particular to the CND, another key barrier is that ***natural forests are as yet undervalued and under-resourced*** despite the fact that they play vital roles in stabilizing natural and human CND ecosystems^{76,77}. The national parks of the CND (Volcanoes, Nyungwe, and Gishwati-Mukura) are globally significant regarding their outstanding species richness, levels of endemism, and presence of endangered and rare species⁷⁸. Although these are well-recognized for their tourism potential, and strong governmental policy and practice exists to protect them, barriers to their full valuation remains: their ***ecosystem services are yet to be fully recognized*** or financially supported, and their ***support is vulnerable to global tourism trends*** (with COVID-reduced tourism as a current example). As climate change considerations enter into landscape management, the values of these natural forests for fog interception, precipitation infiltration, soil retention, drought mitigation, and hydrological balance are only now becoming more prominent. In order to provide such services into the future, Nyungwe and Gishwati-Mukura NPs are in need of restoration and connection^{79,80}. Yet the understanding and capacity for recovering their richness and function is nascent at best, particularly in light of climate change. Activities outlined in **Component 2** are therefore designed with forest sustainability at the core. To build climate considerations into forest planning and management, the Project will facilitate revision of park management plans (2.1.1) that enhance dimensions of fire prevention and management (2.1.3), and will support in-park forest restoration (2.2.2). Establishment of restored forest at park boundaries (2.2.2), riparian linkages outside the parks (2.2.2), and forested “stepping stones” that

⁷⁶ Gatwaza O.C., Wang X., 2021 *Mapping of biodiversity hubs and key ecosystem services as a tool for shaping optimal areas for conservation*. PLOS ONE 16(8): e0253151

⁷⁷ Andrew, G., Masozera, M., 2010. *Payment for Ecosystem Services and Poverty Reduction in Rwanda*. Journal of Sustainable Development in Africa (V12, No.3).

⁷⁸ This area is also significant for global biodiversity – with the Albertine Rift being one of the most biodiverse regions within Africa, including conservative estimates of at least 980 endemic species, 15 Critically Endangered species, 34 Endangered species, and 99 Vulnerable species. Source: Plumptre, A.J., et al. 2016. *Conservation Action Plan for the Albertine Rift*. Wildlife Conservation Society.

⁷⁹ Nyandwi, E., Mukashema, A., 2011. *Excessive Deforestation of Gishwati Mountainous Forest & Biodiversity Changes*. Participatory Geographic Information Systems (P-GIS) for natural resource management and food security in Africa.

⁸⁰ Ordway, E., 2015. *Political shifts and changing forests: Effects of armed conflict on forest conservation in Rwanda*. Global Ecology and Conservation.

allow for forest species to move between parks (2.2.1, 2.2.3) is intended to enhance biodiversity and forest functions across the CND landscape. Fundamental to forest sustainability as well is sustainable long-term financing, which will be addressed by providing technical support from the Project to RBD and REMA to assess various financial instruments (conservation and/or water funds, bonds, revenue streams) to be employed for revenue generation post-GCF (2.1.2).

Outcome 3. Vulnerable rural livelihoods are more climate-resilient, diverse, economically sustainable and nature-positive. The sustainability and climate resilience of the agricultural matrix of this landscape is challenged by additional, multiple barriers. Rural agricultural practices and livelihoods are constrained by the very small size of landholdings (avg. 0.3 ha⁸¹), creating intense pressure to utilize all arable land for food and near-term income-generating products⁵⁶, thus creating **reluctance to incorporate trees or other soil conservation measures on the land**⁸². The resultant **lack of fuelwood in turn continues to drive deforestation of natural forests**¹², **and occupies important time for women and youth** who could instead be engaged in productive and empowering roles. Overall, local communities and extension services⁸³ have **limited skills, information and knowledge to design and implement farming methods** that would enhance their capacity to mitigate the impacts from climate-induced soil erosion, flooding and droughts. Farmers **also lack access to financial know-how**⁵⁶ **and capital, proven alternative livelihoods, and markets** that could incentivize and enable change¹². They are therefore caught in a cycle of immediate need that – without alternatives – forecloses options that could recover and sustain benefits from improved ecosystem services, other forest functions, and new livelihoods. **Component 3** aims to redress these conditions via activities that will demonstrate, incentivize, facilitate, and support actions that maintain and recover ecosystem services, benefit from forest functions and provide alternative on- and off-farm based incomes that reduce pressure on forests. To these ends, the Project will support reafforestation on steep slopes (>55%), on both public and smallholder private lands (Activity 3.1.1). It will develop farm-based agroforestry for climate-resilient species that produce fuelwood and/or market products, while conserving and enriching soils (3.1.2, 3.2.1). The Project will complement land-based production with a scaling-up of alternative value-chain products for smallholders that are nature-positive, including but not limited to tourism-based income, beekeeping, and improved cooking stoves (3.2.2, 3.2.4). Central to these activities will be facilitation of access to market chains (3.2.3) and financial mechanisms for establishing and maintaining businesses (3.3.1, 3.3.2, 3.3.3). Given that these activities are designed at the core to be profitable and promote financial sustainability, it is intended that they be replicable, and sustained once GCF funding expires.

Assumptions. Success in achieving Project outcomes will depend on important assumptions being met, among them that governmental and non-governmental actors will be willing to work in collaboration across sectoral and geographic boundaries; rural adoption of alternative farming and off-farm revenue generation can be scaled up to drive a landscape-level transition to sustainability; financing and revenue-generating mechanisms will be established that assure continuance of initiated actions; and governmental commitment to natural forest restoration and ecosystem-based adaptation continues in national strategies and in meeting global commitments.

The pathways summarized in this ToC are expected to contribute to the overall goal and outcomes of the Project: **natural forests** and **vulnerable communities** will be more resilient and sustainable despite climate impacts, with effective, **coordinated management** across the CND landscape. In doing so, important co-benefits will accrue. Intentional, inclusive engagement in management coordination will **empower women and youth** in environmental thinking and decision making. Forest conservation and restoration, along with agroforestry, will contribute to greater **carbon sequestration and storage**, while fire management and greater cooking efficiency will **reduce GHG emissions**. The globally unique, endemic **biodiversity** of

81 USAID, 2017. LandLinks Country Profile: Rwanda. <https://www.land-links.org/country-profile/rwanda/>

82 Rwanda Water and Forestry Authority, 2017. *Forest Investment Program for Rwanda*.

83 Foster & Graham, 2014. *Connectivity and the Tea Sector in Rwanda: Value Chains and Networks of Connectivity-Based Enterprises in Rwanda*.

Rwanda's forests, from great apes to rare orchids, will recover on the basis of healthy, more connected and resilient forest habitat.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1. Landscape planning, policies & management effective & coordinated across sectors and scales to address climate risks & adaptation benefits	☐	☐	☐	☒	☐	☐	☐	☒
Outcome 2. Natural forests protected, connected, more resilient to climate impacts & risks	☐	☐	☐	☐	☐	☐	☐	☒
Outcome 3. Vulnerable rural livelihoods more climate-resilient, diverse, economically sustainable & nature-positive	☐	☐	☐	☐	☒	☒	☐	☐

Co-benefits number	Co-benefits					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1: Globally unique, endemic biodiversity maintained	☒	☐	☐	☐	☐	☐
Co-benefit 2: Women and Youth empowered	☐	☐	☐	☒	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

Rwanda's CND region is a complex, naturally integrated landscape. Its diverse high elevation forests capture and store water resources, releasing them in a sustained manner throughout the year to surrounding steep hillside and bottomland farms and plantations, before feeding into the national network of wetlands, rivers, and lakes. The Rwanda portion of the CND runs from the Virunga Mountains and Volcanoes National Park (VNP) on the border with Uganda in the north, down through Gishwati Mukura National Park (GMNP), to the south end of Lake Kivu and Nyungwe National Park (NNP) on the southern border with Burundi. These three

national parks contain the country's only remaining montane forests. The boundaries of the CND landscape are defined as all areas greater than 1900m in elevation.

Administratively, the CND overlaps three Provinces (Western, Southern, Northern) and ten Districts (Karongi, Musanze, Ngororero, Nyabihu, Nyamagabe, Nyamasheke, Nyaruguru, Rubavu, Rusizi, Rutsiro).

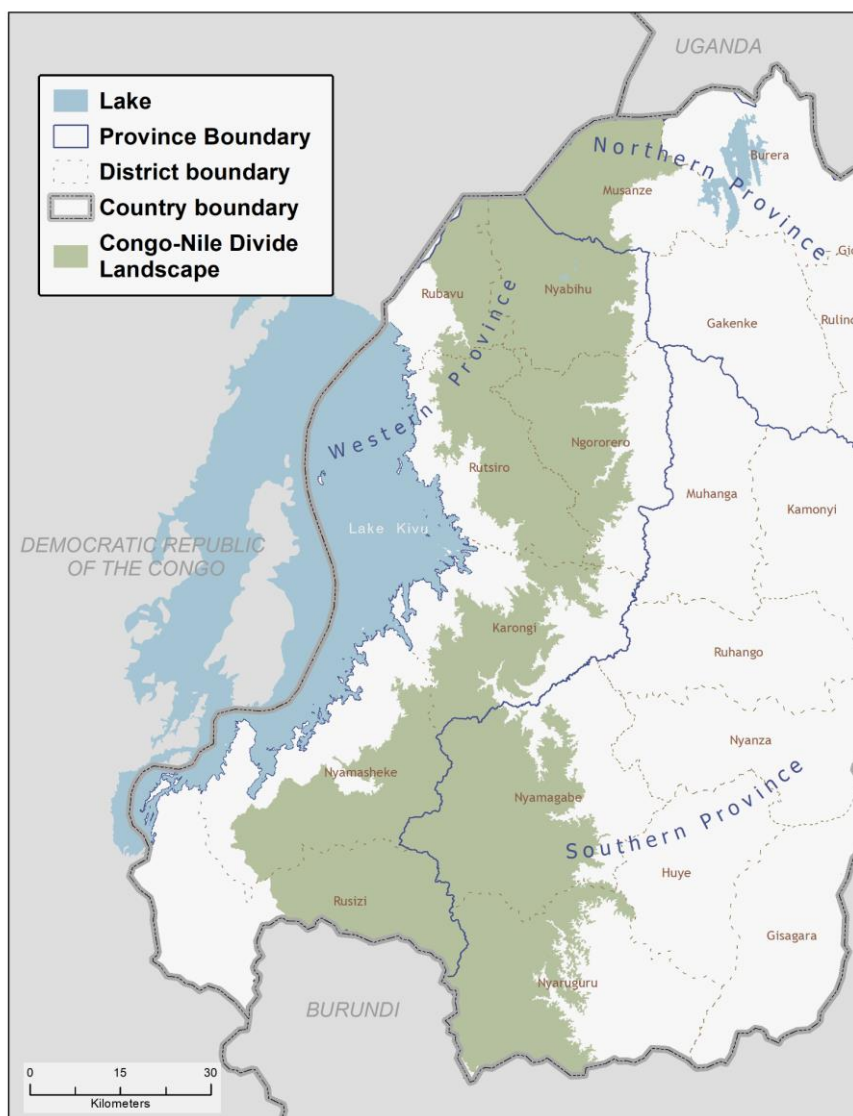


Figure 16. Map of CND Landscape

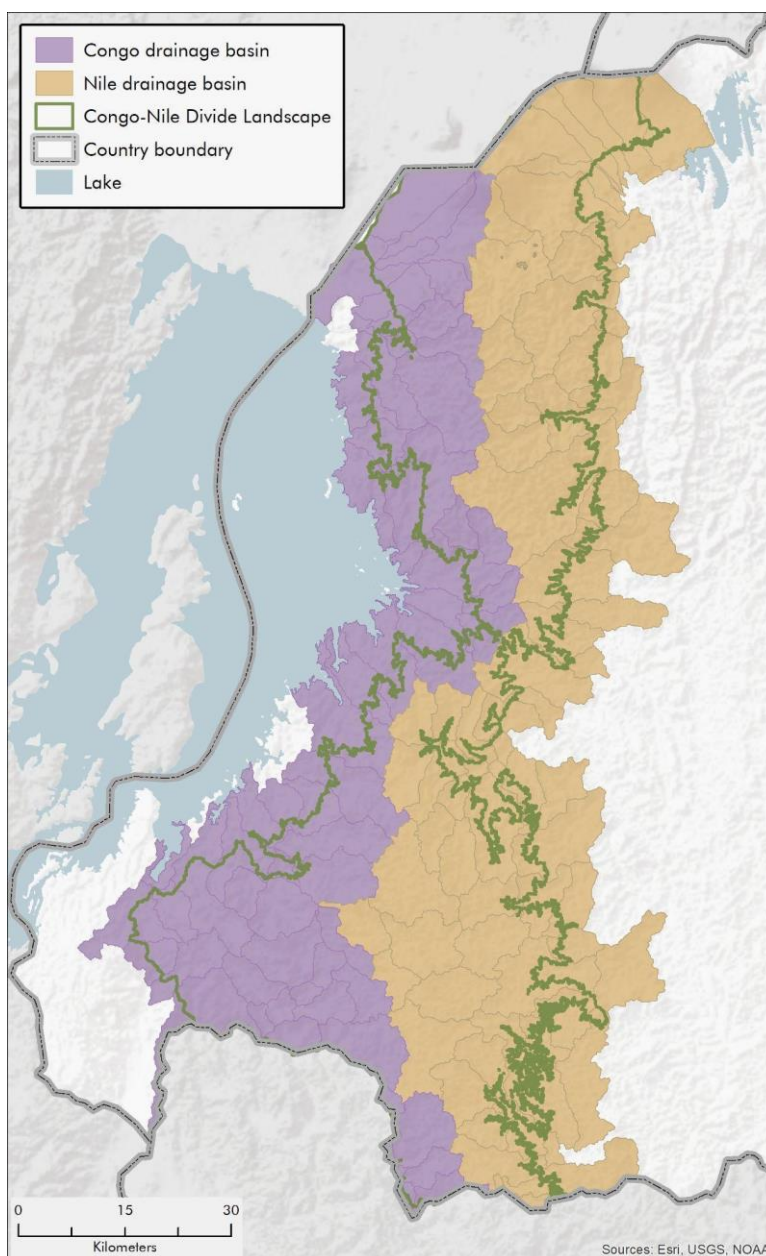


Figure 17. Map of CND Landscape drainage basins

To address the present and intensifying threats from climate change requires that this region also be managed as an integrated and ecologically connected system. This need is recognized in several key policies and plans enacted in recent years by the Rwandan government. However, implementation and integration⁸⁴ of these have lagged, their decentralized application on the ground is largely unrealized, and climate-related technical capacity remains limited in key institutions at all levels. This Project is designed to increase the climate resilience of the CND. Its intended outcomes are:

- The development and implementation of effective landscape planning, policies, and management, coordinated across sectors and scales, to address climate risks and adaptation benefits;

- Improved protection, restoration, and connection of existing natural forests, with enhanced resilience to climate impacts and risks; and
- The transformation of vulnerable rural livelihoods to be more climate-resilient, diverse, economically sustainable, and nature positive.

The Project has three main components directly targeting vulnerabilities within climate impact chains. Detailed analyses of the underlying rationale are provided in the Feasibility Study (Annex 2). Please see section 4.1 of that study, for a list and description of relevant existing and previous projects in the region.

Component 1: Mainstreaming Climate Adaptation into Integrated Land Use Planning

There is an urgent need for more effective collaboration among Rwandan government agencies, civil society, private sector, and diverse partners to balance difficult-yet-inevitable land use trade-offs that will have profound impacts on the climate resilience of both natural systems and people. The GoR recognizes that a piecemeal approach to decision making has exacerbated competition over scarce land resources and reduced the adaptive capacity of both ecosystems and rural populations⁸⁵. Rwanda's Baseline Climate Change Vulnerability Index (2015) recommends establishing a multi-ministry technical climate resilience coordinating committee, including NGOs⁸⁶. In 2017 the MoE developed the Strategic Programme for Climate Resilience (SPCR), to focus on three cross-cutting priorities to achieve climate change resilience: (a) technical capacity building and strengthening institutional coordination; (b) integrated land use and spatial planning; and (c) climate services and disaster risk management⁸⁷. The SPCR is a key step towards ensuring inter-ministerial and multi-sectoral collaboration and integration of climate resilience considerations into development plans and actions. However, there remains a need to comprehensively implement the strategy and mainstream climate change adaptation at the sub-national level, engaging a diverse set of actors and sectors and empowering women to provide input into decision making at district and community levels.

Output 1.1: Landscape-wide land-use plan developed for climate-resilient livelihoods and forest ecosystems, integrating district strategies.

The GoR completed a National Land Use and Development Master Plan (NLUMP 2020) to guide land use planning and regulate permitting processes. However, effective and strategic land use has yet to be applied at a local and district level, and the conversion of plantation and riparian forests to agriculture continues and development projects are approved without sufficient knowledge of climate change risks or the value of existing forests for ecosystem-based adaptation (EbA)⁸⁸. The Project will add climate-change resilience and adaptation content to central government efforts such as NLUMP and the SPCR process by acquiring, synthesizing, and disseminating key information on climate trends, the importance of remaining natural forests, and the value of ecosystem services. This information will also be shared at district and community levels through various outreach networks and workshops, with particular attention to reaching disadvantaged women, youth and vulnerable groups.

Building on the information base outlined above, the Project will work with the National Land Authority (NLA) to establish a working group, with participation from relevant ministries, agencies, and districts, including representatives of women, youth, and vulnerable groups interests, to develop a climate-resilient land-use

⁸⁵ National Land Use Management Plan.

⁸⁶ United Nations, 2015. Economic Commission for Africa; Rwanda Environment Management Authority. Baseline climate change vulnerability index for Rwanda. Kigali. Rwanda Environment Management Authority.

⁸⁷ Rwanda's Green Fund (FONERWA), 2017. Strategic Programme for Climate Resilience (SPCR) Rwanda - December 2017. [Link](#).

⁸⁸ Scarano, F.R. et al, 2017. *Ecosystem-based adaptation to climate change: concept, sustainability and a role for conservation science*, Perspectives in Ecology & Conservation 15(2), 65-73.

plan for the CND landscape. At the national level each relevant ministry and government agency will select and appoint a working group member while at the district level, participants will be selected from the Joint Action Development Forum (JADF). JADF is a multi-stakeholder platform meant to facilitate and promote the full participation of citizens in the decentralized and participatory governance and improve service provision processes with representatives from the public sector, private sector and civil society. The following stakeholders will form the basis of the collaboration: a) the Ministry of Environment represented by four of its agencies: the Rwanda Environment Management Authority (REMA); the National Land Authority (NLA), Rwanda Water Board (RBA) and the Rwanda Forest Authority (RFA); b) the Ministry of Agriculture, including the Rwanda Agriculture Board (RAB); c) the National Agriculture Export Board (NAEB) d) Ministry of Local Government; e) Rwanda Development Board (RDB); f) Gender Monitoring Office (GMO); g) Districts Decentralized Structures – the District Administrative Units, which supervise several technical and administrative activities; h) civil society, international organisations, academia and community based organizations. This list will be discussed during the inception period and expanded as necessary. From this, a spatial framework for district level land use planning will be developed that accounts for current and future climate risks, reconciles cross-sectoral land-use conflicts, and guides development decisions. All the activities and sub activities under this output will be executed by the Rwanda Forest Authority in collaboration with National Land Authority, and districts.

Activity 1.1.1. Synthesize and disseminate information on value of natural forests and ecosystem services

This activity will involve conducting a stakeholder mapping & consultation exercise in order to better understand the use of climate adaptation data in existing sectoral planning processes, and identify cross-sectoral linkages or conflicts. These results, along with a comprehensive literature review, will inform the design of high-level information packages that will synthesize information on climate risks for various sectors (e.g. agriculture, forestry) and highlight the value of forests for increasing resilience of local communities. An information dissemination program (e.g., presentations, workshops, and newsletters) will be implemented to share this knowledge among key ministries as well as government, non-government and community organisations. The Project will use JADF as a platform to disseminate information to key stakeholders. Particular attention will be paid to ensure women, vulnerable and youth groups from Ubudehe categories c, d and e are represented. In addition, the program will use mass media in the form of a radio program and complementary community action campaigns to improve community, women, youth, other vulnerable groups knowledge on climate information, climate risks and climate adaptation options. This will stimulate discussion on issues pertaining to climate risks and other social issues locally and support engagement of each in responding effectively to climate adaptation and resilience options.

Cross-sectoral tradeoffs and climate adaptation solutions with cross-sectoral implications will also be highlighted, in order to begin promoting the benefits of integrated land-use planning. To complement the high-level information dissemination program, a suite of technical training materials will also be developed for delivery to district technical staff. These materials will provide training on climate risks & the need to incorporate climate adaptation into land-use planning, with a focus on the technical skills required for climate-sensitive planning (e.g. sourcing climate data, mapping ecosystem services).

Sub-activities will include:

1.1.1.1 Map the sectors involved in land-use planning in the CND and review how forest ecosystem services and climate resilience are incorporated into each sector's planning process

1.1.1.2 Conduct comprehensive literature review & stakeholder consultation to collect & synthesize information on climate risks for various sectors (e.g. agriculture, forestry), the value of forests for increasing resilience of local communities, and highlight adaptation solutions with cross-sectoral implications.

1.1.1.3 Host workshops & presentations with key ministries, government organizations, NGOs and community organizations to disseminate high-level knowledge on value of forests for increasing resilience of local communities.

1.1.1.4 Develop guidelines for integrating climate risk into land use planning and cross sectoral planning

1.1.1.5 Develop outreach materials on climate and related risks in the CND and the value of forest ecosystems for increasing resilience

1.1.1.6 Implement outreach program tailored to different stakeholders (local government, civil society, communities, women, youth) to enhance capacities for land-use planning, funding mobilization, and delivery of climate adaptation actions

1.1.1.7 Conduct climate literacy seminars for local government and civil society organizations, aimed at increasing women and youth participation.

1.1.1.8 Provide financial and logistical support to trained organizations in grassroots mobilization to increase women and youth participation in climate adaptation planning.

1.1.1.9 Introduce social safeguards at a high level at each meeting, including the GRM, FPIC, and Access Restrictions Mitigations as safeguards to be in place for work with local communities.

Activity 1.1.2. Develop climate-resilient landscape land-use plan

This activity aims to ensure that climate change adaptation requirements are fully integrated into planning processes at local, district and national scales. This will involve i) participatory scenario analysis to understand the problems local communities and district officials face in terms of natural disasters (flooding, soil erosion, forest loss/degradation) and climate change, as well as the future land use plans of the districts, including an assessment of the national consequences of these problems in terms of ecosystem service delivery and climate change resilience and adaptation capacity; ii) risk assessment to estimate the impacts of the planned land use changes, as well as climate change, on flood risk, soil erosion, forest loss/degradation and the well-being of local people, especially any disproportionate impacts on women, youth and vulnerable groups; iii) climate change adaptation and mitigation measures development to decide possible climate actions for both adaptation and mitigation and prioritize these actions according to their feasibility and urgency in consultation with local communities, district leaders, and other stakeholders; and iv) land use plan development based on the recommendations from the previous 3 steps. A comprehensive consultation and feedback process will be undertaken to incorporate views of stakeholders at all levels (e.g. civil society; men, women, youth, and historically marginalized people in local communities; district & national government), upon which the plan will be finalized and approved by the Government of Rwanda.

Sub activities will include:

1.1.2.1 Engage district officials, Joint Action Development Forums (JADF), and community members in the CND (especially women) in reviewing and interpreting the National Land Use and Development Management Plan (NLUMP), to ensure the plan accounts for current and future climate and related risks, while building local support for climate sensitive planning

1.1.2.2 Conduct participatory land-use planning process in communities from village to district level to support integrated climate resilient land use planning

1.1.2.3 Develop Integrated Land-use Plan that supports Resilient Livelihoods and Ecosystems in the CND, ensuring alignment with National Land Use and Development Master Plan and ensuring the CND plan guides the district plan

1.1.2.4 Develop and roll out a series of trainings (virtual sessions and online modules) on gender sensitivity and mainstreaming women and youth into planning

1.1.2.5 Assess specific climate impacts on historically marginalized and Category c,d,e populations through a participatory NR process to ensure the Project components address their needs for adaptation

Output 1.2 Local and national institutional capacities strengthened to integrate biodiversity and climate risks into land use planning and management

Forest data in Rwanda has historically been generated in an ad-hoc fashion⁸⁹ through the support of external agencies. This precludes comprehensive consideration of forest and biodiversity data in land-use planning processes. As Rwanda moves to achieve the goals outlined in the Green Growth and Climate Resilience Strategy and the National Land-Use Development Master Plan, building governmental capacity in generating and utilizing spatial data to inform decision making is essential. The activities under this output will involve recruiting experts in mapping, remote sensing and ecosystem service modeling, in order to establish a formal spatial planning unit within government. These experts will also lead a formal information dissemination and capacity building program, in order to i) promote the importance of considering forest & climate resilience into land-use planning, and ii) deliver detailed training to technical staff on forest mapping, climate and ecosystem service modeling. The Project will also increase the understanding of the importance of forest landscape restoration in securing ecosystem services for local economic development and resilient livelihoods. It will ensure that the public, decision-makers and other stakeholders in the CND landscape have a high level of awareness of the risks to the economy and livelihoods associated with deforestation under current and possible evolution of these risks with the changing climate, and the benefits of reforestation in an integrated landscape approach.

Women's land rights need some key consideration for an effective integration of women in land use planning and management, among which include advocacy for women's knowledge of their rights, women's confidence in the dispute resolution process and outcomes, men's respect of women's rights, and modest improvements in physical and social accessibility of the local justice system for women are most critical avenues for success. These factors will be considered throughout project implementation of the activities proposed within this project using a gender-responsive, participatory and fully transparent approach. For example, under the project trained institutions will hold meetings to train women, youth and marginalized people on forests and mobilize them to participate in planning processes and local institutions will be trained to build capacity of rural people, specifically women and youth, to participate in climate adaptation processes. Program Staff and government officials will be trained understand gender barriers, understand how to mainstream gender equality principles into planning and support women, youth and marginalized participation in meetings.

Furthermore, the Project will support dialogue with the private sector and planning processes needed to mainstream climate change and biodiversity in the tea sector. Projected climate changes predict significant elevation rises in CND agro ecological zones, including the potential for tea cultivation at much higher

⁸⁹ Arakwiye, B., et al, 2021. *Thirty years of forest-cover change in Western Rwanda during periods of wars and environmental policy shifts*. Regional Environmental Change 21(2).

elevations than at present. This is unlikely to threaten Rwanda's existing montane forest parks, as foreign revenues from tourism in these reserves far surpass those from tea. However, tea remains a significant source of foreign revenue and a government target for expansion. Any increased tea production in the CND would therefore most likely affect lands outside of parks: smallholder, cooperative, and district holdings, many of which might also be targets for reforestation, agroforestry, and biodiversity stepping stone initiatives under this Project. The integrated land use planning included in this Project design, will bring together tea industry, agriculture, forestry, biodiversity, and tourism interests to identify and reduce such potential conflicts over use of the CND's limited land resource base. Through this integrated land use planning we aim to establish a formal public-private sector forum at the landscape level to help improve communication across sectors, appreciation of environmental issues, and increase synergies between interventions to avoid conflicting targets and indicators between sectors.

This output will comprise the following activities and sub activities which will be executed by the Rwanda Forestry Agency (RFA) in collaboration with key government agencies, private sector and civil society organizations.

Activity 1.2.1 Create interagency taskforce institutionalizing integrated landscape planning and policy

Insufficient mechanisms for integrated land use planning constitute one of the key barriers for sustainable natural management in Rwanda. Planning processes within government agencies are carried out in silos without paying attention to the impacts a certain land use activity or project will have on other sectors of the economy. This Project aims to facilitate improved collaboration among government agencies by strengthening the government's existing cross-sectoral planning task force. The existing task force was set up to support cross-sectoral implementation of the National Strategy for Economic Transformation, but it suffers a number of weaknesses, including: i) a lack of official mandate for the task force to carry out activities; ii) irregular attendance of members; and iii) lack of clear and regular funding to facilitate meetings. This activity will review the mandate and structure of the cross-sectoral task force and promote solutions for strengthening its operation to RFA (the task force coordinator). It will also facilitate quarterly meetings of the task-force, in particular between institutions in charge of agriculture and agroforestry, to encourage synergies and avoid overlapping mandates and redundancy in different climate resilience interventions. The involved staff at both national and local levels shall be empowered to readily share information and activity plans, and will be offered technical support to generate materials that can inform decision-making (e.g. policy briefs, map, presentations). The Project will support collaborative integrated landscape planning, which secures climate resilience undertaken with appropriate social safeguards.

Sub activities will include:

1.2.1.1 Review & strengthen operationalisation of the current cross-sectoral task force

1.2.1.2 Hold quarterly sectoral planning meetings with both national and district administrations in CND and the private sector involved in mining, agriculture, and livestock production in the landscape

1.2.1.3 Facilitate discussions and provide technical support (e.g. policy briefs) in decision-making for cross-sectoral actions around climate adaptation and forest resilience in the CND

1.2.1.4 Continue to introduce social safeguards at each meeting including the GRM, FPIC, and Access Restrictions Mitigations as safeguards to be in place for work with local communities. For those at the district or community level, provide options for feedback on the process and best ways of communication with local partners.

Activity 1.2.2 Build capacity for spatial planning in national agencies re climate change

Spatial planning is emerging as a valuable tool for the development of evidence-based land use and climate resilience strategies⁹⁰. This subcomponent will develop capacity within the Rwanda's university network and the MoE to use remote sensing, ecosystem service modeling and spatial planning tools (e.g., Marxan, InVEST, RIOS, SWAT and other open access tools such as Open Foris) to incorporate climate risks and climate resilience strategies into development plans and to readily identify and reconcile land use conflicts. A prime example of the latter is the potential for climate-driven shifts in optimal elevation zones for coffee and tea cultivation conflicting with current subsistence agriculture in the case of coffee, and key remaining natural forest fragments with regard to any upward migration of tea. These spatial planning tools will be procured by the project and will be made available to the spatial planning unit.

The Project will assist the GoR in developing a dedicated spatial planning unit (SPU) within an appropriate ministry or cross-cutting agency. The appropriate hosting institution will be determined during the project implementation in consultation with key government agencies operating under the Ministry of Environment. This will include support for training high level Rwandan specialists to provide the technical expertise needed to process and integrate spatial planning into decision-making within and across sectors. A specific contribution of this SPU in coordination with appropriate governmental and non-governmental partners is the development and operationalization of an innovative, scientifically credible, scalable system for monitoring trends in natural forest cover and forest types at multiple scales. This SPU will also allow Rwanda to measure and report on the land use, land use change and forestry (LULUCF) sector as part of a national Greenhouse Gas Inventory (GHG-I). This system will build on the national FMES that was established with the support from ENABEL.

Capacity building is essential for execution of the Project; even more so for the sustainability of Project elements in the long run. To improve Rwanda's collective ability to integrate land use planning in support of climate change resilience and delivery of ecosystem services, these sub-activities are planned:

1.2.2.1 Recruit spatial planning expert/s to support spatial planning unit

1.2.2.2 Assess and identify the institutional home and operationalize the spatial planning unit

1.2.2.3 Conduct a capacity needs assessment and identify the appropriate tools for spatial planning to fit the Rwandan context and conduct familiarization

1.2.2.4 Led by the spatial planning unit, develop capacity within the University of Rwanda and the MoE to use remote sensing, ecosystem service modeling and spatial planning tools (e.g., Marxan, InVEST, RIOS, SWAT) to incorporate climate risks into land use planning processes

1.2.2.5 Deliver training workshops on utilization of earth system models for land use planning purposes, utilization of remote sensing resources (lightning, satellite radiances) for hazards detection and climate monitoring and on regional climate monitoring and applications.

Activity 1.2.3 Develop an effective forest monitoring system to underpin forest management decisions

⁹⁰ Mathias S., et al, 2019. The Spatial Development Framework to facilitate urban management in countries with weak planning systems, International Planning Studies, 24:3-4, 235-254, DOI: 10.1080/13563475.2019.1658571

Within Rwanda, forest-related data is limited in scope and generally outdated⁹¹. Consequently, the current state and trends of Rwanda's forests are not well understood. Efforts to improve the situation through individual studies have been valuable but have not led to systematic increases in spatial information on trends in different forest cover types and resilience at the scales at which land use decisions are being made. For example, the FMES software focuses on forestry and plantation forests, but does not address natural forests. This activity will update and monitor forest cover data in different ecological zones and forest types using high resolution satellite imagery, field work, and remote sensing techniques. In addition to remote sensing techniques, the Project team, once selected, will meet with government and international partners to assess best monitoring tools and platforms, including Open Foris. Close attention will also be given to the system used by Mugabowindekwe et al (2023)⁹² for the most recent, highly detailed forest inventory from 2007 imagery. RFA will recruit a Rwandan expert who will work with WCS experts in applying spatial planning and forest monitoring tools in support of government agencies during the project implementation to strengthen the capacities of RFA, NLA and GIS Center at the University of Rwanda.

An adaptive forest, climate change and land use monitoring system will be established to support climate resilient forest management decisions through the following:

1.2.3.1 Recruit Forest ecologist or remote sensing expert to support the design and implementation of indigenous forest monitoring system

1.2.3.2 Review existing forest mapping data and monitoring software, and assess utility for generating quantitative assessments of indigenous forest cover from local to national scales

1.2.3.3 In collaboration with REMA, MINAGRI, RAB, RDB, RISA, MoE and affiliated agencies and University of Rwanda, design structure and operation of forest monitoring system, including required inputs (data, computing, personnel), desired outputs, and operational structure (e.g. location, reporting structure, funding)

1.2.3.4 Purchase and install 2 new weather automatic stations in the CND

1.2.3.5 Establish & operationalize forest monitoring system

1.2.3.6 Generate updated maps & statistics for indigenous forest cover, forest type etc. using forest monitoring system

1.2.3.7 Develop capacity within hosting institution for continued operation of forest monitoring system

Component 2: Forest and landscape management and restoration

The GoR has prioritized forest management and restoration – including natural forests in protected areas – as a key component of climate change adaptation strategies for the CND landscape.⁹³ Natural forests within the national park system have been generally well protected and managed in recent years, earning high marks for biodiversity conservation while also developing a high quality – and quite lucrative – set of ecotourism attractions. Current park management plans, however, include little attention to the potential effects of projected climate change. Outside of parks, few areas with natural forest cover remain and forest management and restoration approaches are not being implemented at the scale required to significantly improve forest climate resilience. Where forests are restored, most use exotic species and mono-specific

⁹¹ Food and Agriculture Organization of the United Nations, 2017. *Analysis of forests and climate change in Eastern Africa*. Forests and Climate Change Working Paper 16.

⁹² Mugabowindekwe, M., Brandt, M., Chave, J. et al. Nation-wide mapping of tree-level aboveground carbon stocks in Rwanda. *Nat. Clim. Chang.* **13**, 91–97 (2023). <https://doi.org/10.1038/s41558-022-01544-w>

⁹³ IUCN, 2014. *Forest landscape restoration opportunity assessment for Rwanda*. [Link](#).

plantations that are generally incompatible with mixed agricultural uses; they are also vulnerable to diseases and pests with climate warming.

In this component, targeted interventions will integrate climate change awareness and adaptation into national forest park planning, policies and management, as well as restore degraded areas within those parks. These efforts will help sustain the forests' rich biodiversity, the lucrative tourism revenues that flow into local and national economies, and the ecosystem services needed for climate resilience of vulnerable communities. This suite of activities will also identify priority areas outside of parks – in remnant natural stands, around wetlands, and along streams – for protection and restoration as “stepping stones” and linkages for long-term climate change connectivity.

Output 2.1 Protected Area management effectiveness improved re climate risks and adaptation

The remnant natural forests of the CND harbor globally significant biodiversity across multiple taxa, from primates to birds and orchids. More than 10% of these species are regionally endemic⁹⁴ – 47 flowering plants are endemic to Nyungwe Forest alone⁹⁵. Though each has suffered from significant past habitat loss, the Nyungwe and Volcanoes National Parks have been generally well-managed under the Rwanda Development Board (RDB), with assistance from NGOs, PPPs, and the private sector. Rwanda's reputation for ecotourism and nature tourism attractions is world class. It has also earned conservation recognition for the dramatic recovery of its endangered mountain gorillas, sustained protection of its remaining biodiversity, and creation of a significant revenue-sharing program with parks-adjacent communities. The recent addition of Gishwati-Mukura National Park to this mountain forest network underscores the government's commitment to conservation, though both forest blocks require significant restoration for the purposes of biodiversity conservation, connectivity, and ecosystem services enhancement.

In contrast to Rwanda's areas of conservation success, relatively little attention – and less action – has been focused on climate change and its potential impacts on habitats and biodiversity, especially in the climate-adapted mountain forests of the CND. The investment required to build and sustain the institutional and human capacity to achieve Rwanda's recent conservation success now needs to be joined with a parallel investment to ensure that PA management plans, policies, and actions address the realities of climate change both within and outside of their boundaries.

Activity 2.1.1 Facilitate revision of PA management plans to address climate risks

Climate change is increasingly recognized as not only a major future threat to protected areas, but one that will also exacerbate existing threats and vulnerabilities⁹⁶. This is particularly true for the three national parks of the Congo Nile Divide landscape that are surrounded by a large number of smallholder farmers who are dependent on natural resources for their livelihoods. The park planning processes need to be expanded to best meet challenges posed by climate change as well as ensure full incorporation of buffer zone and landscape linkage requirements, and the ability to engage with the larger CND and district landscape management planning processes. The revised management plans will aim to:

- Develop specific goals and objectives for improving climate resilience
- Identify areas and species of particular importance to climate adaptation, mitigation and resilience
- Identify and prioritize threats that exacerbate climate impacts
- Identify and prioritize critical actions for strengthening resilience

⁹⁴ CBD. 2020. Rwanda 6th National Report to the Convention on Biodiversity

⁹⁵ Fischer, E. and Killmann, D., 2008. *Illustrated field guide to the plants of Nyungwe National Park Rwanda*. University of Koblenz-Landau.

⁹⁶ World Bank, 2019. *Rwanda Systematic Country Diagnostic*.

- Identify areas important for climate change adaptation and mitigation into protected area zones and regulations
- Develop indicators of climate resilience

To this end, the Project will undertake these sub-activities:

2.1.1.1 Review and update existing national park management plans to ensure climate and related landscape changes, risks, impacts and required management responses are integrated

2.1.1.2 Provide technical and financial support to planning, research and monitoring as well as community-based conservation units within RDB in managing the parks as part of larger CND landscapes

2.1.1.3 Train PA staff to integrate gender and social inclusion into programming

2.1.1.4 Train PA staff on integrating needs of women, youth, historically marginalized and Ubudehe Categories c,d,e populations into climate adaptation risks and responses

2.1.1.5 Train PA staff on SEAH, GRM, FPIC, and Access Restriction Mitigation Processes to ensure NP climate change responses are undertaken in a way which supports social inclusion and equity

Activity 2.1.2 Establish long-term plans for CND financial sustainability post-GCF

Rwanda has recently developed a Biodiversity Finance Plan⁹⁷ through support from BIOFIN. This plan has identified a number of financing solutions that will not only contribute towards improved biodiversity protection in Rwanda, but also aligns with Rwanda's sustainable development and green growth objectives. These solutions include:

- Introduce a Biodiversity Conservation Fund into Rwanda's National Fund for the Environment (FONERWA) to help streamline and attract domestic and international sources of finance to support conservation investments and thereby reduce transaction costs to increase conservation benefits.
- Improve efficiency and effectiveness of environmental fees and fines to strengthen the effectiveness of the regulatory environment that will ultimately improve ecosystem and biodiversity conservation goals.
- Water User Fees for Catchment Management to secure the financial resources necessary for effective water catchment management.
- Pilot Business Plans for Selected Wetlands to enable the sustainable development of wetlands for ecotourism and biodiversity-friendly enterprises such as handicrafts, fisheries, and sustainable agriculture.
- Promote biodiversity-friendly enterprises in the transition to a green economy and accelerate the transition to a green economy by incentivizing and supporting businesses to adopt sustainable practices and attracting investments in new conservation enterprises.
- Strengthen the tourism revenue sharing scheme to improve biodiversity outcomes and effectively address conservation-development objectives
- Develop a Protected Area Finance Strategy to support the goal of increasing the financial and ecological sustainability of Rwanda's Protected Area System (PAS)

There are also new emerging solutions such as the Wildlife Conservation Bond (WCB) which essentially is a payment for performance-based financial instrument that channels investments to achieve conservation outcomes – measured in this case by specific conservation targets that could be set (e.g., hectares of native forest restored, or an increase in umbrella species populations). In the case of Rwanda, gorillas and

chimpanzees are considered umbrella species that play a crucial role in shaping entire ecosystems on which countless other species depend. For the WCB, a grants-based, outcome payer is needed, which could be the GCF. Through the WCB, and in collaboration with the World Bank (WB), the WB could issue bonds and investors could support the financing of activities to protect and grow a critically endangered species with clear conservation targets, contributing directly to biodiversity and bringing jobs to local communities through the creation of conservation-related employment in rural areas of the CND. A third-party, independent verifier is also needed to evaluate whether conservation targets have been met. A meeting is planned with the WB in the second half of February 2023 in Rwanda to determine the scale and feasibility of the WCB in the CND. The South African Rhino Bond has already attracted foreign investment, and it is a good model Rwanda could emulate⁹⁸. The Government of Rwanda, represented by the Rwanda Development Board, is in advanced stages of negotiation with the World Bank for establishing a wildlife Bond. There is, therefore, clear interest in the bond which suggests that investors still see value in investing in such an instrument.

A technical advisor will be recruited by this project to provide technical support to RDB, REMA, MINECOFIN and MoE and explore these various solutions to identify financial instruments that could be employed for revenue generation post-GCF for sustainable management of the CND landscape. The technical advisor will work closely with the NSC to facilitate buy-in from stakeholders such as RDB and MINECOFIN into compliance with environmental laws through greater emphasis on the links to climate resilience and NDC objectives.

Activity 2.1.3 New fire management curriculum developed and operationalized Given the expected increase in drought, and the natural forest vulnerability already demonstrated in the past, the capacity for effective fire prevention and management in core PAs and adjacent landscapes will be increased.

While current climate projections call for increased rainfall in the CND, they also project higher temperatures and a rise in elevation of the natural forests' cloud cover zone (Annex 2.1). The combination of the latter two trends raises the potential for increased fire risk. In order to sustainably address the threat of fire to the natural forests and regenerated areas in particular, this project will increase the capacity of RDB in the area of fire prevention, with emphasis on education and outreach, monitoring of fire danger conditions across the parks, response plans, and fire suppression. A collaborative effort in fire management planning and implementation in CND districts will be carried out, facilitated by a fire management expert. Based on training needs, the project will also carry out RFA district and sector foresters' training in Integrated Forest Fire Management approaches and activities and firefighting; they will then build the capacity of community fire brigades.

Sub-activities include:

2.1.3.1 Develop a curriculum tailored to needs and capacities of different stakeholders

2.1.3.2 Build the capacity of RDB to manage fire in National Parks

2.1.3.3 Build the capacity of RFA, local authorities and communities to collaboratively manage fire in forests outside Pas

2.1.3.4 Implement a "fire wise" outreach and awareness program for communities and local government (district, sector, cell), as well as Rwanda Forest Authority, surrounding natural forests to reduce the incidence of human-caused fires.

⁹⁸ World Bank, 2022. *Wildlife Conservation Bond Boosts South Africa's Efforts to Protect Black Rhinos and Support Local Communities*. Press Release NO: 2022/059/AFE.

Output 2.2: Natural Forest cover restored, biodiversity connections established.

The natural forests of the CND exist as an archipelago of isolated islands within a surrounding matrix of intensive human land use. Although some have raised the possibility of reconnecting these islands, it is important to understand that their isolation is not a recent phenomenon (with evidence of the initial isolation of the main forest blocks from the pre-colonial era). Furthermore, large-scale proposals for landscape linkages have thus far failed to provide alternatives for the tens of thousands of rural smallholders who would be displaced by such schemes. However, important actions can be taken to restore degraded natural forest lands and protect or reestablish natural “stepping stones” and ensure a functional level of ecological linkage between the existing parks.

In some areas where natural forests have been cleared, the Project will work to restore forest cover. This will be done primarily around the Gishwati and Mukura reserves, using native species and building on nursery and planting techniques already tested in Rwanda. New techniques may need to be developed to expand reforestation to degraded lands on steep slopes and along waterways and wetlands. Tree species will be carefully selected considering the following attributes: pioneer species which grow in full sun, fast growing (height and canopy width), and animal pollinated and dispersed. These “tree islands” serve several purposes. First, by growing quickly and providing canopy cover, they shade out competing vegetation and change the soil microclimate (very important for later successional species establishment). Site-generalist native species such as *Albizia*, *Dombeya*, *Macaranga* and *Polyscias* respond to these characteristics. Secondly, they entice a variety of animal species into the expansion area, thereby dispersing seeds. In certain areas the Project will favor natural regeneration from seed banks.

All work outside of protected areas will require careful attention to private land ownership claims and related social and gender equity issues. Part of the communities’ responsibilities will be afforestation of the degraded forest with indigenous trees and clearing invasive species from the natural forests and buffer zones. Part of the benefits for the communities will be harvesting of non-timber forest products (NTFPs) from the natural forests, under sustainable use plans. The project will provide training on improved harvesting techniques, processing, packaging and marketing, to those engaged in NTFP value chain (financed under outcome 3).

Activities and sub activities under this output will be executed by the Rwanda Forestry Authority (RFA) with the technical support from WCS.

Activity 2.2.1 Secure key remaining natural areas outside PAs

Outside of current protected areas, the Project will identify remaining natural areas and seek to secure their continued protection within district management plans or through other means. These are mostly small stands of natural vegetation on rocky outcroppings or strips of forests along the CND’s myriad of waterways. Although these will not permit large mammals (e.g. elephants or mountain gorillas) to migrate between the forested parks, these small remaining natural areas provide critical intermediate stepping stones for the broader biodiversity elements of the CND forest ecosystems. For the CND’s rich assemblage of bird species, these vegetated patches and gallery groves could prove to be essential stepping stones for connectivity under changing conditions. Small mammals and insects are likely to also directly benefit. And most importantly, the movement of birds and small mammals will support the transfer of seeds between forest patches, greatly improving the ability of forest ecosystems to adapt to changing climates, and continue supplying key ecosystem services to the people of the CND.

Sub-activities include:

2.2.1.1 Raise awareness on remaining protected natural forests in CNDL to secure their protection

2.2.1.2 In collaboration with RFA, REMA, districts identify and implement actions that support conservation and management of remnant protected natural forests in CNDL

Activity 2.2.2 Restore natural forest cover in and outside Protected Areas including riparian linkages

While the boundary integrity of the Nyungwe and Volcanoes NPs has been maintained for several decades now, internal degradation has occurred. Prior to 2004, approximately 12% of Nyungwe National Park had been affected by wildfire, including a catastrophic fire during a period of prolonged drought in 1997.⁹⁹ The natural vegetation was quickly replaced by a carpet of invasive ferns. In recent years, teams of local community workers led by WCS have demonstrated that careful removal of the fern cover reveals still viable native tree seeds and seedlings that respond to this exposure by rapid growth and the formation of new forest stands. The project will expand this labor-intensive process of assisted regeneration to the remaining 4,500 ha of burned lands within the NNP, restoring forest values and generating significant employment for local rural communities.

For the Gishwati Forest, restoration with native species will take place on adjacent, previously cleared lands. This and other restoration efforts should benefit from tree nursery techniques and planting trials already initiated by RFA, RDB, and NGO partners around Nyungwe, Volcanoes, and Gishwati itself. The project will work with the National Tree Seed Centre to provide training on tree propagation and supply of quality seeds/seedlings/propagation materials, especially of indigenous species and plantation species not being sourced locally.

A special element within the CND landscape is its dense network of waterways. Historically forested riparian habitats along these streams and rivers, however, have been largely cleared of their forest cover. Riparian reforestation efforts will restore an important erosion control factor as well as recreate important biodiversity connectivity along linear corridors. RWB will co-finance these riparian restoration efforts. With support from district and sector leaders as well as community representatives, community members will be hired from neighboring villages to the project sites. The project will make efforts to recruit the most vulnerable community members in category c,d and e, women, young landless, and others to carry out these activities.

Key sub-activities include:

2.2.2.1 Review mapping of degraded natural forest areas in core NPs, stepping stones, and unprotected riparian lands using updated imagery and ground truthing

2.2.2.2 Identify parcels for restoration in core PAs, stepping stones, and riparian lands using desktop and field-based assessment

2.2.2.3 Establish indigenous tree seed nurseries in the CND to serve core PAs, stepping stones, riparian land restoration and promotion of indigenous trees on farms and in protective forests

2.2.2.4 Recruit, train (in forest restoration methods), and equip community workers to be involved in core PAs restoration and riparian linkages

2.2.2.5 Assisted rehabilitation of 4,500 ha of indigenous forest in NNP

2.2.2.6 Active restoration of natural forest on 500 ha Gishwati Mukura National Park

2.2.2.7 Restore 1,500 ha of riparian lands

⁹⁹ Masozera, A.B., Mulindahabi, F., 2007. *Post-Fire Regeneration in Nyungwe National Park, Rwanda*. Wildlife Conservation Society.

2.2.2.8 Using permanent plots sampling, collect field monitoring data on tree species recruitment and growth for estimation of biomass, species richness, etc. in restored parcels in Nyungwe and Gishwati-Mukura National Parks as well as on riparian lands to assess success of initiatives, then replicate or adapt as needed

2.2.2.9 Perform Environmental and Social Screening on all positions being created by the project

2.2.2.10 Establish a grievance process for laborers

2.2.2.11 Update emergency and preparedness plan including risk mitigation guidance to local conditions at restoration sites

2.2.2.12 Train all workers on their rights and how to access the GRM

2.2.2.13 Provide code of conduct and emergency preparedness and safety training for all laborers

2.2.2.14 Hire and train labor and safety leads at each site to provide guidance to staff, be available for grievance issues, and monitor health and safety conditions for workers

2.2.2.15 Develop and implement strategy (including social marketing) to ensure that women and youth participate and benefit from forest restoration projects

Activity 2.2.3 Promote silvopastoralism with indigenous trees around Gishwati

Roughly 80% of Rwandans own livestock of some sort. While it is government policy to expand this percentage, conflicts over competing land uses are inevitable, especially for the country's most vulnerable rural populations. In the area around the protected forest of Gishwati, the presence of many slopes greater than 55% opens the possibility for reforestation of larger blocks with tree species compatible with an undergrowth of grass or other forage. In addition, some existing pasturelands could be enriched with tree species. These areas could then provide critical off-farm fodder for goats, sheep, and perhaps even cows, while also serving vital ecosystem services.

Plans for tree planting/restoration on pasture lands will be elaborated in collaboration with the local communities and technical (extension and local government) services. Areas meant to produce fodder and for water retention will be demarcated during community consultations for action plans. Already, important forage grasses, such as *Brachiaria* grass known to increase milk production while reducing methane emission from cattle, have been identified while tree fodder such as *Calliandra*, *Leucaena*, *Acacia*, *Gliricidia* are well accepted but not widely adopted due to local capacity barriers. Improving adoption of better management practices will require technical assistance considering needs of cattle owners, local milk cooperatives, and traders.

Beneficiaries of these activities will be pasture land owners and cattle keepers in the Gishwati area. They will be identified in collaboration with district and sector leaders, RFA, EE and representatives of pasture lands and cattle owners. Special attention will be paid to specific groups such as women-headed households, youth, and historically marginalized groups if they own grazing land in the Gishwati area and for paid work. Awareness and consultations with these pastoral communities will be conducted during project inception to ensure farmers' involvement in identifying and selecting key areas for restoration, suitable species to grow in their pastoral areas, capacity barriers, and associated technical assistance to be provided to adopt better management practices.

Key sub-activities include:

2.2.3.1 Assess the status of pasture lands in the Gishwati area and identify key areas for restoration and the potential to introduce indigenous species

2.2.3.2 In collaboration with landowners, identify suitable indigenous species for fodder trees, shrubs, grasses, and herbaceous legumes that have potential to improve rangelands and increase their climate adaptive capacity

2.2.3.3 Produce and disseminate fodder trees, shrubs, grasses, and herbaceous legumes to project beneficiaries

2.2.3.4 Train beneficiaries on improved livestock and pasture management

Component 3: Enhancing climate adaptation through resilient livelihoods

Increasing the extent of protective forests, enhancing and restoring forest plantations, and increasing agroforestry tree cover are key steps to restoring ecosystem function and services identified in the National Forest Policy Report (NFPR 2017), especially i) reducing the risk of flooding, landslides and soil erosion from extreme climate events and ii) increasing long-term supply of wood fuel resources (the primary energy source for 77% of Rwandans¹⁰⁰). Together with fuel efficient cookstoves, these trees will provide a sustainable source of fuelwood and reduce the amount of time women spend collecting fuelwood, enabling them to participate in other economic activities and programmatic opportunities. These outputs (cookstoves, plantation restoration, and agroforestry) will deliver carbon sequestration benefits totaling 522,606 tCO₂eq over 20 years.

In working to develop a comprehensive land use plan that incorporates climate adaptation, and while ensuring Rwanda's forest ecosystems are strengthened through forest management and restoration, attention must be paid to people who live next to the forests and those who are the most prone to causing degradation of these forest systems. In particular, this includes people who either do not have access to land or whose land holdings are insufficient to meet their daily needs. These people are most likely to enter into and unsustainably use protected and protective forests to help meet their daily needs. Thus, this Project component targets these vulnerable households – youth, people from historically marginalized groups, and farmers with insufficient holdings – to build their capacity in financial literacy and enterprise, in order to strengthen their resilience to economic and climate shocks.

Also, to maximize the opportunities from improved silvicultural practices, it is necessary to improve business practices and market linkages for farming communities. More specifically, organizing and linking farmers to local and national wholesale traders will enable them to sell their produce more efficiently, maximizing profit and reducing wastage. To facilitate these linkages, this component will also build the business capacity and market linkages necessary to support the transition of farmers and their communities away from unsustainable practices that magnify their vulnerability to climate change and towards more secure livelihoods and expanded income opportunities. This includes strengthening capacities, developing inclusive business models, and developing market value chains for livelihoods that do not depend on traditional smallholder farming.

The project will develop alternative sources of income to improve living conditions for people in the CND who depend for a large part of their income on exploitation of the forest and natural vegetation. This approach, focusing on vulnerable groups, aims at real economic empowerment of the concerned families through the realization of their own business plans. Several promising activities related to the use of the forests *aside* from wood-gathering (such as beekeeping, extraction of essential oils, herbs, and other forest

products) will be analyzed. These business plans, tailored to individual circumstances, will be realized with support and tools provided by the project.

From the second year onwards, these vulnerable groups will become members of existing cooperative organizations that will be supported by the project, particularly in value chains important for forest conservation, such as production and export of honey and the extraction of essential oils. These cooperatives provide various services (e.g., information on markets / export, input supply, aggregation of commodities, etc.) and they also play a role in the protection of the interests of their members.

The project will develop the private sector strategy (which will include engagement of impact investors interested in conservation-friendly commodities) for livelihoods implementation in the target areas through the identification of new livelihood opportunities for the CND. We will establish relationships with several buyers fitting our selection criteria, and engage partners such as Blue Ventures to expand market opportunities for products that meet international or local market opportunities. On the producer side, we will build strong relationships with other projects and programmes that support private sector engagement with community organizations, and producer associations to pursue promising market opportunities (local and international, as appropriate).

For component 3 and the project overall, the eligibility criteria for selecting beneficiaries will be:

- Households with limited access to financial resources for implementing climate adaptation measures as determined by Local Authorities
- Women -headed households.
- Households in the CND living in key areas such as those surrounding PAs, identified stepping-stones, landscape linkages, and broader farming mosaic with exposure to erosion.
- Most vulnerable households including people with disabilities
- Land owners where the project intervention will have to intervene

These criteria will be further refined by the project EE in consultation with key stakeholders (MoE, Districts and community representatives) once the project is approved.

Output 3.1 Farming methods enhance productivity, reduce erosion and flooding risks, contribute to ecosystem services, and support connectivity

Soil degradation and loss, largely from the cultivation of steep slopes in the western highlands of the CND, has caused declines in agricultural productivity and serious downstream problems of siltation and flooding. While the former means that smallholder farmers in the hills increasingly struggle to feed their families, the latter poses a threat to the long term viability of water provision services in municipal areas and hydro-power generation¹⁰¹, which is a key pillar for Rwanda's low carbon growth strategy.

This Project will support smallholder farmers in high erosion risk areas to implement Sustainable Land Management (SLM) practices techniques and the active *restoration* of degraded forest lands. By targeting high risk areas the Project can have a significant impact to reduce soil erosion, improving the resilience of agricultural livelihoods in the highlands, while having a positive impact on water flow and water quality in the lowlands.

¹⁰¹ Rwanda Environment Management Authority (REMA), 2009. *Rwanda State of Environment and Outlook Report*, Chapter 8: Energy Resources.

In each selected priority site, the Project will support the implementation of the full range soil erosion control best practices including agroforestry. While the erosion control practices may vary (e.g. radical vs progressive terracing), agroforestry practices will be established across the entire 3,346 ha. RFA will mobilize participating landowners on whose lands soil erosion control best practices including agroforestry shall be implemented. Landowners willing to participate will sign a consent form allowing project activities on their land and will be grouped in vigilance committees. These vigilance committees will sign MoU with Districts and RFA as a commitment to participate and manage sustainably their land and trees with clear role and responsibilities of each (farmers, districts and RFA) specified. The consent forms will be granted to RFA and Districts where landowners are located. The vigilance committee will be responsible for ensuring the sustainability of project interventions implemented on community land. The MoUs signed by Districts, RFA and the vigilance committee will not be legally binding but rather to establish certain understanding in the implementation of the project.

Interventions will be made in collaboration with district officials, according to agreed priorities and objectives, and involving local farmers and contractors. Table 29 in the feasibility study (FS) provides a preliminary list of the fuelwood, fodder, timber and fruit tree species that have been identified as suitable for agroforestry systems in the CND, based on their resilience capacity against climate-driven temperature and precipitation changes, as well as community preferences - particularly the needs of women farmers, historically marginalized households, and youth. They also concur well with recommended species for the North and Western Province as described in the Agroforestry Technical Guide¹⁰².

Activity 3.1.1 Restore high slope areas (>55%) as protective forests

In Western Province, a major part of the CND landscape, 80% of the increase in tree-planting in recent years has come from private landowners seeking to earn revenue or increase their access to wood-based fuel¹⁰³. Beyond this utilitarian interest, however, the NFPR 2017 calls for increasing the extent of protective forests and enhancing and restoring forest plantations as key steps to restoring ecosystem function and services, especially i) reducing the risk of flooding, landslides and soil erosion from extreme climate events and ii) increasing long-term supply of woodfuel resources (the energy source for 77% of Rwandans¹⁰⁴), providing biodiversity habitat, and sequestering 193,472 t CO₂-eq over 20 years.

The current land use master plan discourages cultivation on slopes greater than 55% and encourages protection of existing vegetation and afforestation with indigenous species. This Project will promote indigenous species in rehabilitating existing forests and establishing new ones on bare lands. This activity will result in highly productive, climate-resilient woodlots and forestland with fully restored ecosystem services and significantly increased long-term carbon sequestration. It will afforest/restore 2,500 ha of degraded private smallholder land and district/state land by restoring existing degraded tree stock, afforesting bare areas, promoting good silvicultural practices, and facilitating the adoption of Simplified Forest Management Plans (SFMPs), as recommended by 2013 forest law. In the priority regions selected for afforestation/restoration, awareness and capacity building activities will be undertaken to build support and understanding of the PFMU approach. The objective of PFMUs is to combine economic profitability, preservation of environment, and improvement of living conditions of the local population, as well as to apply a gender-responsive approach to wood resource management. For privately owned land, only small-scale individual private landowners will be included, and this activity will not involve restoration of private institutional plantations. RFA will mobilize participating private landowners on whose lands afforestation

¹⁰² Ministry of Environment. 2018. National Agroforestry Strategy.

¹⁰³ Arakwiye, B., et al, 2021. *Thirty years of forest-cover change in Western Rwanda during periods of wars and environmental policy shifts*. Regional Environmental Change 21(2).

¹⁰⁴ Government of Rwanda, 2021. Rwanda Household Survey 2019/2020. National Institute of Statistics of Rwanda.

trees shall be planted. Landowners willing to participate will sign a consent form allowing project activities on their land. Following this, private woodlot owners will then be organized into local groups (around 40-50 ha of woodlot per group), PFMUs (Private Forest Management Units) and assisted to form and register as cooperatives and elect among themselves two cooperative committees: Administrative committee made of 5 persons (President, Vice president, Secretary and two Advisors); supervisory committee made of 3 persons (President, vice president and secretary). Note that gender consideration is mandatory (at least 30% women).

When the cooperative committees are elected, they are mandated to represent forest landowners at different events and fulfill their interests, especially in seeking the best market for their forest products and seeking for other income generating opportunities. RFA will sign MOUs with different cooperatives specifying each party's respective role and responsibility.

Experiences from pilot PFMUs established under the FMBE project in the districts of Gicumbi, Gakenke and Rulindo Districts of the Northern Province and Rwamagana District of the Eastern province have demonstrated that this approach is helping to achieve economic profitability, preservation of environment and improvement of living conditions of the PFMU members. Forests are sustainably managed, and all PFMU members respect the management plan and planned harvesting period. All silvicultural practices are properly followed because cooperatives supervise all activities with technical guidance from the district technicians. The forests within a PFMU are economically attractive to potential buyers due to good stock at harvesting time. PFMU members have access to small loans while waiting for the forest to mature. This shows that PFMUs established in these areas are also serving as demonstrations to other forest owners supported by other projects in the country. Green AMAYAGA and Green Gicumbi, and TREPA Projects are scaling up this approach in other parts of the country.

On public land, forests will be organized into state/district FMUs to be managed by professional private companies through long-term concession agreements in line with the government strategy. While a concession agreement will be signed between the private forest companies and the MoE, the Project will only support the design of the forest management plans and development of contracting documents. Registration of restored forest parcels will be guided by well-trained district forest officers and forestry sector extensionists, assisted by the new FMES software and related GPS/tablets, which will provide automatic statistics, maps and register owners. For each FMU, a SFMP, reviewed or developed through a participatory process, will outline the basic information of the site (e.g. owner, species etc.), as well as a harvesting plan and schedule for silvicultural operations (thinning, pruning etc.). However, some areas of the state plantations that are vulnerable and exposed to high risk of degradation (high slopes) will require setting up special protective measures. Restoration on identified vulnerable areas will be done by establishing appropriate tree plantations that will offer a better protection and conservation of biodiversity while providing other ecosystem services including tree species that improve forage for bees. The choice of tree species planted will take into consideration women's participation, use and priorities. The project will further strive and advocate for gender equity for selection of plots which will be done in consultation with communities.

Sub-activities include:

3.1.1.1 Introduce and raise awareness of indigenous species to target stakeholders in CND

3.1.1.2 In consultation with RFA, National Land Authority, districts and communities, determine fragile areas (steep slopes > 55%) to be allocated for protective forests and their ownership

3.1.1.3 Assess the current status of the indigenous tree species in selected areas for protective forests and select indigenous tree species appropriate to CNDL

3.1.1.4 Develop restoration plan for protective forests

3.1.1.5 Build capacity of local stakeholders (men and women) on PFMU approach and methods

3.1.1.6 Design and approve SFMPs of private FMUs

3.1.1.7 Ensure consent of smallholders prior to planting trees

3.1.1.8 In collaboration with smallholders reforest/restore 2,500 ha of public or private land with slopes >55% and ensure sustainable management under private FMUs according to approved SFMPs

3.1.1.9 Support monitoring and evaluation of restored private FMUs

3.1.1.10 Assess impacts of exotics on neighboring lands and mitigate their negative impacts

3.1.1.11 Develop and implement strategy (including social marketing) to ensure that women and youth participate and benefit from forest restoration projects

3.1.1.12 Assess the benefits and costs of each proposed tree and plant species and how those benefits affect different population segments

Activity 3.1.2 Develop on-farm agroforestry for high-caloric and indigenous tree species

The extreme land scarcity of the CND precludes smallholder farmers – many of them among the most vulnerable and land-poor of the region – from dedicating any significant proportion of their holdings to tree production. However, the use of certain agroforestry species that both enrich the soil and control erosion when interplanted with ground crops can greatly reduce this conflict.

Under this activity, the Project will facilitate the production of on-farm afforestation plans with action plans for implementation. Formulation of the forest landscape restoration plans will follow the methodology introduced by the World Resources Institute (WRI) and IUCN and already tested in the country by the former Ministry of Natural Resources, as recently modified and applied for the Gatsibo Forest Landscape Restoration baseline conditions assessment. The methodology will involve three simple steps: a) Geospatial analysis to map degraded land that presents an opportunity for forest and landscape restoration (FLR), which will highlight areas with best potential for restoration; b) economic analysis to model the costs and benefits of degraded and restored land; and c) designing a restoration/afforestation action plan, based on an in-depth assessment of the conditions required to implement the FLR in the selected sectors.

For selected areas where Project activities may lead to reduced access to resources, the Project will undertake an in-depth Environmental and Social Impact Assessment (ESIA) in the first year of implementation and design an Environmental and Social Management Plan (ESMP) to guide implementation. This ESMP will include an access restriction mitigation plan, if deemed necessary. The matter of Free, Prior and Informed Consent (FPIC) will be explored during the ESIA and the approach applied if deemed appropriate. These documents will be based on materials contained in Annex 6.

Drawing on experience with high-elevation species identified both in Rwanda and across East Africa, the Project will support the following sub-activities:

3.1.2.1 Identify sub-areas of intervention for agroforestry dissemination in the CND

3.1.2.2 Introduce and raise awareness of agroforestry in target communities

3.1.2.3 Develop the capacity of extension agents at district/sector level and NGOs to support adoption of agroforestry technologies

3.1.2.4 Establish agroforestry/fruit tree nurseries to facilitate access to quality planting material

3.1.2.5 Promote sustainable land management practices by stabilizing existing terraces, and plant high calorific agroforestry species to provide a sustainable source of fuelwood for energy-efficient cookstoves to protect sloping land against severe soil erosion.

3.1.2.6 Train RFA, extension agents, Project participants and community members on specific techniques for identification of and management of invasive pests and pathogens

Participants in sub-activities 3.1.2.3 and 3.1.2.6 will be selected by districts, JADF, RFA and farmers representatives. Special attention will be given to youth, women and other vulnerable groups. RFA and RWB will co-finance sub-activities 3.1.2.4 and 3.1.2.5.

Output 3.2 Rural livelihoods generate alternative incomes & reduce pressure on forests

To respond to the growing pressure on natural resources and increase the local community's resilience to climate change in the CND landscape, the Project will address key drivers of deforestation by creating alternative livelihood pathways for youth and women through ecotourism and income-generating, climate-resilient crops, as well as strengthening supportive financial services.

The setup of interventions in this output will focus on enhancing on-farm production capacities in market-facing promising value chains in agroforestry (especially avocado, macadamia and short-duration crops) for immediate income and beekeeping taking advantage of the agroforestry and the crops; the combination of these will promote ecotourism as supplementary income (avocado, macadamia, forestry, agroforestry, short-term crops, and honey). This output will also provide solutions to the growing households' energy poverty which has exerted pressure on natural resources and eventually exacerbated the local community's exposure to the stress of climate change in the CND landscape.

This project will employ a market systems development approach that encompasses business coaching, value chain development, and access to finance. This entails multi-stakeholder collaboration with business service providers, financial institutions, community-based enterprises, value chain actors, and farmer cooperatives. Ultimately, this project will develop and or promote a financial model that enhances the private sector ecosystem collaboration and partnership to stimulate finance in forestry and agroforestry sectors and support market-oriented access to energy-efficient cook-stoves to reduce pressure on forests.

The value chain development for potential revenue generating streams such as honey, ecotourism, and efficient cookstoves requires grant support of GCF grant to finance the activities that are important for the most vulnerable, as well the drivers for the MSD to take root; these include the development of the identified revenue generating streams, strengthening access to market information, and improving quality of the products and produce aggregation.

To target the poorest and most vulnerable, and empower women and youth towards meeting inclusive development goals, critical support at the nascent stages is essential. The GCF grant will facilitate the integration of the poorest and most vulnerable women and youth in the CND and position for private sector investments that will build a firm back base for sustainable community development. The GCF grant will not only provide financing support but will be critical in creating enabling opportunities for market development beyond the project to catalyze private investments.

While promoting the above-mentioned households cooking solutions and income streams, the selection of the beneficiaries will follow the government' procedures, and in the case of CND will focus on the vulnerable groups of youth, women, and historically marginalized people and will be selected based on an assessment

of poverty levels (also known as Wealth Categories) conducted by the government to ensure the most vulnerable are selected. Selection criteria will further take into account social and cultural factors such as gender and age which intersect to predispose some groups more to vulnerability than others.

Activities and sub activities under this output will be executed by the Rwanda Forestry Authority (RFA) with the technical support from CORDAID.

Activity 3.2.1 Develop agroforestry related value chain for markets access.

Most traditional tree farming in Rwanda is linked to fuel production in terms of firewood, charcoal, and timber for construction. However, to further strengthen community resilience to climate change, there is a need to introduce fruit and nut trees that have high economic value in the agroforestry system to generate more income for smallholder farmers but also increase its adoption, especially women-headed households. Though women-headed households face the same issues and challenges as male-headed households, they are less likely to be included in income-generating activities and are disproportionately impacted by climate change because of prevalent gender inequality issues. In the Feasibility Study, avocado and macadamia have been identified as high economic value crops with high market potential, and the economic benefits of identified fruit trees is paramount to increase farmer adoption and buy-in. Since fruits and nuts take time to mature, short-term crops like chili and French beans will be promoted to support smallholder farmers with limited land resources (less than 0.4 ha) to create additional, more quickly generated income and incentivize adopting agroforestry practices. Regular income will help vulnerable communities to cope with climate variabilities and shocks.

The identified agroforestry value chains have market potential and potential for intercropping with short-term horticultural vegetables and fruits while providing further impacts on carbon sequestration, nutrition value, and market access in the CND. Selected fruit trees are expected to meet the dual purpose of soil protection and climate impact while generating long-term economic benefits to the most vulnerable community members of the CND region.

For each crop value chain identified, specific bottlenecks will be addressed while focusing on specific product-place combinations within the CND. Especially, attention will be on enhancing the quality and quantity of production in sync with particular demands from market niches, development of various models to improve access to finance together with financial institutions, and exploration of market opportunities that can generate off-farm jobs to absorb a good number of landless and other vulnerable groups of youth and women.

These priority crops are:

- Macadamia: a low-volume, high-value export crop; in-country processing creates jobs for women and youth. Two hundred and four trees cover one hectare, and one mature tree produces up to 60 kg of macadamia nuts per year with an income of 25,000 FRW per kilogram. Furthermore, every tree must be separated from the next by 7 meters; thus, many other commercial and consumable crops may be planted and harvested while waiting for macadamia's fruiting maturity.¹⁰⁵ The global demand for macadamia outweighs the available supply, which presents a huge business opportunity for macadamia farmers in Rwanda. For example, the current macadamia supply meets only 48% of global demand and the global market for macadamia is expected to further expand from \$1.17 billion to \$2.36 billion by 2027. The project will build on the government export initiatives to promote

macadamia farming in CND areas through National Export Board (NAEB) and Rwanda Environment Management Authority (REMA) to include macadamia in its tree planting drive in the CND.

- Avocado: The extensive penetration of avocado cultivation amongst smallholder farmers (SHFs) and the favorable growing conditions lead to avocado yields (21MT/ha) that are nearly double those of Kenya. The increasing demand for export, and cash yield/ha are higher than maize or Irish potato. Through linkage to local avocado private value chain actors, packaged avocados will be exported to the European Union, where the channel price for avocados is \$1.75 per kilogram, providing an estimated 32% margin.¹⁰⁶ Avocado production presents numerous opportunities to smallholder farmers in CND: for example, earnings from avocado exports have been increasing from \$1.6 million in 2020/21 to \$4.5 million in 2021/22 and production increased from 1,000 tonnes to 2, 500 tonnes in 2021/22. According to the NAEB, the production is set to double this year and generate \$8 million from exports. To further strengthen its avocado exports the government has launched sea freight, which will not only increase export volumes but also make them competitive in terms of pricing.
- Avocado is projected to be the second most traded tropical fruit, after bananas, by 2030. The United States and the European Union, where consumer interest in avocados is fuelled by the fruit's health benefits, are expected to remain the main importers, with 40% and 31% of global imports in 2030, respectively¹⁰⁷. The project teams in CND will work closely with private sector partners and communities to develop value chain analysis, business development strategies, and business planning. The team, led by our Livelihood Specialists, will work with these private sector partners to prepare action plans detailing technical services we will provide to train producers to supply commodities to markets.
- Vegetable crops such as French beans and chili: It has been demonstrated that these crops can double the income per hectare of maize and potatoes and employ a good percentage of the 16% reported unemployed women and youth¹⁰⁸. Commercial farm wage reported labor in postharvest production handling at 72% women, 20% youth (see Annex 2.3). French beans can potentially deliver a gross margin of 22,040Rwf per acre and a return on labour of 2,755Rwf per day, while chili can offer a gross margin of 46,384Rwf per acre and a return on labour of 7,731Rwf per day.¹⁰⁹

The Project will also enhance ecotourism and honey value chains. Particularly in areas close to national parks, the potential business development of ecotourism can go hand in hand with honey and wax product development. Our analysis shows that the honey value chain already has a sizable production history in Rwanda. Various initiatives exist that have encouraged farmers to invest in modern hives. However, the current sector is constrained by a perception of limited profitability due to low prices, gender patriarchy due to male dominance, and limited market access. The honey (and various by-product) value chain shows a wide range of different consumer target groups, whereas honey is too often marketed as a generic product. This is a crucial bottleneck: while the value chain actors engage in the business, there needs to be more incentives to invest financial resources to improve production or quality, such as through modern beehive technology. It will be essential to link ecotourism destinations and agro-tourism to enhance the potential for an improved business case and help beekeeping cooperatives enhance their processing, quality management, and marketing processes with large scale buyers and exporters such as Api Business Development Company (ABDC) and create a sustainable and expanded source of income especially for

¹⁰⁶ Karuiru, M., 2018. Value chain management and the performance of avocado fruit small scale farmers in Kandara sub county, murang'a County, Kenya. *International Journal of Physical and Social Sciences*, 8(10), pp.24-40.

¹⁰⁷ OECD - FAO Agricultural Outlook. <https://www.fao.org/3/cb5332en/cb5332en.pdf>

¹⁰⁸ Rwanda Development Board, 2020. Labour Market Trends Analysis Brief 2016-2020. <https://rdb.rw/wp-content/uploads/2022/07/Labour-Market-Brief-2020.pdf>

¹⁰⁹ Van Keulen, Rik, Rumenera, P., Banjara, G., Colantuoni, M. and Muthamia-Mwenda, J. 2022. Horticulture value chain analysis – Opportunities for youth employment in Rwanda. Rome, FAO. <https://doi.org/10.4060/cb8266en>

women. Api Business Development Company has applied for EU certification in order to export honey to international markets. Nevertheless, the company—which has a capacity of handling 120 MT of honey per year—still faces the challenge of low production, inadequate even in meeting local demand.

Rwanda has a well-established record of generating off-farm benefits from its protected area ecotourism programs. The Project will build on this by supporting sustainable tourism activities around and between the Gishwati and Mukura Forests, as outlined in the GMNP Tourism Master Plan.¹¹⁰ New trails and base camps outside of the parks will generate employment and income opportunities for rural women and youth, as well as attract tourism revenue to help support forest protection and management. In addition, this Project will initiate and train Community Freelance guides to mutually benefit the parks and local communities neighboring Mukura-Gishwati NPs. A freelance system suits the seasonality of tourism, and the demand and sourcing of guides from the local communities stimulates economic empowerment among the young and unemployed living in rural areas around protected areas. This initiative will enable community members to provide a service and generate income, and benefit from the growing tourism industry in Rwanda, which in turn supports conservation over the long term.

In collaboration with district authorities in the CND, who will provide a list of existing cooperatives, the project will select cooperatives based on value chains, number of women and youth, number of members with vulnerabilities and membership size. Furthermore, we will assess selected cooperatives using the Cooperative Assessment Matrix tool (CAM) to identify capacity gaps. Selected cooperatives, participating financial institutions (PFIs), and value chain actors/small and mid-size enterprises (SMEs) (including seedling producers, aggregators, traders, etc.) will receive capacity-building to improve their business skills and access to finance, and to strengthen value chains.

Women play a crucial role in balancing household demands with market production. This is why it is so important to give women access to financial and value chains related services. The project aims to reach at least 50% of the women beneficiaries in the project area. In areas where women are under-represented in value chains, the project will identify and address gender-specific challenges and barriers related to the different value chains and promote increased female representation and active participation in Producer organizations by creating gender committees that will recruit more women, and support women to be part of the leadership committees.

Key sub activities are:

3.2.1.1 Identify and map of vulnerable groups of youth and women within the community

3.2.1.2 Develop and implement strategies to ensure that women, youth, and other marginalized groups benefit from rural livelihoods and financial services.

3.2.1.3 identify farmers and cooperatives involved in selected value chains in consultation with district/local authorities in CND.

3.2.1.4 Assess capacity gaps of identified farmer organizations to strengthen their capacities

3.2.1.5 Identify and develop Nature-based tourism value chain as a means to generate viable economic alternatives for local people living near areas of high biodiversity in the CND.

RDB will co-finance sub-activity 3.2.1.5.

¹¹⁰ Rwanda Development Board (RDB), Rwanda Environmental Management Authority (REMA), 2018. *Tourism Development Master Plan for Gishwati - Mukura National Park*. [Link](#).

Activity 3.2.2 Facilitate and scale up climate-resilient value chain products

The project will conduct deep analysis for identified value chains, namely macadamia, avocado, chili, and French beans. Such analysis will provide evidence of the opportunities and constraints in the selected value chains and will involve mapping out chain operators and service providers at each segment of the value chain to identify existing capacity gaps and orient overall value chain interventions in the project. Furthermore, value chain analysis will enable the project to clearly understand the characteristics of the selected value chain with attention to smallholder farmers, women, youth and vulnerable groups.

To build strong and sustainable value chains, the project will improve information flows within the chain, by strengthening coordination among value chain actors and supporters. This will involve creating value chain platforms in the CND region to increase transparency and fair business relations. The project will focus on activities that strengthen and scale up value chains: for example, by organizing business-to-business (B2B) events, establishing district value chain interprofessional/platforms, and promoting business development services such as swift, cost-effective, and accessible product certification for agribusiness. The project will play an active role by introducing key supporting functions such as financing, value chain information, certification services, etc., and advocate for fair business relations to ensure farmers are active players in value chains and not passive beneficiaries. The project will help strengthen the capacity of producer organizations in selected value chains. Often farmers fail to participate and benefit in value chain activities due to weak and poor organization, resulting from poor managerial skills and weak internal financial management. The project will use different capacity assessment tools like CAM and Scope insight to identify capacity gaps and develop capacity-building plans to ensure that producer organizations are strong and professional and capable of serving their members and building sustainable value chains. The capacity-building trajectory will also focus on encouraging participation by women, youth, and vulnerable groups and increase their access to services. Activities to strengthen farmers and producer organizations will be based on farming as a business (FAAB) concept to orient farmers toward business-oriented agriculture. Synergy will be built with projects such as [KIWP2](#) (Kayonza irrigation and integrated watershed management), a project funded by IFAD where this module is being applied.

In the selected value chain quality standards are an integral element of the core interventions, due to export market requirements. The project will strengthen the capacity of farmers to increase production in terms of volume and quality but also adhere to market standards (for example, global GAP, maximum residual limits (MRLs), etc.) Another challenge affecting the selected value chains in the CND region is access to inputs, especially seeds, and pesticides. This is exacerbated by the international market which requires certain seed varieties and types of pesticides. The project will strengthen the input supply system in collaboration by National Agricultural export board (NAEB) and selected off takers by establishing and supporting agro-dealers and nursery production through technical support and access to finance, to further access to input the project will also focus on building business models that help farmers access inputs via credit guarantee schemes. The project will create strong linkage and business relations between input suppliers, farmers/producer organizations, and markets/buyers to enhance flow of information among value chain actors. Based on experience, the project will ensure that processors/buyers have a say or influence on type, variety, and quality of inputs used by producers to ensure produce marketability. Furthermore, the project will collaborate with other projects like [PSAC](#) (promoting smallholder agro-export competitiveness project) and NAEB (National agriculture export board) to further build a sustainable input supply system. Most value chains are not well developed or are still nascent in the CND region, based on our experience with agroforestry value chains with the TREPA project for instance. In such situations, off-takers, processors, and traders are few and poorly organized, with minimal outreach to farmers. They normally experience challenges in understanding and meeting market standards (i.e., sanitary and phytosanitary (SPS) standards, Global GAP, hazard critical control point certification (HCCP), etc.) which limit their

competitiveness and market scope. The project will improve their capacity to meet these standards, provide other support to access and integrate cold chains, especially for avocado and French beans, and build out grower schemes in which off takers/exporters play a vital role. For MSMEs and start-ups, the project will strengthen their managerial capability for running successful, efficient businesses. Through organizing value chain linkage events (B2B, market linkage sessions, etc.) the project will enhance buyers to reach more farmers and producer organizations.

Addressing rural poverty and food insecurity using agriculture requires investments in off-farm agricultural activities to complement primary production.¹¹¹ Synergies will be created to reinforce direct linkages of small producers to other value chain actors with inputs (seeds and seedlings, fertilizers) and outputs (processors, wholesalers, and retailers).

The project will use the Market Systems Development (MSD) model to enhance value chain development. This will entail multi-stakeholder collaboration with business service providers, financial institutions, community-based enterprises, value chain actors, and farmer cooperatives. In-depth target area assessments will be conducted to tailor business cases for the selected value chains, particularly focusing on identification and engagement of existing farmer organizations and cooperatives, youth and women-owned businesses, and local small and medium enterprises (SMEs, value chain actors) with the capacity for production aggregation, value addition, and market potential to absorb production from smallholder farmers' cooperatives. This localized stakeholder and value chain exploration will serve as a basis for the organization of market linkages to help value chain actors better understand the market, challenges, and opportunities.

Regarding honey value chains, the project will estimate the number of beekeeping cooperatives operating in the targeted region or its surroundings. Members of these cooperatives will be trained in improved honey production techniques and adding value through wax-based products. This will involve establishing honey and wax storage stations and associated processing facilities. Cooperatives will also be an important vehicle for fostering tree growing for improved bee forage. The project will foster interactions between the cooperatives and private companies through business roundtables, trade fairs, and similar events focused on honey and beeswax products and this activity will be co-financed by RDB.

This activity includes in-depth target area assessments to tailor business cases for the selected value chains, particularly focusing on identification and engagement of existing farmer organizations and cooperatives, youth and women-owned businesses and local small and medium enterprises (SMEs, value chain actors) with capacity for production aggregation, value addition, and market potential to absorb production from smallholder farmers' cooperatives. This localized stakeholder and value chain exploration will serve as a basis for organization of market linkages to help value chain actors better understand the market, challenges and opportunities. Key sub-activities are:

3.2.2.1 Analyze identified value chains and promote them among the CND stakeholder networks

3.2.2.2 Strengthen the capacity of producer organizations to improve managerial capacities, farming practices, and gender inclusion in their organizations

3.2.2.3: Establish and strengthen relevant value chain platforms in the CND area and promote business development services to enhance value chain coordination.

¹¹¹ Selasi, J., 2012. *Making Small Scale Farming Work in Sub-Saharan Africa*. In Global Forum on Food Security and Nutrition "FSN Forum", FAO.

3.2.2.4: Build and strengthen the capacity of off-takers (MSMEs, SMEs, and off-takers aggregators) to increase their competitiveness and comply with various standards

3.2.2.5: Facilitate and strengthen access to quality inputs for farmers and producer organizations in the CND regions

3.2.2.6: Establish “CND brand” for honey and wax products

Activity 3.2.3 Facilitate access to output markets for vulnerable farmers

Lack of access to the market is a challenge to smallholders: only 4.2% of agricultural households have access to contract farming ([Agric HH Survey, 2017](#)). Limited access to markets for smallholder farmers is mainly attributed to lack of access to market information, failure to meet market requirements, or undeveloped value chains. Some of the challenges also stem from poor quality and poor produce aggregation. The project will strengthen the capacity of farmers and producer organizations to improve quality and aggregation capacity, and in creating market linkages with off-takers or buyers through market information. The project will promote contract farming and out-grower schemes, where farmers are required to practice market-oriented farming, with the main objective of establishing long-term business relationships with markets/buyers. A focus on financial literacy and community initiatives will also mobilize domestic investment and expertise, facilitate access to financing, incentivize the private sector to create innovative financial products, and support improved data collection and measurement of outcomes.

Furthermore, through producer organizations, the project will develop an inclusive business model to improve produce aggregation and supply coordination and to increase farmers’ bargaining power and influence in formal value chains, particularly for remote/hard-to-reach women, youth, and vulnerable groups. To further strengthen market access, the project will develop innovative market access approaches such as “it takes two to trade”, which focuses on understanding the challenges and weaknesses that limit demand (traders) and supply (farmers). The project will create interventions to eliminate challenges and improve weaknesses to build sustainable market access. The project will also build on the national digital agriculture strategy (NDAS) to introduce digital market access, where farmers access market information through their mobile phones through USSD or IVR interactive voice response. A similar intervention is currently being implemented in the Kayonza district through an IFAD-funded project: Strengthening Agriculture Resilience through Learning and Innovation ([STARLIT](#)). This approach is particularly customized to poor, rural, and hard-to-reach farmers who do not own smartphones.

To implement this, the project will identify and select two buyers/off-takers for each selected value chain based on market needs and link them to farmers and farmer organizations. The eligibility criteria for buyers/off-takers will be developed by RFA in collaboration with CORDAID during the implementation. The project will organize periodic market linkage events and business-to-business (B2B) workshops through established value chain platforms. Additionally, the project will create a clear link between market access and other value chain supporting functions, such as value chain finance, to ensure that farmers have access to working capital for aggregation and marketing. Various models will be developed including tripartite agreements (buyers, producer organizations, and financial institutions) to increase confidence among partners and financial flow.

The activity will work to scale up the market system developed using the Output 3.2 inception report on beneficiary persona and stakeholder engagements to facilitate market system growth within the selected value chains, including bee products, and other women and youth-focused sectors. The activity will work to scale up the market system developed using the Output 3.2 inception report on beneficiary and stakeholder

engagements to facilitate market system growth within the selected value chains, including bee products, and other women and youth-focused sectors such as adoption and sales of fuel-efficient cookstoves.

The project will also build strong business relationships and linkages between input suppliers, markets, and farming cooperatives. This will build common understanding and smoother flow of information to ensure that input suppliers and farming cooperatives respond swiftly to what markets need. The activity will train vulnerable groups and farming cooperatives on “farming as a business” concepts.

Finally, trade agreements will be created between cooperatives and relevant national and regional companies that buy and sell such products. The Project will support and ensure protection of the interests of the cooperatives and their members while negotiating and signing relevant agreements for win-win situations.

Key sub-activities:

3.2.3.1 Develop an inclusive business model to Integrate vulnerable farmers into formal markets

3.2.3.2 Organize microenterprise training and develop and implement green grants programs targeting landless youth and poor, as well as women who are unable to access direct benefits from land-based project activities.

3.2.3.3 Facilitate market access for women, youth, and historically marginalized communities through market linkage sessions

3.2.3.4 Provide technical support on market standards, contract farming, and financial connectivity.

3.2.3.5 Develop various market access channels including digital market access for rural farmers, especially women and marginalized groups

Activity 3.2.4 Scale up use and sales of clean fuel-efficient cookstoves

The project will actively promote the use of energy-efficient cookstoves to benefit the 90% of households in the CND that depend primarily on firewood to meet their cooking energy needs and thus contributing to accelerated forestry resources degradation while causing profound health hazards to the already impoverished community.

This intervention will reduce pressure on forests and farmland by raising household awareness of the differences between high and low efficiency stoves and by collaboration with the local TVETs institute to manufacture affordable efficient stoves that will address the limited availability of high-efficiency stoves in rural markets. Promoting the local fabrication will reduce dependency on cookstoves made outside the country and therefore make them affordable. Also, promoting locally made cookstoves designed to address local needs will lead to full adoption of improved cookstoves and will contribute to sustainable biomass resource use and prevent overexploitation of forest resources, thereby ensuring the success of the forest landscape restoration activities described in Outputs 2.1, 2.2, and 3.2 above.

The project will adopt a market-oriented approach to strengthen the cookstove supply chain in the CND and create strong linkages and business relationships with several ecosystem including the TVET center for training of youth and women SMEs on cookstove manufacturing and repairing, and microfinance institutions (MFIs) to ensure sustainable access and finance of cookstoves business.

As mentioned above, to ensure product quality and affordability to drive uptake, scalability, and business growth and expansion, the project will partner with integrated polytechnic regional colleges to develop a

Tier 2 (for emissions) prototype compatible with rural settings and affordable to rural communities. Youth-led SMEs will be trained on how to manufacture cookstoves and contracted to produce them as well as operating sales points within the community

The cookstove pricing and financing modality will take into account vulnerable groups, including historically marginalized groups mapped and selected in sub-activity 3.2.1.1 (social wealth categories c, d, e, and people with disabilities). For sustainability, the groups will be organized into saving groups and ensure they lend among themselves with the purpose of purchasing the cookstove, the project subsidies may target women-headed households which has contributed over a percentage of the total cost. Additionally, the product design will incorporate reducing health hazards, such as reducing respiratory diseases risks and others linked to climate impacts, contributing to reduction in morbidity and health treatment cost and enabling the vulnerable to have disposable income for investment into productive ventures and this will ensure sustainability of impacts beyond the GCF financing. Moreover, the project will target women-headed households, recognizing the significant role women and children play in firewood collection and cooking. The project will substantiate the time-saving benefits for fuel collection and links to women's decision-making and empowerment with clear indicators.

Considering the lessons learnt from similar projects where hand-outs lead to poor sustainability, this project will promote alternative income sources from fast-maturing crops such as chili and French beans which have ready market in Rwanda and externally. The members of the saving groups will be able to use income generated from the sale of fast-maturing crops (chili and French beans) coupled with savings accumulated from their groups and save progressively to purchase the cookstoves and replacement. The project's technical support to MFIs will build their capacity to develop tailored financial products for cookstoves with affordable repayment terms aligned to revenue streams from short-term crops and group savings, making it easier for farmers to access loans to buy cookstoves.

GCF funds with co-financing from RDB will provide support for the development of ICS prototypes, youth-led SMEs engaged in cookstove production, subsidies for cookstoves, creation and support of savings groups, and technical support to MFIs to develop tailored financial products to facilitate access to ICS by local communities. The project will ensure that the cookstoves are sustainably sourced and lead to minimal indoor air pollution and related respiratory conditions compared with the cooking methods presently in use both during production of the cookstoves and during usage by the individual households.

Key sub-activities are:

3.2.4.1. Promote local manufacturing through partnership with Integrated Polytechnic Regional Colleges (IPRCs) of Karongi and Musanze to help design prototype Tier 2 ICS to address local constraints.

3.2.4.2. Build business and marketing capacity of the cooperatives and youth led SMEs to act as the marketing channels of energy efficient cookstoves to CND community members based on wealth categories.

3.2.4.3 Create a specialized small business development fund, to help local women and youth groups build innovative business models for accelerated distribution of ICS.

3.2.4.4. Promote demand-through raising public awareness on the benefits of ICS by developing and disseminating messages on mass media and in community meetings.

Output 3.3 Financial services & private sector investment increased

Financing challenges and needs are particularly high in clean cooking, mainly for two reasons: (1) due to lack of income, the upfront purchase of clean cooking equipment presents an affordability challenge to most

households, which adds to the monthly fuel expenditures and (2) businesses operating in the sector are often unable to attract financing for investment in the value chain. Serving low-income households also generally yields low returns due to low customer base.

With a particular focus on marginalized communities, and vulnerable groups of youth and women in the CND, the Project will deploy a FSG model to capacitate the communities' productivity and financial resilience. The project will address barriers to access to financial services, specifically for women and youth by building the financial institutions to develop and financial products that meet the differentiated needs of vulnerable women and men in the CND. The project Gender Expert, Youth Focal Point along with the Community Mobilization Specialist and Community Engagement Specialist will provide support to address barriers to access, ensure women's financial skills are developed and there is community and financial institution awareness and support to ensure marginalized groups, including women and youth can access financial services including credit to invest in their businesses including purchase of the cookstoves. Besides this, to address the significant challenges facing MFIs and Participating Financial Institutes (PFI) such as limited understanding of forestry and agriculture sectors investment and the corresponding risks associated with climate change effects on credit repayment, this Project will coach the participating financial service providers to enhance their on assessing the impact of climate change on their clientele and loan portfolio, and ensure equitable access to financial assets by women, youth, and historically marginalized communities. Furthermore, the Project will facilitate local and international impact investors and PFIs to engage in investment and credit provision for farmer cooperatives and SMEs in the selected value chains.

In collaboration with TVET institutes/ICS youth producers and central (RFA, MININFRA) and local authorities (Districts), CND biomass and micro-finance experts will refine and test, for each type of ICS/fuel to be promoted, the category/profile of households to be targeted and the related subsidy/microcredit schemes to be implemented to facilitate stove adoption with minimum concessionality. These schemes will be designed to align with the procedures and rules of identified local partner financial institutions.

In addition, the financial products will be consistent with climate resilience and capacity-building for macadamia, avocado, vegetables, honey and ecotourism value chain actors under Outputs 3.2.3 and 3.2.4. Techniques will be developed to analyze and score forestry and agroforestry value chains that are climate-resilient.

Activities and sub activities under this output will be executed by the Rwanda Forestry Authority (RFA) with the technical support from WCS.

Activity 3.3.1 Facilitate access to finance & private sector investments

To de-risk the selected forestry and agroforestry-related value chains, the Project will organize sensitization sessions for value chain actors and private financial institutions to build a common understanding with regard to the selected value chains. This will also include introduction of monitoring and measurement systems to support the risk and climate impact assessment on the value chains to be undertaken by the financial service providers.

Communities involved in chosen forestry and agroforestry value chains in selected intervention sub-areas will be selected with a particular focus on landless people, historically marginalized communities, and vulnerable groups of youth and women in collaboration with local authorities (districts & sectors) and community representatives.

Furthermore, the interventions will capitalize on the fast outreach of informal saving groups. Saving groups are self-managed and self-capitalized entities that use members' savings to facilitate lending within the

group. They set up clear rules and regulations which are known and adhered to by all group members. Groups consist of a limited number (10-25) of people who have freely chosen to work together, and offer self-managed savings, insurance, and credit services in remote rural areas. Poorer segments of the community often do not have a regular source of income—however, within a savings group, people can save to purchase agricultural inputs, such as seeds, fertilizer or equipment. Savings groups may also provide mutual assistance for group members, such as medical insurance or loans to members in need.

Saving groups have transformed marginalized communities by mobilizing local savings, which provide members with a means to cope with emergencies, manage household cash-flow, build a capital base, and, crucially, re-build social networks, solidarity and trust. Savings group participants often do not have regular sources of income and commonly come from poorer segments of the community. Women are regularly involved in savings groups and have high participation rates. Saving group participants often come from the same socioeconomic background and know each other.

Activities of saving groups include:

- **SAVING**: The group determines a fixed amount for a given period that each group member will bring to the savings account. The allocation and management of these savings can vary according to the needs of the members or is used to support group activities.
- **LENDING**: The group can grant small internal credits to members. Loans are modest and repayable in the short term. Interest rate is fixed by members. At the request of the group, these credits are renewable almost automatically if there is no problem and if the amounts are within a ceiling set by the group.
- **SOCIAL**: The group not only improves the lives of its members financially but also socially as members are given the opportunity to socialize at each meeting and discuss other issues or concerns. The group further builds a small social fund to help group members in distress. Each group agrees on the amount to be saved, usually a smaller amount than the amount for regular savings, in every meeting for the social fund. This money is not loaned, but instead used to pay group member expenses for things such as funeral costs or medical treatment and are only used if agreed on by all group members.

In case of external loans from financial service providers, which most members can't afford, group members agree to mutually guarantee each other (co-guarantee) when taking a loan. If a member does not pay, the group ensures repayment for him/her. The group uses moral pressure to force repayment and therefore there is no requirement for physical guarantees with third parties such as financial institutions. The internal character assessment for group members is important.

The project will strengthen the capacity of saving groups and link them to more formalized financial institutions. The project will further enhance financial education and client protection principles.

This model is more convenient and more accepted by most women who cannot afford collateral loans. As such, this approach will attract more women and will facilitate their formal access to financing over time. Specific attention will also be paid to women's participation in other project financial products that will be developed, such as asset finance in activity 3.3.2.

The project will digitize saving groups, to reduce time and cost spent to access their savings, thereby facilitating transactions with formal financial institutions. Solutions such as "push and pull technology" (i.e., payments using cell phones, commonly referred to as "mobile money") have proven to be useful and convenient for savings groups to send and receive money.

In activity 3.3.1, communities involved in chosen forestry and agroforestry value chains under this project in selected intervention sub-areas will be selected with a particular focus on landless people, historically marginalized communities, and vulnerable groups of youth and women in collaboration with local authorities (districts & sectors) and community representatives.

Sub-activities include:

3.3.1.1 Develop tailored credit assessment tool for selected value chains, while integrating climate impact data

3.3.1.2 Facilitate access to finance for selected value chains of avocado, macadamia, vegetables and ecotourism that are relevant to build climate resilience

3.3.1.3 Develop tailored financial products to meet the needs of women and youth

Activity 3.3.2 Set up and support savings & loan groups, enhance asset-building

To strengthen the capacity of local smallholder farmers in financial literacy and access to finance, farmers' cooperatives will be assisted and supported in setting up saving groups. Savings groups are a proven and effective approach to the financial inclusion of the most vulnerable and are increasingly recognized as one of the strongest forms of community organization available due to the high levels of social capital generated from its savings approach. Saving groups create a platform for linkage with financial institutions and other business development services. Further, savings groups are recognized for enabling high levels of female inclusion. As groups members co-guarantee each other for accessing a loan and groups are strong, it forms low lending risks for MFIs since members mutually reinforce each other's repayments.

Once grouped, small savings group members, especially women, grow from accessing internal group loans to accessing formal financial institution loans. As members' bankability and trustworthiness increases, members are able to start accessing individual loans, gradually building to larger loans and are able to open individual savings accounts at financial institutions.

Members of a saving group will be self-selected and the saving group will be managed by its members.

Self-selection will be based on the following:

- Knowledge of one another and similar economic background
- Close proximity to where the meetings will be held
- Reputation for honesty and reliability
- Ability to attend all meetings on time and follow rules
- Have a cooperative personality
- Be able to buy at least one share each week (or time set by the group)
- Be able to repay loans on time

Most saving groups are informal but transparent, democratic, and structured groups that have constitutions and clear rules. Members do not enter into contractual arrangements for their membership. Record-keeping is based on simple passbooks in which each member's savings and loans are recorded; all transactions are done in front of the members during group meetings.

Farmer savings groups (FSGs) will be created within cooperatives and farmers will decide which group to join. Groups will set up governing rules and conditions (savings amount, meetings, etc.), and criteria for new members who want to join. Particular attention will be given to social inclusion (including vulnerable communities and landless people).

Youth and women-owned businesses selected for activity 3.2.2 will be linked to targeted FSG beneficiaries for supplying inputs (modern beehives, seedlings, etc.) and energy-efficient cook-stoves. Furthermore, intensive capacity-building and technical support will be delivered to all groups with emphasis on financial literacy, entrepreneurship, and gender mainstreaming. The activity will also help link savings groups to PFIs.

For Activity 3.3.2, in collaboration with local authorities, the project will target existing or potential youth and women-owned businesses that can intervene at any level of selected value chains in activity 3.3.1. The project will ensure the integration of vulnerable and landless people.

Key sub-activity:

3.3.2.1 Establish/form saving groups for access to finance

3.3.2.2 Provide organizational, technical, and financial capacity-building to women, youth, and CSOs (such as cooperatives or VSLA) with focus on financial literacy and provide long-term coaching to beneficiaries

3.3.2.3 Support women, youth, and other marginalized groups to strengthen savings and loan groups and register for MFI accounts

Activity 3.3.3 Build the capacity of financial institutions to serve targeted value chains and communities

Many financial institutions demonstrate little knowledge about the science of agriculture, forestry, climate change and associated risks. A study by the International Food Policy and Research Institute (IFPRI)¹¹², noted that many agricultural activities are located in remote, widely dispersed regions and that financial institutions find it impossible to provide cost-effective and affordable services in this context. Hence for this specific activity, PFIs will be supported to access and adopt a digitalized forestry and agri-loan assessment tool, and to understand climate impacts on their portfolio to actively integrate and promote climate relevant financial products in their product offer.

The Project will work with these institutions by developing and refining appropriate financial products and financing operations that are better understood and user-friendly for smallholder farmers and vulnerable groups. The Project will also collaborate with selected PFIs to build their awareness of prospective farmer and low-income group clients and their exposure to climate impacts.

Key sub-activity:

3.3.3.1 Capacity-building of PFIs

Activity 3.3.4 Facilitate learning & knowledge sharing

To deliver an adaptive program that cooperates with market actors on inclusive and sustainable business models that have a positive impact on smallholder farmers, while generating and sharing lessons on which business models work and which do not, the project will adopt the program-embedded reflection and learning (PERL) model for facilitating learning and knowledge sharing to popularize the CND unique approaches to restoration and environment management.

By **proving**, we seek to know and show what works and what does not work by demonstrating results at all levels, generating deeper insights into farmer benefits and the effects on the wider ecosystem, and building business model sustainability.

¹¹² Kloeppinger-Todd, R. and Sharma, M. eds., 2010. *Innovations in rural and agriculture finance* (Vol. 18). Intl Food Policy Res Inst.

By **improving**, we learn by doing to enhance project adaptation and improvement, build a learning culture through discussion and reflection, generate knowledge to support project effectiveness, make comparisons over time, and enhance data quality procedures.

Through **sharing**, we disseminate high-level results and learnings to relevant audiences, contribute to visibility and communication, showcase successful, sustainable and scalable business models, and generate best practices.

Under this activity we will go through the following actions to prove, improve and share:

Capture - We will establish a KML framework and support case studies and discussion by project staff; improve data collection; collect insights from field visits; write human interest stories; hold quarterly program reviews; measure client satisfaction; and collect information on business models, market uptake, and ecosystem changes.

Synthesize - We will organize discussions and quarterly program reviews with program staff and stakeholders for assessment, interpretation, reflection, and learning from results. Outputs will include reports, briefs, and articles.

Disseminate - We will contribute to and support the project, presenting its achievements in newsletters, conferences, webinars, blogs, publications, media events, partner workshops and sector network presentations.

Take action - We will have regular discussions and reviews in which we revisit the work plan and processes in light of results and insights (e.g., based on outreach data, field visits, case studies, partner feedback, quarterly reports, changes in the ecosystem, etc.), and adapt operational planning when needed (within and outside of the normal project planning and reporting cycle).

Refine agenda - We will help to align the project intervention strategy with a shared vision for success.

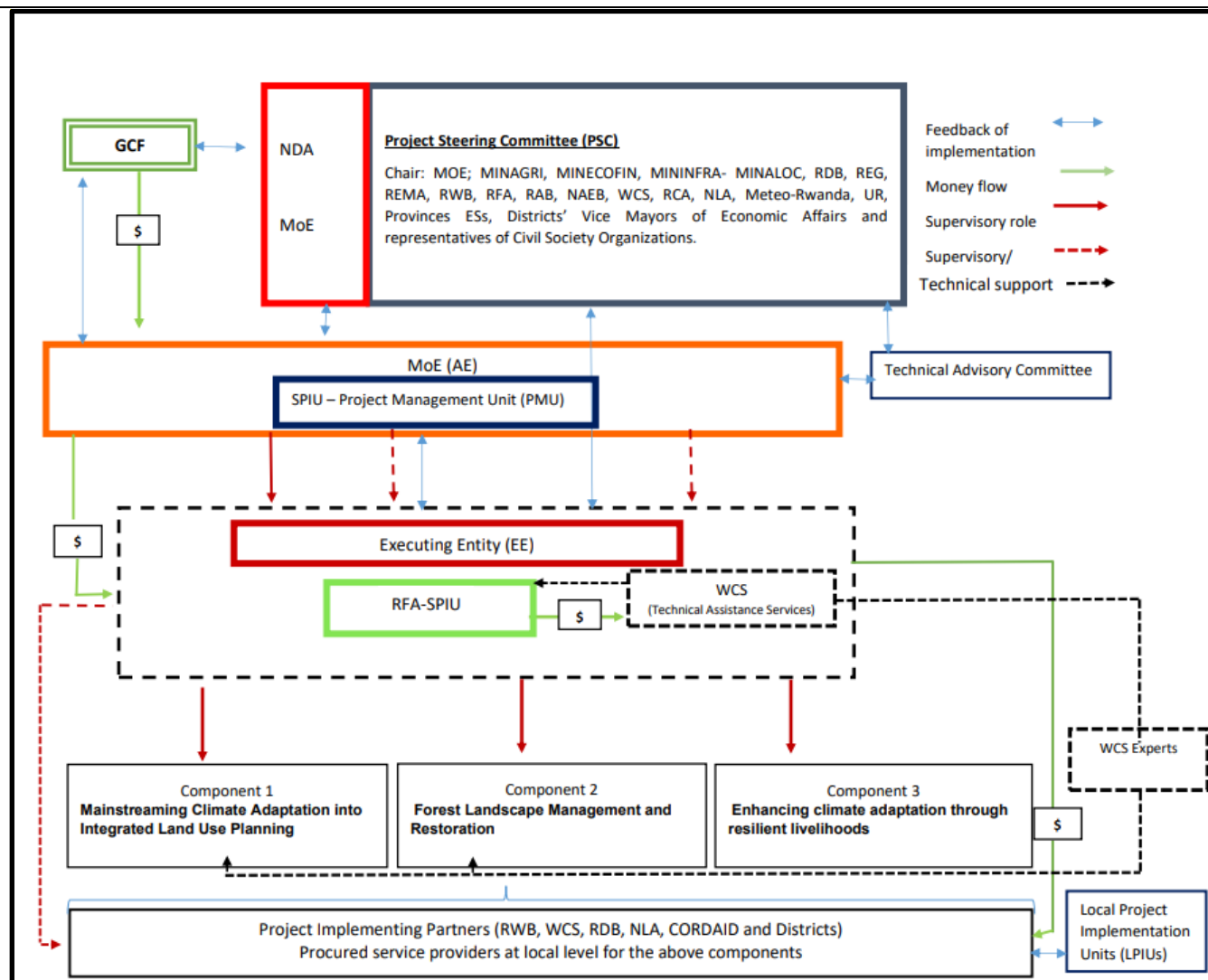
The table below summarizes the project activities to be implemented in Congo Nile Divide.

	Core PA Nodes	Stepping Stones	Landscape Linkages	Broader Farming Mosaic	Across the CND
	Volcanoes NP and Buffer	Gishwati NP and Buffer	Mukura NP to Gishwati NP Linkage	Nyungwe to Mukura Broader Farming Mosaic	
1.1.1 Synthesize & disseminate information on value of natural forests & ecosystem services					
1.1.2 Develop climate-resilient landscape land-use plan					
1.2.1 Create interagency taskforce institutionalizing integrated landscape planning & policy					
1.2.2 Build capacity for spatial planning in national agencies re climate change					
1.2.3 Develop an effective forest monitoring system to underpin forest management decisions					
2.1.1 Facilitate revision of PA management plans to address climate risks					
2.1.2 Establish long-term plans for CND financial sustainability post-GCF					
2.1.3 New fire management curriculum developed and operationalized in PAs and adjacent landscapes					
2.2.1 Secure key remaining natural areas outside PAs					
2.2.2 Restore natural forest cover in & outside PAs including riparian linkages					
2.2.3 Promote silvo-pastoralism with indigenous trees in Gishwati Pasture Stepping Stone areas					
3.1.1 Restore high slope areas (>55%) as protective forests					
3.1.2 Develop on-farm agroforestry for high caloric and indigenous tree species					
3.2.1 Develop agroforestry related value chain for markets access					
3.2.2 Facilitate & scale up capacity, value addition & marketing of select climate-resilient value chain products					
3.2.3 Facilitate access to input & output markets for vulnerable farmers					
3.2.4 Scale up marketing, production, sales, use of fuel-efficient cookstoves					
3.3.1 Facilitate access to finance & private sector investments					
3.3.2 Set up & support savings & loan groups, enhance asset-building					
3.3.3 Build capacity of financial institutions to serve targeted value chains & communities					
3.3.4 Facilitate learning & knowledge sharing					
4.1 - Performance monitoring plan developed					
4.2 - Project monitored and evaluated; lessons learnt integrated into adaptive management processes					
Primary Focus					
Secondary Focus					
Some activity					

Table 4: Summary of project activities (place based and non-place based) to be implemented across the Congo Nile Divide.

B.4. Implementation arrangements

The proposed project will be implemented in five years, from 2024–2029. The Ministry of Environment (MoE) is the Accredited Entity (AE) for the GCF and MoE will enter into a funded activity agreement with the GCF once the project is approved by the Board.



The governance arrangements for the project at the national level are designed to build upon the institutional structure of MOE (the accredited entity) and RFA (the Executing Entity).

Accredited Entity: The MoE on behalf of the Government of Rwanda is a Direct Access Entity of the GCF and will assume the oversight role on implementation, management, monitoring, evaluation and reporting on the CND Program elaborated in the FAA. The MoE will enter into a subsidiary agreement with the Executing Entity (EE) with clear roles and responsibilities for each party. Under the Ministry of Environment there is a Single Project Implementation Unit that has a number of staff charged with resource mobilization and providing oversight over different projects implemented through the Ministry of Environment. The MoE/SPIU has a track record of experience in project oversight, management, monitoring and evaluation of projects and programs including the GCF funded FP073 that is currently at the mid-term level of implementation with over USD 32.8 million, managed over USD 3.8 million for the GCF funded PPFs, overseen the implementation of USD 10 million from Adaptation Fund. The MoE is currently managing the World Bank Funded project preparation advance of USD 6 million and has been implementing since 2018 a program funded by the UNDP focused on Strengthening Capacities of Environment and Natural Resources Sector for a Green Economy Transformation of over USD 6 million. Additionally MoE assumes its oversight role through chairing of the Project Steering Committee for different projects implemented under the Environment and Natural Resources Sector as a whole namely TREPA, a GCF-funded project, Rwanda Urban Development Project Two funded by the World Bank and NDF, an Early Warning System project

(funded by LDCF), the Landscape Approach to Forest Restoration and Conservation (LAFREC) programme (funded by GEF/World Bank), Green Amayaga Project (funded by the GEF) and the “Building resilience of communities living in degraded forests, savannahs and wetlands of Rwanda through an ecosystem management approach” programme (funded by LDCF). A coordination platform between approved GCF projects will be established to identify, maximize linkages and ensure complementarity through learning and sharing best practices and failures arising from the implementation of existing GCF projects. Quarterly meetings will be organized under the supervision of the MoE between project directors and staff responsible for monitoring and evaluation to share lessons learned.

Project Financial Flow and AE Roles

Funds from GCF will be provided to the MoE GCF account (as the AE), under the terms of the FAA. MoE will be responsible for financial management and accountable for the use of GCF resources. Financial transactions will be subject to annual audits undertaken by internationally certified auditors. In addition, as the AE, MoE will: i) ensure that the project is executed in accordance with GCF standards; ii) supervise, oversee and manage the implementation of project interventions; iii) report on Project progress; and iv) ensure that Project activities are well coordinated and aligned with countries' national priorities. MoE will provide oversight to the project consisting of a) entering into subsidiary agreement with the EE; b) managing and disbursing GCF funds to EE on the approved work plans and budgets; c) reviewing financial expenditures and progress reports; d) overseeing Project implementation in accordance with the Project document and Annual Work Plans and Budgets, agreements with co-financiers and executing entity rules and procedures; e) providing technical guidance to ensure that the appropriate technical quality is applied to all Project activities; f) providing financial reports to the GCF for Project funds received; g) ensuring that the project complies with the terms agreed in the Project's respective FAA as well as the AMA signed between MoE and the GCF; and h) undertaking regular annual supervision missions according to the MoE's guidelines and convene Project Steering Committee (PSC) meeting twice a year aimed to review progress and approve work plans. The PMU hosted under the Single Project Implementation Unit of the Ministry of Environment will manage, monitor and report on day-to-day implementation of the Project activities by the EE. The EE will execute the tasks within the overall project management structure: a) implementing day-to-day activities as per the project work plan and budget, including the Environmental and Social Management Plan; b) undertaking procurement activities as per the agreement signed by GCF and AE as well as the sub-agreement signed by the AE and EE; c) managing contracts of suppliers and services providers; d) hiring and managing project staff relevant to the EE-managed project areas; e) implementing activities as per the project work plan; and f) carrying out financial and technical monitoring of activities, including the achievement of outputs and outcomes the EE is in charge of. EE will enter into Memorandum of Understanding with different implementing partner agencies that will provide technical know-how to the implementation of the project activities. The implementing partners including but not limited to RWB, WCS, NLA, RDB, CORDAID and Districts will provide technical and operation services articulated and agreed in respective MoUs. For the Implementing partners RFA will be paying service compensations and at District Level community approach will be used. Procured service providers will be contracted in accordance with MoE rules and regulations of procuring goods and services.

Project Steering Committee (PSC): An inter-ministerial Project Steering Committee (PSC) chaired by the Permanent Secretary will be established by MOE to serve as the project's coordination and decision-making body. The PSC will ensure the project delivers its outputs and achieves its outcomes. The Committee will meet on a semi-annual basis to review project progress and evaluations, facilitate implementation (ensuring the necessary resources and support are provided in a timely manner) and provide guidance to the PMU and EE/PIU. The Steering Committee will also facilitate effective coordination between the key stakeholders at the national and district levels and ensure the project aligns with Government strategies and programs.

The PSC will reflect the multi-sectoral nature of the project and will include senior-level representatives from GoR and other relevant Stakeholders. The PSC will be chaired by the Permanent Secretary, MOE or any Senior Manager from MoE delegated by the Permanent Secretary with representatives from: MINAGRI, MINECOFIN, MININFRA- MINALOC, RDB, REG, REMA, RWB, RFA, RAB, NAEB, WCS, RCA, NLA, UR, Provinces ESs, Districts' Vice Mayors of Economic Affairs and representatives of Civil Society Organizations.

Project Management Unit: It is under the MoE/SPIU that a dedicated Project Management Unit (PMU) of the CND program to supplement the existing MoE/SPIU structure for this complex program will be established for day-to-day management of the program interventions. The Project Management Unit (PMU) will be headed by the Project Coordinator who will serve as the secretary and the rapporteur of the PSC. The Project Coordinator and other PMU staff will be recruited as per the MoE/GoR recruitment practices. The Project Coordinator is responsible for the day-to-day operations of the PMU within approved project funded activity agreement and any other guidance provided by the AE and PSC including the tasks of managing and monitoring project risks initially identified and submitting new risks to the AE and PSC for consideration and decision-making on follow-up actions and updating the status of risks in the project risk log. The PMU Project Coordinator will report to the overall MoE/SPIU- Coordinator as the immediate supervisor and to the Permanent Secretary as the secondary supervisor.

The PMU will provide to the MoE/SPIU coordination and to the Chair of PSC with quarterly progress reports and close its operations when the final project evaluation report and other documentation required by the GCF have been completed and submitted to MoE. The PMU will be staffed with required professional and technically qualified personnel selected and recruited following the MoE/GoR procedures. In all, the PMU will strive to maintain a lean management structure. It will comprise a Project Coordinator, Monitoring & Evaluation (M&E) Specialist, E&S Specialist, Procurement Specialist, Gender Specialist, Financial Management Specialist, Communication and Outreach Specialist, Office Manager, and Driver as a dedicated team to manage the EE for the whole program.

Executing Entity: RFA will serve as the EE for the Project through a subsidiary agreement (SA) signed with MoE. The SA will establish clear roles and responsibilities for both parties for the delivery of the project activities, the schedule and conditions for installments, the determination of the prevailing fiduciary standards and the terms and conditions for arbitrations and termination of contract. The SA will include specific obligations for the EE on project execution, financial management, personnel administration and reporting, as well as on arbitration and liability terms.

Rwanda Forest Authority: RFA was established by the Government of Rwanda by the law No 72/2019 of 29/01/2020 with the mission of ensuring the growth of forest resources, their management and protection for sustainable development purposes. RFA is an affiliated agency under the MoE with a separate legal personality and a budget entity of the Government of Rwanda. RFA is currently implementing TREPA; a GCF funded project as one of the executing entities, COMBIO Project funded by the Swedish government, executed Forest Management and Biomass Energy project (FMBE) implemented in the Northern province; Sustainable forestry, agroforestry and biomass energy management for climate resilience in Gatsibo and Gicumbi districts; and Improving the Efficiency and Sustainability of Charcoal and Wood Fuel Value Chains in the North-western part of Rwanda supported by the Nordic Development Fund. RFA has three main divisions namely Forest Research Division, Forest Management, Administration and Finance and an established Single Project Implementation Unit (SPIU), composed of staff with the required financial management skills where the CND Staff will be hosted. The Director General of RFA as the Chief Budget Manager, the RFA SPIU Coordinator together with Program Manager and a Financial Management Specialist will be responsible for the day-to-day financial management of the Project. RFA SPIU will open a separate account at the National Bank of Rwanda (BNR), from which payments of the Project activities will

be made from. RFA follows the same financial and procurement policy as the one of the Ministry of Environment.

WCS Technical Assistance (TA) Services: As one of the project implementation partners, WCS will provide technical assistance to the EE (RFA) through an MOU signed with RFA. The MOU will detail all the activities that WCS will be directly supporting during the implementation of this project. The TA services will include advising RFA/SPIU, procuring and supervising international experts and consultants, and procuring materials to support the execution activities. Relevant technical experts from WCS will be embedded within RFA to advise SPIU on day-to-day project management to support the implementation of activities under outputs 1.2, 2.1, and 2.2. These experts will support RFA's technical teams and strengthen their capacities in different aspects of the project implementation. Furthermore, WCS technical assistance services will include supporting the development of ToRs and reviewing works of international consultants to ensure high quality deliverables.

CORDAID Technical Assistance Services: CORDAID will provide technical assistance to RFA mainly on output 3.2 and 3.3 through an MOU signed with RFA.

Other Government of Rwanda implementing partners will provide implementation support and RFA will enter into specific MOUs with these partners based on the type of support they will provide.

Table 5: Summary of project outputs. The EE will be responsible for all activities and sub-activities of the project, receiving support from technical implementation partners.

Outputs	Executing Entity per output	Technical Implementation Partners
Output 1.1. Landscape-wide land-use plan developed for climate-resilient livelihoods & forest ecosystems, integrating district development strategies (DDS)	RFA	WCS and NLA
Output 1.2. Local & national institutional capacities strengthened to integrate biodiversity & climate risks into land use planning & management	RFA	WCS and NLA
Output 2.1. Protected areas management effectiveness improved to address climate risks and adaptation	RFA	WCS and RDB
Output 2.2. Natural forest cover restored; biodiversity connections established	RFA	WCS
Output 3.1. Farming methods enhance productivity, reduce erosion & flooding risks, contribute to ecosystem services, and support connectivity	RFA	RWB

Output 3.2. Rural livelihoods generate alternative incomes and reduce pressure on forests	RFA	CORDAID
Output 3.3. Financial services & private sector investment engagement	RFA	CORDAID
Monitoring and Evaluation	PMU	

Technical Advisory Committee (TAC) will be responsible for ensuring technical guidance to both the project execution team and the PSC. The TAC will have technical representatives from all project stakeholders including MINAGRI, MINECOFIN, MININFRA, MINALOC, RDB, REG, REMA, RWB, RFA, RAB, NAEB, WCS, RCA, NLA, UR, Districts representatives and representatives of Civil Society Organizations. The TAC will be engaged in providing technical support and quality assurance of the project implemented activities as well as facilitating effective implementation and mainstreaming of project interventions beyond the life of the project. The institutional stakeholders will nominate relevant technical experts to the TAC, thus bringing expertise from their respective institutions to make technical contributions to the project implementation based on the integrated nature of the project.

The project will use the GoR fund management modalities based on an annual work plan developed by the PMU together with EE and subsequently approved by the PSC. The Annual Work Plan (AWP) indicates which activities should be covered at PMU and EE levels as per the allocated respective budgets. The EE, together with support from implementing partners, will establish Local Project Implementation Committees (LPICs) specific to the zones of intervention in the Western and Southern Provinces of Rwanda. Two LPICs will be established; one for the Southern province and another one for the Western province for the coordination of activities in Western (GMNP) and southern (NNP) Rwanda to ensure synergies and complementarity between partners. These teams will work closely with the Joint Action Development Forums (DJAF) based in each district located in these respective zones. LPIC will be composed of representatives of districts in charge of natural resources management, RDB, CSOs, and conservation organizations, women and youth councils and JADF. This arrangement will ensure that a) project delivery can proceed securely and efficiently despite the complex governance framework; and c) government partners play a leading role in project delivery and capacity development. PSC will provide policy direction and guidance to improve the coherence and efficiency of this innovative approach to be developed as an up-scalable model. At the local level, the EE and its service providers will coordinate closely with districts to ensure smooth local level coordination in project implementation, provide ownership and ensure sustainability. The District Executive Directors (DEDs) are expected to play an active role in project implementation, facilitation and monitoring and play a key role in the grievance redress mechanisms, as described in the Environmental and Social Management Plan of the project.

B.5. Justification for GCF funding request

Rwanda has achieved remarkable social, political, and economic progress over the past two decades. The Rwandan economy has grown well above average compared to its peers and the rest of the world, achieving growth rates of more than 8% regularly, with a 9.4% growth rate in 2019¹¹³. However, the economy which is

dependent on nature-based tourism and agriculture is extremely vulnerable to external shocks, including climate change.

Climate change is likely to increase variability in crop yields and agricultural production, cause severe flood damage to physical infrastructure, reduce labor productivity, and impact demand for tourism¹¹⁴. The recent Country Climate and Development Report (CCDR) estimates that if these risks materialize, Rwanda's GDP levels can drop by 5–7% below baseline in multiple years by 2050¹¹⁵, with negative impacts on private consumption, exports, and government revenues. By strengthening resilience to climate change, Rwanda can better prepare to manage the adverse impacts of climate change. Implementing the commitments articulated in the NDC is estimated to require new investments of US\$11 billion, of which close to US\$7 billion is conditional on new financing.

Rwanda is classified by the U.N. as a Least Developed Country (LDC)¹¹⁶ and lacks sufficient public financial resources to implement its climate change and resilient green development objectives. The implications of climate impact for Rwanda demands concerted action and support both locally and internationally to enable the mainstreaming of climate resilience into public, private and community sectors at the pace and scale required. This Project will allow Rwanda to implement its NDC commitments and pursue its aspirations under its National Forest Policy and Climate Resilient Green Economy plans for reversing trends of high deforestation, allowing for demonstration of sustainable reforestation models and reductions in demand for wood fuel. Rwanda has made an impressive commitment to using forests and forest protection as part of its strategy to mitigate the impacts from climate change, and to provide benefits of biodiversity conservation and livelihood improvements.

Yet, vulnerable citizens' needs, coupled with fragile landscape and forest restoration, will not be met solely with national resources in the rapid timeframe needed to arrest loss of economic assets for households and prevent further degradation of landscapes. Specific investments in awareness and forest governance, training at scale required to increase the resilience of CND communities to the impacts of climate change, and alignment and coordination of support on undertaking climate resilient restoration activities are not currently being developed by the GoR. The GoR does not have the financial means to implement all project activities identified to improve climate resilience and requires technical assistance and support. While project partners are contributing co-finance for this project implementation, this represents only a fraction of the resources required for implementation. The GCF grant investment can help initiate, expedite, and catalyze these critical interventions and assist with alignment of future climate resilience efforts. MoE is requesting grant funding for this Project due to Rwanda's status as an LDC, insufficient alternative finance options, and because the proposed activities will provide a public good on which the livelihoods of many of Rwanda's rural poor depend. The high levels of poverty among project beneficiaries in the CND as identified in the Feasibility Report necessitate the use of grant funds rather than a reimbursable financial option. While the GoR will contribute resources for the implementation and sustaining climate resilience efforts beyond the 5-year Project, this represents only a partial share of implementation resources. The GCF funding will crowd in additional private sector capital, such as efficient cookstoves and forest plantations, and will allow the government to cost-efficiently implement the adaptation capacity-building training and practices to ensure that CND landscape is restored, and climate resilient landscapes maintained.

¹¹⁴ "World Bank Group. 2019. *Rwanda Systematic Country Diagnostic*. World Bank, Washington, DC. World Bank. <https://openknowledge.worldbank.org/handle/10986/32113> License: CC BY 3.0 IGO."

¹¹⁵ World Bank Group. 2022. *Rwanda Country Climate and Development Report*. CCDR Series;. World Bank, Washington, DC. World Bank Group. <https://openknowledge.worldbank.org/handle/10986/38067> License: CC BY-NC-ND 3.0 IGO.

¹¹⁶ United Nations Department of Economic and Social Affairs, 2021. Least Developed Country Category: Rwanda Profile.

The lack of private capital structure, prevalence of subsistence farming, and inadequate market for agriculture and forested products, is the primary cause of the request for grants, not loans as little surplus products are produced and traded, and thus will be inadequate to secure or service any loans. The prevailing market conditions necessitate GCF grant finance, rather than loans, to raise the capacity and bridge the financial and technical gaps until the restoration and value chain models are brought to financial viability that can be sustained by private sector. Furthermore, our Economic and Financial Analysis (EFA) shows that the direct returns on investment in landscape restoration activities are too low to make them financially attractive or feasible for farmers, even if they had access to credit. Without GCF support, lack of private financing, large payback period, uncertainty and transaction cost of getting the loan, and upfront cost and benefits realized over a longer period of time reduces the financial attractiveness of climate resilience investments. It is unlikely that there will be a financial return on investment within the Project time frame to repay loans. The issue of societal benefit investment is accentuated by the fact that there cannot be direct uptake of benefits to communities from restoration within National Parks.

Serving low-income households yields low returns and the community lacks financial education and capacity to link them to formal private institutions. Most formal financial institutions view these as high risk and high cost, as the transactions are often small and the clients usually have neither physical collateral nor the education needed to organize their finances. Without the value chain development for potential revenue generating streams such as honey, tea plantation tourism, and efficient cookstoves, the private sector investments are not occurring. Henceforth, these potential revenue generating streams proposed in the project are not sustainable on their own and the GCF grant will not only provide financing support but will be critical in creating enabling opportunities for market development beyond the project to catalyze private investments. The full concessionality funded by GCF grants is requested so that an enabling environment can be created, and the project can overcome affordability challenges through developing a system of community co-guarantee. This will also go beyond the neoclassical economics model and focus on shifts in social dynamics that move the community as a whole to escape the poverty trap and increase financial resilience. The full concessionality for potential revenue generating streams will be passed on to end beneficiaries in the CND and the project will strengthen the capacity of community organizations on quality improvement and aggregation capacity, create market linkage with off-takers or buyers, and attract finance for development, marketing, and distribution.

Planting indigenous species, well-suited to the climatic conditions of the CND, is likely to increase the health and resilience of forest plantations and thus the erosion control benefits that these forests provide. However, when establishing protective forests, most existing projects in the CND use exotic species because they are fast growing, widely available, and low cost¹¹⁷. GCF grant resources are required to support restoration to help build natural capital (restored forest) which can be used sustainably and provide larger ecosystem benefits in the future.

The financial Net Present Value (NPV) of \$34.48 million for agroforestry, \$2.98 million for protective forests, and \$0.31 million for silvopasture intervention and economic NPV of \$58.93 million for agroforestry, \$30.22 million for protective forests, \$0.97 million for silvopasture, and \$10.59 million for energy efficient cookstoves interventions for 20-year analysis period when GCF support for the Project is considered¹¹⁸. These results are indicative of the investment in time and resources required to increase climate resilience, associated benefits, and restoration of landscape and forest productivity. Furthermore,

¹¹⁷ Mukuralinda, A., et al, 2016. *Taking to Scale Tree-Based Systems in Rwanda to Enhance Food Security, Restore Degraded Land, Improve Resilience to Climate Change and Sequester Carbon*. PROFOR, Washington D.C.

¹¹⁸ The provision of energy efficient cookstoves is a non-income generating intervention and is expected to have no financial cash inflows and no income generating opportunities, then no financial analyses is needed. The cookstoves analyses was pursued only at the economic analysis level.

agroforestry, restoration, and forest management require up-front investments that take 10-30 years to mature fully. However, these future benefits are depressed by the use of a high discount rate that downplays the importance of long-term investments. The discount rate considered in our economic analysis is 13%, which is Rwanda Central Bank's yield for a 20-year treasury bond, as issued in August 2021. This is the rate at which the government can borrow to fund equivalent investments in the absence of grant financing. For financial analysis, we used the discount rate of 15.44%, which is Rwanda Central Bank's lending rate for July 2021. Consequently, the GoR seeks maximum concessionality from the GCF for the urgent adaptation actions proposed under this Project. Without GCF funding, under a business as usual (BAU) scenario, inadequate technical and financial capacity at the national and local level will continue to hinder the efforts of rural communities in the CND to adapt their livelihoods. Consequently, the vicious negative cycle of unsustainable forest use and ecosystem degradation will persist, reducing the supply of goods and services which buffer impacts of climate change on rural communities, compounding their vulnerability. GCF funding is therefore vital to initiate progress towards a complete paradigm shift in GoR's approach to adapting the livelihoods of rural Rwandans to climate change.

The Project cost per direct beneficiary of \$31 (\$39.86 with co-funding) and 20-year economic rate of return of project interventions for agroforestry (49%), protective forests (72%), and silvopasture (51%) highlights its viability. The Economic rate of return for energy efficient cookstoves could not be estimated as there is no change in the sign in the net cash or resource flows from positive to negative over the analytical period; and the economic NPV of \$10,585.675 for this intervention highlights its feasibility. The cost of restoring natural forest for providing both adaptation and mitigation benefits is significantly greater than the cost of simply planting trees for mitigation. Restoration of natural forest in Rwanda is essential for sustaining the critical ecosystem services on which vulnerable rural communities and the national economy depend. The capital market in the country is weak and provides limited opportunities. It has low levels of direct investment which further limits the opportunity for financing. The country's capital markets currently are not able to serve the needs of rural and resource limited beneficiaries. The project addresses market failures and minimizes the risk of disrupting or unduly distorting markets or crowding out private finance, including new entrants, as well as other public investments. Taking the above-mentioned points into account the concessionality that the GCF provides is justified.

B.6. Exit strategy

The Project will establish a new inter-agency and local-to-national-scale spatial planning framework and capacity for an integrated approach to multi-sector analysis and planning. This will enable the analysis of land-use trade-offs that explicitly consider the value of ecosystem services, climate risks and risks of ecosystem degradation. This represents a radical shift away from business-as-usual and will have long-term, far-reaching implications for policy and financial planning in Rwanda. It will provide a spatially explicit framework for mainstreaming climate risk considerations across all sectors while also providing a mechanism for quantifying harmful financial flows to biodiversity from multiple sectors.

In addition, the multi-scale spatial planning system will be able to strengthen and link directly to the system under development by Rwanda's Ministry of Finance and Economic Planning (MINECOFIN) for policy coordination to integrate climate risks into fiscal planning, with support from the IMF Resilience and Sustainability Trust. Further, it will support the ability of both national and local governments to plan and implement the new disaster risk reduction strategy in a spatially explicit way, which previously has not been possible.

Mainstreaming climate change adaptation metrics. The Project will integrate climate resilience metrics into land use planning, district development strategies and annual performance contracts and harmonize cross-

sectoral monitoring and reporting mechanisms. This will lead to a systematic consideration of climate change risks and adaptation in policy planning that will be sustained beyond the Project. An MoU between key government agencies will operationalize the cross-sectoral planning task force, which will be supported by a ministerial order, giving it the mandate and financial support to operate beyond the life of the Project.

Rwanda has adopted a private-public partnership (PPP) business model to make its national parks operations financially viable. However, in Rwanda, as in the rest of the world, national parks and the biodiversity they are intended to protect are threatened by climate change and by the interaction of climate change with other direct anthropogenic threats. By “climate proofing” its national parks—i.e., by establishing a technical unit within RDB, by explicitly incorporating climate change considerations into the management plans of national parks, including climate-wise functional linkages, such as the one this project will create in the CND, and by training rangers and other park personnel to proactively manage climate-change-related threats—Rwanda will ensure that the ecosystem service benefits of the country’s biodiversity, including its national parks, will be sustained beyond the life of the Project. Specific finance instruments—such as payment for ecosystem services (PES) and structured impact conservation bonds to ensure ongoing revenues and benefit-sharing with surrounding communities—will be identified and implemented before the end of the project, in a manner consistent with the Biodiversity Finance Plan that the GoR has recently developed with support from BIOFIN. The Project through RFA will hire a Technical Advisor, to work with RDB, MoE, REMA, FONERWA and MINECOFIN to ensure that a portfolio of sustainable finance mechanisms is designed and implemented before the end of the project period to ensure the long-term sustainability of the Project outputs.

Strategies for Long-term Financial Sustainability. Two specific examples of the Government of Rwanda’s commitment and approaches to financial sustainability that will be key parts of the Exit Strategy are the Tourism Revenue Sharing Program (TRSP) of RDB and the Community Adaptation Fund of FONERWA, the Rwanda Green Fund.

Tourism Revenue Sharing Program. The long-term success of the three protected areas in the CND depends, to a substantial extent, on the support of communities living around protected areas. In recognition of this, in 2005 Rwanda established a Tourism Revenue Sharing Program, allocating 5% of gross tourism revenue to local communities. Currently Rwanda and Uganda are the only two African countries with formal tourism benefit sharing programs. RDB recently completed a 15-year review of its Tourism Revenue Sharing Program (Snyman et al. 2023) and it has increased the revenue sharing allocation from 5% to 10%.

The primary challenges identified in the 15-year review are increasing the effectiveness of community engagement, reaching the most vulnerable community members, and ensuring that the TRSP meets community needs. The GoR is interested in adapting the TRS Policy to more effectively engage the communities living around the PAs in the TRSP and to allocate more funding to directly enhance the lives of Rwandan citizens living around PAs. In this regard the Project will work with RDB and district governments, to increase transparency in the Program, ensure that the most vulnerable community members benefit from the TRSP and to link the cooperatives and youth-led SMEs to the Program. This will include working with RDB to use the insights from this project to revise the TRSP revenue allocation model and eligibility to create greater resilience among communities. The 15-year review recommended that the following percentages be used for the revenue allocation going forward: 70% for livelihood projects; 25% for infrastructure support; and 5% in an emergency fund. However, there is a need for input from this Project to ensure that the specific types of projects to be supported most effectively increase community resilience. This will ensure long-term benefits to communities.

The Community Adaptation Facility (CAF) within FONERWA was developed as part of Rwanda’s NDC Deep Dive: Advancing Financial Innovation to Scale Up Climate Action. The aim of the CAF is to provide grant

funding for community projects that scale up climate resilience. In this regard, the Project will support the cooperatives and youth-led SMEs engaged in Component 3 to prepare business plans and link them to innovative financing for community projects with the CAF. In line with the CAF priorities, the business plans will focus on the investment priorities of the Rwanda Green Fund (FONERWA) targeting conservation and sustainable management of natural resources especially ecosystem rehabilitation, sustainable land management, sustainable forest management and promotion and protection of biodiversity. The Project Technical Assistance will work with FONERWA to design specific long-term funding opportunities for communities in the CND to the CAF.

Additionally, the Project will enhance the environmental, social and economic resilience of communities in the CND by enhancing financial and accounting literacy, marketing and entrepreneurship, and vocational skills of community members (e.g., in native tree propagation, forest restoration, and fuel-efficient stove production). This will mean the Project will invest in businesses that restore or indirectly halt the degradation and loss of native forests by reducing or eliminating local stressors. These businesses and projects are expected to live beyond the lifetime of the project. The value chains, once organized profitably, can continue to grow without further support, while the financial service providers will continue to service savings groups and farmers and SMEs in the value chain with sustainable financial services also after the project ends. By raising the productivity from selected crops, diversifying agriculture towards high-value agriproducts, and strengthening their value chains through trainings in marketing, processing and business, the project will attract investments from the private sector in sustainable agriculture and agroforestry. All economic and environmental benefits from the project will be rigorously monitored, captured, compiled and disseminated as knowledge products, packaged for various stakeholders including investors and MFI, with which the project will engage to secure microcredits to beneficiary farmers.

(c) FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding	Total amount			Currency		
	39,056,421			USD (\$)		
GCF financial instrument	Amount	Tenor		Grace period	Pricing	
Grants	39,056,421	-		-	-	
(b) Co-financing information	Total amount			Currency		
	10,943,579			USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Rwanda Development Board	<u>Grant</u>	<u>5,000,000</u>	<u>USD (\$)</u>	-	-	-
Rwanda Forestry Authority	<u>Grant</u>	<u>831,307</u>	<u>USD (\$)</u>	-	-	-
Rwanda Water Resources Board	<u>Grant</u>	<u>5,000,000</u>	<u>USD (\$)</u>	-	-	-
Ministry of Environment	<u>In-kind</u>	<u>112,272</u>	<u>USD (\$)</u>	-	-	-
(c) Total financing (c) = (a)+(b)	Amount			Currency		
	<u>50,000,000</u>			<u>USD (\$)</u>		
(d) Other financing arrangements and contributions	The Government of Rwanda is providing grant financing via the Rwanda Forest Authority (RFA), Rwanda Water Board (RWB), and Rwanda Development Board (RDB) to support activities in component 3. In addition to their contributions, AE will cover office costs in the form of an in-kind contribution equivalent to \$112,271 for the Project Management Unit which will be hosted within AE premises. The co-financiers have been identified based on their existing mandates and activities they implement in the CND region hence this was the only assessment done and they have been able to commit their involvement in the project implementation. The provided co-financing commitments will serve as a basis of MoE engagement during the project implementation. During implementation of the project, annual work plans will be adjusted together with the executing entity and all the co-financiers to align with everyone’s activities. MoE/AE under its oversight role will ensure that all these stakeholders commit during planning on their co-financing and report on it during Annual Performance Reports submitted to GCF.					

C.2. Financing by component

Component	Output	Indicative cost USD (\$)	GCF financing		Co-financing		
			Amount USD (\$)	Financial Instrument	Amount USD (\$)	Financial Instrument	Name of Institutions
Component 1: Mainstreaming Climate adaptation into integrated land use planning	1.1 Landscape-wide land-use plan developed for climate-resilient livelihoods & forest ecosystems, integrating district strategies	1,729,172	1,729,172	Grant			
	1.2 Local & national institutional capacities strengthened to integrate biodiversity & climate risks into land use planning & management	1,195,652	1,195,652	Grant			
Component	Output	Indicative cost USD (\$)	GCF financing		Co-financing		
			Amount USD (\$)	Financial Instrument	Amount USD (\$)	Financial Instrument	Name of Institutions
Component 2: Forest and landscape management and restoration	2.1 PA management effectiveness improved re climate risks & adaptation	2,051,778	2,051,778	Grant			
	2.2 Natural forest cover restored, biodiversity connections established	16,917,145	14,271,796	Grant	2,645,349		Rwanda Water Board
Component 3: Enhancing climate adaptation through promotion of resilient livelihoods and climate-smart farming methods	3.1 Farming methods enhance productivity, reduce erosion & flooding risks, contribute to ecosystem services, and support connectivity	13,451,014	10,856,387	Grant	478,071	Grant	Rwanda Forestry Authority
	3.2 Rural livelihoods generate alternative incomes & reduce pressure on forests	8,612,188	3,850,283	Grant	4,761,905	Grant	Rwanda Development Board
	3.3 Financial services & private sector investment engage	559,890	559,890	Grant			

Monitoring & Evaluation	4.1 - Performance monitoring plan developed	555,475	555,475	Grant			
	4.2 - Project monitored and evaluated; lessons learnt integrated into adaptive management processes	645,390	645,390	Grant			
Project Management Costs	5.1 Project managed efficiently to achieve its goal to build resilience to climate change in CND	1,901,344	1,480,769	Grant	106,925	In-Kind	Ministry of Environment (AE)
					313,650	Grant	Rwanda Forestry Authority
<u>5% Contingency</u>		2,380,952	1,859,830	Grant	39,586	Grant	Rwanda Forestry Authority
					238,095	Grant	Rwanda Water Board
					238,095	Grant	Rwanda development Board
					5,346	In-Kind	Accredited Entity
Indicative total cost (USD)	50,000,000	39,056,421	10,943,579				

C.3 Capacity building and technology development/transfer

C.3.1 Does GCF funding finance capacity building activities? Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer? Yes ☒ No ☐

This project will build knowledge among land-use planning actors on the value of forests for climate resilience, and institutionalizing cross-sectoral meetings to encourage synergies and avoid overlapping

mandates and redundancy in different climate resilience interventions to ensure climate adaptation and climate resilience are explicitly incorporated into land-use planning processes in the CND, such that trade-offs between different sectors can be resolved while balancing the climate resilience of nature and people. Local land-use planning experts and community members will be trained and engaged to review and interpret the National Land Use and Development Master Plan, to ensure the plan accounts for current and future climate risks, and that these risks are also considered in district level planning schemes. The knowledge and insights gained from these activities will be synthesized and used to inform the design of an Integrated Land-use Plan for Resilient Livelihoods and Ecosystems, which will reconcile the cross-sectoral (e.g., tourism, environment, forestry, food production, cash crops for export) trade-offs necessary to sustain forest ecosystem functions, deliver critical ecosystem services and improve community livelihoods. An innovative scientifically credible system for monitoring trends in natural forest cover and forest types will also be established, and a comprehensive capacity building programme will improve use of remote sensing, ecosystem service modelling and spatial planning tools to inform decision making. **Budget: \$2,000,754**

To enhance forest resilience across the CND, through a combination of increased forest cover, improved forest condition, and improved management to reduce degradation and pressure on forests, the project will increase the extent and condition of natural forest within the CND's national parks, by facilitating regeneration of large areas previously degraded by fires, securing key remaining natural areas outside PAs and developing a financial sustainability plan for the CND. The project will invest in the capacity of government agencies, districts, and communities to manage forest fires. Community members will be trained in and provided technical assistance and equipment for restoration techniques, establishment, and management of indigenous tree nurseries. Management plans for the CND's national parks will also be updated and park staff trained to explicitly consider climate adaptation and gender issues and manage the parks as part of the larger CND landscape. The project will also develop the capacity of key Government ministries and agencies on sustainable conservation finance. **Budget: 5,420,837**

Additionally, this project will strengthen the livelihoods of landless and or the most vulnerable households, who are often most at risk of unsustainable use of their farmlands, protected areas and resources within protective forests. Erosion control measures and on-farm tree cover will be increased across the CND, helping reduce the risk of crop failure from extreme events like landslides, reduce land and water degradation caused by soil erosion, and increase provision of valuable products like wood fuel. To decrease pressure on forests for resource extraction, existing successful programs distributing improved cookstoves will be dramatically scaled up. Community members will be trained in and provided technical assistance for SLM practices, establishment, and management protective forests. Both cooperatives, individuals, participating financial institutions (PFIs), and value chain actors/small and mid-size enterprises (SMEs) will benefit from capacity-building to improve their business skills and access to finance, and to strengthen value chains (including beekeeping, ecotourism, agri value chains, and cookstoves), including the construction of processing facilities and marketing materials. Vocational training schools and local youth cooperatives will be supported to produce, test, and distribute Tier 2 cookstoves. **Budget: 13,893,987**

(d) EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

D.1. Impact potential

Adaptation Impact – This Project is designed to restore, sustain and expand critical forest ecosystem functions in the CND. More specifically, the project will protect, restore and expand natural forest in and around the core national parks; consolidate and expand some smaller but critical "stepping stones" of remaining natural habitat between these national parks (mainly small forest patches and wetlands/ riparian areas); improve the landscape linkage function of key high altitude between the protected areas, while at the same time securing ecosystem services through afforestation and improved native species mix for plantation forests on steep slopes; silvopastoral interventions on cultivated pasture; and agroforestry interventions in regenerative agricultural areas. These interventions will reduce erosion, decrease loss of soil organic matter, and thus increase the water infiltration rate, which increases water recharge which will enhance the resilience of forest ecosystems and smallholder farmers to climate change. The establishment of ecological connectivity between fragmented forests will promote landscape heterogeneity and support species migration to more favorable conditions within this human modified landscape and as a result help build the resilience of biodiversity to climate impacts. Increasing the capacity for species movement across these islands of forest habitat, will improve the adaptive capacity of multiple species that serve as important seed dispersers and insure the long-term viability of forests.

Project interventions will include 10,000 ha of land (including both forest and silvopastoral sites) under improved, climate-resilient management; and improve agroforestry practices on 3,346 ha of on-farm plantations, directly benefiting 1,254,242 people (654,404 women) with more climate resilient livelihoods from reduced exposure to landslides, floods and soil erosion and more knowledge on climate risks, value of forests and forests ecosystems, climate adaptation options, access to indigenous and agroforestry quality materials and improved ecosystem services.

The Project will provide direct benefits to increase the climate resilience of more than 1.5 million of the most vulnerable residents of the CND, including increased adaptive capacity through reduced exposure to flash floods, landslides, and soil erosion which will also increase their crop productivity and livelihoods (direct beneficiaries 52.2% female). Indirectly, 2,464,097 beneficiaries in the CND will receive co-benefits from improved and more sustained ecosystem services, including decreased sedimentation leading to improved water quality downstream and increased hydropower generation. This helps Rwanda produce cleaner hydropower energy at a lower cost, making it preferable to other more detrimental energy sources. The project will also indirectly benefit 9,260,745 people and create approximately 24,212 job opportunities in forest-dependent communities.

The Project will contribute altogether to the restoration of 5,000 ha of intact forest, 2,500 ha of protective forests and 3,346 ha of smallholder farmland, 1000 ha of pastures and 1,500 ha of riparian linkages. Overall, 275,617 ha of the landscape could potentially be positively impacted by the project as land use planning, improved natural forest management, improved buffer management and a stronger agroforestry sector would all contribute to improved outcomes beyond the specific areas directly targeted for restoration or improved management by the project. In particular, stronger capacity for land use planning across the CND and hence improved protection of key areas (e.g. steep slopes, riparian areas and protected area buffers) has strong potential to deliver significant mitigation benefits at a landscape rather than a site level. Although the project is focused on the priority landscapes covering 275,617 ha, it is likely that benefits due to increased adaptive capacity (achieved through improved land use planning and management of key features through

reduced exposure to flash floods, landslides, and soil erosion) could feasibly extend across the full 700,000 ha CND.

Additional adaptation benefits will include the training of 52,589 community members in forest management and restoration for climate resilience, which will be linked with ongoing restoration employment opportunities through other GoR and private sector projects; the strengthening of institutional and regulatory systems to promote climate resilience; and targeted capacity building for certain government institutions and ministries in forest monitoring, remote sensing, spatial planning, and analysis. These beneficiaries will include MoE, REMA, RDB, MINAGRI, and the University of Rwanda.

Complementary activities will address key drivers of deforestation while creating alternative livelihood pathways for youth and women through ecotourism and income-generating, climate-resilient crops, as well as strengthening supportive financial services.

Mitigation Impact – The Project will generate mitigation benefits by delivering sequestration of approximately 224,871 t CO₂-eq during the 5-year Project period, which is about 0.8% of the 28.2M t CO₂ reduction targeted by Rwanda's vision 2050.

- 5,000 ha of critical intact natural forests will be under improved management for climate resilience, accounting for an estimated total of 596,116t CO₂-eq over 20 years.
- 1,000 ha of pasture lands with increased coverage of native trees to secure reasonable landscape connectivity for forest species, accounting for an estimated total of 4,543t CO₂-eq over 20 years.
- 1,500 ha of riparian lands will be afforested which will contribute significantly to overall landscape connectivity, sequestering 110,138 t CO₂-eq over 20 years.
- 2,500 ha of protective forests and plantations will be established on slopes >55%, focusing on areas not currently planted in crops, accounting for an estimated total of 193,472 t CO₂-eq over 20 years.
- Erosion control and agroforestry practices will be implemented over 3,346 ha of smallholder farmland, sequestering 129,006t CO₂-eq over 20 years.
- Improved energy efficient stoves will be adopted by 8,500 households, securing 51,017t CO₂-eq of avoided emissions over 20 years.

Overall, there is potential for a significant contribution to GCF1 Core Indicator 4, hectares of natural resource areas brought under low emission or climate resilient practices. Conservatively, an area of 275,617 ha of the landscape could potentially be positively impacted by the project as land use planning, improved natural forest management, improved buffer management and a stronger agroforestry sector would all contribute to improved outcomes beyond the specific areas directly targeted for restoration or improved management by the project.

D.2. Paradigm shift potential

As described in Section B1, the Project provides a profound opportunity to implement new mechanisms of land-use planning and management that are based on consideration of the CND as a landscape, consisting of an inter-dependent mosaic of forests and farms, the management of each determining the health of the whole. Focusing on this larger scale, with coordination across both geographic and sectoral boundaries, while mainstreaming climate risks into management planning represents a fundamental paradigm shift for land-use planning and management in Rwanda. This approach to land-use planning and management is entirely new for Rwanda and is intended to maintain and recover the ecosystem services that are essential to the resilience of smallholder farming communities within the region and to the globally significant and

unique biodiversity in this part of Rwanda. This represents a reversal of the previous negative feedback loop of ecosystem degradation to a more positive, climate-resilient system.

Potential for scaling-up spatial planning and replication. Spatial planning is recognized by the IPCC, the EU and others as playing a key role in climate change adaptation and climate resilient development, but to date development planning in Rwanda has largely been based on tabular data, rather than on spatial planning. Adoption of spatial planning across ministries and on national regional and local levels, represents a significant paradigm shift for Rwanda. Piecemeal decisions and an ad hoc assemblage of isolated sector-by-sector projects will be replaced with a holistic, climate-smart, evidence-based approach to planning and development – enabling systematic, ecosystem-based, multi-sector analysis of land use tradeoffs. The Project will therefore establish a new inter-agency and local-national model and capacities for holistic planning and implementation – one that can be scaled up for use elsewhere in Rwanda and beyond. This will enable better-informed planning decisions, the identification and valuation of nature-based solutions to climate adaptation and mitigation, and overall will foster climate resilient development pathways in the CND and nationwide. Communication and collaboration with complementary GCF and other projects in Rwanda will accelerate this process.

Potential for knowledge and learning. Component 1 will invest in new technologies and climate-smart knowledge to improve national and district planning, scenario analysis and monitoring capacities. In doing so, new mechanisms of coordination and collaboration will be established, and shared across agencies. Component 2 will invest in best practices in forest management and restoration – some of which will require learning beyond the existing science of indigenous tree propagation and restoration (particularly in re-establishing rain forest habitat, establishing protective forests on slopes of >55%, and enriching farms with indigenous tree species). Component 3 will focus on processes to identify and pilot value-chain products (on-farm and off-farm) not yet in practice. All approaches will include the participation of local farmers, women, and youth to contribute to collective learning processes involving national and local government and vulnerable communities. Knowledge generated – and the processes by which it is generated – will be shared across the region. The project is based on collective learning, knowledge generation and dissemination at community, landscape and national levels. A focus on knowledge management through demonstrations, monitoring, and evaluation across the project activities will inform the implementation of current and future adaptation efforts.

Contribution to the creation of an enabling environment. Strengthening the Task Force in Component 1 will ensure a single, first point of contact for all sectors' planning development and investments in the CND and will ensure that planning for climate resilience is addressed. By establishing a mechanism for coordination and harmonization, it will radically improve current conditions for public and private interventions and will establish partnerships and collaborations that live beyond this Project. Also importantly, promotion of sustainable financing for each component will help to assure that initiatives will either be self-sustaining or have solid plans for continued financial support.

Contribution to the regulatory framework and policies. The Project will finance the development of the *Integrated Land-use Plan* which will be adopted by all ten CND districts as a framework for seeking synergies between rural livelihoods and forest and biodiversity protections, reconciling land use conflicts that arise, and guiding development decisions and climate change adaptation. It will update the National Land Use Master Plan, ensuring it incorporates climate resiliency, and is accurate and relevant to local contexts. As development of an Integrated Land-use Planning approach helps better reconcile cross-sectoral tradeoffs and improve climate adaptation planning in the CND, this approach will prove a successful model for application to other regions of Rwanda and to other forest/farming landscapes beyond. In addition,

revising 2 NP management plans to include climate concerns will serve as a model for the remaining parks of Rwanda, as well as for the East African region.

D.3. Sustainable development (max. 500 words, approximately 1 page)

The proposed Project is aligned with the United Nations 2030 Agenda for Sustainable Development, including SDGs 5, 13 and 15 and will contribute to the achievement of the SDGs 1, 2, 3, 5 and 10. The Project will promote an ecosystem-based approach to forest protection, restoration, and rehabilitation that includes:

- Reduction of erosion, sedimentation, and landslide risk as a result of restoration of degraded lands;
- Increased number of native trees on farms with improved agroforestry systems, improving biological connectivity;
- Increased carbon sequestration potential of plants and soils. While the primary objective of the Project is adaptation, the Project will also produce mitigation benefits of 204,079t CO₂-eq during the 5-year implementation period, and 1,033,275t CO₂-eq over the 20-year Project lifetime.
- Reduced GHG emissions through more fuel efficient cookstove technologies, yielding cumulative savings of approximately 51,017t CO₂-eq during the 20-year Project lifetime.

The Project will enable the transition to more climate resilient and productive livelihood models and empower the rural poor, significantly reducing impacts from extreme weather events and increasing their potential for improved wellbeing. Social benefits will include:

- Raised awareness about climate change effects and adaptation in 1,253,252 people living in the CND (at least 52.2% representing women).
- Capacity building of ~52,589 people (at least 50% women), in a wide range of topics aimed at increasing ecosystem and social resilience.
- Create significant social capital through co-designing and co-managing a range of adaptation strategies, indirectly promoting social cohesiveness among CND villages through joint saving and lending systems.
- Reduced inequality and poverty through increased financial inclusion and improved economic opportunities — especially for women, youth and historically marginalized groups.
- Forest and landscape restoration in the CND will have several economic co-benefits for vulnerable communities, including:
- Soil erosion control measures, protective forests on public lands and agroforestry on private lands, will increase agricultural productivity and associated income.
- The Project will create direct and indirect employment opportunities, benefiting approximately 24,212 households and stimulate the local economy.
- Leveraging private capital to invest in the value chains including cookstoves in the CND. The project will create important social and environmental benefits. However, the project will also have removed a number of risks (notably carbon risk) that worry more traditional (social-impact or sustainability) investors. As a result, the venture becomes an attractive investment for private capital interested in a “triple bottom line”, allowing the business to develop sustainably and scale to more CND areas. During project implementation we will therefore produce a bankable business plan that details the activities and resources required for expanding further.

The Project will result in positive outcomes around access to resources, improved livelihoods, and income generation opportunity and capacity for women. With support from a dedicated Gender Specialist, the Project will engage women in project planning, investment and decision making. With opportunities to generate additional income, women, who traditionally are already further vested in investing in their families, will be more able to respond to incentives that address their families' basic needs, such as better health and nutrition, linking to agriculture and food security improvements. The project will also get support from NGOs that engage men in programming and discussions to increase their participation in household and parental responsibilities and gain greater awareness of the importance of women's fair access and control of resources, including income. Women will benefit from training and educational activities, leadership, entrepreneurship and decision making. Annex 8, which will be further elaborated on at project onset, provides more information on gender-sensitive impacts.

D.4. Needs of recipient

Despite Rwanda's solid economic progress since 2000, poverty remains widespread and pervasive. Overall, 38.2% of the population lives in poverty and 16% in extreme poverty¹¹⁹. Women are more affected by poverty than men, with 47% of female-headed households living in poverty compared with 44.9% of all households. Rural households are more than twice as likely to be in poverty/extreme poverty than urban households. Districts of the Western and Southern Provinces in the CND region have the highest rates of poverty and extreme poverty in the country.

Most rural households are smallholder subsistence farmers and, because of high population density, high relief terrain and dependence on rain fed agriculture, the residents of the CND are highly vulnerable to climate change. These risks include exposure to flash floods¹²⁰, landslides¹²¹ and soil erosion¹²². During stakeholder consultations, communities indicated several socio-economic changes stemming from changing climate conditions. Notably, the longer dry season and unpredictable rains¹²³ have reduced agricultural yield significantly, leading to more widespread malnutrition and extreme poverty, and farmers can barely meet their own families' needs¹²⁴. In particular, women, youth, and historically marginalized people are feeling the impacts. As arable land is increasingly constrained, youth and historically marginalized people have few opportunities and are more focused on survival and short-term gains at the expense of education and sustainable livelihood development (see Annexes 6 and 8 for more detail).

As a result, the GoR has invested significantly in developing an enabling policy and institutional framework for a more climate-resilient development pathway. Yet, while adaptation is a top priority, Rwanda is classified by the U.N. as a LDC and lacks sufficient public financial resources to implement its climate change and resilient green development objectives. Currently there are no alternative funding options of the scale needed to secure Rwanda's priority forests to reduce vulnerability of rural communities to climate risks. The GoR also recognises that a piecemeal approach to decision making has exacerbated competition over scarce land resources and reduced the adaptive capacity of both ecosystems and rural populations. However, many

¹¹⁹ World Bank, 2020. *Poverty and Equity Brief: Rwanda*.

¹²⁰ Karamage, F., et al., 2017. *Modeling Rainfall-Runoff Response to Land Use and Land Cover Change in Rwanda (1990–2016)*. Water 2017, 9(2), 14.

¹²¹ Uwihirwe, J., et al., 2020. *Landslide precipitation thresholds in Rwanda*. Landslides.

¹²² World Bank, 2019. *Rwanda Systematic Country Diagnostic*.

¹²³ World Bank; CIAT. 2015. *Climate-Smart Agriculture in Rwanda*. CSA Country Profiles for Africa, Asia, and Latin America and the Caribbean Series. Washington D.C.: The World Bank Group.

¹²⁴ Clay N., King B., 2019. *Smallholders' uneven capacities to adapt to climate change amid Africa's 'green revolution': Case study of Rwanda's crop intensification program*. World Dev.

agencies operate within siloed environments, focusing on limited scopes framed by individual performance plans, and close coordination is often cited as a key recommendation toward strengthening climate change adaptation measures¹²⁵.

Additionally, district government staff and community members have limited technical understanding or awareness of the complex interactions between forest structure and functions, climate change, and the provision of ecosystem services. Accurate measurement of forest cover and forest type is notoriously difficult in high relief terrain, such as in the CND¹²⁶. Lack of access to and capacity to apply new methods for classifying and monitoring forest change has limited forest monitoring in Rwanda¹²⁷. Consequently, government forest data has historically been limited in scope and infrequently updated, and regional/global maps are too inaccurate to be of use. Capacity building around the technologies and processes to make landscape-scale planning decisions is crucial to the success of Rwanda's climate-resilient development pathway.

D.5. Country ownership

The proposed Project is closely aligned with the NAP's adaptation priorities: 1) promotion of non-agricultural income generating activities, 2) introduction of species resistant to environmental conditions, and 3) development of firewood alternative sources of energy. It is also aligned with several priorities identified in the 2020 NDC: promotion of afforestation/reforestation, improvement of forest management, integrated approach to planning and monitoring for sustainable land management, development of a harmonized and integrated spatial data management system for sustainable land use management. The Project will support implementation of the Green Growth and Climate Resilient Strategy and the Rwanda National Strategy for Transformation–NST1 (2017-2022) and is aligned with the goals of the Rwanda National Forestry Policy 2018, the Forest Sector Strategic Plan 2017-2022, the Strategic Plan for the Environment and Natural Resources Sector (2018-2024), and National Forest Management Plan 2017-2026, reflecting the government's intentions to address climate change impacts and mitigation targets by improving forest management. It is also aligned with the national Biomass Energy Strategy (BEST) 2020 which aims to ensure a more sustainable biomass supply and combustion technologies for homes and small enterprises. This Project aligns with the forthcoming GCF country program which prioritizes forest management through restoration and protection of natural forests, afforestation and agroforestry, as well as distribution of high-quality forestation inputs and capacity development.

While this Project proposal to GCF focuses on a different geographic area - with specific adaptation needs through land restoration, development of markets and finance mechanism, as well as support to mainstreaming climate change into land use planning - the early and ongoing lessons from the experiences from existing GCF-funded projects, "Strengthening Climate Resilience of Rural Communities in Northern Rwanda"¹²⁸ and "Transforming Eastern Province through adaptation" (TREPA)¹²⁹ will be applied during the

¹²⁵ Bagstad, K.J., 2019. *Towards ecosystem accounts for Rwanda: Tracking 25 years of change in flows and potential supply of ecosystem services*. British Ecological Society.

¹²⁶ Arakwiye, B., et al, 2021. *Thirty years of forest-cover change in Western Rwanda during periods of wars and environmental policy shifts*. Regional Environmental Change 21(2).

¹²⁷ Herold, M., Schiller, F., 2009. *An assessment of national forest monitoring capabilities in tropical non-Annex I countries: Recommendations for capacity building*. Global Observation of Forest and Land Cover Dynamics Land Cover Project Office.

¹²⁸ Rwanda Ministry of Environment, 2018. *FP073: Strengthening Climate Resilience of Rural Communities in Northern Rwanda*. Green Climate Fund.

¹²⁹ International Union for Conservation of Nature, 2021. *FP167: Transforming Eastern Province through Adaptation*. Green Climate Fund.

implementation of this Project in the Congo Nile Divide Landscape.

MoE is responsible for ensuring sustainable management of natural resources and for the development of policies, strategies, regulations, and mobilizing resources for the development of the sector. The MoE managed USD 10 million from the Adaptation Fund to implement the Reducing Vulnerability to Climate Change in North West Rwanda through Community Based Adaptation Programme¹³⁰ and has overseen a number of other climate projects including: an early warning system project (funded by LDCF), the Landscape Approach to Forest Restoration and Conservation (LAFREC), and the “Building resilience of communities living in degraded forests, savannahs and wetlands of Rwanda through an ecosystem management approach” programme (funded by LDCF)¹³¹. The OECD rated Rwanda as one of the countries that uses aid most effectively¹³², and Rwanda is on course to achieve the SDGs. The GoR has demonstrated comprehensive country ownership of, and capacity to implement, a funded project. The national designated authority (NDA) is the core interface between a country and the GCF and is mandated to align GCF-supported activities with strategic national objectives and priorities. The NDA coordinates with AEs on proposal development and completes a no-objection letter after it is determined that the proposal is aligned with the national priorities. This Project was initiated by the MoE and supported by the NDA, Rwanda Environment Management Authority, due to the high climate sensitivity of the CND.

Government agencies such as Rwanda Forestry Authority (RFA), Rwanda Development Board (RDB), Rwanda Environment Management Authority (REMA) and National Land Use Authority (NLA) will maintain strong ownership and support activities under component 1 and 2 as aligned with their mandates and the aforementioned plans and strategies.

To better understand the interests of stakeholders and potential impacts of the Project, and as preparation to design future engagement strategies, a comprehensive, gender-responsive, culturally sensitive, non-discriminatory, and inclusive stakeholder analysis was undertaken and is incorporated into the Stakeholder Engagement Plan (Annex 7).

D.6. Efficiency and effectiveness

Financing and concessional: The financial instrument chosen for this Project is grant financing of USD 39.06 million which constitutes roughly 78% of the total Project’s estimated cost. The relatively high level of concessionality reflects the public good character of the investments to be made by the Project, Rwanda’s status as an LDC¹³³ and its high vulnerability to climate change¹³⁴. Furthermore, it must be highlighted that most project expenditures relate to creating public goods through forest and landscape restoration or capacity-building of public institutions. These interventions are primarily non-income-generating; hence very hard to attract private sector investment. Attracting the private sector for public goods investments remains challenging in the context of many countries worldwide, including Rwanda.

Addressing low private sector investment in the forestry sector. One of the major reasons for lack of private-sector investment in Rwanda’s Forestry Sector is that most stands are too small to sustain an

¹³⁰ Adaptation Fund, 2014. Reducing Vulnerability to Climate Change in North West Rwanda through Community Based Adaptation. [Link](#).

¹³¹ UNEP, 2016. *Building resilience of communities living in degraded forests, savannahs and wetlands of Rwanda through an ecosystem management approach*.

¹³² Newfarmer, R., et al., 2013. *Managing Aid for Trade and Development Results: Rwanda Case Study*. OECD.

¹³³ United Nations Department of Economic and Social Affairs, 2021. Least Developed Country Category: Rwanda Profile.

¹³⁴ World Bank, 2021. *Rwanda*. Climate Change Knowledge Portal.

economically viable operation (see the Forest Sector Strategic Plan). By restoring smallholder protective forest plantations and organizing them into Private Forest Management Units (PFMUs), this Project will assist with creation of units that are economically attractive to private investors, helping to facilitate future private investment if landowners desire it. However, no private investors will be subsidized by the Project, and ownership of PFMUs will remain with the state/district governments and smallholders' cooperatives who own forest plantations. RFA in collaboration with RCA and the district government directly implement organization of the cooperatives and conflicts in the cooperatives are managed in accordance with internal rules and regulation governing each cooperative overseen by the Rwanda Cooperative Agency in case of escalated conflicts but specifically in the cooperatives that will be benefiting under this project, RFA and District government will also intervene in conflict settlement.

No excess subsidy will be provided to the private sector, and no private sector investors will be crowded out by these investments. Wherever these are required for institutional and capacity support for market development and crowding in for private investments these are undertaken with concessionality to community in mind. For example, the subsidy/microcredit schemes to facilitate the adoption of efficient cookstoves are undertaken with minimum concessionality. The value chain development for other potential revenue generating streams such as honey undertaken under output 3.2 requires institutional and capacity support funded by GCF grant. In the case of ecotourism for tea plantations, no project concessionality is planned and complete crowding in from the private sector is envisaged. The value chain development of beekeeping is 100% funded by GCF grants so that an enabling environment can be created, and the project can overcome affordability challenges through developing a system such as community co-guarantee, overcome lack of private capital structure and insufficient markets. Because of above mentioned reasons no private sector investors are expected to be crowded out by project investments.

GCF grant as a suitable financing instrument. The GCF grant can help create the enabling environment for the future engagement of the private sector in similar projects. The showcasing of potential revenue-generating ways, thanks to the concessional GCF financing, may help invigorate the potential for future investments in this sector, which in turn might result in a multiplier effect leading to a higher degree of development impact.

Results of the Economic and Financial Analysis (EFA). Following the GCF guidelines for developing the Annex 3 package, the EFA was prepared to assess the economic rationale for the proposed interventions. Consequently, four representative interventions were appraised in the EFA: (i). agroforestry, (ii). protective forests (iii). fuel-efficient cookstoves, (iv). silvo-pastoralism. Best practices in appraisal for public sector projects have been followed. The financial analysis for each intervention was calculated from the private perspective using a discount rate of 15.44% which was the lending rate of Rwanda Central Bank. For the three modeled interventions (see table below) the financial NPV is positive and financial IRR is higher than discount rate signifying that project activities are profitable. The GCF support will result in a positive financial return for farmers or farmers and private actors over a 20-year period and will help overcome high up-front investment costs during the project duration.

Financial analysis (US\$)	Agroforestry	Protective Forest	Silvopasture
FNPV	\$34,483,676	\$2,980,381	\$310,901
FIRR	42%	25%	35%

Aggregate financial analysis of with project (WP) interventions

The obtained ex-ante aggregate economic net present values (ENPVs) for these interventions over the 20 years' timeframe and at a 13% economic discount rate range between USD 0.97million and USD 78.93 million, with economic internal rates of return (EIRRs) between 49% and 72% (see table below).

Economic analysis (US\$)	Agroforestry	Protective Forest	Energy efficient cookstoves	Silvopasture
ENPV	\$58,934,340	\$30,218,468	\$10,585,674.50	\$966,331
EIRR	49 %	72%	N/A	51 %

Aggregate financial analysis of with project (WP) interventions¹³⁵

The sensitivity analysis that was carried out suggests that obtained results are relatively insensitive to increases in costs or decreases in benefits offering the robustness of obtained results. Therefore, the economic sustainability of the Project is likely, and the Project shows the economic rationale for implementation as benefits are likely to outweigh the costs.

Best Practices *Forest Restoration/Tree Planting*: The Project will use Rwanda's best practice guidelines for tree nursery establishment, tree planting and tree management¹³⁶. For assisted natural regeneration, methods developed and proven through multi-year research in Nyungwe and Gishwati-Mukura National Parks will be used¹³⁷. *Forest Monitoring*: The Project will utilize multiple contemporary methods for monitoring the success of project interventions and forest health. Forest plantations will be mapped and monitored using the Rwandan government's newly developed Forest Monitoring and Evaluation Software (FMES)¹³⁸, specifically developed for this purpose. The Project will develop a modern indigenous forest monitoring system, utilize the most high-resolution datasets currently available (<3m) and take advantage of modern remote-sensing advances including cloud computing and machine learning. Cost effective replication for forest restoration and climate resilience throughout Africa and beyond can be anticipated.

¹³⁵ No financial analyses for cookstoves intervention was warranted as energy efficient cookstoves are non-income generating intervention with expectation of no financial cash inflows. Only economic analyses was conducted for cookstoves.

¹³⁶ Unique Ltd. & RNRA. 2015. Tree plantation establishment and management manual for Rwanda. Kigali, Rwanda

¹³⁷ Masozera, A.B., 2004. Regeneration of Burned Forested Areas With Periodic Removal of Pteridium aquilinum. Wildlife Conservation Society Rwanda.

¹³⁸ Rwanda Forest Authority, Enabel, 2021. Technical guidelines on the use of the Forest Monitoring & Evaluation System - FMES. Kigali.

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

- ☐ Reduced emissions (mitigation)
☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Rwanda's natural forests are extremely depleted and degraded. While overall forest cover has increased since the 1980s, this is primarily made up of exotic monocultures, which have very low potential to provide ecosystem services and adaptation benefits. There are small-scale efforts to establish/restore native forests and increase overall forest resilience, but scaling these efforts is hampered by a limited supply of indigenous seedlings.	<u>Low</u>	Paradigm shift would involve a move away from widespread planting of exotic species for fuelwood, as well as improved erosion control measures on farmland and use of improved cookstoves, both of which will reduce pressure on Rwanda's forests. As the benefits of planting indigenous trees become more widely known, behavior changes in non-targeted communities are likely to be catalyzed.	The project is expected to bring 144,058 ha of land under climate-resilient management. This would represent a substantial step towards paradigm shift on increased resilience of forests and vulnerable communities, even before any replication effects, which will likely be catalyzed by project activities focusing on training, capacity building, and peer-to-peer knowledge exchange. The project will also deliver direct carbon reductions of 1,204,448 t CO ₂ -eq over 20 years, and potentially more through increased institutional capacity to mainstream climate adaptation and improve land-use planning.
Replicability	There are a small number of interventions occurring across Rwanda focused on improving resilience of vulnerable communities through forest restoration, agroforestry, livelihood diversification and clean cooking technologies. However, many interventions focus on exotic species which have very low potential to provide ecosystem services and adaptation benefits.	<u>Medium</u>	If increasing the extent and integrity of natural forests and improving capacity for climate-responsive land management can increase climate resilience of vulnerable communities in the CND, then the solution could be replicated throughout Rwanda and across East Africa. Further, if incorporating indigenous species into forest restoration and agroforestry can provide a similar/greater range of adaptation benefits compared to exotic species, and also avoid the drawbacks of exotic monocultures, then the solution could be expanded nationally. If development of an Integrated land use planning approach helps better reconcile cross-sectoral tradeoffs and improve climate adaptation planning	The project will invest in strengthening institutional capacity to promote climate resilience, directly increase the resilience of ecosystems and community livelihoods, and identify options for use of indigenous species in forest restoration and agroforestry, in order to better contribute to climate-resilient livelihoods. These topics have received little attention to date in Rwanda, and lessons learned will inform efforts to improve climate resilience across Rwanda and East Africa.

			in the CND, this approach could be applied to other regions of Rwanda.	
Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Sustainability	The GoR has invested in developing policies and institutional frameworks for a more climate-resilient development pathway. However, domestic funding to implement these policies remains lacking, and as yet they have not translated to wide-scale improvements to climate resilience. Changes in behavioral norms have been limited, and overall uptake of climate adaptation solutions is poor.	<u>Low</u>	Paradigm shift would see Rwanda demonstrate that pursuing development goals need not come at the cost of extensive environmental degradation. Climate-responsive Forest and landscape management will be standard, and government programmes will focus on increasing the resilience of forest ecosystems and enhancing their ecosystem services. Behavior change across the CND would support this, as communities recognise, value, and protect the benefits provided by forests.	The project will work closely with government and communities to build capacity to mainstream climate adaptation into forest and land-use planning and management. The project will facilitate and demonstrate the benefits of climate-resilient agricultural practices, forest management, and alternative livelihoods, raising awareness within and outside target communities.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Mitigation Results Area 4: Forests and land use	Core Indicator 1: GHG emissions reduced, avoided or removed/ sequestered	Greenhouse gas emission assessment	0 tonnes of CO ₂	107,686 tCO ₂ -eq	224,871tCO ₂ -eq (5-year implementation period) 1,084,291tCO ₂ -eq (20-year lifespan period)	Government priorities remain supportive of long-term forest restoration and afforestation plans. Absence of major natural disasters including forest fires and landslides in target areas The economic, social and political context in the country and project areas remain stable.

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Adaptation Results Area 1: Most vulnerable people and communities	Core Indicator 2: Direct and indirect beneficiaries reached	Annual Agricultural Survey ¹³⁹ National census Project Baseline and Completion Reports, including documentation of workers contracted for reforestation/afforestation work, farmers adopting agroforestry activities, good silvo-pastoral practices, restoration of riparian linkages, signed MoUs for Private Forest Management Units, and households receiving improved cookstoves	Direct beneficiaries: 0 Indirect beneficiaries: 0	Direct beneficiaries: 95,873 (50,046 females) Indirect beneficiaries: 2,809,381 (1,466,497 females)	Direct beneficiaries: 108,277 (56,521 females) Indirect beneficiaries: 2,809,381 (1,466,497 females)	Government priorities remain focused on long term forest restoration and afforestation plans Absence of major natural disasters including forest fires and landslides in target areas Rural landowners agree to adoption of agroforestry and reforestation activities
Adaptation Results Area 1: Most vulnerable people and communities	Supplementary Indicator 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options	Project Baseline and Completion Reports, including documentation of beekeepers using modern hives, farmers at different levels of value chains, people employed in ecotourism, have access to	Direct beneficiaries: 0 Indirect beneficiaries: 0	Direct beneficiaries: 26,160 (13,656 females) Indirect beneficiaries:	Direct beneficiaries: 52,319 (27,311 females) Indirect beneficiaries:	

¹³⁹ The National Institute of Statistics of Rwanda conducts the Agricultural Survey yearly. Available at: <https://www.statistics.gov.rw/statistical-publications/subject/agriculture-and-environment>

		finance for other climate-resilient livelihood options		2,809,381 (1,466,497 females)	2,809,381 (1,466,497 females)	
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Adaptation Results Area 4: Ecosystems and ecosystem services	Core Indicator 4: Hectares of natural resource areas brought under improved low emission and/or climate-resilient management practices	Updated National Park Management Plans complete (ha per park) Forest cover statistics from Forest Monitoring and Evaluation System (FMES) GIS Data/Remote Sensing Ranger Based Monitoring data from PAs Vigilance Committees monitoring data	No increase in climate-resilient management practices (o)	42,558 ha of land brought under climate-resilient management	130,104 ¹⁴⁰ ha of land brought under climate-resilient management	Absence of major natural disasters including forest fires in target areas
Adaptation Results Area 2: Health, well-being, food and water security	Supplementary Indicator 2.2: Beneficiaries (female/male) with improved food security	Annual Agricultural Survey Gender responsive Comprehensive Food Security and Vulnerability and Nutrition Analysis Survey (CFSVA) ¹⁴¹	TBD	Direct beneficiaries: 0 Indirect beneficiaries: 0	Direct beneficiaries: 1,598,536 (52.2% female) Indirect beneficiaries: 2,464,097 (52.2% female)	Absence of extreme natural disasters and economic shocks affecting yields and household economy; migration patterns do not significantly affect the number and status of households.

¹⁴⁰ 130,104 ha refer to 121,758 ha of all core PAs and 8,346 ha restored areas outside PAs

¹⁴¹ The National Institute of Statistics of Rwanda conducts the CFSVA every 3 years.

The level of beneficiary of this project are individual households, - who can be in a cooperative or a business community group - who will participate directly in the project, and thus benefit from its implementation. Beneficiaries include all people who own the land where activities will take place and/or those who will be employed by the project (except EE staff, consultants and implementation and contractual organizations). Dependents of household that directly benefit from the project are also considered as direct beneficiaries and were calculated based on 4.2 average household size¹⁴² and 0.37¹⁴³ average land size in the CNDL.

Adaptation Results Area 1, Core Indicator 2 direct beneficiaries are beneficiaries of activities 2.2.2, 2.2.3, 3.1.1, 1.1.2.

Adaptation Results Area 1, Supplementary Indicator 2.1 direct beneficiaries of activity 3.2.1, 3.2.2, 3.2.4 (excluding 8, 500 beneficiaries of cookstoves) and 3.3.2

Adaptation Results Area 2 Supplementary Indicator 2.2 direct beneficiaries are the total population in landscape implementation sectors of this project.

Indirect beneficiaries broadly refer to all other human populations living within the CNDL districts. More details on calculations are provided in annex 23.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)					
Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low-emission and climate-	The GoR is committed to developing an enabling policy and institutional framework for a more climate-resilient development pathway, as evidenced by the large number of government policies and strategies focused on improving climate mitigation and adaptation. However, climate change is not being comprehensively considered by all sectors, cross-sectoral planning is rare, and new institutions are needed to	<u>low</u>	An integrated planning approach is used to reconcile cross-sectoral tradeoffs while explicitly considering climate resilience actions. New institutions have permanent capacity in place to ensure that spatial data on climate, forests and ecosystem services is incorporated into land planning and management.	The project will facilitate development of an integrated land use plan for the CND which will reconcile the cross-sectoral (e.g., tourism, environment, forestry, food production, cash crops for export) trade-offs necessary to sustain forest ecosystem functions, deliver critical ecosystem services and improve community livelihoods.	National level (one country)

¹⁴² National Institute of Statistics of Rwanda (NISR); The Fifth Rwanda Population and Housing Census, Main Indicators Report, February 2023

¹⁴³ National Institute of Statistics of Rwanda (NISR), Agricultural Household Survey 2020 report, December 2021

<i>resilient development pathways in a country-driven manner</i>	provide relevant spatial data to inform planning			The project will facilitate the development and adoption of an integrated landscape level planning taskforce, in line with the JADF to bring the relevant stakeholders (ministries and their agencies, districts, private sector and civil society organizations) on board, to strengthen inter-sectoral collaboration on the planning and implementation of the FLR plans. The project will also support capacity development of relevant technical staff working on FLR and land use planning (from central government agencies, district government, and NGOs) and will assist the GoR in developing a dedicated SPU within an appropriate ministry to provide the technical expertise needed to process and integrate spatial planning into decision-making within and across sectors.	
Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core Indicator 6: <i>Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer, and innovation</i>	Climate-resilient development in Rwanda is hampered by a lack of modern technological advances, both in terms of generating information required for land-use management and decision making, and in deploying climate-resilient practices and technology. Rwanda has only recently developed a system to monitor plantation forests and lacks a robust system to monitor indigenous forests. Similarly, agricultural and forest management techniques are generally rudimentary, and most rural	<u>low</u>	The GoR has an innovative, scientifically credible system for monitoring indigenous forest cover and condition, as well as the technical capacity for incorporating spatial data into land-use management. Improved agricultural practices (e.g. terracing, modern beehives) are widespread, and locally produced improved cookstoves are in use by most households.	The project will design and finance the establishment of a new indigenous forest monitoring system for Rwanda. The project will also establish improved agricultural practices across 3,346 ha, as well as increasing use of modern beehives to reduce forest fire risk. Finally, the project will upskill technical colleges to enable local production and distribution of improved cookstoves to 8,500 households.	National level (one country)

	communities in the CND do not use modern cooking technologies.				
Core Indicator 8: <i>Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards.</i>	While GoR recognizes the need for developing an enabling policy and institutional framework for a more climate-resilient development pathway, the value of forests and ecosystem services for climate adaptation is poorly understood, both in government and by communities. There is no system to regularly produce data on indigenous forest cover and condition, and no specialized government unit dedicated to producing spatial data on climate change, forests or ecosystem services. Agricultural and forest management practices are generally rudimentary.	<u>low</u>	Rwanda has an innovative, scientifically credible system for generating and sharing information on natural forest cover and condition, and a government unit dedicated to ensuring spatial planning knowledge is incorporated into land-use management. Improved agricultural practices (e.g. terracing, modern beehives) are widespread, and degraded forest plantations are restored and managed according to government best-practices.	The project will design and finance the establishment of a new indigenous forest monitoring system for Rwanda, as well as establish a spatial planning unit within government that has the capacity to generate and integrate spatial planning knowledge into decision making. Project activities will also establish best-practice agricultural and forest management practices across 144,058 ha of land, and create community structures (e.g. farmer cooperatives) which facilitate knowledge and information sharing among community members.	National level (one country)

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 1: Landscape planning, policies & management effective & coordinated across sectors and scales to address climate risks & adaptation benefits						
Output 1.1. Landscape-wide land-use plan developed for climate-resilient livelihoods and forest ecosystems, integrating district strategies.	Operationalization of an Integrated Land-use Plan for Resilient Livelihoods and Ecosystems in the CND Climate change and biodiversity reflected in District Development Strategies (DDS)	Interim evaluation and final evaluation report Performance evaluations of the districts	0 integrated plans for the CND Climate change and biodiversity Indicators and targets not included in evaluations	1 draft integrated land use plan developed Climate change and biodiversity Indicators and targets fully integrated in evaluations	1 final integrated land use plan developed, approved and used Climate change and biodiversity Indicators and targets fully integrated in evaluations.	District land-use plans are updated within project timeframe Key stakeholders are supportive of cross sector collaboration. Sectoral ministries incorporate biodiversity and climate change into sectoral plans and programmes.
Output 1.2. Local and national institutional capacities strengthened to integrate biodiversity and climate risks into land use planning and management	District planners, staff within REMA, RDB, NLA, RDB with the skills and capacities to use spatial planning tools, generate data necessary for mainstreaming climate change, understand gender barriers and how to mainstream gender equality principles into planning, including supporting women, youth	Interim evaluation and final evaluation report Pre/post training assessments of unit members	0 units with capacity 0 staff reporting improved awareness and capacity.	Draft structure & function of spatial planning unit developed. Training materials in development	Spatial planning unit formally established. 36 Staff reporting improved awareness and capacity (18 Female, 18 Male)	Lack of leadership of the hosting institution/ government agency could delay the establishment of the unit. RFA is committed to lead and invest in the capacity of its staff to implement the monitoring system.

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	and marginalized groups participation in meetings					Capacity building programs can overcome the effects of high staff turn-over and frequent re-arrangement of the institutions of natural resources management (transfer of departments, merging and creation of new Ministries)
	UR GIS center with the capacity and skills to train students and staff from government agencies on spatial planning and climate modeling	Pre/post capacity assessment of the UR GIS Center	Staff with knowledge in spatial analysis and climate modeling but without quantitative skills to use new technologies/software to generate data for decision making	Training modules developed and staff to be trained identified	10 lectures and assistant lectures reporting improved skills, 20 students trained each year from year 3.	
	A system for monitoring forest trends is established and in use by the GOR	Interim evaluation and final evaluation report	0 systems in place	Operational structure & function of forest monitoring system developed	Forest monitoring system in operation	

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	Rural people, specifically women and youth, are trained and participate in land use planning processes at district level	Interim evaluation and final evaluation report	0	3 women per district 3 youth per district	30 youth 30 women	Rural women and youth groups in targeted districts are motivated to invest time and efforts in the activities.
Outcome 2: Natural forests protected, connected, more resilient to climate impacts & risks						
Output 2.1. Protected Area management effectiveness improved re climate risks and adaptation	Climate adaptation reflected in the new general park management plans	Progress report, mid-term evaluation and final evaluation	0 plans currently contain climate adaptation plans	3 General management plans updated and approved by key stakeholders	Updated general management plans guiding RDB management decisions	General Park management plans are in the process of revision during the project intervention

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	RDB has adopted a landscape approach to conservation. New iterations of park management plans reference integrated CND land use plan A framework for conservation planning at landscape level is operational, and RDB planning and monitoring dept at each park is actively engaging stakeholders	Progress report, mid-term and final evaluations	0 park management plans reference CND land use plan	1 new park management plan referencing CND land use plan RDB proactively engaging stakeholders in the land use planning process of the CND	3 new park management plans referencing CND land use plan	Limited engagement of RDB beyond parks' boundaries could delay the operationalization of the planning framework

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	Number of RDB, RFA, districts staff and local communities trained in fire forecasting, detection and management	Post-training assessments of change in capacity of trained park staff in designated positions on 1) fire detection, early warning methods and fire danger rating system and 2) management of climate information and its relevance for park management Post-training assessments of change in capacity of trained individuals in designated positions	0 staff reporting improved awareness and capacity 0 staff reporting improved capacity for fire management	15 Staff reporting improved awareness and capacity (8 Female, 7 Male) 15 staff reporting improved capacity for fire management	15 Staff reporting improved awareness and capacity (8 Female, 7 Male) 30 staff reporting improved capacity for fire management	Capacity building programs can overcome the effects of high staff turn-over and frequent re-arrangement of the institutions of natural resources management (transfer of departments, merging and creation of new Ministries) All stakeholders embrace the concept of information sharing and learning willingly and enthusiastically.
	Financial instruments to generate revenues post GCF identified and piloted	Interim and final evaluation reports Technical reports	0 financial instruments	1 financial instrument identified	3 financial instruments identified and 1 instrument piloted	GoR commitment to enactment of policies conducive to attracting investment in emerging financing solutions

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Output 2.2. Natural forest cover restored, biodiversity connections established	Area (# ha) of degraded natural forest restored in national parks	GIS mapping of cleared areas using field GPS data	0 ha restored	4,500 ha initially cleared	4,500 ha cleared at least four times, ensuring natural regeneration can take place in Nyungwe NP	Major fire outbreaks during project implementation may prevent reaching the targets
			0 ha restored	500 ha under restoration in GMNP	500 ha restored in GMNP	The park is well managed and there are no major illegal activities
	Area (#ha) of riparian lands restored with indigenous species in the CND	GIS mapping of restored areas using field GPS data	0 ha restored	500 ha	1500 ha restored	District officials are committed to protect and create 10 m buffer around streams/rivers as per land use policy
	Area of remnant natural forest loss avoided	Interim and final evaluation	TBA	Draft strategy and action for the protection and conservation of remnant forests	Strategy and action plan developed and implemented for the conservation of remnant natural forests	RDB, REMA and District officials are committed to protect remnant natural forests

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	Area (ha) of pastoral lands brought under climate resilient silvopastoral packages in GMNP Average tree density/ha in restored silvopastoral lands	GIS mapping Bi-annual technical report Interim Evaluation and Final Evaluation Report Annual forest inventory	TBA TBA TBA	500 ha Density: 50 tree/ha	1000 ha Density: 50 tree/ha	Farmers are willing to integrate trees on their pasture lands
Outcome 3: Vulnerable rural livelihoods more climate-resilient, diverse, economically sustainable & nature-positive						
Output 3.1 Farming methods enhance productivity, reduce erosion and flooding risks, contribute to ecosystem services, and support connectivity.	Area (# ha) of forest restored on slopes >55%	Interim Evaluation and Final Evaluation Report Annual forest cover report ¹⁴⁴ Catalogue of geo-referenced area under restoration	0 hectares	2,500 ha under restoration on slopes >55% planted with seedlings.	2,500 public and private hectares restored and maintained. density: 1,200 trees/ha	Farmers, women, and youth groups in targeted districts/sectors are motivated to invest time and efforts in the activities.

¹⁴⁴ The Annual Forest Report is prepared by Rwanda Forest Authority. The Monitoring system for the preparation of these reports has been developed.

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	% of tree species diversity in restored areas on slopes >50%	Interim Evaluation and Final Evaluation Report Annual forest inventory	0	30% indigenous species and 70% exotic species on private land 70% indigenous species and 30% exotic species on public land	30% indigenous species and 70% exotic species on private land 70% indigenous species and 30% exotic species on public land	
	Number of target beneficiaries (households) in target sites who have adopted agroforestry measures (and report improvements in soil and water management)	Interim Evaluation and Final Evaluation Report Household surveys	0	5,405 households (of which 50% are women headed households) adopted agroforestry measures	5,405 households beneficiaries (of which 50% are female).	
	Average tree density/ha in restored agroforestry lands	Interim Evaluation and Final Evaluation Report Household surveys Catalogue of geo-referenced area under restoration Annual forest inventory	0 TBA	 100 trees/ha	3,346 ha of agroforestry established. 100 trees/ha	

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Output 3.2. Rural livelihoods generate alternative incomes & reduce pressure on forests	Number of value chain products that are viable and marketed (Honey, Avocado, macadamia, horticulture)	Interim evaluation and final evaluation report Annual sales reports	0	3 products developed	5 products developed	Stakeholders in the value chains are collaborative
	Number of businesses/cooperatives with viable business plans established and operational for climate resilient value chains	Bi-annual technical report Interim Evaluation and Final Evaluation Report	37 beekeeping cooperatives with 1,965 members (32% women, 16% youth) 0 seed enterprise, nurseries farmer's cooperatives 0 cooperative of tourism guides	15 new cooperatives (40% women, 50% youth) 5 seed and nursery enterprises owned by youth and women. 2 cooperatives	20 new cooperatives (40% women, 50% youth) 10 seed and nursery enterprises owned by youth and women. 3 cooperatives	Willingness of farmers' representatives of and Rwanda Cooperative Agency (RCA) to participate in the project
	Increase in household income within community associations/cooperatives engaged in the project (disaggregated by gender and social category)	Household surveys	TBA in year 1	20% additional income in participating households in new income generating activities. Targets for health benefits, and time and firewood savings TBD	An additional 10%	Successful rolled out of capacity building and linkage to outside (input-output) market

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	# of target beneficiary households adopting cookstoves	Interim Evaluation and Final Evaluation Report Household surveys	0 target households using Tier 2 cookstoves	4000 target households of which 50% are women-headed households	8,500 target households using Tier 2 cookstoves	
	Number of employments generated from the selected value chains	Interim and final evaluations Annual reports on employees hired per value chain	TBA in year 1	# women:600 # young:400	# women:900 # young:600	No outbreak of a new pandemic Farmers, women and youth groups in targeted districts/sectors are motivated to invest time and efforts in the activities.
Output 3.3. Financial services & private sector investment engage	Agriculture Credit assessment tool upgraded to include forestry	Loan approval reports generated by the tool Interim and final evaluation	0	3 MFIs and PFIs are using the tool to provide loans to smallholder farmers 70% of the smallholder farmers and gradually grow	5 MFIs and PFIs are using the tool to provide loans to smallholder farmers	MFIs and PFIs are willing to participate in this project
	Number of financial products refined, contextualized , and rolled out to improve access to finance for agriproducts for 1) vulnerable groups/ smallholders and 2) horticulture, bee products,	Bi-annual technical report Interim Evaluation and Final Evaluation Report	TBA	4 financial products refined and rolled out . At least 1 for each value chain (4 total)	6 financial products already deployed by PFIs are refined, contextualized and at least 5 rolled out by PFIs for financing	MFIs are willing to contribute in terms of staff and resources

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	energy efficient cookstoves and ecotourism value chains				selected value chains. 10 approved loans' applications	
	Number of people including women and youth with organizational, technical and financial improved capacity	Interim and Final Evaluations	0	7,800 (40% youth, 50% women)	12,445 (40% youth, 50% women)	Farmers, women and youth groups in targeted districts/sectors are motivated to invest time and efforts in the activities.
	Number of men, women and youth who have access to thriving financial services	Interim and Final Evaluations	0	3,000 (40% youth, 50% women)	8,500 (40% youth, 50% women)	Men, women and youth are open to access financial services

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 1.1.1 Synthesize & disseminate information on value of natural forests & ecosystem services	Ensure that the value of natural forests and ecosystem services for supporting climate change resilient livelihoods is fully appreciated by all sectors involved in land use decision making	1.1.1.1: Map the sectors involved in land-use planning in the CND and review how forests, ecosystem services and climate resilience is incorporated into each sector's planning process. 1.1.1.2: Conduct comprehensive literature review & stakeholder consultation to collect & synthesize information on climate risks for various	A map of the sectors involved in the land-use planning in the CND Information package on climate risks, value of forests for increasing resilience of local communities, adaptation options developed and disseminated to ministries, local government and local NGOs

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		sectors (e.g. agriculture, forestry), the value of forests for increasing resilience of local communities, and highlight adaptation solutions with cross-sectoral implications 1.1.1.3: Host workshops & presentations with key ministries, government organizations, NGOs and community organizations to disseminate high-level knowledge on value of forests for increasing resilience of local communities 1.1.1.4: Develop guidelines for integrating climate risk into land use planning and cross sectoral planning 1.1.1.5: Develop outreach materials on climate risk in the CND and the value of forest ecosystems for increasing resilience 1.1.1.6: Implement outreach program tailored to different stakeholders (local government, civil society, communities) to enhance capacities for land-use planning, funding mobilization, and delivery of climate adaptation actions 1.1.1.7 Conduct climate literacy courses for local government and civil society organizations, aimed at increasing women and youth participation 1.1.1.8 Provide financial and logistical support to trained organizations in grassroots mobilization to increase	Guidelines for integrating climate risk into land use planning and cross sectoral plan developed 104 episodes of radio drama to educate communities on climate change and gender issues, climate risks, value of forests and forests ecosystems, climate adaptation options developed Broadcast of 104 radio program episodes on a popular national radio station

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		women and youth participation in climate adaptation planning 1.1.1.9 Introduce social safeguards at a high level at each meeting, including the GRM, FPIC, and Access Restrictions Mitigations as safeguards to be in place for work with local communities	
Activity 1.1.2 Develop climate-resilient landscape land-use plan	Ensure that climate change adaptation requirements are fully integrated into planning processes at local, district and national scales.	1.1.2.1: Engage district officials, JADF, and community members in the CND (especially women) in reviewing and interpreting the NLUDMP, to ensure the plan accounts for current and future climate risks, while building local support for climate sensitive planning. 1.1.2.2: Conduct participatory land-use planning process in communities from village to district level to support integrated climate resilient land use planning 1.1.2.3: Develop Integrated Land-use Plan that supports Resilient Livelihoods and Ecosystems in the CND, ensuring alignment with National Land Use and Development Master Plan and ensuring the CND plan guides the district plan 1.1.2.4: Develop and rollout a series of trainings (virtual sessions and online modules) on gender sensitivity and mainstreaming women and youth into planning	A summary of the current NLUDMP and its recommendations relating to forestry, agriculture and climate adaptation. Reviewed/updated district level planning schemes to ensure they account for current and future climate risks (e.g. DFMPs, District Development Strategies) A community natural resources mapping manual to outline main steps in community land-use planning process A comprehensive review of national and district development plans/strategies, guidelines and projects to understand planned future land-use & climate adaptation projects in the CND and identify synergies and conflicts between land-use & climate adaptation plans from different sectors Integrated Land Use Plan developed

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		1.1.2.5: Assess specific climate impacts on historically marginalized and Category c,d,e populations through a participatory NR process to ensure the component projects address their needs for adaptation	
Activity 1.2.1: Create interagency taskforce institutionalizing integrated landscape planning & policy	Collaborative integrated landscape planning supported, which secures climate resilience undertaken with appropriate social safeguards. This activity aims to strengthen collaborative efforts, in particular between institutions (from local to national level) in charge of natural resources management to encourage synergies and avoid overlapping mandates and redundancy in different climate resilience interventions.	1.2.1.1: Review & strengthen operationalization of the current cross-sectoral task force 1.2.1.2: Hold quarterly sectoral planning meetings with both national and district administrations in CND 1.2.1.3: Facilitate discussions and provide technical support (e.g., policy briefs) in decision-making for cross-sectoral actions around climate adaptation and forest resilience in the CND 1.2.1.4: Continue to introduce social safeguards at each meeting including the GRM, FPIC, and Access Restriction Mitigations as safeguards to be in place for work with local communities. For those at the district or community level, provide options for feedback on the process and best ways of communication with local partners	A cross-sectoral planning task force operational as recognized by a ministerial order
Activity 1.2.2: Build capacity for spatial planning in national agencies re climate change	Capacity for integrated land use planning which supports climate change resilience and delivery of ecosystem services is improved. This activity aims to build the technical capacity of staff in the Ministry	1.2.2.1: Recruit spatial planning expert to support spatial planning unit	Spatial Planning Unit Operational Capacity needs assessment completed

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
	of Environment and University of Rwanda to generate and use the data on forest cover and ecosystem service values necessary for mainstreaming climate adaptation and forest resilience into land-use planning..	1.2.2.2: Assess and identify the institutional home and operationalize the spatial planning unit 1.2.2.3: Conduct a capacity needs assessment and identify the appropriate tools for spatial planning to fit the Rwandan context and conduct familiarization 1.2.2.4: Led by the spatial planning unit, develop capacity within the University of Rwanda and the MoE to use remote sensing, climate change modeling and spatial planning tools (e.g., Marxan, InVEST, RIOS, SWAT) to incorporate climate risks into land use planning processes 1.2.2.5: Deliver training workshops on utilization of earth system models for land use planning purposes, utilization of remote sensing resources (lightning, satellite radiances) for hazards detection and climate monitoring and on regional climate monitoring and applications	Capacity for forest mapping, climate change modeling and spatial analysis built within UoR and MoE
Activity 1.2.3: Develop an effective forest monitoring system to underpin forest management decisions	An effective forest, climate change and land use monitoring system is established to support climate resilient forest management decisions.	1.2.3.1: Recruit Forest ecologist or remote sensing expert to support the design and implementation of indigenous forest monitoring system 1.2.3.2: Review existing forest mapping data and monitoring software, and assess utility for generating quantitative	A report on the status of existing forest mapping data and monitoring software A design of the structure and operation of forest monitoring Two weather stations installed and operational Forest monitoring system operationalized

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		assessments of indigenous forest cover from local to national scales 1.2.3.3: In collaboration with REMA, MINAGRI, RAB, RDB, RISA, MoE and affiliated agencies and University of Rwanda, design structure and operation of forest monitoring system, including required inputs (data, computing, personnel), desired outputs, and operational structure (e.g. location, reporting structure, funding) 1.2.3.4: Purchase and install 2 new weather automatic stations in the CND 1.2.3.5: Establish & operationalize forest monitoring system 1.2.3.6: Generate updated maps & statistics for indigenous forest cover, forest type etc. using forest monitoring system 1.2.3.7: Develop capacity within hosting institution for continued operation of forest monitoring system	Updated maps and statistics for indigenous/natural forest cover and type generated Trainings delivered to develop the capacity of staff within hosting institution in forest monitoring
Activity 2.1.1: Facilitate revision of PA management plans to address climate risks	Park planning process expanded to ensure full incorporation of buffer zone and landscape linkage requirements, and ability to engage with the larger CND and district landscape management planning processes.	2.1.1.1 Review and update existing national park management plans to ensure climate and related landscape changes risks, impacts and required management responses are integrated 2.1.1.2 Provide technical and financial support to planning, research and monitoring as well as community based conservation units within RDB in	General management plans for NNP, GMNP and VNP updated

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		managing the parks as part of larger CND landscapes 2.1.1.3 Train PA staff to integrate gender and social inclusion into programming 2.1.1.4 Train PA staff on integrating needs of women, youth, historically marginalized and Ubudehe Categories c,d,e populations into climate adaptation risks and responses 2.1.1.5 Train PA staff on SEAH, GRM, FPIC, and Access Restriction Mitigation Processes to ensure NP climate change responses are undertaken in a way which supports social inclusion and equity	
Activity 2.1.2 Establish long-term plans for CND financial sustainability post-GCF	Identify financial instruments that could be employed for revenue generation post GCF as part of a broad and comprehensive RDB business planning effort for the sustainable management of the CND landscape	2.1.2.1 RFA to Recruit a Technical Advisor to guide RDB, MoE, REMA, FONERWA and MINECOFIN on sustainable conservation finance.	Technical Advisor recruited CND long term financial sustainability developed
Activity 2.1.3 New fire management curriculum developed and operationalized in PAs	Capacity increased for improved fire management in core PAs and adjacent landscapes. A collaborative effort in fire management planning and implementation in CND districts will be carried out and facilitated.	2.1.3.1 Develop a curriculum tailored to needs and capacities of different stakeholders 2.1.3.2 Build the capacity of RDB to manage fire in National Parks 2.1.3.3 Build the capacity of RFA, local authorities and communities to collaboratively manage fire in forests outside PAs	A fire training program that addresses local needs and skills gaps developed Train the trainer process completed for 25 RDB staff allowing them to instruct future classes without assistance Fire detection, early warning, and fire danger ranger system is established and operationalized

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		2.1.3.4 Implement a “fire wise” outreach and awareness program for communities and local government (district, sector, cell), as well as Rwanda Forest Authority, surrounding natural forests to reduce the incidence of human-caused fires	200 community facilitators (youth) trained to raise awareness on fires in communities surrounding natural forests and forests plantations 10,000 community members reached through social marketing events
Activity 2.2.1 Secure key remaining natural areas outside PAs	Remaining priority natural forests and other ecosystems required for climate change resilience secured through appropriate land use planning processes	2.2.1.1 Raise awareness on remaining protected natural forests in CNDL to secure their protection 2.2.1.2 In collaboration with RFA, REMA, districts identify and implement actions that support conservation and management of remnant protected natural forests in CNDL	Action plan for the conservation of remnant natural forests developed and implemented
Activity 2.2.2 Restore natural forest cover in & outside PAs including riparian linkages	Support natural forest restoration processes in PAs through improved management of alien and invasive species; and actively restore nature forest in priority areas next to core PAs and protected forests to expand the natural forest estate	2.2.2.1 Review mapping of degraded natural forest areas in core NPs, stepping stones and unprotected riparian lands using updated imagery and ground truthing 2.2.2.2 Identify parcels for restoration in core PAs, stepping stones and riparian lands using desktop and field-based assessment 2.2.2.3 Establish indigenous tree seed nurseries in the CND to serve core PAs, stepping stones, riparian land restoration and promotion of indigenous trees on farms and in protective forests 2.2.2.4 Recruit, train in forest restoration methods and equip community workers to	Indigenous tree seed nurseries established and operational 4,500 ha of indigenous forest in NNP restored 500 ha restored in Gishwati Mukura NP 1,500 ha of riparian lands restored

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		be involved in core PAs restoration and riparian linkages 2.2.2.5 Restore 4500 ha of indigenous forest in NNP 2.2.2.6 Restore 500 ha Gishwati Mukura National Park 2.2.2.7 Restore 1,500 ha of riparian lands 2.2.2.8 Using permanent plots sampling, collect field monitoring data on tree species recruitment and growth for estimation of biomass, species richness etc. in restored parcels in Nyungwe and Gishwati-Mukura National Parks in restored parcels in Nyungwe and Gishwati-Mukura National Parks as well as on riparian lands to assess success of initiatives, then replicate or adapt as needed 2.2.2.9 Perform Environmental and Social Screening on all positions being created by the project 2.2.2.10 Establish a grievance process for laborers 2.2.2.11 Update emergency and preparedness plan including risk mitigation guidance to local conditions at restoration sites	
		2.2.2.12 Train all workers on their rights and how to access the GRM	

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		2.2.2.13 Provide code of conduct and emergency preparedness and safety training for all laborers 2.2.2.14 Hire and train labor and safety leads at each site to provide guidance to staff, be available for grievance issues, and monitor health and safety conditions for workers 2.2.2.15 Develop and implement strategy (including social marketing) to ensure that women and youth participate and benefit from forest restoration projects	
Activity 2.2.3 Promote silvopastoralism with indigenous trees in Gishwati Pasture areas	Existing pasture lands will be characterized and livestock farmers clustered according to the size of their grazing lands. Tree species that exist on their grazing land will be identified and grouped according to the level of their resilience to climate change. A participatory approach will be used to assess livestock owners' preferences for indigenous species on their farms. Sites will be identified for indigenous tree nursery establishment and nurseries will be established in project sites and managed by the livestock communities.	2.2.3.1 Assess the status of pasture lands in Gishwati area and identify key areas for restoration and the potential to introduce indigenous species 2.2.3.2 In collaboration with landowners, identify suitable indigenous species for fodder trees, shrubs, grasses, and herbaceous legumes that have potential to improve pasturelands and increase their climate adaptive capacity 2.2.3.3 Produce and disseminate fodder trees, shrubs, grasses, and herbaceous legumes to project beneficiaries 2.2.3.4 Train beneficiaries on improved livestock and pasture management	1,000 ha of pasture land in Gishwati grazing area restored with indigenous tree species

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 3.1.1 Restore high slope areas (>55%) as protective forests	Promotion of protective forest and agroforestry practices to improve landscape connectivity and ability to deliver ecosystem services	3.1.1.1 Introduce and raise awareness of indigenous species to target stakeholders in CND 3.1.1.2 In consultation with RFA, National Land Authority, districts and communities, determine fragile areas (steep slopes > 55%) to be allocated for protective forests and their ownership 3.1.1.3 Assess the current status of the indigenous tree species in selected areas for protective forests and select indigenous tree species appropriate to CNDL 3.1.1.4 Develop restoration plan for protective forests 3.1.1.5 Build capacity of local stakeholders (men and women) on PFMU approach and methods 3.1.1.6 Design and approve SFMPs of private FMUs 3.1.1.7 Ensure consent of smallholders prior to planting trees (SS) 3.1.1.8 In collaboration with smallholders reforest/restore 2,500 ha of public or private land with slopes >55% and ensure sustainable management under private FMUs according to approved SFMPs 3.1.1.9 Support monitoring and evaluation of restored private FMUs	2,500 ha of protective forests on high slope areas restored

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		3.1.1.10 Assess impacts of exotics on neighboring lands and mitigate their negative impacts (SS) 3.1.1.11 Develop and implement strategy (including social marketing) to ensure that women and youth participate and benefit from forest restoration projects (G) 3.1.1.12 Assess the benefits and costs of each proposed tree and plant species and how those benefits affect different population segments (SS)	

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 3.1.2 Develop on-farm agroforestry for high caloric and indigenous tree species	Implement agroforestry/sustainable land management (SLM) over 3,346 ha of smallholder farmland	3.1.2.1 Identify sub-areas of intervention for agroforestry dissemination in the CND 3.1.2.2 Introduce and raise awareness of agroforestry in target communities 3.1.2.3 Develop the capacity of extension agents at district/sector level and NGOs to support adoption of agroforestry technologies 3.1.2.4 Establish agroforestry/fruit tree nurseries to facilitate access to quality planting material 3.1.2.5 Promote sustainable land management practices by stabilizing existing terraces, and plant high calorific agroforestry species to provide a sustainable source of fuelwood for energy efficient cookstoves to protect sloping land against severe soil erosion 3.1.2.6 Train RFA, extension agents, Project participants and community members on specific techniques for identification of and management of invasive pests and pathogens (SS)	SLM/agroforestry established on 3,346 ha of smallholder land

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 3.2.1 Develop forestry and agroforestry-related value chains for market access	Vulnerable groups of youth and women identified and their capacity for production (quality and quantity) enhanced in sync with particular niche markets demands	3.2.1.1 Identification and mapping of vulnerable groups of youth and women within the CND 3.2.1.2 Capacity building for farmers' cooperatives/unions on good agricultural practices (GAP) for market standards in the selected agroforestry-related tree fruits and horticulture 3.2.1.3 Development of Global GAP standards modules and market access linkage for the selected value chains 3.2.1.4 CND Honey value chain capacity production, market, and financial development 3.2.1.5. Tourism value chain development	Vulnerable groups of youth and women within the CND identified and grouped into cooperatives 5 Value chains identified and beneficiaries selected GAP training for 900 people from targeted cooperatives and unions completed Train of trainers completed for 30 cooperative members New tourism trails and base camps outside of Gishwati-Mukura NP developed and operational Community freelance tour guides trained and operational
Activity 3.2.2 Facilitate and scale up climate-resilient value chain products	This activity aims to build the capacity of smallholder farmers in producing, processing and marketing quality products that meet national and international standards to be competitive on the market	3.2.2.1 Value chain contextualization in focus areas in the CND 3.2.2.2 Facilitating market accessibility	25 youth and women cooperatives/enterprises with the capacity to deliver quality products and services
Activity 3.2.3 Facilitate access to markets for vulnerable farmers	This activity will focus on enhancing productivity (quantity, quality, and sustainability at the farm level) and exploring the market linkages model between farmers, suppliers, and buyers.	3.2.3.1 Development of farmer-inclusive business models	900 cooperative members are linked to the market from farming inputs suppliers and production aggregators through inclusive farming-as-a-business.

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 3.2.4 Scale up production, use and sales of fuel-efficient cook-stoves	Evaluate and promote financing models that sustainably foster energy efficient cook-stove access to historically marginalized communities and other vulnerable groups of the CND	3.2.4.1. Identification and production of Energy Efficient Cooking-Stoves 3.2.4.2. Energy efficient cook-stoves distributed to CND community members and categorization based on UBUDEHE Categories	8500 households have access to EES
Activity 3.3.1 Facilitate access to finance & private sector investments	Build common understanding between value chain actors and private financial institutions towards selected value chains	3.3.1.1 Tailoring credit assessment tool for selected value chains, while integrating climate impact data 3.3.1.2 Facilitate access to finance for selected value chains	Agriculture Credit assessment tool upgraded and adopted by PFIs At least 15 projects of the 5 selected value chains funded by PFIs
Activity 3.3.2 Set up & support savings & loan groups, enhance asset-building	Strengthen the capacity of local smallholders' farmers in financial literacy and access to finance	3.3.2.1 Establish/form saving groups for access to finance	Cooperative members are linked to PFIs and at least 70% have accessed to financial service and loan
Activity 3.3.3 Build capacity of financial institutions to serve targeted value chains and communities	Undertake a climate screening of financial service providers using the green index tool and support PFIs to access and adopt a digitalized forestry and agri-loan assessment tool	3.3.3.1 Capacity-building of PFIs	PFIs have developed an environmental policy and strategy for their organization including staff awareness training and integration of climate and livelihood impact measurement systems
Activity 3.3.4 Facilitate learning & knowledge sharing	Produce and share lessons learned during the implementation of this project	3.3.4.1 Programme-embedded learning and reflection	ESMP

E.7. Monitoring, reporting and evaluation arrangements

The Project's logical framework outlines the expected results, indicators, means of verification, baseline values and target values at mid-term and end-term. The Project Coordinator will be responsible for the overall monitoring of Project progress and reporting to the PSC. A national M&E Specialist hosted under the PMU, supported by EE-based Specialists, will be employed to develop practical guidelines for — and operationalize — a performance monitoring framework to track the Project's progress towards achieving its targets. The national M&E Specialist will be responsible for: i) establishing a performance monitoring framework to define biannual targets; ii) measuring the indicators at least 1–2 times per year to evaluate the progress of the Project; iii) reporting to the Project Coordinator and PSC; and iv) participating in the production of reports. RFA M&E Specialist will support the National M&E Specialist by: i) following guidelines developed by the national M&E specialist to conduct performance monitoring; ii) measuring the program indicators at least 1–2 times per year; iii) ensuring all activities comply with the Project's ESMF process and procedures; iv) supporting environmental screening and helping to follow the Environmental Management Plans; and v) participating in production of reports. The M&E Specialists will be supported by a Gender Specialist to oversee and monitor the application of gender disaggregated indicators. The M&E Specialist and Gender Specialist will perform regular visits (at least once/year), randomly selecting villages to monitor implementation. They will use these visits to cross-check the information provided by EE and beneficiaries in monitoring reports. During each visit, the team will collect qualitative data on implementation outcomes and difficulties through focus group discussions with beneficiaries and interviews with key stakeholders. The national M&E Specialist, supported by the EE M&E Specialist will then compile quarterly Project reports outlining progress in each site, allowing for feedback and support from the PMU and PSC in case of delays or challenges. MoE, as the AE, will perform supervision missions from year two onwards, selecting sites to visit each year and assessing the status of implementation. In addition, the MoE procurement office — which oversees the destination and use of funds disbursed to the EE — will monitor the Project's compliance with policies and procedures, as well as alignment between implementation and objectives. The annual performance reports will include a narrative report (with supporting data) on implementation progress based on the logical framework in the Project, including a report on ESMS as well as gender. The report will be aligned with the modalities set out in the GCF results management framework and its performance measurement frameworks.

As per the AMA, an independent consultant will be contracted to conduct an Interim Evaluation and Final Evaluation. Both will comply with GCF evaluation policy and international standards and make use of qualitative and quantitative data analyses, including interviews with relevant stakeholders, data provided by the Project, EE M&E Specialists and Gender Specialist, and participative workshops and focus groups with beneficiaries. The MTE will analyze whether the Project is on track, any problems, and challenges, and required corrective actions. The PSC and PMU will develop a management response to the recommendations along with an implementation plan, and the PMU will monitor implementation of the recommendations. The Final Evaluation will be an independent assessment of Project performance against its contributions to a paradigm shift and enabling environment.

• (f) RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures

Selected Risk Factor 1 - *Lack of coordination, institutional resistance, and regulatory barriers to mainstreaming climate change into land use planning*

Category	Probability	Impact
<u>Macroeconomic/Political</u>	<u>Low</u>	<u>High</u>

Description

Despite its progressive policies, Rwanda has historically had limited capacity for mainstreaming climate change into government/institutional practices and given the pervasive ways in which the impacts of climate change impact across numerous sectors, ministries and government agencies will need to work across any perceived or real institutional barriers to align and mainstream climate change in land use planning. For example, the Rwanda Development Board has historically not worked beyond Park boundaries, and this will be integral for institutionalizing the climate change adaptation processes across the Congo Nile Divide as a large interconnected landscape. Additionally, insufficient coordination, collaboration and information-sharing between relevant government departments, and research institutions may hinder or limit participation of relevant institutions and agencies, reducing the timely delivery and effectiveness of the Project, especially the aspects related to mainstreaming climate change in land use planning and other components of the Project as well. Furthermore, the perception of an uneven playing field, with larger or more well-connected institutions and companies able to evade the payment of environmental fees and fines under existing legislation, may undermine the uptake of the other options for financial sustainability of improved PA and forest ecosystem management.

Mitigation Measure(s)

The Project is part of a larger suite of efforts supported by the multi-agency Project Steering Committee (PSC) to create a spirit of cooperation among the various agencies and institutions charged with addressing climate-related issues. The PSC demonstrates Rwanda's commitment to fulfilling mainstreaming of climate adaptation into regular government processes and procedures. As such numerous government agencies have already had the opportunity to work together on a variety of climate-related issues, align institutional policies and reduce any potential regulatory barriers to ensure successful implementation of projects related to climate change. The PSC will directly oversee the Project Management Unit, and with this larger guidance structure, the EE will communicate to the PMU and the PSC on a regular basis to prompt collaboration and address any unforeseen institutional or regulatory barriers encountered during the life of the Project for immediate attention and remediation. PSC will address the risk of evading the payment of environmental fees by facilitating buy-in from stakeholders such as RDB and MINECOFIN into compliance with environmental laws through greater emphasis on the links to climate resilience and NDC objectives. These measures will reduce the impact to LOW.

Selected Risk Factor 2 - *Extreme weather/climatic events hinder the ability to deliver Project objectives*

Category	Probability	Impact
<u>Other</u>	<u>Low</u>	<u>Medium</u>

Description

The activities that will deliver overall Project success involve dependencies on weather conditions for successful implementation of the projects that rely on accessing steep slopes, planting trees, and establishing beekeeping operations. Extreme events such as heavy rainfall or prolonged drought could severely limit the success of these endeavors making sites inaccessible, creating unfavorable conditions for seedling survival, and decreasing the potential success of new beekeeping operations. This region, given its topography and the expected climatic variability, is likely to experience these types of disruptions. Severe disruptions may impede implementation, including potential implementation delays and fall under GCF's policy on restructuring and cancellation. That said, it is unlikely these extreme conditions will present themselves in the next 5 years for the duration of the project.

Mitigation Measure(s)

As weather is unpredictable, any mitigation measures will not change the probability nor address the impact in the short term. As the entire project is envisioned to improve the resiliency of this landscape to climate variability and promote adaptation, the specific activities will be able to mitigate, and in some cases prevent, longer-term impacts. As our implementation arrangements involve bolstering the climate adaptation capacity of the government and the communities in the CND, the entire project itself will serve as mitigation to this unlikely outcome.

Selected Risk Factor 3 - Project staffing and capacity

Category	Probability	Impact
<u>Operational</u>	<u>Low</u>	<u>Medium</u>

Description

Implementation of the Project hinges on staffing of the Project Management Unit, hiring additional staff within the EE and recruiting thousands of additional laborers in the form of community engagement specialists, contractors for building terraces, individuals to remove ferns, contracts to construct and deliver stoves, consultants to contribute expertise on behavior change, invasive species/pest management, and gender sensitization. Many factors could intervene to slow or thwart hiring progress including the high turnover of staff in the implementing institutions as frequent changes in government bodies could result in a disruption and/or delays in implementation.

Mitigation Measure(s)

Rwanda has a successful track record of implementing large scale projects through government ministries. Competent staff will be available for project implementation, and their abilities will be enhanced through the commitment to training individuals involved in the program. This risk, if realized, may impede implementation including potential implementation delays and fall under GCF's policy on restructuring and cancellation. Therefore, the Project will invest heavily in strengthening institutional capacity to promote climate resilience and apply actions that directly increase the resilience of ecosystems and community livelihoods. The Project will also document decisions, best practices and lessons learned to support the institutional memory to sustain Project activities in case of high staff turnover. Labor and safety as well as community security plans to be established as part of the ESAP (Annex 6) will make the Project a desirable place to work. Mitigation measures reduce impact to LOW.

Selected Risk Factor 4 - Inadequate incorporation of gender, and environmental and social safeguard (ESS) considerations into the implementation of the proposed Project activities as well as its management structures and systems		
Category	Probability	Impact
<u>Technical and Operational</u>	<u>Low</u>	<u>Medium</u>
Description		
Should ESS be inadequately considered in the Project or in the management and governance structures, there could be social and environmental implications. To start there may be a disproportionate distribution of benefits to certain segments of society. Although the Project's primary beneficiaries include women and other most vulnerable people, there is a risk that the Project implementation will experience preferential treatment at sites that favor certain groups over others. For example, despite tremendous policy and legal efforts, a patriarchal culture and persistent disparities in education, employment, land ownership and other factors characterize gender relations in Rwanda and could cloud implementation efforts. Likewise, some vulnerable groups may be unable to benefit from the efforts, such as landless people unable to benefit from trees or having a safe place for a new cookstove. One serious risk afflicting major projects such as that proposed, if not properly addressed, includes sexual exploitation, sexual abuse, and sexual harassment (SEAH) which may be prevalent if preferential treatment occurs (e.g., interfamily SEAH in retaliation for benefits received by women rather than men) or there may be power dynamics with hired staff who work with communities and withhold Project benefits within a SEAH conflicts (e.g. requesting sex for an opportunity to participate in the project, harassment if potential beneficiary refuses). On the environmental side, several environmental risks detailed in Section F and Annex 6 could undermine the success of the Project including topsoil loss and possible collapse of the terraces, unsustainable practices for sourcing clay for the cookstoves, and inadvertent introduction of invasive species, pests or pathogens from the nurseries to the field sites.		
Mitigation Measure(s)		
The Project anticipates these challenges and has envisioned multiple participatory aspects to the Project that are guided by the gender action plan and the ESAP. To mitigate social risks, local community engagement will be facilitated by the EE in cooperation with local community organizations through the integrated landscape delivery teams (ILDT) and the Joint Action Development Forums (DJAF). Safeguards in place to support Labor and Community Health & Safety will require training/sensitization to SEAH and the repercussions for those who engage in such activities (i.e., termination of employment). In the hiring process, staff will be screened for prior infractions. Hired staff will be trained in survivor-centered and gender-responsive approaches to support any victims of SEAH in association with the Project activities. To minimize environmental risk, the PMU and the EE will adhere to the environmental action plan and oversee the hiring of contractors to perform the activities. Multiple consultants or specialists, if not part of subcontracts, will be recruited to train project participants on mitigating measures to reduce these risks. If the proposed ESAP is followed properly the impacts will be LOW.		
Selected Risk Factor 5 - Lack of adoption among target populations for livelihoods activities		
Category	Probability	Impact
<u>Technical and Operational</u>	<u>Medium</u>	<u>High</u>

Description		
The success of the Project depends upon the integration of activities such as beekeeping, improved cookstoves, planting of fruit trees and terrace construction on private lands by the project beneficiaries. If planned and executed incorrectly a myriad of issues could prevent beneficiaries from adopting the activities beyond the life of the Project. This includes but is not limited to overlapping or misalignment between the proposed Project activities and those being implemented by other government departments or non-government organizations in the Project area.		
Mitigation Measure(s)		
Numerous international development projects have experienced mixed results upon completion regarding the uptake of promoted activities into daily lives. This Project has a strong behavior change component based on current best practices in identifying the barriers to behavior change, removing those barriers, and changing norms thus ensuring, through adequate participation and training, that the activities will be adopted and maintained over time. The work plan involves ample collaboration with women and youth groups through the ILDTs to build long-term community ownership for the Project. Further, if there are activities being promoted by the government or other NGOs that either overlap with the proposed activities or contravene with the Project activities, adoption by intended beneficiaries may be disrupted. As such the NSC will serve as the overarching body to ensure such projects within similar geographic regions remain aligned. These mitigation measures reduce the impact to MEDIUM.		
Selected Risk Factor 6 - Unforeseen disruptions due to another COVID variant		
Category	Probability	Impact
<u>Other</u>	<u>Low</u>	<u>Medium</u>
Description		
As demonstrated during the recent years of the COVID pandemic, variants may cause unforeseen disruptions to travel and Project implementation.		
Mitigation Measure(s)		
The Project will likely start well after COVID has run its course; however, in the event of new variants the PMU and EE will be ready to implement necessary COVID-safety practices such as masking, social distancing and reducing travel to areas with low health care options. Mitigation will reduce impact to LOW.		
Selected Risk Factor 7 - Prohibited practices including money laundering and terrorist financing		
Category	Probability	Impact
<u>Prohibited Activities</u>	<u>Low</u>	<u>Medium</u>
Description		
The Project involves a large number of actors for successful implementation including ministries, government agencies, NGOs, sub-grantees, vendors, service providers, community members and beneficiaries. Specifically, there is an intention to distribute or disperse to beneficiaries both directly and indirectly commodities and other items of value such as the distribution of seedlings for trees that will produce fruits or contribute to enhancing biodiversity of farming systems, access to improved cook stoves as well as the		

possibility for distribution of bees to start honey operations. As with any project infusing large sums of money directly or indirectly with multiple actors, corruption could occur.

Mitigation Measure(s)

Since the 1990s Rwanda has set a social foundation for anti-corruption efforts by establishing policies and procedures to prevent corruption/reduce vulnerability to corruption and sanctioning misbehavior through prosecutions. Specifically, a reduction in administrative corruption is of note for a large project such as the one proposed. The systems in place will reduce the potential for prohibited activities. Additionally, commitment to campaigns for individuals to report corruption has created a societal norm for intolerance. As a result, Rwanda moved from a state of high corruption 20 years ago to a level on par with middle income countries. The International Monetary Fund published an assessment of Anti Money Laundering and Combating Financing of Terrorism in 2015 indicating that while the risk of AML/CFT is low, at that time more could be done to bolster the systems in place to address the issue. At present, the GoR has robust measures to fight money laundering and terrorist financing. This was done first by enacting the law N° 001/FIC/2022 of 16/02/2022 on Regulations relating to anti-money laundering, combating the financing of terrorism. The Rwanda government has also put in place a special agency for this purpose named Financial Intelligence Center (FIC) with a clear mission to promote financial system integrity by conducting effective financial intelligence to counter money laundering, the financing of terrorism and related financial crimes in governmental institutions, projects, and parastatal agencies under the supervision of the Ministry of Finance and economic planning. MoE and the GCF Project under its supervision will also comply with FIC regulations. The MoE also has an Internal Audit function as well as regular audit conducted by the Office of the Auditor General to carry out the efficiency and effectiveness of internal controls.

Regarding higher level security threats, at present, there are no UN Security Council Sanctions against Rwanda, though it is noted that there are sanctions in Democratic Republic of Congo, close to the Congo Nile Divide. The UN Security Council Consolidated Sanctions list indicates approximately 20 individuals with ties to Rwanda, a small percentage of whom may live in the country. This list has been circulated to the AEs and will be cross checked when hiring for the project and once project sites are selected to ensure these individuals will not be involved in any manner with the Project or its activities, either as a counterparty, implementer, or beneficiary.

To report and remedy any risks that may occur, the MoE will maintain a Project level grievance redress mechanism to which anyone can report on the occurrence of prohibited activities. All complaints will be investigated following the process identified in Annex 6. Any complaints believed to contribute to prohibited activities will be referred to the appropriate authorities in Rwanda.

Assuming all staff will be made aware of the prohibited activities, all subcontractors/counterparties will be screened for engagement of activities and made aware of their obligations to alert the PMU and/or submit grievances related to such activities occurring as part of the Project, the overall impact of the money laundering and financing of terrorism risk on the Project will be LOW.

Selected Risk Factor 8 - *Political Risk*

Category	Probability	Impact
Other	Low	Low
Description		

Political risk is in part driven by long-standing frictions with neighboring countries and the potential for domestic instability in these countries to spill over into Rwanda. These geopolitical challenges constrain Rwanda's longer-term development as a services hub. Although the government of Rwanda has been effective in its pursuit of social stability through targeted transfers and support and improving accountability at all levels from Villages and to national levels through village councils, JADF and Umushyikirano (National Dialogue Council - an annual forum that brings together leaders and citizens to reflect on the country's progress and share ideas to solve ongoing challenges, this political risk may impede implementation including potential implementation delays and fall under GCF's policy on restructuring and cancellation.

Mitigation Measure(s)

Much of the work of the project will be taking place within a Ministry and through work at the community level. Oftentimes, regardless of political instabilities, communities with devolved decision-making structures and local capacity will have a greater chance to withstand that instability and continue to operate. For this project, much of the work will be turned over to the local community leaders and the laborers hired locally who will be trained and will then own the capacity to do this work in the long-term, regardless of what national leadership is in place. The structures resulting from this workplace citizens participation in decision making which will constitute a buffer against potential domestic instability. Mitigation will reduce impact to LOW.

Selected Risk Factor 9 - Expansion of tea plantations

Category	Probability	Impact
Macroeconomic/Political	Low	High

Description

Projected climate changes predict significant elevation rises in CND agro ecological zones, including the potential for tea cultivation at much higher elevations than at present. This is unlikely to threaten Rwanda's existing montane forest protected areas, as foreign revenues from tourism in these reserves far surpass those from tea. However, tea remains a significant source of foreign revenue and a government target for expansion. Any increased tea production in the CND would therefore most likely affect lands outside of parks: smallholder, cooperative, and district holdings, many of which might also be targets for reforestation, agroforestry, and biodiversity stepping stone initiatives under this Project.

Mitigation Measure(s)

The integrated land use planning included in this Project design, will bring together tea industry, agriculture, forestry, biodiversity, and tourism interests to identify and reduce such potential conflicts over use of the CND's limited land resource base. Through this integrated land use planning we aim to establish a formal public-private sector forum at the landscape level to help improve communication across sectors, appreciation of environmental issues, and increase synergies between interventions to avoid conflicting targets and indicators between sectors and advocate for mainstreaming biodiversity and climate change in different economic sectors.

Selected Risk Factor 10 - Timely provision of Co-financing

Category	Probability	Impact
Other	Low	Low

Description
The Project co-financiers may delay providing their co-financing commitment which can jeopardize implementation and hence project intended objectives could not be met.
Mitigation Measure(s)
The Accredited Entity has already requested for commitments of co-financing from all the project co-financiers which are already Government of Rwanda institutions, and all the committed co-financing are already part of the Institutions Day to day mandates hence there is anticipated low impact on the project implementation

(e) GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment

As proposed, the environmental and social risk category for the Project is **Category B or ‘moderate risk’ according to GCF classification. The project was** partially screened for potential social and environmental risks, following GCF requirements (Annex 6). All preliminary potential negative impacts for activities and sub-activities were determined to be minimal, reversible and readily mitigated through generally accepted mitigation measures and standard international practices, thus in the low to moderate categories. While the impacts will occur within the Congo Nile Divide Project area boundary and implementation zone, the specific locations and communities affected by Project activities will be determined in Project implementation thus an Environmental and Social Management Framework has been developed for a more thorough screening of all activities by the AE/EE.

Summary of Risks	Mitigation Factors / Measure and Management Strategies

Environmental and social risks: For all Project components, disproportionate benefit distribution to some groups and not others including women, youth, historically marginalized people and Rwanda's poverty categories (Ubudehe) C, D, and E is a risk. Such power differentials could exacerbate exposure to sexual abuse, exploitation and harassment, a risk that may affect all Project components if unmitigated. The Project may cause social risks including loss of access to lands due to new land classifications or stricter implementation of land use and park laws, displacement of informal land occupants due to afforestation/reforestation, and loss of access to land for food production. The potential for invasive species spread in the reforestation/afforestation components, water usage during increased seedling production at nurseries, terrace stabilization for planting, loss of topsoil to erosion during planting, disturbance to wildlife during afforestation or reforestation, modification of water flow and topsoil that could increase exposure to disease, increased human-wildlife interactions with increased tree cover and fruit trees, air quality impacts from cookstove production at scale, and supply chain impacts for livelihoods interventions all have associated environmental and social risks to be thoroughly vetted as specific sites and beneficiaries are identified.	The PMU will hire an Environmental and Social Safeguards (ESS) Specialist to oversee the application of the ESMF for a more thorough assessment of impacts by specific activities during Project implementation. The ESS Specialist will also oversee stakeholder engagement, sensitize all relevant staff on social safeguards and community engagement including SEAH training, and ensure Project activities target vulnerable groups to build resilient livelihoods. Consent processes will be established with communities. The Project will have a Project-level grievance redress process available to address complaints that aligns and complements the grievance redress used by the MoE. The PMU will also hire an Environmental Management Specialist (EMS) who will work with the ESS on environmental assessment and implementing proper mitigation processes as required.
Summary of Risks	Mitigation Factors / Measure and Management Strategies
Labor and working conditions: The dangers involve the potential for sexual abuse, exploitation and harassment (SEAH) in the hiring process or while working, forced labor, child labor, and exposure to occupational health and safety risks. These risks include safety of laborers removing ferns and planting trees on steep slopes, exposure to potential disease vectors in standing water or soil and large scale fabrication of stoves, risks using the machinery in the production of stoves, exposure to bees in honey production as well as other similar risks.	The ESS will coordinate with AE/EE human resources staff to conduct a Labor and Working Conditions Risk Assessment that will inform the development of a Labor and Working Conditions Risk Management Plan. The ESS and HR staff will appoint field safety leads within each sector. These individuals will receive training on National Laws governing labor and working conditions, SEAH, and other risks relevant to the activities that involve the hiring of laborers during the course of the Project.

Resource efficiency and pollution prevention: The stabilization of terraces risks topsoil loss and terrace collapse due to deteriorating conditions and improper long-term maintenance and extreme weather events. Pollution/waste may be created across the supply chains for stove production and distribution. Pesticides and fertilizers may be used for seedling production. Water usage in nurseries could divert resources for others. The release of CO2 emissions in the sourcing and distribution of materials in reforestation/afforestation as well as the livelihood enhancement activities could also pose a risk.	The ESS, EMS, and a Procurement Specialist will conduct activity-specific assessment for resource efficiency and pollution control. During implementation they will oversee subcontracts for reforestation/afforestation and all livelihoods enhancement projects. For stabilization of terraces, the subcontractor will have an engineer oversee the technical components of stabilization to ensure best practices and reduction of risks. For livelihoods enhancement projects the ESS and Procurement Specialist will ensure environmentally sourced materials and establish measures to minimize production waste and maximize distribution efficiency of stoves. The EMS will work with nurseries on plans for mitigating pesticides, fertilizers and water use.
Community safety and security: Increased risk of forest fires due to increasing tree coverages, increased exposure to vector-borne diseases, human wildlife conflicts associated with reforestation/afforestation and the influx of international funding that shifts power dynamics and creates environments prone to increased incidents of SEAH. With expected changes in land classifications, interactions among law enforcement and informal land owners or communities who use the resources could become more prevalent and increase risks of harm if law enforcement lacks proper training for working with communities. Work on steep slippery slopes, working with bees, use of new cook stoves and work in remote locations (e.g., fern removal) will require safety considerations for participating community members. Extreme weather or global events (e.g. pandemic) may disrupt safety and security.	The ESS will screen each activity for community, safety and security risks. The EE will have at a minimum, an Emergency Response and Preparedness Plan to address specific local hazards and response at each site; all staff will be trained on the plan as well as on disease prevention and hygiene, SEAH prevention, and addressing human-wildlife conflicts. The PMU will need to address at a high level across the GoR the potential for escalating conflicts among landowners and law enforcement as land classifications change in the course of the Project.
Summary of Risks	Mitigation Factors / Measure and Management Strategies
Resource access/restrictions: Land classification processes could result in changes that affect resource access due to existing land use, occupancy, and land tenure in the CND especially among farmers, settlers, and traditional uses of the land and forest. At a smaller scale with less risk, as tree planting and transitions to native or value-added species (e.g. fruit trees) commence, access to tree resources may be diminished until trees grow larger and become usable.	The ESS will screen all Project activities for the potential to cause displacement, namely access restrictions to resources. The Project is committed to stakeholder engagement and will mitigate access restrictions risks through a process framework and mitigation hierarchy. The AE grievance redress process will manage access complaints.

Biodiversity conservation: The inadvertent introduction of non-native/invasive pest species/diseases during forestry/agroforestry related activities could negatively impact production and forest health. Wildlife may be disturbed in the course of fern removal and/or tree planting as well as other potential livelihood activities that will be implemented. Human-wildlife interactions may increase as a result of increased forest cover and fruit trees being available in the CND. This could cause landowners to retaliate against wildlife.	The ESS will screen all Project activities for biodiversity and conservation risks/impacts. Mitigation measures will be put in place to ensure tree planting, nursery management, and seedling distribution incorporate training by consultants on invasive species recognition and management and that options for managing human-wildlife conflicts are able should issues arise.
Historically marginalized peoples: Rwanda is a nation with a single/common culture, tribe, language and with a national constitution that recognizes all Rwandans are born and remain equal in rights and freedom (article 16 of Rwandan Constitution, 2015). That said, there are historically marginalized groups as well as categories c,d, e, groups in the areas of the project activities. The risks are that the impacts of project activities on these groups are not adequately considered and analyzed and further that the intended benefits for these groups do not materialize given a lack of attention to their particular needs and vulnerabilities. Should this occur, the overall intent of the project could be in jeopardy given the focus on making such communities more resilient in light of climate change uncertainties.	As the project focuses on providing benefits, in particular increased resilience to climate change for vulnerable communities, the ESS will pay particular attention to the screening for activity level impacts on historically marginalized and categories c,d,e. Thus far, 24 focus groups with 4 exclusively historically marginalized, 5 exclusively women and 3 exclusively youth participants informed the project design and provided insights on impacts and improved ways of benefiting individuals in these groups (see Annex 6, section 4). A detailed plan for engaging with women, youth, those below the poverty line and landless groups will be prepared by project staff in collaboration with district, sector, and local ngo leaders in the project implementation phase as specific projects and locations are determined. In addition, all interactions with these groups will include processes for obtaining consent as well as clarity regarding access to the grievance redress process should complaints arise during the project.

Risk details and screening assessment to be conducted in Project implementation are described in the Partial Environmental and Social Impact Assessment and Environmental and Social Management Framework (Annex 6). Positive project impacts include a focus on improving resilience of the most vulnerable communities, communities stewarding/ harnessing the carbon cycle and reducing greenhouse gasses by sequestering carbon (i.e., through conserving indigenous trees, increasing soil organic carbon, and practicing regenerative agriculture); and utilizing efficient cook stoves. The PMU/ESS will be responsible for conducting screenings, including screenings for project-level impacts on historically marginalized groups of each Project activity as specific locations and beneficiaries are determined and with implementing mitigation measures aligned with Rwanda National Laws and GCF policies and practices. These measures include a commitment to training in all relevant EE staff on stakeholder and community engagement; consent with local communities; access restrictions mitigations; SEAH and adhering to the labor and safety laws that meet international best practices. The full Stakeholder Engagement Plan can be found in Annex 7. The EE and the AE will develop a compliance monitoring, supervision and reporting schedule that meets requirements in the GCF Information Disclosure Policy and Environmental and Social Policy. The AE will establish a

Program-level grievance redress mechanism that includes a process of public notification, documentation, investigation and resolution. This GRM will complement the MoE GRM.

G.2. Gender assessment and action plan

Women, youth and marginalized groups in the CND are key stakeholders in the Project and its primary beneficiaries. The Project will be implemented across various sectors along the CND, where natural forests are particularly vulnerable due to high population, increasing dependence upon fuelwood, and illegal use of forests to improve income streams. The Gender Assessment uses a range of primary (24 focus groups and 18 key informant interviews) and secondary (published reports, policies and academic literature) sources to triangulate our understanding of gender in climate resilience and Project design. The Gender Analysis identified and analyzed gender risks and impacts of the program and climate change through four steps: scoping; legislative and policy review; stakeholder consultation and analysis; and baseline analysis. The Action Plan operationalizes the findings from the Assessment, the ESIA, and youth-based information collected during the feasibility studies and stakeholder consultation. Despite tremendous policy and legal efforts, a patriarchal culture and persistent disparities characterize gender relations in Rwanda. Disparities persist in post-primary education; in access to and control of property, including land and economic resources; in employment opportunities and entrepreneurship; in decision-making at household and community levels; in family responsibilities and unpaid care work; and in the experience of violence, harassment, conflict, and insecurity. Sexual and gender-based violence persists at high levels. While women have made impressive political progress, especially at the national level, their local political representation and engagement remains unequal. Economic opportunities remain markedly gender-differentiated. In Rwanda and the CND, structural barriers to economic and political equity and female empowerment have significantly impeded measures to adapt to climate change impacts. While these structural inequalities are lower compared to other countries because of leading gender-considerate policies across sectors, the point remains that women's ability to access, use and control natural resources, infrastructure and services is different and low compared to men. This means the degradation of natural resources and new infrastructure will affect genders differently and result in greater vulnerability for women. However, women are vital change agents and can be powerful leaders from community to global levels in mitigating and adapting to climate change. Three key themes in the Gender and Social Inclusion Action Plan include i) Change negative gender and social norms, ii) Provide access to skills building, leadership and programmatic inclusion, and iii) Facilitate women's financial autonomy. Youth account for over 50% of Rwanda's population. An analysis was done to determine how GCF funding could support activities to align with and strengthen the sustainability of the overall program by supporting youth through the following activity themes: i) Non land-based opportunities, ii) Access to skills building, and iii) Youth leadership and Programmatic Inclusion. Social norms are difficult to change, and deeply entrenched biases will require active engagement by men, women, and youth, in decision-making roles as well as grassroots stakeholders. By working through the themes outlined above, social and technical barriers to participating in decision-making, planning, and activities will be addressed. Specific targets will be developed after a brief inception phase. Further details are in Annex 8.

G.3. Financial management and procurement

Funds from GCF will be provided to the MOE's GCF account (as the AE), under the terms of the FAA. MoE will be responsible for financial management and accountable for the use of GCF resources. Financial transactions will be subject to annual audits undertaken by internationally certified auditors. In addition, as the AE, MoE will: i) ensure that the project is executed in accordance with GCF standards; ii) supervise,

oversee and manage the implementation of project interventions; iii) report on Project progress; and iv) ensure that Project activities are well coordinated and aligned with countries' national priorities. MoE will provide oversight to the project consisting of a) entering into a subsidiary agreement with the EE (RFA); b) managing and disbursing GCF funds to the EE, after providing no objection to work plans and budgets; c) reviewing financial expenditures and progress reports; d) overseeing Project implementation in accordance with the Project document and Annual Work Plans and Budgets, agreements with co-financiers and each executing entity rules and procedures; e) providing technical guidance to ensure that the appropriate technical quality is applied to all Project activities; f) providing financial reports to the GCF for Project funds received; g) ensuring that the project complies with the terms agreed in the Project's respective FAA as well as the AMA signed between MoE and the GCF; and h) undertaking regular annual supervision missions according to the MoE's guidelines and convene Project Steering Committee (PSC) meeting twice a year aimed to review progress and approve work plans. The PMU will manage, monitor and report on day-to-day implementation of the EE. The EE will execute the tasks within the overall project management structure: a) implementing day-to-day activities as per the project work plan and budget, including the Environmental and Social Management Plan; b) undertaking procurement activities as per the agreement signed by GCF and AE as well as the sub-agreement signed by the AE and EE; c) managing contracts of suppliers and services providers; d) hiring and managing project staff relevant to the EE-managed project areas; e) implementing activities as per the project work plan; and f) carrying out financial and technical monitoring of activities, including the achievement of outputs and outcomes the EE are in charge of. The EE will enter into a Memorandum of Understanding with different partner agencies that will provide technical know-how to the implementation of the project activities. Procured parties will be contracted in accordance with MoE rules and regulations of procuring goods and services. The financial management structure of the project is built on the Permanent Secretary of MoE, the Project Coordinator/ Chief of Party and the Finance Management Specialist at the PMU level who will oversee and monitor the project finances with project directors and finance managers from the EE (RFA).

The project will be executed by Rwanda Forestry Authority (RFA), a Rwanda government entity affiliated to the Ministry of Environment with a semi-autonomous legal entity. RFA has transparent and accountable financial management systems in place, which are essential elements for the effective, accountable management of the project. RFA will follow the Government of Rwanda procedures for planning, budgeting, and accounting, which are outlined in the law n° 12/2013/ol of 12/09/2013 on state finances and property, and ministerial order n°001/16/10/tc of 26/01/2016 relating to financials; and supporting operational manuals issued by the Ministry of Finance.

The MoE shall ascertain that the EE has in place efficient internal control systems, review financial reports and proceed funds disbursements and will also periodically inspect the project for compliance as well as coordinate audits exercises for the EE. RFA will enter into agreements with MoE that outline general modalities of the funds management in line with GCF grant requirements and RFA's policies. GCF funds will be transferred to an RFA designated bank account. Funds will be disbursed to the EE based on agreed annual work plans and budgets, and its reporting will be on a quarterly basis. The project financial processes will rely on the EE's charts of accounts, internal control systems, disbursement plans, budgeting, expenditure management and accounting, reporting, and assets management as assessed by MoE. Procurement regulations embed the principles of fairness, transparency, accountability and the promotion of competitive economy and efficiency. The Project will be audited annually and commissioned by the Accredited Entity (MoE) to produce certified financial statements and reports to ascertain that GCF funds are well managed. The project will maintain a regularly updated record of all assets and documents that will be retained for at least 5 years after the end of the project.

G.4. Disclosure of funding proposal

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

(f) ANNEXES		
H.1. Mandatory annexes		
☒	Annex 1	NDA no-objection letter(s)
☒	Annex 2	Feasibility study - and a market study, if applicable
☒	Annex 3	Economic and/or financial analyses in spreadsheet format
☒	Annex 4	Detailed budget plan
☒	Annex 5	Implementation timetable including key project/programme milestones
☒	Annex 6	E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): ☒ Environmental and Social Impact Assessment (ESIA) or ☒ Environmental and Social Management Plan (ESMP) or ☐ Environmental and Social Management System (ESMS) ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
☒	Annex 7	Summary of consultations and stakeholder engagement plan
☒	Annex 8	Gender assessment and project/programme-level action plan
☒	Annex 9	Legal due diligence (regulation, taxation and insurance)
☒	Annex 10	Procurement plan
☒	Annex 11	Monitoring and evaluation plan
☒	Annex 12	AE fee request
☒	Annex 13	Co-financing commitment letter, if applicable
☒	Annex 14	Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule
H.2. Other annexes as applicable		
☒	Annex 15	Evidence of internal approval
☐	Annex 16	Map(s) indicating the location of proposed interventions
☐	Annex 17	Multi-country project/programme information
☐	Annex 18	Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
☒	Annex 19	Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
☐	Annex 20	First level AML/CFT (KYC) assessment
☒	Annex 21	Operations manual (Operations and maintenance)
☒	Annex 22	Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)



Republic of Rwanda

**RWANDA ENVIRONMENT
MANAGEMENT AUTHORITY
(REMA)**



Kigali, on **22 FEB 2022**
N° **0487**/NDA/2022

Deputy Executive Director
Green Climate Fund ("GCF")
Songdo Business District
175 Art Center-Daero
Yeonnsu-gu Incheon 22004
Republic of Korea

Re: Funding proposal for the GCF by the Ministry of Environment regarding "Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration Project"

Dear Sir,

We refer to the project titled "Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide Through Forest and Landscape Restoration Project" in Rwanda as included in the funding proposal submitted by the Ministry of Environment to us on 18 February 2022.

The undersigned is the duly authorized representative of Rwanda Environment Management Authority, the National Designated Authority of Rwanda.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Rwanda has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Rwanda;
- (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards, **FAUSTIN NZAMUKWYIYI** Faustin

For

Juliet KABERA
Director General/National GCF Focal Point



Cc:

-Permanent Secretary/ Ministry of Environment

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	Ministry of Environment (MoE) of the Republic of Rwanda
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Three Provinces of Rwanda (Western, Southern, and Northern) in 10 Districts (Karongi, Musanze, Ngororero, Nyabihu, Nyamagabe, Nyamasheke, Nyaruguru, Rubavu, Rusizi, and Rutsiro).
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Friday, September 22, 2023
Language(s) of disclosure	English and Kinyarwanda
Explanation on language	Kinyarwanda is the national language spoken by all Rwandans.
Link to disclosure	English Kinyarwanda *
Other link(s)	N/A
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the "Environmental and Social Management Framework (ESMF)"
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Friday, September 22, 2023
Language(s) of disclosure	English and Kinyarwanda
Explanation on language	Kinyarwanda is the national language spoken by all Rwandans.
Link to disclosure	English Kinyarwanda *
Other link(s)	N/A
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the "Environmental and Social Management Framework (ESMF)"
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A

Any other relevant ESS reports, Environmental and Social Management Framework (ESMF)	
Description of report/disclosure on accredited entity's website	Indigenous Peoples Planning Framework
Language(s) of disclosure	English and Kinyarwanda
Explanation on language	Kinyarwanda is the national language spoken by all Rwandans.
Link to disclosure	English Kinyarwanda *
Other link(s)	N/A
Remarks	A Historically Marginalized Peoples Plan, which serves as an Indigenous Peoples Planning Framework, is contained as an Annex to the ESMF.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, September 22, 2023
Place	To accommodate for those who may not have access to the Internet, a computer, or a reliable power source, the Ministry of Environment will be sharing the ESMF documents to the following districts: • Karongi district (2.0526° S, 29.4128° E), P.O Box: 23 Kibuye, village: Kamuvunyi, Cell: Gacaca, district: KARongi • Musanze district (1.5020° S, 29.6332° E), P.O Box: 03 Musanze, village: Byimana, Cell: Ruhengeri, District: Musanze • Ngororero district (1.8537° S, 29.6347° E), P.O Box: 103 Muhanga, village: Ngororero, Cell: Kazabe, district: Ngororero • Nyabihu district (1.5911° S, 29.5220° E), P.O Box: 125 Musanze village: Mukamira Cell: Rubaya District: Nyabihu • Nyamagabe district (2.4671° S, 29.5736° E), P.O BOX: 36 Gikongoro, village: Nyarusange, Cell: Nyamugari, district: Nyamagabe • Nyamasheke district (-2.342014193676258, 29.08901503907659), P.O Box: 72 Nyamasheke, village:Gikuyu, Cell: Ninzi, district:Nyamasheke • Nyaruguru district (2.6987° S, 29.5703° E), P.O Box: 215 Butare village: Nyarusovu Cell: Mubuga, district: Nyaruguru • Rubavu district (1.6736° S, 29.2886° E), P.O Box: 173 Gisenyi; village: Nyaburanga, Cell:Nengo, district: Rubavu • Rusizi district (2.4830° S, 28.9054° E), P.O Box: 253 Rusizi, village: Ntwali, Cell:Cyangugu, district: Rubavu • Rutsiro district (1.9617° S, 29.3857° E) P.O Box: 49 Kibuye, village: Nduba, Cell: Congo-Nil district: Rutsiro

	and request them to have physical copies of ESS reports in their local government offices, publish an Executive Summary of the ESMF on their notice board in both Kinyarwanda and English and publish it on their website to ensure it is accessed by all affected people in their location as well as to those who may have access to the Internet, a computer.
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, October 23, 2023

Note: This form was prepared by the accredited entity stated above.

** Subsequent to the disclosure of the ESS report to the Board and active observers on 22 September 2023, the following update has been made: the links to the Environmental & Social Management Framework (ESMF), in which the ESMF document -had been made available, have been updated for both English and Kinyarwanda. The footnote 86 has been removed.*

Secretariat's assessment of FP217

Proposal name:	Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration
Accredited entity:	Ministry of Environment of Rwanda
Country/(ies):	Rwanda
Project/programme size:	Small

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
This Congo Nile Divide project will safeguard Rwanda's biodiversity, enhance climate resilience, and support vulnerable communities.	The project should ensure linkage to other GCF funded projects in the country. In addition, we encourage MOE Rwanda to establish financial mechanisms to upscale across the wider landscape.
The project is located in the Albertine Rift, which is one of the most biologically rich regions in Africa, containing more than half of Africa's birds and 40% of its mammals.	The project should ensure monitoring and evaluation of the project interventions is undertaken.
With more frequent extreme rainfall events in recent years, this agricultural landscape is recognized as the region of Rwanda most vulnerable to floods and landslides.	
Project interventions will include improved management on 250,000 hectares (ha) (including promotion of natural regeneration/planting on 10,000 ha) and improved agroforestry practices on 2,000 ha of on-farm plantations.	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum XXI titled "List of proposed conditions and recommendations", respectively.

II. Summary of the Secretariat's assessment

2.1 Project background

3. The proposed project's goal is to improve the resilience of vulnerable communities to climate variability and shocks by increasing the extent and integrity of natural forests and by strengthening the capacity for climate-responsive forest and landscape management. The project will:

- (a) Strengthen institutional and regulatory frameworks for low-emission climate-resilient development pathways by mainstreaming climate change into integrated land-use planning.
 - (b) Increase the climate resilience of vulnerable rural communities through forest and landscape restoration and reducing demand for fuelwood.
4. The proposed interventions will target districts along the Congo Nile Divide (CND) region. Rwanda's CND, an area of 444,600 hectares (ha), is a high-elevation landscape of remnant natural forest reserves embedded in a mosaic of intensively managed smallholder farms and commercial plantations. The CND separates the drainage basins of the Congo and Nile Rivers. It is part of the Albertine Rift, which is one of the most biologically rich regions in Africa and "one of the most important regions for conservation" on the continent, containing more than half of Africa's birds and 40 per cent of its mammals.
5. Though covering barely 5 per cent of the country, this archipelago of protected forests is of critical importance for:
 - (a) Its globally recognized high biodiversity values;
 - (b) Its role in Rwanda's acclaimed ecotourism offerings, serving as the primary source of Rwanda's foreign revenue and as a major economic development driver;
 - (c) The ecosystem services provided for surrounding subsistence farmlands as well as commercial tea and coffee plantations offering employment opportunities for women while providing the second and third highest sources of foreign revenue;
 - (d) The regulation of water capture and release to growing urban centres on the CND periphery, the nation's extensive river network, and downstream hydroelectric operations; and
 - (e) The sequestration and storage of nearly 60 per cent of Rwanda's national carbon stocks.
6. These values and services are already compromised by a combination of anthropogenic and climatic factors; they will be much more severely threatened by the potential negative feedback loops associated with projected climate change impacts. The project will promote an ecosystem-based approach to forest protection and restoration and will increase the extent and resilience of forest ecosystems and species, including by functionally linking currently isolated forest fragments. Through these approaches, the project will protect, restore and rehabilitate forest ecosystems, thus enhancing biodiversity and reducing key threats while increasing the provision of critical ecosystem services to strengthen the adaptation of vulnerable communities.
7. Project interventions will directly benefit 1,254,242 people (654,404 women). The project will also indirectly benefit 9,260,745 people and create approximately 24,212 job opportunities in forest-dependent communities. The total mitigation impact is estimated to be 1,204,448 tonnes of carbon dioxide equivalent (t CO₂ eq) by 2043.
8. The total project financing is USD 50,000,000 with a request for GCF grants of USD 39,056,422. The Rwanda Forestry Authority (RFA), Rwanda Water Board (RWB), Rwanda Development Board (RDB) and the Ministry of Environment of Rwanda (MOE Rwanda) will provide co-financing of USD 10,831,308 in grants. MOE Rwanda will also provide an in-kind contribution of USD 112,271. The project is submitted under environmental and social safeguards (ESS) category B.

2.2 Component-by-component analysis

Component 1: Mainstreaming climate adaptation into integrated land-use planning (total cost: USD 2.9 million; GCF cost: USD 2.9 million)

9. This component aims to integrate climate change adaptation considerations into land-use planning at the national and subnational levels. The goal is to achieve climate resilience through collaboration among government agencies, civil society, the private sector and diverse partners. The component has two main outputs: The first is to develop a climate-resilient land-use plan for the CND by integrating climate adaptation information into existing land-use planning efforts. The second aims to strengthen institutional capacities in Rwanda to integrate biodiversity and climate risks into land-use planning.

Component 2: Forest and landscape management and restoration (total cost: USD 18.9 million; GCF cost: USD 16.3 million)

10. In this component, targeted interventions will integrate climate change awareness and adaptation into national forest park planning, policies and management, as well as restore degraded areas within those parks. These efforts will help sustain the forests' rich biodiversity, the lucrative tourism revenues that flow into local and national economies, and the ecosystem services needed for the climate resilience of vulnerable communities. This suite of activities will also identify priority areas outside of parks – in remnant natural stands, around wetlands and along streams – for protection and restoration.

Component 3: Enhancing climate adaptation through resilient livelihoods (total cost: USD 22.6 million; GCF cost: USD 15.3 million)

11. The activities under this component aim to combine environmental conservation with socioeconomic development, creating a multifaceted approach that resonates with the global commitment to sustainable growth. They include initiatives such as enhancing and restoring forest plantations, providing sustainable fuel sources, delivering carbon sequestration benefits totalling 522,606 t CO₂ eq over 20 years, and developing comprehensive land-use planning. The other activities focus on rural livelihoods, financial services and sustainable farming methods.

12. The component will support reforestation on steep slopes on both public and smallholder private lands. It will develop farm-based agroforestry for climate-resilient species that produce fuelwood and/or market products, while conserving and enriching soils. The component will complement land-based production with a scaling-up of alternative value-chain products for smallholders that are nature-positive, including, but not limited to, tourism-based income, beekeeping, and improved cooking stoves. Central to these activities will be the facilitation of access to market chains and financial mechanisms for establishing and maintaining businesses. Given that these activities are designed at the core to be profitable and promote financial sustainability, it is intended that they be replicable and sustained once GCF funding expires.

Project management (total cost: USD 3.1 million; GCF cost: USD 2.7 million)

13. The Project Management Unit (PMU) of the CND project will be established under MOE Rwanda/Single Project Implementation Unit (SPIU) to manage day-to-day operations of complex programme interventions. Headed by the Project Coordinator, who will also serve as the secretary and rapporteur of the Project Steering Committee, the PMU will be responsible for tasks such as managing and monitoring identified risks and reporting new risks to the accredited entity (AE) and the Project Steering Committee. The PMU will adhere to MOE Rwanda/Government of Rwanda recruitment practices and provide quarterly progress reports. The Project Coordinator will report to the MOE Rwanda/SPIU Coordinator as the immediate supervisor and to the Permanent Secretary as the secondary supervisor.

14. The PMU's structure will comprise various specialists, including for monitoring and evaluation, procurement, gender, financial management, communication and outreach, along with an office manager and driver. Its staff will be selected and recruited following MOE Rwanda/Government of Rwanda procedures, aiming for a lean management structure. Operations will conclude once the final project evaluation report and other required documents have been submitted to MOE Rwanda, signifying a comprehensive and organized approach to the management of the entire programme.
15. The GCF portion of this project management cost is less than 5 per cent of the total requested GCF funding and is compliant with the GCF Policy on Fees.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: N/A

16. The project has 1,254,24 direct beneficiaries, representing 30.9 per cent of the CND population, and 9,260,745 indirect beneficiaries, representing almost 70 per cent of Rwanda's population. Additional adaptation benefits will include the training of 45,992 community members in forest management and restoration for climate resilience. Altogether, the project will contribute to the restoration of 5,000 ha of intact forest, 2,500 ha of protective forests, 3,346 ha of smallholder farmland, 1,000 ha of pastures and 1,500 ha of riparian linkages. Overall, 275,617 ha of landscape could potentially be positively impacted by the project. Although the project is focused on priority landscapes covering 275,617 ha, it is likely that benefits due to increased adaptive capacity could feasibly extend across the full 700,000 ha CND. The project will generate mitigation benefits by sequestering approximately 250,841 t CO₂ eq during the five-year project period, which is about 0.9 per cent of the 28.2 million t CO₂ eq reduction target in Rwanda's Vision 2050. The total mitigation impact from reduced deforestation, the introduction of forest restoration and adoption of improved cookstoves is estimated to be 1,204,448 t CO₂ eq by 2043.
17. The project contributes to the implementation of Rwanda's updated nationally determined contribution (NDC) (2020) commitments. It is also aligned with the country's Green Growth and Climate Resilience Strategy and other national policies. The proposal objectives are also well aligned with the country's National Forest Policy and related plans for reversing trends of high deforestation.
18. With support from the NDC Partnership, Rwanda now has a single NDC implementation framework that captures both national and subnational climate actions and investments plans and includes a strategic programme for climate resilience, scaling up renewable energy programme, green climate fund strategic framework, public investment priorities, sector strategic plans and district development strategies. Therefore, the proposed project is well supported by this existing NDC implementation framework that provides an enabling environment that will ensure climate actions undertaken are integrated into national planning.
19. The proposed project further strengthens the country's national and subnational adaptation priorities, approaches and plans through promoting ecosystem-based adaptation while utilizing climate science, effective landscape planning, policies, cross-sectoral coordination and management in addressing climate risks and adaptation benefits.
20. The project in general seeks outcomes at a range of levels (from national through to catchment level) across the public sector, while scaling up lessons and initiatives from aligned programming. It will build the capacity of MOE Rwanda (AE) and other related agencies and stakeholders to coordinate and collaborate more effectively to help accelerate subnational and local climate action. It is envisaged that these funds will also crowd in additional private sector

capital (e.g. cookstoves and forest plantations) while enabling the government to implement the adaptation interventions. However, without a clear private sector framework or plan, this might be challenging to implement. Nonetheless, Rwanda has a GCF country programme and strategic framework and has managed GCF funded projects and programmes.

21. Furthermore, the project clearly articulates the alignment, additionality and/or complementarity with several ongoing or planned programmes funded by significant international and multilateral agencies (e.g. World Bank, Global Environment Facility and United Nations Development Programme, etc.), which reflects on the experience and track record of MOE Rwanda to deliver on this project interventions.

3.2 Paradigm shift potential

Scale: N/A

22. By utilizing integrated landscape and ecosystem-based adaptation approaches in addressing climate change impacts in Rwanda, the project offers myriad opportunities for upscaling and replication to similar ecological zones across the country. This approach to land-use planning and management is entirely new for Rwanda and will restore and sustain varied ecosystem services for both adjacent and nationwide populations. The project coordination and collaboration efforts across both geographic and sectoral boundaries, while mainstreaming climate risks in management planning, represents a fundamental paradigm shift for land-use planning and management in Rwanda.

23. The project has the potential to contribute to the regulatory frameworks and policies for driving climate investments in the forest and renewable energy sectors. The project will finance the development of the integrated land-use plan, which will be adopted by all ten CND districts as a framework for seeking synergies between rural livelihoods and forest and biodiversity protections, reconciling land-use conflicts that arise, and guiding development decisions and climate change adaptation. It will also update the National Land Use Master Plan to ensure it incorporates climate resiliency and is accurate and relevant to local contexts. The adoption of an integrated land-use planning approach helps to better reconcile cross-sectoral trade-offs and improve climate adaptation planning in the CND; this approach will prove a successful model for application in other regions of Rwanda and other forest/farming landscapes beyond. In addition, revising two national park management plans to include climate concerns will serve as a model for the remaining parks of Rwanda, as well as for the East Africa region.

24. Overall, the project's efforts in shifting policy, practice and collaborative learning processes symbolize a fundamental change towards climate-resilient development pathways in Rwanda, with implications for broader regional application.

3.3 Sustainable development potential

Scale: N/A

25. The project is closely aligned with several Sustainable Development Goals. Key environmental benefits include restoration of degraded lands to reduce erosion, sedimentation, and landslide risks; the planting of native trees to enhance biological connectivity; and mitigation of greenhouse gas emissions through both increased carbon sequestration and fuel-efficient cookstove technologies.

26. Social benefits extend to more than a million people, focusing on increased awareness of climate change, empowerment of the rural poor (with an emphasis on women's participation), and creation of social capital through collaborative adaptation strategies. The project aims to significantly reduce poverty and inequality. It will provide economic co-benefits such as increased agricultural productivity, employment opportunities for around 24,212 households, modern beekeeping practices yielding significant income, and a net present value (NPV) of USD 369 million over 20 years from forest restoration and other activities. Private capital

investment is also targeted, thus fostering a “triple bottom line” that balances social, environmental and economic benefits. Emphasis on gender-sensitive impacts ensures that women are actively engaged in planning and decision-making, leading to increased income, improved health and nutrition, and strengthened leadership and entrepreneurship skills.

27. Overall, the project offers a comprehensive and strategic approach to sustainable development, thereby promoting climate resilience, economic growth, social cohesion and gender equality within the CND region.

3.4 Needs of the recipient

Scale: N/A

28. The project proposal interventions were designed from key findings and recommendations of a climate change vulnerability assessment of the selected project geographical area (see annex 2). Climate exposure, sensitivity, adaptive capacity, impacts and barriers as well the greenhouse gas emissions profile were established and either quantified or supported with reference to scientific publications. Further, climate change impacts on the target sectors as well as size and economic and social development of the target population, including critical barriers to reduce their vulnerability, are elaborated herein. The proposed interventions are targeted to match the identified barriers. The vulnerability of the target sectors is quantified to the extent possible, however, gender disaggregation is covered in qualitative terms only.

29. The CND landscape districts support nearly one third of Rwanda’s population on barely one fifth of its land base. The total population is 1,598,536, of which women account for 52.2 per cent. Through the project interventions, targeted vulnerable groups will attain more climate-resilient livelihoods from reduced exposure to landslides, floods and soil erosion and more knowledge on climate risks, the value of forests and forest ecosystems, climate adaptation options, access to quality indigenous and agroforestry materials, and improved ecosystem services.

3.5 Country ownership

Scale: N/A

30. MOE Rwanda, which is the AE for this proposal, clearly played a pivotal role in defining and drafting the proposal stakeholder engagement report and plan (see annex 7) as it is aligned with Rwanda’s GCF country programme stakeholder engagement strategic framework. In its country programme, Rwanda articulated how it will conduct stakeholder engagement at the readiness stage and in delivering GCF concepts, projects and programmes. This process has provided Rwanda with an overarching framework that supports its stakeholder engagement processes, which is clearly seen in this proposal. This framework can support the strengthening of local climate action from national to local levels with clear feedback loops and avenues for incorporating learning.

31. The proposal indicates that a detailed stakeholder and gender analysis were conducted, augmented by extensive consultations of various actors from national, subnational and local levels. A range of stakeholders from the public and civil society were consulted, in addition to government agencies at all levels that will be engaged throughout the project cycle, cutting across all key sectors and scale. The general concern relates to local ownership and the transfer of knowledge that can sustain interventions after the project closure. It would be ideal if Indigenous knowledge and technology transfer is well factored into implementation.

3.6 Efficiency and effectiveness

Scale: N/A

32. Utilizing grant financing reflects the public good character of the investments, the project aims to create a model that overcomes barriers to private sector investment in Rwanda's forestry sector. With an economic internal rate of return of 43.32 per cent and a financial internal rate of return of 42 per cent, the project is estimated to generate an NPV of more than USD 120.48 million over 20 years, which translates to USD 64 per direct beneficiary. The mitigation cost is estimated to be USD 43.84 per t CO₂ eq. Implementing best practices for tree planting and utilizing cutting-edge monitoring technologies, the project is aligned with similar initiatives and sets the stage for potential replication in other regions. Its community-focused approach, including measures to develop economically viable units, contributes to a threefold increase in net economic benefits compared to a scenario without the project.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

33. Project background: The project aims to increase resilience of vulnerable rural communities in the Rwandan side of the CND by building capacity for integrated planning and management through forest and landscape restoration and reducing demand for fuelwood. The project will involve the formulation and implementation of plans (i.e. integrated district land-use plans and enhancements of general management plans of national parks), coordination with various agencies, capacity-building, and physical interventions on the ground. This includes afforestation/reforestation of protected forest and private landholdings, installation of soil conservation measures, ecological interventions such as fern removal in natural forests and restoration of degraded forests in Gishwati-Mukura National Park, production and distribution of fuel-efficient cookstoves, beekeeping activities, and support for the value chains of these enterprises. Among the environmental and social (E&S) co-benefits are the enhancement of the unique, endemic biodiversity of the area through decreased sedimentation, improved water quality and recharge, reduced erosion, decrease in loss of soil organic matter as a result of forest restoration and establishment of connectivity between the forests across the CND landscape. The project is also expected to contribute to the empowerment of the women and youth in the target areas.

34. E&S risk category and safeguards instrument: The overall project E&S risk category is Category B (moderate). The Secretariat confirms the proposed E&S risk category since the project is, in itself, an environmental and social enhancement project. The potential adverse E&S risks and impacts will mainly come from the implementation of ground activities, are expected to be mostly minimal and highly localized and can be avoided/mitigated through careful design and the application of generally accepted mitigation measures and standards. The Secretariat conforms to the assigned risk category and it is within the accreditation level of the AE. A partial environmental and social impact assessment and environmental and social management framework (ESMF) have been prepared for the project.

35. Compliance with the GCF ESS standards: The paragraphs below provide a brief description of the programme's compliance with ESS standards.

36. **ESS 1 (Assessment and management of environmental and social risks and impacts):** The ESMF discussed the baseline socioeconomic conditions of the project area. It has also provided a general assessment of the risks and impacts of the project. The assessment included an indication of the positive and negative components of the activities and subactivities, the consequence of the impact, the probability that the impact will occur, the duration for which the impacts are expected to occur, and the overall significance of the impact.

Nevertheless, given that much of the project activities cannot yet be fully specified in terms of location, ownership, design and approach and will only be identified and fully specified through consultations and coordination with various stakeholders during project implementation, the ESMF provided the basis for identifying the risks and impacts of the subprojects and defining the processes and procedures to be applied in assessing and managing the risks and impacts of individual project activities during implementation. The major anticipated negative impacts of the proposed activities and subactivities include, but are not limited to: disproportionate distribution of project benefits, potential for gender-based violence (GBV), risks related to labour and working conditions, community health and safety, potential for introduction of invasive species and pests on planted seedlings, water use at nurseries diverting from other uses, use of pesticides and fertilizers in seedling production, destabilization and erosion during construction/reconstruction of terraces, land classification that impact access and introduce restrictions, increased exposure to disease/vector-borne illnesses, potential for negative human-wildlife interactions, and environmental risks related to the supply chain. Among the mitigating measures include a commitment to stakeholder engagement, a process of consent, training on sexual exploitation, abuse and harassment (SEAH), establishing a grievance redress mechanism (GRM), adhering to the labour and safety laws and international good practices, providing training on invasive species management, providing oversight in the supply chain, and adoption of practices to reduce the use of chemicals and production of wastes.

37. **ESS 2 (Labour and working conditions):** The project will likely require the hiring of professionals, service personnel and assistants, and activities on the ground such as terracing, removal of ferns and reforestation/afforestation activities will require the hiring of field workers and labourers. Risks associated with these activities include, but are not limited to, the potential for SEAH in the hiring process or at work, forced labour, child labour, and exposure to occupational health and safety hazards. Workers may be exposed to hazardous working environments such as steep slopes, disease vectors in standing waters, safety hazards in the fabrication/production of stoves, and exposure to bee stings in honey production, among other things. The project will ensure proper labour and working conditions are provided to workers (e.g. in forest camps), and trainings on labour and occupational health and safety standards will be undertaken. A Code of Conduct as regards worker safety and security will also be reflected in contract documents. Moreover, the project will not support activities that include the use of forced labour, human trafficking and child labour. During implementation, risks and potential impacts related to labour and working conditions will be further assessed and managed through the development of a labour and working conditions risk management plan, including ensuring that workers have access to a GRM.

38. **ESS 3 (Resource efficiency and pollution control):** The nature of the proposed interventions does not entail the use of substantial amounts of fuel, water, electricity or raw materials, except for the operation of nurseries (for its water requirement) and the production and distribution of cookstoves (for raw materials and electricity). The project will however promote the increased use of energy-efficient cooking stoves to lessen the impact on forest resources for fuel. During implementation, further review on resource efficiency and pollution control risks will be undertaken to assess and manage issues as regards water use and water availability, use of pesticides and fertilizers, potential for soil loss, soil erosion and slope destabilization, and use of raw materials and waste generation in the cook stove fabrication/production and distribution. The project will also ensure raw materials are from sustainable sources and will establish measures to minimize production waste and maximize distribution efficiency of stoves. Appropriate management measures will be developed upon screening and assessment of these risks.

39. **ESS 4 (Community health, safety and security):** The project activities could have community health, safety and security issues, which include potential for increased exposure to vector-borne diseases should ecological balance be altered or stagnant water develop as a result of the reforestation/afforestation and fern removal activities in the area (which may increase

the population of certain disease vectors and insect pests). Some of the mitigating measures include proper drainage design of terraces. There could also be possibility for human–wildlife conflict in areas where fruit trees are planted, which may inadvertently increase physical interactions with wildlife that could result in crop loss or physical injury both to humans and wildlife. The risk related to the presence of migrant labour is considered low given that the project will hire residents from surrounding communities. Nevertheless, the activities will be assessed more closely with regard to such risks, with measures to be discussed and determined in consultation with the communities during implementation.

40. **ESS 5 (Land acquisition and involuntary resettlement):** The project does not include any activities that will require resettlement or involuntary taking of land. However, project activities could facilitate a potential loss of access to lands given that the afforestation/reforestation activities could result in stricter enforcement of land use and park laws in changes to rules of access or land classification. To address this, the project will not establish protective forests on lands used for food production. On-farm afforestation will not involve the acquisition of lands, land rights or land-use rights through expropriation or other compulsory procedures and will comply with the country's legal system. Communities will be engaged in an inclusive and participatory manner and will be part of the planning and afforestation processes, and agreements on land rights and responsibilities will be developed with them. An access restrictions mitigation process framework is developed and will be implemented as required.

41. **ESS 6 (Biodiversity conservation and sustainable management of living natural resources):** The project intends to enhance the region's ecosystem to reduce the vulnerability of communities to climate change impacts. For instance, the proposed interventions on protected forests will be restoration (e.g. restoration of the 500 ha of degraded forest in GMNP, and the fern removal covering 4,500 ha of Nyungwe National Park, while the proposed interventions in the Stepping Stones and Landscapes Linkages (located outside of the parks) will be composed of the protection of isolated forest patches and afforestation on steep slopes and riparian areas supplemented with agroforestry activities to enhance ecosystem services. Nevertheless, care should be exercised in implementing these activities to ensure that negative impacts such as potential for the introduction of alien or invasive species, pests and pathogens from poorly managed nurseries are adequately addressed. During project implementation, consultations with communities and stakeholders will aid in determining the selection of appropriate species suitable to the locality.

42. **GCF Indigenous Peoples Policy and ESS 7 (Indigenous Peoples):** The AE recognizes that while there are historically marginalized groups as well as "Categories C, D and E, groups" that may meet the GCF criteria for Indigenous Peoples that could potentially be included in the project area, the screening so far has indicated that Indigenous Peoples have not been found in the project area. Given that specific sites will be selected for project activities during project implementation, each subproject will be subject to more in-depth screening for impacts to historically marginalized peoples, with particular attention to the GCF criteria of Indigenous Peoples, during project implementation. The AE will conduct a screening for subprojects as outlined in the "Screening for Presence of and Risks to Historically Marginalized Groups" to evaluate the characteristics of historically marginalized and Categories C, D and E people. Should the screening process identify Indigenous Peoples, then the subactivity will require a historically marginalized peoples plan. The AE also recognizes that there are potential barriers to integrating historically marginalized peoples into the project. Nevertheless, there is interest in working with historically marginalized groups to identify how traditional livelihoods like beekeeping can be adapted and promoted and how negative impacts on these livelihoods will be mitigated. The AE has included in the project the hiring of an ESS specialist and community engagement specialist to minimize risks as well including a commitment to stakeholder engagement, free, prior and informed consent and the setting up of a project-level GRM. In line with its roles and functions, the GCF Indigenous Peoples Advisory Group (IPAG) is available to

provide advice to the AE, which is also encouraged to share emerging good practices and success stories with IPAG.

43. **ESS 8 (Cultural Heritage):** The physical interventions (i.e. afforestation, reforestation and terracing, etc.) will be undertaken in mountainous areas where there are few human settlements. The preliminary screening of target areas also indicated that it is not expected that the project activities will disturb any cultural heritage sites. Nevertheless, the project has developed a chance find procedure to cover chance encounters of artefacts during project implementation.

44. **Institutional implementation arrangements:** As the AE, MOE Rwanda will oversee and monitor the management and implementation of the project. The Wildlife Conservation Society (WCS) and RFA will serve as EEs for the project through a project cooperation agreement. All work will be coordinated by the PMU and supported by other relevant government agencies, including the National Land Authority and the RWB within MOE Rwanda. The PMU will have oversight of the day-to-day operations of the project, including monitoring and evaluation. It will engage specialists (e.g. Environmental and Social Safeguards Specialist, Invasive Species Specialist, Community Engagement Specialist, Behaviour Change Specialist, Gender Specialist, etc.) that will screen for the potential environmental and social risks and impacts and develop management measures for proposed activities. The PMU will also be responsible for coordination on trainings for all project EEs. As regards safety, a Labour and Safety Coordinator will also oversee working conditions in the field. The project will also engage Community Engagement Specialists and Labour and Safety Officers.

45. **Stakeholder engagement and GRM:** The project benefited substantively from an extensive stakeholder consultation process, which started during the project conceptualization. A Stakeholder Engagement Plan has also been prepared to engage the stakeholders throughout project implementation. The project will have a project-level GRM, which is described in the ESMF. The GRM is independent of governmental judicial procedures. The project also allows access to the GCF independent redress mechanism. In line with the GCF Indigenous Peoples Policy, the GCF Indigenous Peoples' focal point will be available for assistance at any stage, including before a claim has been made.

46. **Sexual exploitation, sexual abuse, and sexual harassment safeguarding:** The SEAH safeguarding approach proposed by the AE is considered by the Secretariat to be in compliance with the SEAH provisions pursuant to the revised GCF Environmental and Social Policy. The project identifies that all project components may exacerbate the existing power differentials and expose the project beneficiaries to potential SEAH risk. The AE will take a number of mitigation measures to prevent and manage SEAH, including ensuring that all staff are trained on survivor-centred approaches to address SEAH, and that the Human Resources and Administration Officer and the ESS Specialist of the project will work together to develop a Security and Risk Management Plan to prevent SEAH incidences. The project's Gender Specialist and Community Engagement Specialist will develop training on SEAH prevention, GBV programming and survivor-centred approach. The project will also safeguard against SEAH through a Code of Conduct that explicitly indicates zero tolerance for GBV. This Code of Conduct will be shared with all personnel and used for training all employees. In addition to the prevalence of GBV in the target country, the project also recognizes that the influx of funds and activities, combined with power dynamics, may disproportionately affect women, youth, historically marginalized people, and others in the poverty categories of Rwanda. As the specific locations and beneficiaries will be determined during full implementation, each subproject will be screened to avoid any subproject that may aggravate risks of GBV and SEAH. The project will respond to SEAH incidents through GRMs at the AE, EE, and project levels. Training will be organized for all project staff, implementing partners, key service providers and other partner organizations to identify and regularly assess SEAH risks along the project cycle as well as

ensure that both the AE and all the EEs respond to SEAH incidents with survivor-centred and gender responsive approaches.

4.2 Gender policy

47. The AE has provided a gender assessment and action plan therefore complies with the requirements of the GCF Gender Policy.

48. The gender assessment provided demonstrates the commitment of Rwanda to advancing gender equality and women's empowerment. The assessment shows that Rwanda has a strong political commitment to promoting gender equality, reflected in its policy and legal framework and various sectoral strategies. It is the first country to have a majority female Parliament, with 61 per cent of these positions being held by women, and has established through its constitution a minimum of 30 per cent female representation in public governance. The country has also made international commitments such as ratifying and implementing the Convention on the Elimination of all Forms of Discrimination against Women, the Beijing Platform for Action and the Maputo Protocol on the Rights of Women in Africa. An institutional framework for coordinating and ensuring oversight of gender equality and women's empowerment is in place, which mainly rests with the Ministry of Gender and Family Promotion. The Ministry is mandated to coordinate the implementation of the national gender policy and advocacy on gender issues, a gender monitoring office and the National Women's Council. All ministries and public institutions have gender focal points. However, there is divergence between the frameworks in place and implementation on the ground due to challenges such as lack of financial resources and technical capacity within the various ministries as well as existing norms and practices that run contrary to equality principles and practices.

49. The gender assessment was conducted through desk review and stakeholder consultation. Additional stakeholder consultation will be conducted at national, district and project site level in order to further adapt project activities to the local context. The assessment indicates that women and men have different roles and responsibilities within society and households. These roles rigidly set women as caregivers and homemakers, while men are considered breadwinners engaging in income-generating activities, wage employment and decision-making. Women own less land, have less access to extension services and financial support (loans), and are less educated and able to gain experience in handling money. Women rely more on informal sources of loans as they do not usually have collateral to put up to access loans from financial institutions. Women engage and represent 66 per cent of the agricultural workforce; however, they occupy 19.7 of the paid jobs. Women's roles in afforestation and agroforestry is also curtailed by their lack of ownership and or land rights. Lack of land rights hinders the financial benefits women can get from resources on land such as trees. Women mostly use trees that provide little commercial value; rather they are focused on fulfilling their household needs through fuelwood, fodder, bean poles and fruits. Discussion with women had indicated that women observed shortages of fuelwood, fodder and wood suitable to their bean poles, and also that there is declining soil fertility and crop yields. Fuelwood also is the main source for cooking, which has health hazards for women, and gaining access to cleaner, improved cook stoves is a challenge. Men usually decide not to spend money to purchase the improved cook stoves that can directly address the multiple challenges faced by women. Female-headed households are identified as much more vulnerable to climate change impacts compared to male-headed households owing to the prevalent gender inequality. Engagement in alternative source of incomes for women poses its own challenges to women, one of which is affordability of required inputs, and this is particularly so for the beekeeping sector. The interest of the government in this area is also high, and investments have taken place, but only 23 per cent of women engaged in the sector.

50. The action plan includes activities that will address the challenges identified through the gender assessment. Activities contain corresponding indicators, targets, gender expertise and budget while also including timelines for each of the actions. The project will direct investments to the areas of equal access to resources (e.g. land), services (finance), capacity development, decision-making and also the participation of women in management, as beneficiaries/partners and as critical stakeholders in processes. Project interventions will respond to and address the growing pressure on natural resources, thus ensuring that the resilience and adaptive capacities of communities, including that of women and disadvantaged people, are built. The gender action plan is focused on ensuring (1) climate adaptation programmes and land-use planning address the priorities of women and youth; (2) women, youth and marginalized households have increased capacity to effectively engage in and benefit from forest management and forest restoration; (3) women, youth and marginalized households have improved capacity to engage in green jobs and livelihood activities; and (4) monitoring and evaluation effectively measures how women, youth and marginalized people are benefiting to enable the adaptive management of the project. A capacity-building component that focuses on strengthening cooperatives and village savings and loans institutions through gender mainstreaming across decision-making, membership, and products and services is included to ensure women are able to benefit from and engage in all aspects of the project. Private landowners who will be willing to engage in land afforestation activities will create private forest management units, which will then be assisted in forming and registering cooperatives that are at least 50 per cent women. Further, the selection of tree species will take into consideration women's usage and priorities. Women will be engaged and supported through knowledge acquisition, trainings, and skills development processes so that they benefit from the project implementation, particularly focusing on critical challenges such as rights to land and access to decision-making positions. Further, the challenge of rights to land will be addressed through advocacy work with men and local judiciary systems. This will be complemented by activities that will advocate for women's land rights and access to redress mechanisms. The project will engage community leaders and women and men community influencers to further promote equal and fair control of resources to mitigate the risk of men taking control over incomes earned by women.

51. There are three recommendations to the AE. The first recommendation is that the AE needs to ensure that gender issues are mainstreamed across all components of the project; this is lacking for component 3 (output 3.1 of the gender action plan), which presents many important opportunities to address gender issues. Secondly, the AE needs to refine and clearly show what outcomes it intends to achieve as a result of the interventions. Activity 3.1.1 in the funding proposal states the following regarding the composition of administrative committees: "Note that gender consideration is mandatory (at least 30% women)". It is not clear if 30 per cent refers to "women in decision-making bodies" or "membership of women in cooperatives", or both. These results are different in nature and should be captured independently. The AE should provide clarification in this regard. Thirdly, as part of its expressed commitment to do so, the AE should consistently collect data and report in a sex-disaggregated manner on the gender action plan.

4.3 Risks

4.3.1. Overall programme assessment (low risk)

4.3.2. Accredited entity/executing entity capability to execute the current programme (low risk)

52. We acknowledge the important role of the Government of Rwanda via MOE Rwanda as the AE (direct access). The AE demonstrated its comparative advantage and capability in

relation to this project with references to its track record in executing projects of comparable size or above (including FP073 and FP167 with GCF) and existing affiliation with the EE, RFA.

53. The AE provided adequate assurance that RFA has the necessary capacity to undertake the role as EE.

4.3.3. Programme-specific execution risks (Medium risk)

54. Co-financing: We acknowledge the sizable amount of co-financing totalling USD 10,493,579 (grants and in-kind) for a co-financing ratio of 1:0.27. If co-financing does not materialize then this may impede project implementation. The AE identified this risk, and the covenants in the term sheet are adequate mitigation measures – noting that the term sheet for the proposed project has not yet been agreed.

55. Project viability (and concessionality): The use of grants (and 100 per cent concessionality) is justified given the country context, the beneficiaries in Rwanda's CND, and the fact that the proposed activities are public goods in nature, including creating enabling opportunities for market development beyond the project to catalyse private investments (see component 3: Enhancing climate adaptation through resilient livelihoods).

56. Capacity-building activities and technology development/transfer (i) represent 45 per cent of total GCF funding requested; (ii) may not result in realizing expected results in case issues arise during implementation; and (iii) are embedded in the project's design and thus crucial to achieving intended objectives and outcomes. On the other hand, we note that identification of risks and the mitigation measures proposed by the AE are adequate.

4.3.4. Compliance risk (medium risk)

57. MOE Rwanda, in its capacity as AE, confirmed that it has conducted a risk assessment of money laundering and terrorist financing (ML/TF) in relation to project implementation. The probability of occurrence was assessed as 'low', and the impact severity in the event of occurrence was assessed as 'medium'.

58. MOE Rwanda confirmed that co-financiers have been identified based on their existing mandates and activities implemented in the CND region. Co-financiers' commitments will serve as a basis of MOE Rwanda engagement during the project implementation. During implementation, annual work plans for the project will be adjusted in collaboration with the EE, and all co-financiers are to align with all activities. MOE Rwanda, per its oversight role, will ensure that all of these stakeholders commit during planning to their co-financing and report this in the annual performance reports submitted to GCF.

59. MOE Rwanda indicated that the Government of Rwanda currently has robust measures to fight ML/TF risks (e.g. law N° 001/FIC/2022 on regulations relating to anti-money laundering and countering the financing terrorism). The Government of Rwanda has also put in place a special agency for this purpose, the Financial Intelligence Center (FIC), which has a clear mission to promote financial system integrity by conducting effective financial intelligence to counter ML/TF and related financial crimes in governmental institutions, projects and parastatal agencies under the supervision of the Ministry of Finance and Economic Planning. Both MOE Rwanda and this project under its supervision will also comply with FIC regulations.

60. MOE Rwanda confirmed that it has an internal audit function as well as regular audits conducted by the Office of the Auditor General to ensure the efficiency and effectiveness of internal controls, which includes handling ML/TF risks. In addition, MOE Rwanda uses segregation of duties within financial management processes.

61. MOE Rwanda confirmed that it has conducted a risk assessment of prohibited practices in relation to project implementation. The probability of occurrence was assessed as Low, and the impact severity in the event of occurrence was assessed as Medium.
62. MOE Rwanda indicated that to ensure materials or technology procured under the project are used only for the purposes intended and not diverted or misused for unauthorized, improper or illicit purposes, a comprehensive set of controls will be used. This includes explicit terms and conditions in all procurement contracts or agreements that clearly (i) state the intended purpose of the materials or technology being provided and explicitly prohibit any unauthorized use or diversion; and (ii) define the individuals that are authorized to access and use the procured materials or technology. Furthermore, this will involve the implementation of strong authentication and access controls to ensure that only approved users have access. This refers to the record of any materials or technology procured under projects within an assets register. The records include assets identification, the person using the material and the location.
63. Monitoring and auditing the usage of the procured materials or technology will be undertaken to detect any anomalies or potential misuse, including periodic site visits. Training sessions will also be conducted to educate all stakeholders involved in the project about the importance of adhering to the intended use of the materials or technology and the consequences of misuse. For technologies that involve data transmission or storage, MOE Rwanda will ensure that appropriate encryption and data protection measures are in place to safeguard sensitive information.
64. MOE Rwanda confirmed that no projects will be undertaken in any jurisdiction or territory that is subject to or affected by sanctions or restrictive measures per the United Nations Security Council (UNSC) resolutions. Additionally, no individual or entity that is listed on any UNSC sanctions list, including the United Nations Consolidated Sanctions list, will be involved in any manner with the project or its activities, either as a counterparty, implementer or beneficiary.
65. MOE Rwanda acknowledged that there are United Nations sanctions in the neighbouring country of the Democratic Republic of the Congo, close to the CND. According to MOE Rwanda, the United Nations Consolidated Sanctions List indicates approximately 20 individuals with ties to Rwanda, a small percentage of whom may live in Rwanda. MOE Rwanda confirmed that the United Nations Consolidated Sanctions List has been circulated internally and will be cross checked when hiring for the project and once project sites are selected to ensure these individuals will not be involved in any manner with the project or its activities, either as a counterparty, implementer or beneficiary.
66. MOE Rwanda indicated that the project involves many actors for successful implementation, including ministries, government agencies, non-governmental organizations, subgrantees, vendors, service providers, community members and beneficiaries. MOE Rwanda confirmed that there is specifically an intention to distribute or disburse to beneficiaries, both directly and indirectly, commodities and other items of value such as (i) the distribution of seedlings for trees that will produce fruits or contribute to enhancing biodiversity of farming systems; (ii) access to improved cook stoves; as well as (iii) the option for the distribution of bees to start honey operations. MOE Rwanda acknowledged that, as with any project infusing large sums of money directly or indirectly with multiple actors, corruption could occur.
67. MOE Rwanda indicated that the distributions will be coordinated through procured contractors. The contractors are hired based on the Government of Rwanda/MOE Rwanda procurement policies, which are aligned with GCF policies and include due diligence in the hiring and contracting process. As for direct community contracting, the intended beneficiaries are screened and certified by local authorities in the area of project interventions, which mitigates the risk of prohibited practices.

68. MOE Rwanda confirmed that for reporting and remedying any risks that may occur, MOE Rwanda will maintain a project-level GRM through which anyone can report the occurrence of prohibited activities. All complaints will be investigated accordingly. Any complaints believed to contribute to prohibited activities will be referred to the appropriate authorities in Rwanda.

69. MOE Rwanda indicated that all staff will be made aware of the prohibited activities, and all subcontractors/counterparties will be screened for current and past engagement in such activities and made aware of their obligations to alert the PMU and/or submit grievances related to such activities occurring as part of the project.

70. **Compliance risk rating:** The Secretariat has conducted a review of the project in accordance with relevant GCF Board-approved policies and does not find any material issue or deviation with respect to compliance issues. Based on available information for this funding proposal, the Secretariat has determined a compliance risk rating of 'Medium' and has no objection to this proposal proceeding.

4.3.5. GCF portfolio concentration risk (low risk)

71. In the case of approval, the impact of this proposal on the GCF portfolio concentration in terms of result area and single proposal is not material.

4.3.6. Recommendation

72. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall programme	Low
Accredited entity/executing entity capability	Low
Project-specific execution	Medium
GCF portfolio concentration	Low
Compliance	Medium

4.4 Fiduciary

73. MOE Rwanda is the AE for the project and will provide oversight to the project consisting of, inter alia: (a) entering into a subsidiary agreement with the EE (RFA); (b) managing and disbursing GCF funds to the EE; (c) reviewing financial expenditures and progress reports; (d) overseeing project implementation following the project document and annual work plans and budgets, agreements with co-financiers and each EE's rules and procedures; and (e) providing technical guidance to ensure that the appropriate technical quality is applied to all project activities.

74. RFA will serve as the EE for the project through a subsidiary agreement signed with MOE Rwanda. RFA will execute the tasks within the overall project management structure as follows: (a) implementing day-to-day activities as per the project work plan and budget, including the environmental and social management plan; (b) undertaking procurement activities as per the signed agreements; and (c) managing contracts of suppliers and service providers. The EE will enter into a memorandum of understanding (MOU) with different partner agencies, including WCS and the Catholic Organisation for Relief and Development, which will provide technical assistance services to implement the project activities. Procured parties will be contracted following MOE Rwanda rules and regulations for procuring goods and services. The PMU will manage, monitor and report on the day-to-day implementation progress of the EE.

75. As one of the project implementation partners, WCS will provide technical assistance to RFA through an MOU signed with RFA. The MOU will detail all the activities that WCS will directly support during this project's implementation. The technical assistance services will include advising RFA/SPIU, procuring and supervising international experts and consultants, and procuring materials to support the execution activities. Relevant technical experts from WCS will be embedded within RFA to advise SPIU on day-to-day project management. At the same time, the Catholic Organisation for Relief and Development will provide technical assistance to RFA through an MOU.

76. The co-financiers (RDB, RFA, RWB and MOE Rwanda) will execute and manage the implementation of their activities.

77. MOE Rwanda shall ascertain that the EE has efficient internal control systems in place, review financial reports and proceed with funds disbursements. It will also periodically inspect the project for compliance and coordinate audit exercises for the EE. RFA will enter into agreements with MOE Rwanda that outline general modalities of the management of funds in line with GCF grant requirements and RFA's policies.

78. GCF funds will be transferred to an RFA-designated bank account. Based on agreed annual work plans and budgets, the funds will be disbursed to the EE, and its reporting will be quarterly. The project's financial processes will rely on the EE's charts of accounts, internal control systems, disbursement plans, budgeting, expenditure management and accounting, reporting, and assets management as assessed by MOE Rwanda.

79. The project will be audited annually and commissioned by MOE Rwanda to produce certified financial statements and reports to ascertain that GCF funds are well managed. The project will maintain a regularly updated record of all assets and documents that will be retained for at least five years after the end of the project.

4.5 Results monitoring and reporting.

80. The Secretariat has provided various rounds of feedback on results monitoring and reporting, which have been generally addressed by the AE. This includes a review of beneficiary numbers, revisions to the logical framework, mapping of project outcomes to GCF results areas, and timelines and resourcing for the mid-term and final evaluations. However, there are some issues related to beneficiary calculation that are pending as of the final review. These should be appropriately addressed as the project is mobilized as follows:

- (a) It would be helpful to provide gender disaggregation for beneficiary numbers wherever possible. This is currently done in section E (logical framework), but not in section A6 and annex 23;
- (b) In section E (logical framework), the AE should also provide a brief description of how beneficiary numbers against GCF results areas have been calculated. These details are currently missing, and while we have mid-term and final targets for various indicators related to beneficiary count, we do not have information to understand how these have been calculated, and how potential double-counting will be avoided. If space is limited in section E, the AE may consider including this information in a clearly marked section in annex 23 related to beneficiary count.

4.6 Legal assessment

81. The original Accreditation Master Agreement was signed with the Accredited Entity on 23 June 2016, and it became effective on 1 August 2017, which agreement was amended and restated pursuant to the first amendment and restatement dated as of May 14, 2019, which became effective on even date. The Accredited Entity's first term of accreditation lapsed on 31 July 2022. The Accredited Entity was recredited by the Board pursuant to Decision B.34/17. The second amendment and restatement agreement amending and restating the accreditation master agreement was signed on 22 November 2022, and the second amended and restated accreditation master agreement (the "AMA") became effective on 26 January 2023.

82. The Accredited Entity has provided a certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

83. The proposed project will be implemented in the Republic of Rwanda. The GCF has signed a bilateral agreement on privileges and immunities with the Government of the Republic of Rwanda. It was signed on 8 November 2022 and became effective on 25 May 2023.

4.7 List of proposed conditions (including legal)

84. To address matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and
- (b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP217

Proposal name:	Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration
Accredited entity:	Ministry of Environment of Rwanda
Country:	Rwanda
Project/programme size:	Small

I. Assessment of the independent Technical Advisory Panel

1. This funding proposal for a small-sized project in environmental and social safeguards (ESS) category B is submitted by the Government of Rwanda as a cross-cutting project contributing to both adaptation (92 per cent) and mitigation (8 per cent) results areas. The Ministry of Environment of Rwanda (MOE Rwanda) (a GCF direct access entity) is the accredited entity (AE) for the proposed project, and the executing entity (EE) is the Rwanda Forestry Agency, with several implementing partners, including other government agencies and the Wildlife Conservation Society (WCS) in Rwanda. The total cost of the funding proposal is USD 50 million, of which USD 39 million is GCF finance and USD 11 million is co-finance from the Government of Rwanda.

2. The project's goal is to apply a landscape approach to reduce the exposure of populations in Rwanda's Congo Nile Divide (CND) landscape to the impacts of unfavourable climatic trends and hazards intensified by climate change, including floods, landslides, soil erosion and wildfires. This is to be achieved through integrated land-use planning and management that allows for protecting, restoring and managing a matrix of interconnected ecosystems, including pristine forest in protected area nodes, important landscape linkages, pockets of forest in stepping stones, and agro-ecosystems in farmlands. Interventions in various spatially defined target areas include forest protection and fire management, assisted natural regeneration, afforestation of steep slopes, and support to silvopastoralism and agroforestry and its value chains.

3. The project has three planned outcomes (see the table below), with outputs under each outcome as well as the parties responsible for executing them (see table 5 in the funding proposal).

Project outcomes and outputs	Executing entity per output	Technical implementation partners
Outcome 1: Landscape planning, policies and management are effective and coordinated across sectors and scales to address climate risks and adaptation benefits		
Output 1.1. Landscape-wide land-use plan developed for climate-resilient livelihoods and forest ecosystems, integrating district development strategies	Rwanda Forest Agency (RFA)	Wildlife Conservation Society (WCS) and National Land Authority (NLA)
Output 1.2. Local and national institutional capacities strengthened to integrate biodiversity and climate risks into land use planning and management	RFA	WCS and NLA

Outcome 2: Natural forests are protected, connected, and more resilient to climate impacts and risks		
Output 2.1. Protected areas management effectiveness improved to address climate risks and adaptation	RFA	WCS and Rwanda Development Board
Output 2.2. Natural forest cover restored; biodiversity connections established	RFA	WCS
Outcome 3: Vulnerable rural livelihoods are more climate-resilient, diverse, economically sustainable and nature-positive		
Output 3.1. Farming methods enhance productivity, reduce erosion and flooding risks, contribute to ecosystem services, and support connectivity	RFA	Rwanda Water Board
Output 3.2. Rural livelihoods generate alternative incomes and reduce pressure on forests	RFA	Cordaid
Output 3.3. Financial services and private sector investment engagement	RFA	Cordaid
Monitoring and evaluation	Project Management Unit, hosted by the Ministry of Environment of Rwanda	

1.1 Impact potential

Scale: N/A

4. The cross-cutting impact potential of the proposed project and the relative weighting of the intended results can be seen in the attribution of GCF and co-financiers' contributions to the relevant GCF results areas as follows:

GCF results area	GCF contribution	Co-financiers' contribution
<u>Mitigation Result Area 4 (MR4)</u> Mitigation through forestry and land use	8%	0%
<u>Adaptation Results Area 1 (AR1)</u> Enhanced resilience of most vulnerable people and communities	13%	49%
<u>Adaptation Results Area 2 (AR2)</u> Enhanced resilience of health and well-being, and food and water security	2%	0%
<u>Adaptation Results Area 4 (AR4)</u> Enhanced resilience of ecosystems and ecosystem services	77%	51%

5. The emphasis on utilizing the GCF investment to enhance the resilience of ecosystems and ecosystem services is reflected in the project's focus on new mechanisms of land-use planning and management that consider the CND as a landscape consisting of an interdependent mosaic of forests and farms, the management of each determining the health of the whole. The independent Technical Advisory Panel (iTAP) believes that such an approach (i.e. targeting project interventions on the ground based on a systematic spatial planning process that emphasizes "ecological infrastructure" functioning as well as biodiversity) represents best practice for enhancing ecosystem resilience. Such an approach ensures that interventions are at sufficient scale to achieve a transformative effect: preserving landscape-wide ecosystem processes and functions that help buffer impacts of climate change. Maximizing connectivity is also important for maintaining the long-term health and stability of forests as carbon sinks.

6. **Adaptation impact potential:** The funding proposal outlines the major climate change-related hazards characterizing the landscape, in particular more frequent intense rainfall events increasing floods and landslides, and worsening soil erosion that is already widespread as a result of direct anthropogenic pressures. The project's main adaptation benefits will accrue to

vulnerable communities primarily engaged in agriculture on the CND's steep terrain, achieved through improved soil management, forestry and agroforestry to reduce catastrophic events affecting production systems, lives and livelihoods. This will be achieved through integrated landscape planning and management to enhance tree cover in crop and grazing lands and improve the connectivity of natural ecosystems and the key services they provide, including regulation of runoff and flows from increasingly extreme rainfall events; and through reduction of localized and downstream flooding, erosion and landslide incidence and risk.

7. Indicators for the main adaptation results are as follows:

- (a) 135,104 hectares (ha) of land to be brought under climate-resilient management through targeted interventions for protection, afforestation and agroforestry (AR4):
 - (i) Including 121,758 ha of core protected areas, linkages and stepping stones that will be brought under climate-risk informed planning and fire management to maintain landscape-scale ecosystem services; and
 - (ii) Including 13,346 ha for focused interventions (i.e. degraded forest in parks to be restored (5,000 ha), riparian lands restored (1,500 ha), pastures brought under silvopastoralism (1,000 ha), steep slopes afforested (2,500 ha), and farmland with new agroforestry practices (3,346 ha));
- (b) 1,254,242 people, resident in the landscape where targeted interventions will be carried out, will benefit directly from reduced exposure to climate-intensified disasters and/or enhanced adaptive capacity (AR 1 and 2); and
- (c) 9,260,745 people, approximately two thirds of the country's population, will benefit indirectly from a wide-scale education and outreach programme, including radio programming, on climate change and gender issues, climate risks, value of forests and forest ecosystems, and climate change adaptation options.

8. **Mitigation impact:** The project seeks to deliver greenhouse gas (GHG) emission reductions and removal benefits equivalent to 224,871 tonnes of carbon dioxide equivalent (t CO₂ eq) during the five-year project implementation period, reaching a total of 1,084,291 t CO₂ eq after 20 years. The estimation of reductions and removals is made based on the difference between the projected emissions with and without the project interventions – through improved cookstoves, and through on-the-ground interventions across a total of 13,346 ha of land. The land-based interventions will have cross-cutting benefits, since they contribute to core indicator 4 on hectares of land to be brought under climate-resilient management, as well as bringing mitigation benefits.

9. The following activities will contribute to achieving this mitigation potential: (i) assisted natural regeneration (ANR) in 4,500 ha of Nyungwe National Park through invasive fern removal; (ii) restoration of 500 ha of degraded forest through ANR in Gishwati-Mukura National Park; (iii) afforestation on 2,500 ha of public and private land with a slope >55 per cent; (iv) planting 1,500 ha of protective forest in riparian zones (public and private); (v) implementing erosion control and agroforestry practices on 3,346 ha of smallholder farmland; (vi) implementing silvopastoralism on 1,000 ha in the Gishwati area; and (vii) adoption of improved cookstoves by 8,500 households. Indicators for achievement of these targets are included in the project logical framework in section E.3 of the funding proposal.

10. The methodology for calculating GHG emission reductions and carbon sinks mirrors two approved consolidated methodologies for the A/R CDM, AR-ACM0003 and AR-AMS0007. This is considered a good practice because the applicability conditions of the methodologies are appropriate to the project conditions. The methodology is clear and conservative, considers the relevant guidance from the Intergovernmental Panel on Climate Change, and was used in an appropriate, clear and transparent manner when estimating the mitigation potential for each activity.

11. With regard to monitoring and reporting mitigation benefits, annex 22 proposes a clear and straightforward approach, but does not clarify the roles and responsibilities of the various actors that will be involved in the process over time, nor the frequency of the steps. Furthermore, the activities as presented in annex 22 are not reflected in annex 11 (monitoring and evaluation plan_rev August 2023). According to its updated nationally determined contribution (NDC) (2020, page 6)¹ Rwanda has already developed a national measurement, reporting and verification (MRV) framework. At the present stage it is not clear how this project will be aligned with the existing arrangements for the national MRV system, and/or how it will support or contribute to enhancing the functioning of this system across the country.
12. The iTAP assesses the impact potential of the project as high.

1.2 Paradigm shift potential

Scale: N/A

13. The proposed project is designed to bring about a paradigm shift away from the current situation of fragmented, unplanned decisions and a collection of isolated sector-by-sector projects that fail to take into account evolving climate risk and allow continued degradation. Instead, a new approach of integrated land use planning and management is promoted that preserves ecological connectivity across the landscape and helps maximize valuable ecosystem services for buffering the impacts of climate change. The new approach is based on international best practice and considers the CND as a landscape consisting of an interdependent mosaic of production and protected lands, the health of which is completely interconnected.
14. The idea behind the project is that a new holistic, climate-smart, evidence-based approach to planning and development will enable systematic, ecosystem-based, multi-sector analysis of land-use tradeoffs. Focusing on this larger scale, with coordination across both geographic and sectoral boundaries, while mainstreaming climate risks in management planning, represents a fundamental paradigm shift for land-use planning and management in Rwanda. Success in putting such a system into place could represent a crucial first step in reversing the previous negative feedback loop of ecosystem degradation to achieve a more positive, climate-resilient system.
15. Utilizing integrated landscape management and ecosystem-based adaptation approaches in addressing climate change impacts in Rwanda offers a myriad of opportunities for scaling up and replication to similar ecological zones across the country. The project also has the potential to contribute to the regulatory frameworks and policies for driving climate investments in the forest and renewable energy sectors. The project will finance the development of an integrated land-use plan, which will be adopted by all 10 CND districts as a framework for seeking synergies between rural livelihoods and forest and biodiversity protection, reconciling land-use conflicts that arise, and guiding development decisions and climate change adaptation. It will also update the National Land Use Master Plan, ensuring it incorporates climate resilience and local plans.
16. **Systematic conservation planning:** Although biodiversity conservation is considered a co-benefit of the project rather than a core focus, the project makes innovative use of systematic conservation planning to identify spatial areas in the landscape that are targeted for restoration, afforestation and agroforestry activities. This includes work to restore national parks as the core of an interconnected system, with landscape linkages and stepping stones identified for interventions to promote connectivity between remaining areas of natural habitat. Optimizing this kind of mosaic of land uses with management actions for protection, restoration and sustainable use helps to maximize the functionality of landscape-wide ecosystem services such

¹ https://unfccc.int/sites/default/files/NDC/2022-06/Rwanda_Updated_NDC_May_2020.pdf.

as nutrient cycling, soil fertility, slope stability, water infiltration, flood management and precipitation.

17. **Innovative financial mechanisms:** The project also includes an output to establish long-term plans for CND financial sustainability post-GCF by employing a technical advisor to support the establishment of innovative mechanisms such as those identified in the Biodiversity Finance Plan and also to refine existing financing mechanisms (for example, integrating climate resilience principles into the existing successful and growing Tourism Revenue Sharing Program). Support will be provided to beneficiaries of the project in applying to the existing Community Adaptation Facility within the Rwanda Green Fund (FONERWA). The Government of Rwanda, represented by the Rwanda Development Board, is also in advanced stages of negotiation with the World Bank for establishing a wildlife bond, potentially a gorilla conservation bond, modelled on the successfully established rhino conservation bond in South Africa, with interest shown by impact investors.

18. Overall, based on the above, the paradigm shift potential of the project is assessed as medium to high.

1.3 Sustainable development potential

Scale: N/A

19. The proposed project is aligned with the United Nations 2030 Agenda for Sustainable Development, including significant contributions towards Rwanda's efforts to meet Sustainable Development Goal (SDG) 5 (Gender Equality), SDG 13 (Climate Action) and SDG 15 (Life on Land), and will contribute to the achievement of SDG 1 (No Poverty), SDG 2 (Zero Hunger), SDG 3 (Good Health and Well-being) and SDG 10 (Reduced Inequalities).

20. **Environmental co-benefits:** Key environmental co-benefits (beyond the core benefits related to ecosystem-based adaptation) include restoration of soil fertility through silvopastoralism and regenerative mixed-use agroforestry, and the enhancement of biodiversity through planting 50 per cent native tree species. Promotion of improved cookstoves using significantly less fuel will contribute to relieving pressure on the remaining indigenous forest.

21. The iTAP was surprised to see the relatively low target of 50 per cent (30 per cent on private land and 70 per cent on public land) in the use of indigenous tree species in afforestation, reforestation and agroforestry interventions. The AE explained that exotic species are more suitable for afforestation of steep slopes, where they can easily establish themselves, "but this does not mean that exclusively exotic species will be used on such sites; there will be a mixture of both wherever applicable. In addition, while establishing blocks in the Participatory Forest Management Units, indigenous species will be used to separate one block from another. Indigenous species will be given due consideration in agroforestry to create positive interactions in the farm as well as promoting native species, avoiding monoculture and benefiting from their positive impacts in terms of resilience."

22. It is understood that there is often a preference for fast-growing, easily available and usually cheaper species like eucalyptus, pine and alnus for urgent needs such as slope stabilization and fuelwood, and also that a balance needs to be struck between promoting indigenous trees species and being responsive to farmers' wishes. At the same time, the project could potentially make more of the opportunity to rekindle traditional knowledge about indigenous species and their uses, and to build the nursery supply chain for quality native planting material given the vast superiority of indigenous tree species in terms of their multiple benefits, non-invasiveness, positive impact on soil and water quality, and resistance to pests and diseases that may worsen with a warming climate.

23. **Social and health co-benefits:** In enabling the transition to more climate-resilient and productive livelihood models, the project aims to increase household incomes and the nutritional status of families. It is expected that significant social capital will be created through

beneficiaries' participation in co-designing and co-managing a range of adaptation strategies, with an emphasis on women's participation. Social cohesiveness will also be enhanced through the strengthening of joint saving and lending systems. Social benefits will also accrue to indirect beneficiaries, for example those listening to radio programming that leads to increased awareness of climate change, or of opportunities for climate-adaptive agroforestry and agri-processing.

24. **Economic co-benefits:** Inequality and poverty will be addressed through increased financial inclusion and improved economic opportunities, especially for women, youth and historically marginalized groups. Forest and landscape restoration in the CND will bring economic benefits, since soil erosion control measures, protective forests on public lands, and agroforestry on private lands will all increase agricultural productivity and associated income. The project will create direct and indirect employment opportunities, benefiting an anticipated 24,212 households and stimulating the local economy. Private capital will be leveraged to invest in value chains, including those for improved cookstoves, irrigation pumps and beehives, as well as honey, woodlots, macadamia, avocado and vegetable production and processing.

25. **Gender benefits:** The project aims to achieve positive outcomes around access to resources, improved livelihoods, and income generation opportunities and capacities for women. With support from a dedicated gender specialist, the project will engage women in project planning, investment and decision-making. It is envisaged that women will respond to incentives that address their families' basic needs, such as better health and nutrition, linked to agriculture and food security improvements. The project will also get support from non-governmental organizations that engage men in programming and discussions to increase their participation in household and parental responsibilities and enhance their awareness of the importance of women's fair access to and control of resources, including income. Women will benefit from training and educational activities, leadership, entrepreneurship and decision-making, as shown in annex 8 (Gender Analysis/Assessment and Gender and Social Inclusion Action Plan).

26. The iTAP assesses sustainable development potential as high.

1.4 Needs of the recipient

Scale: N/A

27. Western Rwanda and its CND region are highly vulnerable to climate change, and are expected to experience a continuing upward trend in average annual temperatures, total annual rainfall (with a longer dry season), and more intense rainfall events and storms and increasing risk of flash floods, landslide, soil erosion and wildfires. Despite Rwanda's solid economic progress since 2000, poverty remains widespread and pervasive. Overall, 38.2 per cent of the country's population lives in poverty and 16 per cent in extreme poverty. Women are more affected by poverty than men, with 47 per cent of female-headed households living in poverty compared with 44.9 per cent of all households.

28. The CND landscape supports nearly one third of Rwanda's population on barely one fifth of its land base. Average population density is 736 people/square kilometre (km²), with some districts now exceeding 1,000 people/km². The highest rates of rural poverty in the country are reported from the CND, including its most vulnerable subsistence farmers. While high rainfall has historically permitted generally sufficient food production, the region's rugged topography requires farming on extremely steep slopes. With more frequent extreme rainfall events in recent years, this agricultural landscape is recognized as the region of Rwanda most vulnerable to flash floods and landslides. Between 2013–2016, heavy rains generated landslides that killed at least 174 people and destroyed the homes of more than 5,000. Landslides wipe out highly productive raised-bed cultivation systems in the CND's valleys and bottomlands, many hectares of crops are washed away, and an estimated 1.5 million tonnes of fertile soil are lost annually to erosion.

29. The funding proposal reports that during stakeholder consultations, communities indicated several socioeconomic changes stemming from changing climate conditions. Notably, the longer dry season and unpredictable rains have reduced agricultural yield significantly, leading to more widespread malnutrition and extreme poverty, and farmers can barely meet their own families' needs. In particular, women, youth, and historically marginalized people are feeling the impacts. As arable land is increasingly constrained, youth and historically marginalized people have few opportunities and are more focused on survival and short-term gains. This becomes a negative feedback loop, in which poverty leads to unsustainable practices such as overharvesting from forests and cultivation on extremely steep slopes, with degraded ecosystems then less able to provide services, worsening the cycle of poverty.

30. As a result, the Government of Rwanda has invested significantly in developing an enabling policy and institutional framework for a more climate-resilient development pathway. Yet, while adaptation is a top priority, as a least developed country, Rwanda lacks sufficient public financial resources to implement its climate change and resilient green development objectives. Currently there are no alternative funding options at the scale needed to secure Rwanda's priority forests to reduce the vulnerability of rural communities to climate risks. Rwanda's debt-to-gross domestic product ratio stands at 64 per cent, making further loans difficult, and the Government of Rwanda seeks maximum concessionality from the GCF for the urgent adaptation actions proposed under the project.

31. The Government of Rwanda recognizes that a piecemeal approach to decision-making has exacerbated competition over scarce land resources and reduced the adaptive capacity of both ecosystems and rural populations. Additionally, district government staff and community members have limited technical understanding of the complex interactions between forest structure and functions, climate change, and the provision of ecosystem services, and limited technical capacity for complex forest monitoring. In the face of worsening climate change impacts, there is an urgent need for a comprehensive approach that crosses siloes of government and achieves integrated land use planning and management informed by the evolving climate and socioeconomic conditions. The proposed interventions are well targeted to address the identified barriers and the great needs of these vulnerable communities.

32. Overall, based on the above, the needs of the recipients are rated as high.

1.5 Country ownership

Scale: N/A

33. **Policy and strategy alignment:** The project is closely aligned with Rwanda's national adaptation plan and three of its adaptation priorities: (i) promotion of non-agricultural income generating activities; (ii) introduction of species resistant to environmental conditions; and (iii) development of alternative sources of energy to firewood. It contributes to several priorities identified in the 2020 NDC: promotion of afforestation/reforestation; improvement of forest management; an integrated approach to planning and monitoring for sustainable land management; and development of a harmonized and integrated spatial data management system for sustainable land use management.

34. The project will support the implementation of Rwanda's Green Growth and Climate Resilient Strategy and the Rwanda National Strategy for Transformation (2017–2022) and is aligned with the goals of the Rwanda National Forestry Policy 2018, the Forest Sector Strategic Plan 2017–2022, the Strategic Plan for the Environment and Natural Resources Sector (2018–2024) and the National Forest Management Plan 2017–2026, thus reflecting the government's intention to address climate change impacts and mitigation targets by improving forest management. It is also aligned with the national Biomass Energy Strategy 2020, which aims to ensure a more sustainable biomass supply and combustion technologies for homes and small enterprises.

35. **GCF country programme:** According to the funding proposal (section D.5), the project aligns with the forthcoming GCF country programme, which “prioritizes forest management through restoration and protection of natural forests, afforestation and agroforestry, as well as distribution of high-quality reforestation inputs and capacity development”. While this project proposal to GCF focuses on a different geographic area, the early and ongoing lessons from the experiences from existing GCF-funded projects “Strengthening Climate Resilience of Rural Communities in Northern Rwanda”² and “Transforming Eastern Province through adaptation”³ will be applied during the implementation of this project in the CND landscape.

36. **Roles of key entities:** The proposed project was initiated by MOE Rwanda and supported by the national designated authority, Rwanda Environment Management Authority, due to the high vulnerability to climate change of the CND region of the country. MOE Rwanda is well placed to oversee the project as AE since the ministry is responsible for (i) ensuring the sustainable management of natural resources and for the development of policies, strategies and regulations; and (ii) mobilizing resources for the development of the sector. MOE Rwanda managed USD 10 million from the Adaptation Fund to implement the Reducing Vulnerability to Climate Change in North West Rwanda through Community Based Adaptation Programme⁴ and has overseen a number of other climate projects, including: an early warning system project (funded by the Least Developed Countries Fund (LDCF)), the Landscape Approach to Forest Restoration and Conservation and the “Building resilience of communities living in degraded forests, savannahs and wetlands of Rwanda through an ecosystem management approach” programme (funded by LDCF)⁵.

37. Given the centrality of forest protection, restoration and management to the project, the RFA is well placed to carry out the role of EE. It is noted that this would be RFA’s first GCF EE role, since MOE Rwanda’s other GCF project is executed by FONERWA. Strong support will be provided by a set of implementing partners – including the WCS and government agencies such as RDB, which are responsible for the national parks, RWB, Rwanda Environment Management Authority and the National Land Use Authority. The project activities overseen by each are shown in the table in the introduction above and are aligned with their mandates, as well as sectoral plans and strategies.

38. **Consultation and engagement with civil society:** To better understand the interests of stakeholders and potential impacts of the proposed project, and as a preparation for designing future engagement strategies, a comprehensive, gender-responsive, culturally sensitive, non-discriminatory, and inclusive stakeholder analysis was undertaken and is reflected in the Stakeholder Engagement Plan (annex 7). In terms of future engagement, the funding proposal indicates a number of specific areas where consultations will be ongoing to inform the rollout of activities, for example, consultations with communities and civil society organizations representing men, women, youth, and historically marginalized people in local communities, as well as district and national government agencies, on the following activities: Activity 1.1.1. Synthesize and disseminate information on value of natural forests and ecosystem services; Activity 1.1.2. Develop climate-resilient landscape land-use plans; Activity 2.2.3 Promote silvopastoralism with indigenous trees around Gishwati; Activity 3.1.1 Restore high slope areas (>55 per cent) as protective forests; and Activity 3.2.1 Develop an agroforestry-related value chain for market access.

²Rwanda Ministry of Environment, 2018. FP073: Strengthening Climate Resilience of Rural Communities in Northern Rwanda. Green Climate Fund.

³International Union for Conservation of Nature, 2021. FP167: Transforming Eastern Province through Adaptation. Green Climate Fund.

⁴Adaptation Fund, 2014. Reducing Vulnerability to Climate Change in North West Rwanda through Community Based Adaptation. [Link](#).

⁵United Nations Environment Programme, 2016. Building resilience of communities living in degraded forests, savannahs and wetlands of Rwanda through an ecosystem management approach.

39. Country ownership of the project is thus seen to be high.

1.6 Efficiency and effectiveness

Scale: N/A

40. Of the project's total proposed budget of USD 50,000,000, 78 per cent (USD 39,056,579) is proposed to be a GCF grant, and 22 per cent (USD 10,943,579) is made up of government co-financing to the project. The Government of Rwanda is providing grant financing via RFA, RWB, and RDB to support activities in component 3. In addition to these contributions totalling USD 10,831,307, the AE will cover office costs in the form of an in-kind contribution equivalent to USD 112,271 for the Project Management Unit, which will be hosted at MOE Rwanda.

41. **Government co-finance:** In response to a question from iTAP about the nature of the co-finance and the extent to which it represents new and additional funds, the AE explained that "[t]he co-financing commitments were provided based on the existing Government commitments provided every year in the project intervention area. These are earmarked funds that go into forest management, community revenue sharing around the parks, monitoring resources for catchment restoration interventions and also different in-kind contributions. All the details are indicated in the Annex 4 of the budget and activities to be covered by co-financing resources and Annex 13 indicates the commitment letters from the co-financiers." It is iTAP's understanding that these funds will not be spent in a business-as-usual manner, but will be newly directed towards climate risk management, through support to the project's adaptation and mitigation interventions and efforts to develop officials' capacity for climate risk-informed integrated land use planning and management.

42. **Role of financial institutions:** Participating financial institutions (PFIs) will be important in two ways – firstly as providers of credit to farmer cooperatives, savings groups and small enterprises in the targeted value chains, and secondly as recipients of training – to enhance their knowledge of good agricultural practices, equip them to assess the impact of climate change on their clientele and loan portfolio, and help ensure equitable access to financial assets by women, youth and historically marginalized communities.

43. In response to a question from iTAP about whether any consultation has happened with the potential PFIs, which are not mentioned in the extensive stakeholder consultations reported in annex 7, the AE stated that "At this stage [neither] Cordaid nor RFA have notified the participating financial institutions. As per the project implementation plan, a 6-month to 9-month inception phase will include in-depth analysis of existing financial institutions including microfinance institutions and saving and credit cooperative societies. They will be scored against a list of criteria to understand their leadership, financial position and financial products portfolio on a case-by-case basis."⁶

44. The iTAP believes this engagement will be crucial to the success of Output 3.3 on "Financial services and private sector investment engagement" in order to ensure sustainability of livelihood interventions, and urges the AE to start dialogue with potential partners as soon as possible since they will have experience and advice that can guide the way this output is approached for maximum effectiveness.

45. **Level of concessionality:** The project's proposed 100 per cent concessionality level is justified because the project is investing in multiple ecosystem goods and services that are public in nature, including landscape mosaics, at the core of which are public assets, in the form of national parks and other protected areas, as well as landscape linkages and stepping stones to promote connectivity and flood management, and steep slopes to be afforested to prevent landslides and erosion. Benefits will accrue also to privately owned small-scale family farms, but these are on a very small scale, and any profits made through processing and sale of

⁶ MOE Rwanda responses to iTAP, 4 September 2023.

agroforestry products will need to be retained at household level to buffer against shocks and stresses to increase and diversify incomes, and to improve families' nutritional status.

46. **Market transformation for cookstoves:** The project aims to avoid challenges which the feasibility study says were experienced with previous improved cookstoves (ICS) programmes in the region, which "have historically been characterized by poorly designed subsidy/incentive programs which in some instances have provided 100% subsidies to end-users" and have sometimes led to "unnecessary long-term dependence and poorly managed exits, which consequently created long-lasting ICS market distortions". Instead, the project adopts a market-oriented approach to strengthening the improved cookstove supply chain in the region, creating "strong linkages and business relationships among the key actors, including cookstove producers/youth small and medium-sized enterprises, farmers/savings groups, and microfinance institutions to ensure sustainable access and finance of cookstoves".

47. In collaboration with technical and vocational education and training institutes, youth producers, central and local authorities, and experts on biomass and micro-finance will refine and test, for each type of ICS/fuel to be promoted, the category/profile of households to be targeted and the related subsidy/microcredit schemes to be implemented to facilitate stove adoption with minimum concessionality. Subsidized cookstoves will be supplied to 4,000 poor households identified as being in a category of vulnerable groups, including youth, women, landless and disabled people. The stoves will be locally manufactured, with a competitive process leading to the award of a supply contract to local women- and youth-owned cooperatives or start-ups. A partnership with two integrated polytechnic regional colleges will enable the design of prototype affordable Tier 2 ICS customized to local conditions and customer preferences.

48. After year 4, the start-ups that will have graduated from the training and incubation programme (also conducted for the manufacture of beehives and irrigation pumps) will be fully empowered to maintain the supply of new equipment and repair of old equipment, as well as to market their products to producer organizations and households. Access to microfinance institutions will be facilitated to allow emerging cookstove business to grow. The members of village savings groups will be able to use income generated from the sale of fast-maturing crops (chili and French beans), coupled with savings accumulated from their groups, to buy new or replacement stoves over time. A specialized small business development fund will be established to support dissemination, accompanied by public awareness-raising of the benefits of ICS.

49. **Return on investment:** The project cost per direct beneficiary of USD 31 (USD 40 with co-finance) and 20-year economic rate of return of project interventions for agroforestry (49 per cent), protective forests (72 per cent), and silvopastoralism (51 per cent) all highlight the project's viability, as does the economic net present value of USD 10,586 for the improved cookstoves intervention. The mitigation cost is high, at an estimated USD 43.84 per t CO₂e, but comes with a multiplicity of adaptation benefits and socioeconomic and environmental co-benefits. The cost of restoring natural forest for providing both adaptation and mitigation benefits is significantly greater than the cost of simply planting trees for mitigation, since a complex multi-strata, multi-trophic level natural ecosystem (like a tropical rainforest) provides more varied and in many cases more robust ecosystem services than a simpler ecosystem (like a monocrop plantation). Restoration of natural forest in Rwanda is essential for sustaining the critical ecosystem services on which vulnerable rural communities and the national economy depend, and are now even more essential for buffering intensifying climate change impacts.

50. **Best practices:** The project will use Rwanda's best practice guidelines for tree nursery establishment, tree planting and tree management. For assisted natural regeneration, methods developed and proven through multi-year research in Nyungwe and Gishwati-Mukura National Parks will be used. The project will use multiple contemporary methods for monitoring the success of project interventions, as well as forest coverage and condition. Forest plantations will

be mapped and monitored using the Rwandan Government's newly developed Forest Monitoring and Evaluation Software specifically developed for this purpose. The project will develop a modern indigenous forest monitoring system, utilize the most high-resolution data sets currently available (<3 metres) and take advantage of recent remote-sensing advances, including cloud computing and machine learning. Although replication of effective and cost-effective project techniques through knowledge-sharing with other countries in Africa is not planned in the project design, this could be a by-product of the project.

51. The project's overall efficiency and effectiveness is rated as medium to high.

II. Overall remarks from the independent Technical Advisory Panel

52. The iTAP believes this cross-cutting public sector project from the Government of Rwanda is an important one that will make significant strides towards effective management and sustainable financing of forest-and-farm landscapes that play a vital role in climate change adaptation and mitigation in the Congo Nile Divide.

53. The iTAP recommends to the AE that they undertake the following actions to strengthen the measurement, reporting and verification (MRV) of the project's core mitigation benefits:

- (a) Clarify the roles and responsibilities of the various actors who will be involved in the steps of the MRV process over time, and the frequency of these steps;
- (b) Capture the MRV activities as presented in annex 22, also in annex 11, such that they become part of the project's overall monitoring and evaluation system; and
- (c) Clarify how this project will be aligned with the existing arrangements for the national MRV system and/or will support or contribute to enhancing the functioning of this system across the country.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP217)

Proposal name:	Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration
Accredited entity:	Ministry of Environment of Rwanda
Country/(ies):	Rwanda
Project/programme size:	Small

Impact potential
The iTAP assessment of the impact potential of the project as high is well acknowledged and no further comments from the AE.
Paradigm shift potential
iTAP assessment as Medium to High is acknowledged by AE. The project is going to utilize integrated landscape management and ecosystem-based adaptation approaches in addressing climate change impacts in Rwanda which offers opportunities for scaling up and replication to similar ecological zones across the country.
Sustainable development potential
iTAP assessment as High is acknowledged by the AE.
Needs of the recipient
iTAP assessment as High is acknowledged by the AE
Country ownership
iTAP assessment as High is acknowledged by the AE
Efficiency and effectiveness
iTAP assessment as Medium to High is acknowledged by the AE. The AE has received the confirmation letter regarding the co-financing from the Rwanda Development Board, Rwanda Water Resources Board and Rwanda Forestry Authority to support complementary activities of this project. The AE will report on the co-financing in line with GCF Policy on co-financing annually in APRs.
Overall remarks from the independent Technical Advisory Panel:
The AE acknowledges iTAP overall assessment and recommendation for the Board approval.

**BUILDING RESILIENCE OF VULNERABLE COMMUNITIES TO
CLIMATE VARIABILITY IN RWANDA'S CONGO NILE DIVIDE
THROUGH FOREST AND LANDSCAPE RESTORATION**

**ANNEX 8 - GENDER ANALYSIS/ASSESSMENT AND GENDER &
SOCIAL INCLUSION ACTION PLAN**

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- **EXECUTIVE SUMMARY**

Since 1994, Rwanda has made significant progress in empowering women and countering the numerous inequities and discrimination to which women were historically subjected. Today Rwanda has one of the most progressive, gender equitable policy frameworks of any country in Africa, and has established and funded the Ministry of Gender and Family Promotion and the Gender Monitoring Office to oversee the implementation of these gender progressive policies.

There is no single definition nor understanding of gender equity and women's empowerment, and the meaning, and reality of these terms differs across communities and cultures and over time. That said, the Government of Rwanda holds the simple belief in the fundamental equality of all people. In a just society, men and women of all ages, at a minimum, should have the same rights to self-determination, should be equally represented at all levels of political leadership, and should not be subject to discrimination or gender-based violence.

Today rural women continue to play a vital role in ensuring the wellbeing and security of their families. But being the caretaker of the family is getting harder for women as farmland becomes increasingly scarce, as soil erosion and declining soil fertility lowers crop yields, and as fuelwood, fodder and clean water take ever longer to find and carry home. Changes in rainfall timing and intensity, driven by climate change, will further exacerbate the challenges women will face to meet the needs of their families.

Women, youth and marginalized groups are the primary targeted beneficiaries of the Green Climate Fund (GCF) proposal "Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide Through Forest and Landscape Restoration." Empowering women, youth and marginalized groups to play an equal role in land and resource use decisions will ensure that their interests and concerns are incorporated into plans designed to increase the wellbeing and security of families in the Congo Nile Divide (CND) by helping them to evolve their livelihood strategies and practices in ways that adapt to climate change.

Investing in women's lives and strengthening their capacity to overcome existing and looming challenges will do much to implement Rwanda's progressive gender policies and help reduce the continuing inequities and discrimination that rural women, in particular, still face in their daily lives. It will also reduce poverty, the incidence of child failure to thrive, and food and income insecurity within families across the CND. By ensuring a robust integration of women and youth in programming, the resilience of households in the Congo-Nile Divide will be strengthened.

The following summarizes the primary desired outcomes for women and youth that will be generated by implementing the four components of the GCF program in the CND.

Component 1: Mainstreaming Climate Adaptation into Integrated Land Use Planning

- Outcome 1: Climate adaptation programs in LUPs reflect and address the priorities of women and youth
- *Component 2: Forest Landscape Management and Restoration*
- Outcome 2: Women, youth and marginalized households have improved capacity to engage in forest management and restoration
- *Component 3: Enhancing climate adaptation through resilient livelihoods*
- Outcome 3: Women, youth and marginalized households have improved capacity to engage in green jobs

Component 4: Monitoring & Evaluation

- Outcome 4: Disaggregated and specialized monitoring generates data showing whether and how women, youth and marginalized people are benefiting from adaptive management systems

- **INTRODUCTION**

Decades of project design, implementation and impact evaluation have demonstrated conclusively that gender equality and women's empowerment are necessary conditions for effective investment and action to achieve environmental conservation, economic development and climate change resilience. This program seeks to ensure that women play equal roles in determining how to improve the wellbeing and security of their families and communities in ways that maintain the agro-ecological systems, biodiversity, and environmental services essential to the economies, nutrition and safety of their families in the Congo-Nile Divide (CND) now and in the future.

Our approach goes beyond trying merely to ensure that women are no longer ignored or marginalized but seeks to empower women so that they can decide how they can best engage in determining what support would best help them and their families to increase food and income security and resilience to climate shocks. To do this we need to complete a thorough gender analysis that identifies the differences in men and women's lives, including those that lead to sociocultural and economic inequalities, and applies this understanding to project development.

This gender analysis attempts to answer four questions:

1. What roles do men and women play in determining wellbeing objectives and the allocation of labor and capital, and how do these differ in male versus female headed households?
2. What are the inequalities, discriminations and rights denials faced by men and women in different social, economic and political contexts? How do these issues intersect with other factors such as age, ethnicity, disability, wealth class, etc.?
3. How might gender inequity impact on the effectiveness and sustainability of the project investments, actions and desired outcomes?
4. What are the risks that project efforts to reduce gender inequality by empowering women, actually exacerbates inequities and further undermines the status, and agency of women?

The purpose of this gender analysis is to ensure that the project proposal design and implementation will be informed by a thorough understanding of gender roles, power relations and dynamics. This assessment provides information to address the relevant environmental and climate change critical issues from a gender perspective. Information generated during the analysis will ensure that sufficient and appropriate attention is paid to gender issues within all CND project interventions.

Furthermore, these gender inclusion methods fall in line with the GCF's gender mainstreaming guidelines, designed to involve women, youth, and the historically marginalized in all climate adaptation, mitigation, and resilience strategies. The gender assessment will provide the foundation for a credible and practical gender action plan to be implemented by the GCF CND project.

1.1 Purpose of the Gender Analysis

Increased greenhouse gas emissions linked to human activity continue to threaten the livelihoods,

safety, and wellbeing of communities worldwide. However, climatic impacts will vary based on numerous factors, including geography, socioeconomic status, age, and gender. Women experience unequal access to resources, land, and power, reducing their ability to adapt to climate change as effectively as men. Women face greater domestic responsibility to maintain the home and raise children, reducing their time for education and careers. As a result, women are more tied to their land and more vulnerable to climatic impacts, yet also represent a potential group of leaders to spur mitigation and adaptation efforts.

Given the potential of women as environmental pioneers, climate resiliency efforts require a thorough understanding of gender roles in the region. MoE recognize the inextricable link between sustainable development of communities through gender planning and the conservation of wild places. Effective conservation strategy requires working to balance gender inequity and providing opportunities for women in the conservation space. Furthermore, the high dependence of women on agriculture as livelihood in Rwanda impacts both conservation and climate change mitigation efforts. Smart conservation programs therefore target agricultural interventions to aid marginalized groups such as women and youth. Such projects offer employment opportunities and reduce the threat of illegal or paramilitary activities that threaten conservation outcomes.

Shifting the imbalance of power and access for women is not only a conservation target, but a human rights issue. Sustainable development at all levels, local, national, and international, requires the empowerment of women. When women receive higher education, they have more flexibility in careers and are more likely to escape poverty. These women strengthen economies through diversified incomes, marry and have babies later, and have greater independence (NISR, 2018). However, climate change is likely to worsen financial stressors, particularly in rural communities, and increase the risk of gender based violence (GBV). As the frequency of extreme weather events increases, understanding how climate change impacts women is essential to combatting the human rights issue of gender equality.

There are numerous international human rights agreements that provide a framework for gender equity. Rwanda has signed on to several of these commitments, including the Convention on Elimination of all Forms of Violence against Women (CEDAW), the Maputo Protocol on the Rights of Women in Africa, and the Convention on Slavery and Repression of Human Trafficking (IUCN, 2021). Participating in these commitments requires periodic progress reporting and helps hold countries accountable.

In Rwanda, the government has made significant strides to address gender disparity following the Genocide. Thirty-four percent of households became run by widows post-Genocide, demonstrating an obvious need for the Rwandan government to address the newfound elevated standing of women (Vanhees, 2014). Several laws help ensure that on paper, Rwandan women have increased access and protection under the law. Such policies include the 1999 Inheritance and Marital Property Law (Law/nº 22/99 of 12/11/1999), the 2008 Girls' Education Policy, and the 2010 National Gender Policy (IUCN, 2021). However, these policies are new and compete with a long history of patriarchal norms in Rwandan society, particularly outside of urban areas. Practice of these policies remains inconsistent, especially when women do not know their rights. Helping men and women understand gender policy is the first step to shifting cultural perceptions. But gender equality cannot be achieved by women alone, it also requires the active participation of men to accept the deconstruction of traditional gender roles and be aware of the intersectionality of women and other groups. For example, considering how to include youth and marginalized communities, especially the women in those groups, in equality strategies will

help lift up the most at-risk groups and reduce illegal activity.

1.2 Methodology

This Gender Assessment uses a range of primary (24 focus groups and 18 key informant interviews) and secondary (published reports, policies and academic literature) information sources to triangulate our understanding of the gender context for climate resilience and the project design.

Key Informant Interviews (KII) were conducted, using purposive sampling, with senior decision makers in governmental organizations at local, district and national levels. Results from these interviews were validated and enriched by 24 Focus Groups (FG) with 58 men and 91 women of different age groups within 21 communities bordering Nyungwe, Gishwati and Volcanoes national parks. Originally a large number of consultations was envisioned, but because of COVID constraints, including limitations on travel between districts, a smaller sample was completed. Additional stakeholder consultations will be conducted at national, district and project site level to inform the implementation of the project and to adapt activity implementation to the local context. These consultations will include the review of Environmental and Social Frameworks included in Annex 6. A desk study of secondary data sources gleaned from government reports and published policies, grey literature produced by civil society groups, and peer-reviewed academic papers has also fed into the report.

The following desired outcomes helped guide our interviews, the desk study and the analysis of all information garnered:

- Gender roles, needs, concerns and aspirations in the context of the proposed project are well understood and documented;
- Root causes of existing gender inequalities in the context of the proposed project are identified;
- Proven options for redressing traditional power imbalances between men and women are identified;
- Increased confidence that gender focused investments and actions proposed by the project will deliver the desired results.

The information generated from the gender analysis and assessment will be considered in all stages of the CND project cycle: design, work planning, implementation, and monitoring and evaluation. In each of these stages, managers will both be enabled and encouraged to view the project through a 'gender lens' so that project investments and actions:

- Reconcile gender inequalities that pre-date and might emerge from the project;
- Ensure the differential needs and concerns of women and men are addressed;
- Ensure women and men have equal access to resources, services, and human capital development;
- Ensure equal participation of women and men in management arrangements and as beneficiaries, partners and key stakeholders; and
- Ensure women's equal participation in decision-making processes.

• SITUATION ANALYSIS

1.3 Policy framework

The implementation of gender focused policy and legislation after 1994 has provided Rwanda with a strong legal framework for advancing women's status and rights. The high prevalence of female and orphan headed households, as a result of the Genocide, required the Rwandan government to reassess the country's property ownership laws. The 1999 Inheritance and Marital Property Law provided women with the right to own and inherit property, protecting widows and orphans from losing their land to male family members (Vanhees, 2014). However, this law only applies to formally registered, monogamous unions, and the children of such unions, denying inheritance rights to women engaged in consensual cohabitation or polygamous relationships and illegitimate children. Full equal rights to land titles and use of land for women were established by Law N° 43/2013 of 16/06/2013 Governing Land in Rwanda (IUCN, 2021). Furthermore, while the 1999 Inheritance and Marital Property Law established equal inheritance rights for male and female children, it was not mandated that all children should inherit equal amounts of property until Law N° 27/2016 of 08/07/2016 Governing Matrimonial Regimes, Donations and Successions (IUCN, 2021).

The Constitution of the Republic of Rwanda of 2003, revised in 2015, outlawed all forms of discrimination, provided legal equality within recognized marriages, and set a minimum of 30% female participation in the Senate and other public governance (East African Community, 2009). These high gender equality standards have helped propel Rwanda to both become the first country with a majority of female representation in Parliament and rank 7th out of 156 countries for the smallest gender gap (World Economic Forum, 2021). The 2010 National Gender Policy functioned to further advance gender mainstreaming and equality, confirming the Rwandan government's stance on gender as a cross-cutting issue that must be addressed for successful development (Republic of Rwanda, 2019).

This foundational principle of gender equality as a driving factor for growth is highlighted in Rwanda Vision 2020 and Vision 2050, which have been used as frameworks for Rwanda's development (Republic of Rwanda, 2000; Republic of Rwanda, 2020). While the primary objectives of these frameworks were to elevate Rwanda to a middle-income country by 2020 and a high-income country by 2050, these goals could not be accomplished without addressing gender inequality. Strategies for mitigating gender inequality included updating existing laws to account for gender aspects, increasing women's representation in decision-making positions, developing gender training programs, increasing women's involvement in associations and cooperatives, and increasing women's access to resources, such as education (IUCN, 2021). The 2008 Girls' Education Policy further aimed to increase girls' access to education by promoting elimination of gender disparities through the development of a 12 year basic education system, a school feeding program, separate girl's bathrooms, and increased infrastructure (Republic of Rwanda, 2019). Separate girl's bathrooms were introduced to limit gender-based violence (GBV) in schools, while the 2011 National Policy against Gender Based Violence aimed to progressively eliminate GBV through the development of a preventive, protective, supportive, and transformational environment (Republic of Rwanda, 2019).

To provide oversight and coordination of the gender policies, the Rwandan government established a number of gender machinery institutions. The Ministry of Gender and Family Promotion (MIGEPROF) implements national gender policy and ensures gender mainstreaming (IUCN, 2021). The main responsibilities of the Ministry are to implement and assess policies and provide resources for programs. The Gender Monitoring Office (GMO) implements gender audits in the public and private sector, and the

National Women's Council (NWC) advocates for women's rights and recruits female participation in development programs (IUCN, 2021). The National Gender Cluster is a forum of representatives from private sector, government, and development groups that meets to discuss the prioritization and coordination of gender equity efforts (IUCN, 2021). The Rwanda Women Parliamentarians oversee legal issues related to gender and help create new gender policy. Together, these groups work to ensure gender policy is not only implemented, but practiced in Rwanda.

At the international level, Rwanda has ratified several humanitarian agreements in support of gender equality. The Convention on Elimination of all Forms of Violence against Women (CEDAW) is an international bill of rights for women. All signatory countries are evaluated for national issues that may conflict with the agreement. In Rwanda, GBV and lower rates of education among women are areas for concern (UN Human Rights, 2017). Almost every country in Africa has committed to the African Charter on human and peoples' rights on the rights of women (ACHPR), which provides a continental framework for human rights. Most recently, Rwanda prioritized several of the United Nations Sustainable Development Goals, namely Quality Education (4), Gender Equality (5) and Reduced Inequalities (10) (United Nations Rwanda, 2021). These international agreements provide Rwanda with a level of accountability to implement and practice national gender policies.

While the Rwandan government has made significant progress to advance women's rights under the law, a gap remains for women not engaged in formal marriages. Approximately 30% of women in Rwanda participate in consensual partnerships but are not legally married, or are in polygamous marriages; neither of which are recognized under the law (Abbott & Malunda, 2016). This creates conflicts regarding land inheritance for women in these situations. Polygamous marriages do take place in Rwanda, and the status of wives is fluid; inheritance (without official marriage) can change based on seniority or who had children in the partnership (Vanhees, 2014). However, the 2008 Gender Based Violence Law does assert that if a person in a polygamous relationship wants to officially marry one of the partners, common belongings must be divided equally between the other spouses (Vanhees, 2014). This law does establish a precedent to address the rights of polygamous women, but remains insufficient. The Rwandan government sought to increase legal protections for women by hosting group weddings in rural areas, but further legislation to recognize women's rights as human rights, regardless of marital status, are necessary (IUCN, 2021).

1.4 National and international legal framework for gender equality

The Government of Rwanda, State of Gender Equity Report (2019) provides a comprehensive summary of constitutional provisions, and national laws that support the principle of gender equality and women's empowerment in Rwanda. Information for the legislative and policy review was drawn primarily from this report. Additional information was gleaned from Rwanda Vision 2020, the National Policy against Gender Based Violence (2011) and the National Gender Policy (2010).

Rwanda is internationally recognized as a world leader in promoting gender equality principles and women's empowerment. In the aftermath of the 1994 genocide, the Government undertook radical and far-reaching reforms to address the political, social, legal, and economic status of women. Legal reforms to give women property rights, and to enable them to inherit property, including land, were especially important. The Constitution, adopted in 2003 and amended in 2015, proactively promotes gender equality. It outlaws any form of gender discrimination, and enshrines the principle of equality within

marriage.

With 53% of the Rwanda population being female, the Vision 2020 national government strategy emphasizes that Gender equality will be one of the driving factors towards achieving rapid growth and sustainable development. Vision 2020 commits to continuously update and adapt laws on gender, strategies for an increased access to productive resources by women, representation in decision-making positions and apply positive discrimination in favour of women.

The National Gender Policy (NGP) of 2010 formulated to support government programs in various sectors that are directly aimed at addressing gender inequalities and women's rights. The NGP seeks to free Rwandan society from all forms of gender based discrimination and create an environment where both men and women equally contribute to and benefit from the national development goals.

The main goal of the NGP is to contribute to reducing gender inequalities in all sectors, as a key component of sustainable development. The NGP was informed by the Economic Development and Poverty Reduction Strategy (EDPRS I) of 2007 It helped to shape theThe Agriculture Gender Strategy (2010), the EDPRS II (2013), the National Strategy for Transformation (2017-2024), the Strategic Plan for the Transformation of Agriculture (PSTA3), the National Information and Communication Technology in Rural Agriculture (ICTRAg) Strategy (2016-2022), the National Policy against Gender- Based Violence (2011), the National Strategy on Climate Change and Low-Carbon Development (2011) and several others.

Constitutional provisions and gender

The Constitution of Rwanda of 4 June 2003, as amended on December 24, 2015, affirms the fundamental rights of all citizens of Rwanda, consistent with the United Nations Declaration of Human Rights and other human rights instruments. According to Article 11 of the Constitution: "All Rwandans are born and remain free and equal in rights and duties. Discrimination of whatever kind based on, inter alia, ethnic origin, tribe, clan, color, sex, region, social origin, religion or faith, opinion, economic status, culture, language, social status, physical or mental disability or any other form of discrimination is prohibited and punishable by law." Article 16 further enshrines the principle of gender equality, and the Government has committed to establishing equity and equality at all levels of society.

Key provisions within the constitution that relate to gender equality include the following:

- The preamble reaffirms Rwanda's adherence to human rights conventions, including the UN Convention on the Elimination of All Forms of Discrimination against Women (CEDAW 1979) and declares Rwanda's commitment to ensure equal rights between women and men;
- Articles 11 and 16 outlaw any form of gender discrimination;
- Articles 9, 76, and 82 mandate a minimum quota of 30% female representation in the Senate and other areas of public governance;
- Article 37 prohibits discrimination in employment; and
- Article 26 enshrines equality within monogamous marriage.

Since 1994, the Government of Rwanda has enacted a series of gender progressive policies. These include the following three key laws:

Gender equity in the right to inherit property

Enacted in 1999 Law n°22/99 (supplements book one of the civil code regarding succession) and revised

in 2016 (Law N°27/2016 of 08/07/2016 Governing Matrimonial Regimes, Donations and Successions) provides that both male and female children have the rights to inherit their parents property. This in terms of that law obviates customary practices where only sons inherit land and tangible assets.

Gender equity in the right to own land

Enacted in 2013, Law N° 43/2013 OF 16/06/2013 Governing Land in Rwanda, ensures that all women have equal rights to land titles registered in their name. This law enables women to secure loans from financial institutions using their land as collateral, which in turn removes a significant barrier from women engaging in commercial enterprise that require capital investment.

Gender-based violence

Enacted in 2008, Law n° 59/2008 of 10/09/2008 on Prevention and Punishment of Gender-Based Violence makes explicit that gender-based violence is a crime that is punishable under Rwanda law.

At the international level, Rwanda is party to key international instruments addressing women's and children's rights, including:

- The Convention on the Elimination of all forms of Discrimination against Women (CEDAW).
- The Convention on Slavery and Repression of Human Trafficking and its Additional Protocol repressing and punishing the sale and trafficking of children and women.
- The UN Convention on the Rights of the Child (UNCRC) and the Optional Protocol within the UNCRC on Child Trafficking, Child Prostitution and Child Pornography.
- The African Charter on Rights and Welfare of the Child (ACRWC)
- The African Charter on Human and Peoples' Rights and the Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa.
- ILO Convention 182 on the Worst Forms of Child Labor.

The government has also ratified and/or implemented numerous international conventions and instruments. Those include:

- The Convention on Elimination of all Forms of Violence against Women (CEDAW),
- The Beijing Platform for Action, the African Charter on human and people's rights on the rights of women (ACHPR),
- The International Convention on Social and Cultural Rights (ICCPR),
- The United Nations Security Council Resolution on Peace and Security (UNSCR 1325 and 1820) and
- The UN declaration on human rights (UDHR).

Lastly, the Government of Rwanda is committed to delivering on the 17 Sustainable Development Goals (SDGs) which includes gender equality. To fast track attainment of the SDGs, the Government of Rwanda argues that policies must be in place and effectively implemented to ensure that both women and men equally participate in and benefit from development investments and processes.

1.5 Institutional framework for implementing gender policies in Rwanda

Key efforts to implement the Government of Rwanda's gender progressive policies are included in the remit of the Agriculture, Justice, Health, Education, and Local Government ministries and their

executing agencies.

To help coordinate and oversee Rwanda's gender equality and women's empowerment efforts and ensure their compliance with national policies and regulations, Rwanda has established four key governmental and civil society entities. These include the Ministry of Gender and Family Promotion (MIGEPROF), the Gender Monitoring Office (GMO), the National Women's Council (NWC) and Rwanda Women Parliamentarians (RWPF/FFRP).

Ministry of Gender and Family Promotion (MIGEPROF)

The Ministry responsible for Gender has the mandate of coordinating the implementation of the national gender policy and advocating on gender issues at local to national levels. MIGEPROF formulates regulations and its staff support gender mainstreaming across the Government of Rwanda.

The Gender Monitoring Office (GMO)

The GMO documents and reports the extent to which a "gender lens" influences the investments and actions of public agencies and private organizations. The GMO undertakes monitoring and evaluation to assess the implementation and social impact of public and private sector implementation of activities that seek to comply with Government gender policies.

The National Women's Council (NWC)

The National Women's Council is a local to national organization of women that advocates for women's rights and the promotion of gender equality. Its members help mobilize women across the nation at all social, economic and political levels to ensure that their voices are heard and acted on in development and conservation programmes and activities. The National Women's Council provides a formal structure to give voice to women and through which women can raise ideas and concerns to inform policy. Lack of financial resources remains a constraint to the active engagement of village women in this process.

Rwanda Women Parliamentarians (RWPF/FFRP)

The RWPF/FFRP works to build the capacity of women Parliamentarians to navigate the halls of government successfully, to represent women's voices in the formulation of all policies not just those related to gender equality, and to hold the parliament and its members accountable for redressing historical inequities in the rights of women and the power that they wield in determining the future of all Rwandans.

The National Gender Cluster

This is a Government convened forum in which the Government of Rwanda, development partners, the Private Sector and Civil Society meet and discuss planning, coordination and prioritization of Gender Equality interventions.

1.6 Poverty and labor force participation

The poverty rate in Rwanda averages 39% (NISR, 2018). The Western (CND) region is the poorest in the nation with a poverty rate that averages 47.7% and ranges from 33.5% in Rusizi to 69.3% in Nyamasheke. Between 2013/14 and 2016/17 (NISR, 2018) 50.5% of female headed households had been poor during that time period, compared to 48.8% of male headed households. Poverty was more prevalent in households with more people and where the household head was older, had little education,

Labor force participation is lower in women (48.2%) than in men (65.6%) 16 years and older (NISR, 2020), and unemployment is higher in women (20.3%) than in men (15.9%). Young women and men are less likely to be in the labor force and more likely to be unemployed. In 2020, women were less likely to be employees and employers, and significantly more likely to be a contributing family worker than were men.

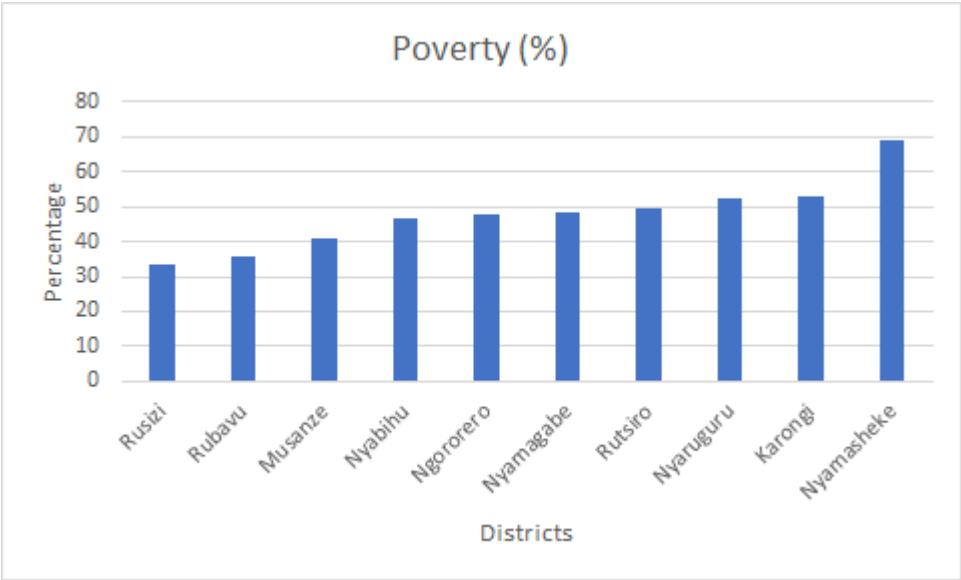


Figure 2: Poverty in districts within Congo Nile Divide (Source: EICV5 NISR, 2018)

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Population 16 years old and over by labour force status, sex, age group, and urban/rural area, RLFS 2020								
	Total	Labour force status				Labour force participation rate (%)	Employment-population ratio (%)	Unemployment rate (%)
		Labour force	Employed	Unemployed	Outside labour force			
Total Population 16 yrs and over	7,472,601	4,212,972	3,460,860	752,112	3,259,630	56.4	46.3	17.9
16-24 yrs	2,309,913	1,080,767	828,459	252,308	1,229,146	46.8	35.9	23.3
25-34 yrs	1,874,035	1,356,005	1,098,949	257,056	518,080	72.4	58.6	19.0
35-54 yrs	2,195,647	1,427,063	1,230,818	196,245	768,584	65.0	56.1	13.8
55-64 yrs	617,306	271,530	233,426	38,104	345,775	44.0	37.8	14.0
65+ yrs	475,700	77,605	69,206	8,399	398,095	16.3	14.5	10.8
Male Pop. 16+ yrs	3,512,698	2,303,622	1,938,268	365,354	1,209,075	65.6	55.2	15.9
16-24 yrs	1,132,034	573,371	456,983	116,388	558,663	50.6	40.4	20.3
25-34 yrs	905,580	755,696	631,419	124,277	149,883	83.4	69.7	16.4
35-54 yrs	1,009,973	782,467	684,302	98,165	227,507	77.5	67.8	12.5
55-64 yrs	267,154	145,126	123,234	21,892	122,028	54.3	46.1	15.1
65+ yrs	197,957	46,963	42,331	4,632	150,994	23.7	21.4	9.9
Female Pop. 16+ yrs	3,959,903	1,909,349	1,522,592	386,757	2,050,554	48.2	38.5	20.3
16-24 yrs	1,177,879	507,396	371,476	135,920	670,483	43.1	31.5	26.8
25-34 yrs	968,455	600,309	467,530	132,779	368,146	62.0	48.3	22.1
35-54 yrs	1,185,674	644,597	546,517	98,080	541,077	54.4	46.1	15.2
55-64 yrs	350,152	126,405	110,193	16,212	223,747	36.1	31.5	12.8
65+ yrs	277,743	30,642	26,875	3,767	247,100	11.0	9.7	12.3
Urban Pop. 16+ yrs	1,581,477	1,060,077	867,565	192,512	521,399	67.0	54.9	18.2
16-24 yrs	501,826	248,767	191,104	57,663	253,060	49.6	38.1	23.2
25-34 yrs	493,474	399,957	321,737	78,220	93,516	81.0	65.2	19.6
35-54 yrs	445,849	357,316	310,082	47,234	88,532	80.1	69.5	13.2
55-64 yrs	78,623	42,609	34,597	8,012	36,014	54.2	44.0	18.8
65+ yrs	61,705	11,428	10,045	1,383	50,277	18.5	16.3	12.1
						0.0	0.0	0.0
Rural Pop. 16+ yrs	5,891,125	3,152,894	2,593,295	559,599	2,738,231	53.5	44.0	17.7
16-24 yrs	1,808,087	832,000	637,355	194,645	976,086	46.0	35.3	23.4
25-34 yrs	1,380,561	956,048	777,212	178,836	424,513	69.3	56.3	18.7
35-54 yrs	1,749,799	1,069,747	920,736	149,011	680,052	61.1	52.6	13.9
55-64 yrs	538,683	228,922	198,830	30,092	309,761	42.5	36.9	13.1
65+ yrs	413,995	66,177	59,162	7,015	347,818	16.0	14.3	10.6

Table 1: Population 16 years old and over by labour force status, sex, age group, and urban/rural area (Source: NISR Labor force survey report 2020)

Employed population by sex, status in employment, and urban/rural area, RLFS 2020			
	Total	Sex	
		Male	Female
Total employed population	3,460,860	56%	44%
Employee, Paid apprentice/intern	2,292,976	57%	43%
Employer	46,221	65%	35%
Own-account worker	968,858	57%	43%
Member of cooperative	5,792	64%	36%
Contributing family worker	147,013	22%	78%

Table 2: Employed population by sex, status in employment, and urban/rural area (Source: NISR Labor force survey report 2020)

1.7 Beliefs, perceptions and cultural norms

Pre-1994 Rwandan families and society in general were characterized by a patriarchal social structure that translated into male dominance and female subordination (Mukabera, 2016). In principle, in traditional Rwandan society, women's responsibilities included raising and educating children, welcoming visitors, managing the household, advising their husbands and maintaining traditions (Mukabera, 2016). In practice, though women were allowed substantial autonomy in their roles as child bearers and food producers men remained the ultimate arbiters of most family decisions.

Traditionally, the Rwandan woman was a symbol of fertility and weakness, while the man was a symbol of strength and protection who makes all major decisions. Rwandan women were valued as child bearers and more specifically the number of boys they gave birth to (Mukabera, 2016). Boys were taught to defend the interests of the family. It was their job to build or repair a house, to engage in activities related to cattle, cut firewood, go to battle, milk the cows, make decisions on farm management, and generate income for the family. Girls were raised to help their mothers with household chores, raise children, and most importantly to be modest, reserved, silent, obedient, and present a submissive attitude toward men (Mukabera, 2016).

These traditions effectively excluded women from accumulating material wealth and participating in economic and political decision making. The Family Code of 1992 precluded women from owning land. If a male head of household died, property passed to male heirs or to the man's brothers (Uwineza, 2009). In the case of divorce or the husband's death, a woman had no claim over the family property if she had not borne children; she would be forced to return to her birth family. Likewise, 'the Commercial Code', dating back to the colonial period, required the written consent of the husband whenever a wife wanted to engage in commercial activities, take up employment outside the household, or obtain credit, or take legal action (Mukabera, 2016).

Prior to 1994 men monopolized the public and political arena. Despite this, there have been some powerful women throughout Rwandan history, particularly the Queen Mothers who held substantial influence as advisers to the King in pre-colonial Rwandan society (Uwineza, 2009). In the customary system of justice, known as *gacaca*, women were not allowed to participate directly in public deliberations. Rather, they were expected to play an indirect role lobbying their husbands (Mukabera, 2016). Women could and did hold powerful positions as high priestesses within the religious realm, but this was undermined with colonization and the introduction of christianity.

Post-1994, the Rwandan government has taken large strides in improving gender equality and women's rights policies, however these policies have yet to reach their full potential in making substantial changes in women's lives. This may be due to gaps in policy and a dearth of resources applied to ensure proper functioning of the legislation and gender machinery institutions. There is a lack of capacity-building strategies to develop gender machinery institution staff's gender based knowledge and skills, and these staff include only a limited number of gender experts that rarely occupy decision-making positions (IUCN, 2021). Strategies to mainstream gender in capacity-building strategies of these institutions are not comprehensive, which contributes to functional issues such as lack of trained staff, staff retention measures, proper orientation for new hires, and clear handover procedures. This suggests that although the Rwandan government has passed strong gender-based legislation, implementation is limited by lack of concrete guidelines and limited resources.

While setbacks to establishing fully functional gender machinery institutions limit the effective execution

of gender equality laws, the current cultural norms and belief systems in place in Rwandan society profoundly diminish the positive effects that these laws have on women's daily lives. Discrimination against women continues to exist in multiple aspects of Rwandan society including economic opportunities, agriculture, politics, and division of labor. The long history of patriarchal dominance in Rwanda has led to the development of robust cultural norms, many of which define men's and women's roles in society, limiting women from taking advantage of the full range of opportunities within their legal rights. Other societal factors, such as the high prevalence of GBV, further limit women's ability to venture outside of their perceived household roles. In a recent study 52% of women surveyed in districts within the CND agreed that men were justified in beating them if they: burned the food, argued with him, went out without telling him, neglected the children, or refused to have sex with him (Government of Rwanda, 2016). Today, male and female policymakers are quick to acknowledge that many aspects of traditional Rwandan culture repressed women and abrogated their human rights (Uwineza, 2009).

1.8 Gender roles, responsibilities and labor allocation

Rwandan cultural norms divide household duties into activities that fall under either men's or women's responsibilities. Men are considered the breadwinners of the family, and engage in significantly more off-farm and wage employment than women (Republic of Rwanda, 2019). Men also hold domain over any valuable property or resources that the family owns or acquires, including cash crops, livestock, and valuable trees (IUCN, 2021; Kiyani et al., 2017). Men possess the majority of the decision-making power within the family, and are largely able to engage in any activities they desire.

Women's role in society is predominantly seen as being a mother and caretaker of both the family and the home. Primary duties include caring for children and elderly, cooking, growing food for the family, and providing labor on their husband's land. While the law allows for women to engage in other endeavors outside of their traditional duties, several factors often limit them from diverting from the cultural norm. The dominant barrier for women is lack of time, as the majority of their time is spent fulfilling their household responsibilities (IUCN, 2021). Women spend anywhere from 1 to 8 hours each day at their cookstove, not including the time it takes to collect the food, water, and fuel wood required for cooking (Campbell et al., 2012). Additionally, women are responsible for growing and tending to subsistence crops for the family, and are often required to take on other informal jobs on their husband's farms. These duties, on top of childcare, take a large toll on women's available time, preventing many from engaging in wage labor, training, and other forms of off-farm activities (Republic of Rwanda, 2019).

Innovations that reduce the time spent on these tasks, such as improved cookstoves and agricultural technologies, may increase women's ability to engage in activities outside of their traditional roles. Furthermore, male involvement in work traditionally delegated to women, such as childcare, would allow women to spend more time outside of the household. However, men are not expected, and are often unwilling to engage in what they perceive as women's work (Abbott & Malunda, 2016). As a result, some Rwandan women believe that government promotion of women's equality has increased their workload, as they are now expected to be bringing home money on top of their childcare, domestic, and farming duties (Abbott & Malunda, 2016). Furthermore, any additional income earned by women outside of traditional female tasks may be taken by the husband or result in increased domestic violence due to a perceived emasculation of the male role as breadwinner.

Violence is an issue for both men and women (30% vs 37%, respectively), but women are more prone to being victims, particularly sexual and gender-based violence (SGBV) (6% for men vs 23% for women,

DHS Survey, 2021). Although there are strong laws and policies against SGVB, trends of violence are not decreasing. Factors that impact higher rates of violence include lower education, earning money, household alcohol levels, and familial historical violence. In seeking support, both men and women turn to neighbors (32% vs 40%) and family (24% vs 33%), whilst some go to the local authorities (17%, 18%), police (17%, 11, ibid). Women also seek out their partners' relatives for support whilst men seek out friends. This demonstrates the importance of social networks much more than government agents.

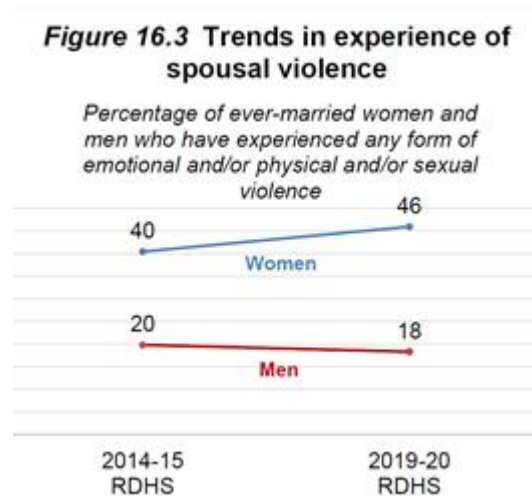


Figure 3, Trends in spousal violence over time. Source: Rwanda Demographic and Health Survey, 2021

1.9 Access to and control over capital and natural assets

The 1994 genocide left behind unprecedented distortions in social relations and exacerbated the already existing gender imbalances and inequalities. This is illustrated by the fact that today, 25% of households in Rwanda are headed by women. Women performing non traditional roles such as decision making, managing financial resources, constructing houses and many others have characterized the post genocide roles of women period in Rwanda.” (Gender and Community Development Analysis in Rwanda, EAC 2009).

Although collection and dissemination of credible data disaggregated by ethnicity is not available.¹ There has been some recognition of some ethnic minorities as Historically Marginalized People (HMPO, see Annex 6 for further details). While there is limited data based on ethnicity of HMPO, which critically impacts the baseline for monitoring activities potential impacts (positive and negative), we will refer to trends aligned with governmental policies supporting pro-poor programming, and rely on data from reports conducted by African Initiative for Mankind Progress Organization (AIMPO), Women’s Organization for Promoting Unity (WOPU) and Minority Rights Group International (MRG) leading a project titled, ‘Strengthening the Capacity of Civil Society Organizations (CSOs), Human Rights Defenders (HRDs) and Media Houses to Challenge Discrimination against HMPO and Promote their

¹ The project will be unable to disaggregate data based on traditionally marginalized people, however, all data collected will be disaggregated by sex and wealth category where possible

Human Rights in Rwanda’.

Women are likely to constitute the majority of genocide survivors which informs a need for intersectional, or the interconnected nature of social categorizations’ integration of history, gender norms, social disparity, power dynamics, rural/urban divide, and most importantly access to land rights. Despite policies and initiatives implemented by the government, other entities and projects, including this project to ensure HMPO are equitably included in project design, implementation and benefits, there are potential barriers which include: (1) Inadequate disaggregated data (2) Limited capacity and resources to analyze intersectionality (3) Limited access to decision-making (women) (4) Social inequality from increased long-term poverty of HMPO (5) Lack of promotion and protection of HMPO rights, including land tenures.

In addition, climate change is adversely impacting CND’s landscape with increasing temperatures and changing rainfall patterns contributing to ecosystem degradation and reduced livelihoods for the most vulnerable populations. The agricultural sector in Rwanda is primarily focused on subsistence farming, and most cultivated land is rain-fed, exposing farmers to significant climate-related food security risk. In addition, Rwanda’s efforts to reduce poverty are affected by high dependence on agriculture, especially when combined with climate change, declining plot sizes, and limited investment in intensification and erosion prevention. Thus, the impacts of climate change is a risk multiplier as it has exacerbated the social, cultural, and economic conditions of HMPO communities in Rwanda. In the face of the climate crisis, culturally-based climate resilient approaches should be adopted in policies along with ways to address the potential barriers listed above.

Compared to men, women are significantly less likely to own land (productive assets), they are less often engaged in income-earning activities, have less access to higher education (Govt of Rwanda, 2018), financial services (though this is rapidly becoming more equitable), health services, employment, direct payments, and non- school training opportunities. But that is changing according to the 2019 State of Gender Equity in Rwanda report. Women-headed households are more likely to use firewood (85% vs 78%), less likely to use charcoal than male-headed households (13% vs 19%), suggesting increased labor burdens for searching for fuelwood. However, women and male-headed households have the same access to water sources (EICV 5, 2018).

There are no significant differences by household’s poverty status [between male and female headed households], education or ownership of land (Abbott 2018). However, as men receive 75% of agricultural loans, and women only 25% this strongly suggests that land owned by married couples is in reality “owned” by the husband and not jointly, and the man in a married household is still culturally considered the head of the household. For married couples it is unclear whether women, on the death of their husbands, inherit their assets or are these, according to tradition, bequeathed to the sons. In addition, should women grow crops or trees, or raise livestock on marriage land do these assets belong to her or him or both. The power of women in illegal but still common polygamous marriages is even less than those in monogamous marriages.

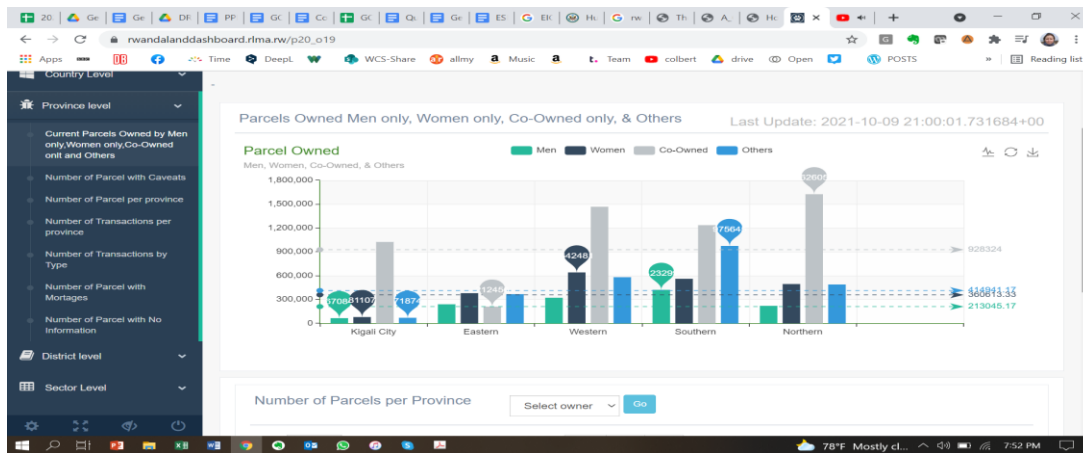


Figure 4. Land ownership by men, women, and co-owned parcels in 2021

(Source: Rwanda Land Dashboard https://rwandalanddashboard.rlma.rw/p20_o19, accessed October 10, 2021)

Though women are not the primary land or livestock owners in Rwanda they still play a key role in food production primarily to feed their family. In fact few Rwandans and almost exclusively men are involved in cash crops as land holdings are typically small and often barely enough to produce food sufficient to meet the nutritional security of the land owners' families.

Access to microloans and credit for women contributes to economic development of Rwandan society as a whole and helps women escape poverty (Mpakaniye, 2016). However, microloans remain difficult for the poor to attain for a number of reasons. Transaction fees reduce the value of already small accounts, while complicated account registration procedures, minimum balance requirements, and illiteracy all hinder access to loans, especially for women, who are more likely to be illiterate or have less experience handling money (Mpakaniye, 2016). Women typically rely on more informal loans from family because they lack collateral to access credit from a financial institution (IUCN, 2021). In 2018, of rural Rwandan women who accessed loans from a financial institution, 44.2% used saving and credits Cooperatives (SACCOs), compared to 39% of rural men (IUCN, 2021). 22% of rural men used a commercial bank, as opposed to only 16% of rural women (IUCN, 2021).

Helping women access loans from formal financial institutions will allow greater independence for women and more economic security for families. Several banks in Rwanda have already signed on to the African Guarantee Fund, aimed to reduce poverty and increase female access to loans (Mpakaniye, 2016). The fund helps those who lack collateral still receive credit (Mpakaniye, 2016). Annually, the number of microloans accessed by women in Rwanda is increasing, but additional support to enhance female financial literacy will help maintain the upward trend (Mpakaniye, 2016).

Agricultural extension services include training to innovate farming practices, and improve profitability and efficiency. The 2010 National Gender Policy (MIGEPROF), the 2018 Strategic Plan for the Transformation of Agriculture (PSTA3) and the 2010 Agriculture Gender Strategy (MINAGRI) have committed to increase extension staff in Rwanda, particularly women (IUCN, 2021). At present, women are significantly underrepresented as extension personnel, which furthers a cycle of reduced female access to extension services; 82% of extension workers are male in Rwanda (IUCN, 2021). Male extension workers are more likely to approach male farmers regarding men's crops, perpetuating the exclusion of women.

Similarly, farm promoters, village leaders who promulgate knowledge from extension services, are also dominated by men (IUCN, 2021). Women tend to be less educated than men and have greater domestic obligations that inhibit them from working with the community as promoters. Promoters are perceived as having elevated social status, leading men to more quickly jump on the opportunity than women (IUCN, 2021). By enabling more women to hold leadership positions as extension workers and farm promoters, more women can subsequently benefit from extension services, training and innovative farming practices.

This sector has vast potential for greater female involvement, but legal frameworks may be necessary to ensure more equal representation. Women represent 66% of the agricultural workforce, but only 19.7% of women in the sector occupy paid jobs (IUCN, 2021). Therefore, enabling women to become agricultural leaders through extension services and farm promoters may offer greater independence, financially and socially. Female-only extension service training may motivate women to participate and bring more attention to female crops, and establishing hiring quotas may enable more female leadership.

Women are equally likely as men to have access to a mobile phone and use mobile financial services. But are less likely to be mobile money account holders, or have access to bank services such as an ATM bank card. This is particularly true for women in rural areas, and ownership of capital goods is much less for women-headed households (see Table 3), demonstrating their marginalization across communities.

Head of household sex	Living room set	% owning at least one					Total surveyed
		Radio	TV	Computer	Mobile phone	Bicycle	
Male	19.6	78.6	11.9	3.6	71.1	16.2	2,031
Female	12.8	59.3	5.7	2.3	54.3	4.9	677

Table 3: Comparison of assets held by women and male-headed households (Data from EICV 5, 2018)

1.10 Women's health and access to health services

In 2010 life expectancy at birth was 71.13 years for women, and only 66.82 years for men (<https://www.statista.com/statistics/971106/life-expectancy-at-birth-in-rwanda-by-gender/>).

Based on the 2019/2020 Demographic and Health Survey report (NISR, 2020), the maternal mortality rate or ratio (MMR), which measures the obstetric risk associated with each live birth and is the age-standardized maternal mortality rate divided by the age-standardized fertility rate, declined from 10.7 per thousand live births in 2000, to 2.03 per 1000 in 2019-2020.

The mortality rate of children under 5 years of age continues to decline from 50 deaths per 1,000 live births in 2014 to 45 per 1,000 in 2020.

Women's health, proxied using Body Mass Index shows that on average 76% of women within CND districts are within the normal BMI range, 19% are considered obese and only 5% fall below 18.5 which is considered nutritionally compromised (Republic of Rwanda, 2016). The averages for Rwanda as a

whole are 73% normal, 21% obese, and 7% too thin. Similarly, only 18% of women exhibit signs of anemia, compared to 19% for the nation. Prevalence of malaria in children and women is low (0.5% and 0.4% respectively) compared to 2.2% and 0.6% for the country as a whole (Republic of Rwanda, 2016).

In 2017 74% of women had health insurance and 94% of women had access to community health services. The percentage of health insured households is lowest in Rubavu district (54%) and highest in Ngororero district (82%) according to the 2014-15 demographic and health survey (Government of Rwanda, 2016).

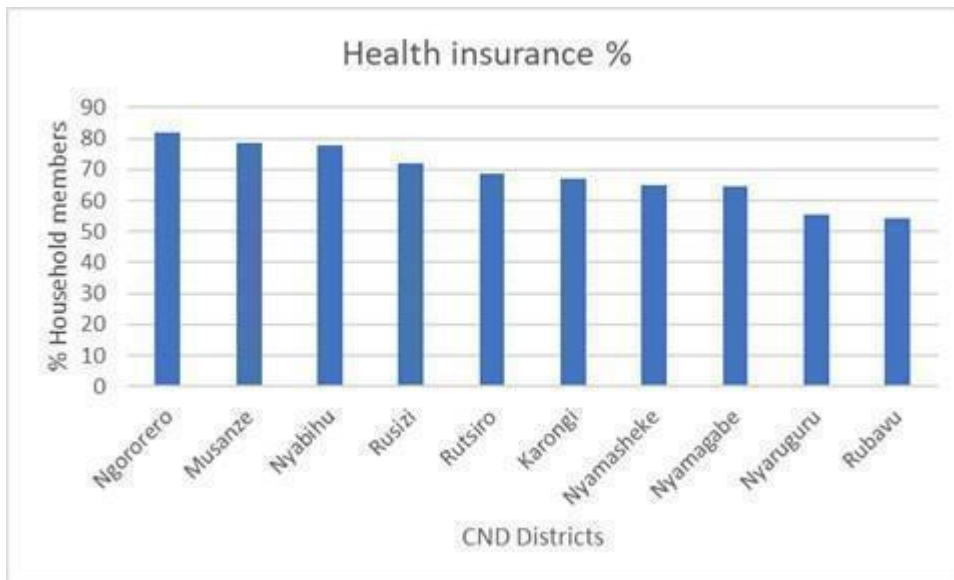


Figure 5: Percent population with health insurance by district (Source: Government of Rwanda, 2016)

Overall access to sanitation facilities is high among Rwandans (IUCN, 2021). However, due to unequal income distribution between men and women, households led by women are less likely to have sanitation access than male led households; 80.6% of female headed households have improved sanitation facilities compared to 88.0% of male headed households (IUCN, 2021). Traditional pit latrines are seen as more cost effective than modern sanitation systems, but pose risks of disease, overflow during the rainy season, groundwater contamination, mosquito breeding, and an increased risk of GBV for women due to distance from the house and night use (Tsinda et al., 2013). Lack of sanitation services poses both a serious social and environmental threat in Rwanda, particularly in informal settlements.

1.11 Women's educational attainment and access to education services

Education can lead to better health and greater access to resources (Allan et al., 2018:26). In Rwanda, the rate of people who have ever attended school is high, at 87%, but rates are higher in urban areas than rural areas (IUCN, 2021). The rates of primary education are approximately equal between boys and girls in Rwanda, and in secondary education women actually attend in higher numbers than men, 53% women to 46% men (Education Statistical Yearbooks, 2018). However, the disparity between men and women shifts with regard to technical training and tertiary education, in which men are more dominant.

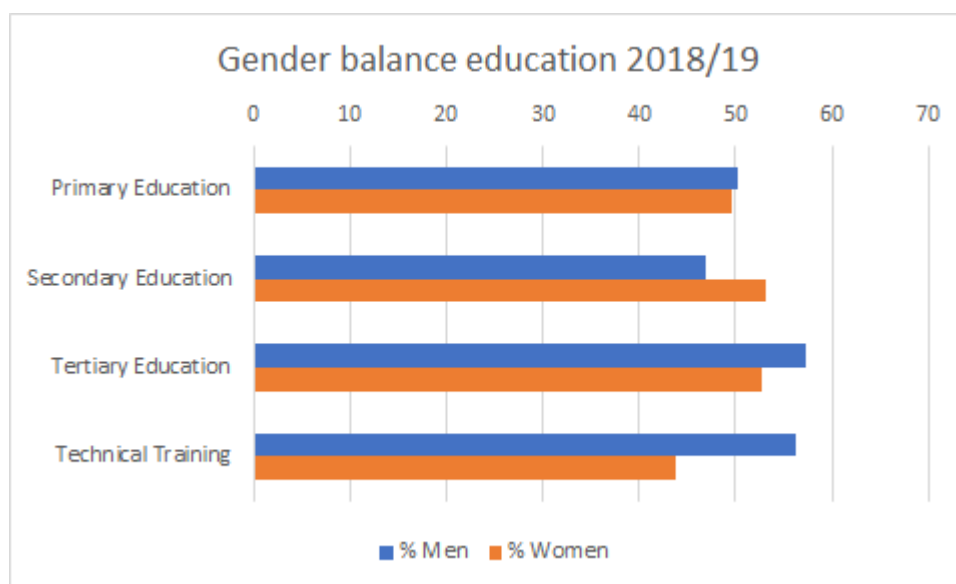


Figure 6: Gender balance in education in Rwanda (Source: State of Gender Equality in Rwanda, 2019)

In 2018, 92% of children (1.64 million boys and 1.68 million girls) were enrolled in pre-nursery to secondary education (Ministry of Education, 2018). Literacy rates continue to climb from 38.2% in 1978 to 73.2% in 2018. By this time youth literacy was at 85%. Girls and women lag behind boys and men. Only 56% of girls, 41% of women, and 28% of elderly women are literate.

Levels	Schools	Males	Females	Total	Percent
Pre-Nursery	96	3,199	3,292	6,491	0.2%
Nursery	3,210	112,044	114,662	226,706	6.3%
Primary	2,909	1,259,344	1,244,361	2,503,705	69.0%
Secondary	1,416	264,782	314,115	578,897	16.0%
TVET	360	57,643	44,842	102,485	2.8%
Tertiary	30	41,458	34,255	75,713	2.1%
Adult literacy	4,991	51,220	81,145	132,365	3.7%
Total	13,012	1,789,690	1,836,672	3,626,362	

Table 4: Levels of schooling in Rwanda (Source: Ministry of Education, 2018 Education Statistics)

The equal rates of primary education for boys and girls in Rwanda suggest a cultural value of education for all children, not only prioritization for boys. A lack of tuition fees for childhood education enables poor families to send boys and girls to school, and increasing rates of education contributes to falling illiteracy rates in Rwanda (IUCN, 2021). Higher levels of girls in secondary education may be attributed to Rwandan policies such as the 2008 Girls' Education Policy and the implementation of a school meal program (IUCN, 2021). Despite lower rates of women enrolled in tertiary education, the percentage is increasing due to flexible programming, such as online learning or night school, which enables women with domestic responsibilities to participate. According to the IUCN, the higher level of education men and women receive, the less likely they are to work in the agriculture sector (IUCN, 2021). In comparison, 92% of women with no education work in agriculture (IUCN, 2021).

Strong gender stereotypes in Rwanda still push girls into "traditional" soft trades including hairdressing, secretarial work and nursing (IUCN, 2021). 87% of women with a university degree work in the service sector (IUCN, 2021). Though women are more likely to be primary school teachers than men, they are rarely secondary, tertiary or technical/vocational teachers. In agriculture, women predominantly occupy informal roles, such as unpaid farming for the family, whereas the men typically hold higher paying positions or manage higher value products such as timber (IUCN, 2021). As a result, the majority of agroforesters are male in Rwanda (Anastase & Nahayo, 2020).

One of the greatest challenges to achieving economic stability lies in challenges in achieving quality educational attainment by female youth which leads to their lower literacy, especially in rural areas. Overall, social factors including traditional gender roles such as domestic chores and family care, historic marginalization of girls and women, patriarchal social traditions, and gender-based violence entrench girls' underperformance throughout their schooling which has a cyclical effect, as low performance in the primary leaving exam results in girls being admitted to lower quality secondary schools, and ultimately into higher education institutions (HIEs) in lower numbers (Huggins & Randell, 2007).

This results in limited access to paid labor and employment in sectors such as services or public government. Similarly, lower literacy has put them at a greater disadvantage in agriculture, as commercialization of farming has made it more difficult to feed families. While there are national laws and policies to protect employment, there is little protection for the vast majority of women who work in informal sectors which significantly affects women's rights within the country (Abbott et. al., 2008). Women in rural areas are primarily care-providers for children and the elderly, which is unpaid labor and invisibilized due to gender norms (Abbot, 2015). Thus, understanding these barriers and challenges and how to overcome them will inform interventions at project-level.

1.12 Patterns of power and decision-making

Rwanda is internationally acclaimed for being the first nation to have a majority female Parliament, with 61% of these positions held by women from 2018-2022 (Republic of Rwanda, 2019). The Constitution of the Republic of Rwanda established a minimum of 30% female representation in public governance, and this high level of female participation in upper levels of governance far exceeds this minimum (East African Community, 2009). However, female participation drastically decreases in many communities'

level governing bodies.

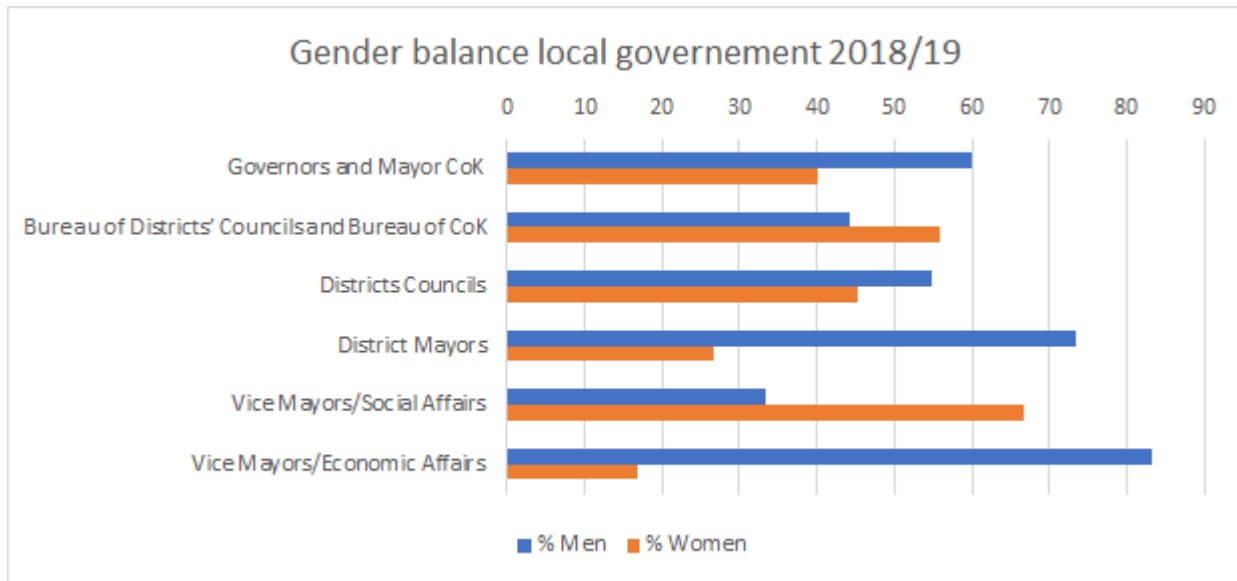


Figure 7: Female and male representation in different levels of local government in Rwanda in 2018-2019 (Republic of Rwanda, 2019)

Although overall female representation in the Rwandan government is increasing, higher levels of female representation in upper levels of governance mirror the disparity between gender equality policy and practice in the country. In higher levels of governance, these policies are implemented well and appear effective, but at community levels, larger gender gaps in prominent roles are clearly evident. Comparable gender gaps exist in central government roles that undergo recruitment processes, with many of these positions not reaching the 30% female representation minimum (Republic of Rwanda, 2019). Furthermore, female participation in the police force is extremely low, although representation is high in most other areas of the justice sector.

Similarly, in the private sector, women hold very few high level positions, particularly at the provincial level. Nationally, 30% of executive committee presidents are women, but there are 0 female presidents at the provincial level (Republic of Rwanda, 2019). Women have high levels of representation in second vice presidency in the private sector, with 70% of national and 80% of provincial second vice presidential roles occupied by women, but have low levels of representation in more powerful executive positions (Republic of Rwanda, 2019). In more local organizations, such as cooperatives, men hold the majority of the decision-making positions, while women tend to hold stereotypically female roles, such as secretary, vice president, and treasurer (IUCN, 2021).

At the household level, women have very little decision-making power. Although under the 1999 *Inheritance and Marital Property Law* women are legally allowed to own land, cultural barriers still inhibit their ability to control how this land is used. These barriers are predominantly based in the belief that men are the head of the household, and thus are the primary owners of all of the family's land and assets (IUCN, 2021). Women's lack of power within their households limits their ability to make decisions about the planting of crops and the use of land as collateral for credit, which would allow increased female engagement in other economic activities. Based on female reports, the majority of households (70%) make decisions jointly, while in 25% of households the husband makes the majority of the decisions

(IUCN, 2021). However, even in households that make joint decisions, women's decisions tend to revolve around day-to-day issues, while men's decisions are primarily focused on productive assets and income.

Women's ability to make decisions within both their own households and broader Rwandan society is further hampered by GBV. The effects of a longstanding patriarchal regime in the country has created and perpetuated the notion that men must be dominant and women must be submissive. Women break these gender norms by asserting their right to be involved in decision-making, and this can be seen as diminishing their husband's masculinity, which can result in violent retribution. In 2010, around 57% of Rwandan women reported experiencing GBV from their partner, and women who bring in more money tend to experience higher levels of GBV (RWAMREC, 2010). This suggests that increasing women's opportunities to make decisions and earn their own income will have a very limited effect on female empowerment until GBV is addressed. Engaging men and boys in gender equality advocacy and action highlights that GBV is not only a women's issue, and is believed to be successful in reducing the prevalence of GBV (UNDP, 2018).

1.13 Social service and support systems that contribute to gender empowerment, wellbeing, and security

Government mandated and running social support programs have reduced the percentage of extremely poor households from 23% and 26% in 2010 to 15% and 18% in 2017 for MHHs and FHHs respectively. Three state managed social support systems have contributed significantly to these observed reductions in poverty and increasing wellbeing and security.

The Girinka Program (One Cow per Family Program) of the Ministry of Agriculture (MINAGRI) started in 2006. By 2017 over 290,000 cows have been provided to vulnerable male and female headed households (approximately 6% of all Rwandan families) as a way to reduce child malnutrition, increase the productivity of agricultural labor, generate a new stream of income through surplus milk sales and strengthen social cohesion amongst Rwandan families and communities as cow owners pass on surplus calves to other needy families. Male headed households were, however, more likely than female headed households to receive a cow through this government program.

The largest social support program of the Government of Rwanda is the Vision 2020 Umurenge Programme (VUP) which was started in 2008 and is managed by the Local Administrative Entities Development Agency (LODA) within the Ministry of Local Government (MINALOC). VUP has four components: wage labor through public works, direct cash transfers, financial services and awareness raising. The majority of public works support agricultural production, land conservation and the maintenance of physical infrastructure. A review by FAO showed that VUP wages paid to women were often captured by men. Retaining control over VUP income was easier for women who owned an individual savings account (e.g., within a Savings and Credit Cooperative - SACCO), though most women only had accounts in Village Savings and Loan Group or informal tontines. Rural women were more disempowered than men through cultural constraints to speaking in public meetings, workloads that limited participation in community meetings and self-help associations, and limited access to credit and savings accounts. The FAO study also found that the majority of both MHHs and FHHs were unaware of the existence of a VUP GRM, or were skeptical about the effectiveness of this complaint process. The most important recommendations were to a) enable women to open their own formal bank accounts to increase their control over their incomes and strengthen their decision-making capacity, and

b) redirect households with high dependency ratios and/or low labor capacity away from public works wage labor and toward VUP direct cash transfers.

MINAGRI further supports poor rural families by offering subsidies for agricultural inputs, improved seeds and irrigation as well as offering credit through the Agriculture Guaranty Fund which provides specific opportunities for women.

● **CONCLUSIONS - ONGOING CHALLENGES TO GENDER EQUITY IN THE CONTEXT OF THE GCF CND PROJECT**

Despite tremendous efforts from the policy and legal perspectives, a patriarchal culture and persistent disparities continue to characterize gender relations in Rwanda. Disparities persist in post-primary education; in access to and control of property including land and economic resources; in employment opportunities and entrepreneurship; in decision-making at household and community levels; in family responsibilities and unpaid care work; and in the experience of violence, harassment, conflict, and insecurity. Sexual and gender-based violence persists at high levels in Rwanda. While women have made impressive political progress, especially at the national level, their progress in terms of local political representation and engagement remains unequal. Economic opportunities remain markedly gender-differentiated.

In Rwanda and in the Congo-Nile Divide, structural barriers to economic and political equity and female empowerment have significantly impeded measures to adapt to climate change impacts. In Rwanda, these structural inequalities are at a lower level compared to other countries in the world, because of leading gender-considerate policies across sectors. However, the point remains that women's ability to access, use and control natural resources, infrastructure and services differently is still low compared to men. This means the degradation of natural resources and new infrastructure will affect women and men differently, and will generally result in greater vulnerability of women. Women are vital agents of change and can be powerful leaders from the community to global level in mitigating and adapting to climate change.

1.14 Afforestation, agroforestry and tree tenure

Kiptot et al (2012) noted that an ICRAF survey conducted across countries in eastern Africa concluded that women rarely have tenure over trees unless they exclusively have tenure over the land. Moreover, they stated that "Women's rights to tree products are usually limited to products that are considered to have little or no commercial value. These products are mainly indigenous fruits....fodder and mulch." They also noted that: a) the survivorship of trees and fodder shrubs planted by men and women was effectively the same, b) low cost/input agroforestry techniques for replenishing soil fertility were attractive to women farmers given that they are both labor and capital limited. Fuelwood, fodder, green manure, and coppicing shrub production substantially increases benefits to women that cannot be captured by men. Women appear to grow trees/shrubs for domestic consumption (fuelwood, fodder, bean poles, fruit) and sell the surplus in local markets. It is unclear whether women who sell wood products from land that is owned by their husbands are able to capture that revenue stream.

Agroforestry offers multiple benefits to Rwandan farmers, in particular, women. Planting trees within farms represents a low effort pathway to high reward. To enhance climate resilience of communities, agroforestry practices sequester carbon, prevent erosion, and enhance soil quality for sustainable

farming. For women, trees provide additional income to elevate socioeconomic status and reduce the risk of GBV by eliminating the need to walk far distances for fuel wood (Anastase & Nahayo, 2020). Despite the majority of farms under female management in Rwanda, women are often excluded from agroforestry, as trees traditionally fall under male domain, women lack the necessary training and have less access to the retail market (Kiptot & Franzel, 2012). In order to better include women in agroforestry, training on forestry and marketing should be targeted to women. Tools and practices that reduce labor requirements for women, such as shorter trees (which decrease the need for male labor) or more modern processing techniques, will save women time and make the field more accessible (Kiptot & Franzel, 2012).

Afforestation and reforestation on public lands should include measures where women and youth formally have the right to capture a proportion of the value of the trees at harvest time. Assuming a harvest cycle of 15-20 years this will provide women with a long-term asset that could be used as retirement income, and/or given as gifts or bequeathed to children. Young men could use the trees as savings to be used to build or buy a house when they marry.

Investments in agroforestry on private farmlands should reflect the different interests and concerns of women and men. During focus groups women reported shortages of fuelwood, fodder and wood suitable for bean poles. They were also concerned about soil erosion on their steeply sloped farmland, and declining soil fertility and crop yields. On-farm tree planting, to address the concerns of women, should focus on fast growing, nitrogen fixing, coppicing species that can be planted in contours between rows of crops. The nutrient rich leaves of leguminous species can be used as fodder for livestock, and as green manure. Cutting the stems to ground level before planting ensures that the trees do not compete with the crops for sunlight and soil nutrients, but the tree roots that remain intact help stabilize the soil and prevent erosion caused by heavy rains. The cut stems can be dried for fuelwood or used as bean poles. Surplus fodder, fuelwood and bean poles can be sold to neighbors or at local markets. Coppicing species readily re-sprout from the roots allowing an annual harvest.

1.15 Improved cook stoves

In rural areas of Rwanda, 98% of households rely on wood as their main cooking fuel. The GoR completed a comprehensive report on Gender in the Energy Sector. The report noted that almost 60% of FHHs spent 1-3 hours per day collecting firewood compared to 36% of MHHs. The latter purchased fuelwood twice as often as FHHs who primarily rely on gathering fuelwood. Over 50% of both Rwandan FHHs and MHHs still cook using the three-stone method, and only 13.5% use efficient cookstoves. Over 50% of FHHs are unaware of the health benefits of improved cookstoves. The report argued for supporting women to transition from fuelwood to Liquid Petroleum Gas (LPG) primarily to reduce smoke inhalation and respiratory disease. The report did not address the fact that in higher altitude areas cooking fires also provide a heating source for the home. The use of clean cookstoves has major positive impacts on gender equality, by reducing household work for women and providing opportunities for women to engage in paid work.

Poor rural families, particularly FHH, are unlikely to be able to afford LPG stoves. Moreover, LPGs do not remain warm after the gas is turned off and thus do not provide a heat source for the home during the night or during cold spells. Tier 1 Canarumwe clay stoves are relatively inexpensive, substantially reduce woodfuel use, and provide a thermal mass that keeps radiating heat long after cooking is

completed. Though imperfect they are an improvement in efficiency compared to the 3 stones cookfire method. As men do not gather firewood nor do they cook they are often unwilling to spend “their” money to purchase an improved stove even though it substantially reduces fuelwood use and the time women must spend gathering fuelwood. Efforts to introduce improved cook stoves into MHHs in the CND must address this barrier.

1.16 Honey production and marketing

The Rwandan government has identified beekeeping as a potential source of additional revenue for farmers (Mushonga et al., 2019). The European demand for honey far outpaces the production in Rwanda, indicating a market incentive to raise bees beyond subsistence level (Mushonga et al., 2019). Furthermore, global decline of bee populations reflects a high biological need for more apiarists in Rwanda. Despite the economic and environmental benefits, along with low startup costs, beekeeping is perceived as a vocation for poor people. Out of 83,000 beekeepers in Rwanda, 86% are adult males and 71% lack primary education (Mushonga et al., 2019). To overcome the negative stigma around beekeeping, materials and training for modern hives need to be provided to women and youth. Modern hives produce a higher yield of honey, have lower risk of causing fires, and do not need to be placed as high up in trees as traditional hives, which deters some women due to cultural restrictions barring women from climbing trees (WCS, 2018). In a WCS study, 83% of beekeepers used traditional hives in Rwanda, demonstrating a need for more modern tools (WCS, 2018). Additionally, marketing training may help women to better sell the honey, as women still have relatively low market and distribution access in Rwanda. Despite the stigma, beekeeping remains a strong potential source of income, which could reduce reliance on more harmful forest livelihoods, such as deforestation and poaching (Fakhry et al., 2010).

A study by Mushonga (2019) suggests that people consider beekeeping for the poor and illiterate are largely the domain of older men no longer able to labor in agricultural production. Women do appear to be engaged in the production and marketing of honey beer. ADF provided \$343,000 in 2017 to strengthen the honey production sector in Rwanda. Production remains relatively low because of dependence on traditional hives rather than modern hives such as Kenya Top Bar or Langstroth (CODIT, 2009). The CODIT report funded by the SNV noted that in 2009, of the estimated 30,293 beekeepers in Rwanda only 23% were women. The report recommended that the GoR do away with subsidies to beekeepers and their co-operatives, focus on addressing the root causes of the prevailing problems not the effects, encouraging beekeepers to actively participate in their own development and advocating for the replication of knowledge and skills from common to individual apiaries.

During focus groups both women and marginalized groups noted that the cost of modern hives and other beekeeping equipment such as smokers was a barrier to entry as was lack of access to credit. A big concern was farmers’ use of pesticides that kill bees, making it impossible to locate apiaries near farmland.

1.17 Coronavirus (COVID-19) implications in advancing gender equity

The global coronavirus (COVID-19) pandemic left no country untouched from its impacts. While research is still emerging, the severity of the pandemic is clear; struggling economies, vast unemployment, and threats to health and wellbeing continue to affect the global community. Women in particular are already

vulnerable to socioeconomic inequity that is currently exacerbated by COVID-19. Therefore, the impact of the pandemic must be considered in gender planning into the future.

Rafaeli and Hutchinson (2020) compared the secondary impacts of COVID-19 to the Ebola crisis on women and girls in Africa, finding that women and girls are more likely to experience higher poverty and school dropout rates, an increase in unintended pregnancies, reduced income and financial empowerment, greater domestic responsibility (increased childcare due to school closures), and food insecurity due to the pandemic. In times of crisis, school closures and financial stress also lead to increased rates of gender based violence (Rafaeli & Hutchinson, 2020). Women face greater exposure to COVID through their roles as caretakers in families and in healthcare, and have less decision-making power, rendering their needs unmet during COVID-19 (Parry & Gordon, 2020). Yet few countries have enacted policies to address the gendered impacts of the pandemic.

In Rwanda, the Ministry of Gender and Family Promotion (MIGEPROF), in partnership with UN WOMEN and UNFPA, conducted a 2020 study on the gendered impacts of COVID-19. The report indicates that 44.2% of female respondents completely stopped their economic activities due to the pandemic, compared to only 17% of men (Bayisenge, Lo & Schreiner, 2020). In rural areas, the percentage was higher for women; 50% stopped their work and thus sources of income, compared to 17.7% of rural men (Bayisenge, Lo & Schreiner, 2020). As women occupy more informal and unsecured jobs in Rwanda, they are at greater risk of income loss than men.

The use of health care and maternal services in Rwanda considerably declined during the COVID-19 outbreak (Wanyana, Wong & Hakizimana, 2020). Declines in child health care services, particularly vaccinations, could be detrimental to avoiding future outbreaks of vaccine-preventable diseases (Wanyana, Wong & Hakizimana, 2020). Many women feared contracting the virus while attending regular checkups, which contributed to the decline of health services in Rwanda (Wanyana, Wong & Hakizimana, 2020).

Recommendations to mitigate the impacts of COVID-19 on women in Rwanda include (Bayisenge, Lo & Schreiner, 2020):

- Implement programs that offer employment or financial security in spite of COVID-19.
- Provide online or socially-distant courses to prevent school dropouts.
- Establish women-specific financial packages in the national economic recovery plan.
- Strengthen the mobilization of men and boys in childcare and domestic responsibilities to offset increased labor caused by the pandemic in parallel with on-going “Men engage” national programmes.
- Accelerate government interventions to address elevated levels of GBV.

1.18 Promoting gender equity within the CND GCF project

Gender equality and women’s empowerment should be considered in all stages of the CND project cycle: design, formulation, implementation, and monitoring and evaluation. In each of these stages, project managers should keep a ‘gender lens’ in mind, looking at ways the project can:

- Address gender inequalities that emerge from the project;
- Ensure the differential needs of women and men are addressed;
- Ensure women and men have equal access to resources, services, and capacity development;
- Ensure equal participation of women and men in management arrangements and as

beneficiaries, partners and key stakeholders; and

- Ensure women's equal participation in decision-making processes.

Based on key expected outcomes, this gender assessment provides the foundation for a realistic gender action plan to be implemented in the CND project.

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● GENDER, YOUTH AND SOCIAL INCLUSION ACTION PLAN

The goal of the programme is to increase climate resilience of vulnerable rural communities in the CND, reduce CO2 emissions, build capacity for integrated spatial planning and increase the extent and integrity of forest ecosystems in the CND. The programme has three components directly targeting vulnerabilities within climate impact chains. Component (1) *Mainstreaming Climate Adaptation into Integrated Land Use Planning and Management* will lead to enhanced adaptive capacity of both ecosystems and rural populations, by strengthening coordination among sectors and integrating climate resilience considerations into planning and development. Component (2) *Landscape Management and Restoration* will implement high priority actions to enhance the resilience of vulnerable communities by increasing the extent and integrity of forest ecosystems and landscapes and by reducing demand for wood fuel. In addition, component (2) is designed to reduce the time women spend collecting and preparing fuelwood, and simultaneously to empower women to provide input to local-level decision making (see also Component (1)). As a co-benefit, greenhouse gas emissions will be reduced through restoration of natural forests, agroforestry, reduced fuelwood demand and fuel efficient cookstoves under components (1 and 2) are expected. Component (3) will focus on *Enhancing climate adaptation through resilient livelihoods*.

Women, youth and marginalized groups, including women-headed households in the CND are key stakeholders in the success of this project, and are the primary target beneficiaries of the project. The project will be implemented across various sectors along the CND, particularly focusing on communities around Nyungwe National Park and Gishwati-Mukura National Park--critical for Rwanda yet highly vulnerable forests due to high population and increasing demography who depend upon fuelwood, and actions taken by poor households that illegally use forests to improve income streams.

1.19 Methods

This Action Plan operationalizes the findings from the Gender Assessment, the ESIA, and youth-based information collected during the feasibility studies and stakeholder engagement during the Project Preparation Facility (PPF) phase. The Assessment used five standard lenses (USAID 2017^[4]) to conduct the analysis:

1. Institutions, Laws and Policies that influence the context in which men and women act and make decisions.
2. Beliefs, Perceptions, and Cultural Norms that shape how men and women, youth and elders are able (or expected) to practice activities, particularly those to be developed by this GCF project.
3. Gender Roles, Responsibilities, and Time Use in reproductive, productive, and community work.
4. Access to and Control over Assets and Resources are different between young and old, men and women.
5. Patterns of Power and Decision-making focused on how individuals are able to be represented and be in decision-making positions.

Based upon this framework of five lenses, the Gender Analysis identified and analyzed the gender risks and impacts of the program and climate change through four main steps:

Scoping: The GYA included a scoping process to review relevant literature and data for the project area, including local reports and data, consult with government and other partners, and conduct field visits to understand the project, identify and characterize the project area, and identify stakeholders,

including program-affected parties and other interested parties, who may have different concerns and priorities about project and climate change impacts, mitigation mechanisms and benefits.

Legislative and policy review: The GYA conducted a review of (1) Rwanda's applicable policy framework, national laws and regulations and institutional capabilities relating to gender and youth, (2) variations in country conditions and program context across the affected area, (3) relevant gender and youth studies, (4) Rwanda's obligations and commitments under relevant international agreements and treaties, (5) applicable requirements under GCF standards and (6) internationally accepted best practices.

Stakeholder consultation and analysis: Based on the identification of stakeholders identified in the scoping exercise, the GYA conducted initial consultations with stakeholders to gather information on concerns and priorities about the project and climate change, analyze the outcomes of the consultations and develop a strategy for continuing stakeholder engagement.

Baseline analysis: Drawing on the scoping exercise, the GYA assembled and analyzed existing information. The information was analyzed in light of the stakeholder consultation and analysis and legislative and policy review, to identify information gaps and ways in which these may be filled.

This information was then used to guide the Gender Youth Social Inclusion Action Plan. The details of the stakeholders consulted, including key informant interviews (18), focus group discussions (24), and district workshops (4) can be found in Annex 7. A literature review is available in the Gender Assessment (above) and ESIA (Annex 6).

1.20 Gender Issues

There are three key gender-based themes that influenced the design of the Gender Action Plan:

a. **Change negative gender and social norms.** The most profound barrier for this project is rural norms about the roles of men and women. Whilst the Government of Rwanda has made many advances and has implemented many policies (see Section 3 of the Gender Assessment) to achieve gender equality, particularly in national-level representation and inclusion, patriarchal privilege and decision-making remains embedded in rural society (IUCN, 2021). The consequences of this for this project is far-reaching. Men are generally seen as the head of household with the decision-making powers over all assets, resulting in male appropriation of household and farm-based assets (IUCN, 2021; Kiyani et al., 2017). Though female headed households face the same issues and challenges as male headed households, they are disproportionately impacted because of prevalent gender inequality issues (GMO 2017). Men in the CND are positioned to be significantly more involved in programs that result in monetary benefits, such as fern clearing, casual labor, and plantation projects. Although under Rwanda law, sons and daughters receive equal inheritance rights (Law/n° 22/99 of 12/11/1999) the PPF focus groups highlighted that due to cultural practices, parents favor sons in land inheritance, leaving young women in a position where they depend on marrying a man to gain access to farmland; without a formal marriage, these women have no land tenure. Broadly, these gender and social norms leave most women with little decision-making power and no real control over assets. The threat of sexual and gender based violence (SGBV) at the household level where women try to assert independence and achievement is high. For this project, the implications are that trees planted by women become men's property; money gained in employment (tree planting, nursery maintenance) or through farmland production (fruit trees, honey, etc) are ultimately controlled by men; and women shouldn't participate in activities without a

husband's permission. Although this seems bleak, this situation is not universal. The 24 CND community focus groups held in the PPF highlighted that many households decide together on how money should be spent; land tenure is broadly seen as joint ownership; and men realize the impact of dwindling fuelwood resources on women and girls. Gender equity is not yet universal and rural culture has a long trajectory to achieving it.

As a result, this program will focus heavily on social and behavior change communications (SBCC) as a means to positively change perceptions towards gender equity. Community members, including community leaders and women and men community influencers specifically will be targeted for sensitization on gender equity in order to shift their attitudes and to build a conducive environment for women and HMPO to be able to participate in project activities. Women have been engaged in the development of this project to ensure time constraints, work burden and reproductive responsibilities do not impede their ability to engage in the various activities designed by the project. Women will be regularly consulted throughout project implementation to ensure they are able to actively participate in project activities and activities take place at times which are convenient for them and at locations that are safe and contextually appropriate. At project start, once communities are selected for implementation, barriers faced by women and HMPO to participate in project activities will be reviewed and solutions to overcome barriers incorporated into programming through an additional screening process outlined in Annex 6. The project ESS Coordinator and Gender Officer will provide support to ensure all project staff understand barriers faced by women and HMPO and programming works to address barriers to ensure their active participation in activities. The project will also get support from NGOs that help men on masculinity and addressing SGVB which will include targeted outreach to mobilize their participation in household and parental responsibilities to ease women's responsibilities and sensitization on the importance of fair access and control of resources, including income. The engagement of community leaders and women and men community influencers to further promote equal and fair control of resources between men and women will further mitigate the risk of men taking over control of incomes earned and targeted for women. Lastly, the project will clearly explain participant selection, intended benefits and how participants can help monitor benefits and risks and reduce risks if necessary, including reducing the risk of men taking over control of incomes targeted for women. The program will have a safe, accessible Grievance Redress Mechanism (GRM) for all project stakeholders, particularly women, to access to pursue redress as needed. The project will also collaborate with existing Isange One Stop Centers in the project area in GBV prevention and immediate assistance to victims. All project staff will be trained and project beneficiaries informed on the purpose and use of the GRM, including reporting gender-based violence. The project will ensure confidentiality of the information and identity of the complainant. The grievance process will include options for victims to remain anonymous which will add a layer of protection for individuals who may be subject to gender-based violence. Throughout project implementation, project staff will disseminate information to beneficiaries on how they can use the GRM and solicit feedback on how the mechanism can be improved if needed.

b. **Provide access to skills building, leadership and programmatic inclusion.** Because of existing gender norms, there is much to do to build technical skills of rural women to be able to participate in the project. In addition to that, finding women leaders and building their leadership skills to be at the forefront of rural gender equity is crucial for addressing the root causes of women's exclusion in land use planning and decision-making. This project however will not just focus on rural women. Ensuring

that women play a strong role at all levels of climate programming in the country is important. Thus developing professional networks and internships to provide opportunities to strengthen leadership and technical skills will be incorporated into the program.

c. **Facilitate women's financial autonomy.** In order to support rural women to capture financial benefits of this project, this project will incorporate three activities: financial skills building, support in joining cooperatives; and facilitating access to entrepreneurship through financial/entrepreneurship training, access to village savings/loans and small grants, and mentoring. The project will ensure all trainings are gender-responsive and inclusive. Gender considerations will be integrated into training curriculum, content and delivery methods. Flexible training schedules and convenient venues will be selected to accommodate women's needs. Staff in charge of gender aspects will ensure gender mainstreaming and ensure the project addresses barriers to women's participation. Training will also use participatory approaches to create an inclusive learning environment and incorporate components on women's empowerment and leadership development, promoting gender equality and active participation. By facilitating independent access to financial services, women's income can be managed independently until shifting norms and household situations reach gender equity for households in adapting to climate change. Within the program, microenterprise, access to finance, beekeeping, and agroforestry/cookstove value chain support were chosen as key activities because they are feasible livelihoods within the target rural households and help strengthen resiliency of the rural economy. Further information on women in the value chains targeted by the project, including gender characteristics, composition of workforce and ownership etc. is detailed in the project Feasibility Study where possible and will be refined during project implementation.

1.21 Youth Analysis and Key Issues

Rwanda is a young country; 82% of the population is under the age of 40 and 51.5% are 19 or under (NISR, EICV5 2018, see Table 5). The sex ratio for youth, classified as people from 14-35 years old, is 92 males for every 100 females. The literacy rate for youth is quite high; averaging 91% in urban areas and 82% in rural areas (ibid). Numeracy rates are linked to wealth, with poorer families having lower capacity (72.4% vs 92.7%). Larger urban-rural discrepancies are found in computer literacy (28.5% vs 7%) and mobile phone ownership (70% vs 41%).

Table 5: Youth population in Rwanda by district. (Source: NISR EICV5 2018)

Youth Population in Rwanda (Source: NISR EICV5)			
District	Youth	Non-Youth	Total Population (000s)
Kigali City	36.5	63.5	1,631
South	24.3	75.7	2,739
West	25.1	74.9	2,685
North	26.8	73.2	1,841
East	24.6	75.4	2,998
All Rwanda	26.6	73.4	11,893

Table 6. Youth Usual Economic Activity by Sex and Province. Source: (NISR EICV5 2018).

Activity Type	Youth Usual Economic Activity by Sex and Province (Data source: EICV5)												
	Kigali City		Southern Province		Western Province		Northern Province		Eastern Province		Total		
	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	All
Worker	70.4	66.5	76	78	78.1	81.8	78.2	81.2	80.5	79.8	76.8	77.6	77.2
Seeking work	5.5	11.7	1	2.3	1.5	1.5	0.9	1.9	0.7	0.9	1.9	3.5	2.7
Student	21.7	15.9	18	14.7	16.8	12.1	17.5	14.6	15.2	14.9	17.7	14.4	16
Domestic	1.8	4.1	2.6	3.5	1.5	2.4	1.2	1.4	1.9	3	1.8	2.9	2.4
Disability	0.3	1	1.4	0.8	0.9	1.3	1.1	0.6	1.2	1.1	1	1	1
Other	0.4	0.7	1	0.7	1.2	0.8	1.2	0.3	0.5	0.3	0.8	0.6	0.7
Total	100	100	100	100	100	100	100	100	100	100	100	100	100
Usually	75.9	78.2	77.1	80.3	79.6	83.3	79.1	83	81.2	80.6	78.6	81.1	79.9
Inactive	24.1	21.8	22.9	19.7	20.4	16.6	20.9	17	18.8	19.4	21.4	18.9	20.1
Total	100	100	100	100	100	100	100	100	100	100	100	100	100

Employment figures for rural youth indicate that 54.7% are engaged in agriculture, and 32.9% in elementary activities. The rest are in services (7.8%), and very few in managerial, professional etc jobs. According to a recent analysis, youth “envision self-sufficiency achieved through economic independence and the ability to provide for and support their families, and they see this path as most available to them through self-employment (USAID 2019)”. The Youth Assessment highlighted key priority areas that youth identified for achieving their goals:

- Livelihoods and skills development for productive self-employment activities
- Education and skills for entering into the formal work sector
- Reproductive autonomy and health to avoid early pregnancy which is a key hurdle towards economic self-sufficiency

Poverty amongst youth has remained stable between EICV4 and EICV5, with about 30% of all youth below the poverty level, and 11.3% in extreme poverty. There are substantial gender differences, with young women increasingly vulnerable as they age (see Figures 7 and 8 with data from NISR’s EICV 5, 2018). Thus, programming ensuring attention on both young men and young women is vital for rural climate resiliency.

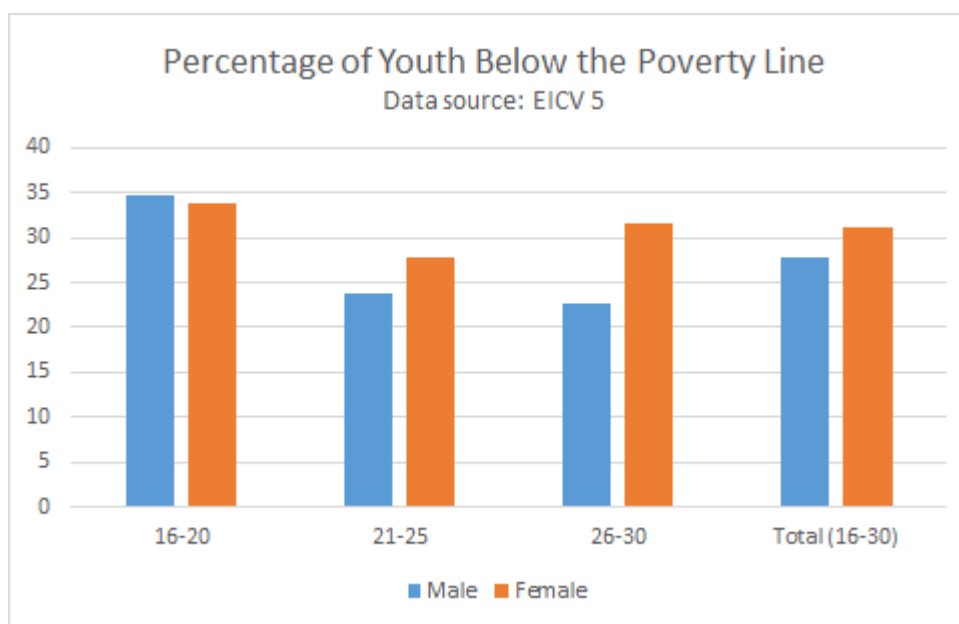


Figure 8: Percentage of youth below the poverty line by age (Source: NISR EICV5 2018)

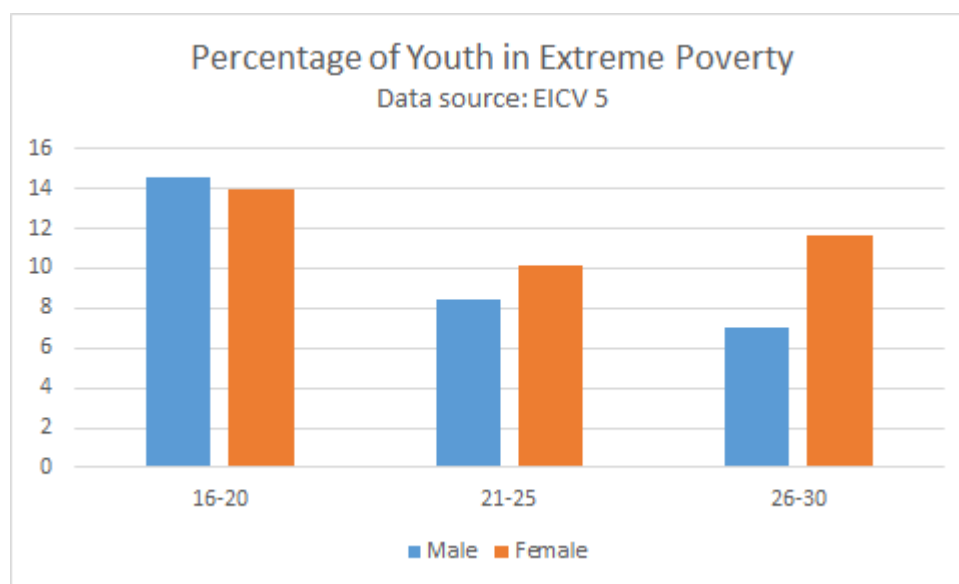


Figure 9: Percentage of youth in extreme poverty line in Rwanda by age. (Source: NISR EICV5 2018)

Translating this information into a climate adaptation program that supports young men and women in the CND, three key topics influenced the action plan:

- a. **Diversified opportunities.** With over 51.9% of the Rwandan population under the age of
- b. 15 and 18% between the ages of 15-24, youth make up a significant percentage of the population. There is already a scarcity of arable land, with many households owning less than what is needed for their subsistence. With the Rwandan population expected to increase from 12 million in 2019 to 20 million in 2042. The future for the upcoming generation of rural youth is worrisome--agricultural

lands will continue to grow scarcer and demand for resources and land increase. The youth interviewed are very cognizant of these constraints, and many are keen to diversify their livelihoods yet they are most interested in short-term gains. This GCF program intends to seek out vulnerable youth in critical areas of the CND to support entrepreneurship and diversified opportunities to reduce pressure on highly marginal lands.

c. **Access to skills building.** Many youth attend schools, but training in practical agroforestry, financial management, and forestry is lacking. This program intends to target youth for diversifying their skill sets and strengthening their resilience to adapt to climate change.

d. **Youth leadership and Programmatic Inclusion.** Many youths voiced their concerns over not being included in decision-making processes and programming, resulting in key stakeholders not being engaged in civil society. In order to engage the new generation of rural farmers and consumers, this project will identify and work with youth leaders in the CND to use SBCC to engage youth to benefit and support long-term tree initiatives. With youth involved in decision-making processes, innovative ideas to involve youth with best practice on government lands and to get youth to appropriate long-term thinking through tree planting as future investments.

1.22 Internal program considerations

In addition to programmatic activities with women and youth, this program will also have an internal focus to ensure best practices. These include:

a. **Avoid perpetuation of gender inequalities** and women's exclusion in the agro/forestry sector. SBCC will be used heavily to shift norms towards inclusivity and skills building to facilitate women and youth to join the sectors. In addition, women need to be engaged in the decision-making bodies of the project itself, including a Steering Committee, which will be at least 50% women, and the Project Management Unit with targets to achieve 50% women by the end of the program as mentoring and training increase capacity. The Gender Monitoring Office will be a part of the Steering Committee and along with the The National Women's Council and The Ministry of Gender and Family Promotion were consulted during proposal development and will continue to be involved at project implementation stage at the national and local level. There will be comprehensive consultation and feedback processes with these stakeholders through meetings and workshops to ensure perpetuating gender inequalities is avoided.

b. **Build partnerships.** This GCF project provides a unique opportunity to engage civil society at national and local levels. The team will work with men's, women's, and youth organizations to build their capacity and facilitate their engagement with project beneficiaries and local government actors.

c. **Gender and Diversity training.** Training will be an important component of a large, 5-year program. Building online training modules that can be viewed across the CND, and using new platforms like WhatsApp to communicate with urban and rural project staff will ensure that static, one-off training is avoided.

d. **Staffing.** Diversity and inclusion are important for the project, and hiring practices at all levels of the program will be reflected in hiring policies. The program management unit will hire a full-time dedicated gender expert and a full-time dedicated youth focal point (both AE staff) and associated costs are included in the project budget. These two staff will participate in all management meetings to support inclusion goals across the project, and will lead and oversee the implementation of the Gender, Youth, and Social Inclusion Action Plan. Another key role of these two positions will be to engage civil society

partners, particularly women's, men's, and youth groups to build capacity and increase the sustainability of achievements during this five-year project. Along with these positions, RFA staff will include a Community Mobilization Specialist and a Community Engagement Specialist who will be in charge of and support project gender mainstreaming. Their responsibilities include reviewing site specific barriers faced by women to participate in project activities, ensuring these barriers are addressed in order to support full participation of women, ensuring gender equity in all project activities and where needed help women access the Grievance Redress Mechanisms to pursue redress and connect women with Isange One Stop Centers in the project area to prevent gender-based violence and provide immediate assistance to victims.

e. **M&E.** Many stakeholder meetings highlighted the lack of community engagement as a key reason for project failures in the past. This project will incorporate suggestions from our stakeholder engagement meetings to ensure best practice is followed. This includes not only deeper local government engagement but also using a rights-based approach to ensure that communities are at the heart of planning our interventions. Targets for youth and gender are incorporated into the Gender and Youth Action Plan below and are set to ensure women and youth are equally targeted where possible and encouraged to actively participate. Exceptions include situations where group composition, such as government agents or community leaders, are outside the project's control or the activity solely targets men, such as masculinity sessions held with men to improve gender equity in program participation and benefits or assisted natural regeneration activities inside the park far from villages that will take male participants several weeks away from their homes. Overall targets for key activities (i.e., G4. 4.1.1.5, G4. 4.1.1.6) have not been set as the target will depend on the number of villages/sites where project activities take place, however disaggregated targets per village/site have been set. A project baseline will be conducted and baselines for all targets will be determined at project onset.

1.23 Gender and Youth Action Plan

Component 1: Mainstreaming Climate Adaptation into Integrated Land Use Planning						
Impact 1: Climate adaptation programming is successfully implemented because it is fundamentally informed by rural women and youth stakeholders <i>Indicator: Equal participation by gender groups in contributing to expected outcomes</i>						
Gender/Youth Outcome 1: Climate adaptation programs in LUPs reflect and address the priorities of women and youth <i>Indicator: % of men, women and youth participating in LUP processes who feel LUPs incorporate their needs as they relate to climate programming</i>						
Activity	Gender/Youth Outputs	Indicators	Targets	Timeline	Responsible	Costs
G1. 1.1.1.7 Conduct climate literacy seminars for local government and civil society organizations, aimed at increasing women and youth participation	Local institutions are trained to build capacity of rural people, specifically women and youth, to participate in climate adaptation processes	# of CSOs and government agencies trained	60 people (50% male, 50% female) from CSOs/Local institutions trained in Y1 and Y3 to increase participation of women/ youth/ marginalized in stakeholder consultations & activities	Y1,3	PMU/RFA	\$10,187
G1. 1.1.1.8 Provide financial and logistical support to trained organizations in grassroots mobilization to increase women and youth participation in climate adaptation planning	Trained institutions hold meetings to train women, youth and marginalized people on forests and mobilize them to participate in planning processes	# of meetings held by trained organizations with local communities on climate adaptation programming	At least 10 grassroots meetings per district are held in Y1 and Y3 in the CND	Y1,3	PMU/RFA/ Local CSOs (TBD)	\$12,262

G1. 1.1.2.4 Develop and rollout a series of trainings (virtual sessions and online modules) on gender sensitivity and mainstreaming women and youth into planning	Program Staff and government officials understand gender barriers, understand how to mainstream gender equality principles into planning and are strong facilitators to support women, youth and marginalized participation in meetings	# of people trained (disaggregated by gender) # project staff trained in social inclusion	- 65 government agents trained per year. - 100% project staff trained	Y1-5	PMU + Local Women's NGO (TBD)	\$40,523
Component 2: Forest Landscape Management and Restoration						
Gender/Youth Impact 2: Sustainable gains in restoration and landscape management are made because women and youth are enabled and motivated to adopt best practice <i>Impact Indicator: Targeted women and youth report an improved enabling environment for their participation in forest management and restoration</i>						
Gender/Youth Outcome 2: Women, youth and marginalized households have improved capacity to engage in forest management and restoration <i>Outcome Indicator: Equal participation by gender groups in forest landscape management and restoration</i>						
G2. 2.2.2.15 & 3.1.1.11 Develop and implement strategy (including social marketing) to ensure that women and youth benefit from forest restoration projects	Women and youth have the capacity, motivation, and opportunity to participate and benefit equitably in restoration programs. Programs will actively seek out direct participation of and benefits for women and youth through paid labor and direct involvement.	% Women beneficiaries % Youth beneficiaries	50% of beneficiaries are women 50% of beneficiaries are youth ²	Y1 Strategy Y2-5 Implementation	PMU, Women/Youth NGOs (TBD)	\$37,735
Component 3: Enhancing climate adaptation through resilient livelihoods						

² The total number of beneficiaries will be determined at project onset

Gender/Youth Impact 3: Women, marginalized and youth are empowered to actively participate in green jobs
Impact indicator: Target groups perception of economic empowerment

Gender/Youth Outcome 3: Women, youth and marginalized households have improved capacity to engage in green jobs
Outcome Indicator: Number of new women, youth and marginalized households that have engaged in green jobs

G3. 3.2.1.2 Develop and implement strategies to ensure that women,youth, and other marginalized groups benefit from rural livelihoods and financial services	Women and youth have the capacity, motivation, and opportunity to participate and benefit equitably in rural livelihoods and financial services	% Women beneficiaries % Youth beneficiaries	50% of beneficiaries are women 50% of beneficiaries are youth ³	Y1 & Y2	RFA & CORDAID	\$53,053
G3. 3.2.3.2 Organize Microenterprise training and develop and implement green grants programs targeting landless youth and poor, as well as women who are unable to access direct benefits from land-based project activities.	Landless poor, youth, and women have enterprise skills and grants to engage in green value chains	# of enterprises formed, beneficiaries disaggregate d by gender and social class	5,600 beneficiaries of which at least 50% are women 500 off-farm jobs created in production quality control	Y2,3 & 4	RFA & CORDAID	\$66,398
G3. 3.3.2.2 Provide organizational/technical/financial capacity building to women, youth, and CSOs (such as cooperatives or VSLA) with focus on financial literacy and provide long-term coaching to beneficiaries	Women and youth groups have the organizational, technical and financial skills to receive support from project	# CSOs/VSLAs trained # beneficiaries receiving coaching	300 CSOs/VSLAs 6,900 of which 40% youth 50% women	Y2,3 & 4	RFA & CORDAID	\$30,040

³ The total number of beneficiaries will be determined at project onset

G3.3.3.2.3. Support women, youth, other marginalized groups strengthen savings and loan groups and register for MFI accounts	Men, women and youth have access to thriving financial services	# people with new bank/MFI accounts	50% women	Y2,3,4	RFA & CORDAID	\$13,713
Component 4: Monitoring and Evaluation						
Gender/Youth Impact 4: Women and youth benefit from an adaptively managed program <i>Impact indicator: Reported attitudes, confidence, and a sense of empowerment towards engaging in climate programming</i>						
Gender/Youth Outcome 4: Disaggregated and specialized monitoring generates data showing whether and how women, youth and marginalized people are benefiting from adaptive management systems <i>Outcome indicator: Gender and youth disaggregated participation data</i>						
G4. 4.1.1.4: Develop an overarching gender and youth program unit within the GCF Project to ensure the project uses adaptive management to ensure strong inclusion of women, youth, and marginalized households such that equitable benefits are shared across the communities.	Gender/Youth program unit provides oversight and support to ensure gender/ youth targets are met	N/A	N/A	Y1-5	PMU	\$3,380
G4. 4.1.1.5: With the local government, project partners, and Youth CSO leaders, identify key male and female youth leaders in each project village to be project champions	Key youth leaders are identified and engaged	Youth leaders identified	2 Youth leaders (50% male, 50% female) per targeted village. ⁴	Y1	PMU, Youth NGO (TBD)	\$88,683

⁴ The number of targeted villages will be determined at project onset and will determine the total number of youth leaders to be reached. 50% male and 50% female youth leaders will be targeted per village.

G4. 4.1.1.6: With local government, project partners and women's civil society, identify key women leaders amongst project beneficiaries to be project champions	Key women leaders are identified and engaged	Women leaders identified	2 women leaders per village ⁵	Y1	PMU, Women's NGO (TBD)	\$87,903
G4. 4.1.1.7: Engage Youth & Women's NGOs to build leadership and coaching skills of identified project champions and facilitate their project involvement in order to advance project goals of greater participation amongst marginalized groups amid greater community-wide support for their participation	Leaders are equipped with skills and financial resources to build empowerment and facilitate women and youth engagement in the project	# of women and youth trained in coaching and leadership skills	200 (50% women, 25% male youth leaders, 25% female youth leaders) women and youth leaders trained as trainers	Y1-2	PMU	\$6,660
G.4. 4.1.1.8: Overarching social behavior change communications strategy and materials are developed to overcome barriers and enhance motivations for women, youth and marginalized people to participate in program activities	Gender sensitive and responsive communications programming supports new norms for social inclusion in project activities	# programs/ materials delivered in communities	At least 10 materials developed and distributed annually	Y1-5	PMU	\$97,552

⁵ The number of targeted villages will be determined at project onset and will determine the total number of women leaders to be reached. 100% of women leaders will be female

G4. 4.1.1.9: Informational campaigns on equity (land rights, access to resources, and decision-making) and programmatic opportunities to showcase the benefits of everyone (women, young men/women, marginalized) participating in programming to improve lives	Gender sensitive and responsive campaigns and informational material on inclusion of different social categories is developed and disseminated	# of people reached by campaigns	At least 200,000 (50% female, 50% male) people reached by campaigns over 5 years	Y1-5	PMU	\$17,030
G4. 4.1.1.10 Engage men's NGO to roll out programming for men as allies in women's economic empowerment and sexual/gender based violence reduction programming	Men in target areas participate in masculinity sessions to improve gender equity in program participation and benefits	# men participating in training sessions	At least 200 men participate in masculinity sessions each year from Y2-5	Y2-5	PMU Men's NGO	\$115,574
G4. 4.1.2.2 Identify and train men and women (adults and youth) to implement key M/E tools in their communities	Women and youth are active participants in M/E at central, sectoral, and community level	# Women and Youth trained on M/E tools	At least 5 women, 5 men and 5 youth (15 people total) per district are trained ⁶	Y1-5	PMU	\$3,380
G4. 4.1.3.2 Engage youth leaders in designing and monitoring targets for youth engagement, participation, and outcomes	Youth targets are identified and incorporated into M/E system	N/A	N/A	Y1-5	PMU	\$ 16,533
G4. 4.1.3.3 Engage women leaders in designing and monitoring targets for women engagement, participation, and outcomes	Women targets are identified and incorporated into M/E system	N/A	N/A	Y1-5	PMU	\$ 16,533

⁶ The number of targeted districts will be determined at project onset and will determine the total number of women, men and youth to be trained. The target for the number of individuals trained overall is 50% female, 50% male.

G4 .4.1.3.4: Monitor program using Gender Checklist to ensure gender is integrated in programming, HR, policy, and activity performance	Project has a strong gender monitoring program	- Project gender checklist developed - Annual checklist scores	Project scores on gender checklist increases yearly	Y1-5	PMU	\$ 16,533
G4. 4.2.2.1 Project partners (government and NGOs) are trained on gender-sensitive approaches	Project staff have skills to monitor inclusion of youth and women into programming	# staff trained	100% of M/E focal points from participating organizations/entities are trained	Y1-5	PMU	\$25,780
Project Gender and Youth staff time is incorporated into each activity cost						
Total Estimated Cost for Implementation of GAP: \$759,452						



**GREEN
CLIMATE
FUND**

Meeting of the Board
23 – 26 October 2023
Tbilisi, Georgia
Provisional agenda item 12

GCF/B.37/02/Add.08

2 October 2023

Consideration of funding proposals – Addendum VIII

Funding proposal package for FP217

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Funding Proposal

Project title:	Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration
Country(ies):	Rwanda
Accredited Entity:	Ministry of Environment
Date of first submission:	2022/02/25
Date of current submission	2023/09/11
Version number	V.6



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(a) PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Project	A.2. Public or private sector	Public	
A.3. Request for Proposals (RFP)	<u>Not applicable</u>			
A.4. Result area(s)			GCF contribution	Co-financiers' contribution
	Mitigation total		8 %	0 %
	☐ Energy generation and access		0 %	0 %
	☐ Low-emission transport		0 %	0 %
	☐ Buildings, cities, industries and appliances		0 %	0 %
	☒ Forestry and land use		8 %	0 %
	Adaptation total		92 %	100 %
	☒ Most vulnerable people and communities		13 %	49 %
	☒ Health and well-being, and food and water security		2 %	0 %
	☐ Infrastructure and built environment		0 %	0 %
☒ Ecosystems and ecosystem services		77 %	51 %	
A.5. Expected mitigation outcome	1,084,291tCO ₂ eq	A.6. Expected adaptation outcome	Direct Beneficiaries: 1,254,242 (654,404 females)	Indirect Beneficiaries: 9,260,745 (4,769,284 females)
			30.9 % of the CND population	69.9% of Rwanda's population
A.7. Total financing (GCF + co-finance)	50,000,000 USD	A.9. Project size	Small (Up to USD 50 million)	
A.8. Total GCF funding requested	39,056,421 USD			
A.10. Financial instrument(s) requested for the GCF funding	☒ Grant 39,056,421 USD ☐ Loan - ☐ Guarantee -		☐ Equity : ☐ Results-based payment :	
A.11. Implementation period	5 years	A.12. Total lifespan	20 years	
A.13. Expected date of AE internal approval	<u>TBD</u>	A.14. ESS category	B	
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐ M	
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐	
A.19. Complementarity and coherence	Yes ☒ No ☐			

A.20. Executing Entity information	Rwanda Forestry Agency (RFA) Country of Registration: Rwanda Ownership Type: Government agency.
A.21. Executive summary	
Rwanda's Congo-Nile Divide (CND), an area of 444,600 hectares, is a high-elevation landscape of remnant natural forest reserves embedded in a mosaic of intensively managed smallholder farms and commercial plantations. Extending along a north-south ridge with an altitudinal gradient from 1,900m to 4,507m, the CND separates the drainage basins of the Congo and Nile Rivers. It is part of the Albertine Rift, which is one of the most biologically rich regions in Africa and "one of the most important regions for conservation" on the continent¹, containing more than half of Africa's birds and 40% of its mammals. At its highest elevations, the region is crowned by the Volcanoes, Gishwati-Mukura, and Nyungwe National Parks, containing Rwanda's only remaining montane forests, one of the most biodiverse ecosystems on the African continent, with high levels of endemism (species found nowhere else on Earth) and dozens of threatened species, including populations of mountain gorillas and chimpanzees. Nyungwe has been recognized as one of the six highest priority protected areas in the Albertine Rift due to its high endemism and globally threatened species (Plumptre et al 2016). Though covering barely 5% of the country, this archipelago of protected forests is of critical importance for: • Its globally recognized high biodiversity values^{2,3, 4, 5}; • Its role in Rwanda's acclaimed ecotourism offerings serving as the primary source of Rwanda's foreign revenue and as a major economic development driver^{6, 7}; • The ecosystem services provided for surrounding subsistence farmlands, as well as for commercial tea and coffee plantations offering employment opportunities for women while providing the 2nd and 3rd highest sources of foreign revenue^{8,9}; • The regulation of water capture and release to growing urban centers on the CND periphery¹⁰, the nation's extensive river network, and downstream hydroelectric operations¹¹; and • The sequestration and storage of nearly 60% of Rwanda's national carbon stocks¹². These values and services are already compromised by a combination of anthropogenic and climatic factors; they will be much more severely threatened by the potential negative feedback loops associated with projected climate change impacts. The Project will promote an ecosystem-based approach to forest protection and restoration and will enhance the extent and resilience of forest ecosystems and species, including functionally linking currently isolated forest fragments. Through these approaches, the Project will protect, restore and rehabilitate forest ecosystems, enhancing biodiversity and reducing key threats, while	

1 Plumptre, A.J., et al. 2007. The biodiversity of the Albertine Rift. *Biological Conservation* 134:178-194.

2 Plumptre, et al. 2016. Conservation Action Plan for the Albertine Rift. Wildlife Conservation Society

3 UN/CBD. 2020. Rwanda 6th National Report to the Convention on Biological Diversity

4 2012. Eastern Afromontane Biodiversity Hotspot. Critical Ecosystems Partnership Fund.

5 2023. Nyungwe National Park. BirdLife International. www.birdlife.org.

6 RDB. 2021. Rwanda Development Board Annual Report

7 2010. *In the Kingdom of Gorillas*. Weber & Vedder.

8 Andrew, G., Masozera, M. 2010. *Payment for Ecosystem Services and Poverty Reduction in Rwanda*. Journal of Sustainable Development in Africa (V12, No.3)

9 RDB. 2021. Annual Report.

10 Karamage, F., et al., 2017. *Modeling Rainfall-Runoff Response to Land Use and Land Cover Change in Rwanda (1990–2016)*. *Water* 2017, 9(2), 14.

11 Rwanda Environment Management Authority (REMA), 2009. Rwanda State of Environment and Outlook Report, Chapter 8: Energy Resources.

12 Mugabowindekwe, M., Brandt, M., Chave, J. et al. Nation-wide mapping of tree-level aboveground carbon stocks in Rwanda. *Nat. Clim. Chang.* 13, 91–97 (2023). <https://doi.org/10.1038/s41558-022-01544-w>

increasing the provision of critical ecosystem services to strengthen adaptation of vulnerable communities.¹³
14

The CND landscape supports nearly one-third of Rwanda's population on barely one-fifth of its land base. Average population density is 736 people/km², with some districts now exceeding 1,000 people/km²¹⁵. The highest rates of rural poverty in the country are reported from the CND, including its most vulnerable subsistence farmers. While high rainfall has historically permitted generally sufficient food production, the region's rugged topography requires farming on extremely steep slopes. With more frequent extreme rainfall events in recent years, this agricultural landscape is recognized as the region of Rwanda most vulnerable to floods and landslides¹⁶. Between 2013-2016 heavy rains generated landslides that killed at least 174 people and destroyed the homes of more than 5,000. Many of these landslides caused flooding of highly productive raised-bed cultivation systems in the CND's many valleys and surrounding bottomlands and many hectares of crops being washed away annually¹⁷. An estimated 1.5 million tons of fertile soil are lost annually to erosion¹⁸.

CND forests are essential to the region's climate resilience. Since independence, the total natural forest cover has declined 40%, from 219,000 ha to 130,700 ha, mostly from episodic planned excisions as well as uncontrolled incursions. The loss around the Gishwati and Mukura core forest area has been most extreme. Of the original 32,098 ha in the 1980s, now only 3,460 ha remains as natural forest: almost a 90% loss (see Annex 2). In comparison, the Volcanoes and Nyungwe boundaries have been stable for decades. The recent creation of the Gishwati-Mukura National Park is an important step toward halting and partly restoring this forest, particularly for its land stabilization and other ecosystem services within the CND's most vulnerable lands and dependent communities.

IPCC models predict rising temperatures and increasingly intense precipitation events for the CND through the rest of this century. The rainfall implications are especially threatening for smallholders farming on steep, heavily deforested slopes, where they are most exposed to landslides, leading to loss of crops, homes, and lives. Yet their adaptive capacity is low because crop yields are vulnerable to rainfall variability and because high population density severely limits their options for relocation in the event of disaster. This is particularly true for women who, until very recently, had fewer legal land rights than men. The expected 2°C increase in temperature would further result in a more than 470m rise in vegetation zone elevations, impacting traditional crop selection and management practices for future farmers. An increase of this scale would also dramatically alter the optimal ranges for tea and coffee production, introducing potential competition between these important export crops and alternative land uses.

This Project is designed to sustain and expand critical forest ecosystem functions in the CND, and ultimately enhance the resilience of the vulnerable communities who depend on them. This will be done by strengthening government capacity for spatial planning, resulting in institutional and regulatory policies to promote and integrate climate adaptation strategies, based on the latest science, through coordinated planning at the national and district levels. It will also protect, restore and expand natural forest in and around

13 2003. Interlinkages Between Biological Diversity and Climate Change. CBD Technical Series Number 10.

14 2022. Ecosystems and ecosystem services Sectoral Guide. Green Climate Fund.

15 National Institute of Statistics of Rwanda (NISR); The Fifth Rwanda Population and Housing Census, Main Indicators Report, February 2023

16 https://www.gfdrr.org/sites/default/files/publication/National_Risk_Atlas_of_Rwanda_electronic_version_0.pdf

17 Nsengiyumva JB, et al., 2018 Landslide Susceptibility Assessment Using Spatial Multi-Criteria Evaluation Model in Rwanda. International Journal of Environmental Research and Public Health. 15(2):243.
<https://doi.org/10.3390/ijerph15020243>

18 Bizimana, H.; Sönmez, O. 2015 *Landslide Occurrences in the Hilly Areas of Rwanda, Their Causes and Protection Measures*. Disaster Sci. Eng. 2015, 1, 1–7

the core national parks; consolidate and expand some smaller but critical "stepping stones"¹⁹ of remaining natural habitat between these national parks (mainly small forest patches and wetlands/ riparian areas); improve the landscape linkage function of key high altitude landscape linkages between the protected areas, while at the same time securing ecosystem services through afforestation and improved native species mix for plantation forests on steep slopes; silvopastoral interventions on cultivated pasture; and agroforestry interventions in regenerative agricultural areas. Complementary activities will address key drivers of deforestation while creating alternative livelihood pathways for youth and women through ecotourism and income-generating, climate-resilient crops, as well as strengthening supportive financial services.

Project interventions will include 10,000 ha of land (including both forest and silvopastoral sites) under improved, climate-resilient management; and improve agroforestry practices on 3,346 ha of on-farm plantations, directly benefiting 1,254,242 people (654,404 women) in CND with more climate resilient livelihoods from reduced exposure to landslides, floods and soil erosion and more knowledge on climate risks, value of forests and forests ecosystems, climate adaptation options, access to indigenous and agroforestry quality materials and improved ecosystem services. The project will also indirectly benefit 9,260,745 people nationally mainly from the education and outreach program on climate change and gender issues, climate risks, value of forests and forest ecosystems, climate adaptation options, and will create approximately 24,212 job opportunities in CND forest-dependent communities. The total mitigation impact from reduced deforestation, forest restoration introduction and adoption of improved cookstoves is estimated to be 1,084,291t CO₂-eq by 2043.

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B.1 (a). Climate context

This section draws from the Climate Change Vulnerability Assessment presented in Annex 2.1, where more comprehensive elaboration can be found. Discussion on climate predictions is based primarily upon multi-model means from the Coupled Model Intercomparison Project's sixth phase (CMIP6)²⁰, used in the Sixth IPCC Assessment Report. Results are shown for paired Shared Socioeconomic Pathways (SSP) and Representative Concentration Pathway (RCP) emission scenarios; for some variables emphasis is placed on a "pessimistic" high SSP-high RCP scenario (SSP5-8.5), for greater distinction between present day and future conditions as a consequence of anthropogenic climate change. Specialized modeling performed at ultra-high resolution for the East African Great Lake Environments project (EAGLE)²¹, and findings from scientific literature are utilized as well. Key findings are outlined in Table 1, and each climate component theme is discussed in the section on projected climate change impacts below.

¹⁹ Bagchi, R., et al. 2018. Forecasting potential routes for movement of endemic birds among important sites for biodiversity in the Albertine Rift under projected climate change. *Ecography*. 41: 401-413.

²⁰ Coupled Model Intercomparison Project Phase 6. [Link](#)

²¹ Grim, J. A., Pinto, J. A., Jensen, A. A., & Seimon, A., 2020: The East African Great Lake Environments (EAGLE) Climate Downscaling Dataset. [Link](#)

Table 1. Principal Climate Hazards of the Congo Nile Divide through Mid-century

Climate component	Character of change	Level of confidence	Near-term concern	Mid-century concern	Project Response
Temperature	Upward trend from global greenhouse gas emissions and land-use change	Very high for +2.6 °C net increase 1970-2040	Elevational range of biota including pests and pathogens increasingly out of balance	Major uphill displacements of biota including cultivars due to ~470 m rise in thermal conditions	Promotion of agroforestry practices: contribute to the conservation of soil moisture and its recharge through the infiltration of rainfall and runoff water, the creation of sheltered microclimatic conditions, and the inclusion into the soil of organic matter that contributes to moisture retention. Promotion of silvopastoral practices: Improve soil properties due to greater uptake of nutrients from deeper soil layers, enhanced availability of nutrients from leaf-litter and increased nitrogen input by N₂-fixing trees.⁸⁵ Moreover, silvopastoral systems enhance the resilience of the soil to degradation, nutrient loss, and climate change, while enhancing water holding and infiltration capacity of the soil which contributes to the regulation of the hydrological cycle by reducing runoff intensity^{22, 23}. Protective restoration measures at erosion-prone areas: Reduce exposure to soil erosion and floods of communities and their assets living in the road and river side. Restoration of degraded natural forests in protected areas Improve protected areas management to reduce potential human threats (forest encroachment, mining, fire). The project will support activities aimed at (1) promote alternative livelihoods that are resilient to climate change (2) decrease of demand and (3) increase of wood supply capacity. Dissemination of Improved Cook Stoves The project will support the development of an integrated
Annual precipitation	Increasing totals	Moderate, but considerable model variation	Low concern, with natural interannual variability still dominant	Moderate concern, with majority of models showing upward trend	
Precipitation intensity	Increasing short-period rainfall and storm totals	Almost certain to occur	Severe landslide hazard already present and increasing, building flash flood potential	Extreme hazard: widespread and frequent landslides and flash floods	
Cloud base height	Rising cloud base and levels of fog immersion in forests	Likely ongoing, will continue with temperature increase	Already significant in deforested highlands with some loss of moisture provision	Of major concern to forest ecology in protected landscapes	
Drought/dry spells	Increasing intensity	Moderate	Some increase in vegetation desiccation potential, possibly enhancing fire risk	Sustained risk of increase in desiccation and fire risk, possibly offset by rainfall increases	Improve protected areas management to reduce potential human threats (forest encroachment, mining, fire). The project will support activities aimed at (1) promote alternative livelihoods that are resilient to climate change (2) decrease of demand and (3) increase of wood supply capacity. Dissemination of Improved Cook Stoves The project will support the development of an integrated
Wildfire	Increasing occurrence and extent in forested areas	Low	Relatively random occurrences, as in the recent past, that could be managed by improved vigilance	If rainfall increases do not occur, increased vegetation desiccation would intensify risk for widespread fire	
Climatic seasonality	Disappearance of mid-year dry season	Low, but explicitly shown in some models	No concern	Increasing convective storm occurrences in June-Aug may eliminate dry season	

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land use plan at landscape level that identifies and integrates the trade-offs necessary to improve delivery of ecosystem services, conserve natural resources and sustain economic growth.

• Climate baseline and current impacts

The climate of western Rwanda and its CND landscape features conditions highly supportive of tropical montane forests, rich biodiversity and human activities alike. Despite being located deep in the tropics just south of the equator, the elevated terrain of the CND landscape, mostly above 2,000 meters, ensures relatively cool temperatures through the year (**Figure 1**).

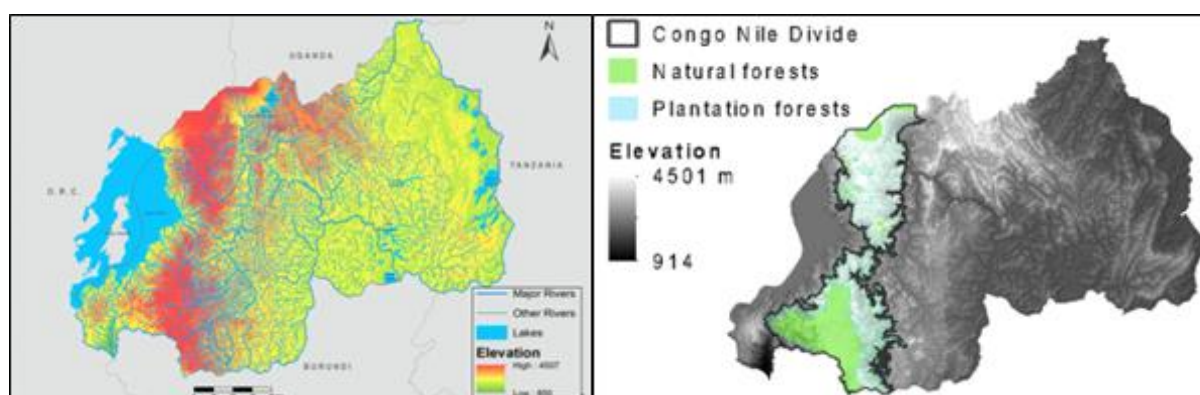


Figure 1 (left) Topographic map of Rwanda with riverine drainage channels overlaid. *Source:* Rwanda (2019)²⁴; **(right)** Congo Nile Divide, identified by the 1,900 m elevation contour and showing the presence of natural and plantation forests.

Rainfall is abundant for nine out of 12 months, and a reliable mid-year dry season adds dynamic character to phenological cycles, pollinator activity and agricultural practices (**Figure 2**). Prior to landscape conversion for agricultural purposes, continuous dense forest cover yielded augmentation of precipitation through fog interception in tree canopies, and near-daily cloud immersion sustained cloud forest ecosystems and species assemblages. The forests also maintained slope stability in steep terrain and yielded percolation and groundwater recharge rather than direct runoff of heavy precipitation. Such conditions are now constrained to within the CND's protected forests within national parks, covering just 30% of the CND landscape and just 5% of Rwanda's total land surface. These are Rwanda's only remaining montane forests yet represent

22 Ibrahim M, Guerra L, Casasola F, Neely N, 2010. Importance of silvopastoral systems for mitigation of climate change and harnessing of environmental benefits. In: Abberton M, Conant R, Batello C (Eds) Grassland carbon sequestration: management, policy and economics. Proceedings of the workshop on the role of grassland carbon sequestration in the mitigation of climate change. Integrated Crop Management, Vol. 11. FAO, Rome, Italy. <http://www.fao.org/docrep/013/i1880e/i1880e09.pdf>.

23 Jose S., 2009. Agroforestry for ecosystem services and environmental benefits: an overview. Agroforest Syst 76 (1):1–10.

24 Vital Signs, 2019: Rwanda Atlas. Environment, Agriculture and Livelihood Options. Vital Signs and WCS Rwanda. [Link](#).

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58.7% of national tree carbon stocks²⁵ and are imperiled by intense population pressures, with 30.7% of the country's population residing within the CND landscape districts.

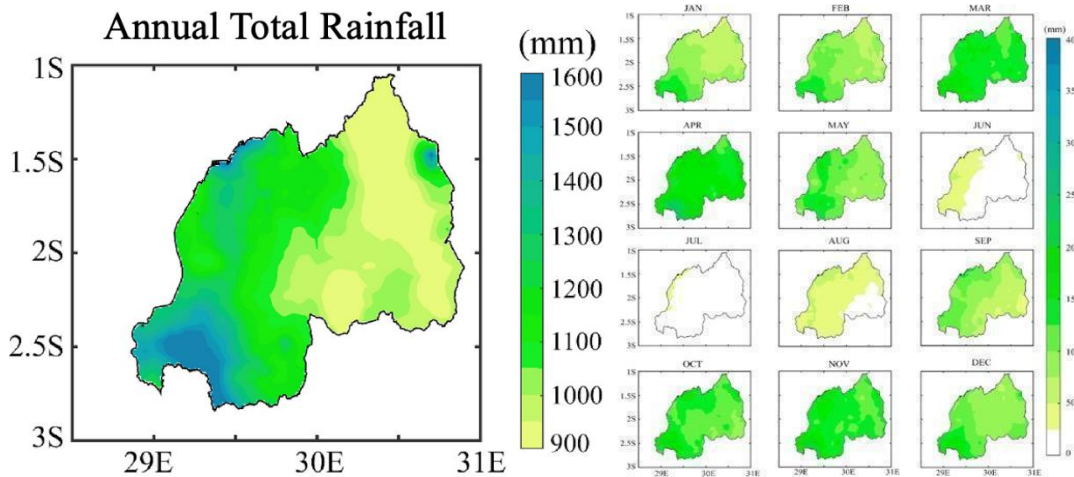


Figure 2. Annual (left) and monthly (right) rainfall (mm) climatology over Rwanda (period of record: 1981–2017), based on algorithms that calibrate satellite-based rainfall estimates with surface observations. Source: Jonah et al., 2021.²⁶

A reconstruction of temperatures for the past 140 years for Rwanda, based on quality controlled instrumental records and centered on Kigali, shows that global climate change already has strong local expression. Rapid temperature increase commenced in the 1970s, with a net gain of approximately +1.1 °C since 1970²⁷ (Figure 3).

²⁵ Mugabowindekwe, M., Brandt, M., Chave, J. et al. Nation-wide mapping of tree-level aboveground carbon stocks in Rwanda. *Nat. Clim. Chang.* 13, 91–97 (2023). <https://doi.org/10.1038/s41558-022-01544-w>

²⁶ Jonah, K., Wen, W., Shahid, S., Ali, M.A., Bilal, M., Habtemicheal, B.A., Iyakaremye, V., Qiu, Z., Almazroui, M., Wang, Y. and Joseph, S.N., 2021. Spatiotemporal variability of rainfall trends and influencing factors in Rwanda. *Journal of Atmospheric and Solar-Terrestrial Physics*, 219, p.105631.

²⁷ Berkeley Earth (website): Rwanda Country Profile. www.berkeleyearth.org

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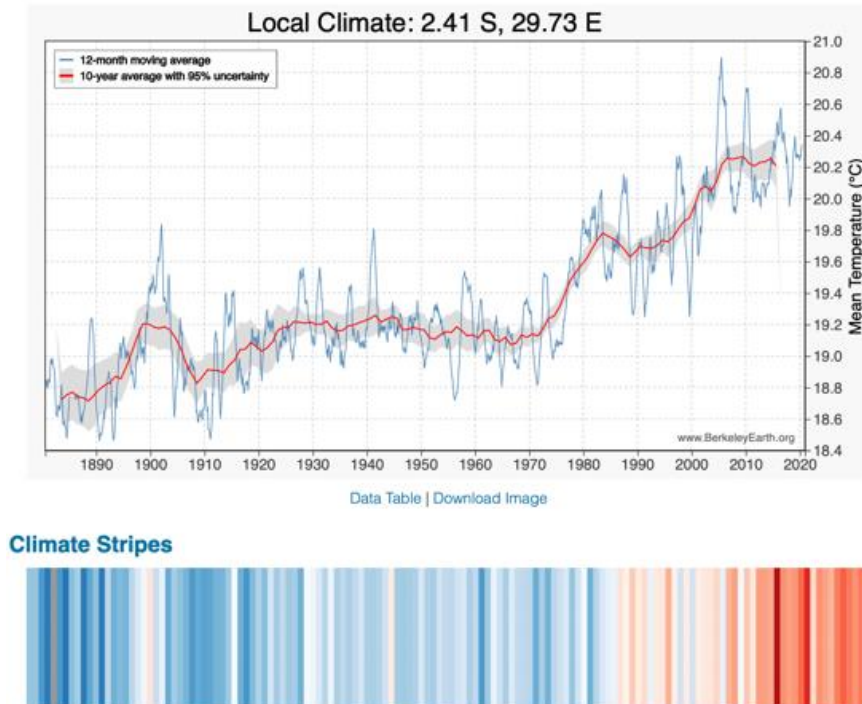


Figure 3. Berkeley Earth temperature reconstruction based on instrumental measurements for Kigali in central Rwanda. The “Climate Stripes” offers a visual display that effectively communicates both individual year temperature anomalies and the overall multi-decadal trend behavior. Source: Berkeley Earth Rwanda Country Profile. www.berkeleyearth.org

The regional warming of the climate over the past half-century coincided with widespread forest conversion to farmland along the CND. As a consequence, at fine spatial scales climatic conditions across the CND at present mimic the landscape mosaic, whereby only the protected forests will retain much of the past climatic character while converted landscapes are likely to be significantly warmer by day, have reduced cloud cover and higher cloud bases, and experience slightly reduced rainfall as well. Due to the increased moisture carrying capacity of warmer air (7% increase per °C of warming)²⁸, ongoing climatic warming promotes increasingly intense short-period rainfall, which acts synergistically with landscape conversion on steep slopes to cause quickflow runoff, yielding large increases in landslide occurrence and soil erosion more generally (**Figure 4**). With its high topographic relief, the CND is highly susceptible to soil erosion. **Throughout Rwanda, soil erosion has increased by 54% since 1990 with the bulk of the increase experienced and highest erosion levels over western Rwanda.**²⁹ Between January and June in 2018, landslides caused more than 200 mortalities, and in May 2020³⁰, 65 deaths from landslides and flash flooding occurred during a single night of storms focused on the eastern slopes of the CND.³¹

28 Trenberth, K.E., 2011. Changes in precipitation with climate change. *Climate Research*, 47, pp.123-138.

29 National Institute of Statistics of Rwanda, 2019: Rwanda Natural Capital Accounts-Ecosystems. p. 23

30 Reuters 2018. <https://www.reuters.com/article/us-rwanda-floods-idUSKBN1I81I>

31 Associated Press 2020. <https://apnews.com/article/b7658b76a403fd9bc8cba249f9d484fd>

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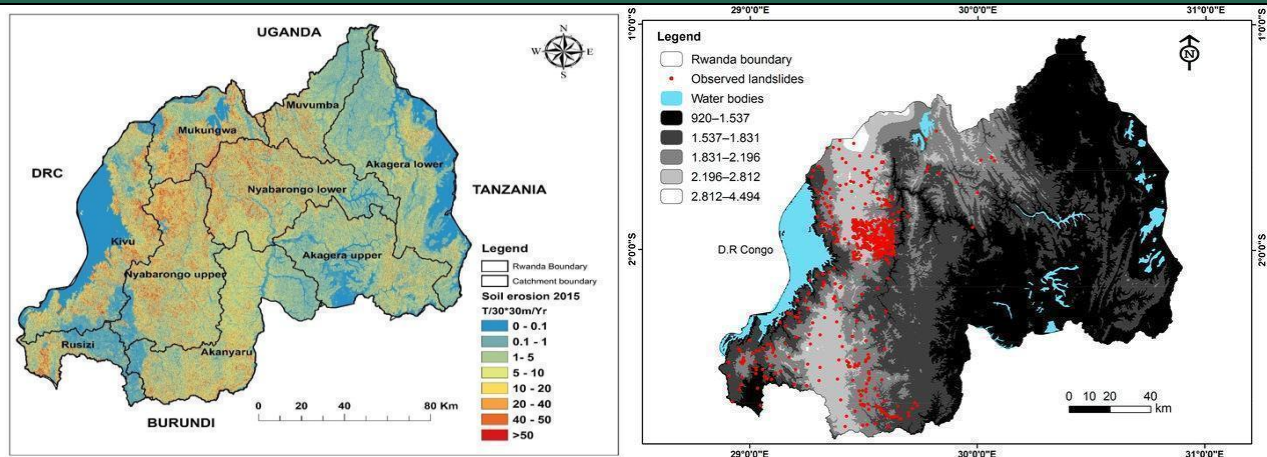


Figure 4. (left) Intensity of soil erosion in Rwanda assessed for the year 2015. Source: NISR: National Institute of Statistics of Rwanda, 2019. **(right)** Observed landslide occurrences in Rwanda against a topographic background, showing intense clusters along both the east and west sides of the CND landscape. Source: Nsengiyumva et al., 2018.³²

Wildfire occurrence in forested lands exacerbated by climatic warming is a rapidly mounting concern globally, and montane forests of the CND carry increasing vulnerability to loss from burning. Absence of fire-adapted flora in CND highland forests, however, suggests that fire has not been a major selective force within the forest ecosystem. The 1997 visitation of weeks-long uncontrolled wildfires in the Nyungwe forest had an ecological imprint that remains evident 25 years later, as burned areas revegetated with dense ferns rather than replacement forests,³³ representing a net loss of 12% of Nyungwe's forest cover and corresponding loss of its carbon endowment. Despite increased propensity for drying of vegetation as the climate warms, forests remain sufficiently moist such that fire causing forest losses are infrequent, and when they occur are usually tied to anomalous dry spells paired with human ignition, as was the case in 1997.³⁴

Cloud base altitude, which is highly influential in CND montane forest ecology and species composition, is likely rising above recent past conditions due to both climatic warming and conversion of forests to agricultural lands. Fog interception by vegetation augments montane forest precipitation in central African highland forests by an estimated 10%³⁵, so highland forest loss by human action therefore represents a local reduction to the ecosystem service of rainwater provision).

Projected climate change impacts

Temperature increases. Localized projections for the CND landscape of climatic changes for coming decades, generated for the most recent IPCC assessment, present a strong consensus on temperature increase for the next two decades. The magnitude of net temperature change by late-century, and the tendency of warming to accelerate, hold steady, decelerate or even begin an overall temperature decline is unknowable at present, but a large number of climate model simulations show common patterns, whereby divergence in possible temperature outcomes will only become apparent around 2040 (**Figure 5**). These

³² Nsengiyumva, J.B., Luo, G., Nahayo, L., Huang, X. and Cai, P., 2018. Landslide susceptibility assessment using spatial multi-criteria evaluation model in Rwanda. *International journal of environmental research and public health*, 15(2), p.243.

³³ Masozera, A.B. and Mulindahabi, F., 2012. Post-fire regeneration in Nyungwe National Park-Rwanda. WCS White paper report.

³⁴ Seimon, A., 2012: Climatology and Potential Climate Change Impacts of the Nyungwe Forest National Park, Rwanda. WCS White paper report, Wildlife Conservation Society, New York, USA, 44 pp.

³⁵ Bruijnzeel, L.A., Mulligan, M. and Scatena, F.N., 2011. Hydrometeorology of tropical montane cloud forests: emerging patterns. *Hydrological Processes*, 25(3), pp.465-498.

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results are also consistent with outputs from the previous CMIP5 model suite for western Rwanda (e.g., IFAD report 2020)³⁶. As such, using the consensus values of such models is a sensible choice for environmental planning, with a +2.0 °C increase over the IPCC-referenced 1995-2014 baseline conditions offering a representative value for warming by mid-century. However, the observed warming of +0.6 °C registered between 1970-95 prior to the IPCC baseline period must be recognized as well, suggesting **net warming of +2.6 °C from a more representative 20th century baseline**.

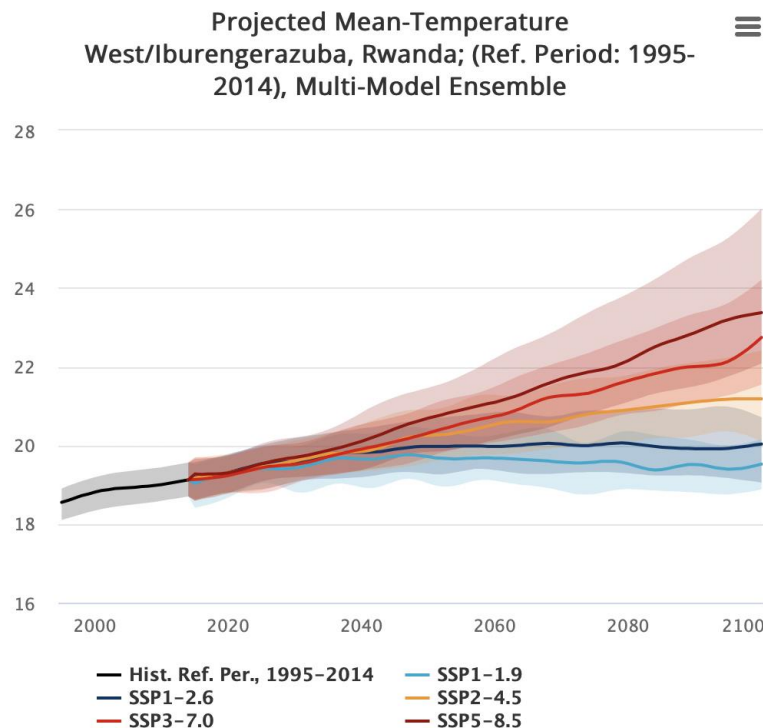


Figure 5. CMIP6 multi-model ensemble predictions for mean annual temperature over western Rwanda across the 21st century under a variety of SSP-RCP combinations. Source: World Bank³⁷

Annual mean precipitation changes. The CMIP6 results for precipitation changes over coming decades are less consistent than the temperature projections. Recent studies focused in East Africa have identified limitations to global model utilization for regionally specific predictions, such as failure to represent observed precipitation magnitude and distribution in the present, and reliance on approximations (i.e., parameterizations) that inadequately represent the role of convective storms in precipitation delivery³⁸. The CMIP6 ensemble means for annual precipitation over western Rwanda have a baseline value of approximately 2,000 mm per year, which is considerably higher than actual values (**Figure 6**; compare values with Figure 2 above). The higher emissions scenarios show increasing totals over time, which is plausible given the increasing water vapor carrying capacity of the warming atmosphere. Strangely, the most optimistic scenario (SSP1-1.9) shows strongly varying trends at decadal scales, increasing far more rapidly

³⁶ Hunter, R., Crespo, O., Coldrey, K., Cronin, K., New, M. 2020. Research Highlights – Climate Change and Future Crop Suitability in Rwanda. International Fund for Agricultural Development (IFAD), Rome.

³⁷ World Bank Climate Change Knowledge Portal. [Link](#)

³⁸ Wainwright, C.M., Marsham, J.H., Rowell, D.P., Finney, D.L. and Black, E., 2021. Future changes in seasonality in East Africa from regional simulations with explicit and parameterized convection. *Journal of Climate*, 34, pp.1367-1385.

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in the near-term than other scenarios. Such inconsistencies reduce confidence in the overall product suite for precipitation predictions.

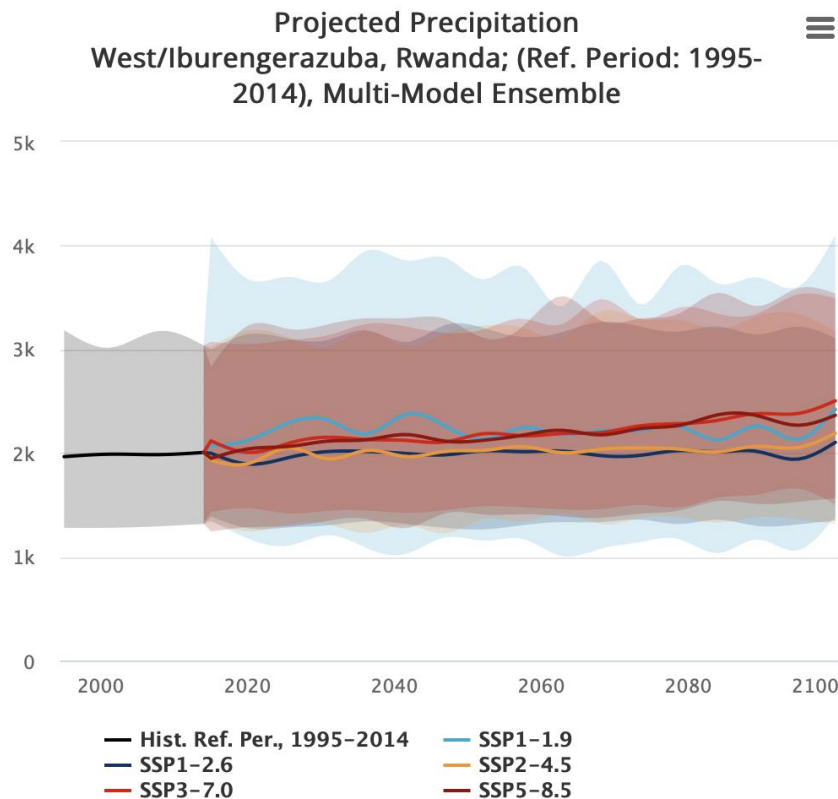


Figure 6. CMIP6 multi-model ensemble prediction for mean annual precipitation over western Rwanda across the 21st century under a variety of SSP-RCP combinations. Left-axis scale is thousands of mm per year. [Source:](#) World Bank³⁹

There is also major inconsistency between CMIP model suites. As assessment for western Rwanda from CMIP5 under RCP 8.5 comparing historical and mid-century outputs suggested rainfall decreases of 5-10% in most months, with net annual decrease of 86 mm⁴⁰; the baseline time periods and geographic domains of these two CMIP ensemble compilations for western Rwanda differ, but the reversal from drier to wetter gives low confidence in the utilization of precipitation predictions specific to the CND if derived from global models.

To generate more refined results, Grim et al., (2020)⁴¹ performed regional simulations of East African climate at ultra-fine, convection-allowing scale (WRF model at 3.3 km model resolution); these were based off the global CESM2 model (part of the CMIP6 suite) boundary conditions under the high emissions RCP8.5. The regional model was configured to optimize parity with regional climate station data. The resulting high spatial resolution output suggests a strongly amplified rainfall signal over the 21st century over western Rwanda relative to the CMIP6 ensemble mean for RCP 8.5 (**Figure 7**). Also notable is that the rest of Rwanda exhibits

³⁹ World Bank Climate Change Knowledge Portal. [Link](#)

⁴⁰ Hunter, R., Crespo, O., Coldrey, K., Cronin, K., New, M. 2020. Research Highlights – Climate Change and Future Crop Suitability in Rwanda. International Fund for Agricultural Development (IFAD), Rome.

⁴¹ Grim, J. A., Pinto, J. A., Jensen, A. A., & Seimon, A. (2020). The East African Great Lake Environments (EAGLE) Climate Downscaling Dataset (NCAR Technical Note TN-563+STR). [Link](#)

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far smaller increases, suggesting that the CND's hydrological significance as a water tower for the nation will be reinforced over time.

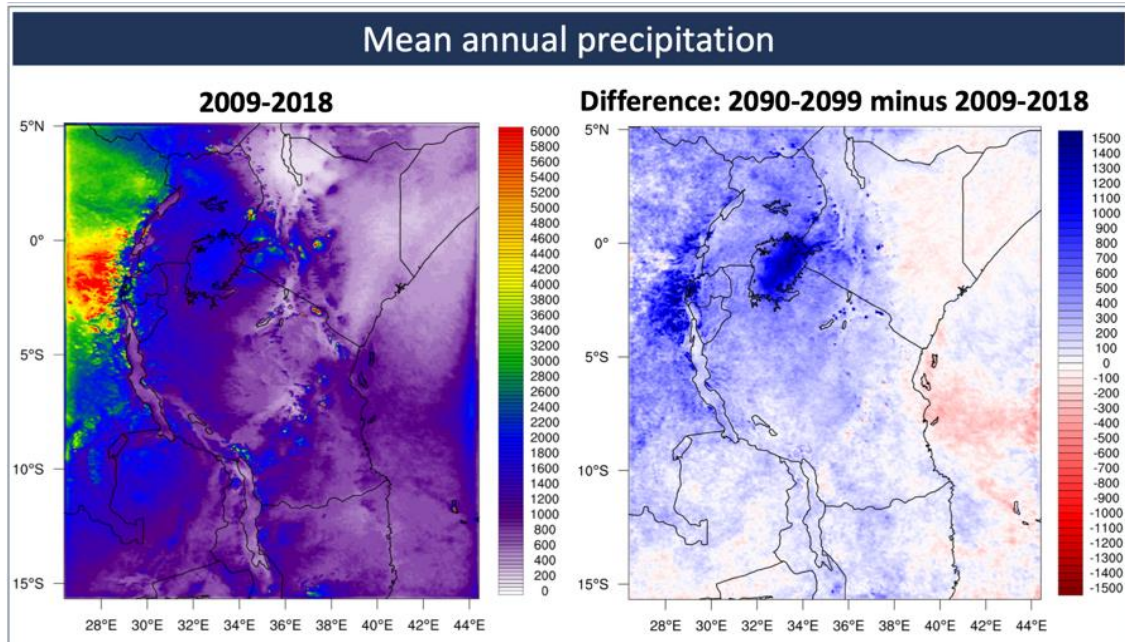


Figure 7. EAGLE projects high-resolution WRF simulation of annual mean precipitation (**left**) and difference between end of century and recent past (**right**) across East Africa, where values are in mm per year. The CND landscape and Lake Kivu show comparable increases to eastern Lake Victoria, and represent the largest rainfall increases across the modeled domain. Source: Grim et al. (2020)⁴²

Precipitation intensity changes. Due to well-understood physical properties of the atmosphere, as warming progresses short-period rainfall rates and individual storm totals, especially for the fraction of higher-end events that can cause flash flooding and promote landslides, will increase and should amplify very significantly over time⁴³. The CMIP6 ensemble means for monthly counts of days with heavy precipitation suggest an upward trend is now underway (**Figure 8**). Forthcoming increases are most pronounced for November through January. Such increases in heavy precipitation events are to be expected under a warming climate and are of major social and ecological concern, being a driver of landslide occurrences and major contributors to soil erosion intensity.

⁴² Grim, J. A., Pinto, J. A., Jensen, A. A., & Seimon, A. (2020). The East African Great Lake Environments (EAGLE) Climate Downscaling Dataset (NCAR Technical Note TN-563+STR). [Link](#)

⁴³ Fowler, H.J., Ali, H., Allan, R.P., Ban, N., Barbero, R., Berg, P., Blenkinsop, S., Cabi, N.S., Chan, S., Dale, M. and Dunn, R.J., 2021. Towards advancing scientific knowledge of climate change impacts on short-duration rainfall extremes. *Philosophical Transactions of the Royal Society A*, 379(2195)

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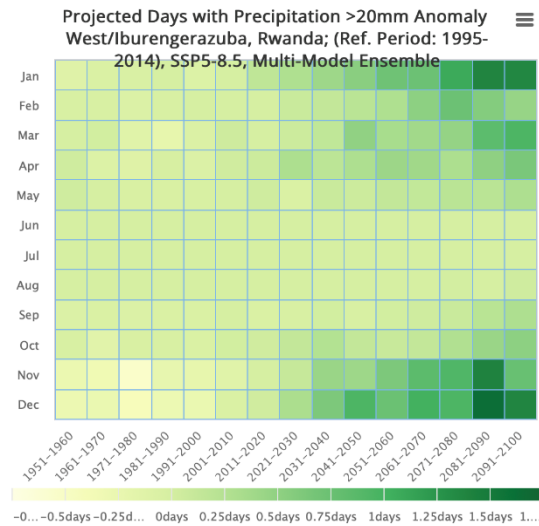


Figure 8. CMIP6 multi-model ensemble prediction for monthly changes in the number of days with rainfall >20 mm over western Rwanda across the 21st century under RCP8.5. Source: World Bank⁴⁴

Drought/dry spells. Warming will also increase the propensity for vegetation to desiccate during dry spells, intensifying wilting of natural vegetation and cultivars. The capacity of air to extract moisture from landscapes through direct evaporation and evapotranspiration of vegetation will also increase exponentially as temperatures rise. This means that land surfaces and vegetation will dry out more quickly between rainfall episodes in the future than in the past. It also means that to maintain hydrological balances that natural and human systems depend upon, total rainfall must increase in proportion to intensifying evaporation rates. The CMIP6 ensemble results for the CND region suggest such a favorable balance may indeed be maintained through at least mid-century (**Figure 9**). Under the full range of emissions scenarios considered, there is almost no change in ensemble means through mid-century.

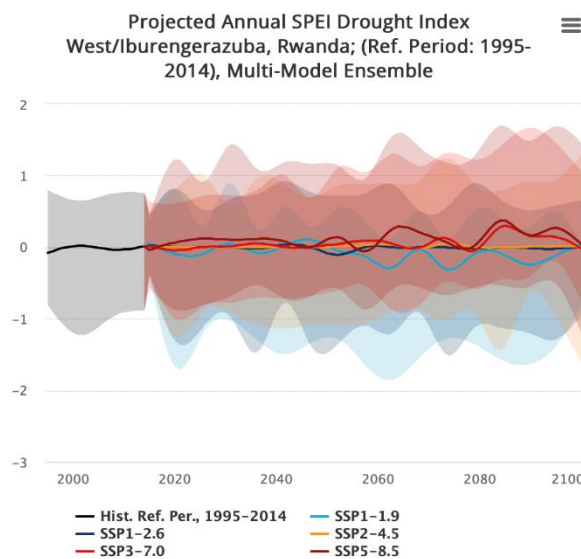


Figure 9. Western Rwanda CMIP6 model-model ensemble mean time series of the Standardized Precipitation Evapotranspiration index (SPEI) for a range of climate change scenarios across the 21st century. The SPEI is a measure of the integrated water deficit

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in a location, taking into account the contribution of temperature dependent evapotranspiration of a 12-month period. Source: World Bank⁴⁵

A more useful variable for assessing drought potential at daily-seasonal times scales is the Vapor Pressure Deficit (VPD). This is an absolute measure of the difference between the water vapor content of the air and its saturation value, and as such, represents a direct metric of the ability of the atmosphere to extract moisture from the land surface⁴⁶. High resolution VPD predictions for present day, mid-century and late-century under boundary conditions prescribed by the CESM model under the high emission RCP 8.5 scenario are shown in **Figure 10**. Annual cycles of VPD for model grid points high in the Nyungwe Forest and at lesser elevation in Kigali east of the CND both show remarkably little variation in VPD over the course of the century. Owing to the cool, persistently moist highland climate that characterize the CND landscape, the VPD values are consistently low in Nyungwe throughout the year for the three decadal time periods 2009-18, 2055-64 and 2090-99. This contrasts significantly with results from Kigali, where VPD registers four times higher than in Nyungwe during the mid-year dry season (20 vs. 5 hectopascal pressure deficit). This represents a favorable outlook over several decades to come for the CND in terms of limiting desiccating conditions that would elevate fire potential, stress crops and rapidly draw down hydrological resources. Importantly, however, not clearly evident in these model outputs is that rising cloud bases and associated warming temperatures increase the rate of drying of vegetation, increasing wildfire risk during rain-free periods, especially in lower reaches of forested terrain bordering settled areas where risks of anthropogenic ignition are highest.

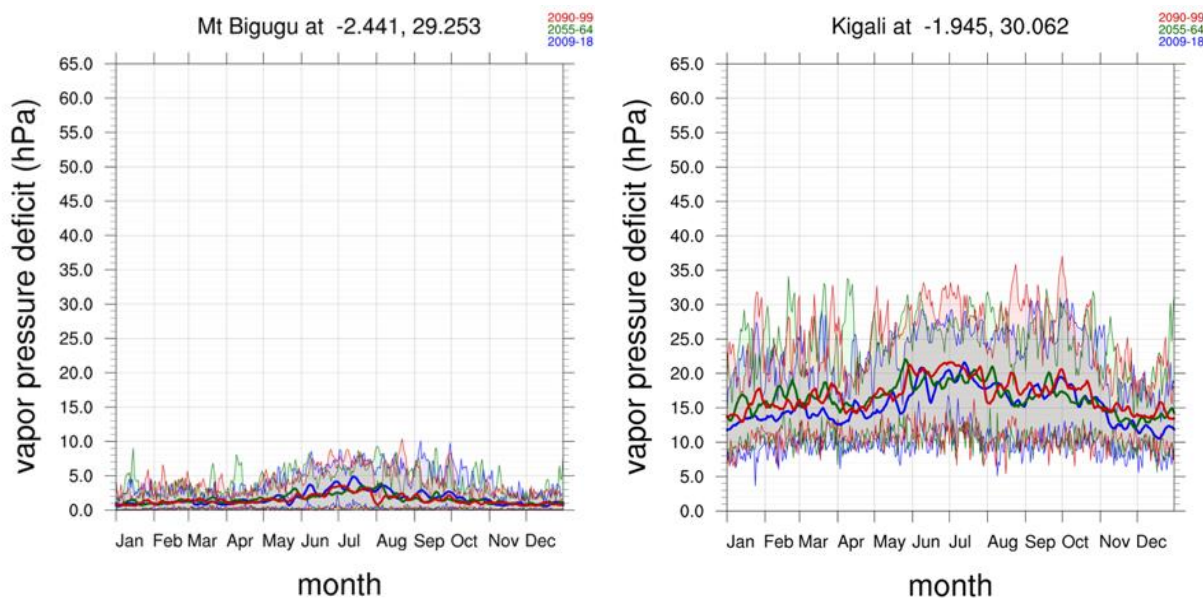


Figure 10. Modeled Vapor Pressure Deficit (VPD) on slope of Mt Bigugu in the Nyungwe Forest (**left**) and Kigali (**right**) for the recent past (2009-18, blue, mid-century (2055-64, green) and late-century (2090-99, red). Source: EAGLE project⁴⁷

Precipitation seasonality. The annual cycle of climate along the CND sees abundant rainfall for much of the year interrupted by a marked dry season in the June-August months when rainfall becomes light and sporadic, and during which hydrological deficits develop as evaporation far exceeds precipitation and resultant runoff. While highland orography largely explains the enhanced rainfall along the CND, the

⁴⁵ World Bank Climate Change Knowledge Portal. [Link](#)

⁴⁶ Seager, R., Hooks, A., Williams, A.P., Cook, B., Nakamura, J. and Henderson, N., 2015. Climatology, variability, and trends in the US vapor pressure deficit, an important fire-related meteorological quantity. *Journal of Applied Meteorology and Climatology*, 54(6), pp.1121-1141.

⁴⁷ East African Great Lake Environments (EAGLE) project. [Link](#)

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occurrence of the dry seasons occurs region wide irrespective of terrain elevation (**Figure 11**). This finds explanation in the seasonal north-south migration of a pan-equatorial band of enhanced cloudiness and rainfall known as the Inter Tropical Convergence Zone (ITCZ)⁴⁸. The slight reduction in rainfall experienced over several weeks along the CND in January and February reflects much more pronounced dryness to the north in Uganda; conversely, the more defined mid-year dry season along the CND spanning June through August is associated with enhanced southeasterly airflows that effectively push the zone of atmospheric convergence promoting rainfall across the equator into Uganda.

In this light, an unusual finding from the EAGLE project's high-resolution modeling studies is that the CND landscape has a heightened risk of major change in climatic seasonality. By the 2055-64 period: the mid-year dry season effectively disappears. Being the output of a single model run under a high-end greenhouse gas emissions scenario may make this seem implausible; It is highly dissimilar from mid-century predictions of the consensus of global models, which sustain the dry season. It finds corroboration, however, in a new study on East African rainfall, which found that the parameterization of convection in global models may be a source of uncertainty for projections of changes in seasonal timing, and that potentially impactful changes in seasonality are therefore quite plausible⁴⁹. Such an eventuality is exceptionally difficult to plan for, and since it would only come to fruition well after the project period, if at all, additional studies could be conducted to provide needed guidance.

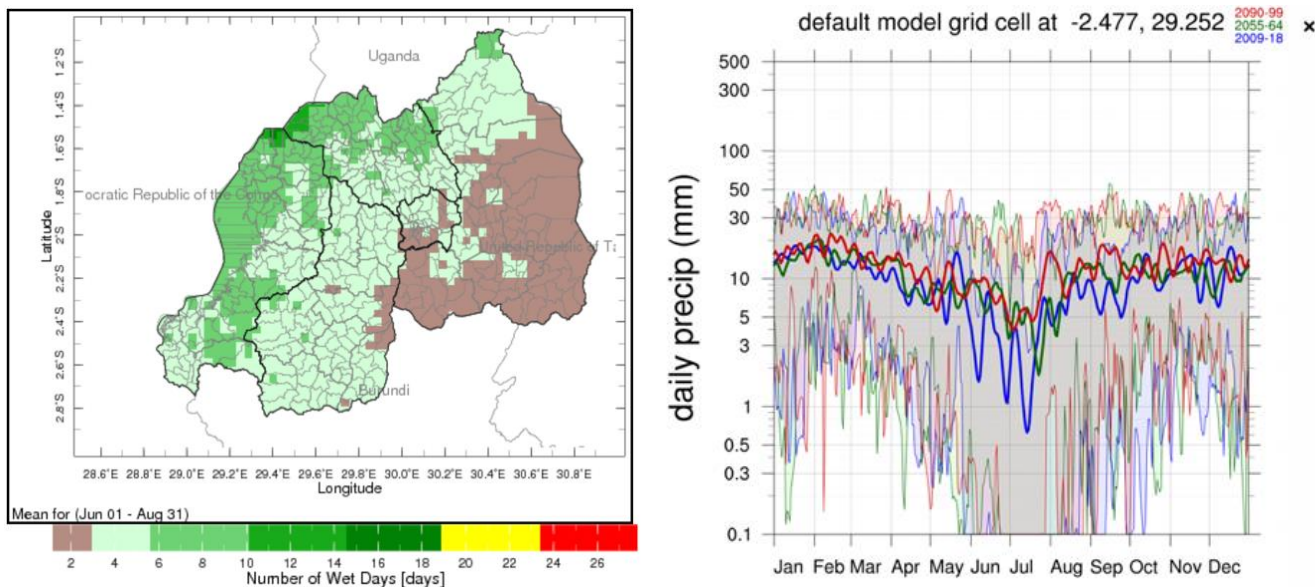


Figure 11. (left) The average number of “wet” days in Rwanda with rainfall exceeding 3 mm for the months of June-August. The wettest parts of the CND only receive significant rainfall on ~10% of days during the dry season. Source: Meteo Rwanda⁵⁰; **(right)** Annual cycle of rainfall rate (mm/day, note logarithmic scale) at Nyungwe Forest National Park showing daily means (thick lines) and extremes (thin lines) for 10-year time periods centered on 2014 (blue), 2060 (green) and 2095 (red). The dry season evident in the 2014 mean reaching below 1 mm/day, effectively disappears by the 2060 time period. Source: EAGLE project⁵¹

⁴⁸ Siebert, A., Dinku, T., Vuguziga, F., Twahirwa, A., Kagabo, D.M., delCorral, J. and Robertson, A.W., 2019. Evaluation of ENACTS-Rwanda: A new multi-decade, high-resolution rainfall and temperature data set—Climatology. *International Journal of Climatology*, 39(6), pp.3104-3120.

⁴⁹ Wainwright, C.M., Marsham, J.H., Rowell, D.P., Finney, D.L. and Black, E., 2021. Future changes in seasonality in East Africa from regional simulations with explicit and parameterized convection. *Journal of Climate*, 34, pp.1367-1385.

⁵⁰ ENACTS online tool at Meteo Rwanda. [Link](#)

⁵¹ EAGLE project: [Link](#)

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• Climate risks, vulnerability and impacts

Without strategic actions to confront mounting stresses borne by climate change interacting with unsustainable land-use practices, the CND would doubtless undergo a profound transformation over coming decades with significant loss of biodiversity and ecosystem services, reducing agricultural potential, severe hazards to humanity and loss of livelihoods. Temperature increase and loss of fog immersion will cause desiccation of the lower margins of protected forests promoting die-offs, arrival of pests, pathogens and invasive species, and increasing fire ignition potential. Current cultivation practices of both subsistence crops and valuable cash crops like tea and coffee will be drawn uphill, adding to pressure to convert protected forests to farmland. Whereas climatic changes will act as inexorably mounting stressors, unsustainable land surface conversion and land use practices will greatly exacerbate their impacts. The steep hillsides that characterize much of western Rwanda will experience intensifying erosion and risk of landslides, while built infrastructure such as road and bridges will be overwhelmed since they were engineered to withstand rainfall intensities expected under past climatic conditions that are sure to be exceeded, and with increasing frequency, as time progresses and temperatures steadily increase. Precipitation intensity will increase for short-period rainfall rates and individual storm totals, especially for the fraction of higher-end events that can cause flash flooding and promote landslides. This is certain to occur, is probably already discernible, and likely to amplify very significantly over time as warming progresses. This ties directly to landslides in particular, which already cause high mortality and significant destruction each year, and the CND's elevated vulnerability due to high topographic relief and deforested slopes makes this hazard of paramount concern to address (**Figure 12**).

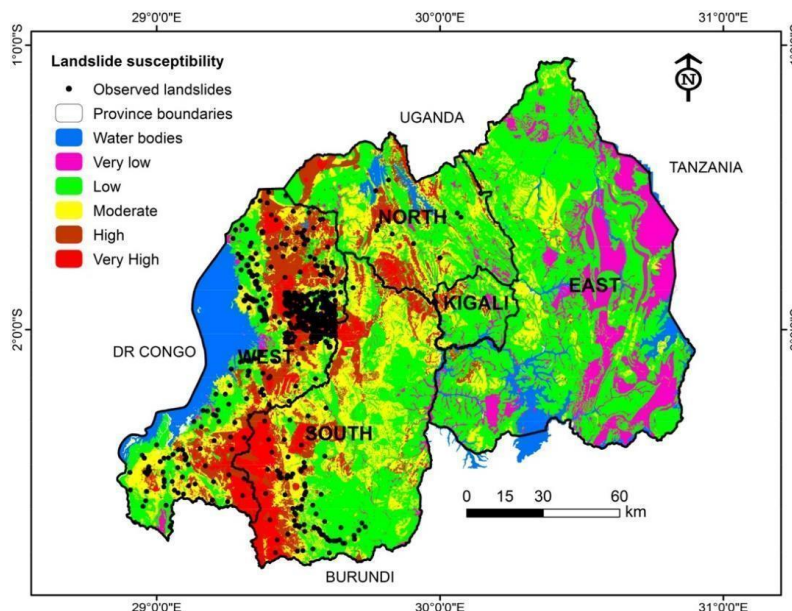


Figure 12. Rwanda landslide susceptibility map showing past observed landslides. Source: Nsengiyumva et al., 2018.⁵²

Significant temperature increases are almost certain to continue for decades to come, and will have mounting impacts on both natural and human systems along the CND, and some are already discernible. Over equatorial Africa observed environmental lapse rates feature temperature decreases by approximately 5.5 °C per kilometer of elevation increase⁵³, so 2 °C of warming expected by mid-century translates to a vertical rise on the order of 473 meters, promoting large uphill displacements of a wide range of organisms,

⁵² Nsengiyumva, J.B., Luo, G., Nahayo, L., Huang, X. and Cai, P., 2018. Landslide susceptibility assessment using spatial multi-criteria evaluation model in Rwanda. *International journal of environmental research and public health*, 15(2), p.243.

⁵³ Camberlin, P., 2018. Climate of Eastern Africa. In *Oxford Research Encyclopedia of Climate Science*. [Link](#)

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ecological processes, cultivars and human activities. While temperatures over the CND region are not expected to rise to levels for heat stress to elevate mortality⁵⁴ or reduce labor productivity, the indirect effect of rising temperatures through increased disease incidence is substantial⁵⁵.

Western Rwanda is the most productive region in the country for the cultivation of both tea and coffee, where conditions are optimized on the CND's sloping terrain. Both are cash crops with high export value, and are the target of active expansion through both governmental programs and private enterprise. Intensifying environmental stresses from climate change particularly due to thermal increases and the arrival of invasive pests from lowland regions, present challenges both for current plantings and in planning for expansion to new areas; the same concerns apply to other cultivars.

Cloud base altitude, which is highly influential in montane forest ecology and species composition, is likely already elevated above recent past conditions due to conversion of forests to agricultural lands, and certain to rise further in proportion to the degree of warming over coming decades. Ecosystems and species assemblages in the lower reaches of protected forests may already be out of balance with this important environmental variable, and this imbalance will only increase over time, promoting rapid species turnovers, enhancing potential for invasives, and act as a strong driver⁵⁶ of upward range extensions. Rising cloud bases in concert with warming temperatures also increase drying of vegetation, enhancing wildfire risk during rain-free periods. Fog interception by vegetation also augments montane forest precipitation by an estimated 10%, so rising cloud bases will act to remove this important hydrological input too. Highland forest loss therefore represents a local loss to the ecosystem service of rainwater provision that could be partially restored through reforestation.

Such challenges highlight how essential highland forests are to the climatic resilience of CND communities, for the ecosystem services and products they provide for both the region's vulnerable communities and the national economy. They also recharge aquifers; regulate water flow; control flooding; retain soil; provide wood fuel energy and timber; underpin the country's tourism, which provides the largest contribution to Rwanda's foreign exchange earnings⁵⁶; and provide wider benefits of atmospheric pollution control that sustain the country's economy and the wellbeing of its people⁵⁷.

Forest conversion to farmland in the CND highlands has until recently served as a release valve for lowland population pressure at the expense of drastically reduced national carbon stocks, diminished resilience to climate change, enhanced potential for destructive outcomes, reduced ecosystem service provision and disrupted biological connectivity. The resultant loss of ecosystem services and functioning and intensifying environmental stresses borne by climate change now require redirection towards forest restoration and other actions to ensure a sustainable future. Safeguarding the Congo-Nile Divide's remaining highland forests, and setting long-term goals of reforesting functional linkages connecting them, therefore serve Rwanda's national long-term interests on lessening the severity of impacts of climate change and sustaining critical ecosystem services both locally and downstream, while contributing to global efforts to draw down greenhouse gas concentrations.

B.1 (b). Summary of spatial biodiversity assessment

The majority of Rwanda's remaining montane forests are restricted to the national parks, which support a variety of threatened and endemic species. A few fragmented forest patches are situated outside of the National Park boundaries. Landcover change (especially for widespread smallholder agriculture), fuelwood

⁵⁴ Asefi-Najafabady, S., K. Vandecar, A. Seimon, P. Lawrence and D. Lawrence, 2018: Climate change, population and poverty: vulnerability and exposure to heat stress in East Africa. *Climatic Change*, 148, 561-573.

⁵⁵ World Bank, 2022: Rwanda Country Climate and Development Report. World Bank, Washington DC. [Link](#)

⁵⁶ Rwanda Development Board 2017 Annual Report [Link](#)

⁵⁷ Andrew, G. and Masozera, M., 2010. Payment for ecosystem services and poverty reduction in Rwanda. *Journal of sustainable development in Africa*, 12(3), pp.122-139.

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harvesting, and human-induced fires, coupled with climate change impacts, especially landslides, erosion and downstream flooding, have compromised the delivery of critical ecosystem services derived from these forests. A detailed spatial analysis of biodiversity in the Congo Nile Divide was conducted to delineate priority areas for the long-term conservation and restoration of forests, and the sustainable management of landscapes, in order to secure the ecosystem services needed to improve the resilience of vulnerable communities to climate change impacts.

The spatial biodiversity assessment for Rwanda's Congo-Nile Divide (CND) (see details in Annex 2 and Annex 2.2) is based on a rapid systematic conservation plan, using MARXAN decision-support software. The key analyses involved identifying and mapping the remaining areas of natural forest and other ecosystem types, modeling bioclimatic change and identifying climate change refugia, a Condatis landscape connectivity and bottlenecks analyses, and evaluating ecosystem threat and protection levels. The analyses thus build in landscape connectivity, climate change refugia, biodiversity values, ecosystem services and social costs.

The final outcome of the MARXAN and Condatis connectivity analyses split the CND domain into four major landscape categories with a set of priority implementation areas, each with their own place-bound project interventions. Interventions range from the restoration of natural forest, establishment and improvement of protective forest on steep slopes and along riparian areas; and to implement biodiversity-friendly agroforestry to reduce landslides, erosion and downstream flooding. Over and above these place-bound interventions are a variety of other mechanisms for promoting the sustainability of rural livelihoods and protecting montane forest in Rwanda's CND landscape.

The Systematic Conservation Plan analyses covered a range of biodiversity features, including:

- Terrestrial ecosystems, including their IUCN Red List threat status and protection level. The analysis focussed on identifying priority remaining intact areas, based on the development of a map of ecological condition.
- Protected Areas, Protected Forests and Protected Wetlands. All the identified ecosystems to be gazetted for protection are included.
- Climate change refugia based on projected changes in bioclimatic envelopes under a range of climate change scenarios.
- Identification of key landscape linkage areas and bottlenecks.
- Hydrological process areas – Rivers and Streams, including buffers.
- Hydrological process areas – Wetlands and Lakes, including buffers.
- Hydrological process areas – Areas with high rainfall.
- Landscape process areas - Steep slopes (over 55%) which are most important for minimizing erosion and landslide risk.

Individual species were not separately considered in the assessment as all key species are tightly linked to their underlying intact habitat (in particular intact forest patches as well as wetlands and riparian areas); and available distribution data for species was much broader than the ecosystem data. In selecting priority conservation areas, the SCP methodology always attempts to be spatially efficient by meeting conservation targets in as small an area as possible, while avoiding conflict with other land users, at the lowest possible cost for other sectors. The following was taken into account in the prioritization:

- Urban and dwellings, tea, roads, plantations, cultivated pasture, coffee, bamboo and agriculture.
- Areas of greatest population density.
- Areas with highly impacted land cover classes.

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Results of the MARXAN spatial analysis:

The MARXAN landscape prioritization (**Figure 13**), which builds in landscape connectivity, climate change refugia, biodiversity values, ecosystem services, and social costs (in terms of avoiding, where possible, areas with highest population density, agriculture etc), splits the landscape into four key planning categories (**Figure 14**):

- **Core Protected Area (PA) Nodes:** National Parks comprise the “Core PA Nodes” that need to be secured and well managed, which include Volcanoes, Gishwati-Mukura and Nyungwe National Parks. Priority activities include strengthening PA management and sustainability, rehabilitation and restoration of natural forests, other conservation-oriented land use activities that reduce stress on PAs and natural forests (e.g. improved wood stove efficiency to reduce pressure on natural forests) and supporting sustainable biodiversity compatible activities (e.g. improved beekeeping). These nodes also include buffer areas around the National Parks.
- **Stepping Stones:** These are priority nodes outside of the current National Parks that are critical for maintaining landscape connectivity, consisting of small, isolated patches of forest, at Dutake and Karehe-Gatuntu Protected Forests and the extensive Gishwati Pastures. These areas would be a sensible focus for some (patches of) forest restoration and protection, beekeeping and energy efficient stoves. The Gishwati Pastures are a focus for agroforestry on pastoral land to increase the coverage of native trees to secure reasonable landscape connectivity for forest species.
- **Landscape linkages:** These are key landscape linkages and knickpoints in the farming landscape that require afforestation on steep slopes and riparian areas to link the CND at a landscape scale. Compatible land use activities include agroforestry, increasing the use of native species, reforestation steep slopes, beekeeping and energy efficient stoves. **Note that the analysis is based on some level of improved connectivity via patches of protective forest and riparian strips, as well as overall improved species composition and tree coverage (within mixed agroforestry systems). It does not imply a continuous natural forest corridor; as this potential no longer exists, and its creation would not be possible in this highly used landscape without unacceptable impacts on livelihoods.**
- **Broader Farming Mosaic:** These are broader areas of moderate priority where conservation interventions can support broader sustainable landscapes and ecosystem service delivery but are likely to be beyond the scope of most project interventions except for those linked to land use planning.

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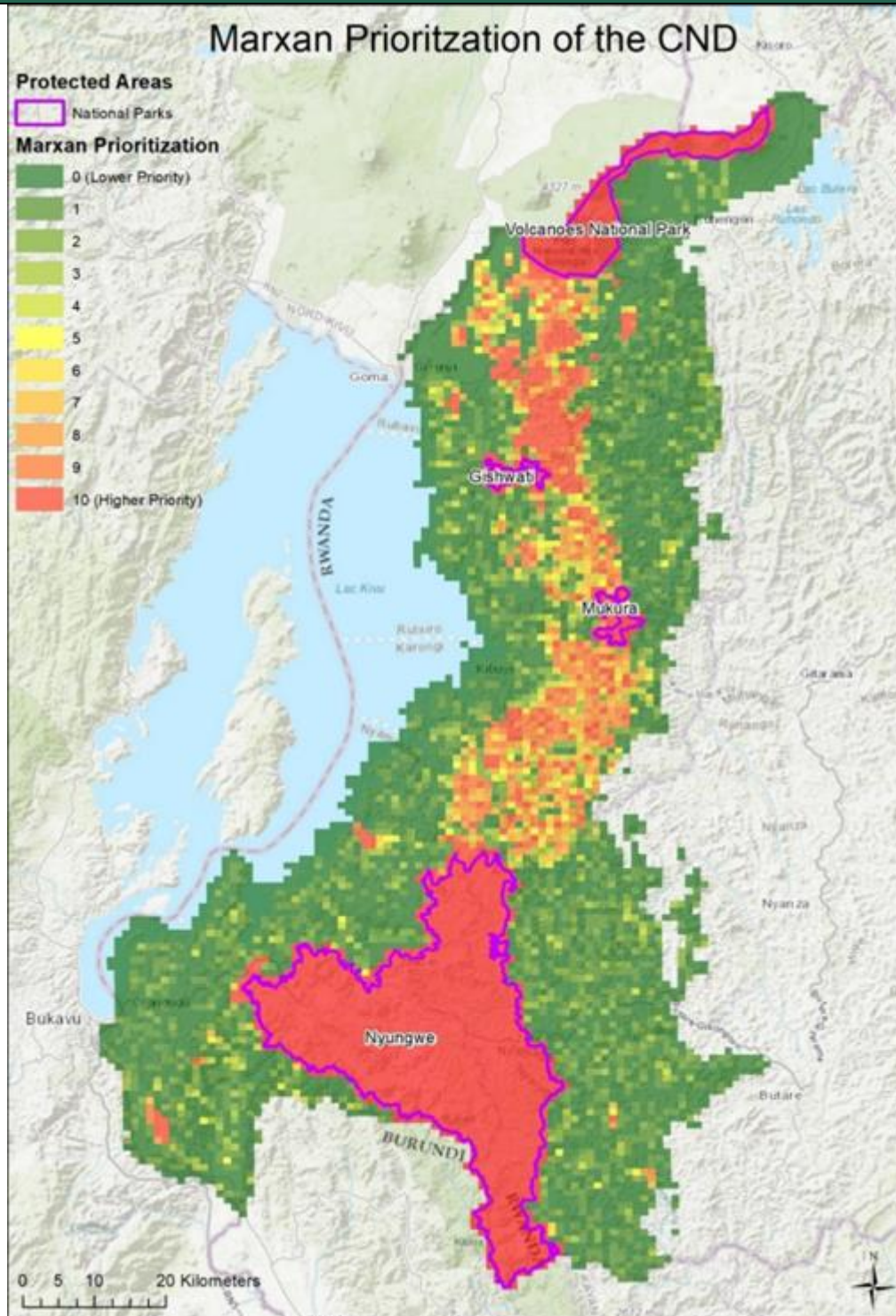


Figure 13. The MARXAN irreplaceability map for the Congo Nile Divide showing the landscape prioritization, ranging from high priority in red to low priority in green.

To aid prioritization and description of the landscape categories above, the four categories were further split into specific planning areas, referred to as “Landscape Implementation Sectors”, as shown **Figure 14** and in **Table 2** with a description of each planning category and associated ideas, required outcomes, core benefits and associated benefits in **Table 3**.

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Table 2. Summary table of landscape categories and associated implementation sectors for the CND.

Landscape Category	Landscape Implementation Sector	Area (ha)
Core PA Nodes	Volcanoes NP and "Buffer" ⁵⁸	19 487,0
	Nyungwe NP and Buffer	116 794,9
	Mukura NP and Buffer ⁵⁹	4 713,5
	Gishwati NP and Buffer ⁵⁹	4 013,7
Stepping Stones	Gishwati Pastures Stepping Stone	15 547,4
	Karehe-Gatuntu Stepping Stone	401,1
	Dutake Stepping Stone	903,0
Landscape Linkages	Nyungwe NP to Mukura NP Linkage	23 375,4
	Gishwati NP to Volcanoes NP Linkage	5 014,8
	Mukura N to Gishwati NP Linkage	7 823,7
Broader Farming Mosaic	Nyungwe to Mukura Broader Farming Mosaic	21 164,4
	Volcanoes Broader Farming Mosaic	11 694,4
	Mukura Broader Farming Mosaic	4 814,0
	Gishwati Broader Farming Mosaic	28 433,2
	Nyungwe Broader Farming Mosaic	11 437,0

Priority Areas for Implementation Activities:

Specific priority areas for a range of interventions were spatially identified, including:

- Natural forest rehabilitation and restoration with the National Parks (and their buffers) and in the Stepping Stones.
- Protective forests for steep slopes and riparian areas, particularly in the Landscape Linkages.
- Agroforestry interventions in key highland Landscape Linkages.
- Specific agroforestry priorities on pastoral land (silvo-pastoral practices).
- Indigenous shade trees for tea and coffee plantations.
- Beekeeping in National Park buffers and Stepping Stone buffers.

⁵⁸Note that buffers refer to both the specific legally designated buffers around Nyungwe and Gishwati-Makura NPs, as well as broader buffer areas adjacent to all these NPs, as well as Volcanoes NP.

⁵⁹In order to allow for specific landscape description, we separately refer to Gishwati NP and Makura NP where necessary, even though these areas are managed as Gishwati-Makura NP.

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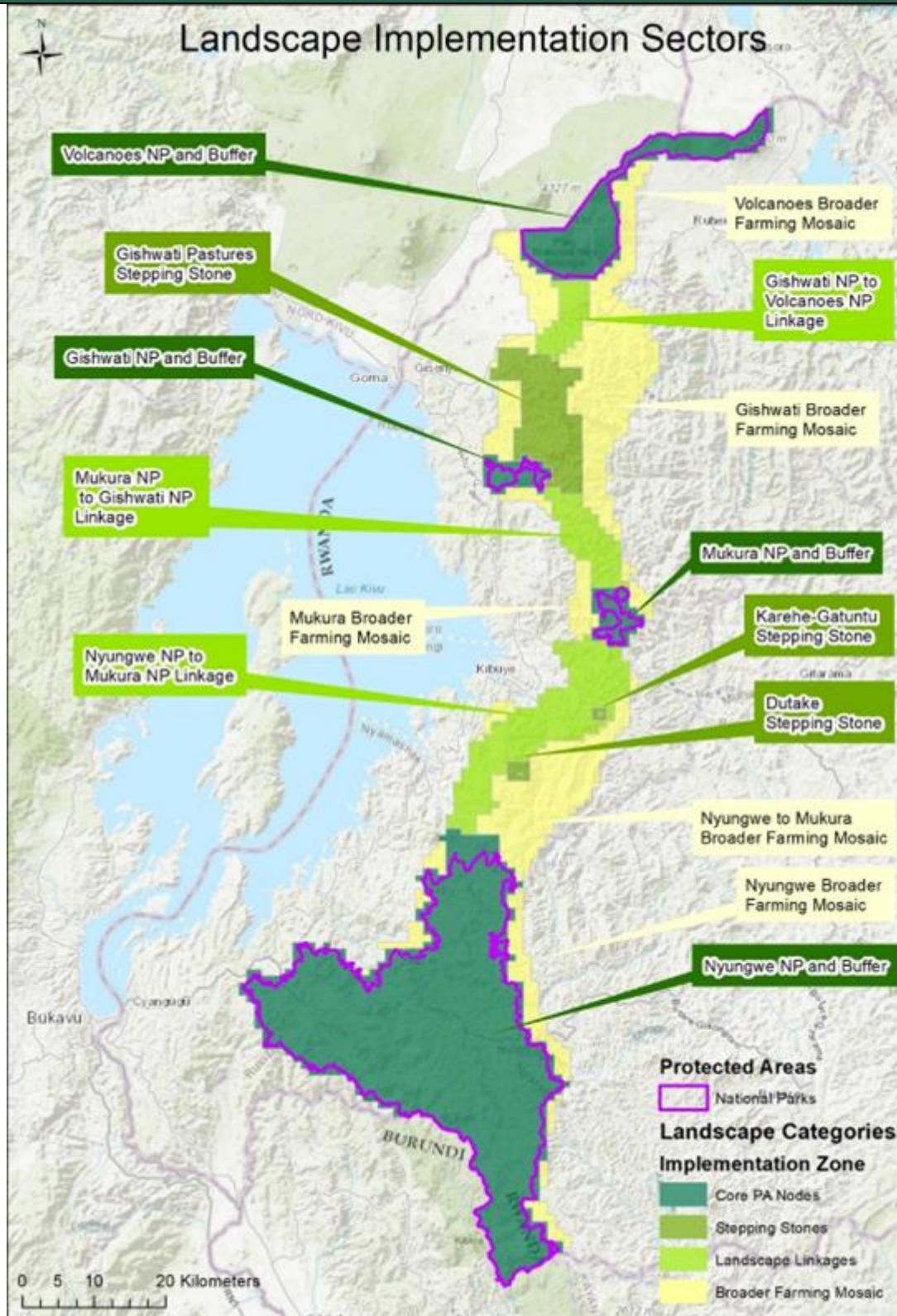


Figure 14. The four major landscape categories (Core PA Nodes, Stepping Stones, Landscape Linkages and the Broader Farming Mosaic) were split into specific areas to aid prioritization and description.

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Table 3. A description of the four key landscape planning categories and associated ideas, required outcomes, core benefits and associated benefits.

Landscape Component	Core PA Nodes The National Parks	Stepping Stones Priority nodes outside of current Pas	Landscape linkages Key landscape linkages and knickpoints in the farming landscape	Broader Farming Mosaic The broader landscape mosaic	Institutional issues
Description	The current core national parks and protected forests.	Critical pieces of biodiversity outside of the PAs required for landscape connectivity, maintenance of biodiversity and delivery of ecosystem services.	The parts of landscape within identified key corridors, where functional connectivity and ability to deliver ecosystem services needs to be urgently maintained or improved.	Remaining farmland areas of the CND.	The non-geographic specific elements of the CND system
Key ideas	Protect and manage for climate resilience.	Restore and protect to ensure landscape connectivity and ecosystem service delivery.	Functional linked farming landscapes delivering ecosystem services.	Diverse climate resilient farmland delivering ecosystem services.	Strong, well-capacitated and equitable environmental governance and land use planning.
Required Outcome	PAs effectively protect and manage natural forests improving resilience to climate change impacts and risks. Natural forests protected, connected, more resilient to climate change impacts and risks.	Critical landscape nodes / stepping-stones are secured and where necessary restored to a natural state.	Priority portions of the farming landscape are specifically managed to improve overall connectivity and ecosystem service delivery.	Sustainably managed farmland landscape is more biodiverse, supports delivery of ecosystem services and is resilient to climate risk.	Government and civil society are well capacitated to ensure robust landscape planning that supports climate resilience.
Core benefit	Maintain globally significant, species-rich natural forests. Core areas secure best possible source and/or refuge areas for species under climate change.	Biodiversity value of critical landscape nodes is maintained. Landscape connectivity supported through retention of key stepping-stones for species movement across the landscape.	Improved connectivity of the landscape ensures long term climate resilience. Value of core PAs and priority nodes is retained (i.e. the inevitable degradation of sites due to isolation is avoided).	Generally improved farmland management ensures rural sustainability and supports livelihoods.	Integrated land use plans, with community participation and spatial planning tools/ monitoring
Associated benefits	Improved delivery of ecosystem services (especially water) and support of rural economies and livelihoods.	Improved delivery of ecosystem services (especially water) and support of rural economies and livelihoods.	Improved delivery of ecosystem services (especially water) and support of rural economies and livelihoods.	Improved delivery of ecosystem services (especially water) and support of rural economies and livelihoods.	Cross sectoral planning and management.

This study was a rapid assessment to support project proposal development and does not replace a full conservation planning process. There are significant additional steps which are required to develop a product that is useful for land use planning during project implementation. These changes include a robust stakeholder engagement process, at a national, district and local scale; incorporation of issues relating to

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land use rights, both of landowners and farm tenants; incorporation of issues relating to social safeguards, especially for marginalized groups; inclusion of issues related to planning processes and strategies, at a national and local scale; finer scale planning (ideally at a 1:50 000 scale); and improved biodiversity data, including revised data on forest degradation, validation of the ecological condition map, and specific species data where possible.

B.2 (a). Theory of change narrative and diagram

The theory of change for this Project is founded on several key principles. Central to all, the CND represents a complex landscape – a geospatial mosaic of forest and farming patches, each dependent on the other. These coupled, biological and socio-economic systems are under great strain due to intense land pressure, which has resulted in a negative cycle of natural resource degradation that undermines the well-being of both natural and socio-economic systems. The advent of climate change is a forceful driver that intensifies this cycle of degradation, with the resultant loss of key ecosystem services that importantly link forests and people. This cycle must be disrupted in order to chart a path toward climate resilience for natural and human systems. Ecological connectivity must be established within this mosaic of land uses to promote landscape heterogeneity to help build resilience to climate change of forest ecosystems and people who depend on them. The approach to be taken by this Project is therefore based on principles of ecosystem-based adaptation: that healthy and resilient ecosystems form an essential foundation for adaptation to climate challenges⁶⁰. Climate-aware management of both natural and human-dominated ecosystems – interacting and integrated at a landscape level – can break the cycle of degradation and instead lead to a positive sequence of mutually reinforcing, climate-smart, sustainable, and resilient benefits..

In sum, the goal statement of the Project is:

IF Rwanda's CND landscape of native forests and neighboring farmlands is effectively managed for climate resilience

THEN the CND will comprise an interconnected and interdependent set of globally significant, species-rich natural forests in a matrix of sustainable farmlands, acting as buffers against damage from extreme weather events: assuring landscape integrity, stability and adaptability, providing vital ecosystem services and improving livelihoods for vulnerable people and the nation at large and contributing importantly to the conservation of biodiversity, the national economy, and mitigation of GHGs

BECAUSE building the capacity and management of integrated, climate-resilient forest and farming systems will ensure natural forest perpetuity and break and reverse the climate change-induced intensifying cycles of drought, flooding, and landslides that lead to degradation of livelihoods and resultant increased pressure on forest resources.

The status and future of the CND's forests and people are tightly intertwined: both heavily dependent on the natural resources and processes that provide a foundation for their ability to thrive. They represent ecological and socio-economic systems that are importantly coupled in a complex landscape, linked in particular by the ecosystem services that flow throughout these systems (hydrological and microclimate regulation, soil conservation, and nature-based tourism). At present these services, and the consequent relationships between forests and people – especially those who are most vulnerable in Rwandan society – are deeply

⁶⁰ Scarano, F.R. et al, 2017. *Ecosystem-based adaptation to climate change: concept, sustainability and a role for conservation science*, Perspectives in Ecology & Conservation 15(2), 65-73.

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strained⁶¹. Forests have been reduced in extent, fragmented, and degraded by fire⁶², overuse of resources⁶³, and introduction of exotic species⁶⁴. The sharply sloped rural lands of the CND have been farmed more and more extensively and intensively on steeper and steeper slopes^{65,66} and throughout riverine bottomlands, further reducing tree cover outside forest blocks: thereby further compromising the ability of the land to retain soils⁶⁷, avoid flooding⁶⁸, provide consistent and healthy water⁴⁴, ameliorate microclimate⁴⁹, store carbon⁶⁹, and remain productive. The coupling of these stressed systems has led to a negative feedback cycle of insecurity and constrained the well-being of farmers. Having little access to alternative sources of livelihoods or capital to enable change⁷⁰, smallholder farmers have consequently increased pressures on natural forests for land, fuel, and other forest resources⁴⁶. Such pressures in turn cause further forest loss and degradation and decline in biodiversity – reducing their ability to provide essential ecosystem services to people living in the landscape and across the nation, including tourism revenues and employment that are of national import. Thus, these systems are caught in a negative cycle that has proven difficult to break, despite significant sector-specific efforts. One such pressure – demand for fuelwood – is quite substantial given that a recent Government of Rwanda (GoR) census found that 77% of Rwandan households use wood as their primary fuel for cooking⁷¹.

With the advent of climate change, these already strained systems are showing clear signs of breakdown. Increased variability in the timing and intensity of rains along with rising temperatures are causing increased landslides⁷² and soil erosion⁵⁰, increased desiccation and drought⁵⁰, and unpredictable timing for agricultural plantings and harvest⁷³. This is reducing farm productivity and security, driving even greater pressures on the remaining forested estate, thereby compromising further the essential ecosystem services that are necessary to such well-being. At a larger scale, climate-induced forest degradation and decline of biodiversity in the CND compromise national-level benefits of water provision and quality¹, nature-based tourism⁷⁴, the supply of hydroelectricity,^{7,75} and carbon storage¹⁰ [see diagram A below].

61 Bagstad, K.J., 2019. *Towards ecosystem accounts for Rwanda: Tracking 25 years of change in flows and potential supply of ecosystem services*. British Ecological Society.

62 Masozera, A.B., Mulindahabi, F., 2007. *Post-Fire Regeneration in Nyungwe National Park*, Rwanda. Wildlife Conservation Society.

63 Republic of Rwanda, 2011. *Green Growth and Climate Resilience: National Strategy for Climate Change and Low Carbon Development*.

64 USAID, 2019. *Rwanda Tropical Forests and Biodiversity Analysis*.

65 Camberlin, P., 2018. *Climate of Eastern Africa*. In Oxford Research Encyclopedia of Climate Science.

66 Seimon, A., 2022. *An Overview of Climate Change and its Impacts along the Congo-Nile Divide in Rwanda*.

67 World Bank, 2019. *Rwanda Systematic Country Diagnostic*.

68 Karamage, F., et al., 2017. *Modeling Rainfall-Runoff Response to Land Use and Land Cover Change in Rwanda (1990–2016)*. *Water* 2017, 9(2), 14.

69 Mugabowindekwe, M., Brandt, M., Chave, J. et al. Nation-wide mapping of tree-level aboveground carbon stocks in Rwanda. *Nat. Clim. Chang.* 13, 91–97 (2023). <https://doi.org/10.1038/s41558-022-01544-w>

70 Clay N., King B., 2019. *Smallholders' uneven capacities to adapt to climate change amid Africa's 'green revolution': Case study of Rwanda's crop intensification program*. World Dev.

71 Government of Rwanda, 2021. *Rwanda Household Survey 2019/2020*. National Institute of Statistics of Rwanda.

72 Uwihirwe, J., et al., 2020. *Landslide precipitation thresholds in Rwanda*. *Landslides*.

73 World Bank; CIAT. 2015. *Climate-Smart Agriculture in Rwanda*. CSA Country Profiles for Africa, Asia, and Latin America and the Caribbean Series. Washington D.C.: The World Bank Group.

74 Rwanda Environment Management Authority (REMA), 2015. *Rwanda State of Environment and Outlook Report*.

75 Rwanda Environment Management Authority (REMA), 2009. *Rwanda State of Environment and Outlook Report*, Chapter 8: Energy Resources.

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Diagram A. Climate impacts & risks intensify negative feedback loop for people & forests

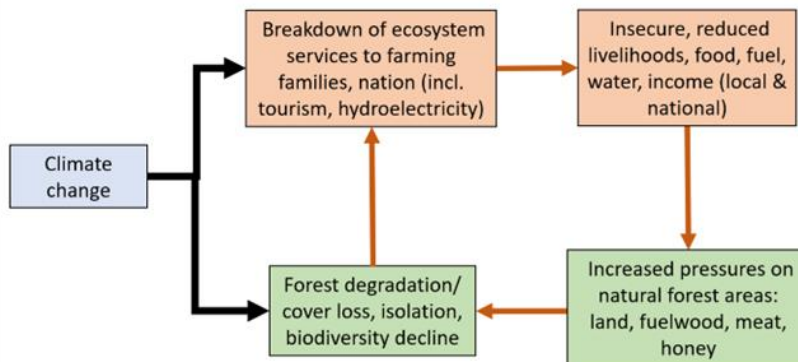
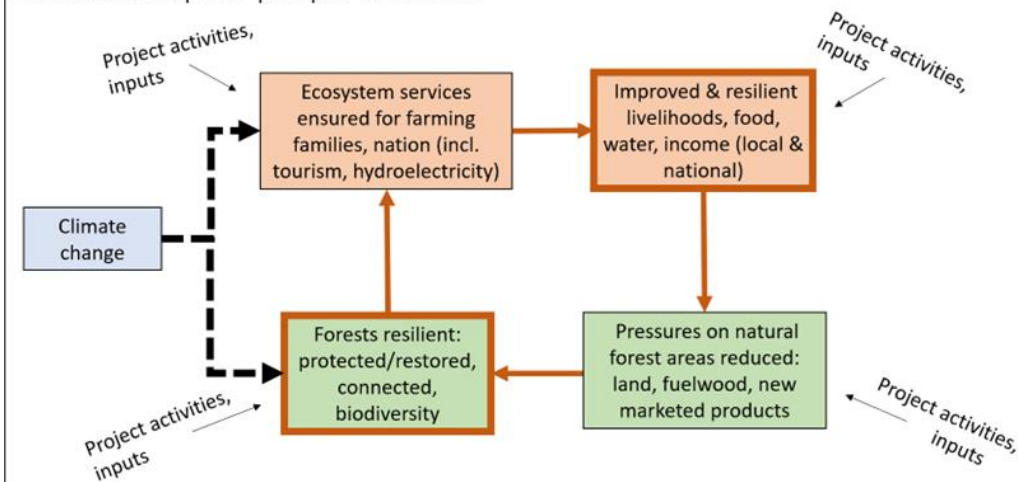


Diagram B. Adaptation to climate impacts & risks promotes positive feedback loop for people & forests



This context calls for new mechanisms of land-use planning and management that are based on consideration of the CND as a landscape, consisting of an inter-dependent mosaic of forests and farms, the management of each determining the health of the whole. Focusing on this larger scale, with coordination across geographic and sectoral boundaries, while mainstreaming climate risks into management planning will be key – and the foundation for a paradigm shift. Use of this framework to design and apply practical, precedent-setting Project actions in-forest and on-farm is expected to shift mindsets and management systems. This, in combination with establishing durable sources of financing, is thereby intended to shift the natural and human ecosystems themselves toward sustainable climate resilience. [see diagram B above, and the ToC (Theory of Change) diagram below.

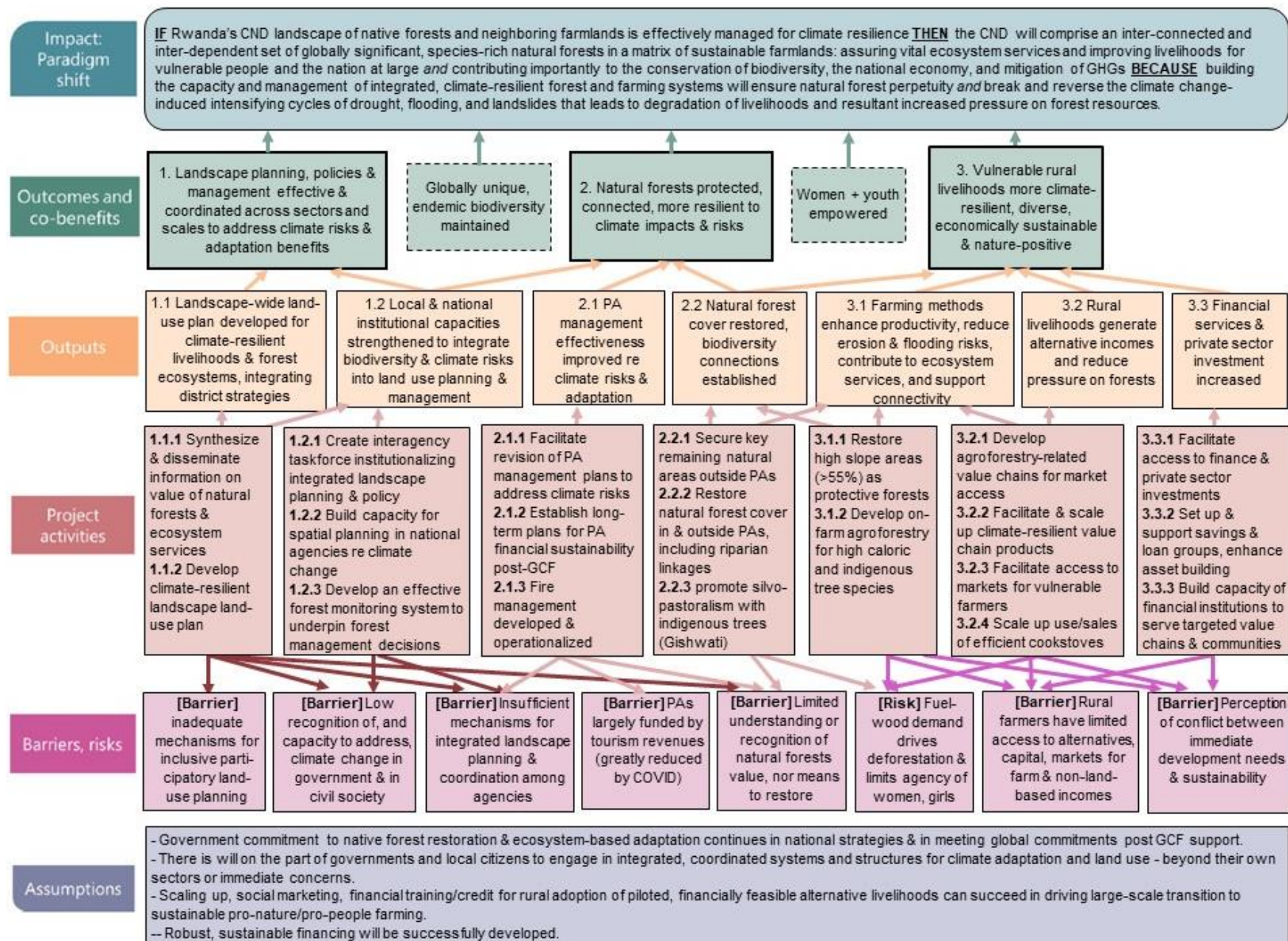


Figure SEQ Figure * ARABIC 15. Theory of change

Outcome 1. Landscape planning, policies and management are effective and coordinated across sectors and scales to address climate risks and adaptation benefits. Major barriers currently stand in the way of moving in this direction, as outlined in the ToC diagram. ***Awareness, understanding, attention, and action regarding climate risks to natural resources are currently insufficient.*** Planning and adaptation to climate impacts are constrained by a lack of information, site-focused modeling, spatial analysis, and real-time monitoring. Where some capacity exists, it ***operates within sector initiatives rather than across them.*** It also proceeds largely within governmental programs, with ***mechanisms for inclusive land-use planning inadequate***, not fully engaging rural peoples – particularly women and youth. Because of these barriers, coordination of landscape planning and management is weak across governmental and non-governmental sectors (agriculture, forestry, national parks, economic development), scales (local, district, national), and actors (women, youth, smallholder farmers, commercial businesses, as well as governmental agencies). Activities outlined in **Component 1**, aimed at integrated landscape land-use planning and the capacity for effective management, are intended to help ensure ecosystem services persist, and thereby align the ties between forests and people as they are faced with climate challenges. Activities under this component will provide the government and public with up-to-date information on climate adaptation (1.1.1), build agency capacities for incorporation into land-use planning – particularly spatial analysis and forest monitoring (1.2.2, 1.2.3), establish mechanisms for integrated landscape planning – including public participation and operations of an inter-agency task force (1.1.2, 1.2.1), and facilitate the creation of a climate-smart, integrated CND land-use plan (1.1.2).

Outcome 2. Natural forests are protected, connected, more resilient to climate impacts and risks. Particular to the CND, another key barrier is that ***natural forests are as yet undervalued and under-resourced*** despite the fact that they play vital roles in stabilizing natural and human CND ecosystems^{76,77}. The national parks of the CND (Volcanoes, Nyungwe, and Gishwati-Mukura) are globally significant regarding their outstanding species richness, levels of endemism, and presence of endangered and rare species⁷⁸. Although these are well-recognized for their tourism potential, and strong governmental policy and practice exists to protect them, barriers to their full valuation remains: their ***ecosystem services are yet to be fully recognized*** or financially supported, and their ***support is vulnerable to global tourism trends*** (with COVID-reduced tourism as a current example). As climate change considerations enter into landscape management, the values of these natural forests for fog interception, precipitation infiltration, soil retention, drought mitigation, and hydrological balance are only now becoming more prominent. In order to provide such services into the future, Nyungwe and Gishwati-Mukura NPs are in need of restoration and connection^{79,80}. Yet the understanding and capacity for recovering their richness and function is nascent at best, particularly in light of climate change. Activities outlined in **Component 2** are therefore designed with forest sustainability at the core. To build climate considerations into forest planning and management, the Project will facilitate revision of park management plans (2.1.1) that enhance dimensions of fire prevention and management (2.1.3), and will support in-park forest restoration (2.2.2). Establishment of restored forest at park boundaries (2.2.2), riparian linkages outside the parks (2.2.2), and forested “stepping stones” that

⁷⁶ Gatwaza O.C., Wang X., 2021 *Mapping of biodiversity hubs and key ecosystem services as a tool for shaping optimal areas for conservation*. PLOS ONE 16(8): e0253151

⁷⁷ Andrew, G., Masozera, M., 2010. *Payment for Ecosystem Services and Poverty Reduction in Rwanda*. Journal of Sustainable Development in Africa (V12, No.3).

⁷⁸ This area is also significant for global biodiversity – with the Albertine Rift being one of the most biodiverse regions within Africa, including conservative estimates of at least 980 endemic species, 15 Critically Endangered species, 34 Endangered species, and 99 Vulnerable species. Source: Plumptre, A.J., et al. 2016. *Conservation Action Plan for the Albertine Rift*. Wildlife Conservation Society.

⁷⁹ Nyandwi, E., Mukashema, A., 2011. *Excessive Deforestation of Gishwati Mountainous Forest & Biodiversity Changes*. Participatory Geographic Information Systems (P-GIS) for natural resource management and food security in Africa.

⁸⁰ Ordway, E., 2015. *Political shifts and changing forests: Effects of armed conflict on forest conservation in Rwanda*. Global Ecology and Conservation.

allow for forest species to move between parks (2.2.1, 2.2.3) is intended to enhance biodiversity and forest functions across the CND landscape. Fundamental to forest sustainability as well is sustainable long-term financing, which will be addressed by providing technical support from the Project to RBD and REMA to assess various financial instruments (conservation and/or water funds, bonds, revenue streams) to be employed for revenue generation post-GCF (2.1.2).

Outcome 3. Vulnerable rural livelihoods are more climate-resilient, diverse, economically sustainable and nature-positive. The sustainability and climate resilience of the agricultural matrix of this landscape is challenged by additional, multiple barriers. Rural agricultural practices and livelihoods are constrained by the very small size of landholdings (avg. 0.3 ha⁸¹), creating intense pressure to utilize all arable land for food and near-term income-generating products⁵⁶, thus creating *reluctance to incorporate trees or other soil conservation measures on the land*⁸². The resultant *lack of fuelwood in turn continues to drive deforestation of natural forests*¹², and *occupies important time for women and youth* who could instead be engaged in productive and empowering roles. Overall, local communities and extension services⁸³ have *limited skills, information and knowledge to design and implement farming methods* that would enhance their capacity to mitigate the impacts from climate-induced soil erosion, flooding and droughts. Farmers *also lack access to financial know-how*⁵⁶ *and capital, proven alternative livelihoods, and markets* that could incentivize and enable change¹². They are therefore caught in a cycle of immediate need that – without alternatives – forecloses options that could recover and sustain benefits from improved ecosystem services, other forest functions, and new livelihoods. **Component 3** aims to redress these conditions via activities that will demonstrate, incentivize, facilitate, and support actions that maintain and recover ecosystem services, benefit from forest functions and provide alternative on- and off-farm based incomes that reduce pressure on forests. To these ends, the Project will support reafforestation on steep slopes (>55%), on both public and smallholder private lands (Activity 3.1.1). It will develop farm-based agroforestry for climate-resilient species that produce fuelwood and/or market products, while conserving and enriching soils (3.1.2, 3.2.1). The Project will complement land-based production with a scaling-up of alternative value-chain products for smallholders that are nature-positive, including but not limited to tourism-based income, beekeeping, and improved cooking stoves (3.2.2, 3.2.4). Central to these activities will be facilitation of access to market chains (3.2.3) and financial mechanisms for establishing and maintaining businesses (3.3.1, 3.3.2, 3.3.3). Given that these activities are designed at the core to be profitable and promote financial sustainability, it is intended that they be replicable, and sustained once GCF funding expires.

Assumptions. Success in achieving Project outcomes will depend on important assumptions being met, among them that governmental and non-governmental actors will be willing to work in collaboration across sectoral and geographic boundaries; rural adoption of alternative farming and off-farm revenue generation can be scaled up to drive a landscape-level transition to sustainability; financing and revenue-generating mechanisms will be established that assure continuance of initiated actions; and governmental commitment to natural forest restoration and ecosystem-based adaptation continues in national strategies and in meeting global commitments.

The pathways summarized in this ToC are expected to contribute to the overall goal and outcomes of the Project: **natural forests** and **vulnerable communities** will be more resilient and sustainable despite climate impacts, with effective, **coordinated management** across the CND landscape. In doing so, important co-benefits will accrue. Intentional, inclusive engagement in management coordination will **empower women and youth** in environmental thinking and decision making. Forest conservation and restoration, along with agroforestry, will contribute to greater **carbon sequestration and storage**, while fire management and greater cooking efficiency will **reduce GHG emissions**. The globally unique, endemic **biodiversity** of

81 USAID, 2017. LandLinks Country Profile: Rwanda. <https://www.land-links.org/country-profile/rwanda/>

82 Rwanda Water and Forestry Authority, 2017. *Forest Investment Program for Rwanda*.

83 Foster & Graham, 2014. *Connectivity and the Tea Sector in Rwanda: Value Chains and Networks of Connectivity-Based Enterprises in Rwanda*.

Rwanda's forests, from great apes to rare orchids, will recover on the basis of healthy, more connected and resilient forest habitat.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1. Landscape planning, policies & management effective & coordinated across sectors and scales to address climate risks & adaptation benefits	☐	☐	☐	☒	☐	☐	☐	☒
Outcome 2. Natural forests protected, connected, more resilient to climate impacts & risks	☐	☐	☐	☐	☐	☐	☐	☒
Outcome 3. Vulnerable rural livelihoods more climate-resilient, diverse, economically sustainable & nature-positive	☐	☐	☐	☐	☒	☒	☐	☐

Co-benefits number	Co-benefits					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1: Globally unique, endemic biodiversity maintained	☒	☐	☐	☐	☐	☐
Co-benefit 2: Women and Youth empowered	☐	☐	☐	☒	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

Rwanda's CND region is a complex, naturally integrated landscape. Its diverse high elevation forests capture and store water resources, releasing them in a sustained manner throughout the year to surrounding steep hillside and bottomland farms and plantations, before feeding into the national network of wetlands, rivers, and lakes. The Rwanda portion of the CND runs from the Virunga Mountains and Volcanoes National Park (VNP) on the border with Uganda in the north, down through Gishwati Mukura National Park (GMNP), to the south end of Lake Kivu and Nyungwe National Park (NNP) on the southern border with Burundi. These three

national parks contain the country's only remaining montane forests. The boundaries of the CND landscape are defined as all areas greater than 1900m in elevation.

Administratively, the CND overlaps three Provinces (Western, Southern, Northern) and ten Districts (Karongi, Musanze, Ngororero, Nyabihu, Nyamagabe, Nyamasheke, Nyaruguru, Rubavu, Rusizi, Rutsiro).

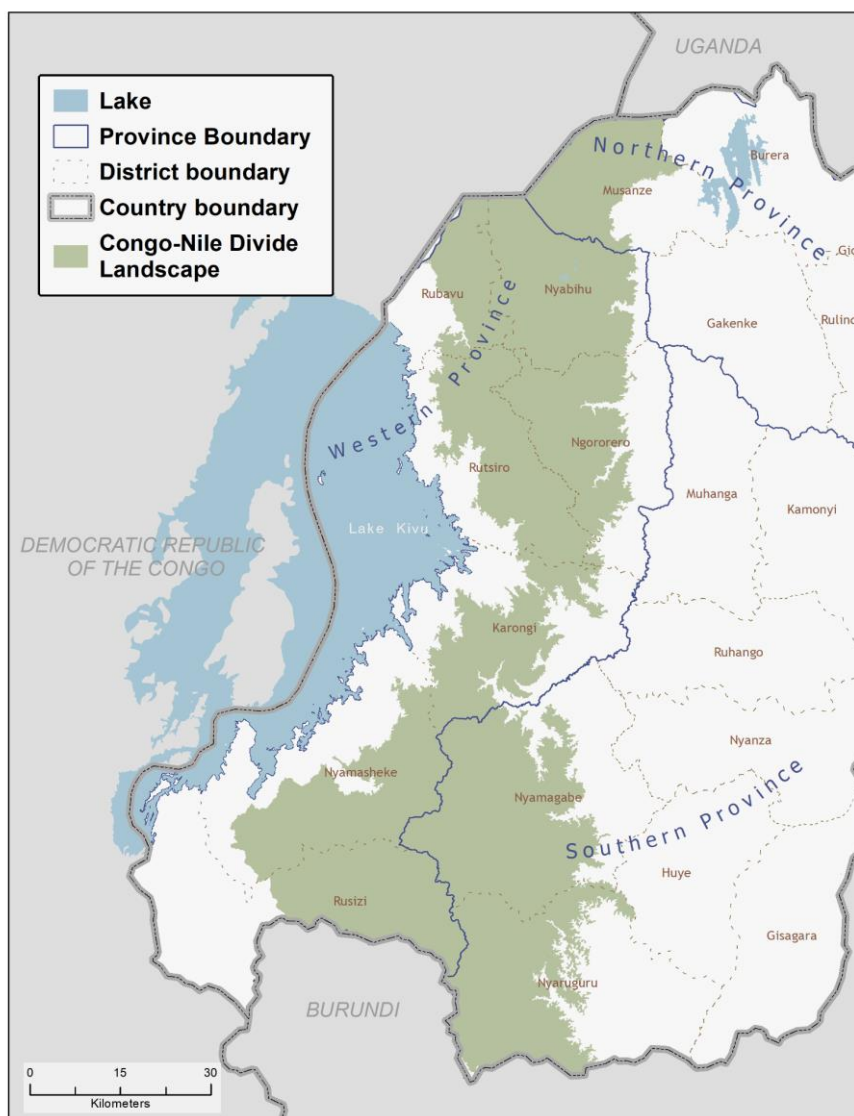


Figure 16. Map of CND Landscape

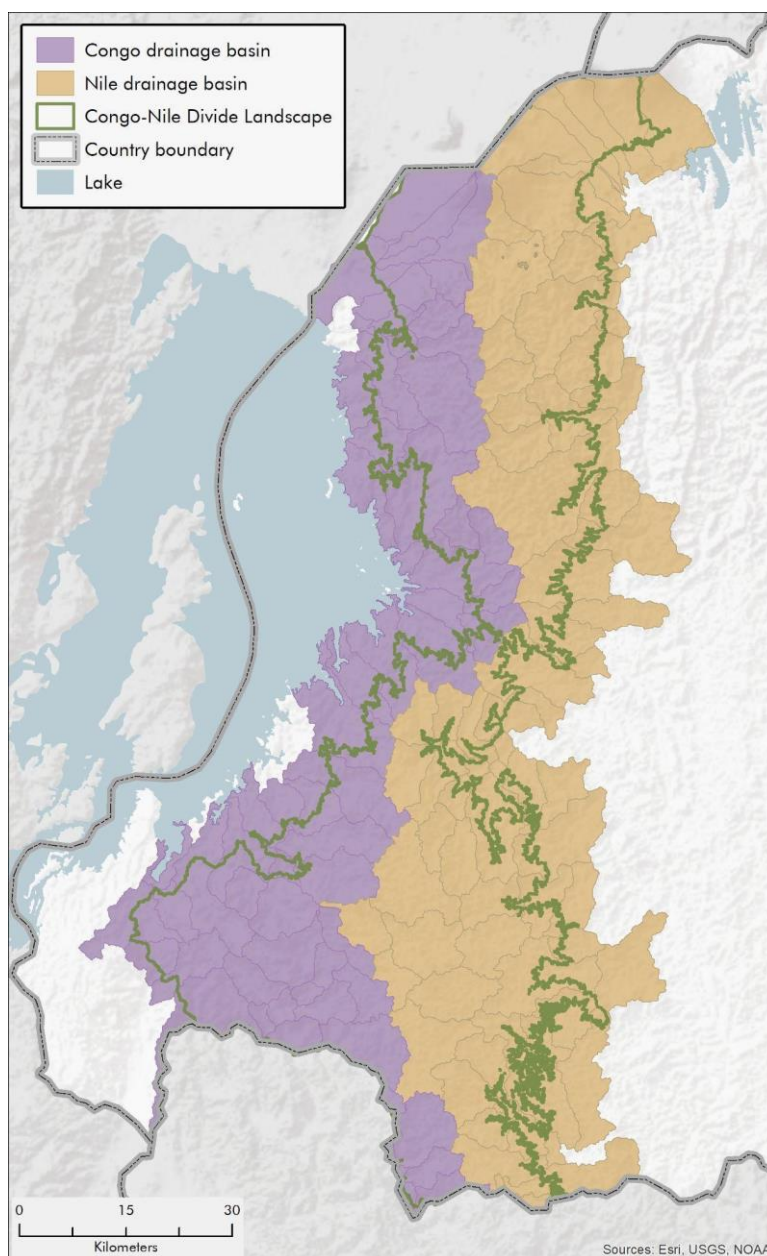


Figure 17. Map of CND Landscape drainage basins

To address the present and intensifying threats from climate change requires that this region also be managed as an integrated and ecologically connected system. This need is recognized in several key policies and plans enacted in recent years by the Rwandan government. However, implementation and integration⁸⁴ of these have lagged, their decentralized application on the ground is largely unrealized, and climate-related technical capacity remains limited in key institutions at all levels. This Project is designed to increase the climate resilience of the CND. Its intended outcomes are:

- The development and implementation of effective landscape planning, policies, and management, coordinated across sectors and scales, to address climate risks and adaptation benefits;

- Improved protection, restoration, and connection of existing natural forests, with enhanced resilience to climate impacts and risks; and
- The transformation of vulnerable rural livelihoods to be more climate-resilient, diverse, economically sustainable, and nature positive.

The Project has three main components directly targeting vulnerabilities within climate impact chains. Detailed analyses of the underlying rationale are provided in the Feasibility Study (Annex 2). Please see section 4.1 of that study, for a list and description of relevant existing and previous projects in the region.

Component 1: Mainstreaming Climate Adaptation into Integrated Land Use Planning

There is an urgent need for more effective collaboration among Rwandan government agencies, civil society, private sector, and diverse partners to balance difficult-yet-inevitable land use trade-offs that will have profound impacts on the climate resilience of both natural systems and people. The GoR recognizes that a piecemeal approach to decision making has exacerbated competition over scarce land resources and reduced the adaptive capacity of both ecosystems and rural populations⁸⁵. Rwanda's Baseline Climate Change Vulnerability Index (2015) recommends establishing a multi-ministry technical climate resilience coordinating committee, including NGOs⁸⁶. In 2017 the MoE developed the Strategic Programme for Climate Resilience (SPCR), to focus on three cross-cutting priorities to achieve climate change resilience: (a) technical capacity building and strengthening institutional coordination; (b) integrated land use and spatial planning; and (c) climate services and disaster risk management⁸⁷. The SPCR is a key step towards ensuring inter-ministerial and multi-sectoral collaboration and integration of climate resilience considerations into development plans and actions. However, there remains a need to comprehensively implement the strategy and mainstream climate change adaptation at the sub-national level, engaging a diverse set of actors and sectors and empowering women to provide input into decision making at district and community levels.

Output 1.1: Landscape-wide land-use plan developed for climate-resilient livelihoods and forest ecosystems, integrating district strategies.

The GoR completed a National Land Use and Development Master Plan (NLUMP 2020) to guide land use planning and regulate permitting processes. However, effective and strategic land use has yet to be applied at a local and district level, and the conversion of plantation and riparian forests to agriculture continues and development projects are approved without sufficient knowledge of climate change risks or the value of existing forests for ecosystem-based adaptation (EbA)⁸⁸. The Project will add climate-change resilience and adaptation content to central government efforts such as NLUMP and the SPCR process by acquiring, synthesizing, and disseminating key information on climate trends, the importance of remaining natural forests, and the value of ecosystem services. This information will also be shared at district and community levels through various outreach networks and workshops, with particular attention to reaching disadvantaged women, youth and vulnerable groups.

Building on the information base outlined above, the Project will work with the National Land Authority (NLA) to establish a working group, with participation from relevant ministries, agencies, and districts, including representatives of women, youth, and vulnerable groups interests, to develop a climate-resilient land-use

⁸⁵ National Land Use Management Plan.

⁸⁶ United Nations, 2015. Economic Commission for Africa; Rwanda Environment Management Authority. Baseline climate change vulnerability index for Rwanda. Kigali. Rwanda Environment Management Authority.

⁸⁷ Rwanda's Green Fund (FONERWA), 2017. Strategic Programme for Climate Resilience (SPCR) Rwanda - December 2017. [Link](#).

⁸⁸ Scarano, F.R. et al, 2017. *Ecosystem-based adaptation to climate change: concept, sustainability and a role for conservation science*, Perspectives in Ecology & Conservation 15(2), 65-73.

plan for the CND landscape. At the national level each relevant ministry and government agency will select and appoint a working group member while at the district level, participants will be selected from the Joint Action Development Forum (JADF). JADF is a multi-stakeholder platform meant to facilitate and promote the full participation of citizens in the decentralized and participatory governance and improve service provision processes with representatives from the public sector, private sector and civil society. The following stakeholders will form the basis of the collaboration: a) the Ministry of Environment represented by four of its agencies: the Rwanda Environment Management Authority (REMA); the National Land Authority (NLA), Rwanda Water Board (RBA) and the Rwanda Forest Authority (RFA); b) the Ministry of Agriculture, including the Rwanda Agriculture Board (RAB); c) the National Agriculture Export Board (NAEB) d) Ministry of Local Government; e) Rwanda Development Board (RDB); f) Gender Monitoring Office (GMO); g) Districts Decentralized Structures – the District Administrative Units, which supervise several technical and administrative activities; h) civil society, international organisations, academia and community based organizations. This list will be discussed during the inception period and expanded as necessary. From this, a spatial framework for district level land use planning will be developed that accounts for current and future climate risks, reconciles cross-sectoral land-use conflicts, and guides development decisions. All the activities and sub activities under this output will be executed by the Rwanda Forest Authority in collaboration with National Land Authority, and districts.

Activity 1.1.1. Synthesize and disseminate information on value of natural forests and ecosystem services

This activity will involve conducting a stakeholder mapping & consultation exercise in order to better understand the use of climate adaptation data in existing sectoral planning processes, and identify cross-sectoral linkages or conflicts. These results, along with a comprehensive literature review, will inform the design of high-level information packages that will synthesize information on climate risks for various sectors (e.g. agriculture, forestry) and highlight the value of forests for increasing resilience of local communities. An information dissemination program (e.g., presentations, workshops, and newsletters) will be implemented to share this knowledge among key ministries as well as government, non-government and community organisations. The Project will use JADF as a platform to disseminate information to key stakeholders. Particular attention will be paid to ensure women, vulnerable and youth groups from Ubudehe categories c, d and e are represented. In addition, the program will use mass media in the form of a radio program and complementary community action campaigns to improve community, women, youth, other vulnerable groups knowledge on climate information, climate risks and climate adaptation options. This will stimulate discussion on issues pertaining to climate risks and other social issues locally and support engagement of each in responding effectively to climate adaptation and resilience options.

Cross-sectoral tradeoffs and climate adaptation solutions with cross-sectoral implications will also be highlighted, in order to begin promoting the benefits of integrated land-use planning. To complement the high-level information dissemination program, a suite of technical training materials will also be developed for delivery to district technical staff. These materials will provide training on climate risks & the need to incorporate climate adaptation into land-use planning, with a focus on the technical skills required for climate-sensitive planning (e.g. sourcing climate data, mapping ecosystem services).

Sub-activities will include:

1.1.1.1 Map the sectors involved in land-use planning in the CND and review how forest ecosystem services and climate resilience are incorporated into each sector's planning process

1.1.1.2 Conduct comprehensive literature review & stakeholder consultation to collect & synthesize information on climate risks for various sectors (e.g. agriculture, forestry), the value of forests for increasing resilience of local communities, and highlight adaptation solutions with cross-sectoral implications.

1.1.1.3 Host workshops & presentations with key ministries, government organizations, NGOs and community organizations to disseminate high-level knowledge on value of forests for increasing resilience of local communities.

1.1.1.4 Develop guidelines for integrating climate risk into land use planning and cross sectoral planning

1.1.1.5 Develop outreach materials on climate and related risks in the CND and the value of forest ecosystems for increasing resilience

1.1.1.6 Implement outreach program tailored to different stakeholders (local government, civil society, communities, women, youth) to enhance capacities for land-use planning, funding mobilization, and delivery of climate adaptation actions

1.1.1.7 Conduct climate literacy seminars for local government and civil society organizations, aimed at increasing women and youth participation.

1.1.1.8 Provide financial and logistical support to trained organizations in grassroots mobilization to increase women and youth participation in climate adaptation planning.

1.1.1.9 Introduce social safeguards at a high level at each meeting, including the GRM, FPIC, and Access Restrictions Mitigations as safeguards to be in place for work with local communities.

Activity 1.1.2. Develop climate-resilient landscape land-use plan

This activity aims to ensure that climate change adaptation requirements are fully integrated into planning processes at local, district and national scales. This will involve i) participatory scenario analysis to understand the problems local communities and district officials face in terms of natural disasters (flooding, soil erosion, forest loss/degradation) and climate change, as well as the future land use plans of the districts, including an assessment of the national consequences of these problems in terms of ecosystem service delivery and climate change resilience and adaptation capacity; ii) risk assessment to estimate the impacts of the planned land use changes, as well as climate change, on flood risk, soil erosion, forest loss/degradation and the well-being of local people, especially any disproportionate impacts on women, youth and vulnerable groups; iii) climate change adaptation and mitigation measures development to decide possible climate actions for both adaptation and mitigation and prioritize these actions according to their feasibility and urgency in consultation with local communities, district leaders, and other stakeholders; and iv) land use plan development based on the recommendations from the previous 3 steps. A comprehensive consultation and feedback process will be undertaken to incorporate views of stakeholders at all levels (e.g. civil society; men, women, youth, and historically marginalized people in local communities; district & national government), upon which the plan will be finalized and approved by the Government of Rwanda.

Sub activities will include:

1.1.2.1 Engage district officials, Joint Action Development Forums (JADF), and community members in the CND (especially women) in reviewing and interpreting the National Land Use and Development Management Plan (NLUMP), to ensure the plan accounts for current and future climate and related risks, while building local support for climate sensitive planning

1.1.2.2 Conduct participatory land-use planning process in communities from village to district level to support integrated climate resilient land use planning

1.1.2.3 Develop Integrated Land-use Plan that supports Resilient Livelihoods and Ecosystems in the CND, ensuring alignment with National Land Use and Development Master Plan and ensuring the CND plan guides the district plan

1.1.2.4 Develop and roll out a series of trainings (virtual sessions and online modules) on gender sensitivity and mainstreaming women and youth into planning

1.1.2.5 Assess specific climate impacts on historically marginalized and Category c,d,e populations through a participatory NR process to ensure the Project components address their needs for adaptation

Output 1.2 Local and national institutional capacities strengthened to integrate biodiversity and climate risks into land use planning and management

Forest data in Rwanda has historically been generated in an ad-hoc fashion⁸⁹ through the support of external agencies. This precludes comprehensive consideration of forest and biodiversity data in land-use planning processes. As Rwanda moves to achieve the goals outlined in the Green Growth and Climate Resilience Strategy and the National Land-Use Development Master Plan, building governmental capacity in generating and utilizing spatial data to inform decision making is essential. The activities under this output will involve recruiting experts in mapping, remote sensing and ecosystem service modeling, in order to establish a formal spatial planning unit within government. These experts will also lead a formal information dissemination and capacity building program, in order to i) promote the importance of considering forest & climate resilience into land-use planning, and ii) deliver detailed training to technical staff on forest mapping, climate and ecosystem service modeling. The Project will also increase the understanding of the importance of forest landscape restoration in securing ecosystem services for local economic development and resilient livelihoods. It will ensure that the public, decision-makers and other stakeholders in the CND landscape have a high level of awareness of the risks to the economy and livelihoods associated with deforestation under current and possible evolution of these risks with the changing climate, and the benefits of reforestation in an integrated landscape approach.

Women's land rights need some key consideration for an effective integration of women in land use planning and management, among which include advocacy for women's knowledge of their rights, women's confidence in the dispute resolution process and outcomes, men's respect of women's rights, and modest improvements in physical and social accessibility of the local justice system for women are most critical avenues for success. These factors will be considered throughout project implementation of the activities proposed within this project using a gender-responsive, participatory and fully transparent approach. For example, under the project trained institutions will hold meetings to train women, youth and marginalized people on forests and mobilize them to participate in planning processes and local institutions will be trained to build capacity of rural people, specifically women and youth, to participate in climate adaptation processes. Program Staff and government officials will be trained understand gender barriers, understand how to mainstream gender equality principles into planning and support women, youth and marginalized participation in meetings.

Furthermore, the Project will support dialogue with the private sector and planning processes needed to mainstream climate change and biodiversity in the tea sector. Projected climate changes predict significant elevation rises in CND agro ecological zones, including the potential for tea cultivation at much higher

⁸⁹ Arakwiye, B., et al, 2021. *Thirty years of forest-cover change in Western Rwanda during periods of wars and environmental policy shifts*. Regional Environmental Change 21(2).

elevations than at present. This is unlikely to threaten Rwanda's existing montane forest parks, as foreign revenues from tourism in these reserves far surpass those from tea. However, tea remains a significant source of foreign revenue and a government target for expansion. Any increased tea production in the CND would therefore most likely affect lands outside of parks: smallholder, cooperative, and district holdings, many of which might also be targets for reforestation, agroforestry, and biodiversity stepping stone initiatives under this Project. The integrated land use planning included in this Project design, will bring together tea industry, agriculture, forestry, biodiversity, and tourism interests to identify and reduce such potential conflicts over use of the CND's limited land resource base. Through this integrated land use planning we aim to establish a formal public-private sector forum at the landscape level to help improve communication across sectors, appreciation of environmental issues, and increase synergies between interventions to avoid conflicting targets and indicators between sectors.

This output will comprise the following activities and sub activities which will be executed by the Rwanda Forestry Agency (RFA) in collaboration with key government agencies, private sector and civil society organizations.

Activity 1.2.1 Create interagency taskforce institutionalizing integrated landscape planning and policy

Insufficient mechanisms for integrated land use planning constitute one of the key barriers for sustainable natural management in Rwanda. Planning processes within government agencies are carried out in silos without paying attention to the impacts a certain land use activity or project will have on other sectors of the economy. This Project aims to facilitate improved collaboration among government agencies by strengthening the government's existing cross-sectoral planning task force. The existing task force was set up to support cross-sectoral implementation of the National Strategy for Economic Transformation, but it suffers a number of weaknesses, including: i) a lack of official mandate for the task force to carry out activities; ii) irregular attendance of members; and iii) lack of clear and regular funding to facilitate meetings. This activity will review the mandate and structure of the cross-sectoral task force and promote solutions for strengthening its operation to RFA (the task force coordinator). It will also facilitate quarterly meetings of the task-force, in particular between institutions in charge of agriculture and agroforestry, to encourage synergies and avoid overlapping mandates and redundancy in different climate resilience interventions. The involved staff at both national and local levels shall be empowered to readily share information and activity plans, and will be offered technical support to generate materials that can inform decision-making (e.g. policy briefs, map, presentations). The Project will support collaborative integrated landscape planning, which secures climate resilience undertaken with appropriate social safeguards.

Sub activities will include:

1.2.1.1 Review & strengthen operationalisation of the current cross-sectoral task force

1.2.1.2 Hold quarterly sectoral planning meetings with both national and district administrations in CND and the private sector involved in mining, agriculture, and livestock production in the landscape

1.2.1.3 Facilitate discussions and provide technical support (e.g. policy briefs) in decision-making for cross-sectoral actions around climate adaptation and forest resilience in the CND

1.2.1.4 Continue to introduce social safeguards at each meeting including the GRM, FPIC, and Access Restrictions Mitigations as safeguards to be in place for work with local communities. For those at the district or community level, provide options for feedback on the process and best ways of communication with local partners.

Activity 1.2.2 Build capacity for spatial planning in national agencies re climate change

Spatial planning is emerging as a valuable tool for the development of evidence-based land use and climate resilience strategies⁹⁰. This subcomponent will develop capacity within the Rwanda's university network and the MoE to use remote sensing, ecosystem service modeling and spatial planning tools (e.g., Marxan, InVEST, RIOS, SWAT and other open access tools such as Open Foris) to incorporate climate risks and climate resilience strategies into development plans and to readily identify and reconcile land use conflicts. A prime example of the latter is the potential for climate-driven shifts in optimal elevation zones for coffee and tea cultivation conflicting with current subsistence agriculture in the case of coffee, and key remaining natural forest fragments with regard to any upward migration of tea. These spatial planning tools will be procured by the project and will be made available to the spatial planning unit.

The Project will assist the GoR in developing a dedicated spatial planning unit (SPU) within an appropriate ministry or cross-cutting agency. The appropriate hosting institution will be determined during the project implementation in consultation with key government agencies operating under the Ministry of Environment. This will include support for training high level Rwandan specialists to provide the technical expertise needed to process and integrate spatial planning into decision-making within and across sectors. A specific contribution of this SPU in coordination with appropriate governmental and non-governmental partners is the development and operationalization of an innovative, scientifically credible, scalable system for monitoring trends in natural forest cover and forest types at multiple scales. This SPU will also allow Rwanda to measure and report on the land use, land use change and forestry (LULUCF) sector as part of a national Greenhouse Gas Inventory (GHG-I). This system will build on the national FMES that was established with the support from ENABEL.

Capacity building is essential for execution of the Project; even more so for the sustainability of Project elements in the long run. To improve Rwanda's collective ability to integrate land use planning in support of climate change resilience and delivery of ecosystem services, these sub-activities are planned:

1.2.2.1 Recruit spatial planning expert/s to support spatial planning unit

1.2.2.2 Assess and identify the institutional home and operationalize the spatial planning unit

1.2.2.3 Conduct a capacity needs assessment and identify the appropriate tools for spatial planning to fit the Rwandan context and conduct familiarization

1.2.2.4 Led by the spatial planning unit, develop capacity within the University of Rwanda and the MoE to use remote sensing, ecosystem service modeling and spatial planning tools (e.g., Marxan, InVEST, RIOS, SWAT) to incorporate climate risks into land use planning processes

1.2.2.5 Deliver training workshops on utilization of earth system models for land use planning purposes, utilization of remote sensing resources (lightning, satellite radiances) for hazards detection and climate monitoring and on regional climate monitoring and applications.

Activity 1.2.3 Develop an effective forest monitoring system to underpin forest management decisions

⁹⁰ Mathias S., et al, 2019. The Spatial Development Framework to facilitate urban management in countries with weak planning systems, International Planning Studies, 24:3-4, 235-254, DOI: 10.1080/13563475.2019.1658571

Within Rwanda, forest-related data is limited in scope and generally outdated⁹¹. Consequently, the current state and trends of Rwanda's forests are not well understood. Efforts to improve the situation through individual studies have been valuable but have not led to systematic increases in spatial information on trends in different forest cover types and resilience at the scales at which land use decisions are being made. For example, the FMES software focuses on forestry and plantation forests, but does not address natural forests. This activity will update and monitor forest cover data in different ecological zones and forest types using high resolution satellite imagery, field work, and remote sensing techniques. In addition to remote sensing techniques, the Project team, once selected, will meet with government and international partners to assess best monitoring tools and platforms, including Open Foris. Close attention will also be given to the system used by Mugabowindekwe et al (2023)⁹² for the most recent, highly detailed forest inventory from 2007 imagery. RFA will recruit a Rwandan expert who will work with WCS experts in applying spatial planning and forest monitoring tools in support of government agencies during the project implementation to strengthen the capacities of RFA, NLA and GIS Center at the University of Rwanda.

An adaptive forest, climate change and land use monitoring system will be established to support climate resilient forest management decisions through the following:

1.2.3.1 Recruit Forest ecologist or remote sensing expert to support the design and implementation of indigenous forest monitoring system

1.2.3.2 Review existing forest mapping data and monitoring software, and assess utility for generating quantitative assessments of indigenous forest cover from local to national scales

1.2.3.3 In collaboration with REMA, MINAGRI, RAB, RDB, RISA, MoE and affiliated agencies and University of Rwanda, design structure and operation of forest monitoring system, including required inputs (data, computing, personnel), desired outputs, and operational structure (e.g. location, reporting structure, funding)

1.2.3.4 Purchase and install 2 new weather automatic stations in the CND

1.2.3.5 Establish & operationalize forest monitoring system

1.2.3.6 Generate updated maps & statistics for indigenous forest cover, forest type etc. using forest monitoring system

1.2.3.7 Develop capacity within hosting institution for continued operation of forest monitoring system

Component 2: Forest and landscape management and restoration

The GoR has prioritized forest management and restoration – including natural forests in protected areas – as a key component of climate change adaptation strategies for the CND landscape.⁹³ Natural forests within the national park system have been generally well protected and managed in recent years, earning high marks for biodiversity conservation while also developing a high quality – and quite lucrative – set of ecotourism attractions. Current park management plans, however, include little attention to the potential effects of projected climate change. Outside of parks, few areas with natural forest cover remain and forest management and restoration approaches are not being implemented at the scale required to significantly improve forest climate resilience. Where forests are restored, most use exotic species and mono-specific

⁹¹ Food and Agriculture Organization of the United Nations, 2017. *Analysis of forests and climate change in Eastern Africa*. Forests and Climate Change Working Paper 16.

⁹² Mugabowindekwe, M., Brandt, M., Chave, J. et al. Nation-wide mapping of tree-level aboveground carbon stocks in Rwanda. *Nat. Clim. Chang.* **13**, 91–97 (2023). <https://doi.org/10.1038/s41558-022-01544-w>

⁹³ IUCN, 2014. *Forest landscape restoration opportunity assessment for Rwanda*. [Link](#).

plantations that are generally incompatible with mixed agricultural uses; they are also vulnerable to diseases and pests with climate warming.

In this component, targeted interventions will integrate climate change awareness and adaptation into national forest park planning, policies and management, as well as restore degraded areas within those parks. These efforts will help sustain the forests' rich biodiversity, the lucrative tourism revenues that flow into local and national economies, and the ecosystem services needed for climate resilience of vulnerable communities. This suite of activities will also identify priority areas outside of parks – in remnant natural stands, around wetlands, and along streams – for protection and restoration as “stepping stones” and linkages for long-term climate change connectivity.

Output 2.1 Protected Area management effectiveness improved re climate risks and adaptation

The remnant natural forests of the CND harbor globally significant biodiversity across multiple taxa, from primates to birds and orchids. More than 10% of these species are regionally endemic⁹⁴ – 47 flowering plants are endemic to Nyungwe Forest alone⁹⁵. Though each has suffered from significant past habitat loss, the Nyungwe and Volcanoes National Parks have been generally well-managed under the Rwanda Development Board (RDB), with assistance from NGOs, PPPs, and the private sector. Rwanda's reputation for ecotourism and nature tourism attractions is world class. It has also earned conservation recognition for the dramatic recovery of its endangered mountain gorillas, sustained protection of its remaining biodiversity, and creation of a significant revenue-sharing program with parks-adjacent communities. The recent addition of Gishwati-Mukura National Park to this mountain forest network underscores the government's commitment to conservation, though both forest blocks require significant restoration for the purposes of biodiversity conservation, connectivity, and ecosystem services enhancement.

In contrast to Rwanda's areas of conservation success, relatively little attention – and less action – has been focused on climate change and its potential impacts on habitats and biodiversity, especially in the climate-adapted mountain forests of the CND. The investment required to build and sustain the institutional and human capacity to achieve Rwanda's recent conservation success now needs to be joined with a parallel investment to ensure that PA management plans, policies, and actions address the realities of climate change both within and outside of their boundaries.

Activity 2.1.1 Facilitate revision of PA management plans to address climate risks

Climate change is increasingly recognized as not only a major future threat to protected areas, but one that will also exacerbate existing threats and vulnerabilities⁹⁶. This is particularly true for the three national parks of the Congo Nile Divide landscape that are surrounded by a large number of smallholder farmers who are dependent on natural resources for their livelihoods. The park planning processes need to be expanded to best meet challenges posed by climate change as well as ensure full incorporation of buffer zone and landscape linkage requirements, and the ability to engage with the larger CND and district landscape management planning processes. The revised management plans will aim to:

- Develop specific goals and objectives for improving climate resilience
- Identify areas and species of particular importance to climate adaptation, mitigation and resilience
- Identify and prioritize threats that exacerbate climate impacts
- Identify and prioritize critical actions for strengthening resilience

⁹⁴ CBD. 2020. Rwanda 6th National Report to the Convention on Biodiversity

⁹⁵ Fischer, E. and Killmann, D., 2008. *Illustrated field guide to the plants of Nyungwe National Park Rwanda*. University of Koblenz-Landau.

⁹⁶ World Bank, 2019. *Rwanda Systematic Country Diagnostic*.

- Identify areas important for climate change adaptation and mitigation into protected area zones and regulations
- Develop indicators of climate resilience

To this end, the Project will undertake these sub-activities:

2.1.1.1 Review and update existing national park management plans to ensure climate and related landscape changes, risks, impacts and required management responses are integrated

2.1.1.2 Provide technical and financial support to planning, research and monitoring as well as community-based conservation units within RDB in managing the parks as part of larger CND landscapes

2.1.1.3 Train PA staff to integrate gender and social inclusion into programming

2.1.1.4 Train PA staff on integrating needs of women, youth, historically marginalized and Ubudehe Categories c,d,e populations into climate adaptation risks and responses

2.1.1.5 Train PA staff on SEAH, GRM, FPIC, and Access Restriction Mitigation Processes to ensure NP climate change responses are undertaken in a way which supports social inclusion and equity

Activity 2.1.2 Establish long-term plans for CND financial sustainability post-GCF

Rwanda has recently developed a Biodiversity Finance Plan⁹⁷ through support from BIOFIN. This plan has identified a number of financing solutions that will not only contribute towards improved biodiversity protection in Rwanda, but also aligns with Rwanda's sustainable development and green growth objectives. These solutions include:

- Introduce a Biodiversity Conservation Fund into Rwanda's National Fund for the Environment (FONERWA) to help streamline and attract domestic and international sources of finance to support conservation investments and thereby reduce transaction costs to increase conservation benefits.
- Improve efficiency and effectiveness of environmental fees and fines to strengthen the effectiveness of the regulatory environment that will ultimately improve ecosystem and biodiversity conservation goals.
- Water User Fees for Catchment Management to secure the financial resources necessary for effective water catchment management.
- Pilot Business Plans for Selected Wetlands to enable the sustainable development of wetlands for ecotourism and biodiversity-friendly enterprises such as handicrafts, fisheries, and sustainable agriculture.
- Promote biodiversity-friendly enterprises in the transition to a green economy and accelerate the transition to a green economy by incentivizing and supporting businesses to adopt sustainable practices and attracting investments in new conservation enterprises.
- Strengthen the tourism revenue sharing scheme to improve biodiversity outcomes and effectively address conservation-development objectives
- Develop a Protected Area Finance Strategy to support the goal of increasing the financial and ecological sustainability of Rwanda's Protected Area System (PAS)

There are also new emerging solutions such as the Wildlife Conservation Bond (WCB) which essentially is a payment for performance-based financial instrument that channels investments to achieve conservation outcomes – measured in this case by specific conservation targets that could be set (e.g., hectares of native forest restored, or an increase in umbrella species populations). In the case of Rwanda, gorillas and

chimpanzees are considered umbrella species that play a crucial role in shaping entire ecosystems on which countless other species depend. For the WCB, a grants-based, outcome payer is needed, which could be the GCF. Through the WCB, and in collaboration with the World Bank (WB), the WB could issue bonds and investors could support the financing of activities to protect and grow a critically endangered species with clear conservation targets, contributing directly to biodiversity and bringing jobs to local communities through the creation of conservation-related employment in rural areas of the CND. A third-party, independent verifier is also needed to evaluate whether conservation targets have been met. A meeting is planned with the WB in the second half of February 2023 in Rwanda to determine the scale and feasibility of the WCB in the CND. The South African Rhino Bond has already attracted foreign investment, and it is a good model Rwanda could emulate⁹⁸. The Government of Rwanda, represented by the Rwanda Development Board, is in advanced stages of negotiation with the World Bank for establishing a wildlife Bond. There is, therefore, clear interest in the bond which suggests that investors still see value in investing in such an instrument.

A technical advisor will be recruited by this project to provide technical support to RDB, REMA, MINECOFIN and MoE and explore these various solutions to identify financial instruments that could be employed for revenue generation post-GCF for sustainable management of the CND landscape. The technical advisor will work closely with the NSC to facilitate buy-in from stakeholders such as RDB and MINECOFIN into compliance with environmental laws through greater emphasis on the links to climate resilience and NDC objectives.

Activity 2.1.3 New fire management curriculum developed and operationalized Given the expected increase in drought, and the natural forest vulnerability already demonstrated in the past, the capacity for effective fire prevention and management in core PAs and adjacent landscapes will be increased.

While current climate projections call for increased rainfall in the CND, they also project higher temperatures and a rise in elevation of the natural forests' cloud cover zone (Annex 2.1). The combination of the latter two trends raises the potential for increased fire risk. In order to sustainably address the threat of fire to the natural forests and regenerated areas in particular, this project will increase the capacity of RDB in the area of fire prevention, with emphasis on education and outreach, monitoring of fire danger conditions across the parks, response plans, and fire suppression. A collaborative effort in fire management planning and implementation in CND districts will be carried out, facilitated by a fire management expert. Based on training needs, the project will also carry out RFA district and sector foresters' training in Integrated Forest Fire Management approaches and activities and firefighting; they will then build the capacity of community fire brigades.

Sub-activities include:

2.1.3.1 Develop a curriculum tailored to needs and capacities of different stakeholders

2.1.3.2 Build the capacity of RDB to manage fire in National Parks

2.1.3.3 Build the capacity of RFA, local authorities and communities to collaboratively manage fire in forests outside Pas

2.1.3.4 Implement a "fire wise" outreach and awareness program for communities and local government (district, sector, cell), as well as Rwanda Forest Authority, surrounding natural forests to reduce the incidence of human-caused fires.

⁹⁸ World Bank, 2022. *Wildlife Conservation Bond Boosts South Africa's Efforts to Protect Black Rhinos and Support Local Communities*. Press Release NO: 2022/059/AFE.

Output 2.2: Natural Forest cover restored, biodiversity connections established.

The natural forests of the CND exist as an archipelago of isolated islands within a surrounding matrix of intensive human land use. Although some have raised the possibility of reconnecting these islands, it is important to understand that their isolation is not a recent phenomenon (with evidence of the initial isolation of the main forest blocks from the pre-colonial era). Furthermore, large-scale proposals for landscape linkages have thus far failed to provide alternatives for the tens of thousands of rural smallholders who would be displaced by such schemes. However, important actions can be taken to restore degraded natural forest lands and protect or reestablish natural “stepping stones” and ensure a functional level of ecological linkage between the existing parks.

In some areas where natural forests have been cleared, the Project will work to restore forest cover. This will be done primarily around the Gishwati and Mukura reserves, using native species and building on nursery and planting techniques already tested in Rwanda. New techniques may need to be developed to expand reforestation to degraded lands on steep slopes and along waterways and wetlands. Tree species will be carefully selected considering the following attributes: pioneer species which grow in full sun, fast growing (height and canopy width), and animal pollinated and dispersed. These “tree islands” serve several purposes. First, by growing quickly and providing canopy cover, they shade out competing vegetation and change the soil microclimate (very important for later successional species establishment). Site-generalist native species such as *Albizia*, *Dombeya*, *Macaranga* and *Polyscias* respond to these characteristics. Secondly, they entice a variety of animal species into the expansion area, thereby dispersing seeds. In certain areas the Project will favor natural regeneration from seed banks.

All work outside of protected areas will require careful attention to private land ownership claims and related social and gender equity issues. Part of the communities’ responsibilities will be afforestation of the degraded forest with indigenous trees and clearing invasive species from the natural forests and buffer zones. Part of the benefits for the communities will be harvesting of non-timber forest products (NTFPs) from the natural forests, under sustainable use plans. The project will provide training on improved harvesting techniques, processing, packaging and marketing, to those engaged in NTFP value chain (financed under outcome 3).

Activities and sub activities under this output will be executed by the Rwanda Forestry Authority (RFA) with the technical support from WCS.

Activity 2.2.1 Secure key remaining natural areas outside PAs

Outside of current protected areas, the Project will identify remaining natural areas and seek to secure their continued protection within district management plans or through other means. These are mostly small stands of natural vegetation on rocky outcroppings or strips of forests along the CND’s myriad of waterways. Although these will not permit large mammals (e.g. elephants or mountain gorillas) to migrate between the forested parks, these small remaining natural areas provide critical intermediate stepping stones for the broader biodiversity elements of the CND forest ecosystems. For the CND’s rich assemblage of bird species, these vegetated patches and gallery groves could prove to be essential stepping stones for connectivity under changing conditions. Small mammals and insects are likely to also directly benefit. And most importantly, the movement of birds and small mammals will support the transfer of seeds between forest patches, greatly improving the ability of forest ecosystems to adapt to changing climates, and continue supplying key ecosystem services to the people of the CND.

Sub-activities include:

2.2.1.1 Raise awareness on remaining protected natural forests in CNDL to secure their protection

2.2.1.2 In collaboration with RFA, REMA, districts identify and implement actions that support conservation and management of remnant protected natural forests in CNDL

Activity 2.2.2 Restore natural forest cover in and outside Protected Areas including riparian linkages

While the boundary integrity of the Nyungwe and Volcanoes NPs has been maintained for several decades now, internal degradation has occurred. Prior to 2004, approximately 12% of Nyungwe National Park had been affected by wildfire, including a catastrophic fire during a period of prolonged drought in 1997.⁹⁹ The natural vegetation was quickly replaced by a carpet of invasive ferns. In recent years, teams of local community workers led by WCS have demonstrated that careful removal of the fern cover reveals still viable native tree seeds and seedlings that respond to this exposure by rapid growth and the formation of new forest stands. The project will expand this labor-intensive process of assisted regeneration to the remaining 4,500 ha of burned lands within the NNP, restoring forest values and generating significant employment for local rural communities.

For the Gishwati Forest, restoration with native species will take place on adjacent, previously cleared lands. This and other restoration efforts should benefit from tree nursery techniques and planting trials already initiated by RFA, RDB, and NGO partners around Nyungwe, Volcanoes, and Gishwati itself. The project will work with the National Tree Seed Centre to provide training on tree propagation and supply of quality seeds/seedlings/propagation materials, especially of indigenous species and plantation species not being sourced locally.

A special element within the CND landscape is its dense network of waterways. Historically forested riparian habitats along these streams and rivers, however, have been largely cleared of their forest cover. Riparian reforestation efforts will restore an important erosion control factor as well as recreate important biodiversity connectivity along linear corridors. RWB will co-finance these riparian restoration efforts. With support from district and sector leaders as well as community representatives, community members will be hired from neighboring villages to the project sites. The project will make efforts to recruit the most vulnerable community members in category c,d and e, women, young landless, and others to carry out these activities.

Key sub-activities include:

2.2.2.1 Review mapping of degraded natural forest areas in core NPs, stepping stones, and unprotected riparian lands using updated imagery and ground truthing

2.2.2.2 Identify parcels for restoration in core PAs, stepping stones, and riparian lands using desktop and field-based assessment

2.2.2.3 Establish indigenous tree seed nurseries in the CND to serve core PAs, stepping stones, riparian land restoration and promotion of indigenous trees on farms and in protective forests

2.2.2.4 Recruit, train (in forest restoration methods), and equip community workers to be involved in core PAs restoration and riparian linkages

2.2.2.5 Assisted rehabilitation of 4,500 ha of indigenous forest in NNP

2.2.2.6 Active restoration of natural forest on 500 ha Gishwati Mukura National Park

2.2.2.7 Restore 1,500 ha of riparian lands

⁹⁹ Masozera, A.B., Mulindahabi, F., 2007. *Post-Fire Regeneration in Nyungwe National Park, Rwanda*. Wildlife Conservation Society.

2.2.2.8 Using permanent plots sampling, collect field monitoring data on tree species recruitment and growth for estimation of biomass, species richness, etc. in restored parcels in Nyungwe and Gishwati-Mukura National Parks as well as on riparian lands to assess success of initiatives, then replicate or adapt as needed

2.2.2.9 Perform Environmental and Social Screening on all positions being created by the project

2.2.2.10 Establish a grievance process for laborers

2.2.2.11 Update emergency and preparedness plan including risk mitigation guidance to local conditions at restoration sites

2.2.2.12 Train all workers on their rights and how to access the GRM

2.2.2.13 Provide code of conduct and emergency preparedness and safety training for all laborers

2.2.2.14 Hire and train labor and safety leads at each site to provide guidance to staff, be available for grievance issues, and monitor health and safety conditions for workers

2.2.2.15 Develop and implement strategy (including social marketing) to ensure that women and youth participate and benefit from forest restoration projects

Activity 2.2.3 Promote silvopastoralism with indigenous trees around Gishwati

Roughly 80% of Rwandans own livestock of some sort. While it is government policy to expand this percentage, conflicts over competing land uses are inevitable, especially for the country's most vulnerable rural populations. In the area around the protected forest of Gishwati, the presence of many slopes greater than 55% opens the possibility for reforestation of larger blocks with tree species compatible with an undergrowth of grass or other forage. In addition, some existing pasturelands could be enriched with tree species. These areas could then provide critical off-farm fodder for goats, sheep, and perhaps even cows, while also serving vital ecosystem services.

Plans for tree planting/restoration on pasture lands will be elaborated in collaboration with the local communities and technical (extension and local government) services. Areas meant to produce fodder and for water retention will be demarcated during community consultations for action plans. Already, important forage grasses, such as *Brachiaria* grass known to increase milk production while reducing methane emission from cattle, have been identified while tree fodder such as *Calliandra*, *Leucaena*, *Acacia*, *Gliricidia* are well accepted but not widely adopted due to local capacity barriers. Improving adoption of better management practices will require technical assistance considering needs of cattle owners, local milk cooperatives, and traders.

Beneficiaries of these activities will be pasture land owners and cattle keepers in the Gishwati area. They will be identified in collaboration with district and sector leaders, RFA, EE and representatives of pasture lands and cattle owners. Special attention will be paid to specific groups such as women-headed households, youth, and historically marginalized groups if they own grazing land in the Gishwati area and for paid work. Awareness and consultations with these pastoral communities will be conducted during project inception to ensure farmers' involvement in identifying and selecting key areas for restoration, suitable species to grow in their pastoral areas, capacity barriers, and associated technical assistance to be provided to adopt better management practices.

Key sub-activities include:

2.2.3.1 Assess the status of pasture lands in the Gishwati area and identify key areas for restoration and the potential to introduce indigenous species

2.2.3.2 In collaboration with landowners, identify suitable indigenous species for fodder trees, shrubs, grasses, and herbaceous legumes that have potential to improve rangelands and increase their climate adaptive capacity

2.2.3.3 Produce and disseminate fodder trees, shrubs, grasses, and herbaceous legumes to project beneficiaries

2.2.3.4 Train beneficiaries on improved livestock and pasture management

Component 3: Enhancing climate adaptation through resilient livelihoods

Increasing the extent of protective forests, enhancing and restoring forest plantations, and increasing agroforestry tree cover are key steps to restoring ecosystem function and services identified in the National Forest Policy Report (NFPR 2017), especially i) reducing the risk of flooding, landslides and soil erosion from extreme climate events and ii) increasing long-term supply of wood fuel resources (the primary energy source for 77% of Rwandans¹⁰⁰). Together with fuel efficient cookstoves, these trees will provide a sustainable source of fuelwood and reduce the amount of time women spend collecting fuelwood, enabling them to participate in other economic activities and programmatic opportunities. These outputs (cookstoves, plantation restoration, and agroforestry) will deliver carbon sequestration benefits totaling 522,606 tCO₂eq over 20 years.

In working to develop a comprehensive land use plan that incorporates climate adaptation, and while ensuring Rwanda's forest ecosystems are strengthened through forest management and restoration, attention must be paid to people who live next to the forests and those who are the most prone to causing degradation of these forest systems. In particular, this includes people who either do not have access to land or whose land holdings are insufficient to meet their daily needs. These people are most likely to enter into and unsustainably use protected and protective forests to help meet their daily needs. Thus, this Project component targets these vulnerable households – youth, people from historically marginalized groups, and farmers with insufficient holdings – to build their capacity in financial literacy and enterprise, in order to strengthen their resilience to economic and climate shocks.

Also, to maximize the opportunities from improved silvicultural practices, it is necessary to improve business practices and market linkages for farming communities. More specifically, organizing and linking farmers to local and national wholesale traders will enable them to sell their produce more efficiently, maximizing profit and reducing wastage. To facilitate these linkages, this component will also build the business capacity and market linkages necessary to support the transition of farmers and their communities away from unsustainable practices that magnify their vulnerability to climate change and towards more secure livelihoods and expanded income opportunities. This includes strengthening capacities, developing inclusive business models, and developing market value chains for livelihoods that do not depend on traditional smallholder farming.

The project will develop alternative sources of income to improve living conditions for people in the CND who depend for a large part of their income on exploitation of the forest and natural vegetation. This approach, focusing on vulnerable groups, aims at real economic empowerment of the concerned families through the realization of their own business plans. Several promising activities related to the use of the forests *aside* from wood-gathering (such as beekeeping, extraction of essential oils, herbs, and other forest

products) will be analyzed. These business plans, tailored to individual circumstances, will be realized with support and tools provided by the project.

From the second year onwards, these vulnerable groups will become members of existing cooperative organizations that will be supported by the project, particularly in value chains important for forest conservation, such as production and export of honey and the extraction of essential oils. These cooperatives provide various services (e.g., information on markets / export, input supply, aggregation of commodities, etc.) and they also play a role in the protection of the interests of their members.

The project will develop the private sector strategy (which will include engagement of impact investors interested in conservation-friendly commodities) for livelihoods implementation in the target areas through the identification of new livelihood opportunities for the CND. We will establish relationships with several buyers fitting our selection criteria, and engage partners such as Blue Ventures to expand market opportunities for products that meet international or local market opportunities. On the producer side, we will build strong relationships with other projects and programmes that support private sector engagement with community organizations, and producer associations to pursue promising market opportunities (local and international, as appropriate).

For component 3 and the project overall, the eligibility criteria for selecting beneficiaries will be:

- Households with limited access to financial resources for implementing climate adaptation measures as determined by Local Authorities
- Women -headed households.
- Households in the CND living in key areas such as those surrounding PAs, identified stepping-stones, landscape linkages, and broader farming mosaic with exposure to erosion.
- Most vulnerable households including people with disabilities
- Land owners where the project intervention will have to intervene

These criteria will be further refined by the project EE in consultation with key stakeholders (MoE, Districts and community representatives) once the project is approved.

Output 3.1 Farming methods enhance productivity, reduce erosion and flooding risks, contribute to ecosystem services, and support connectivity

Soil degradation and loss, largely from the cultivation of steep slopes in the western highlands of the CND, has caused declines in agricultural productivity and serious downstream problems of siltation and flooding. While the former means that smallholder farmers in the hills increasingly struggle to feed their families, the latter poses a threat to the long term viability of water provision services in municipal areas and hydro-power generation¹⁰¹, which is a key pillar for Rwanda's low carbon growth strategy.

This Project will support smallholder farmers in high erosion risk areas to implement Sustainable Land Management (SLM) practices techniques and the active *restoration* of degraded forest lands. By targeting high risk areas the Project can have a significant impact to reduce soil erosion, improving the resilience of agricultural livelihoods in the highlands, while having a positive impact on water flow and water quality in the lowlands.

¹⁰¹ Rwanda Environment Management Authority (REMA), 2009. *Rwanda State of Environment and Outlook Report, Chapter 8: Energy Resources.*

In each selected priority site, the Project will support the implementation of the full range soil erosion control best practices including agroforestry. While the erosion control practices may vary (e.g. radical vs progressive terracing), agroforestry practices will be established across the entire 3,346 ha. RFA will mobilize participating landowners on whose lands soil erosion control best practices including agroforestry shall be implemented. Landowners willing to participate will sign a consent form allowing project activities on their land and will be grouped in vigilance committees. These vigilance committees will sign MoU with Districts and RFA as a commitment to participate and manage sustainably their land and trees with clear role and responsibilities of each (farmers, districts and RFA) specified. The consent forms will be granted to RFA and Districts where landowners are located. The vigilance committee will be responsible for ensuring the sustainability of project interventions implemented on community land. The MoUs signed by Districts, RFA and the vigilance committee will not be legally binding but rather to establish certain understanding in the implementation of the project.

Interventions will be made in collaboration with district officials, according to agreed priorities and objectives, and involving local farmers and contractors. Table 29 in the feasibility study (FS) provides a preliminary list of the fuelwood, fodder, timber and fruit tree species that have been identified as suitable for agroforestry systems in the CND, based on their resilience capacity against climate-driven temperature and precipitation changes, as well as community preferences - particularly the needs of women farmers, historically marginalized households, and youth. They also concur well with recommended species for the North and Western Province as described in the Agroforestry Technical Guide¹⁰².

Activity 3.1.1 Restore high slope areas (>55%) as protective forests

In Western Province, a major part of the CND landscape, 80% of the increase in tree-planting in recent years has come from private landowners seeking to earn revenue or increase their access to wood-based fuel¹⁰³. Beyond this utilitarian interest, however, the NFPR 2017 calls for increasing the extent of protective forests and enhancing and restoring forest plantations as key steps to restoring ecosystem function and services, especially i) reducing the risk of flooding, landslides and soil erosion from extreme climate events and ii) increasing long-term supply of woodfuel resources (the energy source for 77% of Rwandans¹⁰⁴), providing biodiversity habitat, and sequestering 193,472 t CO₂-eq over 20 years.

The current land use master plan discourages cultivation on slopes greater than 55% and encourages protection of existing vegetation and afforestation with indigenous species. This Project will promote indigenous species in rehabilitating existing forests and establishing new ones on bare lands. This activity will result in highly productive, climate-resilient woodlots and forestland with fully restored ecosystem services and significantly increased long-term carbon sequestration. It will afforest/restore 2,500 ha of degraded private smallholder land and district/state land by restoring existing degraded tree stock, afforesting bare areas, promoting good silvicultural practices, and facilitating the adoption of Simplified Forest Management Plans (SFMPs), as recommended by 2013 forest law. In the priority regions selected for afforestation/restoration, awareness and capacity building activities will be undertaken to build support and understanding of the PFMU approach. The objective of PFMUs is to combine economic profitability, preservation of environment, and improvement of living conditions of the local population, as well as to apply a gender-responsive approach to wood resource management. For privately owned land, only small-scale individual private landowners will be included, and this activity will not involve restoration of private institutional plantations. RFA will mobilize participating private landowners on whose lands afforestation

¹⁰² Ministry of Environment. 2018. National Agroforestry Strategy.

¹⁰³ Arakwiye, B., et al, 2021. *Thirty years of forest-cover change in Western Rwanda during periods of wars and environmental policy shifts*. Regional Environmental Change 21(2).

¹⁰⁴ Government of Rwanda, 2021. Rwanda Household Survey 2019/2020. National Institute of Statistics of Rwanda.

trees shall be planted. Landowners willing to participate will sign a consent form allowing project activities on their land. Following this, private woodlot owners will then be organized into local groups (around 40-50 ha of woodlot per group), PFMUs (Private Forest Management Units) and assisted to form and register as cooperatives and elect among themselves two cooperative committees: Administrative committee made of 5 persons (President, Vice president, Secretary and two Advisors); supervisory committee made of 3 persons (President, vice president and secretary). Note that gender consideration is mandatory (at least 30% women).

When the cooperative committees are elected, they are mandated to represent forest landowners at different events and fulfill their interests, especially in seeking the best market for their forest products and seeking for other income generating opportunities. RFA will sign MOUs with different cooperatives specifying each party's respective role and responsibility.

Experiences from pilot PFMUs established under the FMBE project in the districts of Gicumbi, Gakenke and Rulindo Districts of the Northern Province and Rwamagana District of the Eastern province have demonstrated that this approach is helping to achieve economic profitability, preservation of environment and improvement of living conditions of the PFMU members. Forests are sustainably managed, and all PFMU members respect the management plan and planned harvesting period. All silvicultural practices are properly followed because cooperatives supervise all activities with technical guidance from the district technicians. The forests within a PFMU are economically attractive to potential buyers due to good stock at harvesting time. PFMU members have access to small loans while waiting for the forest to mature. This shows that PFMUs established in these areas are also serving as demonstrations to other forest owners supported by other projects in the country. Green AMAYAGA and Green Gicumbi, and TREPA Projects are scaling up this approach in other parts of the country.

On public land, forests will be organized into state/district FMUs to be managed by professional private companies through long-term concession agreements in line with the government strategy. While a concession agreement will be signed between the private forest companies and the MoE, the Project will only support the design of the forest management plans and development of contracting documents. Registration of restored forest parcels will be guided by well-trained district forest officers and forestry sector extensionists, assisted by the new FMES software and related GPS/tablets, which will provide automatic statistics, maps and register owners. For each FMU, a SFMP, reviewed or developed through a participatory process, will outline the basic information of the site (e.g. owner, species etc.), as well as a harvesting plan and schedule for silvicultural operations (thinning, pruning etc.). However, some areas of the state plantations that are vulnerable and exposed to high risk of degradation (high slopes) will require setting up special protective measures. Restoration on identified vulnerable areas will be done by establishing appropriate tree plantations that will offer a better protection and conservation of biodiversity while providing other ecosystem services including tree species that improve forage for bees. The choice of tree species planted will take into consideration women's participation, use and priorities. The project will further strive and advocate for gender equity for selection of plots which will be done in consultation with communities.

Sub-activities include:

3.1.1.1 Introduce and raise awareness of indigenous species to target stakeholders in CND

3.1.1.2 In consultation with RFA, National Land Authority, districts and communities, determine fragile areas (steep slopes > 55%) to be allocated for protective forests and their ownership

3.1.1.3 Assess the current status of the indigenous tree species in selected areas for protective forests and select indigenous tree species appropriate to CNDL

3.1.1.4 Develop restoration plan for protective forests

3.1.1.5 Build capacity of local stakeholders (men and women) on PFMU approach and methods

3.1.1.6 Design and approve SFMPs of private FMUs

3.1.1.7 Ensure consent of smallholders prior to planting trees

3.1.1.8 In collaboration with smallholders reforest/restore 2,500 ha of public or private land with slopes >55% and ensure sustainable management under private FMUs according to approved SFMPs

3.1.1.9 Support monitoring and evaluation of restored private FMUs

3.1.1.10 Assess impacts of exotics on neighboring lands and mitigate their negative impacts

3.1.1.11 Develop and implement strategy (including social marketing) to ensure that women and youth participate and benefit from forest restoration projects

3.1.1.12 Assess the benefits and costs of each proposed tree and plant species and how those benefits affect different population segments

Activity 3.1.2 Develop on-farm agroforestry for high-caloric and indigenous tree species

The extreme land scarcity of the CND precludes smallholder farmers – many of them among the most vulnerable and land-poor of the region – from dedicating any significant proportion of their holdings to tree production. However, the use of certain agroforestry species that both enrich the soil and control erosion when interplanted with ground crops can greatly reduce this conflict.

Under this activity, the Project will facilitate the production of on-farm afforestation plans with action plans for implementation. Formulation of the forest landscape restoration plans will follow the methodology introduced by the World Resources Institute (WRI) and IUCN and already tested in the country by the former Ministry of Natural Resources, as recently modified and applied for the Gatsibo Forest Landscape Restoration baseline conditions assessment. The methodology will involve three simple steps: a) Geospatial analysis to map degraded land that presents an opportunity for forest and landscape restoration (FLR), which will highlight areas with best potential for restoration; b) economic analysis to model the costs and benefits of degraded and restored land; and c) designing a restoration/afforestation action plan, based on an in-depth assessment of the conditions required to implement the FLR in the selected sectors.

For selected areas where Project activities may lead to reduced access to resources, the Project will undertake an in-depth Environmental and Social Impact Assessment (ESIA) in the first year of implementation and design an Environmental and Social Management Plan (ESMP) to guide implementation. This ESMP will include an access restriction mitigation plan, if deemed necessary. The matter of Free, Prior and Informed Consent (FPIC) will be explored during the ESIA and the approach applied if deemed appropriate. These documents will be based on materials contained in Annex 6.

Drawing on experience with high-elevation species identified both in Rwanda and across East Africa, the Project will support the following sub-activities:

3.1.2.1 Identify sub-areas of intervention for agroforestry dissemination in the CND

3.1.2.2 Introduce and raise awareness of agroforestry in target communities

3.1.2.3 Develop the capacity of extension agents at district/sector level and NGOs to support adoption of agroforestry technologies

3.1.2.4 Establish agroforestry/fruit tree nurseries to facilitate access to quality planting material

3.1.2.5 Promote sustainable land management practices by stabilizing existing terraces, and plant high calorific agroforestry species to provide a sustainable source of fuelwood for energy-efficient cookstoves to protect sloping land against severe soil erosion.

3.1.2.6 Train RFA, extension agents, Project participants and community members on specific techniques for identification of and management of invasive pests and pathogens

Participants in sub-activities 3.1.2.3 and 3.1.2.6 will be selected by districts, JADF, RFA and farmers representatives. Special attention will be given to youth, women and other vulnerable groups. RFA and RWB will co-finance sub-activities 3.1.2.4 and 3.1.2.5.

Output 3.2 Rural livelihoods generate alternative incomes & reduce pressure on forests

To respond to the growing pressure on natural resources and increase the local community's resilience to climate change in the CND landscape, the Project will address key drivers of deforestation by creating alternative livelihood pathways for youth and women through ecotourism and income-generating, climate-resilient crops, as well as strengthening supportive financial services.

The setup of interventions in this output will focus on enhancing on-farm production capacities in market-facing promising value chains in agroforestry (especially avocado, macadamia and short-duration crops) for immediate income and beekeeping taking advantage of the agroforestry and the crops; the combination of these will promote ecotourism as supplementary income (avocado, macadamia, forestry, agroforestry, short-term crops, and honey). This output will also provide solutions to the growing households' energy poverty which has exerted pressure on natural resources and eventually exacerbated the local community's exposure to the stress of climate change in the CND landscape.

This project will employ a market systems development approach that encompasses business coaching, value chain development, and access to finance. This entails multi-stakeholder collaboration with business service providers, financial institutions, community-based enterprises, value chain actors, and farmer cooperatives. Ultimately, this project will develop and or promote a financial model that enhances the private sector ecosystem collaboration and partnership to stimulate finance in forestry and agroforestry sectors and support market-oriented access to energy-efficient cook-stoves to reduce pressure on forests.

The value chain development for potential revenue generating streams such as honey, ecotourism, and efficient cookstoves requires grant support of GCF grant to finance the activities that are important for the most vulnerable, as well the drivers for the MSD to take root; these include the development of the identified revenue generating streams, strengthening access to market information, and improving quality of the products and produce aggregation.

To target the poorest and most vulnerable, and empower women and youth towards meeting inclusive development goals, critical support at the nascent stages is essential. The GCF grant will facilitate the integration of the poorest and most vulnerable women and youth in the CND and position for private sector investments that will build a firm back base for sustainable community development. The GCF grant will not only provide financing support but will be critical in creating enabling opportunities for market development beyond the project to catalyze private investments.

While promoting the above-mentioned households cooking solutions and income streams, the selection of the beneficiaries will follow the government's procedures, and in the case of CND will focus on the vulnerable groups of youth, women, and historically marginalized people and will be selected based on an assessment

of poverty levels (also known as Wealth Categories) conducted by the government to ensure the most vulnerable are selected. Selection criteria will further take into account social and cultural factors such as gender and age which intersect to predispose some groups more to vulnerability than others.

Activities and sub activities under this output will be executed by the Rwanda Forestry Authority (RFA) with the technical support from CORDAID.

Activity 3.2.1 Develop agroforestry related value chain for markets access.

Most traditional tree farming in Rwanda is linked to fuel production in terms of firewood, charcoal, and timber for construction. However, to further strengthen community resilience to climate change, there is a need to introduce fruit and nut trees that have high economic value in the agroforestry system to generate more income for smallholder farmers but also increase its adoption, especially women-headed households. Though women-headed households face the same issues and challenges as male-headed households, they are less likely to be included in income-generating activities and are disproportionately impacted by climate change because of prevalent gender inequality issues. In the Feasibility Study, avocado and macadamia have been identified as high economic value crops with high market potential, and the economic benefits of identified fruit trees is paramount to increase farmer adoption and buy-in. Since fruits and nuts take time to mature, short-term crops like chili and French beans will be promoted to support smallholder farmers with limited land resources (less than 0.4 ha) to create additional, more quickly generated income and incentivize adopting agroforestry practices. Regular income will help vulnerable communities to cope with climate variabilities and shocks.

The identified agroforestry value chains have market potential and potential for intercropping with short-term horticultural vegetables and fruits while providing further impacts on carbon sequestration, nutrition value, and market access in the CND. Selected fruit trees are expected to meet the dual purpose of soil protection and climate impact while generating long-term economic benefits to the most vulnerable community members of the CND region.

For each crop value chain identified, specific bottlenecks will be addressed while focusing on specific product-place combinations within the CND. Especially, attention will be on enhancing the quality and quantity of production in sync with particular demands from market niches, development of various models to improve access to finance together with financial institutions, and exploration of market opportunities that can generate off-farm jobs to absorb a good number of landless and other vulnerable groups of youth and women.

These priority crops are:

- Macadamia: a low-volume, high-value export crop; in-country processing creates jobs for women and youth. Two hundred and four trees cover one hectare, and one mature tree produces up to 60 kg of macadamia nuts per year with an income of 25,000 FRW per kilogram. Furthermore, every tree must be separated from the next by 7 meters; thus, many other commercial and consumable crops may be planted and harvested while waiting for macadamia's fruiting maturity.¹⁰⁵ The global demand for macadamia outweighs the available supply, which presents a huge business opportunity for macadamia farmers in Rwanda. For example, the current macadamia supply meets only 48% of global demand and the global market for macadamia is expected to further expand from \$1.17 billion to \$2.36 billion by 2027. The project will build on the government export initiatives to promote

macadamia farming in CND areas through National Export Board (NAEB) and Rwanda Environment Management Authority (REMA) to include macadamia in its tree planting drive in the CND.

- Avocado: The extensive penetration of avocado cultivation amongst smallholder farmers (SHFs) and the favorable growing conditions lead to avocado yields (21MT/ha) that are nearly double those of Kenya. The increasing demand for export, and cash yield/ha are higher than maize or Irish potato. Through linkage to local avocado private value chain actors, packaged avocados will be exported to the European Union, where the channel price for avocados is \$1.75 per kilogram, providing an estimated 32% margin.¹⁰⁶ Avocado production presents numerous opportunities to smallholder farmers in CND: for example, earnings from avocado exports have been increasing from \$1.6 million in 2020/21 to \$4.5 million in 2021/22 and production increased from 1,000 tonnes to 2, 500 tonnes in 2021/22. According to the NAEB, the production is set to double this year and generate \$8 million from exports. To further strengthen its avocado exports the government has launched sea freight, which will not only increase export volumes but also make them competitive in terms of pricing.
- Avocado is projected to be the second most traded tropical fruit, after bananas, by 2030. The United States and the European Union, where consumer interest in avocados is fuelled by the fruit's health benefits, are expected to remain the main importers, with 40% and 31% of global imports in 2030, respectively¹⁰⁷. The project teams in CND will work closely with private sector partners and communities to develop value chain analysis, business development strategies, and business planning. The team, led by our Livelihood Specialists, will work with these private sector partners to prepare action plans detailing technical services we will provide to train producers to supply commodities to markets.
- Vegetable crops such as French beans and chili: It has been demonstrated that these crops can double the income per hectare of maize and potatoes and employ a good percentage of the 16% reported unemployed women and youth¹⁰⁸. Commercial farm wage reported labor in postharvest production handling at 72% women, 20% youth (see Annex 2.3). French beans can potentially deliver a gross margin of 22,040Rwf per acre and a return on labour of 2,755Rwf per day, while chili can offer a gross margin of 46,384Rwf per acre and a return on labour of 7,731Rwf per day.¹⁰⁹

The Project will also enhance ecotourism and honey value chains. Particularly in areas close to national parks, the potential business development of ecotourism can go hand in hand with honey and wax product development. Our analysis shows that the honey value chain already has a sizable production history in Rwanda. Various initiatives exist that have encouraged farmers to invest in modern hives. However, the current sector is constrained by a perception of limited profitability due to low prices, gender patriarchy due to male dominance, and limited market access. The honey (and various by-product) value chain shows a wide range of different consumer target groups, whereas honey is too often marketed as a generic product. This is a crucial bottleneck: while the value chain actors engage in the business, there needs to be more incentives to invest financial resources to improve production or quality, such as through modern beehive technology. It will be essential to link ecotourism destinations and agro-tourism to enhance the potential for an improved business case and help beekeeping cooperatives enhance their processing, quality management, and marketing processes with large scale buyers and exporters such as Api Business Development Company (ABDC) and create a sustainable and expanded source of income especially for

¹⁰⁶ Karuiru, M., 2018. Value chain management and the performance of avocado fruit small scale farmers in Kandara sub county, murang'a County, Kenya. *International Journal of Physical and Social Sciences*, 8(10), pp.24-40.

¹⁰⁷ OECD - FAO Agricultural Outlook. <https://www.fao.org/3/cb5332en/cb5332en.pdf>

¹⁰⁸ Rwanda Development Board, 2020. Labour Market Trends Analysis Brief 2016-2020. <https://rdb.rw/wp-content/uploads/2022/07/Labour-Market-Brief-2020.pdf>

¹⁰⁹ Van Keulen, Rik, Rumenera, P., Banjara, G., Colantuoni, M. and Muthamia-Mwenda, J. 2022. Horticulture value chain analysis – Opportunities for youth employment in Rwanda. Rome, FAO. <https://doi.org/10.4060/cb8266en>

women. Api Business Development Company has applied for EU certification in order to export honey to international markets. Nevertheless, the company—which has a capacity of handling 120 MT of honey per year—still faces the challenge of low production, inadequate even in meeting local demand.

Rwanda has a well-established record of generating off-farm benefits from its protected area ecotourism programs. The Project will build on this by supporting sustainable tourism activities around and between the Gishwati and Mukura Forests, as outlined in the GMNP Tourism Master Plan.¹¹⁰ New trails and base camps outside of the parks will generate employment and income opportunities for rural women and youth, as well as attract tourism revenue to help support forest protection and management. In addition, this Project will initiate and train Community Freelance guides to mutually benefit the parks and local communities neighboring Mukura-Gishwati NPs. A freelance system suits the seasonality of tourism, and the demand and sourcing of guides from the local communities stimulates economic empowerment among the young and unemployed living in rural areas around protected areas. This initiative will enable community members to provide a service and generate income, and benefit from the growing tourism industry in Rwanda, which in turn supports conservation over the long term.

In collaboration with district authorities in the CND, who will provide a list of existing cooperatives, the project will select cooperatives based on value chains, number of women and youth, number of members with vulnerabilities and membership size. Furthermore, we will assess selected cooperatives using the Cooperative Assessment Matrix tool (CAM) to identify capacity gaps. Selected cooperatives, participating financial institutions (PFIs), and value chain actors/small and mid-size enterprises (SMEs) (including seedling producers, aggregators, traders, etc.) will receive capacity-building to improve their business skills and access to finance, and to strengthen value chains.

Women play a crucial role in balancing household demands with market production. This is why it is so important to give women access to financial and value chains related services. The project aims to reach at least 50% of the women beneficiaries in the project area. In areas where women are under-represented in value chains, the project will identify and address gender-specific challenges and barriers related to the different value chains and promote increased female representation and active participation in Producer organizations by creating gender committees that will recruit more women, and support women to be part of the leadership committees.

Key sub activities are:

3.2.1.1 Identify and map of vulnerable groups of youth and women within the community

3.2.1.2 Develop and implement strategies to ensure that women, youth, and other marginalized groups benefit from rural livelihoods and financial services.

3.2.1.3 identify farmers and cooperatives involved in selected value chains in consultation with district/local authorities in CND.

3.2.1.4 Assess capacity gaps of identified farmer organizations to strengthen their capacities

3.2.1.5 Identify and develop Nature-based tourism value chain as a means to generate viable economic alternatives for local people living near areas of high biodiversity in the CND.

RDB will co-finance sub-activity 3.2.1.5.

¹¹⁰ Rwanda Development Board (RDB), Rwanda Environmental Management Authority (REMA), 2018. *Tourism Development Master Plan for Gishwati - Mukura National Park*. [Link](#).

Activity 3.2.2 Facilitate and scale up climate-resilient value chain products

The project will conduct deep analysis for identified value chains, namely macadamia, avocado, chili, and French beans. Such analysis will provide evidence of the opportunities and constraints in the selected value chains and will involve mapping out chain operators and service providers at each segment of the value chain to identify existing capacity gaps and orient overall value chain interventions in the project. Furthermore, value chain analysis will enable the project to clearly understand the characteristics of the selected value chain with attention to smallholder farmers, women, youth and vulnerable groups.

To build strong and sustainable value chains, the project will improve information flows within the chain, by strengthening coordination among value chain actors and supporters. This will involve creating value chain platforms in the CND region to increase transparency and fair business relations. The project will focus on activities that strengthen and scale up value chains: for example, by organizing business-to-business (B2B) events, establishing district value chain interprofessional/platforms, and promoting business development services such as swift, cost-effective, and accessible product certification for agribusiness. The project will play an active role by introducing key supporting functions such as financing, value chain information, certification services, etc., and advocate for fair business relations to ensure farmers are active players in value chains and not passive beneficiaries. The project will help strengthen the capacity of producer organizations in selected value chains. Often farmers fail to participate and benefit in value chain activities due to weak and poor organization, resulting from poor managerial skills and weak internal financial management. The project will use different capacity assessment tools like CAM and Scope insight to identify capacity gaps and develop capacity-building plans to ensure that producer organizations are strong and professional and capable of serving their members and building sustainable value chains. The capacity-building trajectory will also focus on encouraging participation by women, youth, and vulnerable groups and increase their access to services. Activities to strengthen farmers and producer organizations will be based on farming as a business (FAAB) concept to orient farmers toward business-oriented agriculture. Synergy will be built with projects such as [KIWP2](#) (Kayonza irrigation and integrated watershed management), a project funded by IFAD where this module is being applied.

In the selected value chain quality standards are an integral element of the core interventions, due to export market requirements. The project will strengthen the capacity of farmers to increase production in terms of volume and quality but also adhere to market standards (for example, global GAP, maximum residual limits (MRLs), etc.) Another challenge affecting the selected value chains in the CND region is access to inputs, especially seeds, and pesticides. This is exacerbated by the international market which requires certain seed varieties and types of pesticides. The project will strengthen the input supply system in collaboration by National Agricultural export board (NAEB) and selected off takers by establishing and supporting agro-dealers and nursery production through technical support and access to finance, to further access to input the project will also focus on building business models that help farmers access inputs via credit guarantee schemes. The project will create strong linkage and business relations between input suppliers, farmers/producer organizations, and markets/buyers to enhance flow of information among value chain actors. Based on experience, the project will ensure that processors/buyers have a say or influence on type, variety, and quality of inputs used by producers to ensure produce marketability. Furthermore, the project will collaborate with other projects like [PSAC](#) (promoting smallholder agro-export competitiveness project) and NAEB (National agriculture export board) to further build a sustainable input supply system. Most value chains are not well developed or are still nascent in the CND region, based on our experience with agroforestry value chains with the TREPA project for instance. In such situations, off-takers, processors, and traders are few and poorly organized, with minimal outreach to farmers. They normally experience challenges in understanding and meeting market standards (i.e., sanitary and phytosanitary (SPS) standards, Global GAP, hazard critical control point certification (HCCP), etc.) which limit their

competitiveness and market scope. The project will improve their capacity to meet these standards, provide other support to access and integrate cold chains, especially for avocado and French beans, and build out grower schemes in which off takers/exporters play a vital role. For MSMEs and start-ups, the project will strengthen their managerial capability for running successful, efficient businesses. Through organizing value chain linkage events (B2B, market linkage sessions, etc.) the project will enhance buyers to reach more farmers and producer organizations.

Addressing rural poverty and food insecurity using agriculture requires investments in off-farm agricultural activities to complement primary production.¹¹¹ Synergies will be created to reinforce direct linkages of small producers to other value chain actors with inputs (seeds and seedlings, fertilizers) and outputs (processors, wholesalers, and retailers).

The project will use the Market Systems Development (MSD) model to enhance value chain development. This will entail multi-stakeholder collaboration with business service providers, financial institutions, community-based enterprises, value chain actors, and farmer cooperatives. In-depth target area assessments will be conducted to tailor business cases for the selected value chains, particularly focusing on identification and engagement of existing farmer organizations and cooperatives, youth and women-owned businesses, and local small and medium enterprises (SMEs, value chain actors) with the capacity for production aggregation, value addition, and market potential to absorb production from smallholder farmers' cooperatives. This localized stakeholder and value chain exploration will serve as a basis for the organization of market linkages to help value chain actors better understand the market, challenges, and opportunities.

Regarding honey value chains, the project will estimate the number of beekeeping cooperatives operating in the targeted region or its surroundings. Members of these cooperatives will be trained in improved honey production techniques and adding value through wax-based products. This will involve establishing honey and wax storage stations and associated processing facilities. Cooperatives will also be an important vehicle for fostering tree growing for improved bee forage. The project will foster interactions between the cooperatives and private companies through business roundtables, trade fairs, and similar events focused on honey and beeswax products and this activity will be co-financed by RDB.

This activity includes in-depth target area assessments to tailor business cases for the selected value chains, particularly focusing on identification and engagement of existing farmer organizations and cooperatives, youth and women-owned businesses and local small and medium enterprises (SMEs, value chain actors) with capacity for production aggregation, value addition, and market potential to absorb production from smallholder farmers' cooperatives. This localized stakeholder and value chain exploration will serve as a basis for organization of market linkages to help value chain actors better understand the market, challenges and opportunities. Key sub-activities are:

3.2.2.1 Analyze identified value chains and promote them among the CND stakeholder networks

3.2.2.2 Strengthen the capacity of producer organizations to improve managerial capacities, farming practices, and gender inclusion in their organizations

3.2.2.3: Establish and strengthen relevant value chain platforms in the CND area and promote business development services to enhance value chain coordination.

¹¹¹ Selasi, J., 2012. *Making Small Scale Farming Work in Sub-Saharan Africa*. In Global Forum on Food Security and Nutrition "FSN Forum", FAO.

3.2.2.4: Build and strengthen the capacity of off-takers (MSMEs, SMEs, and off-takers aggregators) to increase their competitiveness and comply with various standards

3.2.2.5: Facilitate and strengthen access to quality inputs for farmers and producer organizations in the CND regions

3.2.2.6: Establish “CND brand” for honey and wax products

Activity 3.2.3 Facilitate access to output markets for vulnerable farmers

Lack of access to the market is a challenge to smallholders: only 4.2% of agricultural households have access to contract farming ([Agric HH Survey, 2017](#)). Limited access to markets for smallholder farmers is mainly attributed to lack of access to market information, failure to meet market requirements, or undeveloped value chains. Some of the challenges also stem from poor quality and poor produce aggregation. The project will strengthen the capacity of farmers and producer organizations to improve quality and aggregation capacity, and in creating market linkages with off-takers or buyers through market information. The project will promote contract farming and out-grower schemes, where farmers are required to practice market-oriented farming, with the main objective of establishing long-term business relationships with markets/buyers. A focus on financial literacy and community initiatives will also mobilize domestic investment and expertise, facilitate access to financing, incentivize the private sector to create innovative financial products, and support improved data collection and measurement of outcomes.

Furthermore, through producer organizations, the project will develop an inclusive business model to improve produce aggregation and supply coordination and to increase farmers’ bargaining power and influence in formal value chains, particularly for remote/hard-to-reach women, youth, and vulnerable groups. To further strengthen market access, the project will develop innovative market access approaches such as “it takes two to trade”, which focuses on understanding the challenges and weaknesses that limit demand (traders) and supply (farmers). The project will create interventions to eliminate challenges and improve weaknesses to build sustainable market access. The project will also build on the national digital agriculture strategy (NDAS) to introduce digital market access, where farmers access market information through their mobile phones through USSD or IVR interactive voice response. A similar intervention is currently being implemented in the Kayonza district through an IFAD-funded project: Strengthening Agriculture Resilience through Learning and Innovation ([STARLIT](#)). This approach is particularly customized to poor, rural, and hard-to-reach farmers who do not own smartphones.

To implement this, the project will identify and select two buyers/off-takers for each selected value chain based on market needs and link them to farmers and farmer organizations. The eligibility criteria for buyers/off-takers will be developed by RFA in collaboration with CORDAID during the implementation. The project will organize periodic market linkage events and business-to-business (B2B) workshops through established value chain platforms. Additionally, the project will create a clear link between market access and other value chain supporting functions, such as value chain finance, to ensure that farmers have access to working capital for aggregation and marketing. Various models will be developed including tripartite agreements (buyers, producer organizations, and financial institutions) to increase confidence among partners and financial flow.

The activity will work to scale up the market system developed using the Output 3.2 inception report on beneficiary persona and stakeholder engagements to facilitate market system growth within the selected value chains, including bee products, and other women and youth-focused sectors. The activity will work to scale up the market system developed using the Output 3.2 inception report on beneficiary and stakeholder

engagements to facilitate market system growth within the selected value chains, including bee products, and other women and youth-focused sectors such as adoption and sales of fuel-efficient cookstoves.

The project will also build strong business relationships and linkages between input suppliers, markets, and farming cooperatives. This will build common understanding and smoother flow of information to ensure that input suppliers and farming cooperatives respond swiftly to what markets need. The activity will train vulnerable groups and farming cooperatives on “farming as a business” concepts.

Finally, trade agreements will be created between cooperatives and relevant national and regional companies that buy and sell such products. The Project will support and ensure protection of the interests of the cooperatives and their members while negotiating and signing relevant agreements for win-win situations.

Key sub-activities:

3.2.3.1 Develop an inclusive business model to Integrate vulnerable farmers into formal markets

3.2.3.2 Organize microenterprise training and develop and implement green grants programs targeting landless youth and poor, as well as women who are unable to access direct benefits from land-based project activities.

3.2.3.3 Facilitate market access for women, youth, and historically marginalized communities through market linkage sessions

3.2.3.4 Provide technical support on market standards, contract farming, and financial connectivity.

3.2.3.5 Develop various market access channels including digital market access for rural farmers, especially women and marginalized groups

Activity 3.2.4 Scale up use and sales of clean fuel-efficient cookstoves

The project will actively promote the use of energy-efficient cookstoves to benefit the 90% of households in the CND that depend primarily on firewood to meet their cooking energy needs and thus contributing to accelerated forestry resources degradation while causing profound health hazards to the already impoverished community.

This intervention will reduce pressure on forests and farmland by raising household awareness of the differences between high and low efficiency stoves and by collaboration with the local TVETs institute to manufacture affordable efficient stoves that will address the limited availability of high-efficiency stoves in rural markets. Promoting the local fabrication will reduce dependency on cookstoves made outside the country and therefore make them affordable. Also, promoting locally made cookstoves designed to address local needs will lead to full adoption of improved cookstoves and will contribute to sustainable biomass resource use and prevent overexploitation of forest resources, thereby ensuring the success of the forest landscape restoration activities described in Outputs 2.1, 2.2, and 3.2 above.

The project will adopt a market-oriented approach to strengthen the cookstove supply chain in the CND and create strong linkages and business relationships with several ecosystem including the TVET center for training of youth and women SMEs on cookstove manufacturing and repairing, and microfinance institutions (MFIs) to ensure sustainable access and finance of cookstoves business.

As mentioned above, to ensure product quality and affordability to drive uptake, scalability, and business growth and expansion, the project will partner with integrated polytechnic regional colleges to develop a

Tier 2 (for emissions) prototype compatible with rural settings and affordable to rural communities. Youth-led SMEs will be trained on how to manufacture cookstoves and contracted to produce them as well as operating sales points within the community

The cookstove pricing and financing modality will take into account vulnerable groups, including historically marginalized groups mapped and selected in sub-activity 3.2.1.1 (social wealth categories c, d, e, and people with disabilities). For sustainability, the groups will be organized into saving groups and ensure they lend among themselves with the purpose of purchasing the cookstove, the project subsidies may target women-headed households which has contributed over a percentage of the total cost. Additionally, the product design will incorporate reducing health hazards, such as reducing respiratory diseases risks and others linked to climate impacts, contributing to reduction in morbidity and health treatment cost and enabling the vulnerable to have disposable income for investment into productive ventures and this will ensure sustainability of impacts beyond the GCF financing. Moreover, the project will target women-headed households, recognizing the significant role women and children play in firewood collection and cooking. The project will substantiate the time-saving benefits for fuel collection and links to women's decision-making and empowerment with clear indicators.

Considering the lessons learnt from similar projects where hand-outs lead to poor sustainability, this project will promote alternative income sources from fast-maturing crops such as chili and French beans which have ready market in Rwanda and externally. The members of the saving groups will be able to use income generated from the sale of fast-maturing crops (chili and French beans) coupled with savings accumulated from their groups and save progressively to purchase the cookstoves and replacement. The project's technical support to MFIs will build their capacity to develop tailored financial products for cookstoves with affordable repayment terms aligned to revenue streams from short-term crops and group savings, making it easier for farmers to access loans to buy cookstoves.

GCF funds with co-financing from RDB will provide support for the development of ICS prototypes, youth-led SMEs engaged in cookstove production, subsidies for cookstoves, creation and support of savings groups, and technical support to MFIs to develop tailored financial products to facilitate access to ICS by local communities. The project will ensure that the cookstoves are sustainably sourced and lead to minimal indoor air pollution and related respiratory conditions compared with the cooking methods presently in use both during production of the cookstoves and during usage by the individual households.

Key sub-activities are:

3.2.4.1. Promote local manufacturing through partnership with Integrated Polytechnic Regional Colleges (IPRCs) of Karongi and Musanze to help design prototype Tier 2 ICS to address local constraints.

3.2.4.2. Build business and marketing capacity of the cooperatives and youth led SMEs to act as the marketing channels of energy efficient cookstoves to CND community members based on wealth categories.

3.2.4.3 Create a specialized small business development fund, to help local women and youth groups build innovative business models for accelerated distribution of ICS.

3.2.4.4. Promote demand-through raising public awareness on the benefits of ICS by developing and disseminating messages on mass media and in community meetings.

Output 3.3 Financial services & private sector investment increased

Financing challenges and needs are particularly high in clean cooking, mainly for two reasons: (1) due to lack of income, the upfront purchase of clean cooking equipment presents an affordability challenge to most

households, which adds to the monthly fuel expenditures and (2) businesses operating in the sector are often unable to attract financing for investment in the value chain. Serving low-income households also generally yields low returns due to low customer base.

With a particular focus on marginalized communities, and vulnerable groups of youth and women in the CND, the Project will deploy a FSG model to capacitate the communities' productivity and financial resilience. The project will address barriers to access to financial services, specifically for women and youth by building the financial institutions to develop and financial products that meet the differentiated needs of vulnerable women and men in the CND. The project Gender Expert, Youth Focal Point along with the Community Mobilization Specialist and Community Engagement Specialist will provide support to address barriers to access, ensure women's financial skills are developed and there is community and financial institution awareness and support to ensure marginalized groups, including women and youth can access financial services including credit to invest in their businesses including purchase of the cookstoves. Besides this, to address the significant challenges facing MFIs and Participating Financial Institutes (PFI) such as limited understanding of forestry and agriculture sectors investment and the corresponding risks associated with climate change effects on credit repayment, this Project will coach the participating financial service providers to enhance their on assessing the impact of climate change on their clientele and loan portfolio, and ensure equitable access to financial assets by women, youth, and historically marginalized communities. Furthermore, the Project will facilitate local and international impact investors and PFIs to engage in investment and credit provision for farmer cooperatives and SMEs in the selected value chains.

In collaboration with TVET institutes/ICS youth producers and central (RFA, MININFRA) and local authorities (Districts), CND biomass and micro-finance experts will refine and test, for each type of ICS/fuel to be promoted, the category/profile of households to be targeted and the related subsidy/microcredit schemes to be implemented to facilitate stove adoption with minimum concessionality. These schemes will be designed to align with the procedures and rules of identified local partner financial institutions.

In addition, the financial products will be consistent with climate resilience and capacity-building for macadamia, avocado, vegetables, honey and ecotourism value chain actors under Outputs 3.2.3 and 3.2.4. Techniques will be developed to analyze and score forestry and agroforestry value chains that are climate-resilient.

Activities and sub activities under this output will be executed by the Rwanda Forestry Authority (RFA) with the technical support from WCS.

Activity 3.3.1 Facilitate access to finance & private sector investments

To de-risk the selected forestry and agroforestry-related value chains, the Project will organize sensitization sessions for value chain actors and private financial institutions to build a common understanding with regard to the selected value chains. This will also include introduction of monitoring and measurement systems to support the risk and climate impact assessment on the value chains to be undertaken by the financial service providers.

Communities involved in chosen forestry and agroforestry value chains in selected intervention sub-areas will be selected with a particular focus on landless people, historically marginalized communities, and vulnerable groups of youth and women in collaboration with local authorities (districts & sectors) and community representatives.

Furthermore, the interventions will capitalize on the fast outreach of informal saving groups. Saving groups are self-managed and self-capitalized entities that use members' savings to facilitate lending within the

group. They set up clear rules and regulations which are known and adhered to by all group members. Groups consist of a limited number (10-25) of people who have freely chosen to work together, and offer self-managed savings, insurance, and credit services in remote rural areas. Poorer segments of the community often do not have a regular source of income—however, within a savings group, people can save to purchase agricultural inputs, such as seeds, fertilizer or equipment. Savings groups may also provide mutual assistance for group members, such as medical insurance or loans to members in need.

Saving groups have transformed marginalized communities by mobilizing local savings, which provide members with a means to cope with emergencies, manage household cash-flow, build a capital base, and, crucially, re-build social networks, solidarity and trust. Savings group participants often do not have regular sources of income and commonly come from poorer segments of the community. Women are regularly involved in savings groups and have high participation rates. Saving group participants often come from the same socioeconomic background and know each other.

Activities of saving groups include:

- **SAVING:** The group determines a fixed amount for a given period that each group member will bring to the savings account. The allocation and management of these savings can vary according to the needs of the members or is used to support group activities.
- **LENDING:** The group can grant small internal credits to members. Loans are modest and repayable in the short term. Interest rate is fixed by members. At the request of the group, these credits are renewable almost automatically if there is no problem and if the amounts are within a ceiling set by the group.
- **SOCIAL:** The group not only improves the lives of its members financially but also socially as members are given the opportunity to socialize at each meeting and discuss other issues or concerns. The group further builds a small social fund to help group members in distress. Each group agrees on the amount to be saved, usually a smaller amount than the amount for regular savings, in every meeting for the social fund. This money is not loaned, but instead used to pay group member expenses for things such as funeral costs or medical treatment and are only used if agreed on by all group members.

In case of external loans from financial service providers, which most members can't afford, group members agree to mutually guarantee each other (co-guarantee) when taking a loan. If a member does not pay, the group ensures repayment for him/her. The group uses moral pressure to force repayment and therefore there is no requirement for physical guarantees with third parties such as financial institutions. The internal character assessment for group members is important.

The project will strengthen the capacity of saving groups and link them to more formalized financial institutions. The project will further enhance financial education and client protection principles.

This model is more convenient and more accepted by most women who cannot afford collateral loans. As such, this approach will attract more women and will facilitate their formal access to financing over time. Specific attention will also be paid to women's participation in other project financial products that will be developed, such as asset finance in activity 3.3.2.

The project will digitize saving groups, to reduce time and cost spent to access their savings, thereby facilitating transactions with formal financial institutions. Solutions such as “push and pull technology” (i.e., payments using cell phones, commonly referred to as “mobile money”) have proven to be useful and convenient for savings groups to send and receive money.

In activity 3.3.1, communities involved in chosen forestry and agroforestry value chains under this project in selected intervention sub-areas will be selected with a particular focus on landless people, historically marginalized communities, and vulnerable groups of youth and women in collaboration with local authorities (districts & sectors) and community representatives.

Sub-activities include:

3.3.1.1 Develop tailored credit assessment tool for selected value chains, while integrating climate impact data

3.3.1.2 Facilitate access to finance for selected value chains of avocado, macadamia, vegetables and ecotourism that are relevant to build climate resilience

3.3.1.3 Develop tailored financial products to meet the needs of women and youth

Activity 3.3.2 Set up and support savings & loan groups, enhance asset-building

To strengthen the capacity of local smallholder farmers in financial literacy and access to finance, farmers' cooperatives will be assisted and supported in setting up saving groups. Savings groups are a proven and effective approach to the financial inclusion of the most vulnerable and are increasingly recognized as one of the strongest forms of community organization available due to the high levels of social capital generated from its savings approach. Saving groups create a platform for linkage with financial institutions and other business development services. Further, savings groups are recognized for enabling high levels of female inclusion. As groups members co-guarantee each other for accessing a loan and groups are strong, it forms low lending risks for MFIs since members mutually reinforce each other's repayments.

Once grouped, small savings group members, especially women, grow from accessing internal group loans to accessing formal financial institution loans. As members' bankability and trustworthiness increases, members are able to start accessing individual loans, gradually building to larger loans and are able to open individual savings accounts at financial institutions.

Members of a saving group will be self-selected and the saving group will be managed by its members.

Self-selection will be based on the following:

- Knowledge of one another and similar economic background
- Close proximity to where the meetings will be held
- Reputation for honesty and reliability
- Ability to attend all meetings on time and follow rules
- Have a cooperative personality
- Be able to buy at least one share each week (or time set by the group)
- Be able to repay loans on time

Most saving groups are informal but transparent, democratic, and structured groups that have constitutions and clear rules. Members do not enter into contractual arrangements for their membership. Record-keeping is based on simple passbooks in which each member's savings and loans are recorded; all transactions are done in front of the members during group meetings.

Farmer savings groups (FSGs) will be created within cooperatives and farmers will decide which group to join. Groups will set up governing rules and conditions (savings amount, meetings, etc.), and criteria for new members who want to join. Particular attention will be given to social inclusion (including vulnerable communities and landless people).

Youth and women-owned businesses selected for activity 3.2.2 will be linked to targeted FSG beneficiaries for supplying inputs (modern beehives, seedlings, etc.) and energy-efficient cook-stoves. Furthermore, intensive capacity-building and technical support will be delivered to all groups with emphasis on financial literacy, entrepreneurship, and gender mainstreaming. The activity will also help link savings groups to PFIs.

For Activity 3.3.2, in collaboration with local authorities, the project will target existing or potential youth and women-owned businesses that can intervene at any level of selected value chains in activity 3.3.1. The project will ensure the integration of vulnerable and landless people.

Key sub-activity:

3.3.2.1 Establish/form saving groups for access to finance

3.3.2.2 Provide organizational, technical, and financial capacity-building to women, youth, and CSOs (such as cooperatives or VSLA) with focus on financial literacy and provide long-term coaching to beneficiaries

3.3.2.3 Support women, youth, and other marginalized groups to strengthen savings and loan groups and register for MFI accounts

Activity 3.3.3 Build the capacity of financial institutions to serve targeted value chains and communities

Many financial institutions demonstrate little knowledge about the science of agriculture, forestry, climate change and associated risks. A study by the International Food Policy and Research Institute (IFPRI)¹¹², noted that many agricultural activities are located in remote, widely dispersed regions and that financial institutions find it impossible to provide cost-effective and affordable services in this context. Hence for this specific activity, PFIs will be supported to access and adopt a digitalized forestry and agri-loan assessment tool, and to understand climate impacts on their portfolio to actively integrate and promote climate relevant financial products in their product offer.

The Project will work with these institutions by developing and refining appropriate financial products and financing operations that are better understood and user-friendly for smallholder farmers and vulnerable groups. The Project will also collaborate with selected PFIs to build their awareness of prospective farmer and low-income group clients and their exposure to climate impacts.

Key sub-activity:

3.3.3.1 Capacity-building of PFIs

Activity 3.3.4 Facilitate learning & knowledge sharing

To deliver an adaptive program that cooperates with market actors on inclusive and sustainable business models that have a positive impact on smallholder farmers, while generating and sharing lessons on which business models work and which do not, the project will adopt the program-embedded reflection and learning (PERL) model for facilitating learning and knowledge sharing to popularize the CND unique approaches to restoration and environment management.

By **proving**, we seek to know and show what works and what does not work by demonstrating results at all levels, generating deeper insights into farmer benefits and the effects on the wider ecosystem, and building business model sustainability.

¹¹² Kloeppinger-Todd, R. and Sharma, M. eds., 2010. *Innovations in rural and agriculture finance* (Vol. 18). Intl Food Policy Res Inst.

By **improving**, we learn by doing to enhance project adaptation and improvement, build a learning culture through discussion and reflection, generate knowledge to support project effectiveness, make comparisons over time, and enhance data quality procedures.

Through **sharing**, we disseminate high-level results and learnings to relevant audiences, contribute to visibility and communication, showcase successful, sustainable and scalable business models, and generate best practices.

Under this activity we will go through the following actions to prove, improve and share:

Capture - We will establish a KML framework and support case studies and discussion by project staff; improve data collection; collect insights from field visits; write human interest stories; hold quarterly program reviews; measure client satisfaction; and collect information on business models, market uptake, and ecosystem changes.

Synthesize - We will organize discussions and quarterly program reviews with program staff and stakeholders for assessment, interpretation, reflection, and learning from results. Outputs will include reports, briefs, and articles.

Disseminate - We will contribute to and support the project, presenting its achievements in newsletters, conferences, webinars, blogs, publications, media events, partner workshops and sector network presentations.

Take action - We will have regular discussions and reviews in which we revisit the work plan and processes in light of results and insights (e.g., based on outreach data, field visits, case studies, partner feedback, quarterly reports, changes in the ecosystem, etc.), and adapt operational planning when needed (within and outside of the normal project planning and reporting cycle).

Refine agenda - We will help to align the project intervention strategy with a shared vision for success.

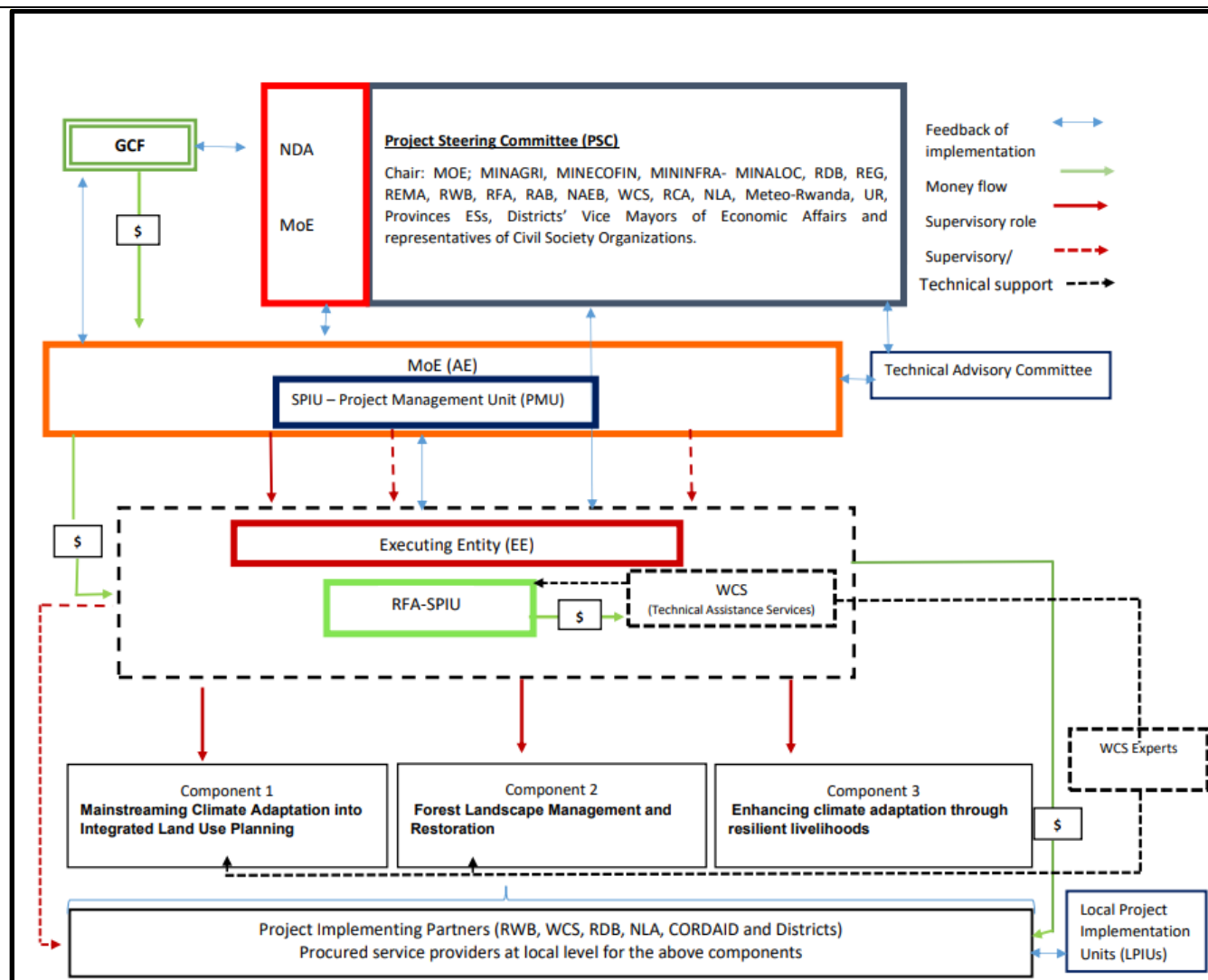
The table below summarizes the project activities to be implemented in Congo Nile Divide.

	Core PA Nodes	Stepping Stones	Landscape Linkages	Broader Farming Mosaic	Across the CND
	Volcanoes NP and Buffer	Gishwati NP and Buffer	Mukura NP to Gishwati NP Linkage	Nyungwe to Mukura Broader Farming Mosaic	
1.1.1 Synthesize & disseminate information on value of natural forests & ecosystem services					
1.1.2 Develop climate-resilient landscape land-use plan					
1.2.1 Create interagency taskforce institutionalizing integrated landscape planning & policy					
1.2.2 Build capacity for spatial planning in national agencies re climate change					
1.2.3 Develop an effective forest monitoring system to underpin forest management decisions					
2.1.1 Facilitate revision of PA management plans to address climate risks					
2.1.2 Establish long-term plans for CND financial sustainability post-GCF					
2.1.3 New fire management curriculum developed and operationalized in PAs and adjacent landscapes					
2.2.1 Secure key remaining natural areas outside PAs					
2.2.2 Restore natural forest cover in & outside PAs including riparian linkages					
2.2.3 Promote silvo-pastoralism with indigenous trees in Gishwati Pasture Stepping Stone areas					
3.1.1 Restore high slope areas (>55%) as protective forests					
3.1.2 Develop on-farm agroforestry for high caloric and indigenous tree species					
3.2.1 Develop agroforestry related value chain for markets access					
3.2.2 Facilitate & scale up capacity, value addition & marketing of select climate-resilient value chain products					
3.2.3 Facilitate access to input & output markets for vulnerable farmers					
3.2.4 Scale up marketing, production, sales, use of fuel-efficient cookstoves					
3.3.1 Facilitate access to finance & private sector investments					
3.3.2 Set up & support savings & loan groups, enhance asset-building					
3.3.3 Build capacity of financial institutions to serve targeted value chains & communities					
3.3.4 Facilitate learning & knowledge sharing					
4.1 - Performance monitoring plan developed					
4.2 - Project monitored and evaluated; lessons learnt integrated into adaptive management processes					
Primary Focus					
Secondary Focus					
Some activity					

Table 4: Summary of project activities (place based and non-place based) to be implemented across the Congo Nile Divide.

B.4. Implementation arrangements

The proposed project will be implemented in five years, from 2024–2029. The Ministry of Environment (MoE) is the Accredited Entity (AE) for the GCF and MoE will enter into a funded activity agreement with the GCF once the project is approved by the Board.



The governance arrangements for the project at the national level are designed to build upon the institutional structure of MOE (the accredited entity) and RFA (the Executing Entity).

Accredited Entity: The MoE on behalf of the Government of Rwanda is a Direct Access Entity of the GCF and will assume the oversight role on implementation, management, monitoring, evaluation and reporting on the CND Program elaborated in the FAA. The MoE will enter into a subsidiary agreement with the Executing Entity (EE) with clear roles and responsibilities for each party. Under the Ministry of Environment there is a Single Project Implementation Unit that has a number of staff charged with resource mobilization and providing oversight over different projects implemented through the Ministry of Environment. The MoE/SPIU has a track record of experience in project oversight, management, monitoring and evaluation of projects and programs including the GCF funded FP073 that is currently at the mid-term level of implementation with over USD 32.8 million, managed over USD 3.8 million for the GCF funded PPFs, overseen the implementation of USD 10 million from Adaptation Fund. The MoE is currently managing the World Bank Funded project preparation advance of USD 6 million and has been implementing since 2018 a program funded by the UNDP focused on Strengthening Capacities of Environment and Natural Resources Sector for a Green Economy Transformation of over USD 6 million. Additionally MoE assumes its oversight role through chairing of the Project Steering Committee for different projects implemented under the Environment and Natural Resources Sector as a whole namely TREPA, a GCF-funded project, Rwanda Urban Development Project Two funded by the World Bank and NDF, an Early Warning System project

(funded by LDCF), the Landscape Approach to Forest Restoration and Conservation (LAFREC) programme (funded by GEF/World Bank), Green Amayaga Project (funded by the GEF) and the “Building resilience of communities living in degraded forests, savannahs and wetlands of Rwanda through an ecosystem management approach” programme (funded by LDCF). A coordination platform between approved GCF projects will be established to identify, maximize linkages and ensure complementarity through learning and sharing best practices and failures arising from the implementation of existing GCF projects. Quarterly meetings will be organized under the supervision of the MoE between project directors and staff responsible for monitoring and evaluation to share lessons learned.

Project Financial Flow and AE Roles

Funds from GCF will be provided to the MoE GCF account (as the AE), under the terms of the FAA. MoE will be responsible for financial management and accountable for the use of GCF resources. Financial transactions will be subject to annual audits undertaken by internationally certified auditors. In addition, as the AE, MoE will: i) ensure that the project is executed in accordance with GCF standards; ii) supervise, oversee and manage the implementation of project interventions; iii) report on Project progress; and iv) ensure that Project activities are well coordinated and aligned with countries' national priorities. MoE will provide oversight to the project consisting of a) entering into subsidiary agreement with the EE; b) managing and disbursing GCF funds to EE on the approved work plans and budgets; c) reviewing financial expenditures and progress reports; d) overseeing Project implementation in accordance with the Project document and Annual Work Plans and Budgets, agreements with co-financiers and executing entity rules and procedures; e) providing technical guidance to ensure that the appropriate technical quality is applied to all Project activities; f) providing financial reports to the GCF for Project funds received; g) ensuring that the project complies with the terms agreed in the Project's respective FAA as well as the AMA signed between MoE and the GCF; and h) undertaking regular annual supervision missions according to the MoE's guidelines and convene Project Steering Committee (PSC) meeting twice a year aimed to review progress and approve work plans. The PMU hosted under the Single Project Implementation Unit of the Ministry of Environment will manage, monitor and report on day-to-day implementation of the Project activities by the EE. The EE will execute the tasks within the overall project management structure: a) implementing day-to-day activities as per the project work plan and budget, including the Environmental and Social Management Plan; b) undertaking procurement activities as per the agreement signed by GCF and AE as well as the sub-agreement signed by the AE and EE; c) managing contracts of suppliers and services providers; d) hiring and managing project staff relevant to the EE-managed project areas; e) implementing activities as per the project work plan; and f) carrying out financial and technical monitoring of activities, including the achievement of outputs and outcomes the EE is in charge of. EE will enter into Memorandum of Understanding with different implementing partner agencies that will provide technical know-how to the implementation of the project activities. The implementing partners including but not limited to RWB, WCS, NLA, RDB, CORDAID and Districts will provide technical and operation services articulated and agreed in respective MoUs. For the Implementing partners RFA will be paying service compensations and at District Level community approach will be used. Procured service providers will be contracted in accordance with MoE rules and regulations of procuring goods and services.

Project Steering Committee (PSC): An inter-ministerial Project Steering Committee (PSC) chaired by the Permanent Secretary will be established by MOE to serve as the project's coordination and decision-making body. The PSC will ensure the project delivers its outputs and achieves its outcomes. The Committee will meet on a semi-annual basis to review project progress and evaluations, facilitate implementation (ensuring the necessary resources and support are provided in a timely manner) and provide guidance to the PMU and EE/PIU. The Steering Committee will also facilitate effective coordination between the key stakeholders at the national and district levels and ensure the project aligns with Government strategies and programs.

The PSC will reflect the multi-sectoral nature of the project and will include senior-level representatives from GoR and other relevant Stakeholders. The PSC will be chaired by the Permanent Secretary, MOE or any Senior Manager from MoE delegated by the Permanent Secretary with representatives from: MINAGRI, MINECOFIN, MININFRA- MINALOC, RDB, REG, REMA, RWB, RFA, RAB, NAEB, WCS, RCA, NLA, UR, Provinces ESs, Districts' Vice Mayors of Economic Affairs and representatives of Civil Society Organizations.

Project Management Unit: It is under the MoE/SPIU that a dedicated Project Management Unit (PMU) of the CND program to supplement the existing MoE/SPIU structure for this complex program will be established for day-to-day management of the program interventions. The Project Management Unit (PMU) will be headed by the Project Coordinator who will serve as the secretary and the rapporteur of the PSC. The Project Coordinator and other PMU staff will be recruited as per the MoE/GoR recruitment practices. The Project Coordinator is responsible for the day-to-day operations of the PMU within approved project funded activity agreement and any other guidance provided by the AE and PSC including the tasks of managing and monitoring project risks initially identified and submitting new risks to the AE and PSC for consideration and decision-making on follow-up actions and updating the status of risks in the project risk log. The PMU Project Coordinator will report to the overall MoE/SPIU- Coordinator as the immediate supervisor and to the Permanent Secretary as the secondary supervisor.

The PMU will provide to the MoE/SPIU coordination and to the Chair of PSC with quarterly progress reports and close its operations when the final project evaluation report and other documentation required by the GCF have been completed and submitted to MoE. The PMU will be staffed with required professional and technically qualified personnel selected and recruited following the MoE/GoR procedures. In all, the PMU will strive to maintain a lean management structure. It will comprise a Project Coordinator, Monitoring & Evaluation (M&E) Specialist, E&S Specialist, Procurement Specialist, Gender Specialist, Financial Management Specialist, Communication and Outreach Specialist, Office Manager, and Driver as a dedicated team to manage the EE for the whole program.

Executing Entity: RFA will serve as the EE for the Project through a subsidiary agreement (SA) signed with MoE. The SA will establish clear roles and responsibilities for both parties for the delivery of the project activities, the schedule and conditions for installments, the determination of the prevailing fiduciary standards and the terms and conditions for arbitrations and termination of contract. The SA will include specific obligations for the EE on project execution, financial management, personnel administration and reporting, as well as on arbitration and liability terms.

Rwanda Forest Authority: RFA was established by the Government of Rwanda by the law No 72/2019 of 29/01/2020 with the mission of ensuring the growth of forest resources, their management and protection for sustainable development purposes. RFA is an affiliated agency under the MoE with a separate legal personality and a budget entity of the Government of Rwanda. RFA is currently implementing TREPA; a GCF funded project as one of the executing entities, COMBIO Project funded by the Swedish government, executed Forest Management and Biomass Energy project (FMBE) implemented in the Northern province; Sustainable forestry, agroforestry and biomass energy management for climate resilience in Gatsibo and Gicumbi districts; and Improving the Efficiency and Sustainability of Charcoal and Wood Fuel Value Chains in the North-western part of Rwanda supported by the Nordic Development Fund. RFA has three main divisions namely Forest Research Division, Forest Management, Administration and Finance and an established Single Project Implementation Unit (SPIU), composed of staff with the required financial management skills where the CND Staff will be hosted. The Director General of RFA as the Chief Budget Manager, the RFA SPIU Coordinator together with Program Manager and a Financial Management Specialist will be responsible for the day-to-day financial management of the Project. RFA SPIU will open a separate account at the National Bank of Rwanda (BNR), from which payments of the Project activities will

be made from. RFA follows the same financial and procurement policy as the one of the Ministry of Environment.

WCS Technical Assistance (TA) Services: As one of the project implementation partners, WCS will provide technical assistance to the EE (RFA) through an MOU signed with RFA. The MOU will detail all the activities that WCS will be directly supporting during the implementation of this project. The TA services will include advising RFA/SPIU, procuring and supervising international experts and consultants, and procuring materials to support the execution activities. Relevant technical experts from WCS will be embedded within RFA to advise SPIU on day-to-day project management to support the implementation of activities under outputs 1.2, 2.1, and 2.2. These experts will support RFA's technical teams and strengthen their capacities in different aspects of the project implementation. Furthermore, WCS technical assistance services will include supporting the development of ToRs and reviewing works of international consultants to ensure high quality deliverables.

CORDAID Technical Assistance Services: CORDAID will provide technical assistance to RFA mainly on output 3.2 and 3.3 through an MOU signed with RFA.

Other Government of Rwanda implementing partners will provide implementation support and RFA will enter into specific MOUs with these partners based on the type of support they will provide.

Table 5: Summary of project outputs. The EE will be responsible for all activities and sub-activities of the project, receiving support from technical implementation partners.

Outputs	Executing Entity per output	Technical Implementation Partners
Output 1.1. Landscape-wide land-use plan developed for climate-resilient livelihoods & forest ecosystems, integrating district development strategies (DDS)	RFA	WCS and NLA
Output 1.2. Local & national institutional capacities strengthened to integrate biodiversity & climate risks into land use planning & management	RFA	WCS and NLA
Output 2.1. Protected areas management effectiveness improved to address climate risks and adaptation	RFA	WCS and RDB
Output 2.2. Natural forest cover restored; biodiversity connections established	RFA	WCS
Output 3.1. Farming methods enhance productivity, reduce erosion & flooding risks, contribute to ecosystem services, and support connectivity	RFA	RWB

Output 3.2. Rural livelihoods generate alternative incomes and reduce pressure on forests	RFA	CORDAID
Output 3.3. Financial services & private sector investment engagement	RFA	CORDAID
Monitoring and Evaluation	PMU	

Technical Advisory Committee (TAC) will be responsible for ensuring technical guidance to both the project execution team and the PSC. The TAC will have technical representatives from all project stakeholders including MINAGRI, MINECOFIN, MININFRA, MINALOC, RDB, REG, REMA, RWB, RFA, RAB, NAEB, WCS, RCA, NLA, UR, Districts representatives and representatives of Civil Society Organizations. The TAC will be engaged in providing technical support and quality assurance of the project implemented activities as well as facilitating effective implementation and mainstreaming of project interventions beyond the life of the project. The institutional stakeholders will nominate relevant technical experts to the TAC, thus bringing expertise from their respective institutions to make technical contributions to the project implementation based on the integrated nature of the project.

The project will use the GoR fund management modalities based on an annual work plan developed by the PMU together with EE and subsequently approved by the PSC. The Annual Work Plan (AWP) indicates which activities should be covered at PMU and EE levels as per the allocated respective budgets. The EE, together with support from implementing partners, will establish Local Project Implementation Committees (LPICs) specific to the zones of intervention in the Western and Southern Provinces of Rwanda. Two LPICs will be established; one for the Southern province and another one for the Western province for the coordination of activities in Western (GMNP) and southern (NNP) Rwanda to ensure synergies and complementarity between partners. These teams will work closely with the Joint Action Development Forums (DJAF) based in each district located in these respective zones. LPIC will be composed of representatives of districts in charge of natural resources management, RDB, CSOs, and conservation organizations, women and youth councils and JADF. This arrangement will ensure that a) project delivery can proceed securely and efficiently despite the complex governance framework; and c) government partners play a leading role in project delivery and capacity development. PSC will provide policy direction and guidance to improve the coherence and efficiency of this innovative approach to be developed as an up-scalable model. At the local level, the EE and its service providers will coordinate closely with districts to ensure smooth local level coordination in project implementation, provide ownership and ensure sustainability. The District Executive Directors (DEDs) are expected to play an active role in project implementation, facilitation and monitoring and play a key role in the grievance redress mechanisms, as described in the Environmental and Social Management Plan of the project.

B.5. Justification for GCF funding request

Rwanda has achieved remarkable social, political, and economic progress over the past two decades. The Rwandan economy has grown well above average compared to its peers and the rest of the world, achieving growth rates of more than 8% regularly, with a 9.4% growth rate in 2019¹¹³. However, the economy which is

dependent on nature-based tourism and agriculture is extremely vulnerable to external shocks, including climate change.

Climate change is likely to increase variability in crop yields and agricultural production, cause severe flood damage to physical infrastructure, reduce labor productivity, and impact demand for tourism¹¹⁴. The recent Country Climate and Development Report (CCDR) estimates that if these risks materialize, Rwanda's GDP levels can drop by 5–7% below baseline in multiple years by 2050¹¹⁵, with negative impacts on private consumption, exports, and government revenues. By strengthening resilience to climate change, Rwanda can better prepare to manage the adverse impacts of climate change. Implementing the commitments articulated in the NDC is estimated to require new investments of US\$11 billion, of which close to US\$7 billion is conditional on new financing.

Rwanda is classified by the U.N. as a Least Developed Country (LDC)¹¹⁶ and lacks sufficient public financial resources to implement its climate change and resilient green development objectives. The implications of climate impact for Rwanda demands concerted action and support both locally and internationally to enable the mainstreaming of climate resilience into public, private and community sectors at the pace and scale required. This Project will allow Rwanda to implement its NDC commitments and pursue its aspirations under its National Forest Policy and Climate Resilient Green Economy plans for reversing trends of high deforestation, allowing for demonstration of sustainable reforestation models and reductions in demand for wood fuel. Rwanda has made an impressive commitment to using forests and forest protection as part of its strategy to mitigate the impacts from climate change, and to provide benefits of biodiversity conservation and livelihood improvements.

Yet, vulnerable citizens' needs, coupled with fragile landscape and forest restoration, will not be met solely with national resources in the rapid timeframe needed to arrest loss of economic assets for households and prevent further degradation of landscapes. Specific investments in awareness and forest governance, training at scale required to increase the resilience of CND communities to the impacts of climate change, and alignment and coordination of support on undertaking climate resilient restoration activities are not currently being developed by the GoR. The GoR does not have the financial means to implement all project activities identified to improve climate resilience and requires technical assistance and support. While project partners are contributing co-finance for this project implementation, this represents only a fraction of the resources required for implementation. The GCF grant investment can help initiate, expedite, and catalyze these critical interventions and assist with alignment of future climate resilience efforts. MoE is requesting grant funding for this Project due to Rwanda's status as an LDC, insufficient alternative finance options, and because the proposed activities will provide a public good on which the livelihoods of many of Rwanda's rural poor depend. The high levels of poverty among project beneficiaries in the CND as identified in the Feasibility Report necessitate the use of grant funds rather than a reimbursable financial option. While the GoR will contribute resources for the implementation and sustaining climate resilience efforts beyond the 5-year Project, this represents only a partial share of implementation resources. The GCF funding will crowd in additional private sector capital, such as efficient cookstoves and forest plantations, and will allow the government to cost-efficiently implement the adaptation capacity-building training and practices to ensure that CND landscape is restored, and climate resilient landscapes maintained.

¹¹⁴ "World Bank Group. 2019. *Rwanda Systematic Country Diagnostic*. World Bank, Washington, DC. World Bank. <https://openknowledge.worldbank.org/handle/10986/32113> License: CC BY 3.0 IGO."

¹¹⁵ World Bank Group. 2022. *Rwanda Country Climate and Development Report*. CCDR Series;. World Bank, Washington, DC. World Bank Group. <https://openknowledge.worldbank.org/handle/10986/38067> License: CC BY-NC-ND 3.0 IGO.

¹¹⁶ United Nations Department of Economic and Social Affairs, 2021. Least Developed Country Category: Rwanda Profile.

The lack of private capital structure, prevalence of subsistence farming, and inadequate market for agriculture and forested products, is the primary cause of the request for grants, not loans as little surplus products are produced and traded, and thus will be inadequate to secure or service any loans. The prevailing market conditions necessitate GCF grant finance, rather than loans, to raise the capacity and bridge the financial and technical gaps until the restoration and value chain models are brought to financial viability that can be sustained by private sector. Furthermore, our Economic and Financial Analysis (EFA) shows that the direct returns on investment in landscape restoration activities are too low to make them financially attractive or feasible for farmers, even if they had access to credit. Without GCF support, lack of private financing, large payback period, uncertainty and transaction cost of getting the loan, and upfront cost and benefits realized over a longer period of time reduces the financial attractiveness of climate resilience investments. It is unlikely that there will be a financial return on investment within the Project time frame to repay loans. The issue of societal benefit investment is accentuated by the fact that there cannot be direct uptake of benefits to communities from restoration within National Parks.

Serving low-income households yields low returns and the community lacks financial education and capacity to link them to formal private institutions. Most formal financial institutions view these as high risk and high cost, as the transactions are often small and the clients usually have neither physical collateral nor the education needed to organize their finances. Without the value chain development for potential revenue generating streams such as honey, tea plantation tourism, and efficient cookstoves, the private sector investments are not occurring. Henceforth, these potential revenue generating streams proposed in the project are not sustainable on their own and the GCF grant will not only provide financing support but will be critical in creating enabling opportunities for market development beyond the project to catalyze private investments. The full concessionality funded by GCF grants is requested so that an enabling environment can be created, and the project can overcome affordability challenges through developing a system of community co-guarantee. This will also go beyond the neoclassical economics model and focus on shifts in social dynamics that move the community as a whole to escape the poverty trap and increase financial resilience. The full concessionality for potential revenue generating streams will be passed on to end beneficiaries in the CND and the project will strengthen the capacity of community organizations on quality improvement and aggregation capacity, create market linkage with off-takers or buyers, and attract finance for development, marketing, and distribution.

Planting indigenous species, well-suited to the climatic conditions of the CND, is likely to increase the health and resilience of forest plantations and thus the erosion control benefits that these forests provide. However, when establishing protective forests, most existing projects in the CND use exotic species because they are fast growing, widely available, and low cost¹¹⁷. GCF grant resources are required to support restoration to help build natural capital (restored forest) which can be used sustainably and provide larger ecosystem benefits in the future.

The financial Net Present Value (NPV) of \$34.48 million for agroforestry, \$2.98 million for protective forests, and \$0.31 million for silvopasture intervention and economic NPV of \$58.93 million for agroforestry, \$30.22 million for protective forests, \$0.97 million for silvopasture, and \$10.59 million for energy efficient cookstoves interventions for 20-year analysis period when GCF support for the Project is considered¹¹⁸. These results are indicative of the investment in time and resources required to increase climate resilience, associated benefits, and restoration of landscape and forest productivity. Furthermore,

¹¹⁷ Mukuralinda, A., et al, 2016. *Taking to Scale Tree-Based Systems in Rwanda to Enhance Food Security, Restore Degraded Land, Improve Resilience to Climate Change and Sequester Carbon*. PROFOR, Washington D.C.

¹¹⁸ The provision of energy efficient cookstoves is a non-income generating intervention and is expected to have no financial cash inflows and no income generating opportunities, then no financial analyses is needed. The cookstoves analyses was pursued only at the economic analysis level.

agroforestry, restoration, and forest management require up-front investments that take 10-30 years to mature fully. However, these future benefits are depressed by the use of a high discount rate that downplays the importance of long-term investments. The discount rate considered in our economic analysis is 13%, which is Rwanda Central Bank's yield for a 20-year treasury bond, as issued in August 2021. This is the rate at which the government can borrow to fund equivalent investments in the absence of grant financing. For financial analysis, we used the discount rate of 15.44%, which is Rwanda Central Bank's lending rate for July 2021. Consequently, the GoR seeks maximum concessionality from the GCF for the urgent adaptation actions proposed under this Project. Without GCF funding, under a business as usual (BAU) scenario, inadequate technical and financial capacity at the national and local level will continue to hinder the efforts of rural communities in the CND to adapt their livelihoods. Consequently, the vicious negative cycle of unsustainable forest use and ecosystem degradation will persist, reducing the supply of goods and services which buffer impacts of climate change on rural communities, compounding their vulnerability. GCF funding is therefore vital to initiate progress towards a complete paradigm shift in GoR's approach to adapting the livelihoods of rural Rwandans to climate change.

The Project cost per direct beneficiary of \$31 (\$39.86 with co-funding) and 20-year economic rate of return of project interventions for agroforestry (49%), protective forests (72%), and silvopasture (51%) highlights its viability. The Economic rate of return for energy efficient cookstoves could not be estimated as there is no change in the sign in the net cash or resource flows from positive to negative over the analytical period; and the economic NPV of \$10,585.675 for this intervention highlights its feasibility. The cost of restoring natural forest for providing both adaptation and mitigation benefits is significantly greater than the cost of simply planting trees for mitigation. Restoration of natural forest in Rwanda is essential for sustaining the critical ecosystem services on which vulnerable rural communities and the national economy depend. The capital market in the country is weak and provides limited opportunities. It has low levels of direct investment which further limits the opportunity for financing. The country's capital markets currently are not able to serve the needs of rural and resource limited beneficiaries. The project addresses market failures and minimizes the risk of disrupting or unduly distorting markets or crowding out private finance, including new entrants, as well as other public investments. Taking the above-mentioned points into account the concessionality that the GCF provides is justified.

B.6. Exit strategy

The Project will establish a new inter-agency and local-to-national-scale spatial planning framework and capacity for an integrated approach to multi-sector analysis and planning. This will enable the analysis of land-use trade-offs that explicitly consider the value of ecosystem services, climate risks and risks of ecosystem degradation. This represents a radical shift away from business-as-usual and will have long-term, far-reaching implications for policy and financial planning in Rwanda. It will provide a spatially explicit framework for mainstreaming climate risk considerations across all sectors while also providing a mechanism for quantifying harmful financial flows to biodiversity from multiple sectors.

In addition, the multi-scale spatial planning system will be able to strengthen and link directly to the system under development by Rwanda's Ministry of Finance and Economic Planning (MINECOFIN) for policy coordination to integrate climate risks into fiscal planning, with support from the IMF Resilience and Sustainability Trust. Further, it will support the ability of both national and local governments to plan and implement the new disaster risk reduction strategy in a spatially explicit way, which previously has not been possible.

Mainstreaming climate change adaptation metrics. The Project will integrate climate resilience metrics into land use planning, district development strategies and annual performance contracts and harmonize cross-

sectoral monitoring and reporting mechanisms. This will lead to a systematic consideration of climate change risks and adaptation in policy planning that will be sustained beyond the Project. An MoU between key government agencies will operationalize the cross-sectoral planning task force, which will be supported by a ministerial order, giving it the mandate and financial support to operate beyond the life of the Project.

Rwanda has adopted a private-public partnership (PPP) business model to make its national parks operations financially viable. However, in Rwanda, as in the rest of the world, national parks and the biodiversity they are intended to protect are threatened by climate change and by the interaction of climate change with other direct anthropogenic threats. By “climate proofing” its national parks—i.e., by establishing a technical unit within RDB, by explicitly incorporating climate change considerations into the management plans of national parks, including climate-wise functional linkages, such as the one this project will create in the CND, and by training rangers and other park personnel to proactively manage climate-change-related threats—Rwanda will ensure that the ecosystem service benefits of the country’s biodiversity, including its national parks, will be sustained beyond the life of the Project. Specific finance instruments—such as payment for ecosystem services (PES) and structured impact conservation bonds to ensure ongoing revenues and benefit-sharing with surrounding communities—will be identified and implemented before the end of the project, in a manner consistent with the Biodiversity Finance Plan that the GoR has recently developed with support from BIOFIN. The Project through RFA will hire a Technical Advisor, to work with RDB, MoE, REMA, FONERWA and MINECOFIN to ensure that a portfolio of sustainable finance mechanisms is designed and implemented before the end of the project period to ensure the long-term sustainability of the Project outputs.

Strategies for Long-term Financial Sustainability. Two specific examples of the Government of Rwanda’s commitment and approaches to financial sustainability that will be key parts of the Exit Strategy are the Tourism Revenue Sharing Program (TRSP) of RDB and the Community Adaptation Fund of FONERWA, the Rwanda Green Fund.

Tourism Revenue Sharing Program. The long-term success of the three protected areas in the CND depends, to a substantial extent, on the support of communities living around protected areas. In recognition of this, in 2005 Rwanda established a Tourism Revenue Sharing Program, allocating 5% of gross tourism revenue to local communities. Currently Rwanda and Uganda are the only two African countries with formal tourism benefit sharing programs. RDB recently completed a 15-year review of its Tourism Revenue Sharing Program (Snyman et al. 2023) and it has increased the revenue sharing allocation from 5% to 10%.

The primary challenges identified in the 15-year review are increasing the effectiveness of community engagement, reaching the most vulnerable community members, and ensuring that the TRSP meets community needs. The GoR is interested in adapting the TRS Policy to more effectively engage the communities living around the PAs in the TRSP and to allocate more funding to directly enhance the lives of Rwandan citizens living around PAs. In this regard the Project will work with RDB and district governments, to increase transparency in the Program, ensure that the most vulnerable community members benefit from the TRSP and to link the cooperatives and youth-led SMEs to the Program. This will include working with RDB to use the insights from this project to revise the TRSP revenue allocation model and eligibility to create greater resilience among communities. The 15-year review recommended that the following percentages be used for the revenue allocation going forward: 70% for livelihood projects; 25% for infrastructure support; and 5% in an emergency fund. However, there is a need for input from this Project to ensure that the specific types of projects to be supported most effectively increase community resilience. This will ensure long-term benefits to communities.

The Community Adaptation Facility (CAF) within FONERWA was developed as part of Rwanda’s NDC Deep Dive: Advancing Financial Innovation to Scale Up Climate Action. The aim of the CAF is to provide grant

funding for community projects that scale up climate resilience. In this regard, the Project will support the cooperatives and youth-led SMEs engaged in Component 3 to prepare business plans and link them to innovative financing for community projects with the CAF. In line with the CAF priorities, the business plans will focus on the investment priorities of the Rwanda Green Fund (FONERWA) targeting conservation and sustainable management of natural resources especially ecosystem rehabilitation, sustainable land management, sustainable forest management and promotion and protection of biodiversity. The Project Technical Assistance will work with FONERWA to design specific long-term funding opportunities for communities in the CND to the CAF.

Additionally, the Project will enhance the environmental, social and economic resilience of communities in the CND by enhancing financial and accounting literacy, marketing and entrepreneurship, and vocational skills of community members (e.g., in native tree propagation, forest restoration, and fuel-efficient stove production). This will mean the Project will invest in businesses that restore or indirectly halt the degradation and loss of native forests by reducing or eliminating local stressors. These businesses and projects are expected to live beyond the lifetime of the project. The value chains, once organized profitably, can continue to grow without further support, while the financial service providers will continue to service savings groups and farmers and SMEs in the value chain with sustainable financial services also after the project ends. By raising the productivity from selected crops, diversifying agriculture towards high-value agriproducts, and strengthening their value chains through trainings in marketing, processing and business, the project will attract investments from the private sector in sustainable agriculture and agroforestry. All economic and environmental benefits from the project will be rigorously monitored, captured, compiled and disseminated as knowledge products, packaged for various stakeholders including investors and MFI, with which the project will engage to secure microcredits to beneficiary farmers.

(c) FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding	Total amount			Currency		
	39,056,421			USD (\$)		
GCF financial instrument	Amount	Tenor		Grace period	Pricing	
Grants	39,056,421	-		-	-	
(b) Co-financing information	Total amount			Currency		
	10,943,579			USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Rwanda Development Board	<u>Grant</u>	<u>5,000,000</u>	<u>USD (\$)</u>	-	-	-
Rwanda Forestry Authority	<u>Grant</u>	<u>831,307</u>	<u>USD (\$)</u>	-	-	-
Rwanda Water Resources Board	<u>Grant</u>	<u>5,000,000</u>	<u>USD (\$)</u>	-	-	-
Ministry of Environment	<u>In-kind</u>	<u>112,272</u>	<u>USD (\$)</u>	-	-	-
(c) Total financing (c) = (a)+(b)	Amount			Currency		
	<u>50,000,000</u>			<u>USD (\$)</u>		
(d) Other financing arrangements and contributions	The Government of Rwanda is providing grant financing via the Rwanda Forest Authority (RFA), Rwanda Water Board (RWB), and Rwanda Development Board (RDB) to support activities in component 3. In addition to their contributions, AE will cover office costs in the form of an in-kind contribution equivalent to \$112,271 for the Project Management Unit which will be hosted within AE premises. The co-financiers have been identified based on their existing mandates and activities they implement in the CND region hence this was the only assessment done and they have been able to commit their involvement in the project implementation. The provided co-financing commitments will serve as a basis of MoE engagement during the project implementation. During implementation of the project, annual work plans will be adjusted together with the executing entity and all the co-financiers to align with everyone’s activities. MoE/AE under its oversight role will ensure that all these stakeholders commit during planning on their co-financing and report on it during Annual Performance Reports submitted to GCF.					

C.2. Financing by component

Component	Output	Indicative cost USD (\$)	GCF financing		Co-financing		
			Amount USD (\$)	Financial Instrument	Amount USD (\$)	Financial Instrument	Name of Institutions
Component 1: Mainstreaming Climate adaptation into integrated land use planning	1.1 Landscape-wide land-use plan developed for climate-resilient livelihoods & forest ecosystems, integrating district strategies	1,729,172	1,729,172	Grant			
	1.2 Local & national institutional capacities strengthened to integrate biodiversity & climate risks into land use planning & management	1,195,652	1,195,652	Grant			
Component	Output	Indicative cost USD (\$)	GCF financing		Co-financing		
			Amount USD (\$)	Financial Instrument	Amount USD (\$)	Financial Instrument	Name of Institutions
Component 2: Forest and landscape management and restoration	2.1 PA management effectiveness improved re climate risks & adaptation	2,051,778	2,051,778	Grant			
	2.2 Natural forest cover restored, biodiversity connections established	16,917,145	14,271,796	Grant	2,645,349		Rwanda Water Board
Component 3: Enhancing climate adaptation through promotion of resilient livelihoods and climate-smart farming methods	3.1 Farming methods enhance productivity, reduce erosion & flooding risks, contribute to ecosystem services, and support connectivity	13,451,014	10,856,387	Grant	478,071 2,116,556	Grant Grant	Rwanda Forestry Authority Rwanda Water Board
	3.2 Rural livelihoods generate alternative incomes & reduce pressure on forests	8,612,188	3,850,283	Grant	4,761,905	Grant	Rwanda Development Board
	3.3 Financial services & private sector investment engage	559,890	559,890	Grant			

Monitoring & Evaluation	4.1 - Performance monitoring plan developed	555,475	555,475	Grant			
	4.2 - Project monitored and evaluated; lessons learnt integrated into adaptive management processes	645,390	645,390	Grant			
Project Management Costs	5.1 Project managed efficiently to achieve its goal to build resilience to climate change in CND	1,901,344	1,480,769	Grant	106,925	In-Kind	Ministry of Environment (AE)
					313,650	Grant	Rwanda Forestry Authority
<u>5% Contingency</u>		2,380,952	1,859,830	Grant	39,586	Grant	Rwanda Forestry Authority
					238,095	Grant	Rwanda Water Board
					238,095	Grant	Rwanda development Board
					5,346	In-Kind	Accredited Entity
Indicative total cost (USD)	50,000,000	39,056,421	10,943,579				

C.3 Capacity building and technology development/transfer

C.3.1 Does GCF funding finance capacity building activities? Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer? Yes ☒ No ☐

This project will build knowledge among land-use planning actors on the value of forests for climate resilience, and institutionalizing cross-sectoral meetings to encourage synergies and avoid overlapping

mandates and redundancy in different climate resilience interventions to ensure climate adaptation and climate resilience are explicitly incorporated into land-use planning processes in the CND, such that trade-offs between different sectors can be resolved while balancing the climate resilience of nature and people. Local land-use planning experts and community members will be trained and engaged to review and interpret the National Land Use and Development Master Plan, to ensure the plan accounts for current and future climate risks, and that these risks are also considered in district level planning schemes. The knowledge and insights gained from these activities will be synthesized and used to inform the design of an Integrated Land-use Plan for Resilient Livelihoods and Ecosystems, which will reconcile the cross-sectoral (e.g., tourism, environment, forestry, food production, cash crops for export) trade-offs necessary to sustain forest ecosystem functions, deliver critical ecosystem services and improve community livelihoods. An innovative scientifically credible system for monitoring trends in natural forest cover and forest types will also be established, and a comprehensive capacity building programme will improve use of remote sensing, ecosystem service modelling and spatial planning tools to inform decision making. **Budget: \$2,000,754**

To enhance forest resilience across the CND, through a combination of increased forest cover, improved forest condition, and improved management to reduce degradation and pressure on forests, the project will increase the extent and condition of natural forest within the CND's national parks, by facilitating regeneration of large areas previously degraded by fires, securing key remaining natural areas outside PAs and developing a financial sustainability plan for the CND. The project will invest in the capacity of government agencies, districts, and communities to manage forest fires. Community members will be trained in and provided technical assistance and equipment for restoration techniques, establishment, and management of indigenous tree nurseries. Management plans for the CND's national parks will also be updated and park staff trained to explicitly consider climate adaptation and gender issues and manage the parks as part of the larger CND landscape. The project will also develop the capacity of key Government ministries and agencies on sustainable conservation finance. **Budget: 5,420,837**

Additionally, this project will strengthen the livelihoods of landless and or the most vulnerable households, who are often most at risk of unsustainable use of their farmlands, protected areas and resources within protective forests. Erosion control measures and on-farm tree cover will be increased across the CND, helping reduce the risk of crop failure from extreme events like landslides, reduce land and water degradation caused by soil erosion, and increase provision of valuable products like wood fuel. To decrease pressure on forests for resource extraction, existing successful programs distributing improved cookstoves will be dramatically scaled up. Community members will be trained in and provided technical assistance for SLM practices, establishment, and management protective forests. Both cooperatives, individuals, participating financial institutions (PFIs), and value chain actors/small and mid-size enterprises (SMEs) will benefit from capacity-building to improve their business skills and access to finance, and to strengthen value chains (including beekeeping, ecotourism, agri value chains, and cookstoves), including the construction of processing facilities and marketing materials. Vocational training schools and local youth cooperatives will be supported to produce, test, and distribute Tier 2 cookstoves. **Budget: 13,893,987**

(d) EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

D.1. Impact potential

Adaptation Impact – This Project is designed to restore, sustain and expand critical forest ecosystem functions in the CND. More specifically, the project will protect, restore and expand natural forest in and around the core national parks; consolidate and expand some smaller but critical "stepping stones" of remaining natural habitat between these national parks (mainly small forest patches and wetlands/ riparian areas); improve the landscape linkage function of key high altitude between the protected areas, while at the same time securing ecosystem services through afforestation and improved native species mix for plantation forests on steep slopes; silvopastoral interventions on cultivated pasture; and agroforestry interventions in regenerative agricultural areas. These interventions will reduce erosion, decrease loss of soil organic matter, and thus increase the water infiltration rate, which increases water recharge which will enhance the resilience of forest ecosystems and smallholder farmers to climate change. The establishment of ecological connectivity between fragmented forests will promote landscape heterogeneity and support species migration to more favorable conditions within this human modified landscape and as a result help build the resilience of biodiversity to climate impacts. Increasing the capacity for species movement across these islands of forest habitat, will improve the adaptive capacity of multiple species that serve as important seed dispersers and insure the long-term viability of forests.

Project interventions will include 10,000 ha of land (including both forest and silvopastoral sites) under improved, climate-resilient management; and improve agroforestry practices on 3,346 ha of on-farm plantations, directly benefiting 1,254,242 people (654,404 women) with more climate resilient livelihoods from reduced exposure to landslides, floods and soil erosion and more knowledge on climate risks, value of forests and forests ecosystems, climate adaptation options, access to indigenous and agroforestry quality materials and improved ecosystem services.

The Project will provide direct benefits to increase the climate resilience of more than 1.5 million of the most vulnerable residents of the CND, including increased adaptive capacity through reduced exposure to flash floods, landslides, and soil erosion which will also increase their crop productivity and livelihoods (direct beneficiaries 52.2% female). Indirectly, 2,464,097 beneficiaries in the CND will receive co-benefits from improved and more sustained ecosystem services, including decreased sedimentation leading to improved water quality downstream and increased hydropower generation. This helps Rwanda produce cleaner hydropower energy at a lower cost, making it preferable to other more detrimental energy sources. The project will also indirectly benefit 9,260,745 people and create approximately 24,212 job opportunities in forest-dependent communities.

The Project will contribute altogether to the restoration of 5,000 ha of intact forest, 2,500 ha of protective forests and 3,346 ha of smallholder farmland, 1000 ha of pastures and 1,500 ha of riparian linkages. Overall, 275,617 ha of the landscape could potentially be positively impacted by the project as land use planning, improved natural forest management, improved buffer management and a stronger agroforestry sector would all contribute to improved outcomes beyond the specific areas directly targeted for restoration or improved management by the project. In particular, stronger capacity for land use planning across the CND and hence improved protection of key areas (e.g. steep slopes, riparian areas and protected area buffers) has strong potential to deliver significant mitigation benefits at a landscape rather than a site level. Although the project is focused on the priority landscapes covering 275,617 ha, it is likely that benefits due to increased adaptive capacity (achieved through improved land use planning and management of key features through

reduced exposure to flash floods, landslides, and soil erosion) could feasibly extend across the full 700,000 ha CND.

Additional adaptation benefits will include the training of 52,589 community members in forest management and restoration for climate resilience, which will be linked with ongoing restoration employment opportunities through other GoR and private sector projects; the strengthening of institutional and regulatory systems to promote climate resilience; and targeted capacity building for certain government institutions and ministries in forest monitoring, remote sensing, spatial planning, and analysis. These beneficiaries will include MoE, REMA, RDB, MINAGRI, and the University of Rwanda.

Complementary activities will address key drivers of deforestation while creating alternative livelihood pathways for youth and women through ecotourism and income-generating, climate-resilient crops, as well as strengthening supportive financial services.

Mitigation Impact – The Project will generate mitigation benefits by delivering sequestration of approximately 224,871 t CO₂-eq during the 5-year Project period, which is about 0.8% of the 28.2M t CO₂ reduction targeted by Rwanda's vision 2050.

- 5,000 ha of critical intact natural forests will be under improved management for climate resilience, accounting for an estimated total of 596,116t CO₂-eq over 20 years.
- 1,000 ha of pasture lands with increased coverage of native trees to secure reasonable landscape connectivity for forest species, accounting for an estimated total of 4,543t CO₂-eq over 20 years.
- 1,500 ha of riparian lands will be afforested which will contribute significantly to overall landscape connectivity, sequestering 110,138 t CO₂-eq over 20 years.
- 2,500 ha of protective forests and plantations will be established on slopes >55%, focusing on areas not currently planted in crops, accounting for an estimated total of 193,472 t CO₂-eq over 20 years.
- Erosion control and agroforestry practices will be implemented over 3,346 ha of smallholder farmland, sequestering 129,006t CO₂-eq over 20 years.
- Improved energy efficient stoves will be adopted by 8,500 households, securing 51,017t CO₂-eq of avoided emissions over 20 years.

Overall, there is potential for a significant contribution to GCF1 Core Indicator 4, hectares of natural resource areas brought under low emission or climate resilient practices. Conservatively, an area of 275,617 ha of the landscape could potentially be positively impacted by the project as land use planning, improved natural forest management, improved buffer management and a stronger agroforestry sector would all contribute to improved outcomes beyond the specific areas directly targeted for restoration or improved management by the project.

D.2. Paradigm shift potential

As described in Section B1, the Project provides a profound opportunity to implement new mechanisms of land-use planning and management that are based on consideration of the CND as a landscape, consisting of an inter-dependent mosaic of forests and farms, the management of each determining the health of the whole. Focusing on this larger scale, with coordination across both geographic and sectoral boundaries, while mainstreaming climate risks into management planning represents a fundamental paradigm shift for land-use planning and management in Rwanda. This approach to land-use planning and management is entirely new for Rwanda and is intended to maintain and recover the ecosystem services that are essential to the resilience of smallholder farming communities within the region and to the globally significant and

unique biodiversity in this part of Rwanda. This represents a reversal of the previous negative feedback loop of ecosystem degradation to a more positive, climate-resilient system.

Potential for scaling-up spatial planning and replication. Spatial planning is recognized by the IPCC, the EU and others as playing a key role in climate change adaptation and climate resilient development, but to date development planning in Rwanda has largely been based on tabular data, rather than on spatial planning. Adoption of spatial planning across ministries and on national regional and local levels, represents a significant paradigm shift for Rwanda. Piecemeal decisions and an ad hoc assemblage of isolated sector-by-sector projects will be replaced with a holistic, climate-smart, evidence-based approach to planning and development – enabling systematic, ecosystem-based, multi-sector analysis of land use tradeoffs. The Project will therefore establish a new inter-agency and local-national model and capacities for holistic planning and implementation – one that can be scaled up for use elsewhere in Rwanda and beyond. This will enable better-informed planning decisions, the identification and valuation of nature-based solutions to climate adaptation and mitigation, and overall will foster climate resilient development pathways in the CND and nationwide. Communication and collaboration with complementary GCF and other projects in Rwanda will accelerate this process.

Potential for knowledge and learning. Component 1 will invest in new technologies and climate-smart knowledge to improve national and district planning, scenario analysis and monitoring capacities. In doing so, new mechanisms of coordination and collaboration will be established, and shared across agencies. Component 2 will invest in best practices in forest management and restoration – some of which will require learning beyond the existing science of indigenous tree propagation and restoration (particularly in re-establishing rain forest habitat, establishing protective forests on slopes of >55%, and enriching farms with indigenous tree species). Component 3 will focus on processes to identify and pilot value-chain products (on-farm and off-farm) not yet in practice. All approaches will include the participation of local farmers, women, and youth to contribute to collective learning processes involving national and local government and vulnerable communities. Knowledge generated – and the processes by which it is generated – will be shared across the region. The project is based on collective learning, knowledge generation and dissemination at community, landscape and national levels. A focus on knowledge management through demonstrations, monitoring, and evaluation across the project activities will inform the implementation of current and future adaptation efforts.

Contribution to the creation of an enabling environment. Strengthening the Task Force in Component 1 will ensure a single, first point of contact for all sectors' planning development and investments in the CND and will ensure that planning for climate resilience is addressed. By establishing a mechanism for coordination and harmonization, it will radically improve current conditions for public and private interventions and will establish partnerships and collaborations that live beyond this Project. Also importantly, promotion of sustainable financing for each component will help to assure that initiatives will either be self-sustaining or have solid plans for continued financial support.

Contribution to the regulatory framework and policies. The Project will finance the development of the *Integrated Land-use Plan* which will be adopted by all ten CND districts as a framework for seeking synergies between rural livelihoods and forest and biodiversity protections, reconciling land use conflicts that arise, and guiding development decisions and climate change adaptation. It will update the National Land Use Master Plan, ensuring it incorporates climate resiliency, and is accurate and relevant to local contexts. As development of an Integrated Land-use Planning approach helps better reconcile cross-sectoral tradeoffs and improve climate adaptation planning in the CND, this approach will prove a successful model for application to other regions of Rwanda and to other forest/farming landscapes beyond. In addition,

revising 2 NP management plans to include climate concerns will serve as a model for the remaining parks of Rwanda, as well as for the East African region.

D.3. Sustainable development (max. 500 words, approximately 1 page)

The proposed Project is aligned with the United Nations 2030 Agenda for Sustainable Development, including SDGs 5, 13 and 15 and will contribute to the achievement of the SDGs 1, 2, 3, 5 and 10. The Project will promote an ecosystem-based approach to forest protection, restoration, and rehabilitation that includes:

- Reduction of erosion, sedimentation, and landslide risk as a result of restoration of degraded lands;
- Increased number of native trees on farms with improved agroforestry systems, improving biological connectivity;
- Increased carbon sequestration potential of plants and soils. While the primary objective of the Project is adaptation, the Project will also produce mitigation benefits of 204,079t CO₂-eq during the 5-year implementation period, and 1,033,275t CO₂-eq over the 20-year Project lifetime.
- Reduced GHG emissions through more fuel efficient cookstove technologies, yielding cumulative savings of approximately 51,017t CO₂-eq during the 20-year Project lifetime.

The Project will enable the transition to more climate resilient and productive livelihood models and empower the rural poor, significantly reducing impacts from extreme weather events and increasing their potential for improved wellbeing. Social benefits will include:

- Raised awareness about climate change effects and adaptation in 1,253,252 people living in the CND (at least 52.2% representing women).
- Capacity building of ~52,589 people (at least 50% women), in a wide range of topics aimed at increasing ecosystem and social resilience.
- Create significant social capital through co-designing and co-managing a range of adaptation strategies, indirectly promoting social cohesiveness among CND villages through joint saving and lending systems.
- Reduced inequality and poverty through increased financial inclusion and improved economic opportunities — especially for women, youth and historically marginalized groups.
- Forest and landscape restoration in the CND will have several economic co-benefits for vulnerable communities, including:
 - Soil erosion control measures, protective forests on public lands and agroforestry on private lands, will increase agricultural productivity and associated income.
- The Project will create direct and indirect employment opportunities, benefiting approximately 24,212 households and stimulate the local economy.
- Leveraging private capital to invest in the value chains including cookstoves in the CND. The project will create important social and environmental benefits. However, the project will also have removed a number of risks (notably carbon risk) that worry more traditional (social-impact or sustainability) investors. As a result, the venture becomes an attractive investment for private capital interested in a “triple bottom line”, allowing the business to develop sustainably and scale to more CND areas. During project implementation we will therefore produce a bankable business plan that details the activities and resources required for expanding further.

The Project will result in positive outcomes around access to resources, improved livelihoods, and income generation opportunity and capacity for women. With support from a dedicated Gender Specialist, the Project will engage women in project planning, investment and decision making. With opportunities to generate additional income, women, who traditionally are already further vested in investing in their families, will be more able to respond to incentives that address their families' basic needs, such as better health and nutrition, linking to agriculture and food security improvements. The project will also get support from NGOs that engage men in programming and discussions to increase their participation in household and parental responsibilities and gain greater awareness of the importance of women's fair access and control of resources, including income. Women will benefit from training and educational activities, leadership, entrepreneurship and decision making. Annex 8, which will be further elaborated on at project onset, provides more information on gender-sensitive impacts.

D.4. Needs of recipient

Despite Rwanda's solid economic progress since 2000, poverty remains widespread and pervasive. Overall, 38.2% of the population lives in poverty and 16% in extreme poverty¹¹⁹. Women are more affected by poverty than men, with 47% of female-headed households living in poverty compared with 44.9% of all households. Rural households are more than twice as likely to be in poverty/extreme poverty than urban households. Districts of the Western and Southern Provinces in the CND region have the highest rates of poverty and extreme poverty in the country.

Most rural households are smallholder subsistence farmers and, because of high population density, high relief terrain and dependence on rain fed agriculture, the residents of the CND are highly vulnerable to climate change. These risks include exposure to flash floods¹²⁰, landslides¹²¹ and soil erosion¹²². During stakeholder consultations, communities indicated several socio-economic changes stemming from changing climate conditions. Notably, the longer dry season and unpredictable rains¹²³ have reduced agricultural yield significantly, leading to more widespread malnutrition and extreme poverty, and farmers can barely meet their own families' needs¹²⁴. In particular, women, youth, and historically marginalized people are feeling the impacts. As arable land is increasingly constrained, youth and historically marginalized people have few opportunities and are more focused on survival and short-term gains at the expense of education and sustainable livelihood development (see Annexes 6 and 8 for more detail).

As a result, the GoR has invested significantly in developing an enabling policy and institutional framework for a more climate-resilient development pathway. Yet, while adaptation is a top priority, Rwanda is classified by the U.N. as a LDC and lacks sufficient public financial resources to implement its climate change and resilient green development objectives. Currently there are no alternative funding options of the scale needed to secure Rwanda's priority forests to reduce vulnerability of rural communities to climate risks. The GoR also recognises that a piecemeal approach to decision making has exacerbated competition over scarce land resources and reduced the adaptive capacity of both ecosystems and rural populations. However, many

¹¹⁹ World Bank, 2020. *Poverty and Equity Brief: Rwanda*.

¹²⁰ Karamage, F., et al., 2017. *Modeling Rainfall-Runoff Response to Land Use and Land Cover Change in Rwanda (1990–2016)*. Water 2017, 9(2), 14.

¹²¹ Uwihirwe, J., et al., 2020. *Landslide precipitation thresholds in Rwanda*. Landslides.

¹²² World Bank, 2019. *Rwanda Systematic Country Diagnostic*.

¹²³ World Bank; CIAT. 2015. *Climate-Smart Agriculture in Rwanda*. CSA Country Profiles for Africa, Asia, and Latin America and the Caribbean Series. Washington D.C.: The World Bank Group.

¹²⁴ Clay N., King B., 2019. *Smallholders' uneven capacities to adapt to climate change amid Africa's 'green revolution': Case study of Rwanda's crop intensification program*. World Dev.

agencies operate within siloed environments, focusing on limited scopes framed by individual performance plans, and close coordination is often cited as a key recommendation toward strengthening climate change adaptation measures¹²⁵.

Additionally, district government staff and community members have limited technical understanding or awareness of the complex interactions between forest structure and functions, climate change, and the provision of ecosystem services. Accurate measurement of forest cover and forest type is notoriously difficult in high relief terrain, such as in the CND¹²⁶. Lack of access to and capacity to apply new methods for classifying and monitoring forest change has limited forest monitoring in Rwanda¹²⁷. Consequently, government forest data has historically been limited in scope and infrequently updated, and regional/global maps are too inaccurate to be of use. Capacity building around the technologies and processes to make landscape-scale planning decisions is crucial to the success of Rwanda's climate-resilient development pathway.

D.5. Country ownership

The proposed Project is closely aligned with the NAP's adaptation priorities: 1) promotion of non-agricultural income generating activities, 2) introduction of species resistant to environmental conditions, and 3) development of firewood alternative sources of energy. It is also aligned with several priorities identified in the 2020 NDC: promotion of afforestation/reforestation, improvement of forest management, integrated approach to planning and monitoring for sustainable land management, development of a harmonized and integrated spatial data management system for sustainable land use management. The Project will support implementation of the Green Growth and Climate Resilient Strategy and the Rwanda National Strategy for Transformation–NST1 (2017-2022) and is aligned with the goals of the Rwanda National Forestry Policy 2018, the Forest Sector Strategic Plan 2017-2022, the Strategic Plan for the Environment and Natural Resources Sector (2018-2024), and National Forest Management Plan 2017-2026, reflecting the government's intentions to address climate change impacts and mitigation targets by improving forest management. It is also aligned with the national Biomass Energy Strategy (BEST) 2020 which aims to ensure a more sustainable biomass supply and combustion technologies for homes and small enterprises. This Project aligns with the forthcoming GCF country program which prioritizes forest management through restoration and protection of natural forests, afforestation and agroforestry, as well as distribution of high-quality forestation inputs and capacity development.

While this Project proposal to GCF focuses on a different geographic area - with specific adaptation needs through land restoration, development of markets and finance mechanism, as well as support to mainstreaming climate change into land use planning - the early and ongoing lessons from the experiences from existing GCF-funded projects, "Strengthening Climate Resilience of Rural Communities in Northern Rwanda"¹²⁸ and "Transforming Eastern Province through adaptation" (TREPA)¹²⁹ will be applied during the

¹²⁵ Bagstad, K.J., 2019. *Towards ecosystem accounts for Rwanda: Tracking 25 years of change in flows and potential supply of ecosystem services*. British Ecological Society.

¹²⁶ Arakwiye, B., et al, 2021. *Thirty years of forest-cover change in Western Rwanda during periods of wars and environmental policy shifts*. Regional Environmental Change 21(2).

¹²⁷ Herold, M., Schiller, F., 2009. *An assessment of national forest monitoring capabilities in tropical non-Annex I countries: Recommendations for capacity building*. Global Observation of Forest and Land Cover Dynamics Land Cover Project Office.

¹²⁸ Rwanda Ministry of Environment, 2018. *FP073: Strengthening Climate Resilience of Rural Communities in Northern Rwanda*. Green Climate Fund.

¹²⁹ International Union for Conservation of Nature, 2021. *FP167: Transforming Eastern Province through Adaptation*. Green Climate Fund.

implementation of this Project in the Congo Nile Divide Landscape.

MoE is responsible for ensuring sustainable management of natural resources and for the development of policies, strategies, regulations, and mobilizing resources for the development of the sector. The MoE managed USD 10 million from the Adaptation Fund to implement the Reducing Vulnerability to Climate Change in North West Rwanda through Community Based Adaptation Programme¹³⁰ and has overseen a number of other climate projects including: an early warning system project (funded by LDCF), the Landscape Approach to Forest Restoration and Conservation (LAFREC), and the “Building resilience of communities living in degraded forests, savannahs and wetlands of Rwanda through an ecosystem management approach” programme (funded by LDCF)¹³¹. The OECD rated Rwanda as one of the countries that uses aid most effectively¹³², and Rwanda is on course to achieve the SDGs. The GoR has demonstrated comprehensive country ownership of, and capacity to implement, a funded project. The national designated authority (NDA) is the core interface between a country and the GCF and is mandated to align GCF-supported activities with strategic national objectives and priorities. The NDA coordinates with AEs on proposal development and completes a no-objection letter after it is determined that the proposal is aligned with the national priorities. This Project was initiated by the MoE and supported by the NDA, Rwanda Environment Management Authority, due to the high climate sensitivity of the CND.

Government agencies such as Rwanda Forestry Authority (RFA), Rwanda Development Board (RDB), Rwanda Environment Management Authority (REMA) and National Land Use Authority (NLA) will maintain strong ownership and support activities under component 1 and 2 as aligned with their mandates and the aforementioned plans and strategies.

To better understand the interests of stakeholders and potential impacts of the Project, and as preparation to design future engagement strategies, a comprehensive, gender-responsive, culturally sensitive, non-discriminatory, and inclusive stakeholder analysis was undertaken and is incorporated into the Stakeholder Engagement Plan (Annex 7).

D.6. Efficiency and effectiveness

Financing and concessional: The financial instrument chosen for this Project is grant financing of USD 39.06 million which constitutes roughly 78% of the total Project’s estimated cost. The relatively high level of concessionality reflects the public good character of the investments to be made by the Project, Rwanda’s status as an LDC¹³³ and its high vulnerability to climate change¹³⁴. Furthermore, it must be highlighted that most project expenditures relate to creating public goods through forest and landscape restoration or capacity-building of public institutions. These interventions are primarily non-income-generating; hence very hard to attract private sector investment. Attracting the private sector for public goods investments remains challenging in the context of many countries worldwide, including Rwanda.

Addressing low private sector investment in the forestry sector. One of the major reasons for lack of private-sector investment in Rwanda’s Forestry Sector is that most stands are too small to sustain an

¹³⁰ Adaptation Fund, 2014. Reducing Vulnerability to Climate Change in North West Rwanda through Community Based Adaptation. [Link](#).

¹³¹ UNEP, 2016. *Building resilience of communities living in degraded forests, savannahs and wetlands of Rwanda through an ecosystem management approach*.

¹³² Newfarmer, R., et al., 2013. *Managing Aid for Trade and Development Results: Rwanda Case Study*. OECD.

¹³³ United Nations Department of Economic and Social Affairs, 2021. Least Developed Country Category: Rwanda Profile.

¹³⁴ World Bank, 2021. *Rwanda*. Climate Change Knowledge Portal.

economically viable operation (see the Forest Sector Strategic Plan). By restoring smallholder protective forest plantations and organizing them into Private Forest Management Units (PFMUs), this Project will assist with creation of units that are economically attractive to private investors, helping to facilitate future private investment if landowners desire it. However, no private investors will be subsidized by the Project, and ownership of PFMUs will remain with the state/district governments and smallholders' cooperatives who own forest plantations. RFA in collaboration with RCA and the district government directly implement organization of the cooperatives and conflicts in the cooperatives are managed in accordance with internal rules and regulation governing each cooperative overseen by the Rwanda Cooperative Agency in case of escalated conflicts but specifically in the cooperatives that will be benefiting under this project, RFA and District government will also intervene in conflict settlement.

No excess subsidy will be provided to the private sector, and no private sector investors will be crowded out by these investments. Wherever these are required for institutional and capacity support for market development and crowding in for private investments these are undertaken with concessionality to community in mind. For example, the subsidy/microcredit schemes to facilitate the adoption of efficient cookstoves are undertaken with minimum concessionality. The value chain development for other potential revenue generating streams such as honey undertaken under output 3.2 requires institutional and capacity support funded by GCF grant. In the case of ecotourism for tea plantations, no project concessionality is planned and complete crowding in from the private sector is envisaged. The value chain development of beekeeping is 100% funded by GCF grants so that an enabling environment can be created, and the project can overcome affordability challenges through developing a system such as community co-guarantee, overcome lack of private capital structure and insufficient markets. Because of above mentioned reasons no private sector investors are expected to be crowded out by project investments.

GCF grant as a suitable financing instrument. The GCF grant can help create the enabling environment for the future engagement of the private sector in similar projects. The showcasing of potential revenue-generating ways, thanks to the concessional GCF financing, may help invigorate the potential for future investments in this sector, which in turn might result in a multiplier effect leading to a higher degree of development impact.

Results of the Economic and Financial Analysis (EFA). Following the GCF guidelines for developing the Annex 3 package, the EFA was prepared to assess the economic rationale for the proposed interventions. Consequently, four representative interventions were appraised in the EFA: (i). agroforestry, (ii). protective forests (iii). fuel-efficient cookstoves, (iv). silvo-pastoralism. Best practices in appraisal for public sector projects have been followed. The financial analysis for each intervention was calculated from the private perspective using a discount rate of 15.44% which was the lending rate of Rwanda Central Bank. For the three modeled interventions (see table below) the financial NPV is positive and financial IRR is higher than discount rate signifying that project activities are profitable. The GCF support will result in a positive financial return for farmers or farmers and private actors over a 20-year period and will help overcome high up-front investment costs during the project duration.

Financial analysis (US\$)	Agroforestry	Protective Forest	Silvopasture
FNPV	\$34,483,676	\$2,980,381	\$310,901
FIRR	42%	25%	35%

Aggregate financial analysis of with project (WP) interventions

The obtained ex-ante aggregate economic net present values (ENPVs) for these interventions over the 20 years' timeframe and at a 13% economic discount rate range between USD 0.97million and USD 78.93 million, with economic internal rates of return (EIRRs) between 49% and 72% (see table below).

Economic analysis (US\$)	Agroforestry	Protective Forest	Energy efficient cookstoves	Silvopasture
ENPV	\$58,934,340	\$30,218,468	\$10,585,674.50	\$966,331
EIRR	49 %	72%	N/A	51 %

Aggregate financial analysis of with project (WP) interventions¹³⁵

The sensitivity analysis that was carried out suggests that obtained results are relatively insensitive to increases in costs or decreases in benefits offering the robustness of obtained results. Therefore, the economic sustainability of the Project is likely, and the Project shows the economic rationale for implementation as benefits are likely to outweigh the costs.

Best Practices *Forest Restoration/Tree Planting*: The Project will use Rwanda's best practice guidelines for tree nursery establishment, tree planting and tree management¹³⁶. For assisted natural regeneration, methods developed and proven through multi-year research in Nyungwe and Gishwati-Mukura National Parks will be used¹³⁷. *Forest Monitoring*: The Project will utilize multiple contemporary methods for monitoring the success of project interventions and forest health. Forest plantations will be mapped and monitored using the Rwandan government's newly developed Forest Monitoring and Evaluation Software (FMES)¹³⁸, specifically developed for this purpose. The Project will develop a modern indigenous forest monitoring system, utilize the most high-resolution datasets currently available (<3m) and take advantage of modern remote-sensing advances including cloud computing and machine learning. Cost effective replication for forest restoration and climate resilience throughout Africa and beyond can be anticipated.

¹³⁵ No financial analyses for cookstoves intervention was warranted as energy efficient cookstoves are non-income generating intervention with expectation of no financial cash inflows. Only economic analyses was conducted for cookstoves.

¹³⁶ Unique Ltd. & RNRA. 2015. Tree plantation establishment and management manual for Rwanda. Kigali, Rwanda

¹³⁷ Masozera, A.B., 2004. Regeneration of Burned Forested Areas With Periodic Removal of Pteridium aquilinum. Wildlife Conservation Society Rwanda.

¹³⁸ Rwanda Forest Authority, Enabel, 2021. Technical guidelines on the use of the Forest Monitoring & Evaluation System - FMES. Kigali.

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

- ☐ Reduced emissions (mitigation)
☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Rwanda's natural forests are extremely depleted and degraded. While overall forest cover has increased since the 1980s, this is primarily made up of exotic monocultures, which have very low potential to provide ecosystem services and adaptation benefits. There are small-scale efforts to establish/restore native forests and increase overall forest resilience, but scaling these efforts is hampered by a limited supply of indigenous seedlings.	<u>Low</u>	Paradigm shift would involve a move away from widespread planting of exotic species for fuelwood, as well as improved erosion control measures on farmland and use of improved cookstoves, both of which will reduce pressure on Rwanda's forests. As the benefits of planting indigenous trees become more widely known, behavior changes in non-targeted communities are likely to be catalyzed.	The project is expected to bring 144,058 ha of land under climate-resilient management. This would represent a substantial step towards paradigm shift on increased resilience of forests and vulnerable communities, even before any replication effects, which will likely be catalyzed by project activities focusing on training, capacity building, and peer-to-peer knowledge exchange. The project will also deliver direct carbon reductions of 1,204,448 t CO ₂ -eq over 20 years, and potentially more through increased institutional capacity to mainstream climate adaptation and improve land-use planning.
Replicability	There are a small number of interventions occurring across Rwanda focused on improving resilience of vulnerable communities through forest restoration, agroforestry, livelihood diversification and clean cooking technologies. However, many interventions focus on exotic species which have very low potential to provide ecosystem services and adaptation benefits.	<u>Medium</u>	If increasing the extent and integrity of natural forests and improving capacity for climate-responsive land management can increase climate resilience of vulnerable communities in the CND, then the solution could be replicated throughout Rwanda and across East Africa. Further, if incorporating indigenous species into forest restoration and agroforestry can provide a similar/greater range of adaptation benefits compared to exotic species, and also avoid the drawbacks of exotic monocultures, then the solution could be expanded nationally. If development of an Integrated land use planning approach helps better reconcile cross-sectoral tradeoffs and improve climate adaptation planning	The project will invest in strengthening institutional capacity to promote climate resilience, directly increase the resilience of ecosystems and community livelihoods, and identify options for use of indigenous species in forest restoration and agroforestry, in order to better contribute to climate-resilient livelihoods. These topics have received little attention to date in Rwanda, and lessons learned will inform efforts to improve climate resilience across Rwanda and East Africa.

			in the CND, this approach could be applied to other regions of Rwanda.	
Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Sustainability	The GoR has invested in developing policies and institutional frameworks for a more climate-resilient development pathway. However, domestic funding to implement these policies remains lacking, and as yet they have not translated to wide-scale improvements to climate resilience. Changes in behavioral norms have been limited, and overall uptake of climate adaptation solutions is poor.	<u>Low</u>	Paradigm shift would see Rwanda demonstrate that pursuing development goals need not come at the cost of extensive environmental degradation. Climate-responsive Forest and landscape management will be standard, and government programmes will focus on increasing the resilience of forest ecosystems and enhancing their ecosystem services. Behavior change across the CND would support this, as communities recognise, value, and protect the benefits provided by forests.	The project will work closely with government and communities to build capacity to mainstream climate adaptation into forest and land-use planning and management. The project will facilitate and demonstrate the benefits of climate-resilient agricultural practices, forest management, and alternative livelihoods, raising awareness within and outside target communities.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Mitigation Results Area 4: Forests and land use	Core Indicator 1: GHG emissions reduced, avoided or removed/ sequestered	Greenhouse gas emission assessment	0 tonnes of CO ₂	107,686 tCO ₂ -eq	224,871tCO ₂ -eq (5-year implementation period) 1,084,291tCO ₂ -eq (20-year lifespan period)	Government priorities remain supportive of long-term forest restoration and afforestation plans. Absence of major natural disasters including forest fires and landslides in target areas The economic, social and political context in the country and project areas remain stable.

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Adaptation Results <i>Area 1: Most vulnerable people and communities</i>	Core Indicator 2: Direct and indirect beneficiaries reached	Annual Agricultural Survey ¹³⁹ National census Project Baseline and Completion Reports, including documentation of workers contracted for reforestation/afforestation work, farmers adopting agroforestry activities, good silvo-pastoral practices, restoration of riparian linkages, signed MoUs for Private Forest Management Units, and households receiving improved cookstoves	Direct beneficiaries: 0 Indirect beneficiaries: 0	Direct beneficiaries: 95,873 (50,046 females) Indirect beneficiaries: 2,809,381 (1,466,497 females)	Direct beneficiaries: 108,277 (56,521 females) Indirect beneficiaries: 2,809,381 (1,466,497 females)	Government priorities remain focused on long term forest restoration and afforestation plans Absence of major natural disasters including forest fires and landslides in target areas Rural landowners agree to adoption of agroforestry and reforestation activities
Adaptation Results <i>Area 1: Most vulnerable people and communities</i>	Supplementary Indicator 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options	Project Baseline and Completion Reports, including documentation of beekeepers using modern hives, farmers at different levels of value chains, people employed in ecotourism, have access to	Direct beneficiaries: 0 Indirect beneficiaries: 0	Direct beneficiaries: 26,160 (13,656 females) Indirect beneficiaries:	Direct beneficiaries: 52,319 (27,311 females) Indirect beneficiaries:	

¹³⁹ The National Institute of Statistics of Rwanda conducts the Agricultural Survey yearly. Available at: <https://www.statistics.gov.rw/statistical-publications/subject/agriculture-and-environment>

		finance for other climate-resilient livelihood options		2,809,381 (1,466,497 females)	2,809,381 (1,466,497 females)	
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Adaptation Results Area 4: Ecosystems and ecosystem services	Core Indicator 4: Hectares of natural resource areas brought under improved low emission and/or climate-resilient management practices	Updated National Park Management Plans complete (ha per park) Forest cover statistics from Forest Monitoring and Evaluation System (FMES) GIS Data/Remote Sensing Ranger Based Monitoring data from PAs Vigilance Committees monitoring data	No increase in climate-resilient management practices (o)	42,558 ha of land brought under climate-resilient management	130,104 ¹⁴⁰ ha of land brought under climate-resilient management	Absence of major natural disasters including forest fires in target areas
Adaptation Results Area 2: Health, well-being, food and water security	Supplementary Indicator 2.2: Beneficiaries (female/male) with improved food security	Annual Agricultural Survey Gender responsive Comprehensive Food Security and Vulnerability and Nutrition Analysis Survey (CFSVA) ¹⁴¹	TBD	Direct beneficiaries: 0 Indirect beneficiaries: 0	Direct beneficiaries: 1,598,536 (52.2% female) Indirect beneficiaries: 2,464,097 (52.2% female)	Absence of extreme natural disasters and economic shocks affecting yields and household economy; migration patterns do not significantly affect the number and status of households.

¹⁴⁰ 130,104 ha refer to 121,758 ha of all core PAs and 8,346 ha restored areas outside PAs

¹⁴¹ The National Institute of Statistics of Rwanda conducts the CFSVA every 3 years.

The level of beneficiary of this project are individual households, - who can be in a cooperative or a business community group - who will participate directly in the project, and thus benefit from its implementation. Beneficiaries include all people who own the land where activities will take place and/or those who will be employed by the project (except EE staff, consultants and implementation and contractual organizations). Dependents of household that directly benefit from the project are also considered as direct beneficiaries and were calculated based on 4.2 average household size¹⁴² and 0.37¹⁴³ average land size in the CNDL.

Adaptation Results Area 1, Core Indicator 2 direct beneficiaries are beneficiaries of activities 2.2.2, 2.2.3, 3.1.1, 1.1.2.

Adaptation Results Area 1, Supplementary Indicator 2.1 direct beneficiaries of activity 3.2.1, 3.2.2, 3.2.4 (excluding 8, 500 beneficiaries of cookstoves) and 3.3.2

Adaptation Results Area 2 Supplementary Indicator 2.2 direct beneficiaries are the total population in landscape implementation sectors of this project.

Indirect beneficiaries broadly refer to all other human populations living within the CNDL districts. More details on calculations are provided in annex 23.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)					
Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low-emission and climate-	The GoR is committed to developing an enabling policy and institutional framework for a more climate-resilient development pathway, as evidenced by the large number of government policies and strategies focused on improving climate mitigation and adaptation. However, climate change is not being comprehensively considered by all sectors, cross-sectoral planning is rare, and new institutions are needed to	<u>low</u>	An integrated planning approach is used to reconcile cross-sectoral tradeoffs while explicitly considering climate resilience actions. New institutions have permanent capacity in place to ensure that spatial data on climate, forests and ecosystem services is incorporated into land planning and management.	The project will facilitate development of an integrated land use plan for the CND which will reconcile the cross-sectoral (e.g., tourism, environment, forestry, food production, cash crops for export) trade-offs necessary to sustain forest ecosystem functions, deliver critical ecosystem services and improve community livelihoods.	National level (one country)

¹⁴² National Institute of Statistics of Rwanda (NISR); The Fifth Rwanda Population and Housing Census, Main Indicators Report, February 2023

¹⁴³ National Institute of Statistics of Rwanda (NISR), Agricultural Household Survey 2020 report, December 2021

<i>resilient development pathways in a country-driven manner</i>	provide relevant spatial data to inform planning			The project will facilitate the development and adoption of an integrated landscape level planning taskforce, in line with the JADF to bring the relevant stakeholders (ministries and their agencies, districts, private sector and civil society organizations) on board, to strengthen inter-sectoral collaboration on the planning and implementation of the FLR plans. The project will also support capacity development of relevant technical staff working on FLR and land use planning (from central government agencies, district government, and NGOs) and will assist the GoR in developing a dedicated SPU within an appropriate ministry to provide the technical expertise needed to process and integrate spatial planning into decision-making within and across sectors.	
Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core Indicator 6: <i>Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer, and innovation</i>	Climate-resilient development in Rwanda is hampered by a lack of modern technological advances, both in terms of generating information required for land-use management and decision making, and in deploying climate-resilient practices and technology. Rwanda has only recently developed a system to monitor plantation forests and lacks a robust system to monitor indigenous forests. Similarly, agricultural and forest management techniques are generally rudimentary, and most rural	<u>low</u>	The GoR has an innovative, scientifically credible system for monitoring indigenous forest cover and condition, as well as the technical capacity for incorporating spatial data into land-use management. Improved agricultural practices (e.g. terracing, modern beehives) are widespread, and locally produced improved cookstoves are in use by most households.	The project will design and finance the establishment of a new indigenous forest monitoring system for Rwanda. The project will also establish improved agricultural practices across 3,346 ha, as well as increasing use of modern beehives to reduce forest fire risk. Finally, the project will upskill technical colleges to enable local production and distribution of improved cookstoves to 8,500 households.	National level (one country)

	communities in the CND do not use modern cooking technologies.				
Core Indicator 8: <i>Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards.</i>	While GoR recognizes the need for developing an enabling policy and institutional framework for a more climate-resilient development pathway, the value of forests and ecosystem services for climate adaptation is poorly understood, both in government and by communities. There is no system to regularly produce data on indigenous forest cover and condition, and no specialized government unit dedicated to producing spatial data on climate change, forests or ecosystem services. Agricultural and forest management practices are generally rudimentary.	<u>low</u>	Rwanda has an innovative, scientifically credible system for generating and sharing information on natural forest cover and condition, and a government unit dedicated to ensuring spatial planning knowledge is incorporated into land-use management. Improved agricultural practices (e.g. terracing, modern beehives) are widespread, and degraded forest plantations are restored and managed according to government best-practices.	The project will design and finance the establishment of a new indigenous forest monitoring system for Rwanda, as well as establish a spatial planning unit within government that has the capacity to generate and integrate spatial planning knowledge into decision making. Project activities will also establish best-practice agricultural and forest management practices across 144,058 ha of land, and create community structures (e.g. farmer cooperatives) which facilitate knowledge and information sharing among community members.	National level (one country)

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 1: Landscape planning, policies & management effective & coordinated across sectors and scales to address climate risks & adaptation benefits						
Output 1.1. Landscape-wide land-use plan developed for climate-resilient livelihoods and forest ecosystems, integrating district strategies.	Operationalization of an Integrated Land-use Plan for Resilient Livelihoods and Ecosystems in the CND Climate change and biodiversity reflected in District Development Strategies (DDS)	Interim evaluation and final evaluation report Performance evaluations of the districts	0 integrated plans for the CND Climate change and biodiversity Indicators and targets not included in evaluations	1 draft integrated land use plan developed Climate change and biodiversity Indicators and targets fully integrated in evaluations	1 final integrated land use plan developed, approved and used Climate change and biodiversity Indicators and targets fully integrated in evaluations.	District land-use plans are updated within project timeframe Key stakeholders are supportive of cross sector collaboration. Sectoral ministries incorporate biodiversity and climate change into sectoral plans and programmes.
Output 1.2. Local and national institutional capacities strengthened to integrate biodiversity and climate risks into land use planning and management	District planners, staff within REMA, RDB, NLA, RDB with the skills and capacities to use spatial planning tools, generate data necessary for mainstreaming climate change, understand gender barriers and how to mainstream gender equality principles into planning, including supporting women, youth	Interim evaluation and final evaluation report Pre/post training assessments of unit members	0 units with capacity 0 staff reporting improved awareness and capacity.	Draft structure & function of spatial planning unit developed. Training materials in development	Spatial planning unit formally established. 36 Staff reporting improved awareness and capacity (18 Female, 18 Male)	Lack of leadership of the hosting institution/ government agency could delay the establishment of the unit. RFA is committed to lead and invest in the capacity of its staff to implement the monitoring system.

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	and marginalized groups participation in meetings					Capacity building programs can overcome the effects of high staff turn-over and frequent re-arrangement of the institutions of natural resources management (transfer of departments, merging and creation of new Ministries)
	UR GIS center with the capacity and skills to train students and staff from government agencies on spatial planning and climate modeling	Pre/post capacity assessment of the UR GIS Center	Staff with knowledge in spatial analysis and climate modeling but without quantitative skills to use new technologies/software to generate data for decision making	Training modules developed and staff to be trained identified	10 lectures and assistant lectures reporting improved skills, 20 students trained each year from year 3.	
	A system for monitoring forest trends is established and in use by the GOR	Interim evaluation and final evaluation report	0 systems in place	Operational structure & function of forest monitoring system developed	Forest monitoring system in operation	

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	Rural people, specifically women and youth, are trained and participate in land use planning processes at district level	Interim evaluation and final evaluation report	0	3 women per district 3 youth per district	30 youth 30 women	Rural women and youth groups in targeted districts are motivated to invest time and efforts in the activities.
Outcome 2: Natural forests protected, connected, more resilient to climate impacts & risks						
Output 2.1. Protected Area management effectiveness improved re climate risks and adaptation	Climate adaptation reflected in the new general park management plans	Progress report, mid-term evaluation and final evaluation	0 plans currently contain climate adaptation plans	3 General management plans updated and approved by key stakeholders	Updated general management plans guiding RDB management decisions	General Park management plans are in the process of revision during the project intervention

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	RDB has adopted a landscape approach to conservation. New iterations of park management plans reference integrated CND land use plan A framework for conservation planning at landscape level is operational, and RDB planning and monitoring dept at each park is actively engaging stakeholders	Progress report, mid-term and final evaluations	0 park management plans reference CND land use plan	1 new park management plan referencing CND land use plan RDB proactively engaging stakeholders in the land use planning process of the CND	3 new park management plans referencing CND land use plan	Limited engagement of RDB beyond parks' boundaries could delay the operationalization of the planning framework

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	Number of RDB, RFA, districts staff and local communities trained in fire forecasting, detection and management	Post-training assessments of change in capacity of trained park staff in designated positions on 1) fire detection, early warning methods and fire danger rating system and 2) management of climate information and its relevance for park management Post-training assessments of change in capacity of trained individuals in designated positions	0 staff reporting improved awareness and capacity 0 staff reporting improved capacity for fire management	15 Staff reporting improved awareness and capacity (8 Female, 7 Male) 15 staff reporting improved capacity for fire management	15 Staff reporting improved awareness and capacity (8 Female, 7 Male) 30 staff reporting improved capacity for fire management	Capacity building programs can overcome the effects of high staff turn-over and frequent re-arrangement of the institutions of natural resources management (transfer of departments, merging and creation of new Ministries) All stakeholders embrace the concept of information sharing and learning willingly and enthusiastically.
	Financial instruments to generate revenues post GCF identified and piloted	Interim and final evaluation reports Technical reports	0 financial instruments	1 financial instrument identified	3 financial instruments identified and 1 instrument piloted	GoR commitment to enactment of policies conducive to attracting investment in emerging financing solutions

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Output 2.2. Natural forest cover restored, biodiversity connections established	Area (# ha) of degraded natural forest restored in national parks	GIS mapping of cleared areas using field GPS data	0 ha restored	4,500 ha initially cleared	4,500 ha cleared at least four times, ensuring natural regeneration can take place in Nyungwe NP	Major fire outbreaks during project implementation may prevent reaching the targets
			0 ha restored	500 ha under restoration in GMNP	500 ha restored in GMNP	The park is well managed and there are no major illegal activities
	Area (#ha) of riparian lands restored with indigenous species in the CND	GIS mapping of restored areas using field GPS data	0 ha restored	500 ha	1500 ha restored	District officials are committed to protect and create 10 m buffer around streams/rivers as per land use policy
	Area of remnant natural forest loss avoided	Interim and final evaluation	TBA	Draft strategy and action for the protection and conservation of remnant forests	Strategy and action plan developed and implemented for the conservation of remnant natural forests	RDB, REMA and District officials are committed to protect remnant natural forests

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	Area (ha) of pastoral lands brought under climate resilient silvopastoral packages in GMNP Average tree density/ha in restored silvopastoral lands	GIS mapping Bi-annual technical report Interim Evaluation and Final Evaluation Report Annual forest inventory	TBA TBA TBA	500 ha Density: 50 tree/ha	1000 ha Density: 50 tree/ha	Farmers are willing to integrate trees on their pasture lands
Outcome 3: Vulnerable rural livelihoods more climate-resilient, diverse, economically sustainable & nature-positive						
Output 3.1 Farming methods enhance productivity, reduce erosion and flooding risks, contribute to ecosystem services, and support connectivity.	Area (# ha) of forest restored on slopes >55%	Interim Evaluation and Final Evaluation Report Annual forest cover report ¹⁴⁴ Catalogue of geo-referenced area under restoration	0 hectares	2,500 ha under restoration on slopes >55% planted with seedlings.	2,500 public and private hectares restored and maintained. density: 1,200 trees/ha	Farmers, women, and youth groups in targeted districts/sectors are motivated to invest time and efforts in the activities.

¹⁴⁴ The Annual Forest Report is prepared by Rwanda Forest Authority. The Monitoring system for the preparation of these reports has been developed.

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	% of tree species diversity in restored areas on slopes >50%	Interim Evaluation and Final Evaluation Report Annual forest inventory	0	30% indigenous species and 70% exotic species on private land 70% indigenous species and 30% exotic species on public land	30% indigenous species and 70% exotic species on private land 70% indigenous species and 30% exotic species on public land	
	Number of target beneficiaries (households) in target sites who have adopted agroforestry measures (and report improvements in soil and water management)	Interim Evaluation and Final Evaluation Report Household surveys	0	5,405 households (of which 50% are women headed households) adopted agroforestry measures	5,405 households beneficiaries (of which 50% are female).	
	Average tree density/ha in restored agroforestry lands	Interim Evaluation and Final Evaluation Report Household surveys Catalogue of geo-referenced area under restoration Annual forest inventory	0 TBA	 100 trees/ha	3,346 ha of agroforestry established. 100 trees/ha	

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Output 3.2. Rural livelihoods generate alternative incomes & reduce pressure on forests	Number of value chain products that are viable and marketed (Honey, Avocado, macadamia, horticulture)	Interim evaluation and final evaluation report Annual sales reports	0	3 products developed	5 products developed	Stakeholders in the value chains are collaborative
	Number of businesses/cooperatives with viable business plans established and operational for climate resilient value chains	Bi-annual technical report Interim Evaluation and Final Evaluation Report	37 beekeeping cooperatives with 1,965 members (32% women, 16% youth) 0 seed enterprise, nurseries farmer's cooperatives 0 cooperative of tourism guides	15 new cooperatives (40% women, 50% youth) 5 seed and nursery enterprises owned by youth and women. 2 cooperatives	20 new cooperatives (40% women, 50% youth) 10 seed and nursery enterprises owned by youth and women. 3 cooperatives	Willingness of farmers' representatives of and Rwanda Cooperative Agency (RCA) to participate in the project
	Increase in household income within community associations/cooperatives engaged in the project (disaggregated by gender and social category)	Household surveys	TBA in year 1	20% additional income in participating households in new income generating activities. Targets for health benefits, and time and firewood savings TBD	An additional 10%	Successful rolled out of capacity building and linkage to outside (input-output) market

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	# of target beneficiary households adopting cookstoves	Interim Evaluation and Final Evaluation Report Household surveys	0 target households using Tier 2 cookstoves	4000 target households of which 50% are women-headed households	8,500 target households using Tier 2 cookstoves	
	Number of employments generated from the selected value chains	Interim and final evaluations Annual reports on employees hired per value chain	TBA in year 1	# women:600 # young:400	# women:900 # young:600	No outbreak of a new pandemic Farmers, women and youth groups in targeted districts/sectors are motivated to invest time and efforts in the activities.
Output 3.3. Financial services & private sector investment engage	Agriculture Credit assessment tool upgraded to include forestry	Loan approval reports generated by the tool Interim and final evaluation	0	3 MFIs and PFIs are using the tool to provide loans to smallholder farmers 70% of the smallholder farmers and gradually grow	5 MFIs and PFIs are using the tool to provide loans to smallholder farmers	MFIs and PFIs are willing to participate in this project
	Number of financial products refined, contextualized , and rolled out to improve access to finance for agriproducts for 1) vulnerable groups/ smallholders and 2) horticulture, bee products,	Bi-annual technical report Interim Evaluation and Final Evaluation Report	TBA	4 financial products refined and rolled out . At least 1 for each value chain (4 total)	6 financial products already deployed by PFIs are refined, contextualized and at least 5 rolled out by PFIs for financing	MFIs are willing to contribute in terms of staff and resources

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	energy efficient cookstoves and ecotourism value chains				selected value chains. 10 approved loans' applications	
	Number of people including women and youth with organizational, technical and financial improved capacity	Interim and Final Evaluations	0	7,800 (40% youth, 50% women)	12,445 (40% youth, 50% women)	Farmers, women and youth groups in targeted districts/sectors are motivated to invest time and efforts in the activities.
	Number of men, women and youth who have access to thriving financial services	Interim and Final Evaluations	0	3,000 (40% youth, 50% women)	8,500 (40% youth, 50% women)	Men, women and youth are open to access financial services

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 1.1.1 Synthesize & disseminate information on value of natural forests & ecosystem services	Ensure that the value of natural forests and ecosystem services for supporting climate change resilient livelihoods is fully appreciated by all sectors involved in land use decision making	1.1.1.1: Map the sectors involved in land-use planning in the CND and review how forests, ecosystem services and climate resilience is incorporated into each sector's planning process. 1.1.1.2: Conduct comprehensive literature review & stakeholder consultation to collect & synthesize information on climate risks for various	A map of the sectors involved in the land-use planning in the CND Information package on climate risks, value of forests for increasing resilience of local communities, adaptation options developed and disseminated to ministries, local government and local NGOs

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		sectors (e.g. agriculture, forestry), the value of forests for increasing resilience of local communities, and highlight adaptation solutions with cross-sectoral implications 1.1.1.3: Host workshops & presentations with key ministries, government organizations, NGOs and community organizations to disseminate high-level knowledge on value of forests for increasing resilience of local communities 1.1.1.4: Develop guidelines for integrating climate risk into land use planning and cross sectoral planning 1.1.1.5: Develop outreach materials on climate risk in the CND and the value of forest ecosystems for increasing resilience 1.1.1.6: Implement outreach program tailored to different stakeholders (local government, civil society, communities) to enhance capacities for land-use planning, funding mobilization, and delivery of climate adaptation actions 1.1.1.7 Conduct climate literacy courses for local government and civil society organizations, aimed at increasing women and youth participation 1.1.1.8 Provide financial and logistical support to trained organizations in grassroots mobilization to increase	Guidelines for integrating climate risk into land use planning and cross sectoral plan developed 104 episodes of radio drama to educate communities on climate change and gender issues, climate risks, value of forests and forests ecosystems, climate adaptation options developed Broadcast of 104 radio program episodes on a popular national radio station

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		women and youth participation in climate adaptation planning 1.1.1.9 Introduce social safeguards at a high level at each meeting, including the GRM, FPIC, and Access Restrictions Mitigations as safeguards to be in place for work with local communities	
Activity 1.1.2 Develop climate-resilient landscape land-use plan	Ensure that climate change adaptation requirements are fully integrated into planning processes at local, district and national scales.	1.1.2.1: Engage district officials, JADF, and community members in the CND (especially women) in reviewing and interpreting the NLUDMP, to ensure the plan accounts for current and future climate risks, while building local support for climate sensitive planning. 1.1.2.2: Conduct participatory land-use planning process in communities from village to district level to support integrated climate resilient land use planning 1.1.2.3: Develop Integrated Land-use Plan that supports Resilient Livelihoods and Ecosystems in the CND, ensuring alignment with National Land Use and Development Master Plan and ensuring the CND plan guides the district plan 1.1.2.4: Develop and rollout a series of trainings (virtual sessions and online modules) on gender sensitivity and mainstreaming women and youth into planning	A summary of the current NLUDMP and its recommendations relating to forestry, agriculture and climate adaptation. Reviewed/updated district level planning schemes to ensure they account for current and future climate risks (e.g. DFMPs, District Development Strategies) A community natural resources mapping manual to outline main steps in community land-use planning process A comprehensive review of national and district development plans/strategies, guidelines and projects to understand planned future land-use & climate adaptation projects in the CND and identify synergies and conflicts between land-use & climate adaptation plans from different sectors Integrated Land Use Plan developed

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		1.1.2.5: Assess specific climate impacts on historically marginalized and Category c,d,e populations through a participatory NR process to ensure the component projects address their needs for adaptation	
Activity 1.2.1: Create interagency taskforce institutionalizing integrated landscape planning & policy	Collaborative integrated landscape planning supported, which secures climate resilience undertaken with appropriate social safeguards. This activity aims to strengthen collaborative efforts, in particular between institutions (from local to national level) in charge of natural resources management to encourage synergies and avoid overlapping mandates and redundancy in different climate resilience interventions.	1.2.1.1: Review & strengthen operationalization of the current cross-sectoral task force 1.2.1.2: Hold quarterly sectoral planning meetings with both national and district administrations in CND 1.2.1.3: Facilitate discussions and provide technical support (e.g., policy briefs) in decision-making for cross-sectoral actions around climate adaptation and forest resilience in the CND 1.2.1.4: Continue to introduce social safeguards at each meeting including the GRM, FPIC, and Access Restriction Mitigations as safeguards to be in place for work with local communities. For those at the district or community level, provide options for feedback on the process and best ways of communication with local partners	A cross-sectoral planning task force operational as recognized by a ministerial order
Activity 1.2.2: Build capacity for spatial planning in national agencies re climate change	Capacity for integrated land use planning which supports climate change resilience and delivery of ecosystem services is improved. This activity aims to build the technical capacity of staff in the Ministry	1.2.2.1: Recruit spatial planning expert to support spatial planning unit	Spatial Planning Unit Operational Capacity needs assessment completed

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
	of Environment and University of Rwanda to generate and use the data on forest cover and ecosystem service values necessary for mainstreaming climate adaptation and forest resilience into land-use planning..	1.2.2.2: Assess and identify the institutional home and operationalize the spatial planning unit 1.2.2.3: Conduct a capacity needs assessment and identify the appropriate tools for spatial planning to fit the Rwandan context and conduct familiarization 1.2.2.4: Led by the spatial planning unit, develop capacity within the University of Rwanda and the MoE to use remote sensing, climate change modeling and spatial planning tools (e.g., Marxan, InVEST, RIOS, SWAT) to incorporate climate risks into land use planning processes 1.2.2.5: Deliver training workshops on utilization of earth system models for land use planning purposes, utilization of remote sensing resources (lightning, satellite radiances) for hazards detection and climate monitoring and on regional climate monitoring and applications	Capacity for forest mapping, climate change modeling and spatial analysis built within UoR and MoE
Activity 1.2.3: Develop an effective forest monitoring system to underpin forest management decisions	An effective forest, climate change and land use monitoring system is established to support climate resilient forest management decisions.	1.2.3.1: Recruit Forest ecologist or remote sensing expert to support the design and implementation of indigenous forest monitoring system 1.2.3.2: Review existing forest mapping data and monitoring software, and assess utility for generating quantitative	A report on the status of existing forest mapping data and monitoring software A design of the structure and operation of forest monitoring Two weather stations installed and operational Forest monitoring system operationalized

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		assessments of indigenous forest cover from local to national scales 1.2.3.3: In collaboration with REMA, MINAGRI, RAB, RDB, RISA, MoE and affiliated agencies and University of Rwanda, design structure and operation of forest monitoring system, including required inputs (data, computing, personnel), desired outputs, and operational structure (e.g. location, reporting structure, funding) 1.2.3.4: Purchase and install 2 new weather automatic stations in the CND 1.2.3.5: Establish & operationalize forest monitoring system 1.2.3.6: Generate updated maps & statistics for indigenous forest cover, forest type etc. using forest monitoring system 1.2.3.7: Develop capacity within hosting institution for continued operation of forest monitoring system	Updated maps and statistics for indigenous/natural forest cover and type generated Trainings delivered to develop the capacity of staff within hosting institution in forest monitoring
Activity 2.1.1: Facilitate revision of PA management plans to address climate risks	Park planning process expanded to ensure full incorporation of buffer zone and landscape linkage requirements, and ability to engage with the larger CND and district landscape management planning processes.	2.1.1.1 Review and update existing national park management plans to ensure climate and related landscape changes risks, impacts and required management responses are integrated 2.1.1.2 Provide technical and financial support to planning, research and monitoring as well as community based conservation units within RDB in	General management plans for NNP, GMNP and VNP updated

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		managing the parks as part of larger CND landscapes 2.1.1.3 Train PA staff to integrate gender and social inclusion into programming 2.1.1.4 Train PA staff on integrating needs of women, youth, historically marginalized and Ubudehe Categories c,d,e populations into climate adaptation risks and responses 2.1.1.5 Train PA staff on SEAH, GRM, FPIC, and Access Restriction Mitigation Processes to ensure NP climate change responses are undertaken in a way which supports social inclusion and equity	
Activity 2.1.2 Establish long-term plans for CND financial sustainability post-GCF	Identify financial instruments that could be employed for revenue generation post GCF as part of a broad and comprehensive RDB business planning effort for the sustainable management of the CND landscape	2.1.2.1 RFA to Recruit a Technical Advisor to guide RDB, MoE, REMA, FONERWA and MINECOFIN on sustainable conservation finance.	Technical Advisor recruited CND long term financial sustainability developed
Activity 2.1.3 New fire management curriculum developed and operationalized in PAs	Capacity increased for improved fire management in core PAs and adjacent landscapes. A collaborative effort in fire management planning and implementation in CND districts will be carried out and facilitated.	2.1.3.1 Develop a curriculum tailored to needs and capacities of different stakeholders 2.1.3.2 Build the capacity of RDB to manage fire in National Parks 2.1.3.3 Build the capacity of RFA, local authorities and communities to collaboratively manage fire in forests outside PAs	A fire training program that addresses local needs and skills gaps developed Train the trainer process completed for 25 RDB staff allowing them to instruct future classes without assistance Fire detection, early warning, and fire danger ranger system is established and operationalized

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		2.1.3.4 Implement a “fire wise” outreach and awareness program for communities and local government (district, sector, cell), as well as Rwanda Forest Authority, surrounding natural forests to reduce the incidence of human-caused fires	200 community facilitators (youth) trained to raise awareness on fires in communities surrounding natural forests and forests plantations 10,000 community members reached through social marketing events
Activity 2.2.1 Secure key remaining natural areas outside PAs	Remaining priority natural forests and other ecosystems required for climate change resilience secured through appropriate land use planning processes	2.2.1.1 Raise awareness on remaining protected natural forests in CNDL to secure their protection 2.2.1.2 In collaboration with RFA, REMA, districts identify and implement actions that support conservation and management of remnant protected natural forests in CNDL	Action plan for the conservation of remnant natural forests developed and implemented
Activity 2.2.2 Restore natural forest cover in & outside PAs including riparian linkages	Support natural forest restoration processes in PAs through improved management of alien and invasive species; and actively restore nature forest in priority areas next to core PAs and protected forests to expand the natural forest estate	2.2.2.1 Review mapping of degraded natural forest areas in core NPs, stepping stones and unprotected riparian lands using updated imagery and ground truthing 2.2.2.2 Identify parcels for restoration in core PAs, stepping stones and riparian lands using desktop and field-based assessment 2.2.2.3 Establish indigenous tree seed nurseries in the CND to serve core PAs, stepping stones, riparian land restoration and promotion of indigenous trees on farms and in protective forests 2.2.2.4 Recruit, train in forest restoration methods and equip community workers to	Indigenous tree seed nurseries established and operational 4,500 ha of indigenous forest in NNP restored 500 ha restored in Gishwati Mukura NP 1,500 ha of riparian lands restored

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		be involved in core PAs restoration and riparian linkages 2.2.2.5 Restore 4500 ha of indigenous forest in NNP 2.2.2.6 Restore 500 ha Gishwati Mukura National Park 2.2.2.7 Restore 1,500 ha of riparian lands 2.2.2.8 Using permanent plots sampling, collect field monitoring data on tree species recruitment and growth for estimation of biomass, species richness etc. in restored parcels in Nyungwe and Gishwati-Mukura National Parks in restored parcels in Nyungwe and Gishwati-Mukura National Parks as well as on riparian lands to assess success of initiatives, then replicate or adapt as needed 2.2.2.9 Perform Environmental and Social Screening on all positions being created by the project 2.2.2.10 Establish a grievance process for laborers 2.2.2.11 Update emergency and preparedness plan including risk mitigation guidance to local conditions at restoration sites	
		2.2.2.12 Train all workers on their rights and how to access the GRM	

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		2.2.2.13 Provide code of conduct and emergency preparedness and safety training for all laborers 2.2.2.14 Hire and train labor and safety leads at each site to provide guidance to staff, be available for grievance issues, and monitor health and safety conditions for workers 2.2.2.15 Develop and implement strategy (including social marketing) to ensure that women and youth participate and benefit from forest restoration projects	
Activity 2.2.3 Promote silvopastoralism with indigenous trees in Gishwati Pasture areas	Existing pasture lands will be characterized and livestock farmers clustered according to the size of their grazing lands. Tree species that exist on their grazing land will be identified and grouped according to the level of their resilience to climate change. A participatory approach will be used to assess livestock owners' preferences for indigenous species on their farms. Sites will be identified for indigenous tree nursery establishment and nurseries will be established in project sites and managed by the livestock communities.	2.2.3.1 Assess the status of pasture lands in Gishwati area and identify key areas for restoration and the potential to introduce indigenous species 2.2.3.2 In collaboration with landowners, identify suitable indigenous species for fodder trees, shrubs, grasses, and herbaceous legumes that have potential to improve pasturelands and increase their climate adaptive capacity 2.2.3.3 Produce and disseminate fodder trees, shrubs, grasses, and herbaceous legumes to project beneficiaries 2.2.3.4 Train beneficiaries on improved livestock and pasture management	1,000 ha of pasture land in Gishwati grazing area restored with indigenous tree species

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 3.1.1 Restore high slope areas (>55%) as protective forests	Promotion of protective forest and agroforestry practices to improve landscape connectivity and ability to deliver ecosystem services	3.1.1.1 Introduce and raise awareness of indigenous species to target stakeholders in CND 3.1.1.2 In consultation with RFA, National Land Authority, districts and communities, determine fragile areas (steep slopes > 55%) to be allocated for protective forests and their ownership 3.1.1.3 Assess the current status of the indigenous tree species in selected areas for protective forests and select indigenous tree species appropriate to CNDL 3.1.1.4 Develop restoration plan for protective forests 3.1.1.5 Build capacity of local stakeholders (men and women) on PFMU approach and methods 3.1.1.6 Design and approve SFMPs of private FMUs 3.1.1.7 Ensure consent of smallholders prior to planting trees (SS) 3.1.1.8 In collaboration with smallholders reforest/restore 2,500 ha of public or private land with slopes >55% and ensure sustainable management under private FMUs according to approved SFMPs 3.1.1.9 Support monitoring and evaluation of restored private FMUs	2,500 ha of protective forests on high slope areas restored

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		3.1.1.10 Assess impacts of exotics on neighboring lands and mitigate their negative impacts (SS) 3.1.1.11 Develop and implement strategy (including social marketing) to ensure that women and youth participate and benefit from forest restoration projects (G) 3.1.1.12 Assess the benefits and costs of each proposed tree and plant species and how those benefits affect different population segments (SS)	

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 3.1.2 Develop on-farm agroforestry for high caloric and indigenous tree species	Implement agroforestry/sustainable land management (SLM) over 3,346 ha of smallholder farmland	3.1.2.1 Identify sub-areas of intervention for agroforestry dissemination in the CND 3.1.2.2 Introduce and raise awareness of agroforestry in target communities 3.1.2.3 Develop the capacity of extension agents at district/sector level and NGOs to support adoption of agroforestry technologies 3.1.2.4 Establish agroforestry/fruit tree nurseries to facilitate access to quality planting material 3.1.2.5 Promote sustainable land management practices by stabilizing existing terraces, and plant high calorific agroforestry species to provide a sustainable source of fuelwood for energy efficient cookstoves to protect sloping land against severe soil erosion 3.1.2.6 Train RFA, extension agents, Project participants and community members on specific techniques for identification of and management of invasive pests and pathogens (SS)	SLM/agroforestry established on 3,346 ha of smallholder land

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 3.2.1 Develop forestry and agroforestry-related value chains for market access	Vulnerable groups of youth and women identified and their capacity for production (quality and quantity) enhanced in sync with particular niche markets demands	3.2.1.1 Identification and mapping of vulnerable groups of youth and women within the CND 3.2.1.2 Capacity building for farmers' cooperatives/unions on good agricultural practices (GAP) for market standards in the selected agroforestry-related tree fruits and horticulture 3.2.1.3 Development of Global GAP standards modules and market access linkage for the selected value chains 3.2.1.4 CND Honey value chain capacity production, market, and financial development 3.2.1.5. Tourism value chain development	Vulnerable groups of youth and women within the CND identified and grouped into cooperatives 5 Value chains identified and beneficiaries selected GAP training for 900 people from targeted cooperatives and unions completed Train of trainers completed for 30 cooperative members New tourism trails and base camps outside of Gishwati-Mukura NP developed and operational Community freelance tour guides trained and operational
Activity 3.2.2 Facilitate and scale up climate-resilient value chain products	This activity aims to build the capacity of smallholder farmers in producing, processing and marketing quality products that meet national and international standards to be competitive on the market	3.2.2.1 Value chain contextualization in focus areas in the CND 3.2.2.2 Facilitating market accessibility	25 youth and women cooperatives/enterprises with the capacity to deliver quality products and services
Activity 3.2.3 Facilitate access to markets for vulnerable farmers	This activity will focus on enhancing productivity (quantity, quality, and sustainability at the farm level) and exploring the market linkages model between farmers, suppliers, and buyers.	3.2.3.1 Development of farmer-inclusive business models	900 cooperative members are linked to the market from farming inputs suppliers and production aggregators through inclusive farming-as-a-business.

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 3.2.4 Scale up production, use and sales of fuel-efficient cook-stoves	Evaluate and promote financing models that sustainably foster energy efficient cook-stove access to historically marginalized communities and other vulnerable groups of the CND	3.2.4.1. Identification and production of Energy Efficient Cooking-Stoves 3.2.4.2. Energy efficient cook-stoves distributed to CND community members and categorization based on UBUDEHE Categories	8500 households have access to EES
Activity 3.3.1 Facilitate access to finance & private sector investments	Build common understanding between value chain actors and private financial institutions towards selected value chains	3.3.1.1 Tailoring credit assessment tool for selected value chains, while integrating climate impact data 3.3.1.2 Facilitate access to finance for selected value chains	Agriculture Credit assessment tool upgraded and adopted by PFIs At least 15 projects of the 5 selected value chains funded by PFIs
Activity 3.3.2 Set up & support savings & loan groups, enhance asset-building	Strengthen the capacity of local smallholders' farmers in financial literacy and access to finance	3.3.2.1 Establish/form saving groups for access to finance	Cooperative members are linked to PFIs and at least 70% have accessed to financial service and loan
Activity 3.3.3 Build capacity of financial institutions to serve targeted value chains and communities	Undertake a climate screening of financial service providers using the green index tool and support PFIs to access and adopt a digitalized forestry and agri-loan assessment tool	3.3.3.1 Capacity-building of PFIs	PFIs have developed an environmental policy and strategy for their organization including staff awareness training and integration of climate and livelihood impact measurement systems
Activity 3.3.4 Facilitate learning & knowledge sharing	Produce and share lessons learned during the implementation of this project	3.3.4.1 Programme-embedded learning and reflection	ESMP

E.7. Monitoring, reporting and evaluation arrangements

The Project's logical framework outlines the expected results, indicators, means of verification, baseline values and target values at mid-term and end-term. The Project Coordinator will be responsible for the overall monitoring of Project progress and reporting to the PSC. A national M&E Specialist hosted under the PMU, supported by EE-based Specialists, will be employed to develop practical guidelines for — and operationalize — a performance monitoring framework to track the Project's progress towards achieving its targets. The national M&E Specialist will be responsible for: i) establishing a performance monitoring framework to define biannual targets; ii) measuring the indicators at least 1–2 times per year to evaluate the progress of the Project; iii) reporting to the Project Coordinator and PSC; and iv) participating in the production of reports. RFA M&E Specialist will support the National M&E Specialist by: i) following guidelines developed by the national M&E specialist to conduct performance monitoring; ii) measuring the program indicators at least 1–2 times per year; iii) ensuring all activities comply with the Project's ESMF process and procedures; iv) supporting environmental screening and helping to follow the Environmental Management Plans; and v) participating in production of reports. The M&E Specialists will be supported by a Gender Specialist to oversee and monitor the application of gender disaggregated indicators. The M&E Specialist and Gender Specialist will perform regular visits (at least once/year), randomly selecting villages to monitor implementation. They will use these visits to cross-check the information provided by EE and beneficiaries in monitoring reports. During each visit, the team will collect qualitative data on implementation outcomes and difficulties through focus group discussions with beneficiaries and interviews with key stakeholders. The national M&E Specialist, supported by the EE M&E Specialist will then compile quarterly Project reports outlining progress in each site, allowing for feedback and support from the PMU and PSC in case of delays or challenges. MoE, as the AE, will perform supervision missions from year two onwards, selecting sites to visit each year and assessing the status of implementation. In addition, the MoE procurement office — which oversees the destination and use of funds disbursed to the EE — will monitor the Project's compliance with policies and procedures, as well as alignment between implementation and objectives. The annual performance reports will include a narrative report (with supporting data) on implementation progress based on the logical framework in the Project, including a report on ESMS as well as gender. The report will be aligned with the modalities set out in the GCF results management framework and its performance measurement frameworks.

As per the AMA, an independent consultant will be contracted to conduct an Interim Evaluation and Final Evaluation. Both will comply with GCF evaluation policy and international standards and make use of qualitative and quantitative data analyses, including interviews with relevant stakeholders, data provided by the Project, EE M&E Specialists and Gender Specialist, and participative workshops and focus groups with beneficiaries. The MTE will analyze whether the Project is on track, any problems, and challenges, and required corrective actions. The PSC and PMU will develop a management response to the recommendations along with an implementation plan, and the PMU will monitor implementation of the recommendations. The Final Evaluation will be an independent assessment of Project performance against its contributions to a paradigm shift and enabling environment.

• (f) RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures

Selected Risk Factor 1 - *Lack of coordination, institutional resistance, and regulatory barriers to mainstreaming climate change into land use planning*

Category	Probability	Impact
<u>Macroeconomic/Political</u>	<u>Low</u>	<u>High</u>

Description

Despite its progressive policies, Rwanda has historically had limited capacity for mainstreaming climate change into government/institutional practices and given the pervasive ways in which the impacts of climate change impact across numerous sectors, ministries and government agencies will need to work across any perceived or real institutional barriers to align and mainstream climate change in land use planning. For example, the Rwanda Development Board has historically not worked beyond Park boundaries, and this will be integral for institutionalizing the climate change adaptation processes across the Congo Nile Divide as a large interconnected landscape. Additionally, insufficient coordination, collaboration and information-sharing between relevant government departments, and research institutions may hinder or limit participation of relevant institutions and agencies, reducing the timely delivery and effectiveness of the Project, especially the aspects related to mainstreaming climate change in land use planning and other components of the Project as well. Furthermore, the perception of an uneven playing field, with larger or more well-connected institutions and companies able to evade the payment of environmental fees and fines under existing legislation, may undermine the uptake of the other options for financial sustainability of improved PA and forest ecosystem management.

Mitigation Measure(s)

The Project is part of a larger suite of efforts supported by the multi-agency Project Steering Committee (PSC) to create a spirit of cooperation among the various agencies and institutions charged with addressing climate-related issues. The PSC demonstrates Rwanda's commitment to fulfilling mainstreaming of climate adaptation into regular government processes and procedures. As such numerous government agencies have already had the opportunity to work together on a variety of climate-related issues, align institutional policies and reduce any potential regulatory barriers to ensure successful implementation of projects related to climate change. The PSC will directly oversee the Project Management Unit, and with this larger guidance structure, the EE will communicate to the PMU and the PSC on a regular basis to prompt collaboration and address any unforeseen institutional or regulatory barriers encountered during the life of the Project for immediate attention and remediation. PSC will address the risk of evading the payment of environmental fees by facilitating buy-in from stakeholders such as RDB and MINECOFIN into compliance with environmental laws through greater emphasis on the links to climate resilience and NDC objectives. These measures will reduce the impact to LOW.

Selected Risk Factor 2 - *Extreme weather/climatic events hinder the ability to deliver Project objectives*

Category	Probability	Impact
<u>Other</u>	<u>Low</u>	<u>Medium</u>

Description

The activities that will deliver overall Project success involve dependencies on weather conditions for successful implementation of the projects that rely on accessing steep slopes, planting trees, and establishing beekeeping operations. Extreme events such as heavy rainfall or prolonged drought could severely limit the success of these endeavors making sites inaccessible, creating unfavorable conditions for seedling survival, and decreasing the potential success of new beekeeping operations. This region, given its topography and the expected climatic variability, is likely to experience these types of disruptions. Severe disruptions may impede implementation, including potential implementation delays and fall under GCF's policy on restructuring and cancellation. That said, it is unlikely these extreme conditions will present themselves in the next 5 years for the duration of the project.

Mitigation Measure(s)

As weather is unpredictable, any mitigation measures will not change the probability nor address the impact in the short term. As the entire project is envisioned to improve the resiliency of this landscape to climate variability and promote adaptation, the specific activities will be able to mitigate, and in some cases prevent, longer-term impacts. As our implementation arrangements involve bolstering the climate adaptation capacity of the government and the communities in the CND, the entire project itself will serve as mitigation to this unlikely outcome.

Selected Risk Factor 3 - Project staffing and capacity

Category	Probability	Impact
<u>Operational</u>	<u>Low</u>	<u>Medium</u>

Description

Implementation of the Project hinges on staffing of the Project Management Unit, hiring additional staff within the EE and recruiting thousands of additional laborers in the form of community engagement specialists, contractors for building terraces, individuals to remove ferns, contracts to construct and deliver stoves, consultants to contribute expertise on behavior change, invasive species/pest management, and gender sensitization. Many factors could intervene to slow or thwart hiring progress including the high turnover of staff in the implementing institutions as frequent changes in government bodies could result in a disruption and/or delays in implementation.

Mitigation Measure(s)

Rwanda has a successful track record of implementing large scale projects through government ministries. Competent staff will be available for project implementation, and their abilities will be enhanced through the commitment to training individuals involved in the program. This risk, if realized, may impede implementation including potential implementation delays and fall under GCF's policy on restructuring and cancellation. Therefore, the Project will invest heavily in strengthening institutional capacity to promote climate resilience and apply actions that directly increase the resilience of ecosystems and community livelihoods. The Project will also document decisions, best practices and lessons learned to support the institutional memory to sustain Project activities in case of high staff turnover. Labor and safety as well as community security plans to be established as part of the ESAP (Annex 6) will make the Project a desirable place to work. Mitigation measures reduce impact to LOW.

Selected Risk Factor 4 - Inadequate incorporation of gender, and environmental and social safeguard (ESS) considerations into the implementation of the proposed Project activities as well as its management structures and systems		
Category	Probability	Impact
<u>Technical and Operational</u>	<u>Low</u>	<u>Medium</u>
Description		
Should ESS be inadequately considered in the Project or in the management and governance structures, there could be social and environmental implications. To start there may be a disproportionate distribution of benefits to certain segments of society. Although the Project's primary beneficiaries include women and other most vulnerable people, there is a risk that the Project implementation will experience preferential treatment at sites that favor certain groups over others. For example, despite tremendous policy and legal efforts, a patriarchal culture and persistent disparities in education, employment, land ownership and other factors characterize gender relations in Rwanda and could cloud implementation efforts. Likewise, some vulnerable groups may be unable to benefit from the efforts, such as landless people unable to benefit from trees or having a safe place for a new cookstove. One serious risk afflicting major projects such as that proposed, if not properly addressed, includes sexual exploitation, sexual abuse, and sexual harassment (SEAH) which may be prevalent if preferential treatment occurs (e.g., interfamily SEAH in retaliation for benefits received by women rather than men) or there may be power dynamics with hired staff who work with communities and withhold Project benefits within a SEAH conflicts (e.g. requesting sex for an opportunity to participate in the project, harassment if potential beneficiary refuses). On the environmental side, several environmental risks detailed in Section F and Annex 6 could undermine the success of the Project including topsoil loss and possible collapse of the terraces, unsustainable practices for sourcing clay for the cookstoves, and inadvertent introduction of invasive species, pests or pathogens from the nurseries to the field sites.		
Mitigation Measure(s)		
The Project anticipates these challenges and has envisioned multiple participatory aspects to the Project that are guided by the gender action plan and the ESAP. To mitigate social risks, local community engagement will be facilitated by the EE in cooperation with local community organizations through the integrated landscape delivery teams (ILDT) and the Joint Action Development Forums (DJAF). Safeguards in place to support Labor and Community Health & Safety will require training/sensitization to SEAH and the repercussions for those who engage in such activities (i.e., termination of employment). In the hiring process, staff will be screened for prior infractions. Hired staff will be trained in survivor-centered and gender-responsive approaches to support any victims of SEAH in association with the Project activities. To minimize environmental risk, the PMU and the EE will adhere to the environmental action plan and oversee the hiring of contractors to perform the activities. Multiple consultants or specialists, if not part of subcontracts, will be recruited to train project participants on mitigating measures to reduce these risks. If the proposed ESAP is followed properly the impacts will be LOW.		
Selected Risk Factor 5 - Lack of adoption among target populations for livelihoods activities		
Category	Probability	Impact
<u>Technical and Operational</u>	<u>Medium</u>	<u>High</u>

Description		
The success of the Project depends upon the integration of activities such as beekeeping, improved cookstoves, planting of fruit trees and terrace construction on private lands by the project beneficiaries. If planned and executed incorrectly a myriad of issues could prevent beneficiaries from adopting the activities beyond the life of the Project. This includes but is not limited to overlapping or misalignment between the proposed Project activities and those being implemented by other government departments or non-government organizations in the Project area.		
Mitigation Measure(s)		
Numerous international development projects have experienced mixed results upon completion regarding the uptake of promoted activities into daily lives. This Project has a strong behavior change component based on current best practices in identifying the barriers to behavior change, removing those barriers, and changing norms thus ensuring, through adequate participation and training, that the activities will be adopted and maintained over time. The work plan involves ample collaboration with women and youth groups through the ILDTs to build long-term community ownership for the Project. Further, if there are activities being promoted by the government or other NGOs that either overlap with the proposed activities or contravene with the Project activities, adoption by intended beneficiaries may be disrupted. As such the NSC will serve as the overarching body to ensure such projects within similar geographic regions remain aligned. These mitigation measures reduce the impact to MEDIUM.		
Selected Risk Factor 6 - Unforeseen disruptions due to another COVID variant		
Category	Probability	Impact
<u>Other</u>	<u>Low</u>	<u>Medium</u>
Description		
As demonstrated during the recent years of the COVID pandemic, variants may cause unforeseen disruptions to travel and Project implementation.		
Mitigation Measure(s)		
The Project will likely start well after COVID has run its course; however, in the event of new variants the PMU and EE will be ready to implement necessary COVID-safety practices such as masking, social distancing and reducing travel to areas with low health care options. Mitigation will reduce impact to LOW.		
Selected Risk Factor 7 - Prohibited practices including money laundering and terrorist financing		
Category	Probability	Impact
<u>Prohibited Activities</u>	<u>Low</u>	<u>Medium</u>
Description		
The Project involves a large number of actors for successful implementation including ministries, government agencies, NGOs, sub-grantees, vendors, service providers, community members and beneficiaries. Specifically, there is an intention to distribute or disperse to beneficiaries both directly and indirectly commodities and other items of value such as the distribution of seedlings for trees that will produce fruits or contribute to enhancing biodiversity of farming systems, access to improved cook stoves as well as the		

possibility for distribution of bees to start honey operations. As with any project infusing large sums of money directly or indirectly with multiple actors, corruption could occur.

Mitigation Measure(s)

Since the 1990s Rwanda has set a social foundation for anti-corruption efforts by establishing policies and procedures to prevent corruption/reduce vulnerability to corruption and sanctioning misbehavior through prosecutions. Specifically, a reduction in administrative corruption is of note for a large project such as the one proposed. The systems in place will reduce the potential for prohibited activities. Additionally, commitment to campaigns for individuals to report corruption has created a societal norm for intolerance. As a result, Rwanda moved from a state of high corruption 20 years ago to a level on par with middle income countries. The International Monetary Fund published an assessment of Anti Money Laundering and Combating Financing of Terrorism in 2015 indicating that while the risk of AML/CFT is low, at that time more could be done to bolster the systems in place to address the issue. At present, the GoR has robust measures to fight money laundering and terrorist financing. This was done first by enacting the law N° 001/FIC/2022 of 16/02/2022 on Regulations relating to anti-money laundering, combating the financing of terrorism. The Rwanda government has also put in place a special agency for this purpose named Financial Intelligence Center (FIC) with a clear mission to promote financial system integrity by conducting effective financial intelligence to counter money laundering, the financing of terrorism and related financial crimes in governmental institutions, projects, and parastatal agencies under the supervision of the Ministry of Finance and economic planning. MoE and the GCF Project under its supervision will also comply with FIC regulations. The MoE also has an Internal Audit function as well as regular audit conducted by the Office of the Auditor General to carry out the efficiency and effectiveness of internal controls.

Regarding higher level security threats, at present, there are no UN Security Council Sanctions against Rwanda, though it is noted that there are sanctions in Democratic Republic of Congo, close to the Congo Nile Divide. The UN Security Council Consolidated Sanctions list indicates approximately 20 individuals with ties to Rwanda, a small percentage of whom may live in the country. This list has been circulated to the AEs and will be cross checked when hiring for the project and once project sites are selected to ensure these individuals will not be involved in any manner with the Project or its activities, either as a counterparty, implementer, or beneficiary.

To report and remedy any risks that may occur, the MoE will maintain a Project level grievance redress mechanism to which anyone can report on the occurrence of prohibited activities. All complaints will be investigated following the process identified in Annex 6. Any complaints believed to contribute to prohibited activities will be referred to the appropriate authorities in Rwanda.

Assuming all staff will be made aware of the prohibited activities, all subcontractors/counterparties will be screened for engagement of activities and made aware of their obligations to alert the PMU and/or submit grievances related to such activities occurring as part of the Project, the overall impact of the money laundering and financing of terrorism risk on the Project will be LOW.

Selected Risk Factor 8 - *Political Risk*

Category	Probability	Impact
Other	Low	Low
Description		

Political risk is in part driven by long-standing frictions with neighboring countries and the potential for domestic instability in these countries to spill over into Rwanda. These geopolitical challenges constrain Rwanda's longer-term development as a services hub. Although the government of Rwanda has been effective in its pursuit of social stability through targeted transfers and support and improving accountability at all levels from Villages and to national levels through village councils, JADF and Umushyikirano (National Dialogue Council - an annual forum that brings together leaders and citizens to reflect on the country's progress and share ideas to solve ongoing challenges, this political risk may impede implementation including potential implementation delays and fall under GCF's policy on restructuring and cancellation.

Mitigation Measure(s)

Much of the work of the project will be taking place within a Ministry and through work at the community level. Oftentimes, regardless of political instabilities, communities with devolved decision-making structures and local capacity will have a greater chance to withstand that instability and continue to operate. For this project, much of the work will be turned over to the local community leaders and the laborers hired locally who will be trained and will then own the capacity to do this work in the long-term, regardless of what national leadership is in place. The structures resulting from this workplace citizens participation in decision making which will constitute a buffer against potential domestic instability. Mitigation will reduce impact to LOW.

Selected Risk Factor 9 - Expansion of tea plantations

Category	Probability	Impact
Macroeconomic/Political	Low	High

Description

Projected climate changes predict significant elevation rises in CND agro ecological zones, including the potential for tea cultivation at much higher elevations than at present. This is unlikely to threaten Rwanda's existing montane forest protected areas, as foreign revenues from tourism in these reserves far surpass those from tea. However, tea remains a significant source of foreign revenue and a government target for expansion. Any increased tea production in the CND would therefore most likely affect lands outside of parks: smallholder, cooperative, and district holdings, many of which might also be targets for reforestation, agroforestry, and biodiversity stepping stone initiatives under this Project.

Mitigation Measure(s)

The integrated land use planning included in this Project design, will bring together tea industry, agriculture, forestry, biodiversity, and tourism interests to identify and reduce such potential conflicts over use of the CND's limited land resource base. Through this integrated land use planning we aim to establish a formal public-private sector forum at the landscape level to help improve communication across sectors, appreciation of environmental issues, and increase synergies between interventions to avoid conflicting targets and indicators between sectors and advocate for mainstreaming biodiversity and climate change in different economic sectors.

Selected Risk Factor 10 - Timely provision of Co-financing

Category	Probability	Impact
Other	Low	Low

Description
The Project co-financiers may delay providing their co-financing commitment which can jeopardize implementation and hence project intended objectives could not be met.
Mitigation Measure(s)
The Accredited Entity has already requested for commitments of co-financing from all the project co-financiers which are already Government of Rwanda institutions, and all the committed co-financing are already part of the Institutions Day to day mandates hence there is anticipated low impact on the project implementation

(e) GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment

As proposed, the environmental and social risk category for the Project is **Category B or ‘moderate risk’ according to GCF classification. The project was** partially screened for potential social and environmental risks, following GCF requirements (Annex 6). All preliminary potential negative impacts for activities and sub-activities were determined to be minimal, reversible and readily mitigated through generally accepted mitigation measures and standard international practices, thus in the low to moderate categories. While the impacts will occur within the Congo Nile Divide Project area boundary and implementation zone, the specific locations and communities affected by Project activities will be determined in Project implementation thus an Environmental and Social Management Framework has been developed for a more thorough screening of all activities by the AE/EE.

Summary of Risks	Mitigation Factors / Measure and Management Strategies

Environmental and social risks: For all Project components, disproportionate benefit distribution to some groups and not others including women, youth, historically marginalized people and Rwanda's poverty categories (Ubudehe) C, D, and E is a risk. Such power differentials could exacerbate exposure to sexual abuse, exploitation and harassment, a risk that may affect all Project components if unmitigated. The Project may cause social risks including loss of access to lands due to new land classifications or stricter implementation of land use and park laws, displacement of informal land occupants due to afforestation/reforestation, and loss of access to land for food production. The potential for invasive species spread in the reforestation/afforestation components, water usage during increased seedling production at nurseries, terrace stabilization for planting, loss of topsoil to erosion during planting, disturbance to wildlife during afforestation or reforestation, modification of water flow and topsoil that could increase exposure to disease, increased human-wildlife interactions with increased tree cover and fruit trees, air quality impacts from cookstove production at scale, and supply chain impacts for livelihoods interventions all have associated environmental and social risks to be thoroughly vetted as specific sites and beneficiaries are identified.	The PMU will hire an Environmental and Social Safeguards (ESS) Specialist to oversee the application of the ESMF for a more thorough assessment of impacts by specific activities during Project implementation. The ESS Specialist will also oversee stakeholder engagement, sensitize all relevant staff on social safeguards and community engagement including SEAH training, and ensure Project activities target vulnerable groups to build resilient livelihoods. Consent processes will be established with communities. The Project will have a Project-level grievance redress process available to address complaints that aligns and complements the grievance redress used by the MoE. The PMU will also hire an Environmental Management Specialist (EMS) who will work with the ESS on environmental assessment and implementing proper mitigation processes as required.
Summary of Risks	Mitigation Factors / Measure and Management Strategies
Labor and working conditions: The dangers involve the potential for sexual abuse, exploitation and harassment (SEAH) in the hiring process or while working, forced labor, child labor, and exposure to occupational health and safety risks. These risks include safety of laborers removing ferns and planting trees on steep slopes, exposure to potential disease vectors in standing water or soil and large scale fabrication of stoves, risks using the machinery in the production of stoves, exposure to bees in honey production as well as other similar risks.	The ESS will coordinate with AE/EE human resources staff to conduct a Labor and Working Conditions Risk Assessment that will inform the development of a Labor and Working Conditions Risk Management Plan. The ESS and HR staff will appoint field safety leads within each sector. These individuals will receive training on National Laws governing labor and working conditions, SEAH, and other risks relevant to the activities that involve the hiring of laborers during the course of the Project.

Resource efficiency and pollution prevention: The stabilization of terraces risks topsoil loss and terrace collapse due to deteriorating conditions and improper long-term maintenance and extreme weather events. Pollution/waste may be created across the supply chains for stove production and distribution. Pesticides and fertilizers may be used for seedling production. Water usage in nurseries could divert resources for others. The release of CO2 emissions in the sourcing and distribution of materials in reforestation/afforestation as well as the livelihood enhancement activities could also pose a risk.	The ESS, EMS, and a Procurement Specialist will conduct activity-specific assessment for resource efficiency and pollution control. During implementation they will oversee subcontracts for reforestation/afforestation and all livelihoods enhancement projects. For stabilization of terraces, the subcontractor will have an engineer oversee the technical components of stabilization to ensure best practices and reduction of risks. For livelihoods enhancement projects the ESS and Procurement Specialist will ensure environmentally sourced materials and establish measures to minimize production waste and maximize distribution efficiency of stoves. The EMS will work with nurseries on plans for mitigating pesticides, fertilizers and water use.
Community safety and security: Increased risk of forest fires due to increasing tree coverages, increased exposure to vector-borne diseases, human wildlife conflicts associated with reforestation/afforestation and the influx of international funding that shifts power dynamics and creates environments prone to increased incidents of SEAH. With expected changes in land classifications, interactions among law enforcement and informal land owners or communities who use the resources could become more prevalent and increase risks of harm if law enforcement lacks proper training for working with communities. Work on steep slippery slopes, working with bees, use of new cook stoves and work in remote locations (e.g., fern removal) will require safety considerations for participating community members. Extreme weather or global events (e.g. pandemic) may disrupt safety and security.	The ESS will screen each activity for community, safety and security risks. The EE will have at a minimum, an Emergency Response and Preparedness Plan to address specific local hazards and response at each site; all staff will be trained on the plan as well as on disease prevention and hygiene, SEAH prevention, and addressing human-wildlife conflicts. The PMU will need to address at a high level across the GoR the potential for escalating conflicts among landowners and law enforcement as land classifications change in the course of the Project.
Summary of Risks	Mitigation Factors / Measure and Management Strategies
Resource access/restrictions: Land classification processes could result in changes that affect resource access due to existing land use, occupancy, and land tenure in the CND especially among farmers, settlers, and traditional uses of the land and forest. At a smaller scale with less risk, as tree planting and transitions to native or value-added species (e.g. fruit trees) commence, access to tree resources may be diminished until trees grow larger and become usable.	The ESS will screen all Project activities for the potential to cause displacement, namely access restrictions to resources. The Project is committed to stakeholder engagement and will mitigate access restrictions risks through a process framework and mitigation hierarchy. The AE grievance redress process will manage access complaints.

Biodiversity conservation: The inadvertent introduction of non-native/invasive pest species/diseases during forestry/agroforestry related activities could negatively impact production and forest health. Wildlife may be disturbed in the course of fern removal and/or tree planting as well as other potential livelihood activities that will be implemented. Human-wildlife interactions may increase as a result of increased forest cover and fruit trees being available in the CND. This could cause landowners to retaliate against wildlife.	The ESS will screen all Project activities for biodiversity and conservation risks/impacts. Mitigation measures will be put in place to ensure tree planting, nursery management, and seedling distribution incorporate training by consultants on invasive species recognition and management and that options for managing human-wildlife conflicts are able should issues arise.
Historically marginalized peoples: Rwanda is a nation with a single/common culture, tribe, language and with a national constitution that recognizes all Rwandans are born and remain equal in rights and freedom (article 16 of Rwandan Constitution, 2015). That said, there are historically marginalized groups as well as categories c,d, e, groups in the areas of the project activities. The risks are that the impacts of project activities on these groups are not adequately considered and analyzed and further that the intended benefits for these groups do not materialize given a lack of attention to their particular needs and vulnerabilities. Should this occur, the overall intent of the project could be in jeopardy given the focus on making such communities more resilient in light of climate change uncertainties.	As the project focuses on providing benefits, in particular increased resilience to climate change for vulnerable communities, the ESS will pay particular attention to the screening for activity level impacts on historically marginalized and categories c,d,e. Thus far, 24 focus groups with 4 exclusively historically marginalized, 5 exclusively women and 3 exclusively youth participants informed the project design and provided insights on impacts and improved ways of benefiting individuals in these groups (see Annex 6, section 4). A detailed plan for engaging with women, youth, those below the poverty line and landless groups will be prepared by project staff in collaboration with district, sector, and local ngo leaders in the project implementation phase as specific projects and locations are determined. In addition, all interactions with these groups will include processes for obtaining consent as well as clarity regarding access to the grievance redress process should complaints arise during the project.

Risk details and screening assessment to be conducted in Project implementation are described in the Partial Environmental and Social Impact Assessment and Environmental and Social Management Framework (Annex 6). Positive project impacts include a focus on improving resilience of the most vulnerable communities, communities stewarding/ harnessing the carbon cycle and reducing greenhouse gasses by sequestering carbon (i.e., through conserving indigenous trees, increasing soil organic carbon, and practicing regenerative agriculture); and utilizing efficient cook stoves. The PMU/ESS will be responsible for conducting screenings, including screenings for project-level impacts on historically marginalized groups of each Project activity as specific locations and beneficiaries are determined and with implementing mitigation measures aligned with Rwanda National Laws and GCF policies and practices. These measures include a commitment to training in all relevant EE staff on stakeholder and community engagement; consent with local communities; access restrictions mitigations; SEAH and adhering to the labor and safety laws that meet international best practices. The full Stakeholder Engagement Plan can be found in Annex 7. The EE and the AE will develop a compliance monitoring, supervision and reporting schedule that meets requirements in the GCF Information Disclosure Policy and Environmental and Social Policy. The AE will establish a

Program-level grievance redress mechanism that includes a process of public notification, documentation, investigation and resolution. This GRM will complement the MoE GRM.

G.2. Gender assessment and action plan

Women, youth and marginalized groups in the CND are key stakeholders in the Project and its primary beneficiaries. The Project will be implemented across various sectors along the CND, where natural forests are particularly vulnerable due to high population, increasing dependence upon fuelwood, and illegal use of forests to improve income streams. The Gender Assessment uses a range of primary (24 focus groups and 18 key informant interviews) and secondary (published reports, policies and academic literature) sources to triangulate our understanding of gender in climate resilience and Project design. The Gender Analysis identified and analyzed gender risks and impacts of the program and climate change through four steps: scoping; legislative and policy review; stakeholder consultation and analysis; and baseline analysis. The Action Plan operationalizes the findings from the Assessment, the ESIA, and youth-based information collected during the feasibility studies and stakeholder consultation. Despite tremendous policy and legal efforts, a patriarchal culture and persistent disparities characterize gender relations in Rwanda. Disparities persist in post-primary education; in access to and control of property, including land and economic resources; in employment opportunities and entrepreneurship; in decision-making at household and community levels; in family responsibilities and unpaid care work; and in the experience of violence, harassment, conflict, and insecurity. Sexual and gender-based violence persists at high levels. While women have made impressive political progress, especially at the national level, their local political representation and engagement remains unequal. Economic opportunities remain markedly gender-differentiated. In Rwanda and the CND, structural barriers to economic and political equity and female empowerment have significantly impeded measures to adapt to climate change impacts. While these structural inequalities are lower compared to other countries because of leading gender-considerate policies across sectors, the point remains that women's ability to access, use and control natural resources, infrastructure and services is different and low compared to men. This means the degradation of natural resources and new infrastructure will affect genders differently and result in greater vulnerability for women. However, women are vital change agents and can be powerful leaders from community to global levels in mitigating and adapting to climate change. Three key themes in the Gender and Social Inclusion Action Plan include i) Change negative gender and social norms, ii) Provide access to skills building, leadership and programmatic inclusion, and iii) Facilitate women's financial autonomy. Youth account for over 50% of Rwanda's population. An analysis was done to determine how GCF funding could support activities to align with and strengthen the sustainability of the overall program by supporting youth through the following activity themes: i) Non land-based opportunities, ii) Access to skills building, and iii) Youth leadership and Programmatic Inclusion. Social norms are difficult to change, and deeply entrenched biases will require active engagement by men, women, and youth, in decision-making roles as well as grassroots stakeholders. By working through the themes outlined above, social and technical barriers to participating in decision-making, planning, and activities will be addressed. Specific targets will be developed after a brief inception phase. Further details are in Annex 8.

G.3. Financial management and procurement

Funds from GCF will be provided to the MOE's GCF account (as the AE), under the terms of the FAA. MoE will be responsible for financial management and accountable for the use of GCF resources. Financial transactions will be subject to annual audits undertaken by internationally certified auditors. In addition, as the AE, MoE will: i) ensure that the project is executed in accordance with GCF standards; ii) supervise,

oversee and manage the implementation of project interventions; iii) report on Project progress; and iv) ensure that Project activities are well coordinated and aligned with countries' national priorities. MoE will provide oversight to the project consisting of a) entering into a subsidiary agreement with the EE (RFA); b) managing and disbursing GCF funds to the EE, after providing no objection to work plans and budgets; c) reviewing financial expenditures and progress reports; d) overseeing Project implementation in accordance with the Project document and Annual Work Plans and Budgets, agreements with co-financiers and each executing entity rules and procedures; e) providing technical guidance to ensure that the appropriate technical quality is applied to all Project activities; f) providing financial reports to the GCF for Project funds received; g) ensuring that the project complies with the terms agreed in the Project's respective FAA as well as the AMA signed between MoE and the GCF; and h) undertaking regular annual supervision missions according to the MoE's guidelines and convene Project Steering Committee (PSC) meeting twice a year aimed to review progress and approve work plans. The PMU will manage, monitor and report on day-to-day implementation of the EE. The EE will execute the tasks within the overall project management structure: a) implementing day-to-day activities as per the project work plan and budget, including the Environmental and Social Management Plan; b) undertaking procurement activities as per the agreement signed by GCF and AE as well as the sub-agreement signed by the AE and EE; c) managing contracts of suppliers and services providers; d) hiring and managing project staff relevant to the EE-managed project areas; e) implementing activities as per the project work plan; and f) carrying out financial and technical monitoring of activities, including the achievement of outputs and outcomes the EE are in charge of. The EE will enter into a Memorandum of Understanding with different partner agencies that will provide technical know-how to the implementation of the project activities. Procured parties will be contracted in accordance with MoE rules and regulations of procuring goods and services. The financial management structure of the project is built on the Permanent Secretary of MoE, the Project Coordinator/ Chief of Party and the Finance Management Specialist at the PMU level who will oversee and monitor the project finances with project directors and finance managers from the EE (RFA).

The project will be executed by Rwanda Forestry Authority (RFA), a Rwanda government entity affiliated to the Ministry of Environment with a semi-autonomous legal entity. RFA has transparent and accountable financial management systems in place, which are essential elements for the effective, accountable management of the project. RFA will follow the Government of Rwanda procedures for planning, budgeting, and accounting, which are outlined in the law n° 12/2013/ol of 12/09/2013 on state finances and property, and ministerial order n°001/16/10/tc of 26/01/2016 relating to financials; and supporting operational manuals issued by the Ministry of Finance.

The MoE shall ascertain that the EE has in place efficient internal control systems, review financial reports and proceed funds disbursements and will also periodically inspect the project for compliance as well as coordinate audits exercises for the EE. RFA will enter into agreements with MoE that outline general modalities of the funds management in line with GCF grant requirements and RFA's policies. GCF funds will be transferred to an RFA designated bank account. Funds will be disbursed to the EE based on agreed annual work plans and budgets, and its reporting will be on a quarterly basis. The project financial processes will rely on the EE's charts of accounts, internal control systems, disbursement plans, budgeting, expenditure management and accounting, reporting, and assets management as assessed by MoE. Procurement regulations embed the principles of fairness, transparency, accountability and the promotion of competitive economy and efficiency. The Project will be audited annually and commissioned by the Accredited Entity (MoE) to produce certified financial statements and reports to ascertain that GCF funds are well managed. The project will maintain a regularly updated record of all assets and documents that will be retained for at least 5 years after the end of the project.

G.4. Disclosure of funding proposal

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

(f) ANNEXES		
H.1. Mandatory annexes		
☒	Annex 1	NDA no-objection letter(s)
☒	Annex 2	Feasibility study - and a market study, if applicable
☒	Annex 3	Economic and/or financial analyses in spreadsheet format
☒	Annex 4	Detailed budget plan
☒	Annex 5	Implementation timetable including key project/programme milestones
☒	Annex 6	E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): ☒ Environmental and Social Impact Assessment (ESIA) or ☒ Environmental and Social Management Plan (ESMP) or ☐ Environmental and Social Management System (ESMS) ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
☒	Annex 7	Summary of consultations and stakeholder engagement plan
☒	Annex 8	Gender assessment and project/programme-level action plan
☒	Annex 9	Legal due diligence (regulation, taxation and insurance)
☒	Annex 10	Procurement plan
☒	Annex 11	Monitoring and evaluation plan
☒	Annex 12	AE fee request
☒	Annex 13	Co-financing commitment letter, if applicable
☒	Annex 14	Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule
H.2. Other annexes as applicable		
☒	Annex 15	Evidence of internal approval
☐	Annex 16	Map(s) indicating the location of proposed interventions
☐	Annex 17	Multi-country project/programme information
☐	Annex 18	Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
☒	Annex 19	Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
☐	Annex 20	First level AML/CFT (KYC) assessment
☒	Annex 21	Operations manual (Operations and maintenance)
☒	Annex 22	Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)



Republic of Rwanda

**RWANDA ENVIRONMENT
MANAGEMENT AUTHORITY
(REMA)**



Kigali, on **22 FEB 2022**
N° **0487**/NDA/2022

Deputy Executive Director
Green Climate Fund ("GCF")
Songdo Business District
175 Art Center-Daero
Yeonnsu-gu Incheon 22004
Republic of Korea

Re: Funding proposal for the GCF by the Ministry of Environment regarding "Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration Project"

Dear Sir,

We refer to the project titled "Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide Through Forest and Landscape Restoration Project" in Rwanda as included in the funding proposal submitted by the Ministry of Environment to us on 18 February 2022.

The undersigned is the duly authorized representative of Rwanda Environment Management Authority, the National Designated Authority of Rwanda.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Rwanda has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Rwanda;
- (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards, **FAUSTIN YAZIKWIYE** Faustin

For

Juliet KABERA
Director General/National GCF Focal Point



Cc:

-Permanent Secretary/ Ministry of Environment

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	Ministry of Environment (MoE) of the Republic of Rwanda
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Three Provinces of Rwanda (Western, Southern, and Northern) in 10 Districts (Karongi, Musanze, Ngororero, Nyabihu, Nyamagabe, Nyamasheke, Nyaruguru, Rubavu, Rusizi, and Rutsiro).
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Friday, September 22, 2023
Language(s) of disclosure	English and Kinyarwanda
Explanation on language	Kinyarwanda is the national language spoken by all Rwandans.
Link to disclosure	English: https://www.environment.gov.rw/index.php?eID=dumpFile&t=f&f=76268&token=0d001d8f3dc586127656b6b8280da983d72dda02 Kinyarwanda*: https://www.environment.gov.rw/index.php?eID=dumpFile&t=f&f=76248&token=a3f7cf9afd24f1eacdd08dc21dd9cd9299ad1e25
Other link(s)	N/A
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the "Environmental and Social Management Framework (ESMF)"
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Friday, September 22, 2023
Language(s) of disclosure	English and Kinyarwanda
Explanation on language	Kinyarwanda is the national language spoken by all Rwandans.
Link to disclosure	English: https://www.environment.gov.rw/index.php?eID=dumpFile&t=f&f=76268&token=0d001d8f3dc586127656b6b8280da983d72dda02 Kinyarwanda*: https://www.environment.gov.rw/index.php?eID=dumpFile&t=f&f=76248&token=a3f7cf9afd24f1eacdd08dc21dd9cd9299ad1e25

Other link(s)	N/A
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the “Environmental and Social Management Framework (ESMF)”
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity’s website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, Environmental and Social Management Framework (ESMF)	
Description of report/disclosure on accredited entity’s website	Indigenous Peoples Planning Framework
Language(s) of disclosure	English and Kinyarwanda
Explanation on language	Kinyarwanda is the national language spoken by all Rwandans.
Link to disclosure	English: https://www.environment.gov.rw/index.php?eID=dumpFile&t=f&f=76268&token=0d001d8f3dc586127656b6b8280da983d72dda02 Kinyarwanda*: https://www.environment.gov.rw/index.php?eID=dumpFile&t=f&f=76248&token=a3f7cf9afd24f1eacdd08dc21dd9cd9299ad1e25
Other link(s)	N/A
Remarks	A Historically Marginalized Peoples Plan, which serves as an Indigenous Peoples Planning Framework, is contained as an Annex to the ESMF.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, September 22, 2023
Place	To accommodate for those who may not have access to the Internet, a computer, or a reliable power source, the Ministry of Environment will be sharing the ESMF documents to the following districts: • Karongi district (2.0526° S, 29.4128° E), P.O Box: 23 Kibuye, village: Kamuvunyi, Cell: Gacaca, district: KARongi • Musanze district (1.5020° S, 29.6332° E), P.O Box: 03 Musanze, village: Byimana, Cell: Ruhengeri, District: Musanze • Ngororero district (1.8537° S, 29.6347° E), P.O Box: 103 Muhanga, village: Ngororero, Cell: Kazabe, district: Ngororero • Nyabihu district (1.5911° S, 29.5220° E), P.O Box: 125 Musanze village: Mukamira Cell: Rubaya District: Nyabihu

	• Nyamagabe district (2.4671° S, 29.5736° E), P.O BOX: 36 Gikongoro, village: Nyarusange, Cell: Nyamugari, district: Nyamagabe • Nyamasheke district (-2.342014193676258, 29.08901503907659), P.O Box: 72 Nyamasheke, village:Gikuyu, Cell: Ninzi, district:Nyamasheke • Nyaruguru district (2.6987° S, 29.5703° E), P.O Box: 215 Butare village: Nyarusovu Cell: Mubuga, district: Nyaruguru • Rubavu district (1.6736° S, 29.2886° E), P.O Box: 173 Gisenyi; village: Nyaburanga, Cell:Nengo, district: Rubavu • Rusizi district (2.4830° S, 28.9054° E), P.O Box: 253 Rusizi, village: Ntwali, Cell:Cyangugu, district: Rubavu • Rutsiro district (1.9617° S, 29.3857° E) P.O Box: 49 Kibuye, village: Nduba, Cell: Congo-Nil district: Rutsiro and request them to have physical copies of ESS reports in their local government offices, publish an Executive Summary of the ESMF on their notice board in both Kinyarwanda and English and publish it on their website to ensure it is accessed by all affected people in their location as well as to those who may have access to the Internet, a computer.
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, October 23, 2023

Note: This form was prepared by the accredited entity stated above.

** Subsequent to the disclosure of the ESS report to the Board and active observers on 22 September 2023, the following update has been made: the links to the Environmental & Social Management Framework (ESMF), in which the ESMF document -had been made available, have been updated for both English and Kinyarwanda. The footnote 86 has been removed.*

Secretariat's assessment of FP217

Proposal name:	Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration
Accredited entity:	Ministry of Environment of Rwanda
Country/(ies):	Rwanda
Project/programme size:	Small

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
This Congo Nile Divide project will safeguard Rwanda's biodiversity, enhance climate resilience, and support vulnerable communities.	The project should ensure linkage to other GCF funded projects in the country. In addition, we encourage MOE Rwanda to establish financial mechanisms to upscale across the wider landscape.
The project is located in the Albertine Rift, which is one of the most biologically rich regions in Africa, containing more than half of Africa's birds and 40% of its mammals.	The project should ensure monitoring and evaluation of the project interventions is undertaken.
With more frequent extreme rainfall events in recent years, this agricultural landscape is recognized as the region of Rwanda most vulnerable to floods and landslides.	
Project interventions will include improved management on 250,000 hectares (ha) (including promotion of natural regeneration/planting on 10,000 ha) and improved agroforestry practices on 2,000 ha of on-farm plantations.	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum XXI titled "List of proposed conditions and recommendations", respectively.

II. Summary of the Secretariat's assessment

2.1 Project background

3. The proposed project's goal is to improve the resilience of vulnerable communities to climate variability and shocks by increasing the extent and integrity of natural forests and by strengthening the capacity for climate-responsive forest and landscape management. The project will:

- (a) Strengthen institutional and regulatory frameworks for low-emission climate-resilient development pathways by mainstreaming climate change into integrated land-use planning.
 - (b) Increase the climate resilience of vulnerable rural communities through forest and landscape restoration and reducing demand for fuelwood.
4. The proposed interventions will target districts along the Congo Nile Divide (CND) region. Rwanda's CND, an area of 444,600 hectares (ha), is a high-elevation landscape of remnant natural forest reserves embedded in a mosaic of intensively managed smallholder farms and commercial plantations. The CND separates the drainage basins of the Congo and Nile Rivers. It is part of the Albertine Rift, which is one of the most biologically rich regions in Africa and "one of the most important regions for conservation" on the continent, containing more than half of Africa's birds and 40 per cent of its mammals.
5. Though covering barely 5 per cent of the country, this archipelago of protected forests is of critical importance for:
- (a) Its globally recognized high biodiversity values;
 - (b) Its role in Rwanda's acclaimed ecotourism offerings, serving as the primary source of Rwanda's foreign revenue and as a major economic development driver;
 - (c) The ecosystem services provided for surrounding subsistence farmlands as well as commercial tea and coffee plantations offering employment opportunities for women while providing the second and third highest sources of foreign revenue;
 - (d) The regulation of water capture and release to growing urban centres on the CND periphery, the nation's extensive river network, and downstream hydroelectric operations; and
 - (e) The sequestration and storage of nearly 60 per cent of Rwanda's national carbon stocks.
6. These values and services are already compromised by a combination of anthropogenic and climatic factors; they will be much more severely threatened by the potential negative feedback loops associated with projected climate change impacts. The project will promote an ecosystem-based approach to forest protection and restoration and will increase the extent and resilience of forest ecosystems and species, including by functionally linking currently isolated forest fragments. Through these approaches, the project will protect, restore and rehabilitate forest ecosystems, thus enhancing biodiversity and reducing key threats while increasing the provision of critical ecosystem services to strengthen the adaptation of vulnerable communities.
7. Project interventions will directly benefit 1,254,242 people (654,404 women). The project will also indirectly benefit 9,260,745 people and create approximately 24,212 job opportunities in forest-dependent communities. The total mitigation impact is estimated to be 1,204,448 tonnes of carbon dioxide equivalent (t CO₂ eq) by 2043.
8. The total project financing is USD 50,000,000 with a request for GCF grants of USD 39,056,422. The Rwanda Forestry Authority (RFA), Rwanda Water Board (RWB), Rwanda Development Board (RDB) and the Ministry of Environment of Rwanda (MOE Rwanda) will provide co-financing of USD 10,831,308 in grants. MOE Rwanda will also provide an in-kind contribution of USD 112,271. The project is submitted under environmental and social safeguards (ESS) category B.

2.2 Component-by-component analysis

Component 1: Mainstreaming climate adaptation into integrated land-use planning (total cost: USD 2.9 million; GCF cost: USD 2.9 million)

9. This component aims to integrate climate change adaptation considerations into land-use planning at the national and subnational levels. The goal is to achieve climate resilience through collaboration among government agencies, civil society, the private sector and diverse partners. The component has two main outputs: The first is to develop a climate-resilient land-use plan for the CND by integrating climate adaptation information into existing land-use planning efforts. The second aims to strengthen institutional capacities in Rwanda to integrate biodiversity and climate risks into land-use planning.

Component 2: Forest and landscape management and restoration (total cost: USD 18.9 million; GCF cost: USD 16.3 million)

10. In this component, targeted interventions will integrate climate change awareness and adaptation into national forest park planning, policies and management, as well as restore degraded areas within those parks. These efforts will help sustain the forests' rich biodiversity, the lucrative tourism revenues that flow into local and national economies, and the ecosystem services needed for the climate resilience of vulnerable communities. This suite of activities will also identify priority areas outside of parks – in remnant natural stands, around wetlands and along streams – for protection and restoration.

Component 3: Enhancing climate adaptation through resilient livelihoods (total cost: USD 22.6 million; GCF cost: USD 15.3 million)

11. The activities under this component aim to combine environmental conservation with socioeconomic development, creating a multifaceted approach that resonates with the global commitment to sustainable growth. They include initiatives such as enhancing and restoring forest plantations, providing sustainable fuel sources, delivering carbon sequestration benefits totalling 522,606 t CO₂ eq over 20 years, and developing comprehensive land-use planning. The other activities focus on rural livelihoods, financial services and sustainable farming methods.

12. The component will support reforestation on steep slopes on both public and smallholder private lands. It will develop farm-based agroforestry for climate-resilient species that produce fuelwood and/or market products, while conserving and enriching soils. The component will complement land-based production with a scaling-up of alternative value-chain products for smallholders that are nature-positive, including, but not limited to, tourism-based income, beekeeping, and improved cooking stoves. Central to these activities will be the facilitation of access to market chains and financial mechanisms for establishing and maintaining businesses. Given that these activities are designed at the core to be profitable and promote financial sustainability, it is intended that they be replicable and sustained once GCF funding expires.

Project management (total cost: USD 3.1 million; GCF cost: USD 2.7 million)

13. The Project Management Unit (PMU) of the CND project will be established under MOE Rwanda/Single Project Implementation Unit (SPIU) to manage day-to-day operations of complex programme interventions. Headed by the Project Coordinator, who will also serve as the secretary and rapporteur of the Project Steering Committee, the PMU will be responsible for tasks such as managing and monitoring identified risks and reporting new risks to the accredited entity (AE) and the Project Steering Committee. The PMU will adhere to MOE Rwanda/Government of Rwanda recruitment practices and provide quarterly progress reports. The Project Coordinator will report to the MOE Rwanda/SPIU Coordinator as the immediate supervisor and to the Permanent Secretary as the secondary supervisor.

14. The PMU's structure will comprise various specialists, including for monitoring and evaluation, procurement, gender, financial management, communication and outreach, along with an office manager and driver. Its staff will be selected and recruited following MOE Rwanda/Government of Rwanda procedures, aiming for a lean management structure. Operations will conclude once the final project evaluation report and other required documents have been submitted to MOE Rwanda, signifying a comprehensive and organized approach to the management of the entire programme.
15. The GCF portion of this project management cost is less than 5 per cent of the total requested GCF funding and is compliant with the GCF Policy on Fees.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: N/A

16. The project has 1,254,24 direct beneficiaries, representing 30.9 per cent of the CND population, and 9,260,745 indirect beneficiaries, representing almost 70 per cent of Rwanda's population. Additional adaptation benefits will include the training of 45,992 community members in forest management and restoration for climate resilience. Altogether, the project will contribute to the restoration of 5,000 ha of intact forest, 2,500 ha of protective forests, 3,346 ha of smallholder farmland, 1,000 ha of pastures and 1,500 ha of riparian linkages. Overall, 275,617 ha of landscape could potentially be positively impacted by the project. Although the project is focused on priority landscapes covering 275,617 ha, it is likely that benefits due to increased adaptive capacity could feasibly extend across the full 700,000 ha CND. The project will generate mitigation benefits by sequestering approximately 250,841 t CO₂ eq during the five-year project period, which is about 0.9 per cent of the 28.2 million t CO₂ eq reduction target in Rwanda's Vision 2050. The total mitigation impact from reduced deforestation, the introduction of forest restoration and adoption of improved cookstoves is estimated to be 1,204,448 t CO₂ eq by 2043.
17. The project contributes to the implementation of Rwanda's updated nationally determined contribution (NDC) (2020) commitments. It is also aligned with the country's Green Growth and Climate Resilience Strategy and other national policies. The proposal objectives are also well aligned with the country's National Forest Policy and related plans for reversing trends of high deforestation.
18. With support from the NDC Partnership, Rwanda now has a single NDC implementation framework that captures both national and subnational climate actions and investments plans and includes a strategic programme for climate resilience, scaling up renewable energy programme, green climate fund strategic framework, public investment priorities, sector strategic plans and district development strategies. Therefore, the proposed project is well supported by this existing NDC implementation framework that provides an enabling environment that will ensure climate actions undertaken are integrated into national planning.
19. The proposed project further strengthens the country's national and subnational adaptation priorities, approaches and plans through promoting ecosystem-based adaptation while utilizing climate science, effective landscape planning, policies, cross-sectoral coordination and management in addressing climate risks and adaptation benefits.
20. The project in general seeks outcomes at a range of levels (from national through to catchment level) across the public sector, while scaling up lessons and initiatives from aligned programming. It will build the capacity of MOE Rwanda (AE) and other related agencies and stakeholders to coordinate and collaborate more effectively to help accelerate subnational and local climate action. It is envisaged that these funds will also crowd in additional private sector

capital (e.g. cookstoves and forest plantations) while enabling the government to implement the adaptation interventions. However, without a clear private sector framework or plan, this might be challenging to implement. Nonetheless, Rwanda has a GCF country programme and strategic framework and has managed GCF funded projects and programmes.

21. Furthermore, the project clearly articulates the alignment, additionality and/or complementarity with several ongoing or planned programmes funded by significant international and multilateral agencies (e.g. World Bank, Global Environment Facility and United Nations Development Programme, etc.), which reflects on the experience and track record of MOE Rwanda to deliver on this project interventions.

3.2 Paradigm shift potential

Scale: N/A

22. By utilizing integrated landscape and ecosystem-based adaptation approaches in addressing climate change impacts in Rwanda, the project offers myriad opportunities for upscaling and replication to similar ecological zones across the country. This approach to land-use planning and management is entirely new for Rwanda and will restore and sustain varied ecosystem services for both adjacent and nationwide populations. The project coordination and collaboration efforts across both geographic and sectoral boundaries, while mainstreaming climate risks in management planning, represents a fundamental paradigm shift for land-use planning and management in Rwanda.

23. The project has the potential to contribute to the regulatory frameworks and policies for driving climate investments in the forest and renewable energy sectors. The project will finance the development of the integrated land-use plan, which will be adopted by all ten CND districts as a framework for seeking synergies between rural livelihoods and forest and biodiversity protections, reconciling land-use conflicts that arise, and guiding development decisions and climate change adaptation. It will also update the National Land Use Master Plan to ensure it incorporates climate resiliency and is accurate and relevant to local contexts. The adoption of an integrated land-use planning approach helps to better reconcile cross-sectoral trade-offs and improve climate adaptation planning in the CND; this approach will prove a successful model for application in other regions of Rwanda and other forest/farming landscapes beyond. In addition, revising two national park management plans to include climate concerns will serve as a model for the remaining parks of Rwanda, as well as for the East Africa region.

24. Overall, the project's efforts in shifting policy, practice and collaborative learning processes symbolize a fundamental change towards climate-resilient development pathways in Rwanda, with implications for broader regional application.

3.3 Sustainable development potential

Scale: N/A

25. The project is closely aligned with several Sustainable Development Goals. Key environmental benefits include restoration of degraded lands to reduce erosion, sedimentation, and landslide risks; the planting of native trees to enhance biological connectivity; and mitigation of greenhouse gas emissions through both increased carbon sequestration and fuel-efficient cookstove technologies.

26. Social benefits extend to more than a million people, focusing on increased awareness of climate change, empowerment of the rural poor (with an emphasis on women's participation), and creation of social capital through collaborative adaptation strategies. The project aims to significantly reduce poverty and inequality. It will provide economic co-benefits such as increased agricultural productivity, employment opportunities for around 24,212 households, modern beekeeping practices yielding significant income, and a net present value (NPV) of USD 369 million over 20 years from forest restoration and other activities. Private capital

investment is also targeted, thus fostering a “triple bottom line” that balances social, environmental and economic benefits. Emphasis on gender-sensitive impacts ensures that women are actively engaged in planning and decision-making, leading to increased income, improved health and nutrition, and strengthened leadership and entrepreneurship skills.

27. Overall, the project offers a comprehensive and strategic approach to sustainable development, thereby promoting climate resilience, economic growth, social cohesion and gender equality within the CND region.

3.4 Needs of the recipient

Scale: N/A

28. The project proposal interventions were designed from key findings and recommendations of a climate change vulnerability assessment of the selected project geographical area (see annex 2). Climate exposure, sensitivity, adaptive capacity, impacts and barriers as well the greenhouse gas emissions profile were established and either quantified or supported with reference to scientific publications. Further, climate change impacts on the target sectors as well as size and economic and social development of the target population, including critical barriers to reduce their vulnerability, are elaborated herein. The proposed interventions are targeted to match the identified barriers. The vulnerability of the target sectors is quantified to the extent possible, however, gender disaggregation is covered in qualitative terms only.

29. The CND landscape districts support nearly one third of Rwanda’s population on barely one fifth of its land base. The total population is 1,598,536, of which women account for 52.2 per cent. Through the project interventions, targeted vulnerable groups will attain more climate-resilient livelihoods from reduced exposure to landslides, floods and soil erosion and more knowledge on climate risks, the value of forests and forest ecosystems, climate adaptation options, access to quality indigenous and agroforestry materials, and improved ecosystem services.

3.5 Country ownership

Scale: N/A

30. MOE Rwanda, which is the AE for this proposal, clearly played a pivotal role in defining and drafting the proposal stakeholder engagement report and plan (see annex 7) as it is aligned with Rwanda’s GCF country programme stakeholder engagement strategic framework. In its country programme, Rwanda articulated how it will conduct stakeholder engagement at the readiness stage and in delivering GCF concepts, projects and programmes. This process has provided Rwanda with an overarching framework that supports its stakeholder engagement processes, which is clearly seen in this proposal. This framework can support the strengthening of local climate action from national to local levels with clear feedback loops and avenues for incorporating learning.

31. The proposal indicates that a detailed stakeholder and gender analysis were conducted, augmented by extensive consultations of various actors from national, subnational and local levels. A range of stakeholders from the public and civil society were consulted, in addition to government agencies at all levels that will be engaged throughout the project cycle, cutting across all key sectors and scale. The general concern relates to local ownership and the transfer of knowledge that can sustain interventions after the project closure. It would be ideal if Indigenous knowledge and technology transfer is well factored into implementation.

3.6 Efficiency and effectiveness

Scale: N/A

32. Utilizing grant financing reflects the public good character of the investments, the project aims to create a model that overcomes barriers to private sector investment in Rwanda's forestry sector. With an economic internal rate of return of 43.32 per cent and a financial internal rate of return of 42 per cent, the project is estimated to generate an NPV of more than USD 120.48 million over 20 years, which translates to USD 64 per direct beneficiary. The mitigation cost is estimated to be USD 43.84 per t CO₂ eq. Implementing best practices for tree planting and utilizing cutting-edge monitoring technologies, the project is aligned with similar initiatives and sets the stage for potential replication in other regions. Its community-focused approach, including measures to develop economically viable units, contributes to a threefold increase in net economic benefits compared to a scenario without the project.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

33. Project background: The project aims to increase resilience of vulnerable rural communities in the Rwandan side of the CND by building capacity for integrated planning and management through forest and landscape restoration and reducing demand for fuelwood. The project will involve the formulation and implementation of plans (i.e. integrated district land-use plans and enhancements of general management plans of national parks), coordination with various agencies, capacity-building, and physical interventions on the ground. This includes afforestation/reforestation of protected forest and private landholdings, installation of soil conservation measures, ecological interventions such as fern removal in natural forests and restoration of degraded forests in Gishwati-Mukura National Park, production and distribution of fuel-efficient cookstoves, beekeeping activities, and support for the value chains of these enterprises. Among the environmental and social (E&S) co-benefits are the enhancement of the unique, endemic biodiversity of the area through decreased sedimentation, improved water quality and recharge, reduced erosion, decrease in loss of soil organic matter as a result of forest restoration and establishment of connectivity between the forests across the CND landscape. The project is also expected to contribute to the empowerment of the women and youth in the target areas.

34. E&S risk category and safeguards instrument: The overall project E&S risk category is Category B (moderate). The Secretariat confirms the proposed E&S risk category since the project is, in itself, an environmental and social enhancement project. The potential adverse E&S risks and impacts will mainly come from the implementation of ground activities, are expected to be mostly minimal and highly localized and can be avoided/mitigated through careful design and the application of generally accepted mitigation measures and standards. The Secretariat conforms to the assigned risk category and it is within the accreditation level of the AE. A partial environmental and social impact assessment and environmental and social management framework (ESMF) have been prepared for the project.

35. Compliance with the GCF ESS standards: The paragraphs below provide a brief description of the programme's compliance with ESS standards.

36. **ESS 1 (Assessment and management of environmental and social risks and impacts):** The ESMF discussed the baseline socioeconomic conditions of the project area. It has also provided a general assessment of the risks and impacts of the project. The assessment included an indication of the positive and negative components of the activities and subactivities, the consequence of the impact, the probability that the impact will occur, the duration for which the impacts are expected to occur, and the overall significance of the impact.

Nevertheless, given that much of the project activities cannot yet be fully specified in terms of location, ownership, design and approach and will only be identified and fully specified through consultations and coordination with various stakeholders during project implementation, the ESMF provided the basis for identifying the risks and impacts of the subprojects and defining the processes and procedures to be applied in assessing and managing the risks and impacts of individual project activities during implementation. The major anticipated negative impacts of the proposed activities and subactivities include, but are not limited to: disproportionate distribution of project benefits, potential for gender-based violence (GBV), risks related to labour and working conditions, community health and safety, potential for introduction of invasive species and pests on planted seedlings, water use at nurseries diverting from other uses, use of pesticides and fertilizers in seedling production, destabilization and erosion during construction/reconstruction of terraces, land classification that impact access and introduce restrictions, increased exposure to disease/vector-borne illnesses, potential for negative human-wildlife interactions, and environmental risks related to the supply chain. Among the mitigating measures include a commitment to stakeholder engagement, a process of consent, training on sexual exploitation, abuse and harassment (SEAH), establishing a grievance redress mechanism (GRM), adhering to the labour and safety laws and international good practices, providing training on invasive species management, providing oversight in the supply chain, and adoption of practices to reduce the use of chemicals and production of wastes.

37. **ESS 2 (Labour and working conditions):** The project will likely require the hiring of professionals, service personnel and assistants, and activities on the ground such as terracing, removal of ferns and reforestation/afforestation activities will require the hiring of field workers and labourers. Risks associated with these activities include, but are not limited to, the potential for SEAH in the hiring process or at work, forced labour, child labour, and exposure to occupational health and safety hazards. Workers may be exposed to hazardous working environments such as steep slopes, disease vectors in standing waters, safety hazards in the fabrication/production of stoves, and exposure to bee stings in honey production, among other things. The project will ensure proper labour and working conditions are provided to workers (e.g. in forest camps), and trainings on labour and occupational health and safety standards will be undertaken. A Code of Conduct as regards worker safety and security will also be reflected in contract documents. Moreover, the project will not support activities that include the use of forced labour, human trafficking and child labour. During implementation, risks and potential impacts related to labour and working conditions will be further assessed and managed through the development of a labour and working conditions risk management plan, including ensuring that workers have access to a GRM.

38. **ESS 3 (Resource efficiency and pollution control):** The nature of the proposed interventions does not entail the use of substantial amounts of fuel, water, electricity or raw materials, except for the operation of nurseries (for its water requirement) and the production and distribution of cookstoves (for raw materials and electricity). The project will however promote the increased use of energy-efficient cooking stoves to lessen the impact on forest resources for fuel. During implementation, further review on resource efficiency and pollution control risks will be undertaken to assess and manage issues as regards water use and water availability, use of pesticides and fertilizers, potential for soil loss, soil erosion and slope destabilization, and use of raw materials and waste generation in the cook stove fabrication/production and distribution. The project will also ensure raw materials are from sustainable sources and will establish measures to minimize production waste and maximize distribution efficiency of stoves. Appropriate management measures will be developed upon screening and assessment of these risks.

39. **ESS 4 (Community health, safety and security):** The project activities could have community health, safety and security issues, which include potential for increased exposure to vector-borne diseases should ecological balance be altered or stagnant water develop as a result of the reforestation/afforestation and fern removal activities in the area (which may increase

the population of certain disease vectors and insect pests). Some of the mitigating measures include proper drainage design of terraces. There could also be possibility for human–wildlife conflict in areas where fruit trees are planted, which may inadvertently increase physical interactions with wildlife that could result in crop loss or physical injury both to humans and wildlife. The risk related to the presence of migrant labour is considered low given that the project will hire residents from surrounding communities. Nevertheless, the activities will be assessed more closely with regard to such risks, with measures to be discussed and determined in consultation with the communities during implementation.

40. **ESS 5 (Land acquisition and involuntary resettlement):** The project does not include any activities that will require resettlement or involuntary taking of land. However, project activities could facilitate a potential loss of access to lands given that the afforestation/reforestation activities could result in stricter enforcement of land use and park laws in changes to rules of access or land classification. To address this, the project will not establish protective forests on lands used for food production. On-farm afforestation will not involve the acquisition of lands, land rights or land-use rights through expropriation or other compulsory procedures and will comply with the country's legal system. Communities will be engaged in an inclusive and participatory manner and will be part of the planning and afforestation processes, and agreements on land rights and responsibilities will be developed with them. An access restrictions mitigation process framework is developed and will be implemented as required.

41. **ESS 6 (Biodiversity conservation and sustainable management of living natural resources):** The project intends to enhance the region's ecosystem to reduce the vulnerability of communities to climate change impacts. For instance, the proposed interventions on protected forests will be restoration (e.g. restoration of the 500 ha of degraded forest in GMNP, and the fern removal covering 4,500 ha of Nyungwe National Park, while the proposed interventions in the Stepping Stones and Landscapes Linkages (located outside of the parks) will be composed of the protection of isolated forest patches and afforestation on steep slopes and riparian areas supplemented with agroforestry activities to enhance ecosystem services. Nevertheless, care should be exercised in implementing these activities to ensure that negative impacts such as potential for the introduction of alien or invasive species, pests and pathogens from poorly managed nurseries are adequately addressed. During project implementation, consultations with communities and stakeholders will aid in determining the selection of appropriate species suitable to the locality.

42. **GCF Indigenous Peoples Policy and ESS 7 (Indigenous Peoples):** The AE recognizes that while there are historically marginalized groups as well as "Categories C, D and E, groups" that may meet the GCF criteria for Indigenous Peoples that could potentially be included in the project area, the screening so far has indicated that Indigenous Peoples have not been found in the project area. Given that specific sites will be selected for project activities during project implementation, each subproject will be subject to more in-depth screening for impacts to historically marginalized peoples, with particular attention to the GCF criteria of Indigenous Peoples, during project implementation. The AE will conduct a screening for subprojects as outlined in the "Screening for Presence of and Risks to Historically Marginalized Groups" to evaluate the characteristics of historically marginalized and Categories C, D and E people. Should the screening process identify Indigenous Peoples, then the subactivity will require a historically marginalized peoples plan. The AE also recognizes that there are potential barriers to integrating historically marginalized peoples into the project. Nevertheless, there is interest in working with historically marginalized groups to identify how traditional livelihoods like beekeeping can be adapted and promoted and how negative impacts on these livelihoods will be mitigated. The AE has included in the project the hiring of an ESS specialist and community engagement specialist to minimize risks as well including a commitment to stakeholder engagement, free, prior and informed consent and the setting up of a project-level GRM. In line with its roles and functions, the GCF Indigenous Peoples Advisory Group (IPAG) is available to

provide advice to the AE, which is also encouraged to share emerging good practices and success stories with IPAG.

43. **ESS 8 (Cultural Heritage):** The physical interventions (i.e. afforestation, reforestation and terracing, etc.) will be undertaken in mountainous areas where there are few human settlements. The preliminary screening of target areas also indicated that it is not expected that the project activities will disturb any cultural heritage sites. Nevertheless, the project has developed a chance find procedure to cover chance encounters of artefacts during project implementation.

44. **Institutional implementation arrangements:** As the AE, MOE Rwanda will oversee and monitor the management and implementation of the project. The Wildlife Conservation Society (WCS) and RFA will serve as EEs for the project through a project cooperation agreement. All work will be coordinated by the PMU and supported by other relevant government agencies, including the National Land Authority and the RWB within MOE Rwanda. The PMU will have oversight of the day-to-day operations of the project, including monitoring and evaluation. It will engage specialists (e.g. Environmental and Social Safeguards Specialist, Invasive Species Specialist, Community Engagement Specialist, Behaviour Change Specialist, Gender Specialist, etc.) that will screen for the potential environmental and social risks and impacts and develop management measures for proposed activities. The PMU will also be responsible for coordination on trainings for all project EEs. As regards safety, a Labour and Safety Coordinator will also oversee working conditions in the field. The project will also engage Community Engagement Specialists and Labour and Safety Officers.

45. **Stakeholder engagement and GRM:** The project benefited substantively from an extensive stakeholder consultation process, which started during the project conceptualization. A Stakeholder Engagement Plan has also been prepared to engage the stakeholders throughout project implementation. The project will have a project-level GRM, which is described in the ESMF. The GRM is independent of governmental judicial procedures. The project also allows access to the GCF independent redress mechanism. In line with the GCF Indigenous Peoples Policy, the GCF Indigenous Peoples' focal point will be available for assistance at any stage, including before a claim has been made.

46. **Sexual exploitation, sexual abuse, and sexual harassment safeguarding:** The SEAH safeguarding approach proposed by the AE is considered by the Secretariat to be in compliance with the SEAH provisions pursuant to the revised GCF Environmental and Social Policy. The project identifies that all project components may exacerbate the existing power differentials and expose the project beneficiaries to potential SEAH risk. The AE will take a number of mitigation measures to prevent and manage SEAH, including ensuring that all staff are trained on survivor-centred approaches to address SEAH, and that the Human Resources and Administration Officer and the ESS Specialist of the project will work together to develop a Security and Risk Management Plan to prevent SEAH incidences. The project's Gender Specialist and Community Engagement Specialist will develop training on SEAH prevention, GBV programming and survivor-centred approach. The project will also safeguard against SEAH through a Code of Conduct that explicitly indicates zero tolerance for GBV. This Code of Conduct will be shared with all personnel and used for training all employees. In addition to the prevalence of GBV in the target country, the project also recognizes that the influx of funds and activities, combined with power dynamics, may disproportionately affect women, youth, historically marginalized people, and others in the poverty categories of Rwanda. As the specific locations and beneficiaries will be determined during full implementation, each subproject will be screened to avoid any subproject that may aggravate risks of GBV and SEAH. The project will respond to SEAH incidents through GRMs at the AE, EE, and project levels. Training will be organized for all project staff, implementing partners, key service providers and other partner organizations to identify and regularly assess SEAH risks along the project cycle as well as

ensure that both the AE and all the EEs respond to SEAH incidents with survivor-centred and gender responsive approaches.

4.2 Gender policy

47. The AE has provided a gender assessment and action plan therefore complies with the requirements of the GCF Gender Policy.

48. The gender assessment provided demonstrates the commitment of Rwanda to advancing gender equality and women's empowerment. The assessment shows that Rwanda has a strong political commitment to promoting gender equality, reflected in its policy and legal framework and various sectoral strategies. It is the first country to have a majority female Parliament, with 61 per cent of these positions being held by women, and has established through its constitution a minimum of 30 per cent female representation in public governance. The country has also made international commitments such as ratifying and implementing the Convention on the Elimination of all Forms of Discrimination against Women, the Beijing Platform for Action and the Maputo Protocol on the Rights of Women in Africa. An institutional framework for coordinating and ensuring oversight of gender equality and women's empowerment is in place, which mainly rests with the Ministry of Gender and Family Promotion. The Ministry is mandated to coordinate the implementation of the national gender policy and advocacy on gender issues, a gender monitoring office and the National Women's Council. All ministries and public institutions have gender focal points. However, there is divergence between the frameworks in place and implementation on the ground due to challenges such as lack of financial resources and technical capacity within the various ministries as well as existing norms and practices that run contrary to equality principles and practices.

49. The gender assessment was conducted through desk review and stakeholder consultation. Additional stakeholder consultation will be conducted at national, district and project site level in order to further adapt project activities to the local context. The assessment indicates that women and men have different roles and responsibilities within society and households. These roles rigidly set women as caregivers and homemakers, while men are considered breadwinners engaging in income-generating activities, wage employment and decision-making. Women own less land, have less access to extension services and financial support (loans), and are less educated and able to gain experience in handling money. Women rely more on informal sources of loans as they do not usually have collateral to put up to access loans from financial institutions. Women engage and represent 66 per cent of the agricultural workforce; however, they occupy 19.7 of the paid jobs. Women's roles in afforestation and agroforestry is also curtailed by their lack of ownership and or land rights. Lack of land rights hinders the financial benefits women can get from resources on land such as trees. Women mostly use trees that provide little commercial value; rather they are focused on fulfilling their household needs through fuelwood, fodder, bean poles and fruits. Discussion with women had indicated that women observed shortages of fuelwood, fodder and wood suitable to their bean poles, and also that there is declining soil fertility and crop yields. Fuelwood also is the main source for cooking, which has health hazards for women, and gaining access to cleaner, improved cook stoves is a challenge. Men usually decide not to spend money to purchase the improved cook stoves that can directly address the multiple challenges faced by women. Female-headed households are identified as much more vulnerable to climate change impacts compared to male-headed households owing to the prevalent gender inequality. Engagement in alternative source of incomes for women poses its own challenges to women, one of which is affordability of required inputs, and this is particularly so for the beekeeping sector. The interest of the government in this area is also high, and investments have taken place, but only 23 per cent of women engaged in the sector.

50. The action plan includes activities that will address the challenges identified through the gender assessment. Activities contain corresponding indicators, targets, gender expertise and budget while also including timelines for each of the actions. The project will direct investments to the areas of equal access to resources (e.g. land), services (finance), capacity development, decision-making and also the participation of women in management, as beneficiaries/partners and as critical stakeholders in processes. Project interventions will respond to and address the growing pressure on natural resources, thus ensuring that the resilience and adaptive capacities of communities, including that of women and disadvantaged people, are built. The gender action plan is focused on ensuring (1) climate adaptation programmes and land-use planning address the priorities of women and youth; (2) women, youth and marginalized households have increased capacity to effectively engage in and benefit from forest management and forest restoration; (3) women, youth and marginalized households have improved capacity to engage in green jobs and livelihood activities; and (4) monitoring and evaluation effectively measures how women, youth and marginalized people are benefiting to enable the adaptive management of the project. A capacity-building component that focuses on strengthening cooperatives and village savings and loans institutions through gender mainstreaming across decision-making, membership, and products and services is included to ensure women are able to benefit from and engage in all aspects of the project. Private landowners who will be willing to engage in land afforestation activities will create private forest management units, which will then be assisted in forming and registering cooperatives that are at least 50 per cent women. Further, the selection of tree species will take into consideration women's usage and priorities. Women will be engaged and supported through knowledge acquisition, trainings, and skills development processes so that they benefit from the project implementation, particularly focusing on critical challenges such as rights to land and access to decision-making positions. Further, the challenge of rights to land will be addressed through advocacy work with men and local judiciary systems. This will be complemented by activities that will advocate for women's land rights and access to redress mechanisms. The project will engage community leaders and women and men community influencers to further promote equal and fair control of resources to mitigate the risk of men taking control over incomes earned by women.

51. There are three recommendations to the AE. The first recommendation is that the AE needs to ensure that gender issues are mainstreamed across all components of the project; this is lacking for component 3 (output 3.1 of the gender action plan), which presents many important opportunities to address gender issues. Secondly, the AE needs to refine and clearly show what outcomes it intends to achieve as a result of the interventions. Activity 3.1.1 in the funding proposal states the following regarding the composition of administrative committees: "Note that gender consideration is mandatory (at least 30% women)". It is not clear if 30 per cent refers to "women in decision-making bodies" or "membership of women in cooperatives", or both. These results are different in nature and should be captured independently. The AE should provide clarification in this regard. Thirdly, as part of its expressed commitment to do so, the AE should consistently collect data and report in a sex-disaggregated manner on the gender action plan.

4.3 Risks

4.3.1. Overall programme assessment (low risk)

4.3.2. Accredited entity/executing entity capability to execute the current programme (low risk)

52. We acknowledge the important role of the Government of Rwanda via MOE Rwanda as the AE (direct access). The AE demonstrated its comparative advantage and capability in

relation to this project with references to its track record in executing projects of comparable size or above (including FP073 and FP167 with GCF) and existing affiliation with the EE, RFA.

53. The AE provided adequate assurance that RFA has the necessary capacity to undertake the role as EE.

4.3.3. Programme-specific execution risks (Medium risk)

54. Co-financing: We acknowledge the sizable amount of co-financing totalling USD 10,493,579 (grants and in-kind) for a co-financing ratio of 1:0.27. If co-financing does not materialize then this may impede project implementation. The AE identified this risk, and the covenants in the term sheet are adequate mitigation measures – noting that the term sheet for the proposed project has not yet been agreed.

55. Project viability (and concessionality): The use of grants (and 100 per cent concessionality) is justified given the country context, the beneficiaries in Rwanda's CND, and the fact that the proposed activities are public goods in nature, including creating enabling opportunities for market development beyond the project to catalyse private investments (see component 3: Enhancing climate adaptation through resilient livelihoods).

56. Capacity-building activities and technology development/transfer (i) represent 45 per cent of total GCF funding requested; (ii) may not result in realizing expected results in case issues arise during implementation; and (iii) are embedded in the project's design and thus crucial to achieving intended objectives and outcomes. On the other hand, we note that identification of risks and the mitigation measures proposed by the AE are adequate.

4.3.4. Compliance risk (medium risk)

57. MOE Rwanda, in its capacity as AE, confirmed that it has conducted a risk assessment of money laundering and terrorist financing (ML/TF) in relation to project implementation. The probability of occurrence was assessed as 'low', and the impact severity in the event of occurrence was assessed as 'medium'.

58. MOE Rwanda confirmed that co-financiers have been identified based on their existing mandates and activities implemented in the CND region. Co-financiers' commitments will serve as a basis of MOE Rwanda engagement during the project implementation. During implementation, annual work plans for the project will be adjusted in collaboration with the EE, and all co-financiers are to align with all activities. MOE Rwanda, per its oversight role, will ensure that all of these stakeholders commit during planning to their co-financing and report this in the annual performance reports submitted to GCF.

59. MOE Rwanda indicated that the Government of Rwanda currently has robust measures to fight ML/TF risks (e.g. law N° 001/FIC/2022 on regulations relating to anti-money laundering and countering the financing terrorism). The Government of Rwanda has also put in place a special agency for this purpose, the Financial Intelligence Center (FIC), which has a clear mission to promote financial system integrity by conducting effective financial intelligence to counter ML/TF and related financial crimes in governmental institutions, projects and parastatal agencies under the supervision of the Ministry of Finance and Economic Planning. Both MOE Rwanda and this project under its supervision will also comply with FIC regulations.

60. MOE Rwanda confirmed that it has an internal audit function as well as regular audits conducted by the Office of the Auditor General to ensure the efficiency and effectiveness of internal controls, which includes handling ML/TF risks. In addition, MOE Rwanda uses segregation of duties within financial management processes.

61. MOE Rwanda confirmed that it has conducted a risk assessment of prohibited practices in relation to project implementation. The probability of occurrence was assessed as Low, and the impact severity in the event of occurrence was assessed as Medium.
62. MOE Rwanda indicated that to ensure materials or technology procured under the project are used only for the purposes intended and not diverted or misused for unauthorized, improper or illicit purposes, a comprehensive set of controls will be used. This includes explicit terms and conditions in all procurement contracts or agreements that clearly (i) state the intended purpose of the materials or technology being provided and explicitly prohibit any unauthorized use or diversion; and (ii) define the individuals that are authorized to access and use the procured materials or technology. Furthermore, this will involve the implementation of strong authentication and access controls to ensure that only approved users have access. This refers to the record of any materials or technology procured under projects within an assets register. The records include assets identification, the person using the material and the location.
63. Monitoring and auditing the usage of the procured materials or technology will be undertaken to detect any anomalies or potential misuse, including periodic site visits. Training sessions will also be conducted to educate all stakeholders involved in the project about the importance of adhering to the intended use of the materials or technology and the consequences of misuse. For technologies that involve data transmission or storage, MOE Rwanda will ensure that appropriate encryption and data protection measures are in place to safeguard sensitive information.
64. MOE Rwanda confirmed that no projects will be undertaken in any jurisdiction or territory that is subject to or affected by sanctions or restrictive measures per the United Nations Security Council (UNSC) resolutions. Additionally, no individual or entity that is listed on any UNSC sanctions list, including the United Nations Consolidated Sanctions list, will be involved in any manner with the project or its activities, either as a counterparty, implementer or beneficiary.
65. MOE Rwanda acknowledged that there are United Nations sanctions in the neighbouring country of the Democratic Republic of the Congo, close to the CND. According to MOE Rwanda, the United Nations Consolidated Sanctions List indicates approximately 20 individuals with ties to Rwanda, a small percentage of whom may live in Rwanda. MOE Rwanda confirmed that the United Nations Consolidated Sanctions List has been circulated internally and will be cross checked when hiring for the project and once project sites are selected to ensure these individuals will not be involved in any manner with the project or its activities, either as a counterparty, implementer or beneficiary.
66. MOE Rwanda indicated that the project involves many actors for successful implementation, including ministries, government agencies, non-governmental organizations, subgrantees, vendors, service providers, community members and beneficiaries. MOE Rwanda confirmed that there is specifically an intention to distribute or disburse to beneficiaries, both directly and indirectly, commodities and other items of value such as (i) the distribution of seedlings for trees that will produce fruits or contribute to enhancing biodiversity of farming systems; (ii) access to improved cook stoves; as well as (iii) the option for the distribution of bees to start honey operations. MOE Rwanda acknowledged that, as with any project infusing large sums of money directly or indirectly with multiple actors, corruption could occur.
67. MOE Rwanda indicated that the distributions will be coordinated through procured contractors. The contractors are hired based on the Government of Rwanda/MOE Rwanda procurement policies, which are aligned with GCF policies and include due diligence in the hiring and contracting process. As for direct community contracting, the intended beneficiaries are screened and certified by local authorities in the area of project interventions, which mitigates the risk of prohibited practices.

68. MOE Rwanda confirmed that for reporting and remedying any risks that may occur, MOE Rwanda will maintain a project-level GRM through which anyone can report the occurrence of prohibited activities. All complaints will be investigated accordingly. Any complaints believed to contribute to prohibited activities will be referred to the appropriate authorities in Rwanda.

69. MOE Rwanda indicated that all staff will be made aware of the prohibited activities, and all subcontractors/counterparties will be screened for current and past engagement in such activities and made aware of their obligations to alert the PMU and/or submit grievances related to such activities occurring as part of the project.

70. **Compliance risk rating:** The Secretariat has conducted a review of the project in accordance with relevant GCF Board-approved policies and does not find any material issue or deviation with respect to compliance issues. Based on available information for this funding proposal, the Secretariat has determined a compliance risk rating of 'Medium' and has no objection to this proposal proceeding.

4.3.5. GCF portfolio concentration risk (low risk)

71. In the case of approval, the impact of this proposal on the GCF portfolio concentration in terms of result area and single proposal is not material.

4.3.6. Recommendation

72. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall programme	Low
Accredited entity/executing entity capability	Low
Project-specific execution	Medium
GCF portfolio concentration	Low
Compliance	Medium

4.4 Fiduciary

73. MOE Rwanda is the AE for the project and will provide oversight to the project consisting of, inter alia: (a) entering into a subsidiary agreement with the EE (RFA); (b) managing and disbursing GCF funds to the EE; (c) reviewing financial expenditures and progress reports; (d) overseeing project implementation following the project document and annual work plans and budgets, agreements with co-financiers and each EE's rules and procedures; and (e) providing technical guidance to ensure that the appropriate technical quality is applied to all project activities.

74. RFA will serve as the EE for the project through a subsidiary agreement signed with MOE Rwanda. RFA will execute the tasks within the overall project management structure as follows: (a) implementing day-to-day activities as per the project work plan and budget, including the environmental and social management plan; (b) undertaking procurement activities as per the signed agreements; and (c) managing contracts of suppliers and service providers. The EE will enter into a memorandum of understanding (MOU) with different partner agencies, including WCS and the Catholic Organisation for Relief and Development, which will provide technical assistance services to implement the project activities. Procured parties will be contracted following MOE Rwanda rules and regulations for procuring goods and services. The PMU will manage, monitor and report on the day-to-day implementation progress of the EE.

75. As one of the project implementation partners, WCS will provide technical assistance to RFA through an MOU signed with RFA. The MOU will detail all the activities that WCS will directly support during this project's implementation. The technical assistance services will include advising RFA/SPIU, procuring and supervising international experts and consultants, and procuring materials to support the execution activities. Relevant technical experts from WCS will be embedded within RFA to advise SPIU on day-to-day project management. At the same time, the Catholic Organisation for Relief and Development will provide technical assistance to RFA through an MOU.

76. The co-financiers (RDB, RFA, RWB and MOE Rwanda) will execute and manage the implementation of their activities.

77. MOE Rwanda shall ascertain that the EE has efficient internal control systems in place, review financial reports and proceed with funds disbursements. It will also periodically inspect the project for compliance and coordinate audit exercises for the EE. RFA will enter into agreements with MOE Rwanda that outline general modalities of the management of funds in line with GCF grant requirements and RFA's policies.

78. GCF funds will be transferred to an RFA-designated bank account. Based on agreed annual work plans and budgets, the funds will be disbursed to the EE, and its reporting will be quarterly. The project's financial processes will rely on the EE's charts of accounts, internal control systems, disbursement plans, budgeting, expenditure management and accounting, reporting, and assets management as assessed by MOE Rwanda.

79. The project will be audited annually and commissioned by MOE Rwanda to produce certified financial statements and reports to ascertain that GCF funds are well managed. The project will maintain a regularly updated record of all assets and documents that will be retained for at least five years after the end of the project.

4.5 Results monitoring and reporting.

80. The Secretariat has provided various rounds of feedback on results monitoring and reporting, which have been generally addressed by the AE. This includes a review of beneficiary numbers, revisions to the logical framework, mapping of project outcomes to GCF results areas, and timelines and resourcing for the mid-term and final evaluations. However, there are some issues related to beneficiary calculation that are pending as of the final review. These should be appropriately addressed as the project is mobilized as follows:

- (a) It would be helpful to provide gender disaggregation for beneficiary numbers wherever possible. This is currently done in section E (logical framework), but not in section A6 and annex 23;
- (b) In section E (logical framework), the AE should also provide a brief description of how beneficiary numbers against GCF results areas have been calculated. These details are currently missing, and while we have mid-term and final targets for various indicators related to beneficiary count, we do not have information to understand how these have been calculated, and how potential double-counting will be avoided. If space is limited in section E, the AE may consider including this information in a clearly marked section in annex 23 related to beneficiary count.

4.6 Legal assessment

81. The original Accreditation Master Agreement was signed with the Accredited Entity on 23 June 2016, and it became effective on 1 August 2017, which agreement was amended and restated pursuant to the first amendment and restatement dated as of May 14, 2019, which became effective on even date. The Accredited Entity's first term of accreditation lapsed on 31 July 2022. The Accredited Entity was recredited by the Board pursuant to Decision B.34/17. The second amendment and restatement agreement amending and restating the accreditation master agreement was signed on 22 November 2022, and the second amended and restated accreditation master agreement (the "AMA") became effective on 26 January 2023.

82. The Accredited Entity has provided a certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

83. The proposed project will be implemented in the Republic of Rwanda. The GCF has signed a bilateral agreement on privileges and immunities with the Government of the Republic of Rwanda. It was signed on 8 November 2022 and became effective on 25 May 2023.

4.7 List of proposed conditions (including legal)

84. To address matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and
- (b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP217

Proposal name:	Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration
Accredited entity:	Ministry of Environment of Rwanda
Country:	Rwanda
Project/programme size:	Small

I. Assessment of the independent Technical Advisory Panel

1. This funding proposal for a small-sized project in environmental and social safeguards (ESS) category B is submitted by the Government of Rwanda as a cross-cutting project contributing to both adaptation (92 per cent) and mitigation (8 per cent) results areas. The Ministry of Environment of Rwanda (MOE Rwanda) (a GCF direct access entity) is the accredited entity (AE) for the proposed project, and the executing entity (EE) is the Rwanda Forestry Agency, with several implementing partners, including other government agencies and the Wildlife Conservation Society (WCS) in Rwanda. The total cost of the funding proposal is USD 50 million, of which USD 39 million is GCF finance and USD 11 million is co-finance from the Government of Rwanda.
2. The project's goal is to apply a landscape approach to reduce the exposure of populations in Rwanda's Congo Nile Divide (CND) landscape to the impacts of unfavourable climatic trends and hazards intensified by climate change, including floods, landslides, soil erosion and wildfires. This is to be achieved through integrated land-use planning and management that allows for protecting, restoring and managing a matrix of interconnected ecosystems, including pristine forest in protected area nodes, important landscape linkages, pockets of forest in stepping stones, and agro-ecosystems in farmlands. Interventions in various spatially defined target areas include forest protection and fire management, assisted natural regeneration, afforestation of steep slopes, and support to silvopastoralism and agroforestry and its value chains.
3. The project has three planned outcomes (see the table below), with outputs under each outcome as well as the parties responsible for executing them (see table 5 in the funding proposal).

Project outcomes and outputs	Executing entity per output	Technical implementation partners
Outcome 1: Landscape planning, policies and management are effective and coordinated across sectors and scales to address climate risks and adaptation benefits		
Output 1.1. Landscape-wide land-use plan developed for climate-resilient livelihoods and forest ecosystems, integrating district development strategies	Rwanda Forest Agency (RFA)	Wildlife Conservation Society (WCS) and National Land Authority (NLA)
Output 1.2. Local and national institutional capacities strengthened to integrate biodiversity and climate risks into land use planning and management	RFA	WCS and NLA

Outcome 2: Natural forests are protected, connected, and more resilient to climate impacts and risks		
Output 2.1. Protected areas management effectiveness improved to address climate risks and adaptation	RFA	WCS and Rwanda Development Board
Output 2.2. Natural forest cover restored; biodiversity connections established	RFA	WCS
Outcome 3: Vulnerable rural livelihoods are more climate-resilient, diverse, economically sustainable and nature-positive		
Output 3.1. Farming methods enhance productivity, reduce erosion and flooding risks, contribute to ecosystem services, and support connectivity	RFA	Rwanda Water Board
Output 3.2. Rural livelihoods generate alternative incomes and reduce pressure on forests	RFA	Cordaid
Output 3.3. Financial services and private sector investment engagement	RFA	Cordaid
Monitoring and evaluation	Project Management Unit, hosted by the Ministry of Environment of Rwanda	

1.1 Impact potential

Scale: N/A

4. The cross-cutting impact potential of the proposed project and the relative weighting of the intended results can be seen in the attribution of GCF and co-financiers' contributions to the relevant GCF results areas as follows:

GCF results area	GCF contribution	Co-financiers' contribution
<u>Mitigation Result Area 4 (MR4)</u> Mitigation through forestry and land use	8%	0%
<u>Adaptation Results Area 1 (AR1)</u> Enhanced resilience of most vulnerable people and communities	13%	49%
<u>Adaptation Results Area 2 (AR2)</u> Enhanced resilience of health and well-being, and food and water security	2%	0%
<u>Adaptation Results Area 4 (AR4)</u> Enhanced resilience of ecosystems and ecosystem services	77%	51%

5. The emphasis on utilizing the GCF investment to enhance the resilience of ecosystems and ecosystem services is reflected in the project's focus on new mechanisms of land-use planning and management that consider the CND as a landscape consisting of an interdependent mosaic of forests and farms, the management of each determining the health of the whole. The independent Technical Advisory Panel (iTAP) believes that such an approach (i.e. targeting project interventions on the ground based on a systematic spatial planning process that emphasizes "ecological infrastructure" functioning as well as biodiversity) represents best practice for enhancing ecosystem resilience. Such an approach ensures that interventions are at sufficient scale to achieve a transformative effect: preserving landscape-wide ecosystem processes and functions that help buffer impacts of climate change. Maximizing connectivity is also important for maintaining the long-term health and stability of forests as carbon sinks.

6. **Adaptation impact potential:** The funding proposal outlines the major climate change-related hazards characterizing the landscape, in particular more frequent intense rainfall events increasing floods and landslides, and worsening soil erosion that is already widespread as a result of direct anthropogenic pressures. The project's main adaptation benefits will accrue to

vulnerable communities primarily engaged in agriculture on the CND's steep terrain, achieved through improved soil management, forestry and agroforestry to reduce catastrophic events affecting production systems, lives and livelihoods. This will be achieved through integrated landscape planning and management to enhance tree cover in crop and grazing lands and improve the connectivity of natural ecosystems and the key services they provide, including regulation of runoff and flows from increasingly extreme rainfall events; and through reduction of localized and downstream flooding, erosion and landslide incidence and risk.

7. Indicators for the main adaptation results are as follows:

- (a) 135,104 hectares (ha) of land to be brought under climate-resilient management through targeted interventions for protection, afforestation and agroforestry (AR4):
 - (i) Including 121,758 ha of core protected areas, linkages and stepping stones that will be brought under climate-risk informed planning and fire management to maintain landscape-scale ecosystem services; and
 - (ii) Including 13,346 ha for focused interventions (i.e. degraded forest in parks to be restored (5,000 ha), riparian lands restored (1,500 ha), pastures brought under silvopastoralism (1,000 ha), steep slopes afforested (2,500 ha), and farmland with new agroforestry practices (3,346 ha));
- (b) 1,254,242 people, resident in the landscape where targeted interventions will be carried out, will benefit directly from reduced exposure to climate-intensified disasters and/or enhanced adaptive capacity (AR 1 and 2); and
- (c) 9,260,745 people, approximately two thirds of the country's population, will benefit indirectly from a wide-scale education and outreach programme, including radio programming, on climate change and gender issues, climate risks, value of forests and forest ecosystems, and climate change adaptation options.

8. **Mitigation impact:** The project seeks to deliver greenhouse gas (GHG) emission reductions and removal benefits equivalent to 224,871 tonnes of carbon dioxide equivalent (t CO₂ eq) during the five-year project implementation period, reaching a total of 1,084,291 t CO₂ eq after 20 years. The estimation of reductions and removals is made based on the difference between the projected emissions with and without the project interventions – through improved cookstoves, and through on-the-ground interventions across a total of 13,346 ha of land. The land-based interventions will have cross-cutting benefits, since they contribute to core indicator 4 on hectares of land to be brought under climate-resilient management, as well as bringing mitigation benefits.

9. The following activities will contribute to achieving this mitigation potential: (i) assisted natural regeneration (ANR) in 4,500 ha of Nyungwe National Park through invasive fern removal; (ii) restoration of 500 ha of degraded forest through ANR in Gishwati-Mukura National Park; (iii) afforestation on 2,500 ha of public and private land with a slope >55 per cent; (iv) planting 1,500 ha of protective forest in riparian zones (public and private); (v) implementing erosion control and agroforestry practices on 3,346 ha of smallholder farmland; (vi) implementing silvopastoralism on 1,000 ha in the Gishwati area; and (vii) adoption of improved cookstoves by 8,500 households. Indicators for achievement of these targets are included in the project logical framework in section E.3 of the funding proposal.

10. The methodology for calculating GHG emission reductions and carbon sinks mirrors two approved consolidated methodologies for the A/R CDM, AR-ACM0003 and AR-AMS0007. This is considered a good practice because the applicability conditions of the methodologies are appropriate to the project conditions. The methodology is clear and conservative, considers the relevant guidance from the Intergovernmental Panel on Climate Change, and was used in an appropriate, clear and transparent manner when estimating the mitigation potential for each activity.

11. With regard to monitoring and reporting mitigation benefits, annex 22 proposes a clear and straightforward approach, but does not clarify the roles and responsibilities of the various actors that will be involved in the process over time, nor the frequency of the steps. Furthermore, the activities as presented in annex 22 are not reflected in annex 11 (monitoring and evaluation plan_rev August 2023). According to its updated nationally determined contribution (NDC) (2020, page 6)¹ Rwanda has already developed a national measurement, reporting and verification (MRV) framework. At the present stage it is not clear how this project will be aligned with the existing arrangements for the national MRV system, and/or how it will support or contribute to enhancing the functioning of this system across the country.
12. The iTAP assesses the impact potential of the project as high.

1.2 Paradigm shift potential

Scale: N/A

13. The proposed project is designed to bring about a paradigm shift away from the current situation of fragmented, unplanned decisions and a collection of isolated sector-by-sector projects that fail to take into account evolving climate risk and allow continued degradation. Instead, a new approach of integrated land use planning and management is promoted that preserves ecological connectivity across the landscape and helps maximize valuable ecosystem services for buffering the impacts of climate change. The new approach is based on international best practice and considers the CND as a landscape consisting of an interdependent mosaic of production and protected lands, the health of which is completely interconnected.
14. The idea behind the project is that a new holistic, climate-smart, evidence-based approach to planning and development will enable systematic, ecosystem-based, multi-sector analysis of land-use tradeoffs. Focusing on this larger scale, with coordination across both geographic and sectoral boundaries, while mainstreaming climate risks in management planning, represents a fundamental paradigm shift for land-use planning and management in Rwanda. Success in putting such a system into place could represent a crucial first step in reversing the previous negative feedback loop of ecosystem degradation to achieve a more positive, climate-resilient system.
15. Utilizing integrated landscape management and ecosystem-based adaptation approaches in addressing climate change impacts in Rwanda offers a myriad of opportunities for scaling up and replication to similar ecological zones across the country. The project also has the potential to contribute to the regulatory frameworks and policies for driving climate investments in the forest and renewable energy sectors. The project will finance the development of an integrated land-use plan, which will be adopted by all 10 CND districts as a framework for seeking synergies between rural livelihoods and forest and biodiversity protection, reconciling land-use conflicts that arise, and guiding development decisions and climate change adaptation. It will also update the National Land Use Master Plan, ensuring it incorporates climate resilience and local plans.
16. **Systematic conservation planning:** Although biodiversity conservation is considered a co-benefit of the project rather than a core focus, the project makes innovative use of systematic conservation planning to identify spatial areas in the landscape that are targeted for restoration, afforestation and agroforestry activities. This includes work to restore national parks as the core of an interconnected system, with landscape linkages and stepping stones identified for interventions to promote connectivity between remaining areas of natural habitat. Optimizing this kind of mosaic of land uses with management actions for protection, restoration and sustainable use helps to maximize the functionality of landscape-wide ecosystem services such

¹ https://unfccc.int/sites/default/files/NDC/2022-06/Rwanda_Updated_NDC_May_2020.pdf.

as nutrient cycling, soil fertility, slope stability, water infiltration, flood management and precipitation.

17. **Innovative financial mechanisms:** The project also includes an output to establish long-term plans for CND financial sustainability post-GCF by employing a technical advisor to support the establishment of innovative mechanisms such as those identified in the Biodiversity Finance Plan and also to refine existing financing mechanisms (for example, integrating climate resilience principles into the existing successful and growing Tourism Revenue Sharing Program). Support will be provided to beneficiaries of the project in applying to the existing Community Adaptation Facility within the Rwanda Green Fund (FONERWA). The Government of Rwanda, represented by the Rwanda Development Board, is also in advanced stages of negotiation with the World Bank for establishing a wildlife bond, potentially a gorilla conservation bond, modelled on the successfully established rhino conservation bond in South Africa, with interest shown by impact investors.

18. Overall, based on the above, the paradigm shift potential of the project is assessed as medium to high.

1.3 Sustainable development potential

Scale: N/A

19. The proposed project is aligned with the United Nations 2030 Agenda for Sustainable Development, including significant contributions towards Rwanda's efforts to meet Sustainable Development Goal (SDG) 5 (Gender Equality), SDG 13 (Climate Action) and SDG 15 (Life on Land), and will contribute to the achievement of SDG 1 (No Poverty), SDG 2 (Zero Hunger), SDG 3 (Good Health and Well-being) and SDG 10 (Reduced Inequalities).

20. **Environmental co-benefits:** Key environmental co-benefits (beyond the core benefits related to ecosystem-based adaptation) include restoration of soil fertility through silvopastoralism and regenerative mixed-use agroforestry, and the enhancement of biodiversity through planting 50 per cent native tree species. Promotion of improved cookstoves using significantly less fuel will contribute to relieving pressure on the remaining indigenous forest.

21. The iTAP was surprised to see the relatively low target of 50 per cent (30 per cent on private land and 70 per cent on public land) in the use of indigenous tree species in afforestation, reforestation and agroforestry interventions. The AE explained that exotic species are more suitable for afforestation of steep slopes, where they can easily establish themselves, "but this does not mean that exclusively exotic species will be used on such sites; there will be a mixture of both wherever applicable. In addition, while establishing blocks in the Participatory Forest Management Units, indigenous species will be used to separate one block from another. Indigenous species will be given due consideration in agroforestry to create positive interactions in the farm as well as promoting native species, avoiding monoculture and benefiting from their positive impacts in terms of resilience."

22. It is understood that there is often a preference for fast-growing, easily available and usually cheaper species like eucalyptus, pine and alnus for urgent needs such as slope stabilization and fuelwood, and also that a balance needs to be struck between promoting indigenous trees species and being responsive to farmers' wishes. At the same time, the project could potentially make more of the opportunity to rekindle traditional knowledge about indigenous species and their uses, and to build the nursery supply chain for quality native planting material given the vast superiority of indigenous tree species in terms of their multiple benefits, non-invasiveness, positive impact on soil and water quality, and resistance to pests and diseases that may worsen with a warming climate.

23. **Social and health co-benefits:** In enabling the transition to more climate-resilient and productive livelihood models, the project aims to increase household incomes and the nutritional status of families. It is expected that significant social capital will be created through

beneficiaries' participation in co-designing and co-managing a range of adaptation strategies, with an emphasis on women's participation. Social cohesiveness will also be enhanced through the strengthening of joint saving and lending systems. Social benefits will also accrue to indirect beneficiaries, for example those listening to radio programming that leads to increased awareness of climate change, or of opportunities for climate-adaptive agroforestry and agri-processing.

24. **Economic co-benefits:** Inequality and poverty will be addressed through increased financial inclusion and improved economic opportunities, especially for women, youth and historically marginalized groups. Forest and landscape restoration in the CND will bring economic benefits, since soil erosion control measures, protective forests on public lands, and agroforestry on private lands will all increase agricultural productivity and associated income. The project will create direct and indirect employment opportunities, benefiting an anticipated 24,212 households and stimulating the local economy. Private capital will be leveraged to invest in value chains, including those for improved cookstoves, irrigation pumps and beehives, as well as honey, woodlots, macadamia, avocado and vegetable production and processing.

25. **Gender benefits:** The project aims to achieve positive outcomes around access to resources, improved livelihoods, and income generation opportunities and capacities for women. With support from a dedicated gender specialist, the project will engage women in project planning, investment and decision-making. It is envisaged that women will respond to incentives that address their families' basic needs, such as better health and nutrition, linked to agriculture and food security improvements. The project will also get support from non-governmental organizations that engage men in programming and discussions to increase their participation in household and parental responsibilities and enhance their awareness of the importance of women's fair access to and control of resources, including income. Women will benefit from training and educational activities, leadership, entrepreneurship and decision-making, as shown in annex 8 (Gender Analysis/Assessment and Gender and Social Inclusion Action Plan).

26. The iTAP assesses sustainable development potential as high.

1.4 Needs of the recipient

Scale: N/A

27. Western Rwanda and its CND region are highly vulnerable to climate change, and are expected to experience a continuing upward trend in average annual temperatures, total annual rainfall (with a longer dry season), and more intense rainfall events and storms and increasing risk of flash floods, landslide, soil erosion and wildfires. Despite Rwanda's solid economic progress since 2000, poverty remains widespread and pervasive. Overall, 38.2 per cent of the country's population lives in poverty and 16 per cent in extreme poverty. Women are more affected by poverty than men, with 47 per cent of female-headed households living in poverty compared with 44.9 per cent of all households.

28. The CND landscape supports nearly one third of Rwanda's population on barely one fifth of its land base. Average population density is 736 people/square kilometre (km²), with some districts now exceeding 1,000 people/km². The highest rates of rural poverty in the country are reported from the CND, including its most vulnerable subsistence farmers. While high rainfall has historically permitted generally sufficient food production, the region's rugged topography requires farming on extremely steep slopes. With more frequent extreme rainfall events in recent years, this agricultural landscape is recognized as the region of Rwanda most vulnerable to flash floods and landslides. Between 2013–2016, heavy rains generated landslides that killed at least 174 people and destroyed the homes of more than 5,000. Landslides wipe out highly productive raised-bed cultivation systems in the CND's valleys and bottomlands, many hectares of crops are washed away, and an estimated 1.5 million tonnes of fertile soil are lost annually to erosion.

29. The funding proposal reports that during stakeholder consultations, communities indicated several socioeconomic changes stemming from changing climate conditions. Notably, the longer dry season and unpredictable rains have reduced agricultural yield significantly, leading to more widespread malnutrition and extreme poverty, and farmers can barely meet their own families' needs. In particular, women, youth, and historically marginalized people are feeling the impacts. As arable land is increasingly constrained, youth and historically marginalized people have few opportunities and are more focused on survival and short-term gains. This becomes a negative feedback loop, in which poverty leads to unsustainable practices such as overharvesting from forests and cultivation on extremely steep slopes, with degraded ecosystems then less able to provide services, worsening the cycle of poverty.

30. As a result, the Government of Rwanda has invested significantly in developing an enabling policy and institutional framework for a more climate-resilient development pathway. Yet, while adaptation is a top priority, as a least developed country, Rwanda lacks sufficient public financial resources to implement its climate change and resilient green development objectives. Currently there are no alternative funding options at the scale needed to secure Rwanda's priority forests to reduce the vulnerability of rural communities to climate risks. Rwanda's debt-to-gross domestic product ratio stands at 64 per cent, making further loans difficult, and the Government of Rwanda seeks maximum concessionality from the GCF for the urgent adaptation actions proposed under the project.

31. The Government of Rwanda recognizes that a piecemeal approach to decision-making has exacerbated competition over scarce land resources and reduced the adaptive capacity of both ecosystems and rural populations. Additionally, district government staff and community members have limited technical understanding of the complex interactions between forest structure and functions, climate change, and the provision of ecosystem services, and limited technical capacity for complex forest monitoring. In the face of worsening climate change impacts, there is an urgent need for a comprehensive approach that crosses siloes of government and achieves integrated land use planning and management informed by the evolving climate and socioeconomic conditions. The proposed interventions are well targeted to address the identified barriers and the great needs of these vulnerable communities.

32. Overall, based on the above, the needs of the recipients are rated as high.

1.5 Country ownership

Scale: N/A

33. **Policy and strategy alignment:** The project is closely aligned with Rwanda's national adaptation plan and three of its adaptation priorities: (i) promotion of non-agricultural income generating activities; (ii) introduction of species resistant to environmental conditions; and (iii) development of alternative sources of energy to firewood. It contributes to several priorities identified in the 2020 NDC: promotion of afforestation/reforestation; improvement of forest management; an integrated approach to planning and monitoring for sustainable land management; and development of a harmonized and integrated spatial data management system for sustainable land use management.

34. The project will support the implementation of Rwanda's Green Growth and Climate Resilient Strategy and the Rwanda National Strategy for Transformation (2017–2022) and is aligned with the goals of the Rwanda National Forestry Policy 2018, the Forest Sector Strategic Plan 2017–2022, the Strategic Plan for the Environment and Natural Resources Sector (2018–2024) and the National Forest Management Plan 2017–2026, thus reflecting the government's intention to address climate change impacts and mitigation targets by improving forest management. It is also aligned with the national Biomass Energy Strategy 2020, which aims to ensure a more sustainable biomass supply and combustion technologies for homes and small enterprises.

35. **GCF country programme:** According to the funding proposal (section D.5), the project aligns with the forthcoming GCF country programme, which “prioritizes forest management through restoration and protection of natural forests, afforestation and agroforestry, as well as distribution of high-quality reforestation inputs and capacity development”. While this project proposal to GCF focuses on a different geographic area, the early and ongoing lessons from the experiences from existing GCF-funded projects “Strengthening Climate Resilience of Rural Communities in Northern Rwanda”² and “Transforming Eastern Province through adaptation”³ will be applied during the implementation of this project in the CND landscape.

36. **Roles of key entities:** The proposed project was initiated by MOE Rwanda and supported by the national designated authority, Rwanda Environment Management Authority, due to the high vulnerability to climate change of the CND region of the country. MOE Rwanda is well placed to oversee the project as AE since the ministry is responsible for (i) ensuring the sustainable management of natural resources and for the development of policies, strategies and regulations; and (ii) mobilizing resources for the development of the sector. MOE Rwanda managed USD 10 million from the Adaptation Fund to implement the Reducing Vulnerability to Climate Change in North West Rwanda through Community Based Adaptation Programme⁴ and has overseen a number of other climate projects, including: an early warning system project (funded by the Least Developed Countries Fund (LDCF)), the Landscape Approach to Forest Restoration and Conservation and the “Building resilience of communities living in degraded forests, savannahs and wetlands of Rwanda through an ecosystem management approach” programme (funded by LDCF)⁵.

37. Given the centrality of forest protection, restoration and management to the project, the RFA is well placed to carry out the role of EE. It is noted that this would be RFA’s first GCF EE role, since MOE Rwanda’s other GCF project is executed by FONERWA. Strong support will be provided by a set of implementing partners – including the WCS and government agencies such as RDB, which are responsible for the national parks, RWB, Rwanda Environment Management Authority and the National Land Use Authority. The project activities overseen by each are shown in the table in the introduction above and are aligned with their mandates, as well as sectoral plans and strategies.

38. **Consultation and engagement with civil society:** To better understand the interests of stakeholders and potential impacts of the proposed project, and as a preparation for designing future engagement strategies, a comprehensive, gender-responsive, culturally sensitive, non-discriminatory, and inclusive stakeholder analysis was undertaken and is reflected in the Stakeholder Engagement Plan (annex 7). In terms of future engagement, the funding proposal indicates a number of specific areas where consultations will be ongoing to inform the rollout of activities, for example, consultations with communities and civil society organizations representing men, women, youth, and historically marginalized people in local communities, as well as district and national government agencies, on the following activities: Activity 1.1.1. Synthesize and disseminate information on value of natural forests and ecosystem services; Activity 1.1.2. Develop climate-resilient landscape land-use plans; Activity 2.2.3 Promote silvopastoralism with indigenous trees around Gishwati; Activity 3.1.1 Restore high slope areas (>55 per cent) as protective forests; and Activity 3.2.1 Develop an agroforestry-related value chain for market access.

²Rwanda Ministry of Environment, 2018. FP073: Strengthening Climate Resilience of Rural Communities in Northern Rwanda. Green Climate Fund.

³International Union for Conservation of Nature, 2021. FP167: Transforming Eastern Province through Adaptation. Green Climate Fund.

⁴Adaptation Fund, 2014. Reducing Vulnerability to Climate Change in North West Rwanda through Community Based Adaptation. [Link](#).

⁵United Nations Environment Programme, 2016. Building resilience of communities living in degraded forests, savannahs and wetlands of Rwanda through an ecosystem management approach.

39. Country ownership of the project is thus seen to be high.

1.6 Efficiency and effectiveness

Scale: N/A

40. Of the project's total proposed budget of USD 50,000,000, 78 per cent (USD 39,056,579) is proposed to be a GCF grant, and 22 per cent (USD 10,943,579) is made up of government co-financing to the project. The Government of Rwanda is providing grant financing via RFA, RWB, and RDB to support activities in component 3. In addition to these contributions totalling USD 10,831,307, the AE will cover office costs in the form of an in-kind contribution equivalent to USD 112,271 for the Project Management Unit, which will be hosted at MOE Rwanda.

41. **Government co-finance:** In response to a question from iTAP about the nature of the co-finance and the extent to which it represents new and additional funds, the AE explained that "[t]he co-financing commitments were provided based on the existing Government commitments provided every year in the project intervention area. These are earmarked funds that go into forest management, community revenue sharing around the parks, monitoring resources for catchment restoration interventions and also different in-kind contributions. All the details are indicated in the Annex 4 of the budget and activities to be covered by co-financing resources and Annex 13 indicates the commitment letters from the co-financiers." It is iTAP's understanding that these funds will not be spent in a business-as-usual manner, but will be newly directed towards climate risk management, through support to the project's adaptation and mitigation interventions and efforts to develop officials' capacity for climate risk-informed integrated land use planning and management.

42. **Role of financial institutions:** Participating financial institutions (PFIs) will be important in two ways – firstly as providers of credit to farmer cooperatives, savings groups and small enterprises in the targeted value chains, and secondly as recipients of training – to enhance their knowledge of good agricultural practices, equip them to assess the impact of climate change on their clientele and loan portfolio, and help ensure equitable access to financial assets by women, youth and historically marginalized communities.

43. In response to a question from iTAP about whether any consultation has happened with the potential PFIs, which are not mentioned in the extensive stakeholder consultations reported in annex 7, the AE stated that "At this stage [neither] Cordaid nor RFA have notified the participating financial institutions. As per the project implementation plan, a 6-month to 9-month inception phase will include in-depth analysis of existing financial institutions including microfinance institutions and saving and credit cooperative societies. They will be scored against a list of criteria to understand their leadership, financial position and financial products portfolio on a case-by-case basis."⁶

44. The iTAP believes this engagement will be crucial to the success of Output 3.3 on "Financial services and private sector investment engagement" in order to ensure sustainability of livelihood interventions, and urges the AE to start dialogue with potential partners as soon as possible since they will have experience and advice that can guide the way this output is approached for maximum effectiveness.

45. **Level of concessionality:** The project's proposed 100 per cent concessionality level is justified because the project is investing in multiple ecosystem goods and services that are public in nature, including landscape mosaics, at the core of which are public assets, in the form of national parks and other protected areas, as well as landscape linkages and stepping stones to promote connectivity and flood management, and steep slopes to be afforested to prevent landslides and erosion. Benefits will accrue also to privately owned small-scale family farms, but these are on a very small scale, and any profits made through processing and sale of

⁶ MOE Rwanda responses to iTAP, 4 September 2023.

agroforestry products will need to be retained at household level to buffer against shocks and stresses to increase and diversify incomes, and to improve families' nutritional status.

46. **Market transformation for cookstoves:** The project aims to avoid challenges which the feasibility study says were experienced with previous improved cookstoves (ICS) programmes in the region, which "have historically been characterized by poorly designed subsidy/incentive programs which in some instances have provided 100% subsidies to end-users" and have sometimes led to "unnecessary long-term dependence and poorly managed exits, which consequently created long-lasting ICS market distortions". Instead, the project adopts a market-oriented approach to strengthening the improved cookstove supply chain in the region, creating "strong linkages and business relationships among the key actors, including cookstove producers/youth small and medium-sized enterprises, farmers/savings groups, and microfinance institutions to ensure sustainable access and finance of cookstoves".

47. In collaboration with technical and vocational education and training institutes, youth producers, central and local authorities, and experts on biomass and micro-finance will refine and test, for each type of ICS/fuel to be promoted, the category/profile of households to be targeted and the related subsidy/microcredit schemes to be implemented to facilitate stove adoption with minimum concessionality. Subsidized cookstoves will be supplied to 4,000 poor households identified as being in a category of vulnerable groups, including youth, women, landless and disabled people. The stoves will be locally manufactured, with a competitive process leading to the award of a supply contract to local women- and youth-owned cooperatives or start-ups. A partnership with two integrated polytechnic regional colleges will enable the design of prototype affordable Tier 2 ICS customized to local conditions and customer preferences.

48. After year 4, the start-ups that will have graduated from the training and incubation programme (also conducted for the manufacture of beehives and irrigation pumps) will be fully empowered to maintain the supply of new equipment and repair of old equipment, as well as to market their products to producer organizations and households. Access to microfinance institutions will be facilitated to allow emerging cookstove business to grow. The members of village savings groups will be able to use income generated from the sale of fast-maturing crops (chili and French beans), coupled with savings accumulated from their groups, to buy new or replacement stoves over time. A specialized small business development fund will be established to support dissemination, accompanied by public awareness-raising of the benefits of ICS.

49. **Return on investment:** The project cost per direct beneficiary of USD 31 (USD 40 with co-finance) and 20-year economic rate of return of project interventions for agroforestry (49 per cent), protective forests (72 per cent), and silvopastoralism (51 per cent) all highlight the project's viability, as does the economic net present value of USD 10,586 for the improved cookstoves intervention. The mitigation cost is high, at an estimated USD 43.84 per t CO₂e, but comes with a multiplicity of adaptation benefits and socioeconomic and environmental co-benefits. The cost of restoring natural forest for providing both adaptation and mitigation benefits is significantly greater than the cost of simply planting trees for mitigation, since a complex multi-strata, multi-trophic level natural ecosystem (like a tropical rainforest) provides more varied and in many cases more robust ecosystem services than a simpler ecosystem (like a monocrop plantation). Restoration of natural forest in Rwanda is essential for sustaining the critical ecosystem services on which vulnerable rural communities and the national economy depend, and are now even more essential for buffering intensifying climate change impacts.

50. **Best practices:** The project will use Rwanda's best practice guidelines for tree nursery establishment, tree planting and tree management. For assisted natural regeneration, methods developed and proven through multi-year research in Nyungwe and Gishwati-Mukura National Parks will be used. The project will use multiple contemporary methods for monitoring the success of project interventions, as well as forest coverage and condition. Forest plantations will

be mapped and monitored using the Rwandan Government's newly developed Forest Monitoring and Evaluation Software specifically developed for this purpose. The project will develop a modern indigenous forest monitoring system, utilize the most high-resolution data sets currently available (<3 metres) and take advantage of recent remote-sensing advances, including cloud computing and machine learning. Although replication of effective and cost-effective project techniques through knowledge-sharing with other countries in Africa is not planned in the project design, this could be a by-product of the project.

51. The project's overall efficiency and effectiveness is rated as medium to high.

II. Overall remarks from the independent Technical Advisory Panel

52. The iTAP believes this cross-cutting public sector project from the Government of Rwanda is an important one that will make significant strides towards effective management and sustainable financing of forest-and-farm landscapes that play a vital role in climate change adaptation and mitigation in the Congo Nile Divide.

53. The iTAP recommends to the AE that they undertake the following actions to strengthen the measurement, reporting and verification (MRV) of the project's core mitigation benefits:

- (a) Clarify the roles and responsibilities of the various actors who will be involved in the steps of the MRV process over time, and the frequency of these steps;
- (b) Capture the MRV activities as presented in annex 22, also in annex 11, such that they become part of the project's overall monitoring and evaluation system; and
- (c) Clarify how this project will be aligned with the existing arrangements for the national MRV system and/or will support or contribute to enhancing the functioning of this system across the country.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP217)

Proposal name:	Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration
Accredited entity:	Ministry of Environment of Rwanda
Country/(ies):	Rwanda
Project/programme size:	Small

Impact potential
The iTAP assessment of the impact potential of the project as high is well acknowledged and no further comments from the AE.
Paradigm shift potential
iTAP assessment as Medium to High is acknowledged by AE. The project is going to utilize integrated landscape management and ecosystem-based adaptation approaches in addressing climate change impacts in Rwanda which offers opportunities for scaling up and replication to similar ecological zones across the country.
Sustainable development potential
iTAP assessment as High is acknowledged by the AE.
Needs of the recipient
iTAP assessment as High is acknowledged by the AE
Country ownership
iTAP assessment as High is acknowledged by the AE
Efficiency and effectiveness
iTAP assessment as Medium to High is acknowledged by the AE. The AE has received the confirmation letter regarding the co-financing from the Rwanda Development Board, Rwanda Water Resources Board and Rwanda Forestry Authority to support complementary activities of this project. The AE will report on the co-financing in line with GCF Policy on co-financing annually in APRs.
Overall remarks from the independent Technical Advisory Panel:
The AE acknowledges iTAP overall assessment and recommendation for the Board approval.

**BUILDING RESILIENCE OF VULNERABLE COMMUNITIES TO
CLIMATE VARIABILITY IN RWANDA'S CONGO NILE DIVIDE
THROUGH FOREST AND LANDSCAPE RESTORATION**

**ANNEX 8 - GENDER ANALYSIS/ASSESSMENT AND GENDER &
SOCIAL INCLUSION ACTION PLAN**

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- **EXECUTIVE SUMMARY**

Since 1994, Rwanda has made significant progress in empowering women and countering the numerous inequities and discrimination to which women were historically subjected. Today Rwanda has one of the most progressive, gender equitable policy frameworks of any country in Africa, and has established and funded the Ministry of Gender and Family Promotion and the Gender Monitoring Office to oversee the implementation of these gender progressive policies.

There is no single definition nor understanding of gender equity and women's empowerment, and the meaning, and reality of these terms differs across communities and cultures and over time. That said, the Government of Rwanda holds the simple belief in the fundamental equality of all people. In a just society, men and women of all ages, at a minimum, should have the same rights to self-determination, should be equally represented at all levels of political leadership, and should not be subject to discrimination or gender-based violence.

Today rural women continue to play a vital role in ensuring the wellbeing and security of their families. But being the caretaker of the family is getting harder for women as farmland becomes increasingly scarce, as soil erosion and declining soil fertility lowers crop yields, and as fuelwood, fodder and clean water take ever longer to find and carry home. Changes in rainfall timing and intensity, driven by climate change, will further exacerbate the challenges women will face to meet the needs of their families.

Women, youth and marginalized groups are the primary targeted beneficiaries of the Green Climate Fund (GCF) proposal "Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide Through Forest and Landscape Restoration." Empowering women, youth and marginalized groups to play an equal role in land and resource use decisions will ensure that their interests and concerns are incorporated into plans designed to increase the wellbeing and security of families in the Congo Nile Divide (CND) by helping them to evolve their livelihood strategies and practices in ways that adapt to climate change.

Investing in women's lives and strengthening their capacity to overcome existing and looming challenges will do much to implement Rwanda's progressive gender policies and help reduce the continuing inequities and discrimination that rural women, in particular, still face in their daily lives. It will also reduce poverty, the incidence of child failure to thrive, and food and income insecurity within families across the CND. By ensuring a robust integration of women and youth in programming, the resilience of households in the Congo-Nile Divide will be strengthened.

The following summarizes the primary desired outcomes for women and youth that will be generated by implementing the four components of the GCF program in the CND.

Component 1: Mainstreaming Climate Adaptation into Integrated Land Use Planning

- Outcome 1: Climate adaptation programs in LUPs reflect and address the priorities of women and youth
- *Component 2: Forest Landscape Management and Restoration*
- Outcome 2: Women, youth and marginalized households have improved capacity to engage in forest management and restoration
- *Component 3: Enhancing climate adaptation through resilient livelihoods*
- Outcome 3: Women, youth and marginalized households have improved capacity to engage in green jobs

Component 4: Monitoring & Evaluation

- Outcome 4: Disaggregated and specialized monitoring generates data showing whether and how women, youth and marginalized people are benefiting from adaptive management systems

- **INTRODUCTION**

Decades of project design, implementation and impact evaluation have demonstrated conclusively that gender equality and women's empowerment are necessary conditions for effective investment and action to achieve environmental conservation, economic development and climate change resilience. This program seeks to ensure that women play equal roles in determining how to improve the wellbeing and security of their families and communities in ways that maintain the agro-ecological systems, biodiversity, and environmental services essential to the economies, nutrition and safety of their families in the Congo-Nile Divide (CND) now and in the future.

Our approach goes beyond trying merely to ensure that women are no longer ignored or marginalized but seeks to empower women so that they can decide how they can best engage in determining what support would best help them and their families to increase food and income security and resilience to climate shocks. To do this we need to complete a thorough gender analysis that identifies the differences in men and women's lives, including those that lead to sociocultural and economic inequalities, and applies this understanding to project development.

This gender analysis attempts to answer four questions:

1. What roles do men and women play in determining wellbeing objectives and the allocation of labor and capital, and how do these differ in male versus female headed households?
2. What are the inequalities, discriminations and rights denials faced by men and women in different social, economic and political contexts? How do these issues intersect with other factors such as age, ethnicity, disability, wealth class, etc.?
3. How might gender inequity impact on the effectiveness and sustainability of the project investments, actions and desired outcomes?
4. What are the risks that project efforts to reduce gender inequality by empowering women, actually exacerbates inequities and further undermines the status, and agency of women?

The purpose of this gender analysis is to ensure that the project proposal design and implementation will be informed by a thorough understanding of gender roles, power relations and dynamics. This assessment provides information to address the relevant environmental and climate change critical issues from a gender perspective. Information generated during the analysis will ensure that sufficient and appropriate attention is paid to gender issues within all CND project interventions.

Furthermore, these gender inclusion methods fall in line with the GCF's gender mainstreaming guidelines, designed to involve women, youth, and the historically marginalized in all climate adaptation, mitigation, and resilience strategies. The gender assessment will provide the foundation for a credible and practical gender action plan to be implemented by the GCF CND project.

1.1 Purpose of the Gender Analysis

Increased greenhouse gas emissions linked to human activity continue to threaten the livelihoods,

safety, and wellbeing of communities worldwide. However, climatic impacts will vary based on numerous factors, including geography, socioeconomic status, age, and gender. Women experience unequal access to resources, land, and power, reducing their ability to adapt to climate change as effectively as men. Women face greater domestic responsibility to maintain the home and raise children, reducing their time for education and careers. As a result, women are more tied to their land and more vulnerable to climatic impacts, yet also represent a potential group of leaders to spur mitigation and adaptation efforts.

Given the potential of women as environmental pioneers, climate resiliency efforts require a thorough understanding of gender roles in the region. MoE recognize the inextricable link between sustainable development of communities through gender planning and the conservation of wild places. Effective conservation strategy requires working to balance gender inequity and providing opportunities for women in the conservation space. Furthermore, the high dependence of women on agriculture as livelihood in Rwanda impacts both conservation and climate change mitigation efforts. Smart conservation programs therefore target agricultural interventions to aid marginalized groups such as women and youth. Such projects offer employment opportunities and reduce the threat of illegal or paramilitary activities that threaten conservation outcomes.

Shifting the imbalance of power and access for women is not only a conservation target, but a human rights issue. Sustainable development at all levels, local, national, and international, requires the empowerment of women. When women receive higher education, they have more flexibility in careers and are more likely to escape poverty. These women strengthen economies through diversified incomes, marry and have babies later, and have greater independence (NISR, 2018). However, climate change is likely to worsen financial stressors, particularly in rural communities, and increase the risk of gender based violence (GBV). As the frequency of extreme weather events increases, understanding how climate change impacts women is essential to combatting the human rights issue of gender equality.

There are numerous international human rights agreements that provide a framework for gender equity. Rwanda has signed on to several of these commitments, including the Convention on Elimination of all Forms of Violence against Women (CEDAW), the Maputo Protocol on the Rights of Women in Africa, and the Convention on Slavery and Repression of Human Trafficking (IUCN, 2021). Participating in these commitments requires periodic progress reporting and helps hold countries accountable.

In Rwanda, the government has made significant strides to address gender disparity following the Genocide. Thirty-four percent of households became run by widows post-Genocide, demonstrating an obvious need for the Rwandan government to address the newfound elevated standing of women (Vanhees, 2014). Several laws help ensure that on paper, Rwandan women have increased access and protection under the law. Such policies include the 1999 Inheritance and Marital Property Law (Law/n^o 22/99 of 12/11/1999), the 2008 Girls' Education Policy, and the 2010 National Gender Policy (IUCN, 2021). However, these policies are new and compete with a long history of patriarchal norms in Rwandan society, particularly outside of urban areas. Practice of these policies remains inconsistent, especially when women do not know their rights. Helping men and women understand gender policy is the first step to shifting cultural perceptions. But gender equality cannot be achieved by women alone, it also requires the active participation of men to accept the deconstruction of traditional gender roles and be aware of the intersectionality of women and other groups. For example, considering how to include youth and marginalized communities, especially the women in those groups, in equality strategies will

help lift up the most at-risk groups and reduce illegal activity.

1.2 Methodology

This Gender Assessment uses a range of primary (24 focus groups and 18 key informant interviews) and secondary (published reports, policies and academic literature) information sources to triangulate our understanding of the gender context for climate resilience and the project design.

Key Informant Interviews (KII) were conducted, using purposive sampling, with senior decision makers in governmental organizations at local, district and national levels. Results from these interviews were validated and enriched by 24 Focus Groups (FG) with 58 men and 91 women of different age groups within 21 communities bordering Nyungwe, Gishwati and Volcanoes national parks. Originally a large number of consultations was envisioned, but because of COVID constraints, including limitations on travel between districts, a smaller sample was completed. Additional stakeholder consultations will be conducted at national, district and project site level to inform the implementation of the project and to adapt activity implementation to the local context. These consultations will include the review of Environmental and Social Frameworks included in Annex 6. A desk study of secondary data sources gleaned from government reports and published policies, grey literature produced by civil society groups, and peer-reviewed academic papers has also fed into the report.

The following desired outcomes helped guide our interviews, the desk study and the analysis of all information garnered:

- Gender roles, needs, concerns and aspirations in the context of the proposed project are well understood and documented;
- Root causes of existing gender inequalities in the context of the proposed project are identified;
- Proven options for redressing traditional power imbalances between men and women are identified;
- Increased confidence that gender focused investments and actions proposed by the project will deliver the desired results.

The information generated from the gender analysis and assessment will be considered in all stages of the CND project cycle: design, work planning, implementation, and monitoring and evaluation. In each of these stages, managers will both be enabled and encouraged to view the project through a 'gender lens' so that project investments and actions:

- Reconcile gender inequalities that pre-date and might emerge from the project;
- Ensure the differential needs and concerns of women and men are addressed;
- Ensure women and men have equal access to resources, services, and human capital development;
- Ensure equal participation of women and men in management arrangements and as beneficiaries, partners and key stakeholders; and
- Ensure women's equal participation in decision-making processes.

• SITUATION ANALYSIS

1.3 Policy framework

The implementation of gender focused policy and legislation after 1994 has provided Rwanda with a strong legal framework for advancing women's status and rights. The high prevalence of female and orphan headed households, as a result of the Genocide, required the Rwandan government to reassess the country's property ownership laws. The 1999 Inheritance and Marital Property Law provided women with the right to own and inherit property, protecting widows and orphans from losing their land to male family members (Vanhees, 2014). However, this law only applies to formally registered, monogamous unions, and the children of such unions, denying inheritance rights to women engaged in consensual cohabitation or polygamous relationships and illegitimate children. Full equal rights to land titles and use of land for women were established by Law N° 43/2013 of 16/06/2013 Governing Land in Rwanda (IUCN, 2021). Furthermore, while the 1999 Inheritance and Marital Property Law established equal inheritance rights for male and female children, it was not mandated that all children should inherit equal amounts of property until Law N° 27/2016 of 08/07/2016 Governing Matrimonial Regimes, Donations and Successions (IUCN, 2021).

The Constitution of the Republic of Rwanda of 2003, revised in 2015, outlawed all forms of discrimination, provided legal equality within recognized marriages, and set a minimum of 30% female participation in the Senate and other public governance (East African Community, 2009). These high gender equality standards have helped propel Rwanda to both become the first country with a majority of female representation in Parliament and rank 7th out of 156 countries for the smallest gender gap (World Economic Forum, 2021). The 2010 National Gender Policy functioned to further advance gender mainstreaming and equality, confirming the Rwandan government's stance on gender as a cross-cutting issue that must be addressed for successful development (Republic of Rwanda, 2019).

This foundational principle of gender equality as a driving factor for growth is highlighted in Rwanda Vision 2020 and Vision 2050, which have been used as frameworks for Rwanda's development (Republic of Rwanda, 2000; Republic of Rwanda, 2020). While the primary objectives of these frameworks were to elevate Rwanda to a middle-income country by 2020 and a high-income country by 2050, these goals could not be accomplished without addressing gender inequality. Strategies for mitigating gender inequality included updating existing laws to account for gender aspects, increasing women's representation in decision-making positions, developing gender training programs, increasing women's involvement in associations and cooperatives, and increasing women's access to resources, such as education (IUCN, 2021). The 2008 Girls' Education Policy further aimed to increase girls' access to education by promoting elimination of gender disparities through the development of a 12 year basic education system, a school feeding program, separate girl's bathrooms, and increased infrastructure (Republic of Rwanda, 2019). Separate girl's bathrooms were introduced to limit gender-based violence (GBV) in schools, while the 2011 National Policy against Gender Based Violence aimed to progressively eliminate GBV through the development of a preventive, protective, supportive, and transformational environment (Republic of Rwanda, 2019).

To provide oversight and coordination of the gender policies, the Rwandan government established a number of gender machinery institutions. The Ministry of Gender and Family Promotion (MIGEPROF) implements national gender policy and ensures gender mainstreaming (IUCN, 2021). The main responsibilities of the Ministry are to implement and assess policies and provide resources for programs. The Gender Monitoring Office (GMO) implements gender audits in the public and private sector, and the

National Women's Council (NWC) advocates for women's rights and recruits female participation in development programs (IUCN, 2021). The National Gender Cluster is a forum of representatives from private sector, government, and development groups that meets to discuss the prioritization and coordination of gender equity efforts (IUCN, 2021). The Rwanda Women Parliamentarians oversee legal issues related to gender and help create new gender policy. Together, these groups work to ensure gender policy is not only implemented, but practiced in Rwanda.

At the international level, Rwanda has ratified several humanitarian agreements in support of gender equality. The Convention on Elimination of all Forms of Violence against Women (CEDAW) is an international bill of rights for women. All signatory countries are evaluated for national issues that may conflict with the agreement. In Rwanda, GBV and lower rates of education among women are areas for concern (UN Human Rights, 2017). Almost every country in Africa has committed to the African Charter on human and peoples' rights on the rights of women (ACHPR), which provides a continental framework for human rights. Most recently, Rwanda prioritized several of the United Nations Sustainable Development Goals, namely Quality Education (4), Gender Equality (5) and Reduced Inequalities (10) (United Nations Rwanda, 2021). These international agreements provide Rwanda with a level of accountability to implement and practice national gender policies.

While the Rwandan government has made significant progress to advance women's rights under the law, a gap remains for women not engaged in formal marriages. Approximately 30% of women in Rwanda participate in consensual partnerships but are not legally married, or are in polygamous marriages; neither of which are recognized under the law (Abbott & Malunda, 2016). This creates conflicts regarding land inheritance for women in these situations. Polygamous marriages do take place in Rwanda, and the status of wives is fluid; inheritance (without official marriage) can change based on seniority or who had children in the partnership (Vanhees, 2014). However, the 2008 Gender Based Violence Law does assert that if a person in a polygamous relationship wants to officially marry one of the partners, common belongings must be divided equally between the other spouses (Vanhees, 2014). This law does establish a precedent to address the rights of polygamous women, but remains insufficient. The Rwandan government sought to increase legal protections for women by hosting group weddings in rural areas, but further legislation to recognize women's rights as human rights, regardless of marital status, are necessary (IUCN, 2021).

1.4 National and international legal framework for gender equality

The Government of Rwanda, State of Gender Equity Report (2019) provides a comprehensive summary of constitutional provisions, and national laws that support the principle of gender equality and women's empowerment in Rwanda. Information for the legislative and policy review was drawn primarily from this report. Additional information was gleaned from Rwanda Vision 2020, the National Policy against Gender Based Violence (2011) and the National Gender Policy (2010).

Rwanda is internationally recognized as a world leader in promoting gender equality principles and women's empowerment. In the aftermath of the 1994 genocide, the Government undertook radical and far-reaching reforms to address the political, social, legal, and economic status of women. Legal reforms to give women property rights, and to enable them to inherit property, including land, were especially important. The Constitution, adopted in 2003 and amended in 2015, proactively promotes gender equality. It outlaws any form of gender discrimination, and enshrines the principle of equality within

marriage.

With 53% of the Rwanda population being female, the Vision 2020 national government strategy emphasizes that Gender equality will be one of the driving factors towards achieving rapid growth and sustainable development. Vision 2020 commits to continuously update and adapt laws on gender, strategies for an increased access to productive resources by women, representation in decision-making positions and apply positive discrimination in favour of women.

The National Gender Policy (NGP) of 2010 formulated to support government programs in various sectors that are directly aimed at addressing gender inequalities and women's rights. The NGP seeks to free Rwandan society from all forms of gender based discrimination and create an environment where both men and women equally contribute to and benefit from the national development goals.

The main goal of the NGP is to contribute to reducing gender inequalities in all sectors, as a key component of sustainable development. The NGP was informed by the Economic Development and Poverty Reduction Strategy (EDPRS I) of 2007 It helped to shape theThe Agriculture Gender Strategy (2010), the EDPRS II (2013), the National Strategy for Transformation (2017-2024), the Strategic Plan for the Transformation of Agriculture (PSTA3), the National Information and Communication Technology in Rural Agriculture (ICTRAg) Strategy (2016-2022), the National Policy against Gender- Based Violence (2011), the National Strategy on Climate Change and Low-Carbon Development (2011) and several others.

Constitutional provisions and gender

The Constitution of Rwanda of 4 June 2003, as amended on December 24, 2015, affirms the fundamental rights of all citizens of Rwanda, consistent with the United Nations Declaration of Human Rights and other human rights instruments. According to Article 11 of the Constitution: "All Rwandans are born and remain free and equal in rights and duties. Discrimination of whatever kind based on, inter alia, ethnic origin, tribe, clan, color, sex, region, social origin, religion or faith, opinion, economic status, culture, language, social status, physical or mental disability or any other form of discrimination is prohibited and punishable by law." Article 16 further enshrines the principle of gender equality, and the Government has committed to establishing equity and equality at all levels of society.

Key provisions within the constitution that relate to gender equality include the following:

- The preamble reaffirms Rwanda's adherence to human rights conventions, including the UN Convention on the Elimination of All Forms of Discrimination against Women (CEDAW 1979) and declares Rwanda's commitment to ensure equal rights between women and men;
- Articles 11 and 16 outlaw any form of gender discrimination;
- Articles 9, 76, and 82 mandate a minimum quota of 30% female representation in the Senate and other areas of public governance;
- Article 37 prohibits discrimination in employment; and
- Article 26 enshrines equality within monogamous marriage.

Since 1994, the Government of Rwanda has enacted a series of gender progressive policies. These include the following three key laws:

Gender equity in the right to inherit property

Enacted in 1999 Law n°22/99 (supplements book one of the civil code regarding succession) and revised

in 2016 (Law N°27/2016 of 08/07/2016 Governing Matrimonial Regimes, Donations and Successions) provides that both male and female children have the rights to inherit their parents property. This in terms of that law obviates customary practices where only sons inherit land and tangible assets.

Gender equity in the right to own land

Enacted in 2013, Law N° 43/2013 OF 16/06/2013 Governing Land in Rwanda, ensures that all women have equal rights to land titles registered in their name. This law enables women to secure loans from financial institutions using their land as collateral, which in turn removes a significant barrier from women engaging in commercial enterprise that require capital investment.

Gender-based violence

Enacted in 2008, Law n° 59/2008 of 10/09/2008 on Prevention and Punishment of Gender-Based Violence makes explicit that gender-based violence is a crime that is punishable under Rwanda law.

At the international level, Rwanda is party to key international instruments addressing women's and children's rights, including:

- The Convention on the Elimination of all forms of Discrimination against Women (CEDAW).
- The Convention on Slavery and Repression of Human Trafficking and its Additional Protocol repressing and punishing the sale and trafficking of children and women.
- The UN Convention on the Rights of the Child (UNCRC) and the Optional Protocol within the UNCRC on Child Trafficking, Child Prostitution and Child Pornography.
- The African Charter on Rights and Welfare of the Child (ACRWC)
- The African Charter on Human and Peoples' Rights and the Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa.
- ILO Convention 182 on the Worst Forms of Child Labor.

The government has also ratified and/or implemented numerous international conventions and instruments. Those include:

- The Convention on Elimination of all Forms of Violence against Women (CEDAW),
- The Beijing Platform for Action, the African Charter on human and people's rights on the rights of women (ACHPR),
- The International Convention on Social and Cultural Rights (ICCPR),
- The United Nations Security Council Resolution on Peace and Security (UNSCR 1325 and 1820) and
- The UN declaration on human rights (UDHR).

Lastly, the Government of Rwanda is committed to delivering on the 17 Sustainable Development Goals (SDGs) which includes gender equality. To fast track attainment of the SDGs, the Government of Rwanda argues that policies must be in place and effectively implemented to ensure that both women and men equally participate in and benefit from development investments and processes.

1.5 Institutional framework for implementing gender policies in Rwanda

Key efforts to implement the Government of Rwanda's gender progressive policies are included in the remit of the Agriculture, Justice, Health, Education, and Local Government ministries and their

executing agencies.

To help coordinate and oversee Rwanda's gender equality and women's empowerment efforts and ensure their compliance with national policies and regulations, Rwanda has established four key governmental and civil society entities. These include the Ministry of Gender and Family Promotion (MIGEPROF), the Gender Monitoring Office (GMO), the National Women's Council (NWC) and Rwanda Women Parliamentarians (RWPF/FFRP).

Ministry of Gender and Family Promotion (MIGEPROF)

The Ministry responsible for Gender has the mandate of coordinating the implementation of the national gender policy and advocating on gender issues at local to national levels. MIGEPROF formulates regulations and its staff support gender mainstreaming across the Government of Rwanda.

The Gender Monitoring Office (GMO)

The GMO documents and reports the extent to which a "gender lens" influences the investments and actions of public agencies and private organizations. The GMO undertakes monitoring and evaluation to assess the implementation and social impact of public and private sector implementation of activities that seek to comply with Government gender policies.

The National Women's Council (NWC)

The National Women's Council is a local to national organization of women that advocates for women's rights and the promotion of gender equality. Its members help mobilize women across the nation at all social, economic and political levels to ensure that their voices are heard and acted on in development and conservation programmes and activities. The National Women's Council provides a formal structure to give voice to women and through which women can raise ideas and concerns to inform policy. Lack of financial resources remains a constraint to the active engagement of village women in this process.

Rwanda Women Parliamentarians (RWPF/FFRP)

The RWPF/FFRP works to build the capacity of women Parliamentarians to navigate the halls of government successfully, to represent women's voices in the formulation of all policies not just those related to gender equality, and to hold the parliament and its members accountable for redressing historical inequities in the rights of women and the power that they wield in determining the future of all Rwandans.

The National Gender Cluster

This is a Government convened forum in which the Government of Rwanda, development partners, the Private Sector and Civil Society meet and discuss planning, coordination and prioritization of Gender Equality interventions.

1.6 Poverty and labor force participation

The poverty rate in Rwanda averages 39% (NISR, 2018). The Western (CND) region is the poorest in the nation with a poverty rate that averages 47.7% and ranges from 33.5% in Rusizi to 69.3% in Nyamasheke. Between 2013/14 and 2016/17 (NISR, 2018) 50.5% of female headed households had been poor during that time period, compared to 48.8% of male headed households. Poverty was more prevalent in households with more people and where the household head was older, had little education,

Labor force participation is lower in women (48.2%) than in men (65.6%) 16 years and older (NISR, 2020), and unemployment is higher in women (20.3%) than in men (15.9%). Young women and men are less likely to be in the labor force and more likely to be unemployed. In 2020, women were less likely to be employees and employers, and significantly more likely to be a contributing family worker than were men.

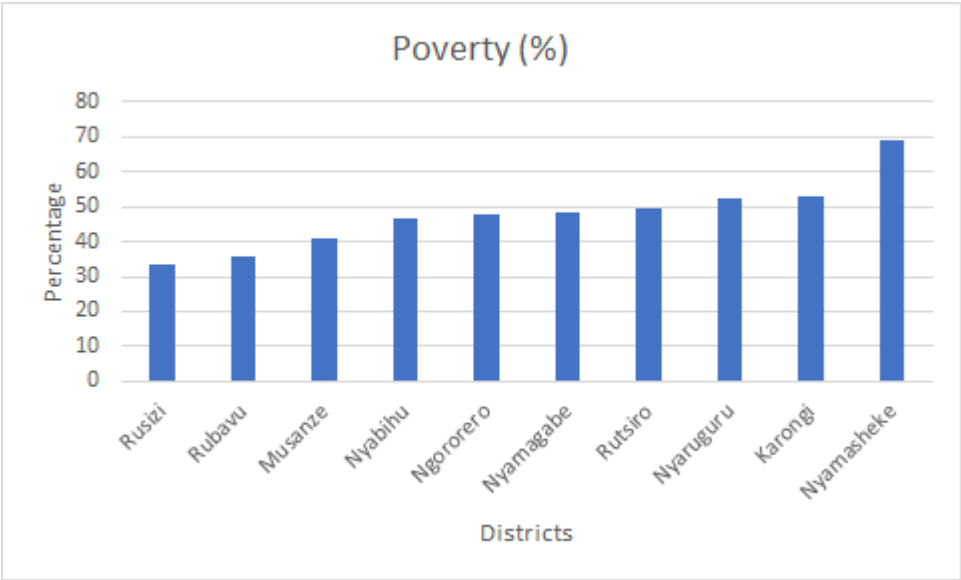


Figure 2: Poverty in districts within Congo Nile Divide (Source: EICV5 NISR, 2018)

12

Population 16 years old and over by labour force status, sex, age group, and urban/rural area, RLFS 2020								
	Total	Labour force status				Labour force participation rate (%)	Employment-population ratio (%)	Unemployment rate (%)
		Labour force	Employed	Unemployed	Outside labour force			
Total Population 16 yrs and over	7,472,601	4,212,972	3,460,860	752,112	3,259,630	56.4	46.3	17.9
16-24 yrs	2,309,913	1,080,767	828,459	252,308	1,229,146	46.8	35.9	23.3
25-34 yrs	1,874,035	1,356,005	1,098,949	257,056	518,080	72.4	58.6	19.0
35-54 yrs	2,195,647	1,427,063	1,230,818	196,245	768,584	65.0	56.1	13.8
55-64 yrs	617,306	271,530	233,426	38,104	345,775	44.0	37.8	14.0
65+ yrs	475,700	77,605	69,206	8,399	398,095	16.3	14.5	10.8
Male Pop. 16+ yrs	3,512,698	2,303,622	1,938,268	365,354	1,209,075	65.6	55.2	15.9
16-24 yrs	1,132,034	573,371	456,983	116,388	558,663	50.6	40.4	20.3
25-34 yrs	905,580	755,696	631,419	124,277	149,883	83.4	69.7	16.4
35-54 yrs	1,009,973	782,467	684,302	98,165	227,507	77.5	67.8	12.5
55-64 yrs	267,154	145,126	123,234	21,892	122,028	54.3	46.1	15.1
65+ yrs	197,957	46,963	42,331	4,632	150,994	23.7	21.4	9.9
Female Pop. 16+ yrs	3,959,903	1,909,349	1,522,592	386,757	2,050,554	48.2	38.5	20.3
16-24 yrs	1,177,879	507,396	371,476	135,920	670,483	43.1	31.5	26.8
25-34 yrs	968,455	600,309	467,530	132,779	368,146	62.0	48.3	22.1
35-54 yrs	1,185,674	644,597	546,517	98,080	541,077	54.4	46.1	15.2
55-64 yrs	350,152	126,405	110,193	16,212	223,747	36.1	31.5	12.8
65+ yrs	277,743	30,642	26,875	3,767	247,100	11.0	9.7	12.3
Urban Pop. 16+ yrs	1,581,477	1,060,077	867,565	192,512	521,399	67.0	54.9	18.2
16-24 yrs	501,826	248,767	191,104	57,663	253,060	49.6	38.1	23.2
25-34 yrs	493,474	399,957	321,737	78,220	93,516	81.0	65.2	19.6
35-54 yrs	445,849	357,316	310,082	47,234	88,532	80.1	69.5	13.2
55-64 yrs	78,623	42,609	34,597	8,012	36,014	54.2	44.0	18.8
65+ yrs	61,705	11,428	10,045	1,383	50,277	18.5	16.3	12.1
						0.0	0.0	0.0
Rural Pop. 16+ yrs	5,891,125	3,152,894	2,593,295	559,599	2,738,231	53.5	44.0	17.7
16-24 yrs	1,808,087	832,000	637,355	194,645	976,086	46.0	35.3	23.4
25-34 yrs	1,380,561	956,048	777,212	178,836	424,513	69.3	56.3	18.7
35-54 yrs	1,749,799	1,069,747	920,736	149,011	680,052	61.1	52.6	13.9
55-64 yrs	538,683	228,922	198,830	30,092	309,761	42.5	36.9	13.1
65+ yrs	413,995	66,177	59,162	7,015	347,818	16.0	14.3	10.6

Table 1: Population 16 years old and over by labour force status, sex, age group, and urban/rural area (Source: NISR Labor force survey report 2020)

Employed population by sex, status in employment, and urban/rural area, RLFS 2020			
	Total	Sex	
		Male	Female
Total employed population	3,460,860	56%	44%
Employee, Paid apprentice/intern	2,292,976	57%	43%
Employer	46,221	65%	35%
Own-account worker	968,858	57%	43%
Member of cooperative	5,792	64%	36%
Contributing family worker	147,013	22%	78%

Table 2: Employed population by sex, status in employment, and urban/rural area (Source: NISR Labor force survey report 2020)

1.7 Beliefs, perceptions and cultural norms

Pre-1994 Rwandan families and society in general were characterized by a patriarchal social structure that translated into male dominance and female subordination (Mukabera, 2016). In principle, in traditional Rwandan society, women's responsibilities included raising and educating children, welcoming visitors, managing the household, advising their husbands and maintaining traditions (Mukabera, 2016). In practice, though women were allowed substantial autonomy in their roles as child bearers and food producers men remained the ultimate arbiters of most family decisions.

Traditionally, the Rwandan woman was a symbol of fertility and weakness, while the man was a symbol of strength and protection who makes all major decisions. Rwandan women were valued as child bearers and more specifically the number of boys they gave birth to (Mukabera, 2016). Boys were taught to defend the interests of the family. It was their job to build or repair a house, to engage in activities related to cattle, cut firewood, go to battle, milk the cows, make decisions on farm management, and generate income for the family. Girls were raised to help their mothers with household chores, raise children, and most importantly to be modest, reserved, silent, obedient, and present a submissive attitude toward men (Mukabera, 2016).

These traditions effectively excluded women from accumulating material wealth and participating in economic and political decision making. The Family Code of 1992 precluded women from owning land. If a male head of household died, property passed to male heirs or to the man's brothers (Uwineza, 2009). In the case of divorce or the husband's death, a woman had no claim over the family property if she had not borne children; she would be forced to return to her birth family. Likewise, 'the Commercial Code', dating back to the colonial period, required the written consent of the husband whenever a wife wanted to engage in commercial activities, take up employment outside the household, or obtain credit, or take legal action (Mukabera, 2016).

Prior to 1994 men monopolized the public and political arena. Despite this, there have been some powerful women throughout Rwandan history, particularly the Queen Mothers who held substantial influence as advisers to the King in pre-colonial Rwandan society (Uwineza, 2009). In the customary system of justice, known as *gacaca*, women were not allowed to participate directly in public deliberations. Rather, they were expected to play an indirect role lobbying their husbands (Mukabera, 2016). Women could and did hold powerful positions as high priestesses within the religious realm, but this was undermined with colonization and the introduction of christianity.

Post-1994, the Rwandan government has taken large strides in improving gender equality and women's rights policies, however these policies have yet to reach their full potential in making substantial changes in women's lives. This may be due to gaps in policy and a dearth of resources applied to ensure proper functioning of the legislation and gender machinery institutions. There is a lack of capacity-building strategies to develop gender machinery institution staff's gender based knowledge and skills, and these staff include only a limited number of gender experts that rarely occupy decision-making positions (IUCN, 2021). Strategies to mainstream gender in capacity-building strategies of these institutions are not comprehensive, which contributes to functional issues such as lack of trained staff, staff retention measures, proper orientation for new hires, and clear handover procedures. This suggests that although the Rwandan government has passed strong gender-based legislation, implementation is limited by lack of concrete guidelines and limited resources.

While setbacks to establishing fully functional gender machinery institutions limit the effective execution

of gender equality laws, the current cultural norms and belief systems in place in Rwandan society profoundly diminish the positive effects that these laws have on women's daily lives. Discrimination against women continues to exist in multiple aspects of Rwandan society including economic opportunities, agriculture, politics, and division of labor. The long history of patriarchal dominance in Rwanda has led to the development of robust cultural norms, many of which define men's and women's roles in society, limiting women from taking advantage of the full range of opportunities within their legal rights. Other societal factors, such as the high prevalence of GBV, further limit women's ability to venture outside of their perceived household roles. In a recent study 52% of women surveyed in districts within the CND agreed that men were justified in beating them if they: burned the food, argued with him, went out without telling him, neglected the children, or refused to have sex with him (Government of Rwanda, 2016). Today, male and female policymakers are quick to acknowledge that many aspects of traditional Rwandan culture repressed women and abrogated their human rights (Uwineza, 2009).

1.8 Gender roles, responsibilities and labor allocation

Rwandan cultural norms divide household duties into activities that fall under either men's or women's responsibilities. Men are considered the breadwinners of the family, and engage in significantly more off-farm and wage employment than women (Republic of Rwanda, 2019). Men also hold domain over any valuable property or resources that the family owns or acquires, including cash crops, livestock, and valuable trees (IUCN, 2021; Kiyani et al., 2017). Men possess the majority of the decision-making power within the family, and are largely able to engage in any activities they desire.

Women's role in society is predominantly seen as being a mother and caretaker of both the family and the home. Primary duties include caring for children and elderly, cooking, growing food for the family, and providing labor on their husband's land. While the law allows for women to engage in other endeavors outside of their traditional duties, several factors often limit them from diverting from the cultural norm. The dominant barrier for women is lack of time, as the majority of their time is spent fulfilling their household responsibilities (IUCN, 2021). Women spend anywhere from 1 to 8 hours each day at their cookstove, not including the time it takes to collect the food, water, and fuel wood required for cooking (Campbell et al., 2012). Additionally, women are responsible for growing and tending to subsistence crops for the family, and are often required to take on other informal jobs on their husband's farms. These duties, on top of childcare, take a large toll on women's available time, preventing many from engaging in wage labor, training, and other forms of off-farm activities (Republic of Rwanda, 2019).

Innovations that reduce the time spent on these tasks, such as improved cookstoves and agricultural technologies, may increase women's ability to engage in activities outside of their traditional roles. Furthermore, male involvement in work traditionally delegated to women, such as childcare, would allow women to spend more time outside of the household. However, men are not expected, and are often unwilling to engage in what they perceive as women's work (Abbott & Malunda, 2016). As a result, some Rwandan women believe that government promotion of women's equality has increased their workload, as they are now expected to be bringing home money on top of their childcare, domestic, and farming duties (Abbott & Malunda, 2016). Furthermore, any additional income earned by women outside of traditional female tasks may be taken by the husband or result in increased domestic violence due to a perceived emasculation of the male role as breadwinner.

Violence is an issue for both men and women (30% vs 37%, respectively), but women are more prone to being victims, particularly sexual and gender-based violence (SGBV) (6% for men vs 23% for women,

DHS Survey, 2021). Although there are strong laws and policies against SGVB, trends of violence are not decreasing. Factors that impact higher rates of violence include lower education, earning money, household alcohol levels, and familial historical violence. In seeking support, both men and women turn to neighbors (32% vs 40%) and family (24% vs 33%), whilst some go to the local authorities (17%, 18%), police (17%, 11, ibid). Women also seek out their partners' relatives for support whilst men seek out friends. This demonstrates the importance of social networks much more than government agents.

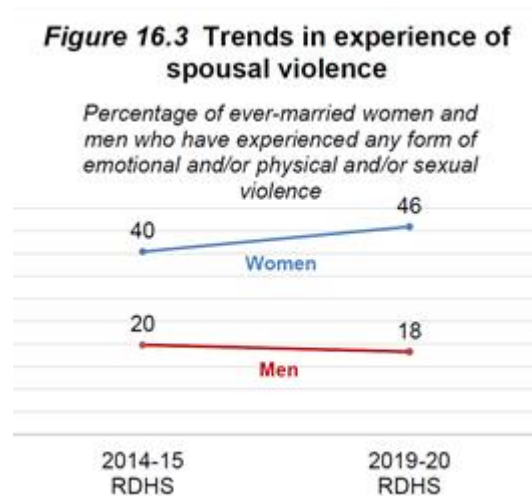


Figure 3, Trends in spousal violence over time. Source: Rwanda Demographic and Health Survey, 2021

1.9 Access to and control over capital and natural assets

The 1994 genocide left behind unprecedented distortions in social relations and exacerbated the already existing gender imbalances and inequalities. This is illustrated by the fact that today, 25% of households in Rwanda are headed by women. Women performing non traditional roles such as decision making, managing financial resources, constructing houses and many others have characterized the post genocide roles of women period in Rwanda.” (Gender and Community Development Analysis in Rwanda, EAC 2009).

Although collection and dissemination of credible data disaggregated by ethnicity is not available.¹ There has been some recognition of some ethnic minorities as Historically Marginalized People (HMPO, see Annex 6 for further details). While there is limited data based on ethnicity of HMPO, which critically impacts the baseline for monitoring activities potential impacts (positive and negative), we will refer to trends aligned with governmental policies supporting pro-poor programming, and rely on data from reports conducted by African Initiative for Mankind Progress Organization (AIMPO), Women’s Organization for Promoting Unity (WOPU) and Minority Rights Group International (MRG) leading a project titled, ‘Strengthening the Capacity of Civil Society Organizations (CSOs), Human Rights Defenders (HRDs) and Media Houses to Challenge Discrimination against HMPO and Promote their

¹ The project will be unable to disaggregate data based on traditionally marginalized people, however, all data collected will be disaggregated by sex and wealth category where possible

Human Rights in Rwanda’.

Women are likely to constitute the majority of genocide survivors which informs a need for intersectional, or the interconnected nature of social categorizations’ integration of history, gender norms, social disparity, power dynamics, rural/urban divide, and most importantly access to land rights. Despite policies and initiatives implemented by the government, other entities and projects, including this project to ensure HMPO are equitably included in project design, implementation and benefits, there are potential barriers which include: (1) Inadequate disaggregated data (2) Limited capacity and resources to analyze intersectionality (3) Limited access to decision-making (women) (4) Social inequality from increased long-term poverty of HMPO (5) Lack of promotion and protection of HMPO rights, including land tenures.

In addition, climate change is adversely impacting CND’s landscape with increasing temperatures and changing rainfall patterns contributing to ecosystem degradation and reduced livelihoods for the most vulnerable populations. The agricultural sector in Rwanda is primarily focused on subsistence farming, and most cultivated land is rain-fed, exposing farmers to significant climate-related food security risk. In addition, Rwanda’s efforts to reduce poverty are affected by high dependence on agriculture, especially when combined with climate change, declining plot sizes, and limited investment in intensification and erosion prevention. Thus, the impacts of climate change is a risk multiplier as it has exacerbated the social, cultural, and economic conditions of HMPO communities in Rwanda. In the face of the climate crisis, culturally-based climate resilient approaches should be adopted in policies along with ways to address the potential barriers listed above.

Compared to men, women are significantly less likely to own land (productive assets), they are less often engaged in income-earning activities, have less access to higher education (Govt of Rwanda, 2018), financial services (though this is rapidly becoming more equitable), health services, employment, direct payments, and non- school training opportunities. But that is changing according to the 2019 State of Gender Equity in Rwanda report. Women-headed households are more likely to use firewood (85% vs 78%), less likely to use charcoal than male-headed households (13% vs 19%), suggesting increased labor burdens for searching for fuelwood. However, women and male-headed households have the same access to water sources (EICV 5, 2018).

There are no significant differences by household’s poverty status [between male and female headed households], education or ownership of land (Abbott 2018). However, as men receive 75% of agricultural loans, and women only 25% this strongly suggests that land owned by married couples is in reality “owned” by the husband and not jointly, and the man in a married household is still culturally considered the head of the household. For married couples it is unclear whether women, on the death of their husbands, inherit their assets or are these, according to tradition, bequeathed to the sons. In addition, should women grow crops or trees, or raise livestock on marriage land do these assets belong to her or him or both. The power of women in illegal but still common polygamous marriages is even less than those in monogamous marriages.

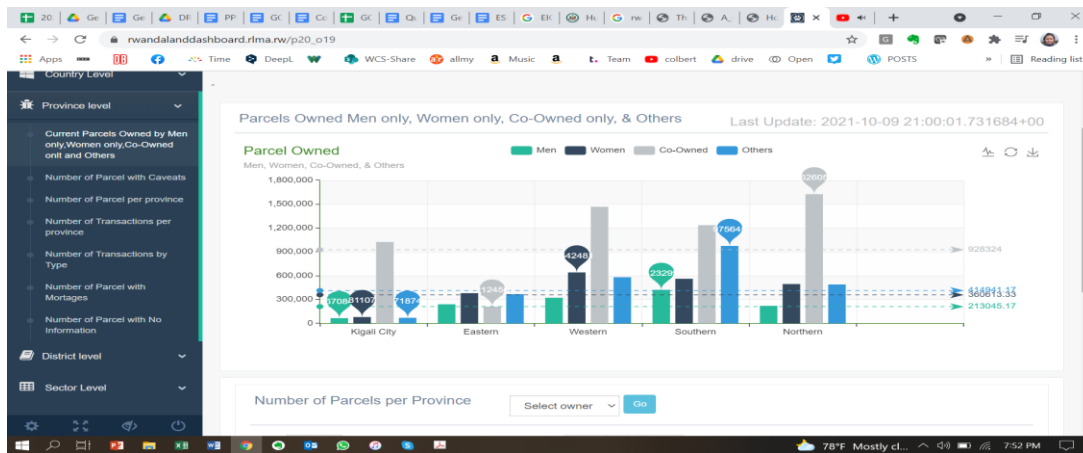


Figure 4. Land ownership by men, women, and co-owned parcels in 2021

(Source: Rwanda Land Dashboard https://rwandalanddashboard.rlma.rw/p20_o19, accessed October 10, 2021)

Though women are not the primary land or livestock owners in Rwanda they still play a key role in food production primarily to feed their family. In fact few Rwandans and almost exclusively men are involved in cash crops as land holdings are typically small and often barely enough to produce food sufficient to meet the nutritional security of the land owners' families.

Access to microloans and credit for women contributes to economic development of Rwandan society as a whole and helps women escape poverty (Mpakaniye, 2016). However, microloans remain difficult for the poor to attain for a number of reasons. Transaction fees reduce the value of already small accounts, while complicated account registration procedures, minimum balance requirements, and illiteracy all hinder access to loans, especially for women, who are more likely to be illiterate or have less experience handling money (Mpakaniye, 2016). Women typically rely on more informal loans from family because they lack collateral to access credit from a financial institution (IUCN, 2021). In 2018, of rural Rwandan women who accessed loans from a financial institution, 44.2% used saving and credits Cooperatives (SACCOs), compared to 39% of rural men (IUCN, 2021). 22% of rural men used a commercial bank, as opposed to only 16% of rural women (IUCN, 2021).

Helping women access loans from formal financial institutions will allow greater independence for women and more economic security for families. Several banks in Rwanda have already signed on to the African Guarantee Fund, aimed to reduce poverty and increase female access to loans (Mpakaniye, 2016). The fund helps those who lack collateral still receive credit (Mpakaniye, 2016). Annually, the number of microloans accessed by women in Rwanda is increasing, but additional support to enhance female financial literacy will help maintain the upward trend (Mpakaniye, 2016).

Agricultural extension services include training to innovate farming practices, and improve profitability and efficiency. The 2010 National Gender Policy (MIGEPROF), the 2018 Strategic Plan for the Transformation of Agriculture (PSTA3) and the 2010 Agriculture Gender Strategy (MINAGRI) have committed to increase extension staff in Rwanda, particularly women (IUCN, 2021). At present, women are significantly underrepresented as extension personnel, which furthers a cycle of reduced female access to extension services; 82% of extension workers are male in Rwanda (IUCN, 2021). Male extension workers are more likely to approach male farmers regarding men's crops, perpetuating the exclusion of women.

Similarly, farm promoters, village leaders who promulgate knowledge from extension services, are also dominated by men (IUCN, 2021). Women tend to be less educated than men and have greater domestic obligations that inhibit them from working with the community as promoters. Promoters are perceived as having elevated social status, leading men to more quickly jump on the opportunity than women (IUCN, 2021). By enabling more women to hold leadership positions as extension workers and farm promoters, more women can subsequently benefit from extension services, training and innovative farming practices.

This sector has vast potential for greater female involvement, but legal frameworks may be necessary to ensure more equal representation. Women represent 66% of the agricultural workforce, but only 19.7% of women in the sector occupy paid jobs (IUCN, 2021). Therefore, enabling women to become agricultural leaders through extension services and farm promoters may offer greater independence, financially and socially. Female-only extension service training may motivate women to participate and bring more attention to female crops, and establishing hiring quotas may enable more female leadership.

Women are equally likely as men to have access to a mobile phone and use mobile financial services. But are less likely to be mobile money account holders, or have access to bank services such as an ATM bank card. This is particularly true for women in rural areas, and ownership of capital goods is much less for women-headed households (see Table 3), demonstrating their marginalization across communities.

Head of household sex	Living room set	% owning at least one					Total surveyed
		Radio	TV	Computer	Mobile phone	Bicycle	
Male	19.6	78.6	11.9	3.6	71.1	16.2	2,031
Female	12.8	59.3	5.7	2.3	54.3	4.9	677

Table 3: Comparison of assets held by women and male-headed households (Data from EICV 5, 2018)

1.10 Women's health and access to health services

In 2010 life expectancy at birth was 71.13 years for women, and only 66.82 years for men (<https://www.statista.com/statistics/971106/life-expectancy-at-birth-in-rwanda-by-gender/>).

Based on the 2019/2020 Demographic and Health Survey report (NISR, 2020), the maternal mortality rate or ratio (MMR), which measures the obstetric risk associated with each live birth and is the age-standardized maternal mortality rate divided by the age-standardized fertility rate, declined from 10.7 per thousand live births in 2000, to 2.03 per 1000 in 2019-2020.

The mortality rate of children under 5 years of age continues to decline from 50 deaths per 1,000 live births in 2014 to 45 per 1,000 in 2020.

Women's health, proxied using Body Mass Index shows that on average 76% of women within CND districts are within the normal BMI range, 19% are considered obese and only 5% fall below 18.5 which is considered nutritionally compromised (Republic of Rwanda, 2016). The averages for Rwanda as a

whole are 73% normal, 21% obese, and 7% too thin. Similarly, only 18% of women exhibit signs of anemia, compared to 19% for the nation. Prevalence of malaria in children and women is low (0.5% and 0.4% respectively) compared to 2.2% and 0.6% for the country as a whole (Republic of Rwanda, 2016).

In 2017 74% of women had health insurance and 94% of women had access to community health services. The percentage of health insured households is lowest in Rubavu district (54%) and highest in Ngororero district (82%) according to the 2014-15 demographic and health survey (Government of Rwanda, 2016).

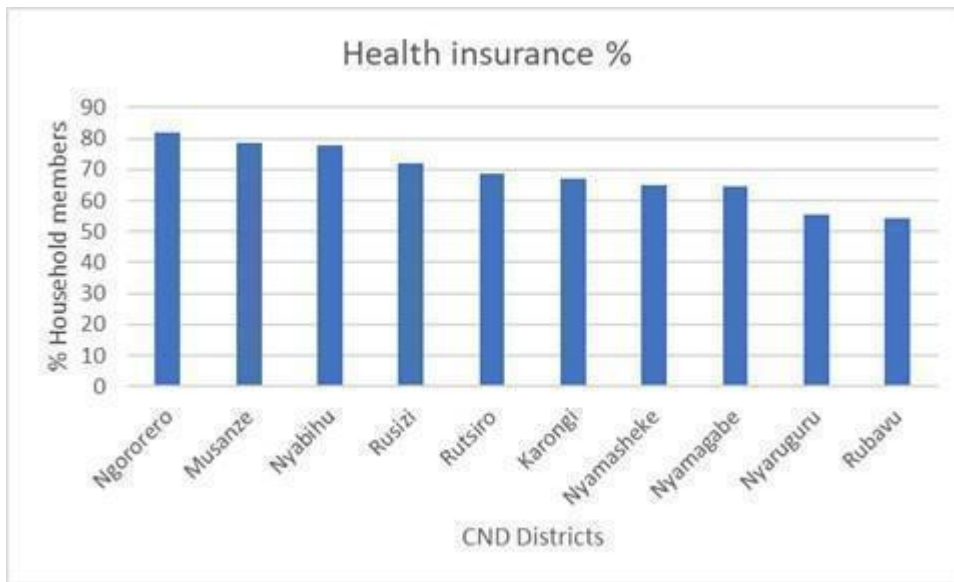


Figure 5: Percent population with health insurance by district (Source: Government of Rwanda, 2016)

Overall access to sanitation facilities is high among Rwandans (IUCN, 2021). However, due to unequal income distribution between men and women, households led by women are less likely to have sanitation access than male led households; 80.6% of female headed households have improved sanitation facilities compared to 88.0% of male headed households (IUCN, 2021). Traditional pit latrines are seen as more cost effective than modern sanitation systems, but pose risks of disease, overflow during the rainy season, groundwater contamination, mosquito breeding, and an increased risk of GBV for women due to distance from the house and night use (Tsinda et al., 2013). Lack of sanitation services poses both a serious social and environmental threat in Rwanda, particularly in informal settlements.

1.11 Women's educational attainment and access to education services

Education can lead to better health and greater access to resources (Allan et al., 2018:26). In Rwanda, the rate of people who have ever attended school is high, at 87%, but rates are higher in urban areas than rural areas (IUCN, 2021). The rates of primary education are approximately equal between boys and girls in Rwanda, and in secondary education women actually attend in higher numbers than men, 53% women to 46% men (Education Statistical Yearbooks, 2018). However, the disparity between men and women shifts with regard to technical training and tertiary education, in which men are more dominant.

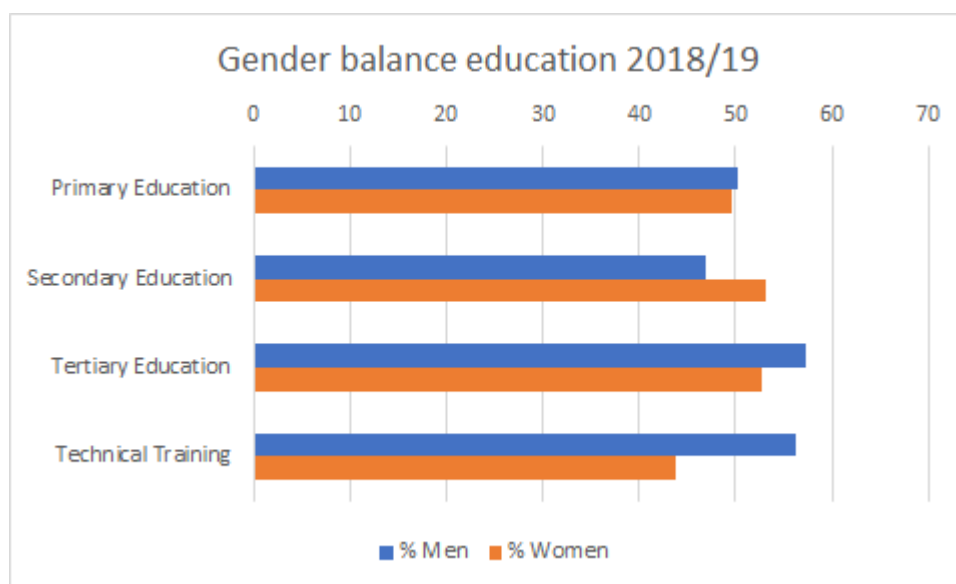


Figure 6: Gender balance in education in Rwanda (Source: State of Gender Equality in Rwanda, 2019)

In 2018, 92% of children (1.64 million boys and 1.68 million girls) were enrolled in pre-nursery to secondary education (Ministry of Education, 2018). Literacy rates continue to climb from 38.2% in 1978 to 73.2% in 2018. By this time youth literacy was at 85%. Girls and women lag behind boys and men. Only 56% of girls, 41% of women, and 28% of elderly women are literate.

Levels	Schools	Males	Females	Total	Percent
Pre-Nursery	96	3,199	3,292	6,491	0.2%
Nursery	3,210	112,044	114,662	226,706	6.3%
Primary	2,909	1,259,344	1,244,361	2,503,705	69.0%
Secondary	1,416	264,782	314,115	578,897	16.0%
TVET	360	57,643	44,842	102,485	2.8%
Tertiary	30	41,458	34,255	75,713	2.1%
Adult literacy	4,991	51,220	81,145	132,365	3.7%
Total	13,012	1,789,690	1,836,672	3,626,362	

Table 4: Levels of schooling in Rwanda (Source: Ministry of Education, 2018 Education Statistics)

The equal rates of primary education for boys and girls in Rwanda suggest a cultural value of education for all children, not only prioritization for boys. A lack of tuition fees for childhood education enables poor families to send boys and girls to school, and increasing rates of education contributes to falling illiteracy rates in Rwanda (IUCN, 2021). Higher levels of girls in secondary education may be attributed to Rwandan policies such as the 2008 Girls' Education Policy and the implementation of a school meal program (IUCN, 2021). Despite lower rates of women enrolled in tertiary education, the percentage is increasing due to flexible programming, such as online learning or night school, which enables women with domestic responsibilities to participate. According to the IUCN, the higher level of education men and women receive, the less likely they are to work in the agriculture sector (IUCN, 2021). In comparison, 92% of women with no education work in agriculture (IUCN, 2021).

Strong gender stereotypes in Rwanda still push girls into "traditional" soft trades including hairdressing, secretarial work and nursing (IUCN, 2021). 87% of women with a university degree work in the service sector (IUCN, 2021). Though women are more likely to be primary school teachers than men, they are rarely secondary, tertiary or technical/vocational teachers. In agriculture, women predominantly occupy informal roles, such as unpaid farming for the family, whereas the men typically hold higher paying positions or manage higher value products such as timber (IUCN, 2021). As a result, the majority of agroforesters are male in Rwanda (Anastase & Nahayo, 2020).

One of the greatest challenges to achieving economic stability lies in challenges in achieving quality educational attainment by female youth which leads to their lower literacy, especially in rural areas. Overall, social factors including traditional gender roles such as domestic chores and family care, historic marginalization of girls and women, patriarchal social traditions, and gender-based violence entrench girls' underperformance throughout their schooling which has a cyclical effect, as low performance in the primary leaving exam results in girls being admitted to lower quality secondary schools, and ultimately into higher education institutions (HIEs) in lower numbers (Huggins & Randell, 2007).

This results in limited access to paid labor and employment in sectors such as services or public government. Similarly, lower literacy has put them at a greater disadvantage in agriculture, as commercialization of farming has made it more difficult to feed families. While there are national laws and policies to protect employment, there is little protection for the vast majority of women who work in informal sectors which significantly affects women's rights within the country (Abbott et. al., 2008). Women in rural areas are primarily care-providers for children and the elderly, which is unpaid labor and invisibilized due to gender norms (Abbot, 2015). Thus, understanding these barriers and challenges and how to overcome them will inform interventions at project-level.

1.12 Patterns of power and decision-making

Rwanda is internationally acclaimed for being the first nation to have a majority female Parliament, with 61% of these positions held by women from 2018-2022 (Republic of Rwanda, 2019). The Constitution of the Republic of Rwanda established a minimum of 30% female representation in public governance, and this high level of female participation in upper levels of governance far exceeds this minimum (East African Community, 2009). However, female participation drastically decreases in many communities'

level governing bodies.

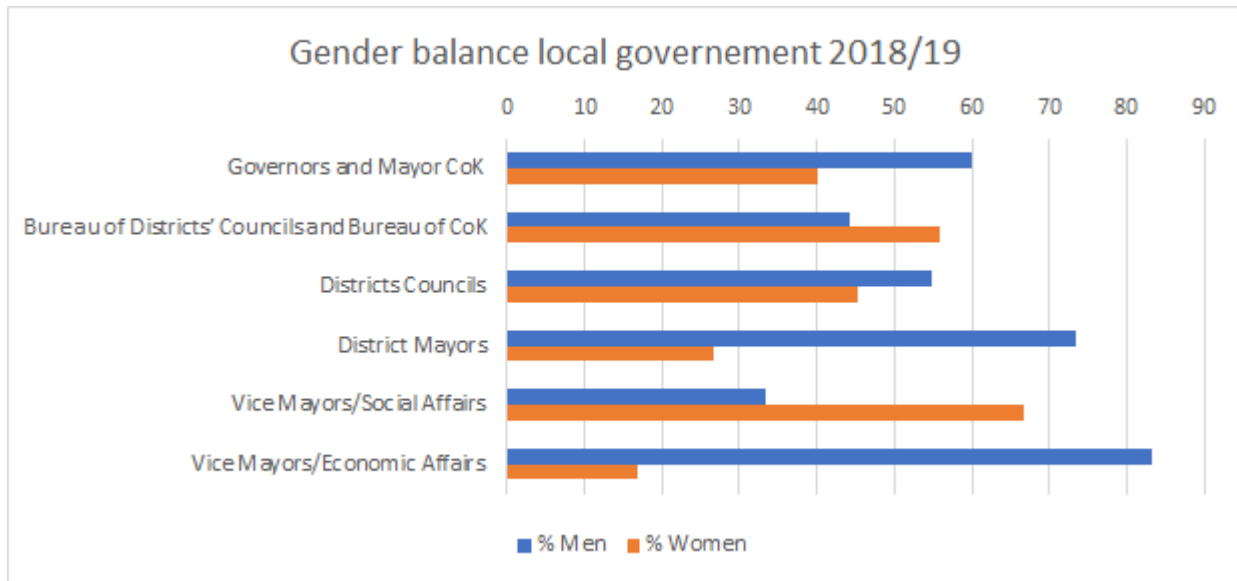


Figure 7: Female and male representation in different levels of local government in Rwanda in 2018-2019 (Republic of Rwanda, 2019)

Although overall female representation in the Rwandan government is increasing, higher levels of female representation in upper levels of governance mirror the disparity between gender equality policy and practice in the country. In higher levels of governance, these policies are implemented well and appear effective, but at community levels, larger gender gaps in prominent roles are clearly evident. Comparable gender gaps exist in central government roles that undergo recruitment processes, with many of these positions not reaching the 30% female representation minimum (Republic of Rwanda, 2019). Furthermore, female participation in the police force is extremely low, although representation is high in most other areas of the justice sector.

Similarly, in the private sector, women hold very few high level positions, particularly at the provincial level. Nationally, 30% of executive committee presidents are women, but there are 0 female presidents at the provincial level (Republic of Rwanda, 2019). Women have high levels of representation in second vice presidency in the private sector, with 70% of national and 80% of provincial second vice presidential roles occupied by women, but have low levels of representation in more powerful executive positions (Republic of Rwanda, 2019). In more local organizations, such as cooperatives, men hold the majority of the decision-making positions, while women tend to hold stereotypically female roles, such as secretary, vice president, and treasurer (IUCN, 2021).

At the household level, women have very little decision-making power. Although under the 1999 *Inheritance and Marital Property Law* women are legally allowed to own land, cultural barriers still inhibit their ability to control how this land is used. These barriers are predominantly based in the belief that men are the head of the household, and thus are the primary owners of all of the family's land and assets (IUCN, 2021). Women's lack of power within their households limits their ability to make decisions about the planting of crops and the use of land as collateral for credit, which would allow increased female engagement in other economic activities. Based on female reports, the majority of households (70%) make decisions jointly, while in 25% of households the husband makes the majority of the decisions

(IUCN, 2021). However, even in households that make joint decisions, women's decisions tend to revolve around day-to-day issues, while men's decisions are primarily focused on productive assets and income.

Women's ability to make decisions within both their own households and broader Rwandan society is further hampered by GBV. The effects of a longstanding patriarchal regime in the country has created and perpetuated the notion that men must be dominant and women must be submissive. Women break these gender norms by asserting their right to be involved in decision-making, and this can be seen as diminishing their husband's masculinity, which can result in violent retribution. In 2010, around 57% of Rwandan women reported experiencing GBV from their partner, and women who bring in more money tend to experience higher levels of GBV (RWAMREC, 2010). This suggests that increasing women's opportunities to make decisions and earn their own income will have a very limited effect on female empowerment until GBV is addressed. Engaging men and boys in gender equality advocacy and action highlights that GBV is not only a women's issue, and is believed to be successful in reducing the prevalence of GBV (UNDP, 2018).

1.13 Social service and support systems that contribute to gender empowerment, wellbeing, and security

Government mandated and running social support programs have reduced the percentage of extremely poor households from 23% and 26% in 2010 to 15% and 18% in 2017 for MHHs and FHHs respectively. Three state managed social support systems have contributed significantly to these observed reductions in poverty and increasing wellbeing and security.

The Girinka Program (One Cow per Family Program) of the Ministry of Agriculture (MINAGRI) started in 2006. By 2017 over 290,000 cows have been provided to vulnerable male and female headed households (approximately 6% of all Rwandan families) as a way to reduce child malnutrition, increase the productivity of agricultural labor, generate a new stream of income through surplus milk sales and strengthen social cohesion amongst Rwandan families and communities as cow owners pass on surplus calves to other needy families. Male headed households were, however, more likely than female headed households to receive a cow through this government program.

The largest social support program of the Government of Rwanda is the Vision 2020 Umurenge Programme (VUP) which was started in 2008 and is managed by the Local Administrative Entities Development Agency (LODA) within the Ministry of Local Government (MINALOC). VUP has four components: wage labor through public works, direct cash transfers, financial services and awareness raising. The majority of public works support agricultural production, land conservation and the maintenance of physical infrastructure. A review by FAO showed that VUP wages paid to women were often captured by men. Retaining control over VUP income was easier for women who owned an individual savings account (e.g., within a Savings and Credit Cooperative - SACCO), though most women only had accounts in Village Savings and Loan Group or informal tontines. Rural women were more disempowered than men through cultural constraints to speaking in public meetings, workloads that limited participation in community meetings and self-help associations, and limited access to credit and savings accounts. The FAO study also found that the majority of both MHHs and FHHs were unaware of the existence of a VUP GRM, or were skeptical about the effectiveness of this complaint process. The most important recommendations were to a) enable women to open their own formal bank accounts to increase their control over their incomes and strengthen their decision-making capacity, and

b) redirect households with high dependency ratios and/or low labor capacity away from public works wage labor and toward VUP direct cash transfers.

MINAGRI further supports poor rural families by offering subsidies for agricultural inputs, improved seeds and irrigation as well as offering credit through the Agriculture Guaranty Fund which provides specific opportunities for women.

● **CONCLUSIONS - ONGOING CHALLENGES TO GENDER EQUITY IN THE CONTEXT OF THE GCF CND PROJECT**

Despite tremendous efforts from the policy and legal perspectives, a patriarchal culture and persistent disparities continue to characterize gender relations in Rwanda. Disparities persist in post-primary education; in access to and control of property including land and economic resources; in employment opportunities and entrepreneurship; in decision-making at household and community levels; in family responsibilities and unpaid care work; and in the experience of violence, harassment, conflict, and insecurity. Sexual and gender-based violence persists at high levels in Rwanda. While women have made impressive political progress, especially at the national level, their progress in terms of local political representation and engagement remains unequal. Economic opportunities remain markedly gender-differentiated.

In Rwanda and in the Congo-Nile Divide, structural barriers to economic and political equity and female empowerment have significantly impeded measures to adapt to climate change impacts. In Rwanda, these structural inequalities are at a lower level compared to other countries in the world, because of leading gender-considerate policies across sectors. However, the point remains that women's ability to access, use and control natural resources, infrastructure and services differently is still low compared to men. This means the degradation of natural resources and new infrastructure will affect women and men differently, and will generally result in greater vulnerability of women. Women are vital agents of change and can be powerful leaders from the community to global level in mitigating and adapting to climate change.

1.14 Afforestation, agroforestry and tree tenure

Kiptot et al (2012) noted that an ICRAF survey conducted across countries in eastern Africa concluded that women rarely have tenure over trees unless they exclusively have tenure over the land. Moreover, they stated that "Women's rights to tree products are usually limited to products that are considered to have little or no commercial value. These products are mainly indigenous fruits....fodder and mulch." They also noted that: a) the survivorship of trees and fodder shrubs planted by men and women was effectively the same, b) low cost/input agroforestry techniques for replenishing soil fertility were attractive to women farmers given that they are both labor and capital limited. Fuelwood, fodder, green manure, and coppicing shrub production substantially increases benefits to women that cannot be captured by men. Women appear to grow trees/shrubs for domestic consumption (fuelwood, fodder, bean poles, fruit) and sell the surplus in local markets. It is unclear whether women who sell wood products from land that is owned by their husbands are able to capture that revenue stream.

Agroforestry offers multiple benefits to Rwandan farmers, in particular, women. Planting trees within farms represents a low effort pathway to high reward. To enhance climate resilience of communities, agroforestry practices sequester carbon, prevent erosion, and enhance soil quality for sustainable

farming. For women, trees provide additional income to elevate socioeconomic status and reduce the risk of GBV by eliminating the need to walk far distances for fuel wood (Anastase & Nahayo, 2020). Despite the majority of farms under female management in Rwanda, women are often excluded from agroforestry, as trees traditionally fall under male domain, women lack the necessary training and have less access to the retail market (Kiptot & Franzel, 2012). In order to better include women in agroforestry, training on forestry and marketing should be targeted to women. Tools and practices that reduce labor requirements for women, such as shorter trees (which decrease the need for male labor) or more modern processing techniques, will save women time and make the field more accessible (Kiptot & Franzel, 2012).

Afforestation and reforestation on public lands should include measures where women and youth formally have the right to capture a proportion of the value of the trees at harvest time. Assuming a harvest cycle of 15-20 years this will provide women with a long-term asset that could be used as retirement income, and/or given as gifts or bequeathed to children. Young men could use the trees as savings to be used to build or buy a house when they marry.

Investments in agroforestry on private farmlands should reflect the different interests and concerns of women and men. During focus groups women reported shortages of fuelwood, fodder and wood suitable for bean poles. They were also concerned about soil erosion on their steeply sloped farmland, and declining soil fertility and crop yields. On-farm tree planting, to address the concerns of women, should focus on fast growing, nitrogen fixing, coppicing species that can be planted in contours between rows of crops. The nutrient rich leaves of leguminous species can be used as fodder for livestock, and as green manure. Cutting the stems to ground level before planting ensures that the trees do not compete with the crops for sunlight and soil nutrients, but the tree roots that remain intact help stabilize the soil and prevent erosion caused by heavy rains. The cut stems can be dried for fuelwood or used as bean poles. Surplus fodder, fuelwood and bean poles can be sold to neighbors or at local markets. Coppicing species readily re-sprout from the roots allowing an annual harvest.

1.15 Improved cook stoves

In rural areas of Rwanda, 98% of households rely on wood as their main cooking fuel. The GoR completed a comprehensive report on Gender in the Energy Sector. The report noted that almost 60% of FHHs spent 1-3 hours per day collecting firewood compared to 36% of MHHs. The latter purchased fuelwood twice as often as FHHs who primarily rely on gathering fuelwood. Over 50% of both Rwandan FHHs and MHHs still cook using the three-stone method, and only 13.5% use efficient cookstoves. Over 50% of FHHs are unaware of the health benefits of improved cookstoves. The report argued for supporting women to transition from fuelwood to Liquid Petroleum Gas (LPG) primarily to reduce smoke inhalation and respiratory disease. The report did not address the fact that in higher altitude areas cooking fires also provide a heating source for the home. The use of clean cookstoves has major positive impacts on gender equality, by reducing household work for women and providing opportunities for women to engage in paid work.

Poor rural families, particularly FHH, are unlikely to be able to afford LPG stoves. Moreover, LPGs do not remain warm after the gas is turned off and thus do not provide a heat source for the home during the night or during cold spells. Tier 1 Canarumwe clay stoves are relatively inexpensive, substantially reduce woodfuel use, and provide a thermal mass that keeps radiating heat long after cooking is

completed. Though imperfect they are an improvement in efficiency compared to the 3 stones cookfire method. As men do not gather firewood nor do they cook they are often unwilling to spend “their” money to purchase an improved stove even though it substantially reduces fuelwood use and the time women must spend gathering fuelwood. Efforts to introduce improved cook stoves into MHHs in the CND must address this barrier.

1.16 Honey production and marketing

The Rwandan government has identified beekeeping as a potential source of additional revenue for farmers (Mushonga et al., 2019). The European demand for honey far outpaces the production in Rwanda, indicating a market incentive to raise bees beyond subsistence level (Mushonga et al., 2019). Furthermore, global decline of bee populations reflects a high biological need for more apiarists in Rwanda. Despite the economic and environmental benefits, along with low startup costs, beekeeping is perceived as a vocation for poor people. Out of 83,000 beekeepers in Rwanda, 86% are adult males and 71% lack primary education (Mushonga et al., 2019). To overcome the negative stigma around beekeeping, materials and training for modern hives need to be provided to women and youth. Modern hives produce a higher yield of honey, have lower risk of causing fires, and do not need to be placed as high up in trees as traditional hives, which deters some women due to cultural restrictions barring women from climbing trees (WCS, 2018). In a WCS study, 83% of beekeepers used traditional hives in Rwanda, demonstrating a need for more modern tools (WCS, 2018). Additionally, marketing training may help women to better sell the honey, as women still have relatively low market and distribution access in Rwanda. Despite the stigma, beekeeping remains a strong potential source of income, which could reduce reliance on more harmful forest livelihoods, such as deforestation and poaching (Fakhry et al., 2010).

A study by Mushonga (2019) suggests that people consider beekeeping for the poor and illiterate are largely the domain of older men no longer able to labor in agricultural production. Women do appear to be engaged in the production and marketing of honey beer. ADF provided \$343,000 in 2017 to strengthen the honey production sector in Rwanda. Production remains relatively low because of dependence on traditional hives rather than modern hives such as Kenya Top Bar or Langstroth (CODIT, 2009). The CODIT report funded by the SNV noted that in 2009, of the estimated 30,293 beekeepers in Rwanda only 23% were women. The report recommended that the GoR do away with subsidies to beekeepers and their co-operatives, focus on addressing the root causes of the prevailing problems not the effects, encouraging beekeepers to actively participate in their own development and advocating for the replication of knowledge and skills from common to individual apiaries.

During focus groups both women and marginalized groups noted that the cost of modern hives and other beekeeping equipment such as smokers was a barrier to entry as was lack of access to credit. A big concern was farmers’ use of pesticides that kill bees, making it impossible to locate apiaries near farmland.

1.17 Coronavirus (COVID-19) implications in advancing gender equity

The global coronavirus (COVID-19) pandemic left no country untouched from its impacts. While research is still emerging, the severity of the pandemic is clear; struggling economies, vast unemployment, and threats to health and wellbeing continue to affect the global community. Women in particular are already

vulnerable to socioeconomic inequity that is currently exacerbated by COVID-19. Therefore, the impact of the pandemic must be considered in gender planning into the future.

Rafaeli and Hutchinson (2020) compared the secondary impacts of COVID-19 to the Ebola crisis on women and girls in Africa, finding that women and girls are more likely to experience higher poverty and school dropout rates, an increase in unintended pregnancies, reduced income and financial empowerment, greater domestic responsibility (increased childcare due to school closures), and food insecurity due to the pandemic. In times of crisis, school closures and financial stress also lead to increased rates of gender based violence (Rafaeli & Hutchinson, 2020). Women face greater exposure to COVID through their roles as caretakers in families and in healthcare, and have less decision-making power, rendering their needs unmet during COVID-19 (Parry & Gordon, 2020). Yet few countries have enacted policies to address the gendered impacts of the pandemic.

In Rwanda, the Ministry of Gender and Family Promotion (MIGEPROF), in partnership with UN WOMEN and UNFPA, conducted a 2020 study on the gendered impacts of COVID-19. The report indicates that 44.2% of female respondents completely stopped their economic activities due to the pandemic, compared to only 17% of men (Bayisenge, Lo & Schreiner, 2020). In rural areas, the percentage was higher for women; 50% stopped their work and thus sources of income, compared to 17.7% of rural men (Bayisenge, Lo & Schreiner, 2020). As women occupy more informal and unsecured jobs in Rwanda, they are at greater risk of income loss than men.

The use of health care and maternal services in Rwanda considerably declined during the COVID-19 outbreak (Wanyana, Wong & Hakizimana, 2020). Declines in child health care services, particularly vaccinations, could be detrimental to avoiding future outbreaks of vaccine-preventable diseases (Wanyana, Wong & Hakizimana, 2020). Many women feared contracting the virus while attending regular checkups, which contributed to the decline of health services in Rwanda (Wanyana, Wong & Hakizimana, 2020).

Recommendations to mitigate the impacts of COVID-19 on women in Rwanda include (Bayisenge, Lo & Schreiner, 2020):

- Implement programs that offer employment or financial security in spite of COVID-19.
- Provide online or socially-distant courses to prevent school dropouts.
- Establish women-specific financial packages in the national economic recovery plan.
- Strengthen the mobilization of men and boys in childcare and domestic responsibilities to offset increased labor caused by the pandemic in parallel with on-going “Men engage” national programmes.
- Accelerate government interventions to address elevated levels of GBV.

1.18 Promoting gender equity within the CND GCF project

Gender equality and women’s empowerment should be considered in all stages of the CND project cycle: design, formulation, implementation, and monitoring and evaluation. In each of these stages, project managers should keep a ‘gender lens’ in mind, looking at ways the project can:

- Address gender inequalities that emerge from the project;
- Ensure the differential needs of women and men are addressed;
- Ensure women and men have equal access to resources, services, and capacity development;
- Ensure equal participation of women and men in management arrangements and as

beneficiaries, partners and key stakeholders; and

- Ensure women's equal participation in decision-making processes.

Based on key expected outcomes, this gender assessment provides the foundation for a realistic gender action plan to be implemented in the CND project.

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● GENDER, YOUTH AND SOCIAL INCLUSION ACTION PLAN

The goal of the programme is to increase climate resilience of vulnerable rural communities in the CND, reduce CO2 emissions, build capacity for integrated spatial planning and increase the extent and integrity of forest ecosystems in the CND. The programme has three components directly targeting vulnerabilities within climate impact chains. Component (1) *Mainstreaming Climate Adaptation into Integrated Land Use Planning and Management* will lead to enhanced adaptive capacity of both ecosystems and rural populations, by strengthening coordination among sectors and integrating climate resilience considerations into planning and development. Component (2) *Landscape Management and Restoration* will implement high priority actions to enhance the resilience of vulnerable communities by increasing the extent and integrity of forest ecosystems and landscapes and by reducing demand for wood fuel. In addition, component (2) is designed to reduce the time women spend collecting and preparing fuelwood, and simultaneously to empower women to provide input to local-level decision making (see also Component (1)). As a co-benefit, greenhouse gas emissions will be reduced through restoration of natural forests, agroforestry, reduced fuelwood demand and fuel efficient cookstoves under components (1 and 2) are expected. Component (3) will focus on *Enhancing climate adaptation through resilient livelihoods*.

Women, youth and marginalized groups, including women-headed households in the CND are key stakeholders in the success of this project, and are the primary target beneficiaries of the project. The project will be implemented across various sectors along the CND, particularly focusing on communities around Nyungwe National Park and Gishwati-Mukura National Park--critical for Rwanda yet highly vulnerable forests due to high population and increasing demography who depend upon fuelwood, and actions taken by poor households that illegally use forests to improve income streams.

1.19 Methods

This Action Plan operationalizes the findings from the Gender Assessment, the ESIA, and youth-based information collected during the feasibility studies and stakeholder engagement during the Project Preparation Facility (PPF) phase. The Assessment used five standard lenses (USAID 2017^[4]) to conduct the analysis:

1. Institutions, Laws and Policies that influence the context in which men and women act and make decisions.
2. Beliefs, Perceptions, and Cultural Norms that shape how men and women, youth and elders are able (or expected) to practice activities, particularly those to be developed by this GCF project.
3. Gender Roles, Responsibilities, and Time Use in reproductive, productive, and community work.
4. Access to and Control over Assets and Resources are different between young and old, men and women.
5. Patterns of Power and Decision-making focused on how individuals are able to be represented and be in decision-making positions.

Based upon this framework of five lenses, the Gender Analysis identified and analyzed the gender risks and impacts of the program and climate change through four main steps:

Scoping: The GYA included a scoping process to review relevant literature and data for the project area, including local reports and data, consult with government and other partners, and conduct field visits to understand the project, identify and characterize the project area, and identify stakeholders,

including program-affected parties and other interested parties, who may have different concerns and priorities about project and climate change impacts, mitigation mechanisms and benefits.

Legislative and policy review: The GYA conducted a review of (1) Rwanda's applicable policy framework, national laws and regulations and institutional capabilities relating to gender and youth, (2) variations in country conditions and program context across the affected area, (3) relevant gender and youth studies, (4) Rwanda's obligations and commitments under relevant international agreements and treaties, (5) applicable requirements under GCF standards and (6) internationally accepted best practices.

Stakeholder consultation and analysis: Based on the identification of stakeholders identified in the scoping exercise, the GYA conducted initial consultations with stakeholders to gather information on concerns and priorities about the project and climate change, analyze the outcomes of the consultations and develop a strategy for continuing stakeholder engagement.

Baseline analysis: Drawing on the scoping exercise, the GYA assembled and analyzed existing information. The information was analyzed in light of the stakeholder consultation and analysis and legislative and policy review, to identify information gaps and ways in which these may be filled.

This information was then used to guide the Gender Youth Social Inclusion Action Plan. The details of the stakeholders consulted, including key informant interviews (18), focus group discussions (24), and district workshops (4) can be found in Annex 7. A literature review is available in the Gender Assessment (above) and ESIA (Annex 6).

1.20 Gender Issues

There are three key gender-based themes that influenced the design of the Gender Action Plan:

a. **Change negative gender and social norms.** The most profound barrier for this project is rural norms about the roles of men and women. Whilst the Government of Rwanda has made many advances and has implemented many policies (see Section 3 of the Gender Assessment) to achieve gender equality, particularly in national-level representation and inclusion, patriarchal privilege and decision-making remains embedded in rural society (IUCN, 2021). The consequences of this for this project is far-reaching. Men are generally seen as the head of household with the decision-making powers over all assets, resulting in male appropriation of household and farm-based assets (IUCN, 2021; Kiyani et al., 2017). Though female headed households face the same issues and challenges as male headed households, they are disproportionately impacted because of prevalent gender inequality issues (GMO 2017). Men in the CND are positioned to be significantly more involved in programs that result in monetary benefits, such as fern clearing, casual labor, and plantation projects. Although under Rwanda law, sons and daughters receive equal inheritance rights (Law/n° 22/99 of 12/11/1999) the PPF focus groups highlighted that due to cultural practices, parents favor sons in land inheritance, leaving young women in a position where they depend on marrying a man to gain access to farmland; without a formal marriage, these women have no land tenure. Broadly, these gender and social norms leave most women with little decision-making power and no real control over assets. The threat of sexual and gender based violence (SGBV) at the household level where women try to assert independence and achievement is high. For this project, the implications are that trees planted by women become men's property; money gained in employment (tree planting, nursery maintenance) or through farmland production (fruit trees, honey, etc) are ultimately controlled by men; and women shouldn't participate in activities without a

husband's permission. Although this seems bleak, this situation is not universal. The 24 CND community focus groups held in the PPF highlighted that many households decide together on how money should be spent; land tenure is broadly seen as joint ownership; and men realize the impact of dwindling fuelwood resources on women and girls. Gender equity is not yet universal and rural culture has a long trajectory to achieving it.

As a result, this program will focus heavily on social and behavior change communications (SBCC) as a means to positively change perceptions towards gender equity. Community members, including community leaders and women and men community influencers specifically will be targeted for sensitization on gender equity in order to shift their attitudes and to build a conducive environment for women and HMPO to be able to participate in project activities. Women have been engaged in the development of this project to ensure time constraints, work burden and reproductive responsibilities do not impede their ability to engage in the various activities designed by the project. Women will be regularly consulted throughout project implementation to ensure they are able to actively participate in project activities and activities take place at times which are convenient for them and at locations that are safe and contextually appropriate. At project start, once communities are selected for implementation, barriers faced by women and HMPO to participate in project activities will be reviewed and solutions to overcome barriers incorporated into programming through an additional screening process outlined in Annex 6. The project ESS Coordinator and Gender Officer will provide support to ensure all project staff understand barriers faced by women and HMPO and programming works to address barriers to ensure their active participation in activities. The project will also get support from NGOs that help men on masculinity and addressing SGVB which will include targeted outreach to mobilize their participation in household and parental responsibilities to ease women's responsibilities and sensitization on the importance of fair access and control of resources, including income. The engagement of community leaders and women and men community influencers to further promote equal and fair control of resources between men and women will further mitigate the risk of men taking over control of incomes earned and targeted for women. Lastly, the project will clearly explain participant selection, intended benefits and how participants can help monitor benefits and risks and reduce risks if necessary, including reducing the risk of men taking over control of incomes targeted for women. The program will have a safe, accessible Grievance Redress Mechanism (GRM) for all project stakeholders, particularly women, to access to pursue redress as needed. The project will also collaborate with existing Isange One Stop Centers in the project area in GBV prevention and immediate assistance to victims. All project staff will be trained and project beneficiaries informed on the purpose and use of the GRM, including reporting gender-based violence. The project will ensure confidentiality of the information and identity of the complainant. The grievance process will include options for victims to remain anonymous which will add a layer of protection for individuals who may be subject to gender-based violence. Throughout project implementation, project staff will disseminate information to beneficiaries on how they can use the GRM and solicit feedback on how the mechanism can be improved if needed.

b. **Provide access to skills building, leadership and programmatic inclusion.** Because of existing gender norms, there is much to do to build technical skills of rural women to be able to participate in the project. In addition to that, finding women leaders and building their leadership skills to be at the forefront of rural gender equity is crucial for addressing the root causes of women's exclusion in land use planning and decision-making. This project however will not just focus on rural women. Ensuring

that women play a strong role at all levels of climate programming in the country is important. Thus developing professional networks and internships to provide opportunities to strengthen leadership and technical skills will be incorporated into the program.

c. **Facilitate women's financial autonomy.** In order to support rural women to capture financial benefits of this project, this project will incorporate three activities: financial skills building, support in joining cooperatives; and facilitating access to entrepreneurship through financial/entrepreneurship training, access to village savings/loans and small grants, and mentoring. The project will ensure all trainings are gender-responsive and inclusive. Gender considerations will be integrated into training curriculum, content and delivery methods. Flexible training schedules and convenient venues will be selected to accommodate women's needs. Staff in charge of gender aspects will ensure gender mainstreaming and ensure the project addresses barriers to women's participation. Training will also use participatory approaches to create an inclusive learning environment and incorporate components on women's empowerment and leadership development, promoting gender equality and active participation. By facilitating independent access to financial services, women's income can be managed independently until shifting norms and household situations reach gender equity for households in adapting to climate change. Within the program, microenterprise, access to finance, beekeeping, and agroforestry/cookstove value chain support were chosen as key activities because they are feasible livelihoods within the target rural households and help strengthen resiliency of the rural economy. Further information on women in the value chains targeted by the project, including gender characteristics, composition of workforce and ownership etc. is detailed in the project Feasibility Study where possible and will be refined during project implementation.

1.21 Youth Analysis and Key Issues

Rwanda is a young country; 82% of the population is under the age of 40 and 51.5% are 19 or under (NISR, EICV5 2018, see Table 5). The sex ratio for youth, classified as people from 14-35 years old, is 92 males for every 100 females. The literacy rate for youth is quite high; averaging 91% in urban areas and 82% in rural areas (ibid). Numeracy rates are linked to wealth, with poorer families having lower capacity (72.4% vs 92.7%). Larger urban-rural discrepancies are found in computer literacy (28.5% vs 7%) and mobile phone ownership (70% vs 41%).

Table 5: Youth population in Rwanda by district. (Source: NISR EICV5 2018)

Youth Population in Rwanda (Source: NISR EICV5)			
District	Youth	Non-Youth	Total Population (000s)
Kigali City	36.5	63.5	1,631
South	24.3	75.7	2,739
West	25.1	74.9	2,685
North	26.8	73.2	1,841
East	24.6	75.4	2,998
All Rwanda	26.6	73.4	11,893

Table 6. Youth Usual Economic Activity by Sex and Province. Source: (NISR EICV5 2018).

Activity Type	Youth Usual Economic Activity by Sex and Province (Data source: EICV5)												
	Kigali City		Southern Province		Western Province		Northern Province		Eastern Province		Total		
	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	All
Worker	70.4	66.5	76	78	78.1	81.8	78.2	81.2	80.5	79.8	76.8	77.6	77.2
Seeking work	5.5	11.7	1	2.3	1.5	1.5	0.9	1.9	0.7	0.9	1.9	3.5	2.7
Student	21.7	15.9	18	14.7	16.8	12.1	17.5	14.6	15.2	14.9	17.7	14.4	16
Domestic	1.8	4.1	2.6	3.5	1.5	2.4	1.2	1.4	1.9	3	1.8	2.9	2.4
Disability	0.3	1	1.4	0.8	0.9	1.3	1.1	0.6	1.2	1.1	1	1	1
Other	0.4	0.7	1	0.7	1.2	0.8	1.2	0.3	0.5	0.3	0.8	0.6	0.7
Total	100	100	100	100	100	100	100	100	100	100	100	100	100
Usually	75.9	78.2	77.1	80.3	79.6	83.3	79.1	83	81.2	80.6	78.6	81.1	79.9
Inactive	24.1	21.8	22.9	19.7	20.4	16.6	20.9	17	18.8	19.4	21.4	18.9	20.1
Total	100	100	100	100	100	100	100	100	100	100	100	100	100

Employment figures for rural youth indicate that 54.7% are engaged in agriculture, and 32.9% in elementary activities. The rest are in services (7.8%), and very few in managerial, professional etc jobs. According to a recent analysis, youth “envision self-sufficiency achieved through economic independence and the ability to provide for and support their families, and they see this path as most available to them through self-employment (USAID 2019)”. The Youth Assessment highlighted key priority areas that youth identified for achieving their goals:

- Livelihoods and skills development for productive self-employment activities
- Education and skills for entering into the formal work sector
- Reproductive autonomy and health to avoid early pregnancy which is a key hurdle towards economic self-sufficiency

Poverty amongst youth has remained stable between EICV4 and EICV5, with about 30% of all youth below the poverty level, and 11.3% in extreme poverty. There are substantial gender differences, with young women increasingly vulnerable as they age (see Figures 7 and 8 with data from NISR’s EICV 5, 2018). Thus, programming ensuring attention on both young men and young women is vital for rural climate resiliency.

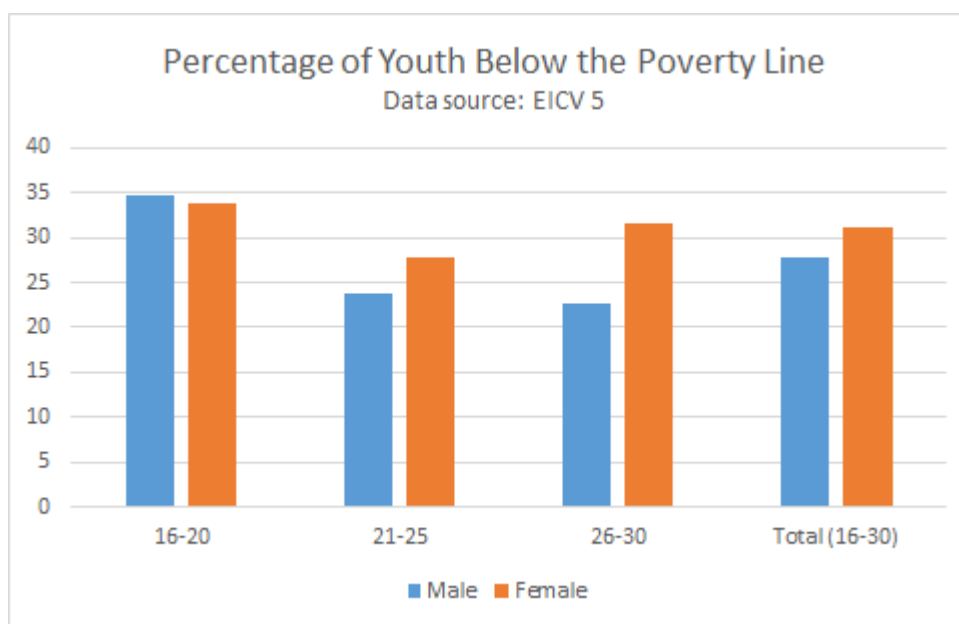


Figure 8: Percentage of youth below the poverty line by age (Source: NISR EICV5 2018)

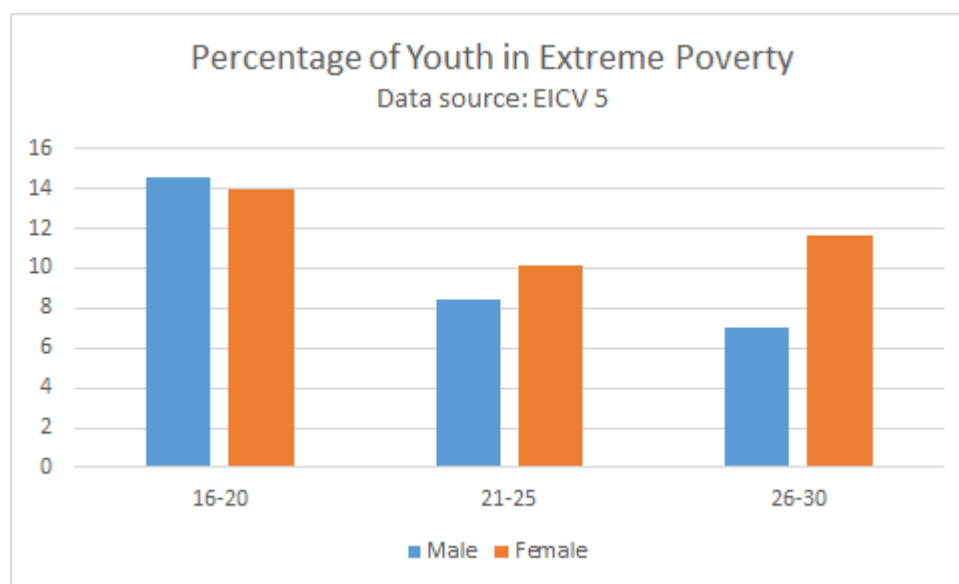


Figure 9: Percentage of youth in extreme poverty line in Rwanda by age. (Source: NISR EICV5 2018)

Translating this information into a climate adaptation program that supports young men and women in the CND, three key topics influenced the action plan:

- a. **Diversified opportunities.** With over 51.9% of the Rwandan population under the age of
- b. 15 and 18% between the ages of 15-24, youth make up a significant percentage of the population. There is already a scarcity of arable land, with many households owning less than what is needed for their subsistence. With the Rwandan population expected to increase from 12 million in 2019 to 20 million in 2042. The future for the upcoming generation of rural youth is worrisome--agricultural

lands will continue to grow scarcer and demand for resources and land increase. The youth interviewed are very cognizant of these constraints, and many are keen to diversify their livelihoods yet they are most interested in short-term gains. This GCF program intends to seek out vulnerable youth in critical areas of the CND to support entrepreneurship and diversified opportunities to reduce pressure on highly marginal lands.

c. **Access to skills building.** Many youth attend schools, but training in practical agroforestry, financial management, and forestry is lacking. This program intends to target youth for diversifying their skill sets and strengthening their resilience to adapt to climate change.

d. **Youth leadership and Programmatic Inclusion.** Many youths voiced their concerns over not being included in decision-making processes and programming, resulting in key stakeholders not being engaged in civil society. In order to engage the new generation of rural farmers and consumers, this project will identify and work with youth leaders in the CND to use SBCC to engage youth to benefit and support long-term tree initiatives. With youth involved in decision-making processes, innovative ideas to involve youth with best practice on government lands and to get youth to appropriate long-term thinking through tree planting as future investments.

1.22 Internal program considerations

In addition to programmatic activities with women and youth, this program will also have an internal focus to ensure best practices. These include:

a. **Avoid perpetuation of gender inequalities** and women's exclusion in the agro/forestry sector. SBCC will be used heavily to shift norms towards inclusivity and skills building to facilitate women and youth to join the sectors. In addition, women need to be engaged in the decision-making bodies of the project itself, including a Steering Committee, which will be at least 50% women, and the Project Management Unit with targets to achieve 50% women by the end of the program as mentoring and training increase capacity. The Gender Monitoring Office will be a part of the Steering Committee and along with the The National Women's Council and The Ministry of Gender and Family Promotion were consulted during proposal development and will continue to be involved at project implementation stage at the national and local level. There will be comprehensive consultation and feedback processes with these stakeholders through meetings and workshops to ensure perpetuating gender inequalities is avoided.

b. **Build partnerships.** This GCF project provides a unique opportunity to engage civil society at national and local levels. The team will work with men's, women's, and youth organizations to build their capacity and facilitate their engagement with project beneficiaries and local government actors.

c. **Gender and Diversity training.** Training will be an important component of a large, 5-year program. Building online training modules that can be viewed across the CND, and using new platforms like WhatsApp to communicate with urban and rural project staff will ensure that static, one-off training is avoided.

d. **Staffing.** Diversity and inclusion are important for the project, and hiring practices at all levels of the program will be reflected in hiring policies. The program management unit will hire a full-time dedicated gender expert and a full-time dedicated youth focal point (both AE staff) and associated costs are included in the project budget. These two staff will participate in all management meetings to support inclusion goals across the project, and will lead and oversee the implementation of the Gender, Youth, and Social Inclusion Action Plan. Another key role of these two positions will be to engage civil society

partners, particularly women's, men's, and youth groups to build capacity and increase the sustainability of achievements during this five-year project. Along with these positions, RFA staff will include a Community Mobilization Specialist and a Community Engagement Specialist who will be in charge of and support project gender mainstreaming. Their responsibilities include reviewing site specific barriers faced by women to participate in project activities, ensuring these barriers are addressed in order to support full participation of women, ensuring gender equity in all project activities and where needed help women access the Grievance Redress Mechanisms to pursue redress and connect women with Isange One Stop Centers in the project area to prevent gender-based violence and provide immediate assistance to victims.

e. **M&E.** Many stakeholder meetings highlighted the lack of community engagement as a key reason for project failures in the past. This project will incorporate suggestions from our stakeholder engagement meetings to ensure best practice is followed. This includes not only deeper local government engagement but also using a rights-based approach to ensure that communities are at the heart of planning our interventions. Targets for youth and gender are incorporated into the Gender and Youth Action Plan below and are set to ensure women and youth are equally targeted where possible and encouraged to actively participate. Exceptions include situations where group composition, such as government agents or community leaders, are outside the project's control or the activity solely targets men, such as masculinity sessions held with men to improve gender equity in program participation and benefits or assisted natural regeneration activities inside the park far from villages that will take male participants several weeks away from their homes. Overall targets for key activities (i.e., G4. 4.1.1.5, G4. 4.1.1.6) have not been set as the target will depend on the number of villages/sites where project activities take place, however disaggregated targets per village/site have been set. A project baseline will be conducted and baselines for all targets will be determined at project onset.

1.23 Gender and Youth Action Plan

Component 1: Mainstreaming Climate Adaptation into Integrated Land Use Planning						
Impact 1: Climate adaptation programming is successfully implemented because it is fundamentally informed by rural women and youth stakeholders <i>Indicator: Equal participation by gender groups in contributing to expected outcomes</i>						
Gender/Youth Outcome 1: Climate adaptation programs in LUPs reflect and address the priorities of women and youth <i>Indicator: % of men, women and youth participating in LUP processes who feel LUPs incorporate their needs as they relate to climate programming</i>						
Activity	Gender/Youth Outputs	Indicators	Targets	Timeline	Responsible	Costs
G1. 1.1.1.7 Conduct climate literacy seminars for local government and civil society organizations, aimed at increasing women and youth participation	Local institutions are trained to build capacity of rural people, specifically women and youth, to participate in climate adaptation processes	# of CSOs and government agencies trained	60 people (50% male, 50% female) from CSOs/Local institutions trained in Y1 and Y3 to increase participation of women/ youth/ marginalized in stakeholder consultations & activities	Y1,3	PMU/RFA	\$10,187
G1. 1.1.1.8 Provide financial and logistical support to trained organizations in grassroots mobilization to increase women and youth participation in climate adaptation planning	Trained institutions hold meetings to train women, youth and marginalized people on forests and mobilize them to participate in planning processes	# of meetings held by trained organizations with local communities on climate adaptation programming	At least 10 grassroots meetings per district are held in Y1 and Y3 in the CND	Y1,3	PMU/RFA/ Local CSOs (TBD)	\$12,262

G1. 1.1.2.4 Develop and rollout a series of trainings (virtual sessions and online modules) on gender sensitivity and mainstreaming women and youth into planning	Program Staff and government officials understand gender barriers, understand how to mainstream gender equality principles into planning and are strong facilitators to support women, youth and marginalized participation in meetings	# of people trained (disaggregated by gender) # project staff trained in social inclusion	- 65 government agents trained per year. - 100% project staff trained	Y1-5	PMU + Local Women's NGO (TBD)	\$40,523
Component 2: Forest Landscape Management and Restoration						
Gender/Youth Impact 2: Sustainable gains in restoration and landscape management are made because women and youth are enabled and motivated to adopt best practice <i>Impact Indicator: Targeted women and youth report an improved enabling environment for their participation in forest management and restoration</i>						
Gender/Youth Outcome 2: Women, youth and marginalized households have improved capacity to engage in forest management and restoration <i>Outcome Indicator: Equal participation by gender groups in forest landscape management and restoration</i>						
G2. 2.2.2.15 & 3.1.1.11 Develop and implement strategy (including social marketing) to ensure that women and youth benefit from forest restoration projects	Women and youth have the capacity, motivation, and opportunity to participate and benefit equitably in restoration programs. Programs will actively seek out direct participation of and benefits for women and youth through paid labor and direct involvement.	% Women beneficiaries % Youth beneficiaries	50% of beneficiaries are women 50% of beneficiaries are youth ²	Y1 Strategy Y2-5 Implementation	PMU, Women/Youth NGOs (TBD)	\$37,735
Component 3: Enhancing climate adaptation through resilient livelihoods						

² The total number of beneficiaries will be determined at project onset

Gender/Youth Impact 3: Women, marginalized and youth are empowered to actively participate in green jobs
Impact indicator: Target groups perception of economic empowerment

Gender/Youth Outcome 3: Women, youth and marginalized households have improved capacity to engage in green jobs
Outcome Indicator: Number of new women, youth and marginalized households that have engaged in green jobs

G3. 3.2.1.2 Develop and implement strategies to ensure that women,youth, and other marginalized groups benefit from rural livelihoods and financial services	Women and youth have the capacity, motivation, and opportunity to participate and benefit equitably in rural livelihoods and financial services	% Women beneficiaries % Youth beneficiaries	50% of beneficiaries are women 50% of beneficiaries are youth ³	Y1 & Y2	RFA & CORDAID	\$53,053
G3. 3.2.3.2 Organize Microenterprise training and develop and implement green grants programs targeting landless youth and poor, as well as women who are unable to access direct benefits from land-based project activities.	Landless poor, youth, and women have enterprise skills and grants to engage in green value chains	# of enterprises formed, beneficiaries disaggregate d by gender and social class	5,600 beneficiaries of which at least 50% are women 500 off-farm jobs created in production quality control	Y2,3 & 4	RFA & CORDAID	\$66,398
G3. 3.3.2.2 Provide organizational/technical/financial capacity building to women, youth, and CSOs (such as cooperatives or VSLA) with focus on financial literacy and provide long-term coaching to beneficiaries	Women and youth groups have the organizational, technical and financial skills to receive support from project	# CSOs/VSLAs trained # beneficiaries receiving coaching	300 CSOs/VSLAs 6,900 of which 40% youth 50% women	Y2,3 & 4	RFA & CORDAID	\$30,040

³ The total number of beneficiaries will be determined at project onset

G3.3.3.2.3. Support women, youth, other marginalized groups strengthen savings and loan groups and register for MFI accounts	Men, women and youth have access to thriving financial services	# people with new bank/MFI accounts	50% women	Y2,3,4	RFA & CORDAID	\$13,713
Component 4: Monitoring and Evaluation						
Gender/Youth Impact 4: Women and youth benefit from an adaptively managed program <i>Impact indicator: Reported attitudes, confidence, and a sense of empowerment towards engaging in climate programming</i>						
Gender/Youth Outcome 4: Disaggregated and specialized monitoring generates data showing whether and how women, youth and marginalized people are benefiting from adaptive management systems <i>Outcome indicator: Gender and youth disaggregated participation data</i>						
G4. 4.1.1.4: Develop an overarching gender and youth program unit within the GCF Project to ensure the project uses adaptive management to ensure strong inclusion of women, youth, and marginalized households such that equitable benefits are shared across the communities.	Gender/Youth program unit provides oversight and support to ensure gender/ youth targets are met	N/A	N/A	Y1-5	PMU	\$3,380
G4. 4.1.1.5: With the local government, project partners, and Youth CSO leaders, identify key male and female youth leaders in each project village to be project champions	Key youth leaders are identified and engaged	Youth leaders identified	2 Youth leaders (50% male, 50% female) per targeted village. ⁴	Y1	PMU, Youth NGO (TBD)	\$88,683

⁴ The number of targeted villages will be determined at project onset and will determine the total number of youth leaders to be reached. 50% male and 50% female youth leaders will be targeted per village.

G4. 4.1.1.6: With local government, project partners and women's civil society, identify key women leaders amongst project beneficiaries to be project champions	Key women leaders are identified and engaged	Women leaders identified	2 women leaders per village ⁵	Y1	PMU, Women's NGO (TBD)	\$87,903
G4. 4.1.1.7: Engage Youth & Women's NGOs to build leadership and coaching skills of identified project champions and facilitate their project involvement in order to advance project goals of greater participation amongst marginalized groups amid greater community-wide support for their participation	Leaders are equipped with skills and financial resources to build empowerment and facilitate women and youth engagement in the project	# of women and youth trained in coaching and leadership skills	200 (50% women, 25% male youth leaders, 25% female youth leaders) women and youth leaders trained as trainers	Y1-2	PMU	\$6,660
G.4. 4.1.1.8: Overarching social behavior change communications strategy and materials are developed to overcome barriers and enhance motivations for women, youth and marginalized people to participate in program activities	Gender sensitive and responsive communications programming supports new norms for social inclusion in project activities	# programs/ materials delivered in communities	At least 10 materials developed and distributed annually	Y1-5	PMU	\$97,552

⁵ The number of targeted villages will be determined at project onset and will determine the total number of women leaders to be reached. 100% of women leaders will be female

G4. 4.1.1.9: Informational campaigns on equity (land rights, access to resources, and decision-making) and programmatic opportunities to showcase the benefits of everyone (women, young men/women, marginalized) participating in programming to improve lives	Gender sensitive and responsive campaigns and informational material on inclusion of different social categories is developed and disseminated	# of people reached by campaigns	At least 200,000 (50% female, 50% male) people reached by campaigns over 5 years	Y1-5	PMU	\$17,030
G4. 4.1.1.10 Engage men's NGO to roll out programming for men as allies in women's economic empowerment and sexual/gender based violence reduction programming	Men in target areas participate in masculinity sessions to improve gender equity in program participation and benefits	# men participating in training sessions	At least 200 men participate in masculinity sessions each year from Y2-5	Y2-5	PMU Men's NGO	\$115,574
G4. 4.1.2.2 Identify and train men and women (adults and youth) to implement key M/E tools in their communities	Women and youth are active participants in M/E at central, sectoral, and community level	# Women and Youth trained on M/E tools	At least 5 women, 5 men and 5 youth (15 people total) per district are trained ⁶	Y1-5	PMU	\$3,380
G4. 4.1.3.2 Engage youth leaders in designing and monitoring targets for youth engagement, participation, and outcomes	Youth targets are identified and incorporated into M/E system	N/A	N/A	Y1-5	PMU	\$ 16,533
G4. 4.1.3.3 Engage women leaders in designing and monitoring targets for women engagement, participation, and outcomes	Women targets are identified and incorporated into M/E system	N/A	N/A	Y1-5	PMU	\$ 16,533

⁶ The number of targeted districts will be determined at project onset and will determine the total number of women, men and youth to be trained. The target for the number of individuals trained overall is 50% female, 50% male.

G4 .4.1.3.4: Monitor program using Gender Checklist to ensure gender is integrated in programming, HR, policy, and activity performance	Project has a strong gender monitoring program	- Project gender checklist developed - Annual checklist scores	Project scores on gender checklist increases yearly	Y1-5	PMU	\$ 16,533
G4. 4.2.2.1 Project partners (government and NGOs) are trained on gender-sensitive approaches	Project staff have skills to monitor inclusion of youth and women into programming	# staff trained	100% of M/E focal points from participating organizations/entities are trained	Y1-5	PMU	\$25,780
Project Gender and Youth staff time is incorporated into each activity cost						
Total Estimated Cost for Implementation of GAP: \$759,452						



**GREEN
CLIMATE
FUND**

Meeting of the Board
23 – 26 October 2023
Tbilisi, Georgia
Provisional agenda item 12

GCF/B.37/02/Add.08

2 October 2023

Consideration of funding proposals – Addendum VIII

Funding proposal package for FP217

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Funding Proposal

Project title:	Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration
Country(ies):	Rwanda
Accredited Entity:	Ministry of Environment
Date of first submission:	2022/02/25
Date of current submission	2023/09/11
Version number	V.6



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(a) PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Project	A.2. Public or private sector	Public	
A.3. Request for Proposals (RFP)	<u>Not applicable</u>			
A.4. Result area(s)			GCF contribution	Co-financiers' contribution
	Mitigation total		8 %	0 %
	☐ Energy generation and access		0 %	0 %
	☐ Low-emission transport		0 %	0 %
	☐ Buildings, cities, industries and appliances		0 %	0 %
	☒ Forestry and land use		8 %	0 %
	Adaptation total		92 %	100 %
	☒ Most vulnerable people and communities		13 %	49 %
	☒ Health and well-being, and food and water security		2 %	0 %
	☐ Infrastructure and built environment		0 %	0 %
☒ Ecosystems and ecosystem services		77 %	51 %	
A.5. Expected mitigation outcome	1,084,291tCO ₂ eq	A.6. Expected adaptation outcome	Direct Beneficiaries: 1,254,242 (654,404 females)	Indirect Beneficiaries: 9,260,745 (4,769,284 females)
			30.9 % of the CND population	69.9% of Rwanda's population
A.7. Total financing (GCF + co-finance)	50,000,000 USD	A.9. Project size	Small (Up to USD 50 million)	
A.8. Total GCF funding requested	39,056,421 USD			
A.10. Financial instrument(s) requested for the GCF funding	☒ Grant 39,056,421 USD ☐ Loan- ☐ Guarantee -		☐ Equity : ☐ Results-based payment :	
A.11. Implementation period	5 years	A.12. Total lifespan	20 years	
A.13. Expected date of AE internal approval	<u>TBD</u>	A.14. ESS category	B	
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐ M	
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐	
A.19. Complementarity and coherence	Yes ☒ No ☐			

A.20. Executing Entity information	Rwanda Forestry Agency (RFA) Country of Registration: Rwanda Ownership Type: Government agency.
A.21. Executive summary	
Rwanda's Congo-Nile Divide (CND), an area of 444,600 hectares, is a high-elevation landscape of remnant natural forest reserves embedded in a mosaic of intensively managed smallholder farms and commercial plantations. Extending along a north-south ridge with an altitudinal gradient from 1,900m to 4,507m, the CND separates the drainage basins of the Congo and Nile Rivers. It is part of the Albertine Rift, which is one of the most biologically rich regions in Africa and "one of the most important regions for conservation" on the continent¹, containing more than half of Africa's birds and 40% of its mammals. At its highest elevations, the region is crowned by the Volcanoes, Gishwati-Mukura, and Nyungwe National Parks, containing Rwanda's only remaining montane forests, one of the most biodiverse ecosystems on the African continent, with high levels of endemism (species found nowhere else on Earth) and dozens of threatened species, including populations of mountain gorillas and chimpanzees. Nyungwe has been recognized as one of the six highest priority protected areas in the Albertine Rift due to its high endemism and globally threatened species (Plumptre et al 2016). Though covering barely 5% of the country, this archipelago of protected forests is of critical importance for: • Its globally recognized high biodiversity values^{2,3, 4, 5}; • Its role in Rwanda's acclaimed ecotourism offerings serving as the primary source of Rwanda's foreign revenue and as a major economic development driver^{6, 7}; • The ecosystem services provided for surrounding subsistence farmlands, as well as for commercial tea and coffee plantations offering employment opportunities for women while providing the 2nd and 3rd highest sources of foreign revenue^{8,9}; • The regulation of water capture and release to growing urban centers on the CND periphery¹⁰, the nation's extensive river network, and downstream hydroelectric operations¹¹; and • The sequestration and storage of nearly 60% of Rwanda's national carbon stocks¹². These values and services are already compromised by a combination of anthropogenic and climatic factors; they will be much more severely threatened by the potential negative feedback loops associated with projected climate change impacts. The Project will promote an ecosystem-based approach to forest protection and restoration and will enhance the extent and resilience of forest ecosystems and species, including functionally linking currently isolated forest fragments. Through these approaches, the Project will protect, restore and rehabilitate forest ecosystems, enhancing biodiversity and reducing key threats, while	

1 Plumptre, A.J., et al. 2007. The biodiversity of the Albertine Rift. *Biological Conservation* 134:178-194.

2 Plumptre, et al. 2016. Conservation Action Plan for the Albertine Rift. Wildlife Conservation Society

3 UN/CBD. 2020. Rwanda 6th National Report to the Convention on Biological Diversity

4 2012. Eastern Afromontane Biodiversity Hotspot. Critical Ecosystems Partnership Fund.

5 2023. Nyungwe National Park. BirdLife International. www.birdlife.org.

6 RDB. 2021. Rwanda Development Board Annual Report

7 2010. *In the Kingdom of Gorillas*. Weber & Vedder.

8 Andrew, G., Masozera, M. 2010. *Payment for Ecosystem Services and Poverty Reduction in Rwanda*. Journal of Sustainable Development in Africa (V12, No.3)

9 RDB. 2021. Annual Report.

10 Karamage, F., et al., 2017. *Modeling Rainfall-Runoff Response to Land Use and Land Cover Change in Rwanda (1990–2016)*. *Water* 2017, 9(2), 14.

11 Rwanda Environment Management Authority (REMA), 2009. Rwanda State of Environment and Outlook Report, Chapter 8: Energy Resources.

12 Mugabowindekwe, M., Brandt, M., Chave, J. et al. Nation-wide mapping of tree-level aboveground carbon stocks in Rwanda. *Nat. Clim. Chang.* 13, 91–97 (2023). <https://doi.org/10.1038/s41558-022-01544-w>

increasing the provision of critical ecosystem services to strengthen adaptation of vulnerable communities.¹³
14

The CND landscape supports nearly one-third of Rwanda's population on barely one-fifth of its land base. Average population density is 736 people/km², with some districts now exceeding 1,000 people/km²¹⁵. The highest rates of rural poverty in the country are reported from the CND, including its most vulnerable subsistence farmers. While high rainfall has historically permitted generally sufficient food production, the region's rugged topography requires farming on extremely steep slopes. With more frequent extreme rainfall events in recent years, this agricultural landscape is recognized as the region of Rwanda most vulnerable to floods and landslides¹⁶. Between 2013-2016 heavy rains generated landslides that killed at least 174 people and destroyed the homes of more than 5,000. Many of these landslides caused flooding of highly productive raised-bed cultivation systems in the CND's many valleys and surrounding bottomlands and many hectares of crops being washed away annually¹⁷. An estimated 1.5 million tons of fertile soil are lost annually to erosion¹⁸.

CND forests are essential to the region's climate resilience. Since independence, the total natural forest cover has declined 40%, from 219,000 ha to 130,700 ha, mostly from episodic planned excisions as well as uncontrolled incursions. The loss around the Gishwati and Mukura core forest area has been most extreme. Of the original 32,098 ha in the 1980s, now only 3,460 ha remains as natural forest: almost a 90% loss (see Annex 2). In comparison, the Volcanoes and Nyungwe boundaries have been stable for decades. The recent creation of the Gishwati-Mukura National Park is an important step toward halting and partly restoring this forest, particularly for its land stabilization and other ecosystem services within the CND's most vulnerable lands and dependent communities.

IPCC models predict rising temperatures and increasingly intense precipitation events for the CND through the rest of this century. The rainfall implications are especially threatening for smallholders farming on steep, heavily deforested slopes, where they are most exposed to landslides, leading to loss of crops, homes, and lives. Yet their adaptive capacity is low because crop yields are vulnerable to rainfall variability and because high population density severely limits their options for relocation in the event of disaster. This is particularly true for women who, until very recently, had fewer legal land rights than men. The expected 2°C increase in temperature would further result in a more than 470m rise in vegetation zone elevations, impacting traditional crop selection and management practices for future farmers. An increase of this scale would also dramatically alter the optimal ranges for tea and coffee production, introducing potential competition between these important export crops and alternative land uses.

This Project is designed to sustain and expand critical forest ecosystem functions in the CND, and ultimately enhance the resilience of the vulnerable communities who depend on them. This will be done by strengthening government capacity for spatial planning, resulting in institutional and regulatory policies to promote and integrate climate adaptation strategies, based on the latest science, through coordinated planning at the national and district levels. It will also protect, restore and expand natural forest in and around

13 2003. Interlinkages Between Biological Diversity and Climate Change. CBD Technical Series Number 10.

14 2022. Ecosystems and ecosystem services Sectoral Guide. Green Climate Fund.

15 National Institute of Statistics of Rwanda (NISR); The Fifth Rwanda Population and Housing Census, Main Indicators Report, February 2023

16 https://www.gfdr.org/sites/default/files/publication/National_Risk_Atlas_of_Rwanda_electronic_version_0.pdf

17 Nsengiyumva JB, et al., 2018 Landslide Susceptibility Assessment Using Spatial Multi-Criteria Evaluation Model in Rwanda. International Journal of Environmental Research and Public Health. 15(2):243.
<https://doi.org/10.3390/ijerph15020243>

18 Bizimana, H.; Sönmez, O. 2015 *Landslide Occurrences in the Hilly Areas of Rwanda, Their Causes and Protection Measures*. Disaster Sci. Eng. 2015, 1, 1–7

the core national parks; consolidate and expand some smaller but critical "stepping stones"¹⁹ of remaining natural habitat between these national parks (mainly small forest patches and wetlands/ riparian areas); improve the landscape linkage function of key high altitude landscape linkages between the protected areas, while at the same time securing ecosystem services through afforestation and improved native species mix for plantation forests on steep slopes; silvopastoral interventions on cultivated pasture; and agroforestry interventions in regenerative agricultural areas. Complementary activities will address key drivers of deforestation while creating alternative livelihood pathways for youth and women through ecotourism and income-generating, climate-resilient crops, as well as strengthening supportive financial services.

Project interventions will include 10,000 ha of land (including both forest and silvopastoral sites) under improved, climate-resilient management; and improve agroforestry practices on 3,346 ha of on-farm plantations, directly benefiting 1,254,242 people (654,404 women) in CND with more climate resilient livelihoods from reduced exposure to landslides, floods and soil erosion and more knowledge on climate risks, value of forests and forests ecosystems, climate adaptation options, access to indigenous and agroforestry quality materials and improved ecosystem services. The project will also indirectly benefit 9,260,745 people nationally mainly from the education and outreach program on climate change and gender issues, climate risks, value of forests and forest ecosystems, climate adaptation options, and will create approximately 24,212 job opportunities in CND forest-dependent communities. The total mitigation impact from reduced deforestation, forest restoration introduction and adoption of improved cookstoves is estimated to be 1,084,291t CO₂-eq by 2043.

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B.1 (a). Climate context

This section draws from the Climate Change Vulnerability Assessment presented in Annex 2.1, where more comprehensive elaboration can be found. Discussion on climate predictions is based primarily upon multi-model means from the Coupled Model Intercomparison Project's sixth phase (CMIP6)²⁰, used in the Sixth IPCC Assessment Report. Results are shown for paired Shared Socioeconomic Pathways (SSP) and Representative Concentration Pathway (RCP) emission scenarios; for some variables emphasis is placed on a "pessimistic" high SSP-high RCP scenario (SSP5-8.5), for greater distinction between present day and future conditions as a consequence of anthropogenic climate change. Specialized modeling performed at ultra-high resolution for the East African Great Lake Environments project (EAGLE)²¹, and findings from scientific literature are utilized as well. Key findings are outlined in Table 1, and each climate component theme is discussed in the section on projected climate change impacts below.

¹⁹ Bagchi, R., et al. 2018. Forecasting potential routes for movement of endemic birds among important sites for biodiversity in the Albertine Rift under projected climate change. *Ecography*. 41: 401-413.

²⁰ Coupled Model Intercomparison Project Phase 6. [Link](#)

²¹ Grim, J. A., Pinto, J. A., Jensen, A. A., & Seimon, A., 2020: The East African Great Lake Environments (EAGLE) Climate Downscaling Dataset. [Link](#)

Table 1. Principal Climate Hazards of the Congo Nile Divide through Mid-century

Climate component	Character of change	Level of confidence	Near-term concern	Mid-century concern	Project Response
Temperature	Upward trend from global greenhouse gas emissions and land-use change	Very high for +2.6 °C net increase 1970-2040	Elevational range of biota including pests and pathogens increasingly out of balance	Major uphill displacements of biota including cultivars due to ~470 m rise in thermal conditions	Promotion of agroforestry practices: contribute to the conservation of soil moisture and its recharge through the infiltration of rainfall and runoff water, the creation of sheltered microclimatic conditions, and the inclusion into the soil of organic matter that contributes to moisture retention. Promotion of silvopastoral practices: Improve soil properties due to greater uptake of nutrients from deeper soil layers, enhanced availability of nutrients from leaf-litter and increased nitrogen input by N₂-fixing trees.⁸⁵ Moreover, silvopastoral systems enhance the resilience of the soil to degradation, nutrient loss, and climate change, while enhancing water holding and infiltration capacity of the soil which contributes to the regulation of the hydrological cycle by reducing runoff intensity^{22, 23}. Protective restoration measures at erosion-prone areas: Reduce exposure to soil erosion and floods of communities and their assets living in the road and river side. Restoration of degraded natural forests in protected areas Improve protected areas management to reduce potential human threats (forest encroachment, mining, fire). The project will support activities aimed at (1) promote alternative livelihoods that are resilient to climate change (2) decrease of demand and (3) increase of wood supply capacity. Dissemination of Improved Cook Stoves The project will support the development of an integrated
Annual precipitation	Increasing totals	Moderate, but considerable model variation	Low concern, with natural interannual variability still dominant	Moderate concern, with majority of models showing upward trend	
Precipitation intensity	Increasing short-period rainfall and storm totals	Almost certain to occur	Severe landslide hazard already present and increasing, building flash flood potential	Extreme hazard: widespread and frequent landslides and flash floods	
Cloud base height	Rising cloud base and levels of fog immersion in forests	Likely ongoing, will continue with temperature increase	Already significant in deforested highlands with some loss of moisture provision	Of major concern to forest ecology in protected landscapes	
Drought/dry spells	Increasing intensity	Moderate	Some increase in vegetation desiccation potential, possibly enhancing fire risk	Sustained risk of increase in desiccation and fire risk, possibly offset by rainfall increases	Improve protected areas management to reduce potential human threats (forest encroachment, mining, fire). The project will support activities aimed at (1) promote alternative livelihoods that are resilient to climate change (2) decrease of demand and (3) increase of wood supply capacity. Dissemination of Improved Cook Stoves The project will support the development of an integrated
Wildfire	Increasing occurrence and extent in forested areas	Low	Relatively random occurrences, as in the recent past, that could be managed by improved vigilance	If rainfall increases do not occur, increased vegetation desiccation would intensify risk for widespread fire	
Climatic seasonality	Disappearance of mid-year dry season	Low, but explicitly shown in some models	No concern	Increasing convective storm occurrences in June-Aug may eliminate dry season	

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					land use plan at landscape level that identifies and integrates the trade-offs necessary to improve delivery of ecosystem services, conserve natural resources and sustain economic growth.
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• Climate baseline and current impacts

The climate of western Rwanda and its CND landscape features conditions highly supportive of tropical montane forests, rich biodiversity and human activities alike. Despite being located deep in the tropics just south of the equator, the elevated terrain of the CND landscape, mostly above 2,000 meters, ensures relatively cool temperatures through the year (**Figure 1**).

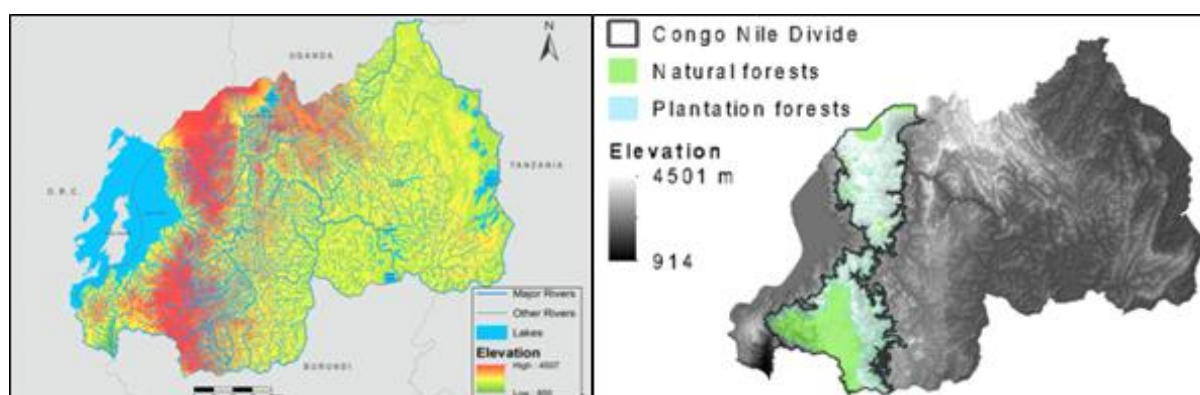


Figure 1 (left) Topographic map of Rwanda with riverine drainage channels overlaid. *Source: Rwanda (2019)²⁴*; **(right)** Congo Nile Divide, identified by the 1,900 m elevation contour and showing the presence of natural and plantation forests.

Rainfall is abundant for nine out of 12 months, and a reliable mid-year dry season adds dynamic character to phenological cycles, pollinator activity and agricultural practices (**Figure 2**). Prior to landscape conversion for agricultural purposes, continuous dense forest cover yielded augmentation of precipitation through fog interception in tree canopies, and near-daily cloud immersion sustained cloud forest ecosystems and species assemblages. The forests also maintained slope stability in steep terrain and yielded percolation and groundwater recharge rather than direct runoff of heavy precipitation. Such conditions are now constrained to within the CND's protected forests within national parks, covering just 30% of the CND landscape and just 5% of Rwanda's total land surface. These are Rwanda's only remaining montane forests yet represent

22 Ibrahim M, Guerra L, Casasola F, Neely N, 2010. Importance of silvopastoral systems for mitigation of climate change and harnessing of environmental benefits. In: Abberton M, Conant R, Batello C (Eds) Grassland carbon sequestration: management, policy and economics. Proceedings of the workshop on the role of grassland carbon sequestration in the mitigation of climate change. Integrated Crop Management, Vol. 11. FAO, Rome, Italy. <http://www.fao.org/docrep/013/i1880e/i1880e09.pdf>.

23 Jose S., 2009. Agroforestry for ecosystem services and environmental benefits: an overview. Agroforest Syst 76 (1):1–10.

24 Vital Signs, 2019: Rwanda Atlas. Environment, Agriculture and Livelihood Options. Vital Signs and WCS Rwanda. [Link](#).

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58.7% of national tree carbon stocks²⁵ and are imperiled by intense population pressures, with 30.7% of the country's population residing within the CND landscape districts.

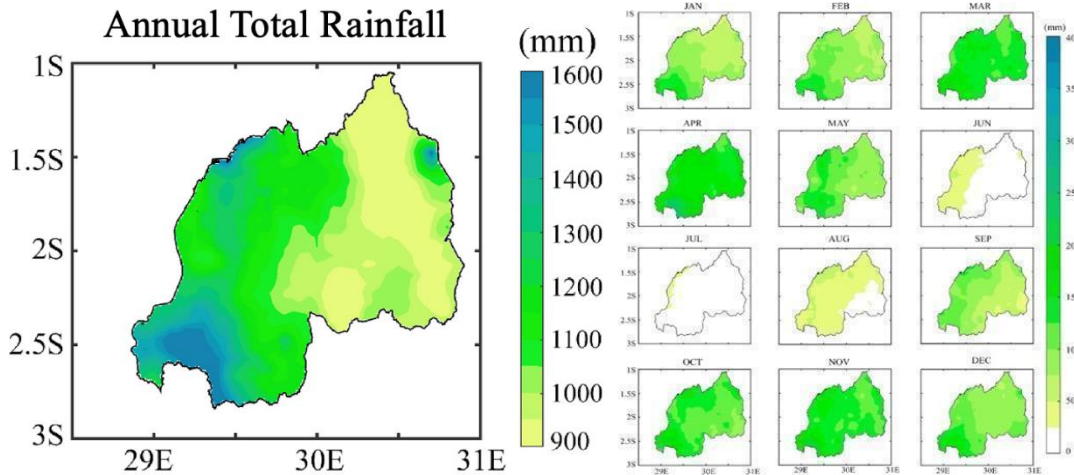


Figure 2. Annual (left) and monthly (right) rainfall (mm) climatology over Rwanda (period of record: 1981–2017), based on algorithms that calibrate satellite-based rainfall estimates with surface observations. Source: Jonah et al., 2021.²⁶

A reconstruction of temperatures for the past 140 years for Rwanda, based on quality controlled instrumental records and centered on Kigali, shows that global climate change already has strong local expression. Rapid temperature increase commenced in the 1970s, with a net gain of approximately +1.1 °C since 1970²⁷ (Figure 3).

²⁵ Mugabowindekwe, M., Brandt, M., Chave, J. et al. Nation-wide mapping of tree-level aboveground carbon stocks in Rwanda. *Nat. Clim. Chang.* 13, 91–97 (2023). <https://doi.org/10.1038/s41558-022-01544-w>

²⁶ Jonah, K., Wen, W., Shahid, S., Ali, M.A., Bilal, M., Habtemicheal, B.A., Iyakaremye, V., Qiu, Z., Almazroui, M., Wang, Y. and Joseph, S.N., 2021. Spatiotemporal variability of rainfall trends and influencing factors in Rwanda. *Journal of Atmospheric and Solar-Terrestrial Physics*, 219, p.105631.

²⁷ Berkeley Earth (website): Rwanda Country Profile. www.berkeleyearth.org

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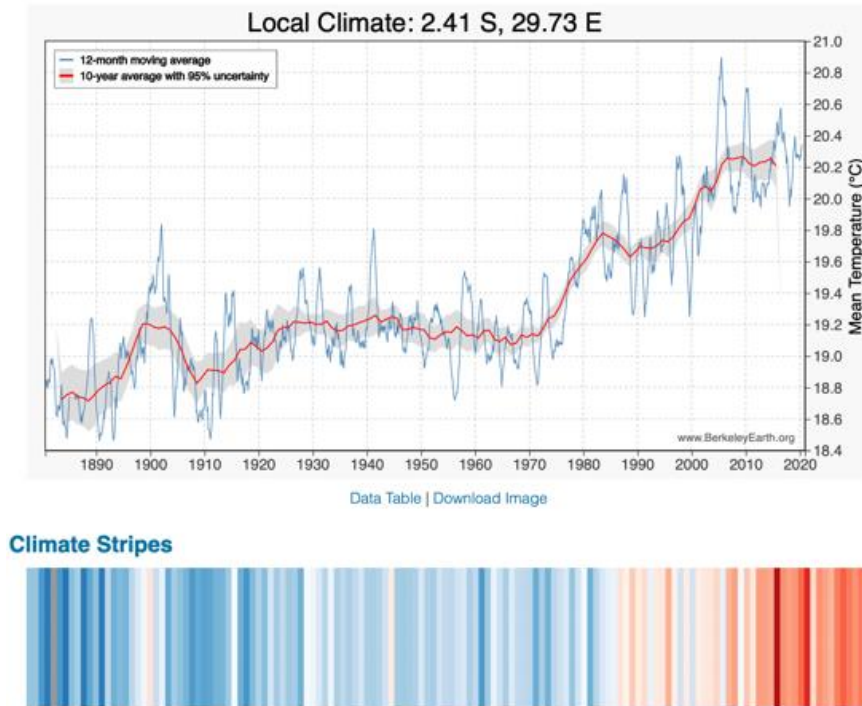


Figure 3. Berkeley Earth temperature reconstruction based on instrumental measurements for Kigali in central Rwanda. The “Climate Stripes” offers a visual display that effectively communicates both individual year temperature anomalies and the overall multi-decadal trend behavior. Source: Berkeley Earth Rwanda Country Profile. www.berkeleyearth.org

The regional warming of the climate over the past half-century coincided with widespread forest conversion to farmland along the CND. As a consequence, at fine spatial scales climatic conditions across the CND at present mimic the landscape mosaic, whereby only the protected forests will retain much of the past climatic character while converted landscapes are likely to be significantly warmer by day, have reduced cloud cover and higher cloud bases, and experience slightly reduced rainfall as well. Due to the increased moisture carrying capacity of warmer air (7% increase per °C of warming)²⁸, ongoing climatic warming promotes increasingly intense short-period rainfall, which acts synergistically with landscape conversion on steep slopes to cause quickflow runoff, yielding large increases in landslide occurrence and soil erosion more generally (**Figure 4**). With its high topographic relief, the CND is highly susceptible to soil erosion. **Throughout Rwanda, soil erosion has increased by 54% since 1990 with the bulk of the increase experienced and highest erosion levels over western Rwanda.**²⁹ Between January and June in 2018, landslides caused more than 200 mortalities, and in May 2020³⁰, 65 deaths from landslides and flash flooding occurred during a single night of storms focused on the eastern slopes of the CND.³¹

28 Trenberth, K.E., 2011. Changes in precipitation with climate change. *Climate Research*, 47, pp.123-138.

29 National Institute of Statistics of Rwanda, 2019: Rwanda Natural Capital Accounts-Ecosystems. p. 23

30 Reuters 2018. <https://www.reuters.com/article/us-rwanda-floods-idUSKBN1I81I>

31 Associated Press 2020. <https://apnews.com/article/b7658b76a403fd9bc8cba249f9d484fd>

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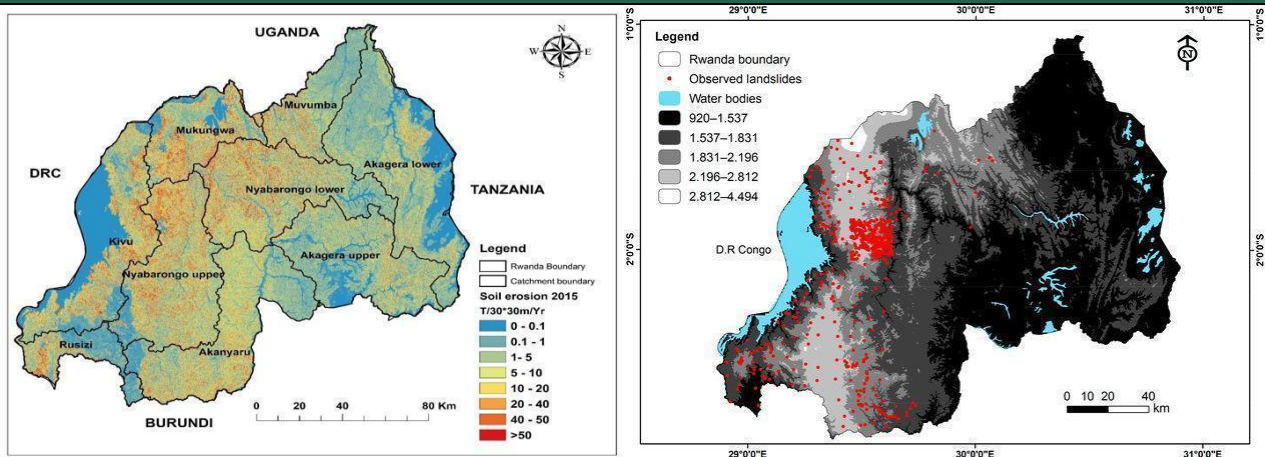


Figure 4. (left) Intensity of soil erosion in Rwanda assessed for the year 2015. Source: NISR: National Institute of Statistics of Rwanda, 2019. **(right)** Observed landslide occurrences in Rwanda against a topographic background, showing intense clusters along both the east and west sides of the CND landscape. Source: Nsengiyumva et al., 2018.³²

Wildfire occurrence in forested lands exacerbated by climatic warming is a rapidly mounting concern globally, and montane forests of the CND carry increasing vulnerability to loss from burning. Absence of fire-adapted flora in CND highland forests, however, suggests that fire has not been a major selective force within the forest ecosystem. The 1997 visitation of weeks-long uncontrolled wildfires in the Nyungwe forest had an ecological imprint that remains evident 25 years later, as burned areas revegetated with dense ferns rather than replacement forests,³³ representing a net loss of 12% of Nyungwe's forest cover and corresponding loss of its carbon endowment. Despite increased propensity for drying of vegetation as the climate warms, forests remain sufficiently moist such that fire causing forest losses are infrequent, and when they occur are usually tied to anomalous dry spells paired with human ignition, as was the case in 1997.³⁴

Cloud base altitude, which is highly influential in CND montane forest ecology and species composition, is likely rising above recent past conditions due to both climatic warming and conversion of forests to agricultural lands. Fog interception by vegetation augments montane forest precipitation in central African highland forests by an estimated 10%³⁵, so highland forest loss by human action therefore represents a local reduction to the ecosystem service of rainwater provision).

Projected climate change impacts

Temperature increases. Localized projections for the CND landscape of climatic changes for coming decades, generated for the most recent IPCC assessment, present a strong consensus on temperature increase for the next two decades. The magnitude of net temperature change by late-century, and the tendency of warming to accelerate, hold steady, decelerate or even begin an overall temperature decline is unknowable at present, but a large number of climate model simulations show common patterns, whereby divergence in possible temperature outcomes will only become apparent around 2040 (**Figure 5**). These

³² Nsengiyumva, J.B., Luo, G., Nahayo, L., Huang, X. and Cai, P., 2018. Landslide susceptibility assessment using spatial multi-criteria evaluation model in Rwanda. *International journal of environmental research and public health*, 15(2), p.243.

³³ Masozera, A.B. and Mulindahabi, F., 2012. Post-fire regeneration in Nyungwe National Park-Rwanda. WCS White paper report.

³⁴ Seimon, A., 2012: Climatology and Potential Climate Change Impacts of the Nyungwe Forest National Park, Rwanda. WCS White paper report, Wildlife Conservation Society, New York, USA, 44 pp.

³⁵ Bruijnzeel, L.A., Mulligan, M. and Scatena, F.N., 2011. Hydrometeorology of tropical montane cloud forests: emerging patterns. *Hydrological Processes*, 25(3), pp.465-498.

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results are also consistent with outputs from the previous CMIP5 model suite for western Rwanda (e.g., IFAD report 2020)³⁶. As such, using the consensus values of such models is a sensible choice for environmental planning, with a +2.0 °C increase over the IPCC-referenced 1995-2014 baseline conditions offering a representative value for warming by mid-century. However, the observed warming of +0.6 °C registered between 1970-95 prior to the IPCC baseline period must be recognized as well, suggesting **net warming of +2.6 °C from a more representative 20th century baseline**.

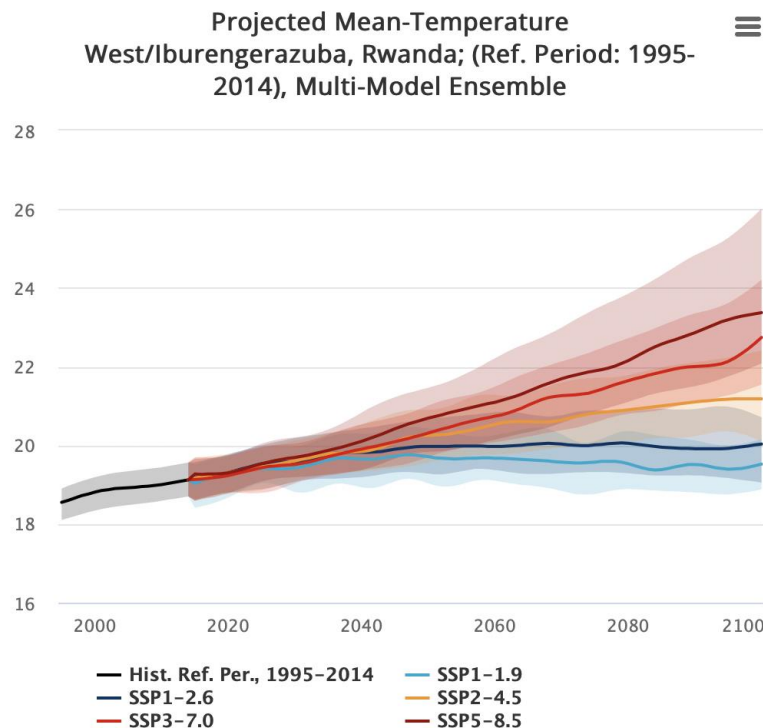


Figure 5. CMIP6 multi-model ensemble predictions for mean annual temperature over western Rwanda across the 21st century under a variety of SSP-RCP combinations. Source: World Bank³⁷

Annual mean precipitation changes. The CMIP6 results for precipitation changes over coming decades are less consistent than the temperature projections. Recent studies focused in East Africa have identified limitations to global model utilization for regionally specific predictions, such as failure to represent observed precipitation magnitude and distribution in the present, and reliance on approximations (i.e., parameterizations) that inadequately represent the role of convective storms in precipitation delivery³⁸. The CMIP6 ensemble means for annual precipitation over western Rwanda have a baseline value of approximately 2,000 mm per year, which is considerably higher than actual values (**Figure 6**; compare values with Figure 2 above). The higher emissions scenarios show increasing totals over time, which is plausible given the increasing water vapor carrying capacity of the warming atmosphere. Strangely, the most optimistic scenario (SSP1-1.9) shows strongly varying trends at decadal scales, increasing far more rapidly

³⁶ Hunter, R., Crespo, O., Coldrey, K., Cronin, K., New, M. 2020. Research Highlights – Climate Change and Future Crop Suitability in Rwanda. International Fund for Agricultural Development (IFAD), Rome.

³⁷ World Bank Climate Change Knowledge Portal. [Link](#)

³⁸ Wainwright, C.M., Marsham, J.H., Rowell, D.P., Finney, D.L. and Black, E., 2021. Future changes in seasonality in East Africa from regional simulations with explicit and parameterized convection. *Journal of Climate*, 34, pp.1367-1385.

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in the near-term than other scenarios. Such inconsistencies reduce confidence in the overall product suite for precipitation predictions.

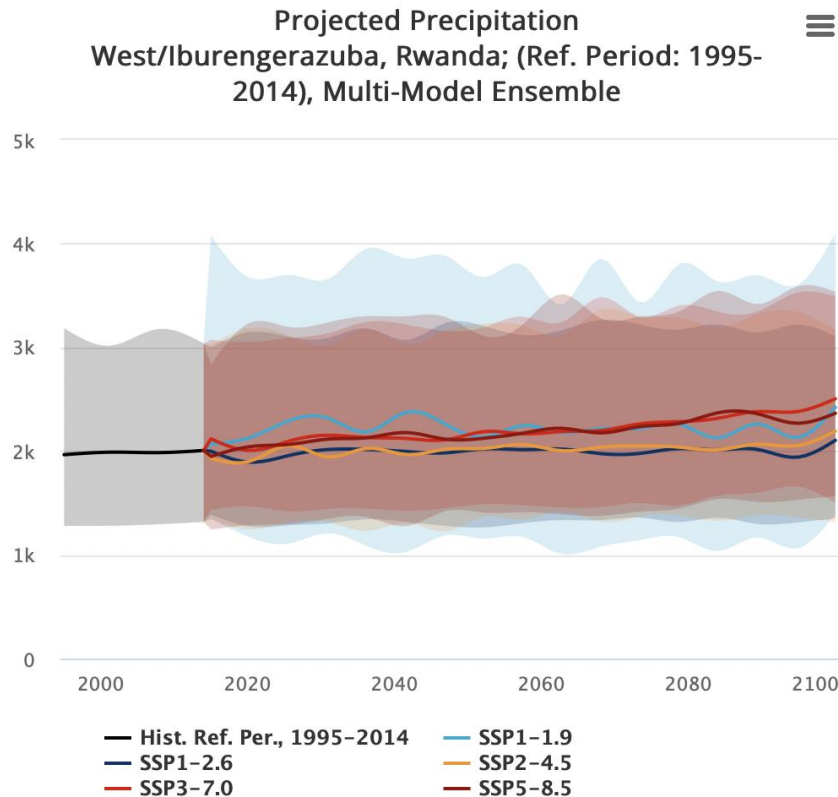


Figure 6. CMIP6 multi-model ensemble prediction for mean annual precipitation over western Rwanda across the 21st century under a variety of SSP-RCP combinations. Left-axis scale is thousands of mm per year. [Source:](#) World Bank³⁹

There is also major inconsistency between CMIP model suites. As assessment for western Rwanda from CMIP5 under RCP 8.5 comparing historical and mid-century outputs suggested rainfall decreases of 5-10% in most months, with net annual decrease of 86 mm⁴⁰; the baseline time periods and geographic domains of these two CMIP ensemble compilations for western Rwanda differ, but the reversal from drier to wetter gives low confidence in the utilization of precipitation predictions specific to the CND if derived from global models.

To generate more refined results, Grim et al., (2020)⁴¹ performed regional simulations of East African climate at ultra-fine, convection-allowing scale (WRF model at 3.3 km model resolution); these were based off the global CESM2 model (part of the CMIP6 suite) boundary conditions under the high emissions RCP8.5. The regional model was configured to optimize parity with regional climate station data. The resulting high spatial resolution output suggests a strongly amplified rainfall signal over the 21st century over western Rwanda relative to the CMIP6 ensemble mean for RCP 8.5 (**Figure 7**). Also notable is that the rest of Rwanda exhibits

³⁹ World Bank Climate Change Knowledge Portal. [Link](#)

⁴⁰ Hunter, R., Crespo, O., Coldrey, K., Cronin, K., New, M. 2020. Research Highlights – Climate Change and Future Crop Suitability in Rwanda. International Fund for Agricultural Development (IFAD), Rome.

⁴¹ Grim, J. A., Pinto, J. A., Jensen, A. A., & Seimon, A. (2020). The East African Great Lake Environments (EAGLE) Climate Downscaling Dataset (NCAR Technical Note TN-563+STR). [Link](#)

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far smaller increases, suggesting that the CND's hydrological significance as a water tower for the nation will be reinforced over time.

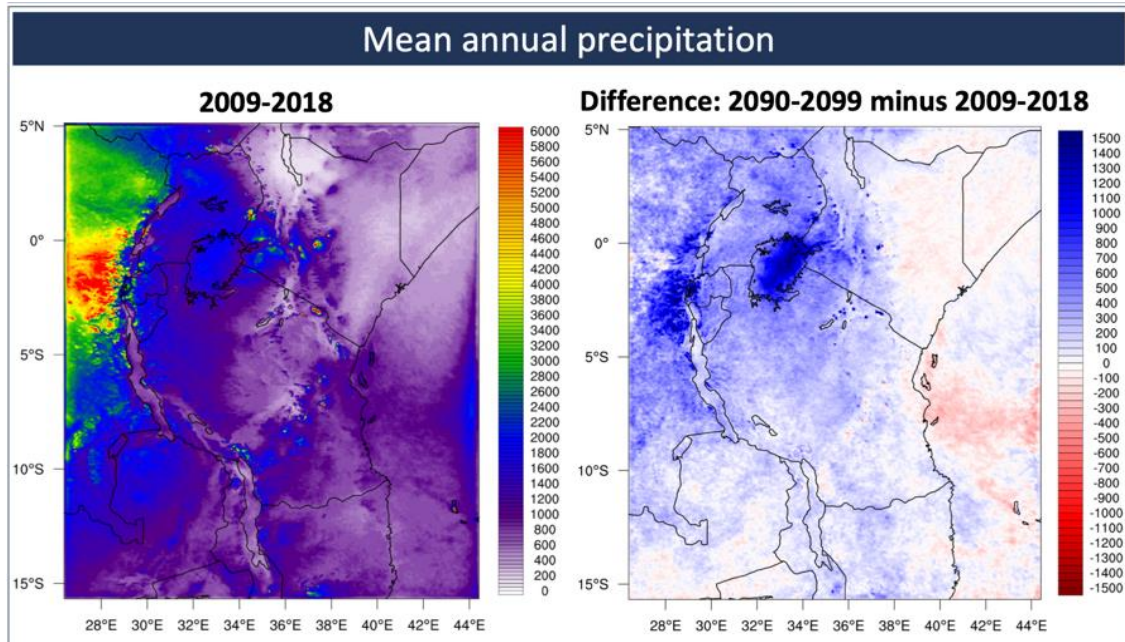


Figure 7. EAGLE projects high-resolution WRF simulation of annual mean precipitation (**left**) and difference between end of century and recent past (**right**) across East Africa, where values are in mm per year. The CND landscape and Lake Kivu show comparable increases to eastern Lake Victoria, and represent the largest rainfall increases across the modeled domain. Source: Grim et al. (2020)⁴²

Precipitation intensity changes. Due to well-understood physical properties of the atmosphere, as warming progresses short-period rainfall rates and individual storm totals, especially for the fraction of higher-end events that can cause flash flooding and promote landslides, will increase and should amplify very significantly over time⁴³. The CMIP6 ensemble means for monthly counts of days with heavy precipitation suggest an upward trend is now underway (**Figure 8**). Forthcoming increases are most pronounced for November through January. Such increases in heavy precipitation events are to be expected under a warming climate and are of major social and ecological concern, being a driver of landslide occurrences and major contributors to soil erosion intensity.

⁴² Grim, J. A., Pinto, J. A., Jensen, A. A., & Seimon, A. (2020). The East African Great Lake Environments (EAGLE) Climate Downscaling Dataset (NCAR Technical Note TN-563+STR). [Link](#)

⁴³ Fowler, H.J., Ali, H., Allan, R.P., Ban, N., Barbero, R., Berg, P., Blenkinsop, S., Cabi, N.S., Chan, S., Dale, M. and Dunn, R.J., 2021. Towards advancing scientific knowledge of climate change impacts on short-duration rainfall extremes. *Philosophical Transactions of the Royal Society A*, 379(2195)

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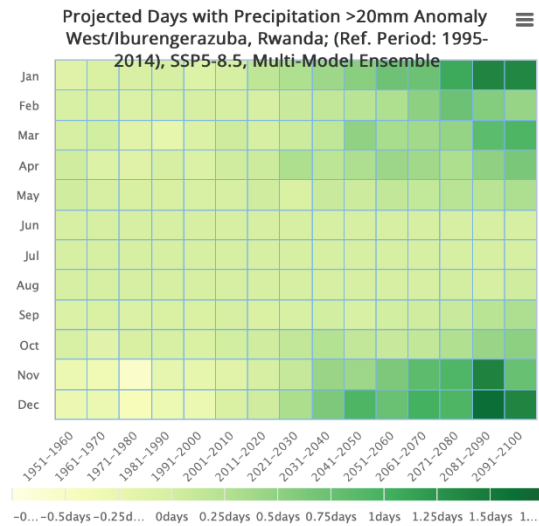


Figure 8. CMIP6 multi-model ensemble prediction for monthly changes in the number of days with rainfall >20 mm over western Rwanda across the 21st century under RCP8.5. Source: World Bank⁴⁴

Drought/dry spells. Warming will also increase the propensity for vegetation to desiccate during dry spells, intensifying wilting of natural vegetation and cultivars. The capacity of air to extract moisture from landscapes through direct evaporation and evapotranspiration of vegetation will also increase exponentially as temperatures rise. This means that land surfaces and vegetation will dry out more quickly between rainfall episodes in the future than in the past. It also means that to maintain hydrological balances that natural and human systems depend upon, total rainfall must increase in proportion to intensifying evaporation rates. The CMIP6 ensemble results for the CND region suggest such a favorable balance may indeed be maintained through at least mid-century (**Figure 9**). Under the full range of emissions scenarios considered, there is almost no change in ensemble means through mid-century.

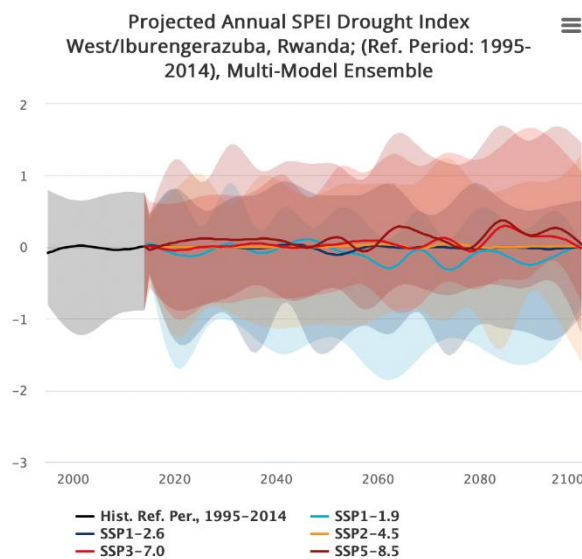


Figure 9. Western Rwanda CMIP6 model-model ensemble mean time series of the Standardized Precipitation Evapotranspiration index (SPEI) for a range of climate change scenarios across the 21st century. The SPEI is a measure of the integrated water deficit

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in a location, taking into account the contribution of temperature dependent evapotranspiration of a 12-month period. Source: World Bank⁴⁵

A more useful variable for assessing drought potential at daily-seasonal times scales is the Vapor Pressure Deficit (VPD). This is an absolute measure of the difference between the water vapor content of the air and its saturation value, and as such, represents a direct metric of the ability of the atmosphere to extract moisture from the land surface⁴⁶. High resolution VPD predictions for present day, mid-century and late-century under boundary conditions prescribed by the CESM model under the high emission RCP 8.5 scenario are shown in **Figure 10**. Annual cycles of VPD for model grid points high in the Nyungwe Forest and at lesser elevation in Kigali east of the CND both show remarkably little variation in VPD over the course of the century. Owing to the cool, persistently moist highland climate that characterize the CND landscape, the VPD values are consistently low in Nyungwe throughout the year for the three decadal time periods 2009-18, 2055-64 and 2090-99. This contrasts significantly with results from Kigali, where VPD registers four times higher than in Nyungwe during the mid-year dry season (20 vs. 5 hectopascal pressure deficit). This represents a favorable outlook over several decades to come for the CND in terms of limiting desiccating conditions that would elevate fire potential, stress crops and rapidly draw down hydrological resources. Importantly, however, not clearly evident in these model outputs is that rising cloud bases and associated warming temperatures increase the rate of drying of vegetation, increasing wildfire risk during rain-free periods, especially in lower reaches of forested terrain bordering settled areas where risks of anthropogenic ignition are highest.

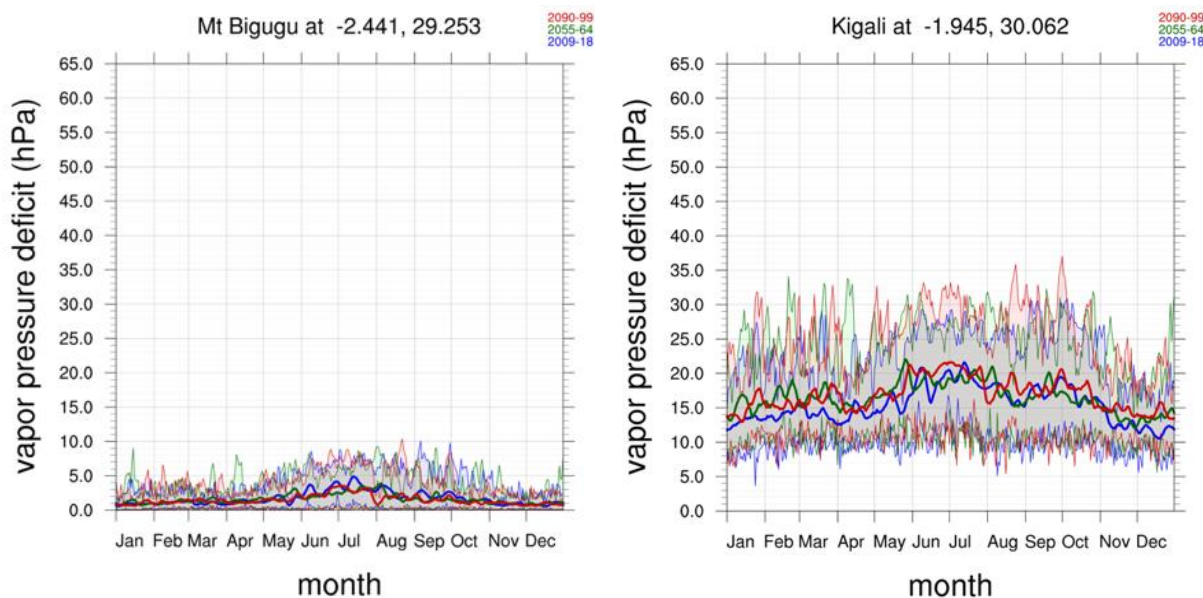


Figure 10. Modeled Vapor Pressure Deficit (VPD) on slope of Mt Bigugu in the Nyungwe Forest (**left**) and Kigali (**right**) for the recent past (2009-18, blue, mid-century (2055-64, green) and late-century (2090-99, red). Source: EAGLE project⁴⁷

Precipitation seasonality. The annual cycle of climate along the CND sees abundant rainfall for much of the year interrupted by a marked dry season in the June-August months when rainfall becomes light and sporadic, and during which hydrological deficits develop as evaporation far exceeds precipitation and resultant runoff. While highland orography largely explains the enhanced rainfall along the CND, the

⁴⁵ World Bank Climate Change Knowledge Portal. [Link](#)

⁴⁶ Seager, R., Hooks, A., Williams, A.P., Cook, B., Nakamura, J. and Henderson, N., 2015. Climatology, variability, and trends in the US vapor pressure deficit, an important fire-related meteorological quantity. *Journal of Applied Meteorology and Climatology*, 54(6), pp.1121-1141.

⁴⁷ East African Great Lake Environments (EAGLE) project. [Link](#)

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occurrence of the dry seasons occurs region wide irrespective of terrain elevation (**Figure 11**). This finds explanation in the seasonal north-south migration of a pan-equatorial band of enhanced cloudiness and rainfall known as the Inter Tropical Convergence Zone (ITCZ)⁴⁸. The slight reduction in rainfall experienced over several weeks along the CND in January and February reflects much more pronounced dryness to the north in Uganda; conversely, the more defined mid-year dry season along the CND spanning June through August is associated with enhanced southeasterly airflows that effectively push the zone of atmospheric convergence promoting rainfall across the equator into Uganda.

In this light, an unusual finding from the EAGLE project's high-resolution modeling studies is that the CND landscape has a heightened risk of major change in climatic seasonality. By the 2055-64 period: the mid-year dry season effectively disappears. Being the output of a single model run under a high-end greenhouse gas emissions scenario may make this seem implausible; It is highly dissimilar from mid-century predictions of the consensus of global models, which sustain the dry season. It finds corroboration, however, in a new study on East African rainfall, which found that the parameterization of convection in global models may be a source of uncertainty for projections of changes in seasonal timing, and that potentially impactful changes in seasonality are therefore quite plausible⁴⁹. Such an eventuality is exceptionally difficult to plan for, and since it would only come to fruition well after the project period, if at all, additional studies could be conducted to provide needed guidance.

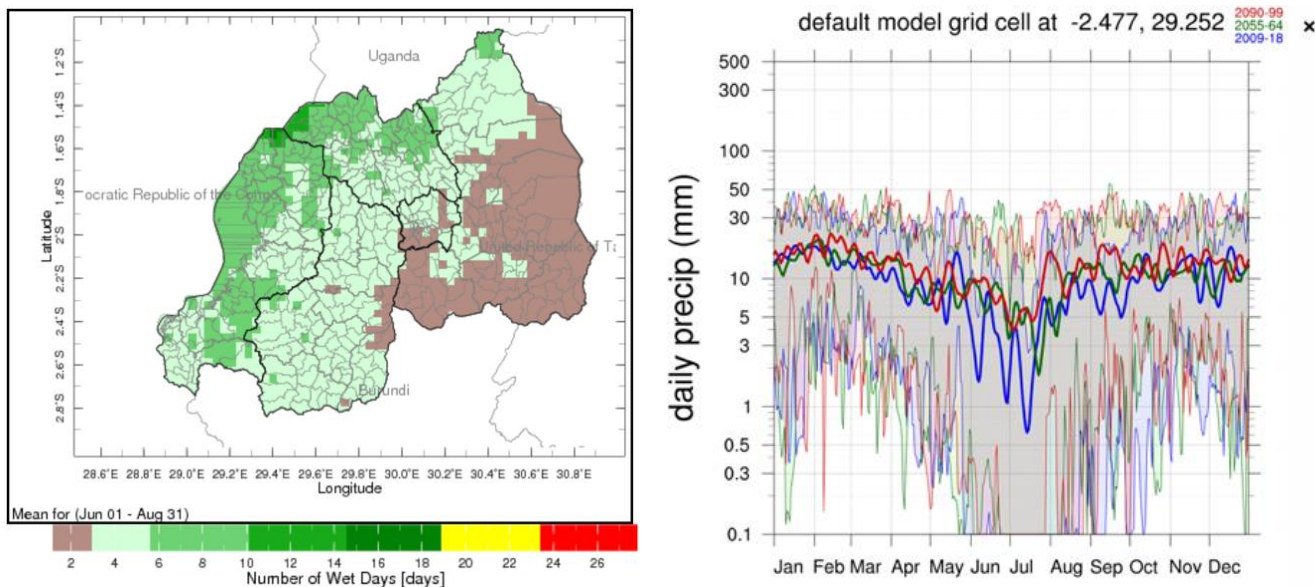


Figure 11. (left) The average number of “wet” days in Rwanda with rainfall exceeding 3 mm for the months of June-August. The wettest parts of the CND only receive significant rainfall on ~10% of days during the dry season. Source: Meteo Rwanda⁵⁰; **(right)** Annual cycle of rainfall rate (mm/day, note logarithmic scale) at Nyungwe Forest National Park showing daily means (thick lines) and extremes (thin lines) for 10-year time periods centered on 2014 (blue), 2060 (green) and 2095 (red). The dry season evident in the 2014 mean reaching below 1 mm/day, effectively disappears by the 2060 time period. Source: EAGLE project⁵¹

⁴⁸ Siebert, A., Dinku, T., Vuguziga, F., Twahirwa, A., Kagabo, D.M., delCorral, J. and Robertson, A.W., 2019. Evaluation of ENACTS-Rwanda: A new multi-decade, high-resolution rainfall and temperature data set—Climatology. *International Journal of Climatology*, 39(6), pp.3104-3120.

⁴⁹ Wainwright, C.M., Marsham, J.H., Rowell, D.P., Finney, D.L. and Black, E., 2021. Future changes in seasonality in East Africa from regional simulations with explicit and parameterized convection. *Journal of Climate*, 34, pp.1367-1385.

⁵⁰ ENACTS online tool at Meteo Rwanda. [Link](#)

⁵¹ EAGLE project: [Link](#)

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• Climate risks, vulnerability and impacts

Without strategic actions to confront mounting stresses borne by climate change interacting with unsustainable land-use practices, the CND would doubtless undergo a profound transformation over coming decades with significant loss of biodiversity and ecosystem services, reducing agricultural potential, severe hazards to humanity and loss of livelihoods. Temperature increase and loss of fog immersion will cause desiccation of the lower margins of protected forests promoting die-offs, arrival of pests, pathogens and invasive species, and increasing fire ignition potential. Current cultivation practices of both subsistence crops and valuable cash crops like tea and coffee will be drawn uphill, adding to pressure to convert protected forests to farmland. Whereas climatic changes will act as inexorably mounting stressors, unsustainable land surface conversion and land use practices will greatly exacerbate their impacts. The steep hillsides that characterize much of western Rwanda will experience intensifying erosion and risk of landslides, while built infrastructure such as road and bridges will be overwhelmed since they were engineered to withstand rainfall intensities expected under past climatic conditions that are sure to be exceeded, and with increasing frequency, as time progresses and temperatures steadily increase. Precipitation intensity will increase for short-period rainfall rates and individual storm totals, especially for the fraction of higher-end events that can cause flash flooding and promote landslides. This is certain to occur, is probably already discernible, and likely to amplify very significantly over time as warming progresses. This ties directly to landslides in particular, which already cause high mortality and significant destruction each year, and the CND's elevated vulnerability due to high topographic relief and deforested slopes makes this hazard of paramount concern to address (**Figure 12**).

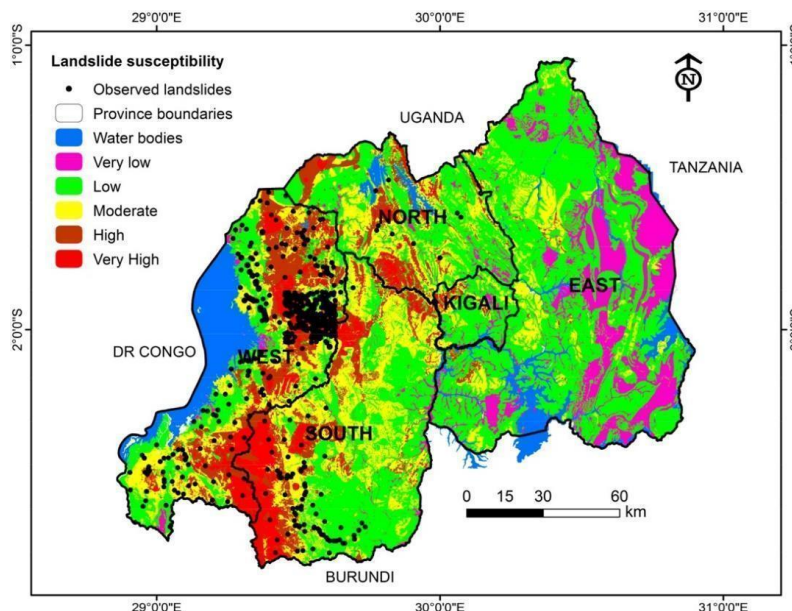


Figure 12. Rwanda landslide susceptibility map showing past observed landslides. Source: Nsengiyumva et al., 2018.⁵²

Significant temperature increases are almost certain to continue for decades to come, and will have mounting impacts on both natural and human systems along the CND, and some are already discernible. Over equatorial Africa observed environmental lapse rates feature temperature decreases by approximately 5.5 °C per kilometer of elevation increase⁵³, so 2 °C of warming expected by mid-century translates to a vertical rise on the order of 473 meters, promoting large uphill displacements of a wide range of organisms,

⁵² Nsengiyumva, J.B., Luo, G., Nahayo, L., Huang, X. and Cai, P., 2018. Landslide susceptibility assessment using spatial multi-criteria evaluation model in Rwanda. *International journal of environmental research and public health*, 15(2), p.243.

⁵³ Camberlin, P., 2018. Climate of Eastern Africa. In *Oxford Research Encyclopedia of Climate Science*. [Link](#)

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ecological processes, cultivars and human activities. While temperatures over the CND region are not expected to rise to levels for heat stress to elevate mortality⁵⁴ or reduce labor productivity, the indirect effect of rising temperatures through increased disease incidence is substantial⁵⁵.

Western Rwanda is the most productive region in the country for the cultivation of both tea and coffee, where conditions are optimized on the CND's sloping terrain. Both are cash crops with high export value, and are the target of active expansion through both governmental programs and private enterprise. Intensifying environmental stresses from climate change particularly due to thermal increases and the arrival of invasive pests from lowland regions, present challenges both for current plantings and in planning for expansion to new areas; the same concerns apply to other cultivars.

Cloud base altitude, which is highly influential in montane forest ecology and species composition, is likely already elevated above recent past conditions due to conversion of forests to agricultural lands, and certain to rise further in proportion to the degree of warming over coming decades. Ecosystems and species assemblages in the lower reaches of protected forests may already be out of balance with this important environmental variable, and this imbalance will only increase over time, promoting rapid species turnovers, enhancing potential for invasives, and act as a strong driver⁵⁶ of upward range extensions. Rising cloud bases in concert with warming temperatures also increase drying of vegetation, enhancing wildfire risk during rain-free periods. Fog interception by vegetation also augments montane forest precipitation by an estimated 10%, so rising cloud bases will act to remove this important hydrological input too. Highland forest loss therefore represents a local loss to the ecosystem service of rainwater provision that could be partially restored through reforestation.

Such challenges highlight how essential highland forests are to the climatic resilience of CND communities, for the ecosystem services and products they provide for both the region's vulnerable communities and the national economy. They also recharge aquifers; regulate water flow; control flooding; retain soil; provide wood fuel energy and timber; underpin the country's tourism, which provides the largest contribution to Rwanda's foreign exchange earnings⁵⁶; and provide wider benefits of atmospheric pollution control that sustain the country's economy and the wellbeing of its people⁵⁷.

Forest conversion to farmland in the CND highlands has until recently served as a release valve for lowland population pressure at the expense of drastically reduced national carbon stocks, diminished resilience to climate change, enhanced potential for destructive outcomes, reduced ecosystem service provision and disrupted biological connectivity. The resultant loss of ecosystem services and functioning and intensifying environmental stresses borne by climate change now require redirection towards forest restoration and other actions to ensure a sustainable future. Safeguarding the Congo-Nile Divide's remaining highland forests, and setting long-term goals of reforesting functional linkages connecting them, therefore serve Rwanda's national long-term interests on lessening the severity of impacts of climate change and sustaining critical ecosystem services both locally and downstream, while contributing to global efforts to draw down greenhouse gas concentrations.

B.1 (b). Summary of spatial biodiversity assessment

The majority of Rwanda's remaining montane forests are restricted to the national parks, which support a variety of threatened and endemic species. A few fragmented forest patches are situated outside of the National Park boundaries. Landcover change (especially for widespread smallholder agriculture), fuelwood

⁵⁴ Asefi-Najafabady, S., K. Vandecar, A. Seimon, P. Lawrence and D. Lawrence, 2018: Climate change, population and poverty: vulnerability and exposure to heat stress in East Africa. *Climatic Change*, 148, 561-573.

⁵⁵ World Bank, 2022: Rwanda Country Climate and Development Report. World Bank, Washington DC. [Link](#)

⁵⁶ Rwanda Development Board 2017 Annual Report [Link](#)

⁵⁷ Andrew, G. and Masozera, M., 2010. Payment for ecosystem services and poverty reduction in Rwanda. *Journal of sustainable development in Africa*, 12(3), pp.122-139.

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harvesting, and human-induced fires, coupled with climate change impacts, especially landslides, erosion and downstream flooding, have compromised the delivery of critical ecosystem services derived from these forests. A detailed spatial analysis of biodiversity in the Congo Nile Divide was conducted to delineate priority areas for the long-term conservation and restoration of forests, and the sustainable management of landscapes, in order to secure the ecosystem services needed to improve the resilience of vulnerable communities to climate change impacts.

The spatial biodiversity assessment for Rwanda's Congo-Nile Divide (CND) (see details in Annex 2 and Annex 2.2) is based on a rapid systematic conservation plan, using MARXAN decision-support software. The key analyses involved identifying and mapping the remaining areas of natural forest and other ecosystem types, modeling bioclimatic change and identifying climate change refugia, a Condatis landscape connectivity and bottlenecks analyses, and evaluating ecosystem threat and protection levels. The analyses thus build in landscape connectivity, climate change refugia, biodiversity values, ecosystem services and social costs.

The final outcome of the MARXAN and Condatis connectivity analyses split the CND domain into four major landscape categories with a set of priority implementation areas, each with their own place-bound project interventions. Interventions range from the restoration of natural forest, establishment and improvement of protective forest on steep slopes and along riparian areas; and to implement biodiversity-friendly agroforestry to reduce landslides, erosion and downstream flooding. Over and above these place-bound interventions are a variety of other mechanisms for promoting the sustainability of rural livelihoods and protecting montane forest in Rwanda's CND landscape.

The Systematic Conservation Plan analyses covered a range of biodiversity features, including:

- Terrestrial ecosystems, including their IUCN Red List threat status and protection level. The analysis focussed on identifying priority remaining intact areas, based on the development of a map of ecological condition.
- Protected Areas, Protected Forests and Protected Wetlands. All the identified ecosystems to be gazetted for protection are included.
- Climate change refugia based on projected changes in bioclimatic envelopes under a range of climate change scenarios.
- Identification of key landscape linkage areas and bottlenecks.
- Hydrological process areas – Rivers and Streams, including buffers.
- Hydrological process areas – Wetlands and Lakes, including buffers.
- Hydrological process areas – Areas with high rainfall.
- Landscape process areas - Steep slopes (over 55%) which are most important for minimizing erosion and landslide risk.

Individual species were not separately considered in the assessment as all key species are tightly linked to their underlying intact habitat (in particular intact forest patches as well as wetlands and riparian areas); and available distribution data for species was much broader than the ecosystem data. In selecting priority conservation areas, the SCP methodology always attempts to be spatially efficient by meeting conservation targets in as small an area as possible, while avoiding conflict with other land users, at the lowest possible cost for other sectors. The following was taken into account in the prioritization:

- Urban and dwellings, tea, roads, plantations, cultivated pasture, coffee, bamboo and agriculture.
- Areas of greatest population density.
- Areas with highly impacted land cover classes.

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Results of the MARXAN spatial analysis:

The MARXAN landscape prioritization (**Figure 13**), which builds in landscape connectivity, climate change refugia, biodiversity values, ecosystem services, and social costs (in terms of avoiding, where possible, areas with highest population density, agriculture etc), splits the landscape into four key planning categories (**Figure 14**):

- **Core Protected Area (PA) Nodes:** National Parks comprise the “Core PA Nodes” that need to be secured and well managed, which include Volcanoes, Gishwati-Mukura and Nyungwe National Parks. Priority activities include strengthening PA management and sustainability, rehabilitation and restoration of natural forests, other conservation-oriented land use activities that reduce stress on PAs and natural forests (e.g. improved wood stove efficiency to reduce pressure on natural forests) and supporting sustainable biodiversity compatible activities (e.g. improved beekeeping). These nodes also include buffer areas around the National Parks.
- **Stepping Stones:** These are priority nodes outside of the current National Parks that are critical for maintaining landscape connectivity, consisting of small, isolated patches of forest, at Dutake and Karehe-Gatuntu Protected Forests and the extensive Gishwati Pastures. These areas would be a sensible focus for some (patches of) forest restoration and protection, beekeeping and energy efficient stoves. The Gishwati Pastures are a focus for agroforestry on pastoral land to increase the coverage of native trees to secure reasonable landscape connectivity for forest species.
- **Landscape linkages:** These are key landscape linkages and knickpoints in the farming landscape that require afforestation on steep slopes and riparian areas to link the CND at a landscape scale. Compatible land use activities include agroforestry, increasing the use of native species, reforesting steep slopes, beekeeping and energy efficient stoves. **Note that the analysis is based on some level of improved connectivity via patches of protective forest and riparian strips, as well as overall improved species composition and tree coverage (within mixed agroforestry systems). It does not imply a continuous natural forest corridor; as this potential no longer exists, and its creation would not be possible in this highly used landscape without unacceptable impacts on livelihoods.**
- **Broader Farming Mosaic:** These are broader areas of moderate priority where conservation interventions can support broader sustainable landscapes and ecosystem service delivery but are likely to be beyond the scope of most project interventions except for those linked to land use planning.

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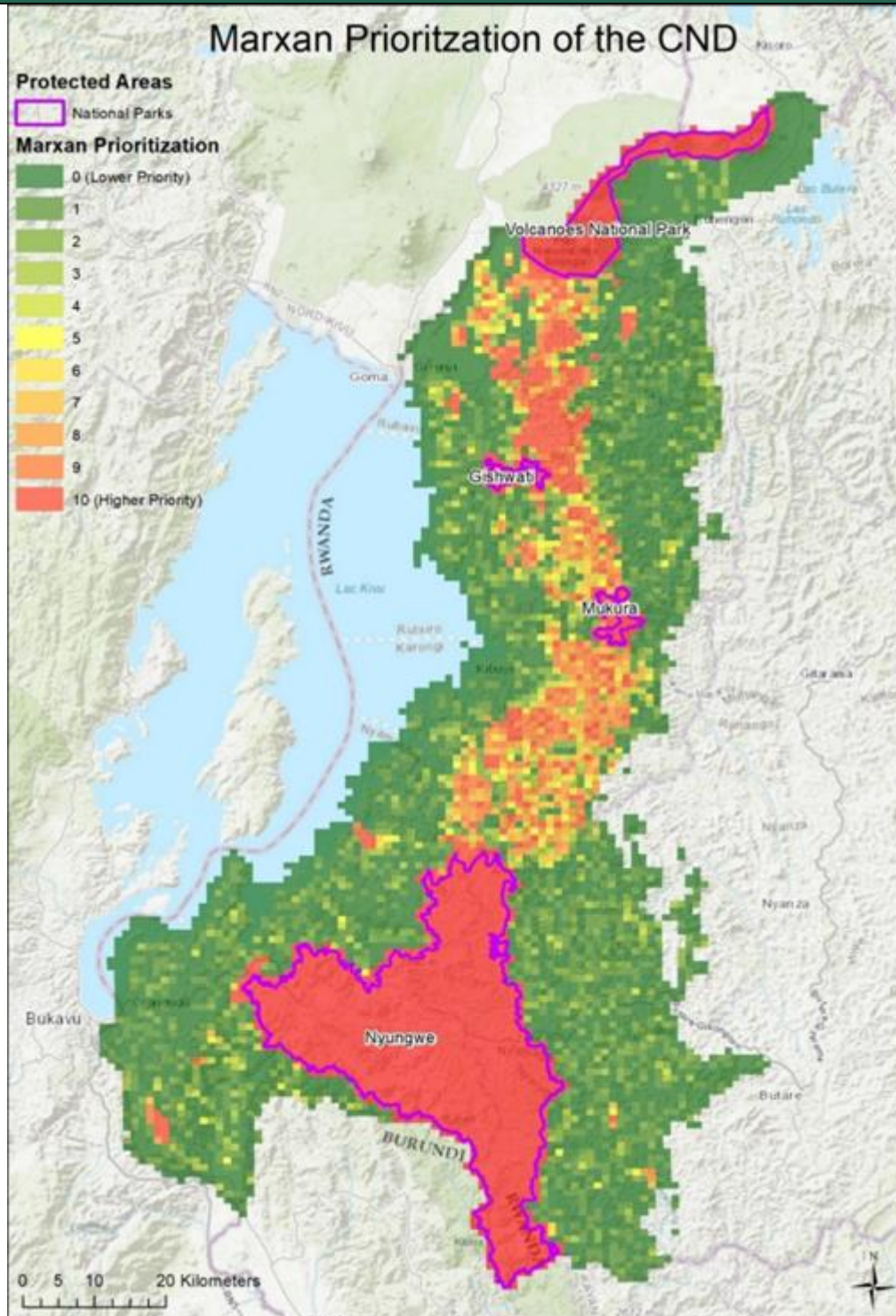


Figure 13. The MARXAN irreplaceability map for the Congo Nile Divide showing the landscape prioritization, ranging from high priority in red to low priority in green.

To aid prioritization and description of the landscape categories above, the four categories were further split into specific planning areas, referred to as “Landscape Implementation Sectors”, as shown **Figure 14** and in **Table 2** with a description of each planning category and associated ideas, required outcomes, core benefits and associated benefits in **Table 3**.

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Table 2. Summary table of landscape categories and associated implementation sectors for the CND.

Landscape Category	Landscape Implementation Sector	Area (ha)
Core PA Nodes	Volcanoes NP and "Buffer" ⁵⁸	19 487,0
	Nyungwe NP and Buffer	116 794,9
	Mukura NP and Buffer ⁵⁹	4 713,5
	Gishwati NP and Buffer ⁵⁹	4 013,7
Stepping Stones	Gishwati Pastures Stepping Stone	15 547,4
	Karehe-Gatuntu Stepping Stone	401,1
	Dutake Stepping Stone	903,0
Landscape Linkages	Nyungwe NP to Mukura NP Linkage	23 375,4
	Gishwati NP to Volcanoes NP Linkage	5 014,8
	Mukura N to Gishwati NP Linkage	7 823,7
Broader Farming Mosaic	Nyungwe to Mukura Broader Farming Mosaic	21 164,4
	Volcanoes Broader Farming Mosaic	11 694,4
	Mukura Broader Farming Mosaic	4 814,0
	Gishwati Broader Farming Mosaic	28 433,2
	Nyungwe Broader Farming Mosaic	11 437,0

Priority Areas for Implementation Activities:

Specific priority areas for a range of interventions were spatially identified, including:

- Natural forest rehabilitation and restoration with the National Parks (and their buffers) and in the Stepping Stones.
- Protective forests for steep slopes and riparian areas, particularly in the Landscape Linkages.
- Agroforestry interventions in key highland Landscape Linkages.
- Specific agroforestry priorities on pastoral land (silvo-pastoral practices).
- Indigenous shade trees for tea and coffee plantations.
- Beekeeping in National Park buffers and Stepping Stone buffers.

⁵⁸Note that buffers refer to both the specific legally designated buffers around Nyungwe and Gishwati-Makura NPs, as well as broader buffer areas adjacent to all these NPs, as well as Volcanoes NP.

⁵⁹In order to allow for specific landscape description, we separately refer to Gishwati NP and Makura NP where necessary, even though these areas are managed as Gishwati-Makura NP.

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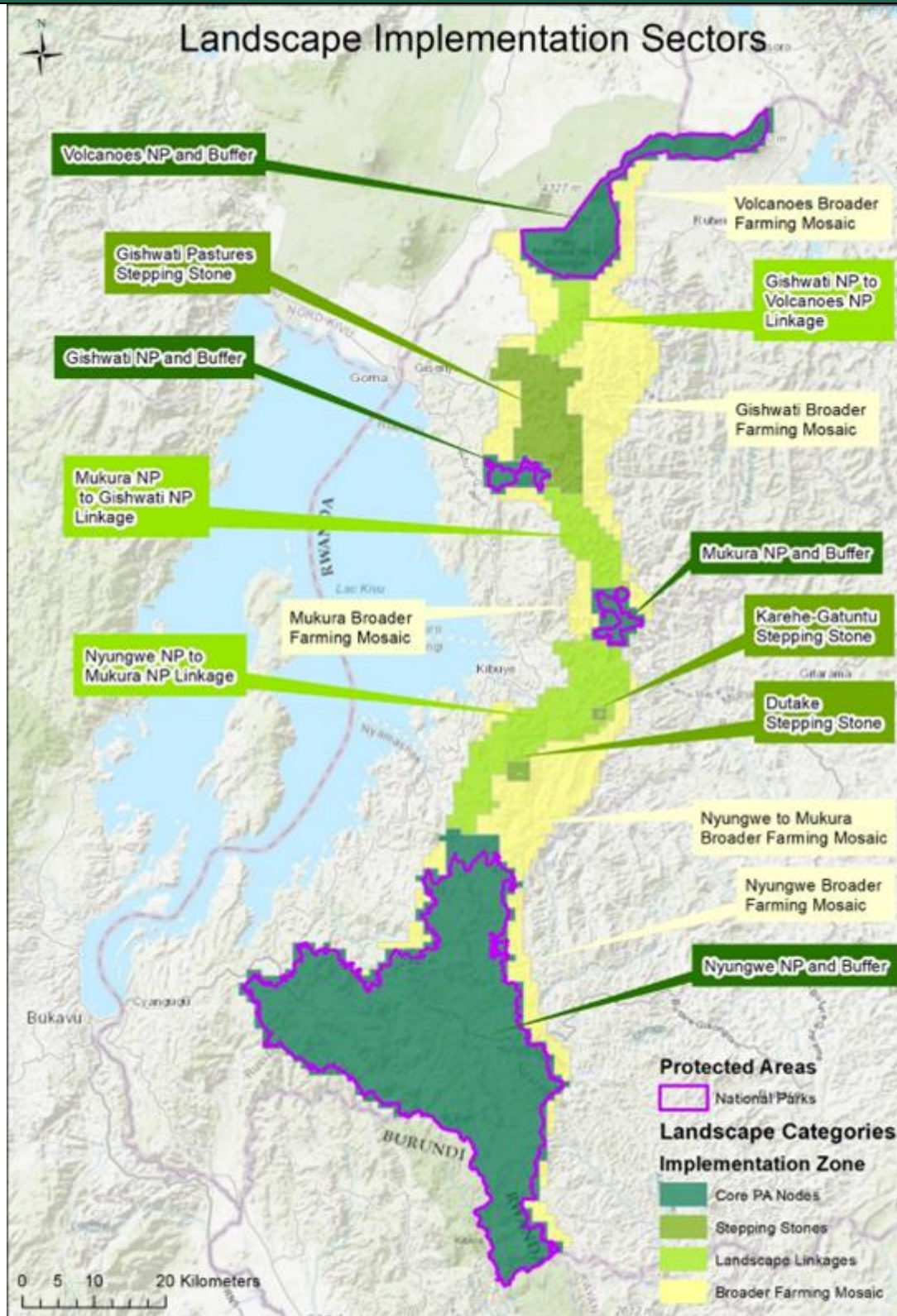


Figure 14. The four major landscape categories (Core PA Nodes, Stepping Stones, Landscape Linkages and the Broader Farming Mosaic) were split into specific areas to aid prioritization and description.

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Table 3. A description of the four key landscape planning categories and associated ideas, required outcomes, core benefits and associated benefits.

Landscape Component	Core PA Nodes The National Parks	Stepping Stones Priority nodes outside of current Pas	Landscape linkages Key landscape linkages and knickpoints in the farming landscape	Broader Farming Mosaic The broader landscape mosaic	Institutional issues
Description	The current core national parks and protected forests.	Critical pieces of biodiversity outside of the PAs required for landscape connectivity, maintenance of biodiversity and delivery of ecosystem services.	The parts of landscape within identified key corridors, where functional connectivity and ability to deliver ecosystem services needs to be urgently maintained or improved.	Remaining farmland areas of the CND.	The non-geographic specific elements of the CND system
Key ideas	Protect and manage for climate resilience.	Restore and protect to ensure landscape connectivity and ecosystem service delivery.	Functional linked farming landscapes delivering ecosystem services.	Diverse climate resilient farmland delivering ecosystem services.	Strong, well-capacitated and equitable environmental governance and land use planning.
Required Outcome	PAs effectively protect and manage natural forests improving resilience to climate change impacts and risks. Natural forests protected, connected, more resilient to climate change impacts and risks.	Critical landscape nodes / stepping-stones are secured and where necessary restored to a natural state.	Priority portions of the farming landscape are specifically managed to improve overall connectivity and ecosystem service delivery.	Sustainably managed farmland landscape is more biodiverse, supports delivery of ecosystem services and is resilient to climate risk.	Government and civil society are well capacitated to ensure robust landscape planning that supports climate resilience.
Core benefit	Maintain globally significant, species-rich natural forests. Core areas secure best possible source and/or refuge areas for species under climate change.	Biodiversity value of critical landscape nodes is maintained. Landscape connectivity supported through retention of key stepping-stones for species movement across the landscape.	Improved connectivity of the landscape ensures long term climate resilience. Value of core PAs and priority nodes is retained (i.e. the inevitable degradation of sites due to isolation is avoided).	Generally improved farmland management ensures rural sustainability and supports livelihoods.	Integrated land use plans, with community participation and spatial planning tools/ monitoring
Associated benefits	Improved delivery of ecosystem services (especially water) and support of rural economies and livelihoods.	Improved delivery of ecosystem services (especially water) and support of rural economies and livelihoods.	Improved delivery of ecosystem services (especially water) and support of rural economies and livelihoods.	Improved delivery of ecosystem services (especially water) and support of rural economies and livelihoods.	Cross sectoral planning and management.

This study was a rapid assessment to support project proposal development and does not replace a full conservation planning process. There are significant additional steps which are required to develop a product that is useful for land use planning during project implementation. These changes include a robust stakeholder engagement process, at a national, district and local scale; incorporation of issues relating to

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land use rights, both of landowners and farm tenants; incorporation of issues relating to social safeguards, especially for marginalized groups; inclusion of issues related to planning processes and strategies, at a national and local scale; finer scale planning (ideally at a 1:50 000 scale); and improved biodiversity data, including revised data on forest degradation, validation of the ecological condition map, and specific species data where possible.

B.2 (a). Theory of change narrative and diagram

The theory of change for this Project is founded on several key principles. Central to all, the CND represents a complex landscape – a geospatial mosaic of forest and farming patches, each dependent on the other. These coupled, biological and socio-economic systems are under great strain due to intense land pressure, which has resulted in a negative cycle of natural resource degradation that undermines the well-being of both natural and socio-economic systems. The advent of climate change is a forceful driver that intensifies this cycle of degradation, with the resultant loss of key ecosystem services that importantly link forests and people. This cycle must be disrupted in order to chart a path toward climate resilience for natural and human systems. Ecological connectivity must be established within this mosaic of land uses to promote landscape heterogeneity to help build resilience to climate change of forest ecosystems and people who depend on them. The approach to be taken by this Project is therefore based on principles of ecosystem-based adaptation: that healthy and resilient ecosystems form an essential foundation for adaptation to climate challenges⁶⁰. Climate-aware management of both natural and human-dominated ecosystems – interacting and integrated at a landscape level – can break the cycle of degradation and instead lead to a positive sequence of mutually reinforcing, climate-smart, sustainable, and resilient benefits..

In sum, the goal statement of the Project is:

IF Rwanda's CND landscape of native forests and neighboring farmlands is effectively managed for climate resilience

THEN the CND will comprise an interconnected and interdependent set of globally significant, species-rich natural forests in a matrix of sustainable farmlands, acting as buffers against damage from extreme weather events: assuring landscape integrity, stability and adaptability, providing vital ecosystem services and improving livelihoods for vulnerable people and the nation at large and contributing importantly to the conservation of biodiversity, the national economy, and mitigation of GHGs

BECAUSE building the capacity and management of integrated, climate-resilient forest and farming systems will ensure natural forest perpetuity and break and reverse the climate change-induced intensifying cycles of drought, flooding, and landslides that lead to degradation of livelihoods and resultant increased pressure on forest resources.

The status and future of the CND's forests and people are tightly intertwined: both heavily dependent on the natural resources and processes that provide a foundation for their ability to thrive. They represent ecological and socio-economic systems that are importantly coupled in a complex landscape, linked in particular by the ecosystem services that flow throughout these systems (hydrological and microclimate regulation, soil conservation, and nature-based tourism). At present these services, and the consequent relationships between forests and people – especially those who are most vulnerable in Rwandan society – are deeply

⁶⁰ Scarano, F.R. et al, 2017. *Ecosystem-based adaptation to climate change: concept, sustainability and a role for conservation science*, Perspectives in Ecology & Conservation 15(2), 65-73.

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strained⁶¹. Forests have been reduced in extent, fragmented, and degraded by fire⁶², overuse of resources⁶³, and introduction of exotic species⁶⁴. The sharply sloped rural lands of the CND have been farmed more and more extensively and intensively on steeper and steeper slopes^{65,66} and throughout riverine bottomlands, further reducing tree cover outside forest blocks: thereby further compromising the ability of the land to retain soils⁶⁷, avoid flooding⁶⁸, provide consistent and healthy water⁴⁴, ameliorate microclimate⁴⁹, store carbon⁶⁹, and remain productive. The coupling of these stressed systems has led to a negative feedback cycle of insecurity and constrained the well-being of farmers. Having little access to alternative sources of livelihoods or capital to enable change⁷⁰, smallholder farmers have consequently increased pressures on natural forests for land, fuel, and other forest resources⁴⁶. Such pressures in turn cause further forest loss and degradation and decline in biodiversity – reducing their ability to provide essential ecosystem services to people living in the landscape and across the nation, including tourism revenues and employment that are of national import. Thus, these systems are caught in a negative cycle that has proven difficult to break, despite significant sector-specific efforts. One such pressure – demand for fuelwood – is quite substantial given that a recent Government of Rwanda (GoR) census found that 77% of Rwandan households use wood as their primary fuel for cooking⁷¹.

With the advent of climate change, these already strained systems are showing clear signs of breakdown. Increased variability in the timing and intensity of rains along with rising temperatures are causing increased landslides⁷² and soil erosion⁵⁰, increased desiccation and drought⁵⁰, and unpredictable timing for agricultural plantings and harvest⁷³. This is reducing farm productivity and security, driving even greater pressures on the remaining forested estate, thereby compromising further the essential ecosystem services that are necessary to such well-being. At a larger scale, climate-induced forest degradation and decline of biodiversity in the CND compromise national-level benefits of water provision and quality¹, nature-based tourism⁷⁴, the supply of hydroelectricity,^{7,75} and carbon storage¹⁰ [see diagram A below].

61 Bagstad, K.J., 2019. *Towards ecosystem accounts for Rwanda: Tracking 25 years of change in flows and potential supply of ecosystem services*. British Ecological Society.

62 Masozera, A.B., Mulindahabi, F., 2007. *Post-Fire Regeneration in Nyungwe National Park*, Rwanda. Wildlife Conservation Society.

63 Republic of Rwanda, 2011. *Green Growth and Climate Resilience: National Strategy for Climate Change and Low Carbon Development*.

64 USAID, 2019. *Rwanda Tropical Forests and Biodiversity Analysis*.

65 Camberlin, P., 2018. *Climate of Eastern Africa*. In Oxford Research Encyclopedia of Climate Science.

66 Seimon, A., 2022. *An Overview of Climate Change and its Impacts along the Congo-Nile Divide in Rwanda*.

67 World Bank, 2019. *Rwanda Systematic Country Diagnostic*.

68 Karamage, F., et al., 2017. *Modeling Rainfall-Runoff Response to Land Use and Land Cover Change in Rwanda (1990–2016)*. *Water* 2017, 9(2), 14.

69 Mugabowindekwe, M., Brandt, M., Chave, J. et al. Nation-wide mapping of tree-level aboveground carbon stocks in Rwanda. *Nat. Clim. Chang.* 13, 91–97 (2023). <https://doi.org/10.1038/s41558-022-01544-w>

70 Clay N., King B., 2019. *Smallholders' uneven capacities to adapt to climate change amid Africa's 'green revolution': Case study of Rwanda's crop intensification program*. World Dev.

71 Government of Rwanda, 2021. *Rwanda Household Survey 2019/2020*. National Institute of Statistics of Rwanda.

72 Uwihirwe, J., et al., 2020. *Landslide precipitation thresholds in Rwanda*. *Landslides*.

73 World Bank; CIAT. 2015. *Climate-Smart Agriculture in Rwanda*. CSA Country Profiles for Africa, Asia, and Latin America and the Caribbean Series. Washington D.C.: The World Bank Group.

74 Rwanda Environment Management Authority (REMA), 2015. *Rwanda State of Environment and Outlook Report*.

75 Rwanda Environment Management Authority (REMA), 2009. *Rwanda State of Environment and Outlook Report*, Chapter 8: Energy Resources.

(b) PROJECT/PROGRAMME INFORMATION

Diagram A. Climate impacts & risks intensify negative feedback loop for people & forests

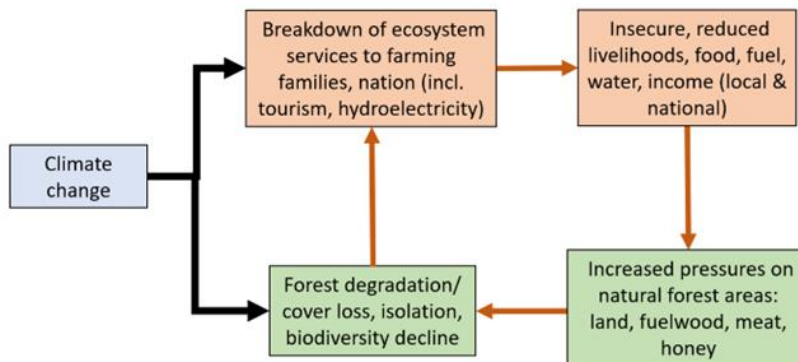
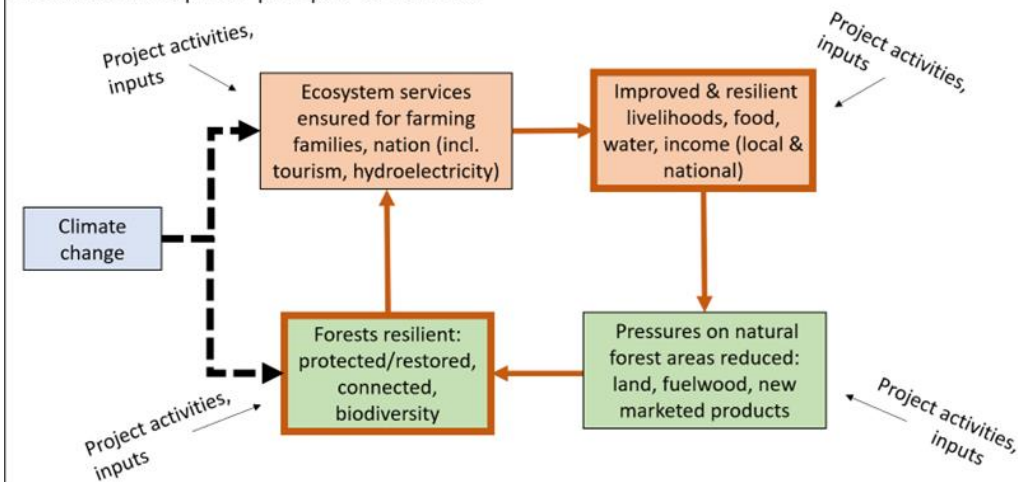


Diagram B. Adaptation to climate impacts & risks promotes positive feedback loop for people & forests



This context calls for new mechanisms of land-use planning and management that are based on consideration of the CND as a landscape, consisting of an inter-dependent mosaic of forests and farms, the management of each determining the health of the whole. Focusing on this larger scale, with coordination across geographic and sectoral boundaries, while mainstreaming climate risks into management planning will be key – and the foundation for a paradigm shift. Use of this framework to design and apply practical, precedent-setting Project actions in-forest and on-farm is expected to shift mindsets and management systems. This, in combination with establishing durable sources of financing, is thereby intended to shift the natural and human ecosystems themselves toward sustainable climate resilience. [see diagram B above, and the ToC (Theory of Change) diagram below.

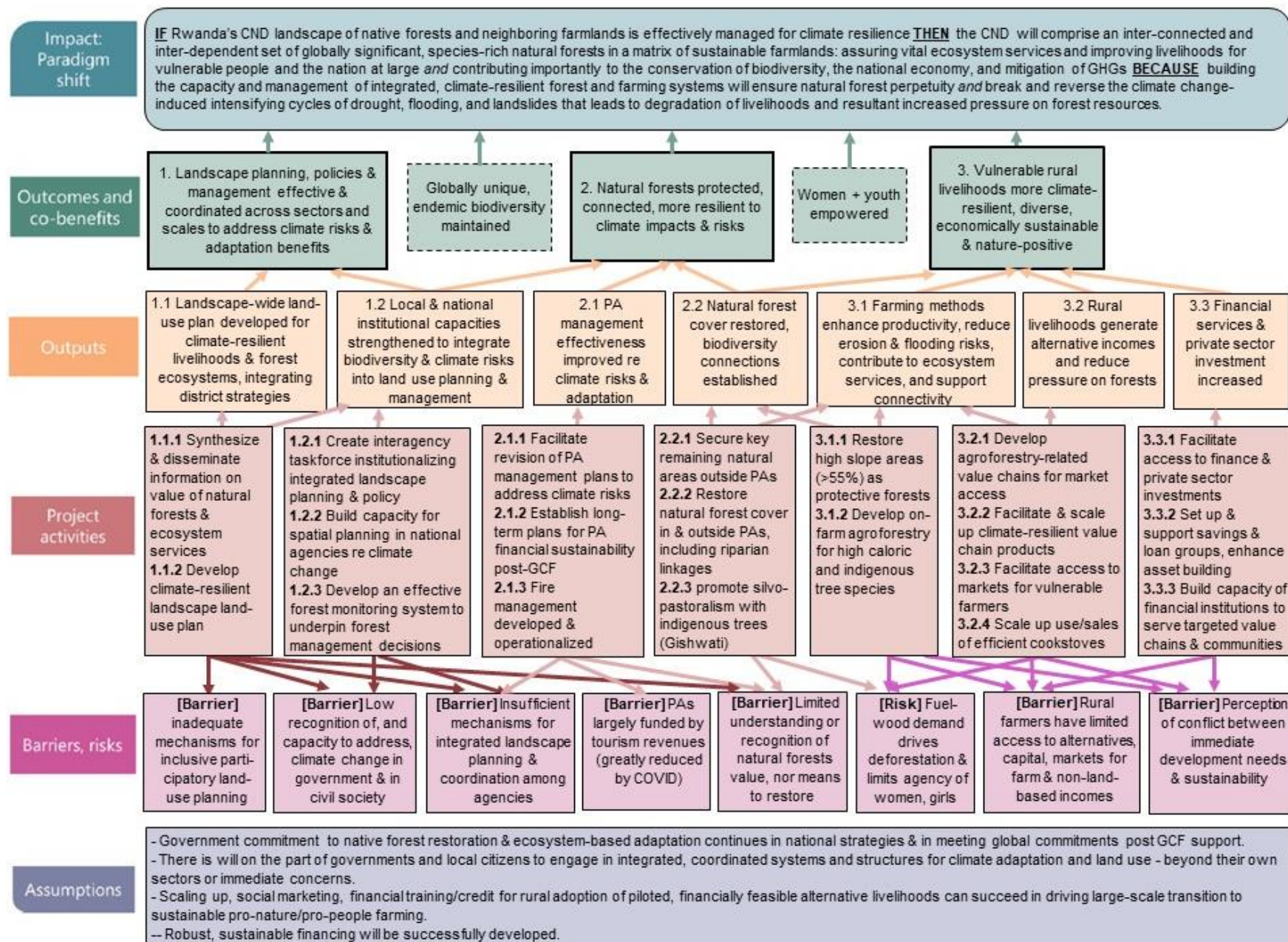


Figure SEQ Figure * ARABIC 15. Theory of change

Outcome 1. Landscape planning, policies and management are effective and coordinated across sectors and scales to address climate risks and adaptation benefits. Major barriers currently stand in the way of moving in this direction, as outlined in the ToC diagram. ***Awareness, understanding, attention, and action regarding climate risks to natural resources are currently insufficient.*** Planning and adaptation to climate impacts are constrained by a lack of information, site-focused modeling, spatial analysis, and real-time monitoring. Where some capacity exists, it ***operates within sector initiatives rather than across them.*** It also proceeds largely within governmental programs, with ***mechanisms for inclusive land-use planning inadequate***, not fully engaging rural peoples – particularly women and youth. Because of these barriers, coordination of landscape planning and management is weak across governmental and non-governmental sectors (agriculture, forestry, national parks, economic development), scales (local, district, national), and actors (women, youth, smallholder farmers, commercial businesses, as well as governmental agencies). Activities outlined in **Component 1**, aimed at integrated landscape land-use planning and the capacity for effective management, are intended to help ensure ecosystem services persist, and thereby align the ties between forests and people as they are faced with climate challenges. Activities under this component will provide the government and public with up-to-date information on climate adaptation (1.1.1), build agency capacities for incorporation into land-use planning – particularly spatial analysis and forest monitoring (1.2.2, 1.2.3), establish mechanisms for integrated landscape planning – including public participation and operations of an inter-agency task force (1.1.2, 1.2.1), and facilitate the creation of a climate-smart, integrated CND land-use plan (1.1.2).

Outcome 2. Natural forests are protected, connected, more resilient to climate impacts and risks. Particular to the CND, another key barrier is that ***natural forests are as yet undervalued and under-resourced*** despite the fact that they play vital roles in stabilizing natural and human CND ecosystems^{76,77}. The national parks of the CND (Volcanoes, Nyungwe, and Gishwati-Mukura) are globally significant regarding their outstanding species richness, levels of endemism, and presence of endangered and rare species⁷⁸. Although these are well-recognized for their tourism potential, and strong governmental policy and practice exists to protect them, barriers to their full valuation remains: their ***ecosystem services are yet to be fully recognized*** or financially supported, and their ***support is vulnerable to global tourism trends*** (with COVID-reduced tourism as a current example). As climate change considerations enter into landscape management, the values of these natural forests for fog interception, precipitation infiltration, soil retention, drought mitigation, and hydrological balance are only now becoming more prominent. In order to provide such services into the future, Nyungwe and Gishwati-Mukura NPs are in need of restoration and connection^{79,80}. Yet the understanding and capacity for recovering their richness and function is nascent at best, particularly in light of climate change. Activities outlined in **Component 2** are therefore designed with forest sustainability at the core. To build climate considerations into forest planning and management, the Project will facilitate revision of park management plans (2.1.1) that enhance dimensions of fire prevention and management (2.1.3), and will support in-park forest restoration (2.2.2). Establishment of restored forest at park boundaries (2.2.2), riparian linkages outside the parks (2.2.2), and forested “stepping stones” that

⁷⁶ Gatwaza O.C., Wang X., 2021 *Mapping of biodiversity hubs and key ecosystem services as a tool for shaping optimal areas for conservation*. PLOS ONE 16(8): e0253151

⁷⁷ Andrew, G., Masozera, M., 2010. *Payment for Ecosystem Services and Poverty Reduction in Rwanda*. Journal of Sustainable Development in Africa (V12, No.3).

⁷⁸ This area is also significant for global biodiversity – with the Albertine Rift being one of the most biodiverse regions within Africa, including conservative estimates of at least 980 endemic species, 15 Critically Endangered species, 34 Endangered species, and 99 Vulnerable species. Source: Plumptre, A.J., et al. 2016. *Conservation Action Plan for the Albertine Rift*. Wildlife Conservation Society.

⁷⁹ Nyandwi, E., Mukashema, A., 2011. *Excessive Deforestation of Gishwati Mountainous Forest & Biodiversity Changes*. Participatory Geographic Information Systems (P-GIS) for natural resource management and food security in Africa.

⁸⁰ Ordway, E., 2015. *Political shifts and changing forests: Effects of armed conflict on forest conservation in Rwanda*. Global Ecology and Conservation.

allow for forest species to move between parks (2.2.1, 2.2.3) is intended to enhance biodiversity and forest functions across the CND landscape. Fundamental to forest sustainability as well is sustainable long-term financing, which will be addressed by providing technical support from the Project to RBD and REMA to assess various financial instruments (conservation and/or water funds, bonds, revenue streams) to be employed for revenue generation post-GCF (2.1.2).

Outcome 3. Vulnerable rural livelihoods are more climate-resilient, diverse, economically sustainable and nature-positive. The sustainability and climate resilience of the agricultural matrix of this landscape is challenged by additional, multiple barriers. Rural agricultural practices and livelihoods are constrained by the very small size of landholdings (avg. 0.3 ha⁸¹), creating intense pressure to utilize all arable land for food and near-term income-generating products⁵⁶, thus creating **reluctance to incorporate trees or other soil conservation measures on the land**⁸². The resultant **lack of fuelwood in turn continues to drive deforestation of natural forests**¹², **and occupies important time for women and youth** who could instead be engaged in productive and empowering roles. Overall, local communities and extension services⁸³ have **limited skills, information and knowledge to design and implement farming methods** that would enhance their capacity to mitigate the impacts from climate-induced soil erosion, flooding and droughts. Farmers **also lack access to financial know-how**⁵⁶ **and capital, proven alternative livelihoods, and markets** that could incentivize and enable change¹². They are therefore caught in a cycle of immediate need that – without alternatives – forecloses options that could recover and sustain benefits from improved ecosystem services, other forest functions, and new livelihoods. **Component 3** aims to redress these conditions via activities that will demonstrate, incentivize, facilitate, and support actions that maintain and recover ecosystem services, benefit from forest functions and provide alternative on- and off-farm based incomes that reduce pressure on forests. To these ends, the Project will support reafforestation on steep slopes (>55%), on both public and smallholder private lands (Activity 3.1.1). It will develop farm-based agroforestry for climate-resilient species that produce fuelwood and/or market products, while conserving and enriching soils (3.1.2, 3.2.1). The Project will complement land-based production with a scaling-up of alternative value-chain products for smallholders that are nature-positive, including but not limited to tourism-based income, beekeeping, and improved cooking stoves (3.2.2, 3.2.4). Central to these activities will be facilitation of access to market chains (3.2.3) and financial mechanisms for establishing and maintaining businesses (3.3.1, 3.3.2, 3.3.3). Given that these activities are designed at the core to be profitable and promote financial sustainability, it is intended that they be replicable, and sustained once GCF funding expires.

Assumptions. Success in achieving Project outcomes will depend on important assumptions being met, among them that governmental and non-governmental actors will be willing to work in collaboration across sectoral and geographic boundaries; rural adoption of alternative farming and off-farm revenue generation can be scaled up to drive a landscape-level transition to sustainability; financing and revenue-generating mechanisms will be established that assure continuance of initiated actions; and governmental commitment to natural forest restoration and ecosystem-based adaptation continues in national strategies and in meeting global commitments.

The pathways summarized in this ToC are expected to contribute to the overall goal and outcomes of the Project: **natural forests** and **vulnerable communities** will be more resilient and sustainable despite climate impacts, with effective, **coordinated management** across the CND landscape. In doing so, important co-benefits will accrue. Intentional, inclusive engagement in management coordination will **empower women and youth** in environmental thinking and decision making. Forest conservation and restoration, along with agroforestry, will contribute to greater **carbon sequestration and storage**, while fire management and greater cooking efficiency will **reduce GHG emissions**. The globally unique, endemic **biodiversity** of

81 USAID, 2017. LandLinks Country Profile: Rwanda. <https://www.land-links.org/country-profile/rwanda/>

82 Rwanda Water and Forestry Authority, 2017. *Forest Investment Program for Rwanda*.

83 Foster & Graham, 2014. *Connectivity and the Tea Sector in Rwanda: Value Chains and Networks of Connectivity-Based Enterprises in Rwanda*.

Rwanda's forests, from great apes to rare orchids, will recover on the basis of healthy, more connected and resilient forest habitat.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1. Landscape planning, policies & management effective & coordinated across sectors and scales to address climate risks & adaptation benefits	☐	☐	☐	☒	☐	☐	☐	☒
Outcome 2. Natural forests protected, connected, more resilient to climate impacts & risks	☐	☐	☐	☐	☐	☐	☐	☒
Outcome 3. Vulnerable rural livelihoods more climate-resilient, diverse, economically sustainable & nature-positive	☐	☐	☐	☐	☒	☒	☐	☐

Co-benefits number	Co-benefits					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1: Globally unique, endemic biodiversity maintained	☒	☐	☐	☐	☐	☐
Co-benefit 2: Women and Youth empowered	☐	☐	☐	☒	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

Rwanda's CND region is a complex, naturally integrated landscape. Its diverse high elevation forests capture and store water resources, releasing them in a sustained manner throughout the year to surrounding steep hillside and bottomland farms and plantations, before feeding into the national network of wetlands, rivers, and lakes. The Rwanda portion of the CND runs from the Virunga Mountains and Volcanoes National Park (VNP) on the border with Uganda in the north, down through Gishwati Mukura National Park (GMNP), to the south end of Lake Kivu and Nyungwe National Park (NNP) on the southern border with Burundi. These three

national parks contain the country's only remaining montane forests. The boundaries of the CND landscape are defined as all areas greater than 1900m in elevation.

Administratively, the CND overlaps three Provinces (Western, Southern, Northern) and ten Districts (Karongi, Musanze, Ngororero, Nyabihu, Nyamagabe, Nyamasheke, Nyaruguru, Rubavu, Rusizi, Rutsiro).

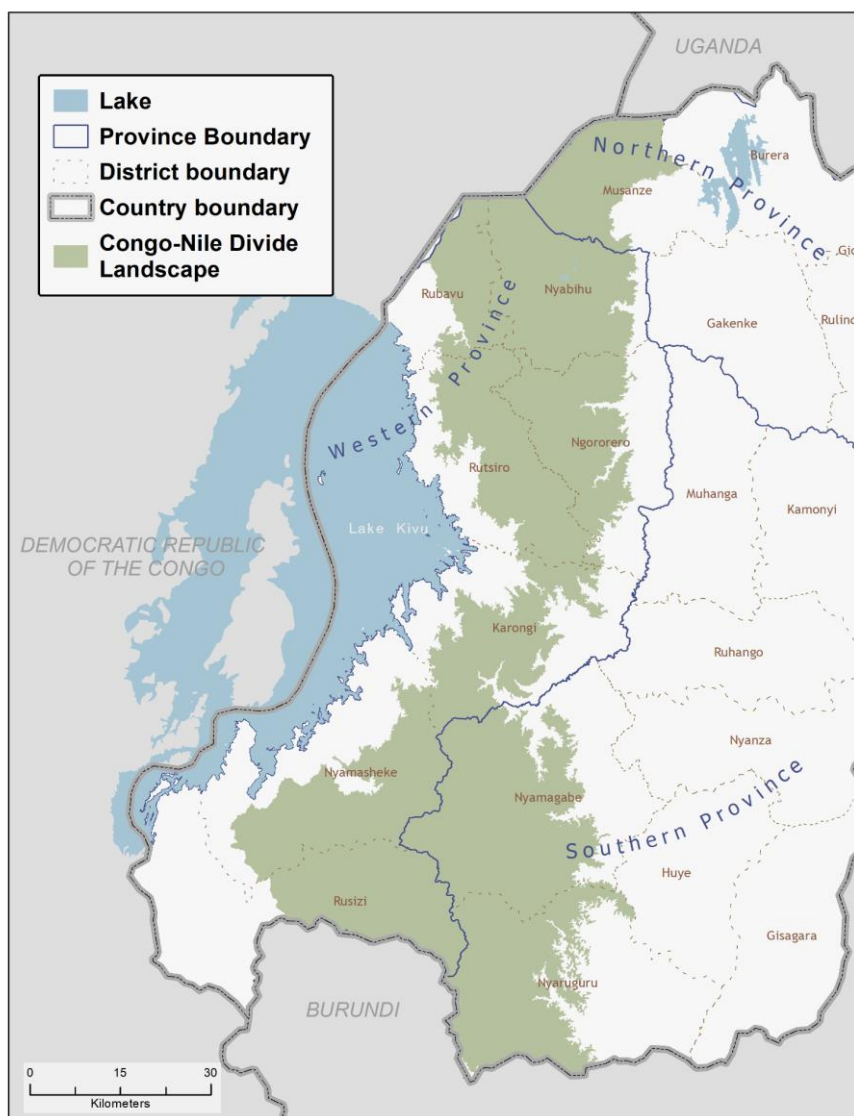


Figure 16. Map of CND Landscape

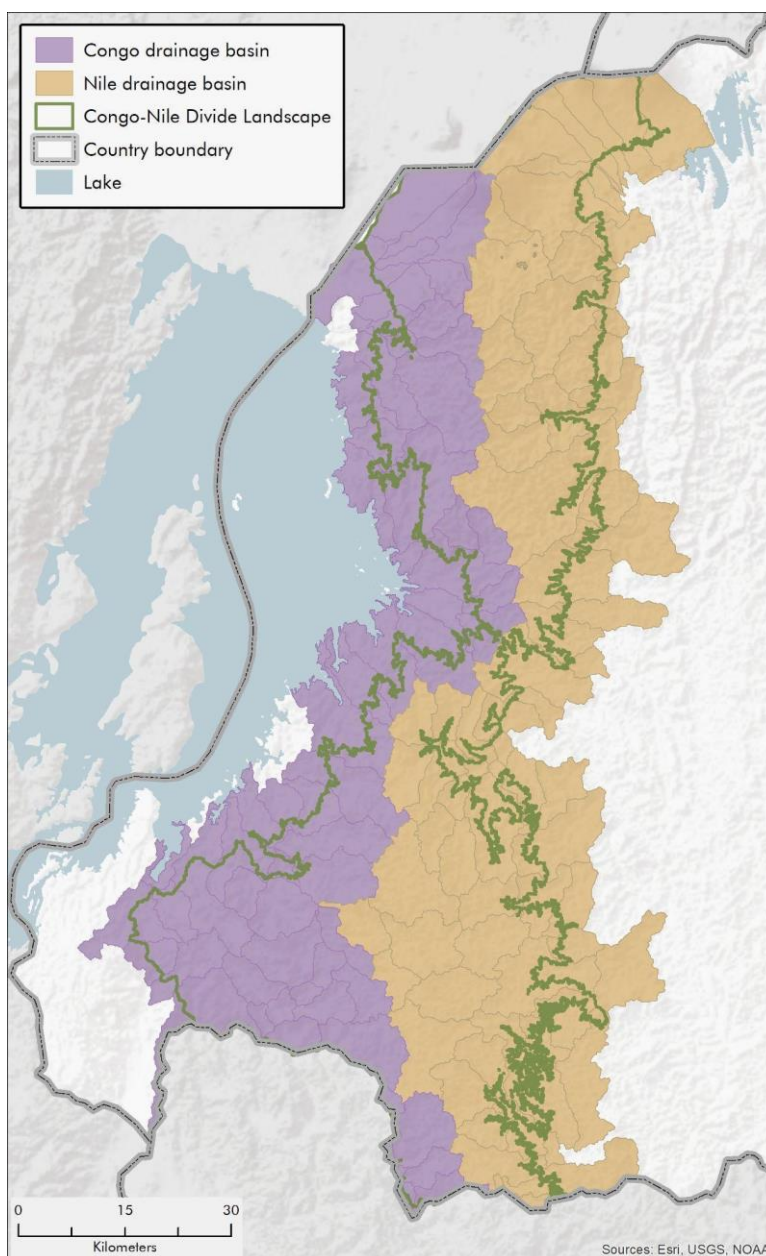


Figure 17. Map of CND Landscape drainage basins

To address the present and intensifying threats from climate change requires that this region also be managed as an integrated and ecologically connected system. This need is recognized in several key policies and plans enacted in recent years by the Rwandan government. However, implementation and integration⁸⁴ of these have lagged, their decentralized application on the ground is largely unrealized, and climate-related technical capacity remains limited in key institutions at all levels. This Project is designed to increase the climate resilience of the CND. Its intended outcomes are:

- The development and implementation of effective landscape planning, policies, and management, coordinated across sectors and scales, to address climate risks and adaptation benefits;

⁸⁴ Bagstad, K.J., 2019. *Towards ecosystem accounts for Rwanda: Tracking 25 years of change in flows and potential supply of ecosystem services*. British Ecological Society.

- Improved protection, restoration, and connection of existing natural forests, with enhanced resilience to climate impacts and risks; and
- The transformation of vulnerable rural livelihoods to be more climate-resilient, diverse, economically sustainable, and nature positive.

The Project has three main components directly targeting vulnerabilities within climate impact chains. Detailed analyses of the underlying rationale are provided in the Feasibility Study (Annex 2). Please see section 4.1 of that study, for a list and description of relevant existing and previous projects in the region.

Component 1: Mainstreaming Climate Adaptation into Integrated Land Use Planning

There is an urgent need for more effective collaboration among Rwandan government agencies, civil society, private sector, and diverse partners to balance difficult-yet-inevitable land use trade-offs that will have profound impacts on the climate resilience of both natural systems and people. The GoR recognizes that a piecemeal approach to decision making has exacerbated competition over scarce land resources and reduced the adaptive capacity of both ecosystems and rural populations⁸⁵. Rwanda's Baseline Climate Change Vulnerability Index (2015) recommends establishing a multi-ministry technical climate resilience coordinating committee, including NGOs⁸⁶. In 2017 the MoE developed the Strategic Programme for Climate Resilience (SPCR), to focus on three cross-cutting priorities to achieve climate change resilience: (a) technical capacity building and strengthening institutional coordination; (b) integrated land use and spatial planning; and (c) climate services and disaster risk management⁸⁷. The SPCR is a key step towards ensuring inter-ministerial and multi-sectoral collaboration and integration of climate resilience considerations into development plans and actions. However, there remains a need to comprehensively implement the strategy and mainstream climate change adaptation at the sub-national level, engaging a diverse set of actors and sectors and empowering women to provide input into decision making at district and community levels.

Output 1.1: Landscape-wide land-use plan developed for climate-resilient livelihoods and forest ecosystems, integrating district strategies.

The GoR completed a National Land Use and Development Master Plan (NLUMP 2020) to guide land use planning and regulate permitting processes. However, effective and strategic land use has yet to be applied at a local and district level, and the conversion of plantation and riparian forests to agriculture continues and development projects are approved without sufficient knowledge of climate change risks or the value of existing forests for ecosystem-based adaptation (EbA)⁸⁸. The Project will add climate-change resilience and adaptation content to central government efforts such as NLUMP and the SPCR process by acquiring, synthesizing, and disseminating key information on climate trends, the importance of remaining natural forests, and the value of ecosystem services. This information will also be shared at district and community levels through various outreach networks and workshops, with particular attention to reaching disadvantaged women, youth and vulnerable groups.

Building on the information base outlined above, the Project will work with the National Land Authority (NLA) to establish a working group, with participation from relevant ministries, agencies, and districts, including representatives of women, youth, and vulnerable groups interests, to develop a climate-resilient land-use

⁸⁵ National Land Use Management Plan.

⁸⁶ United Nations, 2015. Economic Commission for Africa; Rwanda Environment Management Authority. Baseline climate change vulnerability index for Rwanda. Kigali. Rwanda Environment Management Authority.

⁸⁷ Rwanda's Green Fund (FONERWA), 2017. Strategic Programme for Climate Resilience (SPCR) Rwanda - December 2017. [Link](#).

⁸⁸ Scarano, F.R. et al, 2017. *Ecosystem-based adaptation to climate change: concept, sustainability and a role for conservation science*, Perspectives in Ecology & Conservation 15(2), 65-73.

plan for the CND landscape. At the national level each relevant ministry and government agency will select and appoint a working group member while at the district level, participants will be selected from the Joint Action Development Forum (JADF). JADF is a multi-stakeholder platform meant to facilitate and promote the full participation of citizens in the decentralized and participatory governance and improve service provision processes with representatives from the public sector, private sector and civil society. The following stakeholders will form the basis of the collaboration: a) the Ministry of Environment represented by four of its agencies: the Rwanda Environment Management Authority (REMA); the National Land Authority (NLA), Rwanda Water Board (RBA) and the Rwanda Forest Authority (RFA); b) the Ministry of Agriculture, including the Rwanda Agriculture Board (RAB); c) the National Agriculture Export Board (NAEB) d) Ministry of Local Government; e) Rwanda Development Board (RDB); f) Gender Monitoring Office (GMO); g) Districts Decentralized Structures – the District Administrative Units, which supervise several technical and administrative activities; h) civil society, international organisations, academia and community based organizations. This list will be discussed during the inception period and expanded as necessary. From this, a spatial framework for district level land use planning will be developed that accounts for current and future climate risks, reconciles cross-sectoral land-use conflicts, and guides development decisions. All the activities and sub activities under this output will be executed by the Rwanda Forest Authority in collaboration with National Land Authority, and districts.

Activity 1.1.1. Synthesize and disseminate information on value of natural forests and ecosystem services

This activity will involve conducting a stakeholder mapping & consultation exercise in order to better understand the use of climate adaptation data in existing sectoral planning processes, and identify cross-sectoral linkages or conflicts. These results, along with a comprehensive literature review, will inform the design of high-level information packages that will synthesize information on climate risks for various sectors (e.g. agriculture, forestry) and highlight the value of forests for increasing resilience of local communities. An information dissemination program (e.g., presentations, workshops, and newsletters) will be implemented to share this knowledge among key ministries as well as government, non-government and community organisations. The Project will use JADF as a platform to disseminate information to key stakeholders. Particular attention will be paid to ensure women, vulnerable and youth groups from Ubudehe categories c, d and e are represented. In addition, the program will use mass media in the form of a radio program and complementary community action campaigns to improve community, women, youth, other vulnerable groups knowledge on climate information, climate risks and climate adaptation options. This will stimulate discussion on issues pertaining to climate risks and other social issues locally and support engagement of each in responding effectively to climate adaptation and resilience options.

Cross-sectoral tradeoffs and climate adaptation solutions with cross-sectoral implications will also be highlighted, in order to begin promoting the benefits of integrated land-use planning. To complement the high-level information dissemination program, a suite of technical training materials will also be developed for delivery to district technical staff. These materials will provide training on climate risks & the need to incorporate climate adaptation into land-use planning, with a focus on the technical skills required for climate-sensitive planning (e.g. sourcing climate data, mapping ecosystem services).

Sub-activities will include:

1.1.1.1 Map the sectors involved in land-use planning in the CND and review how forest ecosystem services and climate resilience are incorporated into each sector's planning process

1.1.1.2 Conduct comprehensive literature review & stakeholder consultation to collect & synthesize information on climate risks for various sectors (e.g. agriculture, forestry), the value of forests for increasing resilience of local communities, and highlight adaptation solutions with cross-sectoral implications.

1.1.1.3 Host workshops & presentations with key ministries, government organizations, NGOs and community organizations to disseminate high-level knowledge on value of forests for increasing resilience of local communities.

1.1.1.4 Develop guidelines for integrating climate risk into land use planning and cross sectoral planning

1.1.1.5 Develop outreach materials on climate and related risks in the CND and the value of forest ecosystems for increasing resilience

1.1.1.6 Implement outreach program tailored to different stakeholders (local government, civil society, communities, women, youth) to enhance capacities for land-use planning, funding mobilization, and delivery of climate adaptation actions

1.1.1.7 Conduct climate literacy seminars for local government and civil society organizations, aimed at increasing women and youth participation.

1.1.1.8 Provide financial and logistical support to trained organizations in grassroots mobilization to increase women and youth participation in climate adaptation planning.

1.1.1.9 Introduce social safeguards at a high level at each meeting, including the GRM, FPIC, and Access Restrictions Mitigations as safeguards to be in place for work with local communities.

Activity 1.1.2. Develop climate-resilient landscape land-use plan

This activity aims to ensure that climate change adaptation requirements are fully integrated into planning processes at local, district and national scales. This will involve i) participatory scenario analysis to understand the problems local communities and district officials face in terms of natural disasters (flooding, soil erosion, forest loss/degradation) and climate change, as well as the future land use plans of the districts, including an assessment of the national consequences of these problems in terms of ecosystem service delivery and climate change resilience and adaptation capacity; ii) risk assessment to estimate the impacts of the planned land use changes, as well as climate change, on flood risk, soil erosion, forest loss/degradation and the well-being of local people, especially any disproportionate impacts on women, youth and vulnerable groups; iii) climate change adaptation and mitigation measures development to decide possible climate actions for both adaptation and mitigation and prioritize these actions according to their feasibility and urgency in consultation with local communities, district leaders, and other stakeholders; and iv) land use plan development based on the recommendations from the previous 3 steps. A comprehensive consultation and feedback process will be undertaken to incorporate views of stakeholders at all levels (e.g. civil society; men, women, youth, and historically marginalized people in local communities; district & national government), upon which the plan will be finalized and approved by the Government of Rwanda.

Sub activities will include:

1.1.2.1 Engage district officials, Joint Action Development Forums (JADF), and community members in the CND (especially women) in reviewing and interpreting the National Land Use and Development Management Plan (NLUMP), to ensure the plan accounts for current and future climate and related risks, while building local support for climate sensitive planning

1.1.2.2 Conduct participatory land-use planning process in communities from village to district level to support integrated climate resilient land use planning

1.1.2.3 Develop Integrated Land-use Plan that supports Resilient Livelihoods and Ecosystems in the CND, ensuring alignment with National Land Use and Development Master Plan and ensuring the CND plan guides the district plan

1.1.2.4 Develop and roll out a series of trainings (virtual sessions and online modules) on gender sensitivity and mainstreaming women and youth into planning

1.1.2.5 Assess specific climate impacts on historically marginalized and Category c,d,e populations through a participatory NR process to ensure the Project components address their needs for adaptation

Output 1.2 Local and national institutional capacities strengthened to integrate biodiversity and climate risks into land use planning and management

Forest data in Rwanda has historically been generated in an ad-hoc fashion⁸⁹ through the support of external agencies. This precludes comprehensive consideration of forest and biodiversity data in land-use planning processes. As Rwanda moves to achieve the goals outlined in the Green Growth and Climate Resilience Strategy and the National Land-Use Development Master Plan, building governmental capacity in generating and utilizing spatial data to inform decision making is essential. The activities under this output will involve recruiting experts in mapping, remote sensing and ecosystem service modeling, in order to establish a formal spatial planning unit within government. These experts will also lead a formal information dissemination and capacity building program, in order to i) promote the importance of considering forest & climate resilience into land-use planning, and ii) deliver detailed training to technical staff on forest mapping, climate and ecosystem service modeling. The Project will also increase the understanding of the importance of forest landscape restoration in securing ecosystem services for local economic development and resilient livelihoods. It will ensure that the public, decision-makers and other stakeholders in the CND landscape have a high level of awareness of the risks to the economy and livelihoods associated with deforestation under current and possible evolution of these risks with the changing climate, and the benefits of reforestation in an integrated landscape approach.

Women's land rights need some key consideration for an effective integration of women in land use planning and management, among which include advocacy for women's knowledge of their rights, women's confidence in the dispute resolution process and outcomes, men's respect of women's rights, and modest improvements in physical and social accessibility of the local justice system for women are most critical avenues for success. These factors will be considered throughout project implementation of the activities proposed within this project using a gender-responsive, participatory and fully transparent approach. For example, under the project trained institutions will hold meetings to train women, youth and marginalized people on forests and mobilize them to participate in planning processes and local institutions will be trained to build capacity of rural people, specifically women and youth, to participate in climate adaptation processes. Program Staff and government officials will be trained understand gender barriers, understand how to mainstream gender equality principles into planning and support women, youth and marginalized participation in meetings.

Furthermore, the Project will support dialogue with the private sector and planning processes needed to mainstream climate change and biodiversity in the tea sector. Projected climate changes predict significant elevation rises in CND agro ecological zones, including the potential for tea cultivation at much higher

⁸⁹ Arakwiye, B., et al, 2021. *Thirty years of forest-cover change in Western Rwanda during periods of wars and environmental policy shifts*. Regional Environmental Change 21(2).

elevations than at present. This is unlikely to threaten Rwanda's existing montane forest parks, as foreign revenues from tourism in these reserves far surpass those from tea. However, tea remains a significant source of foreign revenue and a government target for expansion. Any increased tea production in the CND would therefore most likely affect lands outside of parks: smallholder, cooperative, and district holdings, many of which might also be targets for reforestation, agroforestry, and biodiversity stepping stone initiatives under this Project. The integrated land use planning included in this Project design, will bring together tea industry, agriculture, forestry, biodiversity, and tourism interests to identify and reduce such potential conflicts over use of the CND's limited land resource base. Through this integrated land use planning we aim to establish a formal public-private sector forum at the landscape level to help improve communication across sectors, appreciation of environmental issues, and increase synergies between interventions to avoid conflicting targets and indicators between sectors.

This output will comprise the following activities and sub activities which will be executed by the Rwanda Forestry Agency (RFA) in collaboration with key government agencies, private sector and civil society organizations.

Activity 1.2.1 Create interagency taskforce institutionalizing integrated landscape planning and policy

Insufficient mechanisms for integrated land use planning constitute one of the key barriers for sustainable natural management in Rwanda. Planning processes within government agencies are carried out in silos without paying attention to the impacts a certain land use activity or project will have on other sectors of the economy. This Project aims to facilitate improved collaboration among government agencies by strengthening the government's existing cross-sectoral planning task force. The existing task force was set up to support cross-sectoral implementation of the National Strategy for Economic Transformation, but it suffers a number of weaknesses, including: i) a lack of official mandate for the task force to carry out activities; ii) irregular attendance of members; and iii) lack of clear and regular funding to facilitate meetings. This activity will review the mandate and structure of the cross-sectoral task force and promote solutions for strengthening its operation to RFA (the task force coordinator). It will also facilitate quarterly meetings of the task-force, in particular between institutions in charge of agriculture and agroforestry, to encourage synergies and avoid overlapping mandates and redundancy in different climate resilience interventions. The involved staff at both national and local levels shall be empowered to readily share information and activity plans, and will be offered technical support to generate materials that can inform decision-making (e.g. policy briefs, map, presentations). The Project will support collaborative integrated landscape planning, which secures climate resilience undertaken with appropriate social safeguards.

Sub activities will include:

1.2.1.1 Review & strengthen operationalisation of the current cross-sectoral task force

1.2.1.2 Hold quarterly sectoral planning meetings with both national and district administrations in CND and the private sector involved in mining, agriculture, and livestock production in the landscape

1.2.1.3 Facilitate discussions and provide technical support (e.g. policy briefs) in decision-making for cross-sectoral actions around climate adaptation and forest resilience in the CND

1.2.1.4 Continue to introduce social safeguards at each meeting including the GRM, FPIC, and Access Restrictions Mitigations as safeguards to be in place for work with local communities. For those at the district or community level, provide options for feedback on the process and best ways of communication with local partners.

Activity 1.2.2 Build capacity for spatial planning in national agencies re climate change

Spatial planning is emerging as a valuable tool for the development of evidence-based land use and climate resilience strategies⁹⁰. This subcomponent will develop capacity within the Rwanda's university network and the MoE to use remote sensing, ecosystem service modeling and spatial planning tools (e.g., Marxan, InVEST, RIOS, SWAT and other open access tools such as Open Foris) to incorporate climate risks and climate resilience strategies into development plans and to readily identify and reconcile land use conflicts. A prime example of the latter is the potential for climate-driven shifts in optimal elevation zones for coffee and tea cultivation conflicting with current subsistence agriculture in the case of coffee, and key remaining natural forest fragments with regard to any upward migration of tea. These spatial planning tools will be procured by the project and will be made available to the spatial planning unit.

The Project will assist the GoR in developing a dedicated spatial planning unit (SPU) within an appropriate ministry or cross-cutting agency. The appropriate hosting institution will be determined during the project implementation in consultation with key government agencies operating under the Ministry of Environment. This will include support for training high level Rwandan specialists to provide the technical expertise needed to process and integrate spatial planning into decision-making within and across sectors. A specific contribution of this SPU in coordination with appropriate governmental and non-governmental partners is the development and operationalization of an innovative, scientifically credible, scalable system for monitoring trends in natural forest cover and forest types at multiple scales. This SPU will also allow Rwanda to measure and report on the land use, land use change and forestry (LULUCF) sector as part of a national Greenhouse Gas Inventory (GHG-I). This system will build on the national FMES that was established with the support from ENABEL.

Capacity building is essential for execution of the Project; even more so for the sustainability of Project elements in the long run. To improve Rwanda's collective ability to integrate land use planning in support of climate change resilience and delivery of ecosystem services, these sub-activities are planned:

1.2.2.1 Recruit spatial planning expert/s to support spatial planning unit

1.2.2.2 Assess and identify the institutional home and operationalize the spatial planning unit

1.2.2.3 Conduct a capacity needs assessment and identify the appropriate tools for spatial planning to fit the Rwandan context and conduct familiarization

1.2.2.4 Led by the spatial planning unit, develop capacity within the University of Rwanda and the MoE to use remote sensing, ecosystem service modeling and spatial planning tools (e.g., Marxan, InVEST, RIOS, SWAT) to incorporate climate risks into land use planning processes

1.2.2.5 Deliver training workshops on utilization of earth system models for land use planning purposes, utilization of remote sensing resources (lightning, satellite radiances) for hazards detection and climate monitoring and on regional climate monitoring and applications.

Activity 1.2.3 Develop an effective forest monitoring system to underpin forest management decisions

⁹⁰ Mathias S., et al, 2019. The Spatial Development Framework to facilitate urban management in countries with weak planning systems, International Planning Studies, 24:3-4, 235-254, DOI: 10.1080/13563475.2019.1658571

Within Rwanda, forest-related data is limited in scope and generally outdated⁹¹. Consequently, the current state and trends of Rwanda's forests are not well understood. Efforts to improve the situation through individual studies have been valuable but have not led to systematic increases in spatial information on trends in different forest cover types and resilience at the scales at which land use decisions are being made. For example, the FMES software focuses on forestry and plantation forests, but does not address natural forests. This activity will update and monitor forest cover data in different ecological zones and forest types using high resolution satellite imagery, field work, and remote sensing techniques. In addition to remote sensing techniques, the Project team, once selected, will meet with government and international partners to assess best monitoring tools and platforms, including Open Foris. Close attention will also be given to the system used by Mugabowindekwe et al (2023)⁹² for the most recent, highly detailed forest inventory from 2007 imagery. RFA will recruit a Rwandan expert who will work with WCS experts in applying spatial planning and forest monitoring tools in support of government agencies during the project implementation to strengthen the capacities of RFA, NLA and GIS Center at the University of Rwanda.

An adaptive forest, climate change and land use monitoring system will be established to support climate resilient forest management decisions through the following:

1.2.3.1 Recruit Forest ecologist or remote sensing expert to support the design and implementation of indigenous forest monitoring system

1.2.3.2 Review existing forest mapping data and monitoring software, and assess utility for generating quantitative assessments of indigenous forest cover from local to national scales

1.2.3.3 In collaboration with REMA, MINAGRI, RAB, RDB, RISA, MoE and affiliated agencies and University of Rwanda, design structure and operation of forest monitoring system, including required inputs (data, computing, personnel), desired outputs, and operational structure (e.g. location, reporting structure, funding)

1.2.3.4 Purchase and install 2 new weather automatic stations in the CND

1.2.3.5 Establish & operationalize forest monitoring system

1.2.3.6 Generate updated maps & statistics for indigenous forest cover, forest type etc. using forest monitoring system

1.2.3.7 Develop capacity within hosting institution for continued operation of forest monitoring system

Component 2: Forest and landscape management and restoration

The GoR has prioritized forest management and restoration – including natural forests in protected areas – as a key component of climate change adaptation strategies for the CND landscape.⁹³ Natural forests within the national park system have been generally well protected and managed in recent years, earning high marks for biodiversity conservation while also developing a high quality – and quite lucrative – set of ecotourism attractions. Current park management plans, however, include little attention to the potential effects of projected climate change. Outside of parks, few areas with natural forest cover remain and forest management and restoration approaches are not being implemented at the scale required to significantly improve forest climate resilience. Where forests are restored, most use exotic species and mono-specific

⁹¹ Food and Agriculture Organization of the United Nations, 2017. *Analysis of forests and climate change in Eastern Africa*. Forests and Climate Change Working Paper 16.

⁹² Mugabowindekwe, M., Brandt, M., Chave, J. et al. Nation-wide mapping of tree-level aboveground carbon stocks in Rwanda. *Nat. Clim. Chang.* **13**, 91–97 (2023). <https://doi.org/10.1038/s41558-022-01544-w>

⁹³ IUCN, 2014. *Forest landscape restoration opportunity assessment for Rwanda*. [Link](#).

plantations that are generally incompatible with mixed agricultural uses; they are also vulnerable to diseases and pests with climate warming.

In this component, targeted interventions will integrate climate change awareness and adaptation into national forest park planning, policies and management, as well as restore degraded areas within those parks. These efforts will help sustain the forests' rich biodiversity, the lucrative tourism revenues that flow into local and national economies, and the ecosystem services needed for climate resilience of vulnerable communities. This suite of activities will also identify priority areas outside of parks – in remnant natural stands, around wetlands, and along streams – for protection and restoration as “stepping stones” and linkages for long-term climate change connectivity.

Output 2.1 Protected Area management effectiveness improved re climate risks and adaptation

The remnant natural forests of the CND harbor globally significant biodiversity across multiple taxa, from primates to birds and orchids. More than 10% of these species are regionally endemic⁹⁴ – 47 flowering plants are endemic to Nyungwe Forest alone⁹⁵. Though each has suffered from significant past habitat loss, the Nyungwe and Volcanoes National Parks have been generally well-managed under the Rwanda Development Board (RDB), with assistance from NGOs, PPPs, and the private sector. Rwanda's reputation for ecotourism and nature tourism attractions is world class. It has also earned conservation recognition for the dramatic recovery of its endangered mountain gorillas, sustained protection of its remaining biodiversity, and creation of a significant revenue-sharing program with parks-adjacent communities. The recent addition of Gishwati-Mukura National Park to this mountain forest network underscores the government's commitment to conservation, though both forest blocks require significant restoration for the purposes of biodiversity conservation, connectivity, and ecosystem services enhancement.

In contrast to Rwanda's areas of conservation success, relatively little attention – and less action – has been focused on climate change and its potential impacts on habitats and biodiversity, especially in the climate-adapted mountain forests of the CND. The investment required to build and sustain the institutional and human capacity to achieve Rwanda's recent conservation success now needs to be joined with a parallel investment to ensure that PA management plans, policies, and actions address the realities of climate change both within and outside of their boundaries.

Activity 2.1.1 Facilitate revision of PA management plans to address climate risks

Climate change is increasingly recognized as not only a major future threat to protected areas, but one that will also exacerbate existing threats and vulnerabilities⁹⁶. This is particularly true for the three national parks of the Congo Nile Divide landscape that are surrounded by a large number of smallholder farmers who are dependent on natural resources for their livelihoods. The park planning processes need to be expanded to best meet challenges posed by climate change as well as ensure full incorporation of buffer zone and landscape linkage requirements, and the ability to engage with the larger CND and district landscape management planning processes. The revised management plans will aim to:

- Develop specific goals and objectives for improving climate resilience
- Identify areas and species of particular importance to climate adaptation, mitigation and resilience
- Identify and prioritize threats that exacerbate climate impacts
- Identify and prioritize critical actions for strengthening resilience

⁹⁴ CBD. 2020. Rwanda 6th National Report to the Convention on Biodiversity

⁹⁵ Fischer, E. and Killmann, D., 2008. *Illustrated field guide to the plants of Nyungwe National Park Rwanda*. University of Koblenz-Landau.

⁹⁶ World Bank, 2019. *Rwanda Systematic Country Diagnostic*.

- Identify areas important for climate change adaptation and mitigation into protected area zones and regulations
- Develop indicators of climate resilience

To this end, the Project will undertake these sub-activities:

2.1.1.1 Review and update existing national park management plans to ensure climate and related landscape changes, risks, impacts and required management responses are integrated

2.1.1.2 Provide technical and financial support to planning, research and monitoring as well as community-based conservation units within RDB in managing the parks as part of larger CND landscapes

2.1.1.3 Train PA staff to integrate gender and social inclusion into programming

2.1.1.4 Train PA staff on integrating needs of women, youth, historically marginalized and Ubudehe Categories c,d,e populations into climate adaptation risks and responses

2.1.1.5 Train PA staff on SEAH, GRM, FPIC, and Access Restriction Mitigation Processes to ensure NP climate change responses are undertaken in a way which supports social inclusion and equity

Activity 2.1.2 Establish long-term plans for CND financial sustainability post-GCF

Rwanda has recently developed a Biodiversity Finance Plan⁹⁷ through support from BIOFIN. This plan has identified a number of financing solutions that will not only contribute towards improved biodiversity protection in Rwanda, but also aligns with Rwanda's sustainable development and green growth objectives. These solutions include:

- Introduce a Biodiversity Conservation Fund into Rwanda's National Fund for the Environment (FONERWA) to help streamline and attract domestic and international sources of finance to support conservation investments and thereby reduce transaction costs to increase conservation benefits.
- Improve efficiency and effectiveness of environmental fees and fines to strengthen the effectiveness of the regulatory environment that will ultimately improve ecosystem and biodiversity conservation goals.
- Water User Fees for Catchment Management to secure the financial resources necessary for effective water catchment management.
- Pilot Business Plans for Selected Wetlands to enable the sustainable development of wetlands for ecotourism and biodiversity-friendly enterprises such as handicrafts, fisheries, and sustainable agriculture.
- Promote biodiversity-friendly enterprises in the transition to a green economy and accelerate the transition to a green economy by incentivizing and supporting businesses to adopt sustainable practices and attracting investments in new conservation enterprises.
- Strengthen the tourism revenue sharing scheme to improve biodiversity outcomes and effectively address conservation-development objectives
- Develop a Protected Area Finance Strategy to support the goal of increasing the financial and ecological sustainability of Rwanda's Protected Area System (PAS)

There are also new emerging solutions such as the Wildlife Conservation Bond (WCB) which essentially is a payment for performance-based financial instrument that channels investments to achieve conservation outcomes – measured in this case by specific conservation targets that could be set (e.g., hectares of native forest restored, or an increase in umbrella species populations). In the case of Rwanda, gorillas and

chimpanzees are considered umbrella species that play a crucial role in shaping entire ecosystems on which countless other species depend. For the WCB, a grants-based, outcome payer is needed, which could be the GCF. Through the WCB, and in collaboration with the World Bank (WB), the WB could issue bonds and investors could support the financing of activities to protect and grow a critically endangered species with clear conservation targets, contributing directly to biodiversity and bringing jobs to local communities through the creation of conservation-related employment in rural areas of the CND. A third-party, independent verifier is also needed to evaluate whether conservation targets have been met. A meeting is planned with the WB in the second half of February 2023 in Rwanda to determine the scale and feasibility of the WCB in the CND. The South African Rhino Bond has already attracted foreign investment, and it is a good model Rwanda could emulate⁹⁸. The Government of Rwanda, represented by the Rwanda Development Board, is in advanced stages of negotiation with the World Bank for establishing a wildlife Bond. There is, therefore, clear interest in the bond which suggests that investors still see value in investing in such an instrument.

A technical advisor will be recruited by this project to provide technical support to RDB, REMA, MINECOFIN and MoE and explore these various solutions to identify financial instruments that could be employed for revenue generation post-GCF for sustainable management of the CND landscape. The technical advisor will work closely with the NSC to facilitate buy-in from stakeholders such as RDB and MINECOFIN into compliance with environmental laws through greater emphasis on the links to climate resilience and NDC objectives.

Activity 2.1.3 New fire management curriculum developed and operationalized Given the expected increase in drought, and the natural forest vulnerability already demonstrated in the past, the capacity for effective fire prevention and management in core PAs and adjacent landscapes will be increased.

While current climate projections call for increased rainfall in the CND, they also project higher temperatures and a rise in elevation of the natural forests' cloud cover zone (Annex 2.1). The combination of the latter two trends raises the potential for increased fire risk. In order to sustainably address the threat of fire to the natural forests and regenerated areas in particular, this project will increase the capacity of RDB in the area of fire prevention, with emphasis on education and outreach, monitoring of fire danger conditions across the parks, response plans, and fire suppression. A collaborative effort in fire management planning and implementation in CND districts will be carried out, facilitated by a fire management expert. Based on training needs, the project will also carry out RFA district and sector foresters' training in Integrated Forest Fire Management approaches and activities and firefighting; they will then build the capacity of community fire brigades.

Sub-activities include:

2.1.3.1 Develop a curriculum tailored to needs and capacities of different stakeholders

2.1.3.2 Build the capacity of RDB to manage fire in National Parks

2.1.3.3 Build the capacity of RFA, local authorities and communities to collaboratively manage fire in forests outside Pas

2.1.3.4 Implement a "fire wise" outreach and awareness program for communities and local government (district, sector, cell), as well as Rwanda Forest Authority, surrounding natural forests to reduce the incidence of human-caused fires.

⁹⁸ World Bank, 2022. *Wildlife Conservation Bond Boosts South Africa's Efforts to Protect Black Rhinos and Support Local Communities*. Press Release NO: 2022/059/AFE.

Output 2.2: Natural Forest cover restored, biodiversity connections established.

The natural forests of the CND exist as an archipelago of isolated islands within a surrounding matrix of intensive human land use. Although some have raised the possibility of reconnecting these islands, it is important to understand that their isolation is not a recent phenomenon (with evidence of the initial isolation of the main forest blocks from the pre-colonial era). Furthermore, large-scale proposals for landscape linkages have thus far failed to provide alternatives for the tens of thousands of rural smallholders who would be displaced by such schemes. However, important actions can be taken to restore degraded natural forest lands and protect or reestablish natural “stepping stones” and ensure a functional level of ecological linkage between the existing parks.

In some areas where natural forests have been cleared, the Project will work to restore forest cover. This will be done primarily around the Gishwati and Mukura reserves, using native species and building on nursery and planting techniques already tested in Rwanda. New techniques may need to be developed to expand reforestation to degraded lands on steep slopes and along waterways and wetlands. Tree species will be carefully selected considering the following attributes: pioneer species which grow in full sun, fast growing (height and canopy width), and animal pollinated and dispersed. These “tree islands” serve several purposes. First, by growing quickly and providing canopy cover, they shade out competing vegetation and change the soil microclimate (very important for later successional species establishment). Site-generalist native species such as *Albizia*, *Dombeya*, *Macaranga* and *Polyscias* respond to these characteristics. Secondly, they entice a variety of animal species into the expansion area, thereby dispersing seeds. In certain areas the Project will favor natural regeneration from seed banks.

All work outside of protected areas will require careful attention to private land ownership claims and related social and gender equity issues. Part of the communities’ responsibilities will be afforestation of the degraded forest with indigenous trees and clearing invasive species from the natural forests and buffer zones. Part of the benefits for the communities will be harvesting of non-timber forest products (NTFPs) from the natural forests, under sustainable use plans. The project will provide training on improved harvesting techniques, processing, packaging and marketing, to those engaged in NTFP value chain (financed under outcome 3).

Activities and sub activities under this output will be executed by the Rwanda Forestry Authority (RFA) with the technical support from WCS.

Activity 2.2.1 Secure key remaining natural areas outside PAs

Outside of current protected areas, the Project will identify remaining natural areas and seek to secure their continued protection within district management plans or through other means. These are mostly small stands of natural vegetation on rocky outcroppings or strips of forests along the CND’s myriad of waterways. Although these will not permit large mammals (e.g. elephants or mountain gorillas) to migrate between the forested parks, these small remaining natural areas provide critical intermediate stepping stones for the broader biodiversity elements of the CND forest ecosystems. For the CND’s rich assemblage of bird species, these vegetated patches and gallery groves could prove to be essential stepping stones for connectivity under changing conditions. Small mammals and insects are likely to also directly benefit. And most importantly, the movement of birds and small mammals will support the transfer of seeds between forest patches, greatly improving the ability of forest ecosystems to adapt to changing climates, and continue supplying key ecosystem services to the people of the CND.

Sub-activities include:

2.2.1.1 Raise awareness on remaining protected natural forests in CNDL to secure their protection

2.2.1.2 In collaboration with RFA, REMA, districts identify and implement actions that support conservation and management of remnant protected natural forests in CNDL

Activity 2.2.2 Restore natural forest cover in and outside Protected Areas including riparian linkages

While the boundary integrity of the Nyungwe and Volcanoes NPs has been maintained for several decades now, internal degradation has occurred. Prior to 2004, approximately 12% of Nyungwe National Park had been affected by wildfire, including a catastrophic fire during a period of prolonged drought in 1997.⁹⁹ The natural vegetation was quickly replaced by a carpet of invasive ferns. In recent years, teams of local community workers led by WCS have demonstrated that careful removal of the fern cover reveals still viable native tree seeds and seedlings that respond to this exposure by rapid growth and the formation of new forest stands. The project will expand this labor-intensive process of assisted regeneration to the remaining 4,500 ha of burned lands within the NNP, restoring forest values and generating significant employment for local rural communities.

For the Gishwati Forest, restoration with native species will take place on adjacent, previously cleared lands. This and other restoration efforts should benefit from tree nursery techniques and planting trials already initiated by RFA, RDB, and NGO partners around Nyungwe, Volcanoes, and Gishwati itself. The project will work with the National Tree Seed Centre to provide training on tree propagation and supply of quality seeds/seedlings/propagation materials, especially of indigenous species and plantation species not being sourced locally.

A special element within the CND landscape is its dense network of waterways. Historically forested riparian habitats along these streams and rivers, however, have been largely cleared of their forest cover. Riparian reforestation efforts will restore an important erosion control factor as well as recreate important biodiversity connectivity along linear corridors. RWB will co-finance these riparian restoration efforts. With support from district and sector leaders as well as community representatives, community members will be hired from neighboring villages to the project sites. The project will make efforts to recruit the most vulnerable community members in category c,d and e, women, young landless, and others to carry out these activities.

Key sub-activities include:

2.2.2.1 Review mapping of degraded natural forest areas in core NPs, stepping stones, and unprotected riparian lands using updated imagery and ground truthing

2.2.2.2 Identify parcels for restoration in core PAs, stepping stones, and riparian lands using desktop and field-based assessment

2.2.2.3 Establish indigenous tree seed nurseries in the CND to serve core PAs, stepping stones, riparian land restoration and promotion of indigenous trees on farms and in protective forests

2.2.2.4 Recruit, train (in forest restoration methods), and equip community workers to be involved in core PAs restoration and riparian linkages

2.2.2.5 Assisted rehabilitation of 4,500 ha of indigenous forest in NNP

2.2.2.6 Active restoration of natural forest on 500 ha Gishwati Mukura National Park

2.2.2.7 Restore 1,500 ha of riparian lands

⁹⁹ Masozera, A.B., Mulindahabi, F., 2007. *Post-Fire Regeneration in Nyungwe National Park, Rwanda*. Wildlife Conservation Society.

2.2.2.8 Using permanent plots sampling, collect field monitoring data on tree species recruitment and growth for estimation of biomass, species richness, etc. in restored parcels in Nyungwe and Gishwati-Mukura National Parks as well as on riparian lands to assess success of initiatives, then replicate or adapt as needed

2.2.2.9 Perform Environmental and Social Screening on all positions being created by the project

2.2.2.10 Establish a grievance process for laborers

2.2.2.11 Update emergency and preparedness plan including risk mitigation guidance to local conditions at restoration sites

2.2.2.12 Train all workers on their rights and how to access the GRM

2.2.2.13 Provide code of conduct and emergency preparedness and safety training for all laborers

2.2.2.14 Hire and train labor and safety leads at each site to provide guidance to staff, be available for grievance issues, and monitor health and safety conditions for workers

2.2.2.15 Develop and implement strategy (including social marketing) to ensure that women and youth participate and benefit from forest restoration projects

Activity 2.2.3 Promote silvopastoralism with indigenous trees around Gishwati

Roughly 80% of Rwandans own livestock of some sort. While it is government policy to expand this percentage, conflicts over competing land uses are inevitable, especially for the country's most vulnerable rural populations. In the area around the protected forest of Gishwati, the presence of many slopes greater than 55% opens the possibility for reforestation of larger blocks with tree species compatible with an undergrowth of grass or other forage. In addition, some existing pasturelands could be enriched with tree species. These areas could then provide critical off-farm fodder for goats, sheep, and perhaps even cows, while also serving vital ecosystem services.

Plans for tree planting/restoration on pasture lands will be elaborated in collaboration with the local communities and technical (extension and local government) services. Areas meant to produce fodder and for water retention will be demarcated during community consultations for action plans. Already, important forage grasses, such as *Brachiaria* grass known to increase milk production while reducing methane emission from cattle, have been identified while tree fodder such as *Calliandra*, *Leucaena*, *Acacia*, *Gliricidia* are well accepted but not widely adopted due to local capacity barriers. Improving adoption of better management practices will require technical assistance considering needs of cattle owners, local milk cooperatives, and traders.

Beneficiaries of these activities will be pasture land owners and cattle keepers in the Gishwati area. They will be identified in collaboration with district and sector leaders, RFA, EE and representatives of pasture lands and cattle owners. Special attention will be paid to specific groups such as women-headed households, youth, and historically marginalized groups if they own grazing land in the Gishwati area and for paid work. Awareness and consultations with these pastoral communities will be conducted during project inception to ensure farmers' involvement in identifying and selecting key areas for restoration, suitable species to grow in their pastoral areas, capacity barriers, and associated technical assistance to be provided to adopt better management practices.

Key sub-activities include:

2.2.3.1 Assess the status of pasture lands in the Gishwati area and identify key areas for restoration and the potential to introduce indigenous species

2.2.3.2 In collaboration with landowners, identify suitable indigenous species for fodder trees, shrubs, grasses, and herbaceous legumes that have potential to improve rangelands and increase their climate adaptive capacity

2.2.3.3 Produce and disseminate fodder trees, shrubs, grasses, and herbaceous legumes to project beneficiaries

2.2.3.4 Train beneficiaries on improved livestock and pasture management

Component 3: Enhancing climate adaptation through resilient livelihoods

Increasing the extent of protective forests, enhancing and restoring forest plantations, and increasing agroforestry tree cover are key steps to restoring ecosystem function and services identified in the National Forest Policy Report (NFPR 2017), especially i) reducing the risk of flooding, landslides and soil erosion from extreme climate events and ii) increasing long-term supply of wood fuel resources (the primary energy source for 77% of Rwandans¹⁰⁰). Together with fuel efficient cookstoves, these trees will provide a sustainable source of fuelwood and reduce the amount of time women spend collecting fuelwood, enabling them to participate in other economic activities and programmatic opportunities. These outputs (cookstoves, plantation restoration, and agroforestry) will deliver carbon sequestration benefits totaling 522,606 tCO₂eq over 20 years.

In working to develop a comprehensive land use plan that incorporates climate adaptation, and while ensuring Rwanda's forest ecosystems are strengthened through forest management and restoration, attention must be paid to people who live next to the forests and those who are the most prone to causing degradation of these forest systems. In particular, this includes people who either do not have access to land or whose land holdings are insufficient to meet their daily needs. These people are most likely to enter into and unsustainably use protected and protective forests to help meet their daily needs. Thus, this Project component targets these vulnerable households – youth, people from historically marginalized groups, and farmers with insufficient holdings – to build their capacity in financial literacy and enterprise, in order to strengthen their resilience to economic and climate shocks.

Also, to maximize the opportunities from improved silvicultural practices, it is necessary to improve business practices and market linkages for farming communities. More specifically, organizing and linking farmers to local and national wholesale traders will enable them to sell their produce more efficiently, maximizing profit and reducing wastage. To facilitate these linkages, this component will also build the business capacity and market linkages necessary to support the transition of farmers and their communities away from unsustainable practices that magnify their vulnerability to climate change and towards more secure livelihoods and expanded income opportunities. This includes strengthening capacities, developing inclusive business models, and developing market value chains for livelihoods that do not depend on traditional smallholder farming.

The project will develop alternative sources of income to improve living conditions for people in the CND who depend for a large part of their income on exploitation of the forest and natural vegetation. This approach, focusing on vulnerable groups, aims at real economic empowerment of the concerned families through the realization of their own business plans. Several promising activities related to the use of the forests *aside* from wood-gathering (such as beekeeping, extraction of essential oils, herbs, and other forest

products) will be analyzed. These business plans, tailored to individual circumstances, will be realized with support and tools provided by the project.

From the second year onwards, these vulnerable groups will become members of existing cooperative organizations that will be supported by the project, particularly in value chains important for forest conservation, such as production and export of honey and the extraction of essential oils. These cooperatives provide various services (e.g., information on markets / export, input supply, aggregation of commodities, etc.) and they also play a role in the protection of the interests of their members.

The project will develop the private sector strategy (which will include engagement of impact investors interested in conservation-friendly commodities) for livelihoods implementation in the target areas through the identification of new livelihood opportunities for the CND. We will establish relationships with several buyers fitting our selection criteria, and engage partners such as Blue Ventures to expand market opportunities for products that meet international or local market opportunities. On the producer side, we will build strong relationships with other projects and programmes that support private sector engagement with community organizations, and producer associations to pursue promising market opportunities (local and international, as appropriate).

For component 3 and the project overall, the eligibility criteria for selecting beneficiaries will be:

- Households with limited access to financial resources for implementing climate adaptation measures as determined by Local Authorities
- Women -headed households.
- Households in the CND living in key areas such as those surrounding PAs, identified stepping-stones, landscape linkages, and broader farming mosaic with exposure to erosion.
- Most vulnerable households including people with disabilities
- Land owners where the project intervention will have to intervene

These criteria will be further refined by the project EE in consultation with key stakeholders (MoE, Districts and community representatives) once the project is approved.

Output 3.1 Farming methods enhance productivity, reduce erosion and flooding risks, contribute to ecosystem services, and support connectivity

Soil degradation and loss, largely from the cultivation of steep slopes in the western highlands of the CND, has caused declines in agricultural productivity and serious downstream problems of siltation and flooding. While the former means that smallholder farmers in the hills increasingly struggle to feed their families, the latter poses a threat to the long term viability of water provision services in municipal areas and hydro-power generation¹⁰¹, which is a key pillar for Rwanda's low carbon growth strategy.

This Project will support smallholder farmers in high erosion risk areas to implement Sustainable Land Management (SLM) practices techniques and the active *restoration* of degraded forest lands. By targeting high risk areas the Project can have a significant impact to reduce soil erosion, improving the resilience of agricultural livelihoods in the highlands, while having a positive impact on water flow and water quality in the lowlands.

¹⁰¹ Rwanda Environment Management Authority (REMA), 2009. *Rwanda State of Environment and Outlook Report*, Chapter 8: Energy Resources.

In each selected priority site, the Project will support the implementation of the full range soil erosion control best practices including agroforestry. While the erosion control practices may vary (e.g. radical vs progressive terracing), agroforestry practices will be established across the entire 3,346 ha. RFA will mobilize participating landowners on whose lands soil erosion control best practices including agroforestry shall be implemented. Landowners willing to participate will sign a consent form allowing project activities on their land and will be grouped in vigilance committees. These vigilance committees will sign MoU with Districts and RFA as a commitment to participate and manage sustainably their land and trees with clear role and responsibilities of each (farmers, districts and RFA) specified. The consent forms will be granted to RFA and Districts where landowners are located. The vigilance committee will be responsible for ensuring the sustainability of project interventions implemented on community land. The MoUs signed by Districts, RFA and the vigilance committee will not be legally binding but rather to establish certain understanding in the implementation of the project.

Interventions will be made in collaboration with district officials, according to agreed priorities and objectives, and involving local farmers and contractors. Table 29 in the feasibility study (FS) provides a preliminary list of the fuelwood, fodder, timber and fruit tree species that have been identified as suitable for agroforestry systems in the CND, based on their resilience capacity against climate-driven temperature and precipitation changes, as well as community preferences - particularly the needs of women farmers, historically marginalized households, and youth. They also concur well with recommended species for the North and Western Province as described in the Agroforestry Technical Guide¹⁰².

Activity 3.1.1 Restore high slope areas (>55%) as protective forests

In Western Province, a major part of the CND landscape, 80% of the increase in tree-planting in recent years has come from private landowners seeking to earn revenue or increase their access to wood-based fuel¹⁰³. Beyond this utilitarian interest, however, the NFPR 2017 calls for increasing the extent of protective forests and enhancing and restoring forest plantations as key steps to restoring ecosystem function and services, especially i) reducing the risk of flooding, landslides and soil erosion from extreme climate events and ii) increasing long-term supply of woodfuel resources (the energy source for 77% of Rwandans¹⁰⁴), providing biodiversity habitat, and sequestering 193,472 t CO₂-eq over 20 years.

The current land use master plan discourages cultivation on slopes greater than 55% and encourages protection of existing vegetation and afforestation with indigenous species. This Project will promote indigenous species in rehabilitating existing forests and establishing new ones on bare lands. This activity will result in highly productive, climate-resilient woodlots and forestland with fully restored ecosystem services and significantly increased long-term carbon sequestration. It will afforest/restore 2,500 ha of degraded private smallholder land and district/state land by restoring existing degraded tree stock, afforesting bare areas, promoting good silvicultural practices, and facilitating the adoption of Simplified Forest Management Plans (SFMPs), as recommended by 2013 forest law. In the priority regions selected for afforestation/restoration, awareness and capacity building activities will be undertaken to build support and understanding of the PFMU approach. The objective of PFMUs is to combine economic profitability, preservation of environment, and improvement of living conditions of the local population, as well as to apply a gender-responsive approach to wood resource management. For privately owned land, only small-scale individual private landowners will be included, and this activity will not involve restoration of private institutional plantations. RFA will mobilize participating private landowners on whose lands afforestation

¹⁰² Ministry of Environment. 2018. National Agroforestry Strategy.

¹⁰³ Arakwiye, B., et al, 2021. *Thirty years of forest-cover change in Western Rwanda during periods of wars and environmental policy shifts*. Regional Environmental Change 21(2).

¹⁰⁴ Government of Rwanda, 2021. Rwanda Household Survey 2019/2020. National Institute of Statistics of Rwanda.

trees shall be planted. Landowners willing to participate will sign a consent form allowing project activities on their land. Following this, private woodlot owners will then be organized into local groups (around 40-50 ha of woodlot per group), PFMUs (Private Forest Management Units) and assisted to form and register as cooperatives and elect among themselves two cooperative committees: Administrative committee made of 5 persons (President, Vice president, Secretary and two Advisors); supervisory committee made of 3 persons (President, vice president and secretary). Note that gender consideration is mandatory (at least 30% women).

When the cooperative committees are elected, they are mandated to represent forest landowners at different events and fulfill their interests, especially in seeking the best market for their forest products and seeking for other income generating opportunities. RFA will sign MOUs with different cooperatives specifying each party's respective role and responsibility.

Experiences from pilot PFMUs established under the FMBE project in the districts of Gicumbi, Gakenke and Rulindo Districts of the Northern Province and Rwamagana District of the Eastern province have demonstrated that this approach is helping to achieve economic profitability, preservation of environment and improvement of living conditions of the PFMU members. Forests are sustainably managed, and all PFMU members respect the management plan and planned harvesting period. All silvicultural practices are properly followed because cooperatives supervise all activities with technical guidance from the district technicians. The forests within a PFMU are economically attractive to potential buyers due to good stock at harvesting time. PFMU members have access to small loans while waiting for the forest to mature. This shows that PFMUs established in these areas are also serving as demonstrations to other forest owners supported by other projects in the country. Green AMAYAGA and Green Gicumbi, and TREPA Projects are scaling up this approach in other parts of the country.

On public land, forests will be organized into state/district FMUs to be managed by professional private companies through long-term concession agreements in line with the government strategy. While a concession agreement will be signed between the private forest companies and the MoE, the Project will only support the design of the forest management plans and development of contracting documents. Registration of restored forest parcels will be guided by well-trained district forest officers and forestry sector extensionists, assisted by the new FMES software and related GPS/tablets, which will provide automatic statistics, maps and register owners. For each FMU, a SFMP, reviewed or developed through a participatory process, will outline the basic information of the site (e.g. owner, species etc.), as well as a harvesting plan and schedule for silvicultural operations (thinning, pruning etc.). However, some areas of the state plantations that are vulnerable and exposed to high risk of degradation (high slopes) will require setting up special protective measures. Restoration on identified vulnerable areas will be done by establishing appropriate tree plantations that will offer a better protection and conservation of biodiversity while providing other ecosystem services including tree species that improve forage for bees. The choice of tree species planted will take into consideration women's participation, use and priorities. The project will further strive and advocate for gender equity for selection of plots which will be done in consultation with communities.

Sub-activities include:

3.1.1.1 Introduce and raise awareness of indigenous species to target stakeholders in CND

3.1.1.2 In consultation with RFA, National Land Authority, districts and communities, determine fragile areas (steep slopes > 55%) to be allocated for protective forests and their ownership

3.1.1.3 Assess the current status of the indigenous tree species in selected areas for protective forests and select indigenous tree species appropriate to CNDL

3.1.1.4 Develop restoration plan for protective forests

3.1.1.5 Build capacity of local stakeholders (men and women) on PFMU approach and methods

3.1.1.6 Design and approve SFMPs of private FMUs

3.1.1.7 Ensure consent of smallholders prior to planting trees

3.1.1.8 In collaboration with smallholders reforest/restore 2,500 ha of public or private land with slopes >55% and ensure sustainable management under private FMUs according to approved SFMPs

3.1.1.9 Support monitoring and evaluation of restored private FMUs

3.1.1.10 Assess impacts of exotics on neighboring lands and mitigate their negative impacts

3.1.1.11 Develop and implement strategy (including social marketing) to ensure that women and youth participate and benefit from forest restoration projects

3.1.1.12 Assess the benefits and costs of each proposed tree and plant species and how those benefits affect different population segments

Activity 3.1.2 Develop on-farm agroforestry for high-caloric and indigenous tree species

The extreme land scarcity of the CND precludes smallholder farmers – many of them among the most vulnerable and land-poor of the region – from dedicating any significant proportion of their holdings to tree production. However, the use of certain agroforestry species that both enrich the soil and control erosion when interplanted with ground crops can greatly reduce this conflict.

Under this activity, the Project will facilitate the production of on-farm afforestation plans with action plans for implementation. Formulation of the forest landscape restoration plans will follow the methodology introduced by the World Resources Institute (WRI) and IUCN and already tested in the country by the former Ministry of Natural Resources, as recently modified and applied for the Gatsibo Forest Landscape Restoration baseline conditions assessment. The methodology will involve three simple steps: a) Geospatial analysis to map degraded land that presents an opportunity for forest and landscape restoration (FLR), which will highlight areas with best potential for restoration; b) economic analysis to model the costs and benefits of degraded and restored land; and c) designing a restoration/afforestation action plan, based on an in-depth assessment of the conditions required to implement the FLR in the selected sectors.

For selected areas where Project activities may lead to reduced access to resources, the Project will undertake an in-depth Environmental and Social Impact Assessment (ESIA) in the first year of implementation and design an Environmental and Social Management Plan (ESMP) to guide implementation. This ESMP will include an access restriction mitigation plan, if deemed necessary. The matter of Free, Prior and Informed Consent (FPIC) will be explored during the ESIA and the approach applied if deemed appropriate. These documents will be based on materials contained in Annex 6.

Drawing on experience with high-elevation species identified both in Rwanda and across East Africa, the Project will support the following sub-activities:

3.1.2.1 Identify sub-areas of intervention for agroforestry dissemination in the CND

3.1.2.2 Introduce and raise awareness of agroforestry in target communities

3.1.2.3 Develop the capacity of extension agents at district/sector level and NGOs to support adoption of agroforestry technologies

3.1.2.4 Establish agroforestry/fruit tree nurseries to facilitate access to quality planting material

3.1.2.5 Promote sustainable land management practices by stabilizing existing terraces, and plant high calorific agroforestry species to provide a sustainable source of fuelwood for energy-efficient cookstoves to protect sloping land against severe soil erosion.

3.1.2.6 Train RFA, extension agents, Project participants and community members on specific techniques for identification of and management of invasive pests and pathogens

Participants in sub-activities 3.1.2.3 and 3.1.2.6 will be selected by districts, JADF, RFA and farmers representatives. Special attention will be given to youth, women and other vulnerable groups. RFA and RWB will co-finance sub-activities 3.1.2.4 and 3.1.2.5.

Output 3.2 Rural livelihoods generate alternative incomes & reduce pressure on forests

To respond to the growing pressure on natural resources and increase the local community's resilience to climate change in the CND landscape, the Project will address key drivers of deforestation by creating alternative livelihood pathways for youth and women through ecotourism and income-generating, climate-resilient crops, as well as strengthening supportive financial services.

The setup of interventions in this output will focus on enhancing on-farm production capacities in market-facing promising value chains in agroforestry (especially avocado, macadamia and short-duration crops) for immediate income and beekeeping taking advantage of the agroforestry and the crops; the combination of these will promote ecotourism as supplementary income (avocado, macadamia, forestry, agroforestry, short-term crops, and honey). This output will also provide solutions to the growing households' energy poverty which has exerted pressure on natural resources and eventually exacerbated the local community's exposure to the stress of climate change in the CND landscape.

This project will employ a market systems development approach that encompasses business coaching, value chain development, and access to finance. This entails multi-stakeholder collaboration with business service providers, financial institutions, community-based enterprises, value chain actors, and farmer cooperatives. Ultimately, this project will develop and or promote a financial model that enhances the private sector ecosystem collaboration and partnership to stimulate finance in forestry and agroforestry sectors and support market-oriented access to energy-efficient cook-stoves to reduce pressure on forests.

The value chain development for potential revenue generating streams such as honey, ecotourism, and efficient cookstoves requires grant support of GCF grant to finance the activities that are important for the most vulnerable, as well the drivers for the MSD to take root; these include the development of the identified revenue generating streams, strengthening access to market information, and improving quality of the products and produce aggregation.

To target the poorest and most vulnerable, and empower women and youth towards meeting inclusive development goals, critical support at the nascent stages is essential. The GCF grant will facilitate the integration of the poorest and most vulnerable women and youth in the CND and position for private sector investments that will build a firm back base for sustainable community development. The GCF grant will not only provide financing support but will be critical in creating enabling opportunities for market development beyond the project to catalyze private investments.

While promoting the above-mentioned households cooking solutions and income streams, the selection of the beneficiaries will follow the government's procedures, and in the case of CND will focus on the vulnerable groups of youth, women, and historically marginalized people and will be selected based on an assessment

of poverty levels (also known as Wealth Categories) conducted by the government to ensure the most vulnerable are selected. Selection criteria will further take into account social and cultural factors such as gender and age which intersect to predispose some groups more to vulnerability than others.

Activities and sub activities under this output will be executed by the Rwanda Forestry Authority (RFA) with the technical support from CORDAID.

Activity 3.2.1 Develop agroforestry related value chain for markets access.

Most traditional tree farming in Rwanda is linked to fuel production in terms of firewood, charcoal, and timber for construction. However, to further strengthen community resilience to climate change, there is a need to introduce fruit and nut trees that have high economic value in the agroforestry system to generate more income for smallholder farmers but also increase its adoption, especially women-headed households. Though women-headed households face the same issues and challenges as male-headed households, they are less likely to be included in income-generating activities and are disproportionately impacted by climate change because of prevalent gender inequality issues. In the Feasibility Study, avocado and macadamia have been identified as high economic value crops with high market potential, and the economic benefits of identified fruit trees is paramount to increase farmer adoption and buy-in. Since fruits and nuts take time to mature, short-term crops like chili and French beans will be promoted to support smallholder farmers with limited land resources (less than 0.4 ha) to create additional, more quickly generated income and incentivize adopting agroforestry practices. Regular income will help vulnerable communities to cope with climate variabilities and shocks.

The identified agroforestry value chains have market potential and potential for intercropping with short-term horticultural vegetables and fruits while providing further impacts on carbon sequestration, nutrition value, and market access in the CND. Selected fruit trees are expected to meet the dual purpose of soil protection and climate impact while generating long-term economic benefits to the most vulnerable community members of the CND region.

For each crop value chain identified, specific bottlenecks will be addressed while focusing on specific product-place combinations within the CND. Especially, attention will be on enhancing the quality and quantity of production in sync with particular demands from market niches, development of various models to improve access to finance together with financial institutions, and exploration of market opportunities that can generate off-farm jobs to absorb a good number of landless and other vulnerable groups of youth and women.

These priority crops are:

- Macadamia: a low-volume, high-value export crop; in-country processing creates jobs for women and youth. Two hundred and four trees cover one hectare, and one mature tree produces up to 60 kg of macadamia nuts per year with an income of 25,000 FRW per kilogram. Furthermore, every tree must be separated from the next by 7 meters; thus, many other commercial and consumable crops may be planted and harvested while waiting for macadamia's fruiting maturity.¹⁰⁵ The global demand for macadamia outweighs the available supply, which presents a huge business opportunity for macadamia farmers in Rwanda. For example, the current macadamia supply meets only 48% of global demand and the global market for macadamia is expected to further expand from \$1.17 billion to \$2.36 billion by 2027. The project will build on the government export initiatives to promote

macadamia farming in CND areas through National Export Board (NAEB) and Rwanda Environment Management Authority (REMA) to include macadamia in its tree planting drive in the CND.

- Avocado: The extensive penetration of avocado cultivation amongst smallholder farmers (SHFs) and the favorable growing conditions lead to avocado yields (21MT/ha) that are nearly double those of Kenya. The increasing demand for export, and cash yield/ha are higher than maize or Irish potato. Through linkage to local avocado private value chain actors, packaged avocados will be exported to the European Union, where the channel price for avocados is \$1.75 per kilogram, providing an estimated 32% margin.¹⁰⁶ Avocado production presents numerous opportunities to smallholder farmers in CND: for example, earnings from avocado exports have been increasing from \$1.6 million in 2020/21 to \$4.5 million in 2021/22 and production increased from 1,000 tonnes to 2, 500 tonnes in 2021/22. According to the NAEB, the production is set to double this year and generate \$8 million from exports. To further strengthen its avocado exports the government has launched sea freight, which will not only increase export volumes but also make them competitive in terms of pricing.
- Avocado is projected to be the second most traded tropical fruit, after bananas, by 2030. The United States and the European Union, where consumer interest in avocados is fuelled by the fruit's health benefits, are expected to remain the main importers, with 40% and 31% of global imports in 2030, respectively¹⁰⁷. The project teams in CND will work closely with private sector partners and communities to develop value chain analysis, business development strategies, and business planning. The team, led by our Livelihood Specialists, will work with these private sector partners to prepare action plans detailing technical services we will provide to train producers to supply commodities to markets.
- Vegetable crops such as French beans and chili: It has been demonstrated that these crops can double the income per hectare of maize and potatoes and employ a good percentage of the 16% reported unemployed women and youth¹⁰⁸. Commercial farm wage reported labor in postharvest production handling at 72% women, 20% youth (see Annex 2.3). French beans can potentially deliver a gross margin of 22,040Rwf per acre and a return on labour of 2,755Rwf per day, while chili can offer a gross margin of 46,384Rwf per acre and a return on labour of 7,731Rwf per day.¹⁰⁹

The Project will also enhance ecotourism and honey value chains. Particularly in areas close to national parks, the potential business development of ecotourism can go hand in hand with honey and wax product development. Our analysis shows that the honey value chain already has a sizable production history in Rwanda. Various initiatives exist that have encouraged farmers to invest in modern hives. However, the current sector is constrained by a perception of limited profitability due to low prices, gender patriarchy due to male dominance, and limited market access. The honey (and various by-product) value chain shows a wide range of different consumer target groups, whereas honey is too often marketed as a generic product. This is a crucial bottleneck: while the value chain actors engage in the business, there needs to be more incentives to invest financial resources to improve production or quality, such as through modern beehive technology. It will be essential to link ecotourism destinations and agro-tourism to enhance the potential for an improved business case and help beekeeping cooperatives enhance their processing, quality management, and marketing processes with large scale buyers and exporters such as Api Business Development Company (ABDC) and create a sustainable and expanded source of income especially for

¹⁰⁶ Karuiru, M., 2018. Value chain management and the performance of avocado fruit small scale farmers in Kandara sub county, murang'a County, Kenya. *International Journal of Physical and Social Sciences*, 8(10), pp.24-40.

¹⁰⁷ OECD - FAO Agricultural Outlook. <https://www.fao.org/3/cb5332en/cb5332en.pdf>

¹⁰⁸ Rwanda Development Board, 2020. Labour Market Trends Analysis Brief 2016-2020. <https://rdb.rw/wp-content/uploads/2022/07/Labour-Market-Brief-2020.pdf>

¹⁰⁹ Van Keulen, Rik, Rumenera, P., Banjara, G., Colantuoni, M. and Muthamia-Mwenda, J. 2022. Horticulture value chain analysis – Opportunities for youth employment in Rwanda. Rome, FAO. <https://doi.org/10.4060/cb8266en>

women. Api Business Development Company has applied for EU certification in order to export honey to international markets. Nevertheless, the company—which has a capacity of handling 120 MT of honey per year—still faces the challenge of low production, inadequate even in meeting local demand.

Rwanda has a well-established record of generating off-farm benefits from its protected area ecotourism programs. The Project will build on this by supporting sustainable tourism activities around and between the Gishwati and Mukura Forests, as outlined in the GMNP Tourism Master Plan.¹¹⁰ New trails and base camps outside of the parks will generate employment and income opportunities for rural women and youth, as well as attract tourism revenue to help support forest protection and management. In addition, this Project will initiate and train Community Freelance guides to mutually benefit the parks and local communities neighboring Mukura-Gishwati NPs. A freelance system suits the seasonality of tourism, and the demand and sourcing of guides from the local communities stimulates economic empowerment among the young and unemployed living in rural areas around protected areas. This initiative will enable community members to provide a service and generate income, and benefit from the growing tourism industry in Rwanda, which in turn supports conservation over the long term.

In collaboration with district authorities in the CND, who will provide a list of existing cooperatives, the project will select cooperatives based on value chains, number of women and youth, number of members with vulnerabilities and membership size. Furthermore, we will assess selected cooperatives using the Cooperative Assessment Matrix tool (CAM) to identify capacity gaps. Selected cooperatives, participating financial institutions (PFIs), and value chain actors/small and mid-size enterprises (SMEs) (including seedling producers, aggregators, traders, etc.) will receive capacity-building to improve their business skills and access to finance, and to strengthen value chains.

Women play a crucial role in balancing household demands with market production. This is why it is so important to give women access to financial and value chains related services. The project aims to reach at least 50% of the women beneficiaries in the project area. In areas where women are under-represented in value chains, the project will identify and address gender-specific challenges and barriers related to the different value chains and promote increased female representation and active participation in Producer organizations by creating gender committees that will recruit more women, and support women to be part of the leadership committees.

Key sub activities are:

3.2.1.1 Identify and map of vulnerable groups of youth and women within the community

3.2.1.2 Develop and implement strategies to ensure that women, youth, and other marginalized groups benefit from rural livelihoods and financial services.

3.2.1.3 identify farmers and cooperatives involved in selected value chains in consultation with district/local authorities in CND.

3.2.1.4 Assess capacity gaps of identified farmer organizations to strengthen their capacities

3.2.1.5 Identify and develop Nature-based tourism value chain as a means to generate viable economic alternatives for local people living near areas of high biodiversity in the CND.

RDB will co-finance sub-activity 3.2.1.5.

¹¹⁰ Rwanda Development Board (RDB), Rwanda Environmental Management Authority (REMA), 2018. *Tourism Development Master Plan for Gishwati - Mukura National Park*. [Link](#).

Activity 3.2.2 Facilitate and scale up climate-resilient value chain products

The project will conduct deep analysis for identified value chains, namely macadamia, avocado, chili, and French beans. Such analysis will provide evidence of the opportunities and constraints in the selected value chains and will involve mapping out chain operators and service providers at each segment of the value chain to identify existing capacity gaps and orient overall value chain interventions in the project. Furthermore, value chain analysis will enable the project to clearly understand the characteristics of the selected value chain with attention to smallholder farmers, women, youth and vulnerable groups.

To build strong and sustainable value chains, the project will improve information flows within the chain, by strengthening coordination among value chain actors and supporters. This will involve creating value chain platforms in the CND region to increase transparency and fair business relations. The project will focus on activities that strengthen and scale up value chains: for example, by organizing business-to-business (B2B) events, establishing district value chain interprofessional/platforms, and promoting business development services such as swift, cost-effective, and accessible product certification for agribusiness. The project will play an active role by introducing key supporting functions such as financing, value chain information, certification services, etc., and advocate for fair business relations to ensure farmers are active players in value chains and not passive beneficiaries. The project will help strengthen the capacity of producer organizations in selected value chains. Often farmers fail to participate and benefit in value chain activities due to weak and poor organization, resulting from poor managerial skills and weak internal financial management. The project will use different capacity assessment tools like CAM and Scope insight to identify capacity gaps and develop capacity-building plans to ensure that producer organizations are strong and professional and capable of serving their members and building sustainable value chains. The capacity-building trajectory will also focus on encouraging participation by women, youth, and vulnerable groups and increase their access to services. Activities to strengthen farmers and producer organizations will be based on farming as a business (FAAB) concept to orient farmers toward business-oriented agriculture. Synergy will be built with projects such as [KIWP2](#) (Kayonza irrigation and integrated watershed management), a project funded by IFAD where this module is being applied.

In the selected value chain quality standards are an integral element of the core interventions, due to export market requirements. The project will strengthen the capacity of farmers to increase production in terms of volume and quality but also adhere to market standards (for example, global GAP, maximum residual limits (MRLs), etc.) Another challenge affecting the selected value chains in the CND region is access to inputs, especially seeds, and pesticides. This is exacerbated by the international market which requires certain seed varieties and types of pesticides. The project will strengthen the input supply system in collaboration by National Agricultural export board (NAEB) and selected off takers by establishing and supporting agro-dealers and nursery production through technical support and access to finance, to further access to input the project will also focus on building business models that help farmers access inputs via credit guarantee schemes. The project will create strong linkage and business relations between input suppliers, farmers/producer organizations, and markets/buyers to enhance flow of information among value chain actors. Based on experience, the project will ensure that processors/buyers have a say or influence on type, variety, and quality of inputs used by producers to ensure produce marketability. Furthermore, the project will collaborate with other projects like [PSAC](#) (promoting smallholder agro-export competitiveness project) and NAEB (National agriculture export board) to further build a sustainable input supply system. Most value chains are not well developed or are still nascent in the CND region, based on our experience with agroforestry value chains with the TREPA project for instance. In such situations, off-takers, processors, and traders are few and poorly organized, with minimal outreach to farmers. They normally experience challenges in understanding and meeting market standards (i.e., sanitary and phytosanitary (SPS) standards, Global GAP, hazard critical control point certification (HCCP), etc.) which limit their

competitiveness and market scope. The project will improve their capacity to meet these standards, provide other support to access and integrate cold chains, especially for avocado and French beans, and build out grower schemes in which off takers/exporters play a vital role. For MSMEs and start-ups, the project will strengthen their managerial capability for running successful, efficient businesses. Through organizing value chain linkage events (B2B, market linkage sessions, etc.) the project will enhance buyers to reach more farmers and producer organizations.

Addressing rural poverty and food insecurity using agriculture requires investments in off-farm agricultural activities to complement primary production.¹¹¹ Synergies will be created to reinforce direct linkages of small producers to other value chain actors with inputs (seeds and seedlings, fertilizers) and outputs (processors, wholesalers, and retailers).

The project will use the Market Systems Development (MSD) model to enhance value chain development. This will entail multi-stakeholder collaboration with business service providers, financial institutions, community-based enterprises, value chain actors, and farmer cooperatives. In-depth target area assessments will be conducted to tailor business cases for the selected value chains, particularly focusing on identification and engagement of existing farmer organizations and cooperatives, youth and women-owned businesses, and local small and medium enterprises (SMEs, value chain actors) with the capacity for production aggregation, value addition, and market potential to absorb production from smallholder farmers' cooperatives. This localized stakeholder and value chain exploration will serve as a basis for the organization of market linkages to help value chain actors better understand the market, challenges, and opportunities.

Regarding honey value chains, the project will estimate the number of beekeeping cooperatives operating in the targeted region or its surroundings. Members of these cooperatives will be trained in improved honey production techniques and adding value through wax-based products. This will involve establishing honey and wax storage stations and associated processing facilities. Cooperatives will also be an important vehicle for fostering tree growing for improved bee forage. The project will foster interactions between the cooperatives and private companies through business roundtables, trade fairs, and similar events focused on honey and beeswax products and this activity will be co-financed by RDB.

This activity includes in-depth target area assessments to tailor business cases for the selected value chains, particularly focusing on identification and engagement of existing farmer organizations and cooperatives, youth and women-owned businesses and local small and medium enterprises (SMEs, value chain actors) with capacity for production aggregation, value addition, and market potential to absorb production from smallholder farmers' cooperatives. This localized stakeholder and value chain exploration will serve as a basis for organization of market linkages to help value chain actors better understand the market, challenges and opportunities. Key sub-activities are:

3.2.2.1 Analyze identified value chains and promote them among the CND stakeholder networks

3.2.2.2 Strengthen the capacity of producer organizations to improve managerial capacities, farming practices, and gender inclusion in their organizations

3.2.2.3: Establish and strengthen relevant value chain platforms in the CND area and promote business development services to enhance value chain coordination.

¹¹¹ Selasi, J., 2012. *Making Small Scale Farming Work in Sub-Saharan Africa*. In Global Forum on Food Security and Nutrition "FSN Forum", FAO.

3.2.2.4: Build and strengthen the capacity of off-takers (MSMEs, SMEs, and off-takers aggregators) to increase their competitiveness and comply with various standards

3.2.2.5: Facilitate and strengthen access to quality inputs for farmers and producer organizations in the CND regions

3.2.2.6: Establish “CND brand” for honey and wax products

Activity 3.2.3 Facilitate access to output markets for vulnerable farmers

Lack of access to the market is a challenge to smallholders: only 4.2% of agricultural households have access to contract farming ([Agric HH Survey, 2017](#)). Limited access to markets for smallholder farmers is mainly attributed to lack of access to market information, failure to meet market requirements, or undeveloped value chains. Some of the challenges also stem from poor quality and poor produce aggregation. The project will strengthen the capacity of farmers and producer organizations to improve quality and aggregation capacity, and in creating market linkages with off-takers or buyers through market information. The project will promote contract farming and out-grower schemes, where farmers are required to practice market-oriented farming, with the main objective of establishing long-term business relationships with markets/buyers. A focus on financial literacy and community initiatives will also mobilize domestic investment and expertise, facilitate access to financing, incentivize the private sector to create innovative financial products, and support improved data collection and measurement of outcomes.

Furthermore, through producer organizations, the project will develop an inclusive business model to improve produce aggregation and supply coordination and to increase farmers’ bargaining power and influence in formal value chains, particularly for remote/hard-to-reach women, youth, and vulnerable groups. To further strengthen market access, the project will develop innovative market access approaches such as “it takes two to trade”, which focuses on understanding the challenges and weaknesses that limit demand (traders) and supply (farmers). The project will create interventions to eliminate challenges and improve weaknesses to build sustainable market access. The project will also build on the national digital agriculture strategy (NDAS) to introduce digital market access, where farmers access market information through their mobile phones through USSD or IVR interactive voice response. A similar intervention is currently being implemented in the Kayonza district through an IFAD-funded project: Strengthening Agriculture Resilience through Learning and Innovation ([STARLIT](#)). This approach is particularly customized to poor, rural, and hard-to-reach farmers who do not own smartphones.

To implement this, the project will identify and select two buyers/off-takers for each selected value chain based on market needs and link them to farmers and farmer organizations. The eligibility criteria for buyers/off-takers will be developed by RFA in collaboration with CORDAID during the implementation. The project will organize periodic market linkage events and business-to-business (B2B) workshops through established value chain platforms. Additionally, the project will create a clear link between market access and other value chain supporting functions, such as value chain finance, to ensure that farmers have access to working capital for aggregation and marketing. Various models will be developed including tripartite agreements (buyers, producer organizations, and financial institutions) to increase confidence among partners and financial flow.

The activity will work to scale up the market system developed using the Output 3.2 inception report on beneficiary persona and stakeholder engagements to facilitate market system growth within the selected value chains, including bee products, and other women and youth-focused sectors. The activity will work to scale up the market system developed using the Output 3.2 inception report on beneficiary and stakeholder

engagements to facilitate market system growth within the selected value chains, including bee products, and other women and youth-focused sectors such as adoption and sales of fuel-efficient cookstoves.

The project will also build strong business relationships and linkages between input suppliers, markets, and farming cooperatives. This will build common understanding and smoother flow of information to ensure that input suppliers and farming cooperatives respond swiftly to what markets need. The activity will train vulnerable groups and farming cooperatives on “farming as a business” concepts.

Finally, trade agreements will be created between cooperatives and relevant national and regional companies that buy and sell such products. The Project will support and ensure protection of the interests of the cooperatives and their members while negotiating and signing relevant agreements for win-win situations.

Key sub-activities:

3.2.3.1 Develop an inclusive business model to Integrate vulnerable farmers into formal markets

3.2.3.2 Organize microenterprise training and develop and implement green grants programs targeting landless youth and poor, as well as women who are unable to access direct benefits from land-based project activities.

3.2.3.3 Facilitate market access for women, youth, and historically marginalized communities through market linkage sessions

3.2.3.4 Provide technical support on market standards, contract farming, and financial connectivity.

3.2.3.5 Develop various market access channels including digital market access for rural farmers, especially women and marginalized groups

Activity 3.2.4 Scale up use and sales of clean fuel-efficient cookstoves

The project will actively promote the use of energy-efficient cookstoves to benefit the 90% of households in the CND that depend primarily on firewood to meet their cooking energy needs and thus contributing to accelerated forestry resources degradation while causing profound health hazards to the already impoverished community.

This intervention will reduce pressure on forests and farmland by raising household awareness of the differences between high and low efficiency stoves and by collaboration with the local TVETs institute to manufacture affordable efficient stoves that will address the limited availability of high-efficiency stoves in rural markets. Promoting the local fabrication will reduce dependency on cookstoves made outside the country and therefore make them affordable. Also, promoting locally made cookstoves designed to address local needs will lead to full adoption of improved cookstoves and will contribute to sustainable biomass resource use and prevent overexploitation of forest resources, thereby ensuring the success of the forest landscape restoration activities described in Outputs 2.1, 2.2, and 3.2 above.

The project will adopt a market-oriented approach to strengthen the cookstove supply chain in the CND and create strong linkages and business relationships with several ecosystem including the TVET center for training of youth and women SMEs on cookstove manufacturing and repairing, and microfinance institutions (MFIs) to ensure sustainable access and finance of cookstoves business.

As mentioned above, to ensure product quality and affordability to drive uptake, scalability, and business growth and expansion, the project will partner with integrated polytechnic regional colleges to develop a

Tier 2 (for emissions) prototype compatible with rural settings and affordable to rural communities. Youth-led SMEs will be trained on how to manufacture cookstoves and contracted to produce them as well as operating sales points within the community

The cookstove pricing and financing modality will take into account vulnerable groups, including historically marginalized groups mapped and selected in sub-activity 3.2.1.1 (social wealth categories c, d, e, and people with disabilities). For sustainability, the groups will be organized into saving groups and ensure they lend among themselves with the purpose of purchasing the cookstove, the project subsidies may target women-headed households which has contributed over a percentage of the total cost. Additionally, the product design will incorporate reducing health hazards, such as reducing respiratory diseases risks and others linked to climate impacts, contributing to reduction in morbidity and health treatment cost and enabling the vulnerable to have disposable income for investment into productive ventures and this will ensure sustainability of impacts beyond the GCF financing. Moreover, the project will target women-headed households, recognizing the significant role women and children play in firewood collection and cooking. The project will substantiate the time-saving benefits for fuel collection and links to women's decision-making and empowerment with clear indicators.

Considering the lessons learnt from similar projects where hand-outs lead to poor sustainability, this project will promote alternative income sources from fast-maturing crops such as chili and French beans which have ready market in Rwanda and externally. The members of the saving groups will be able to use income generated from the sale of fast-maturing crops (chili and French beans) coupled with savings accumulated from their groups and save progressively to purchase the cookstoves and replacement. The project's technical support to MFIs will build their capacity to develop tailored financial products for cookstoves with affordable repayment terms aligned to revenue streams from short-term crops and group savings, making it easier for farmers to access loans to buy cookstoves.

GCF funds with co-financing from RDB will provide support for the development of ICS prototypes, youth-led SMEs engaged in cookstove production, subsidies for cookstoves, creation and support of savings groups, and technical support to MFIs to develop tailored financial products to facilitate access to ICS by local communities. The project will ensure that the cookstoves are sustainably sourced and lead to minimal indoor air pollution and related respiratory conditions compared with the cooking methods presently in use both during production of the cookstoves and during usage by the individual households.

Key sub-activities are:

3.2.4.1. Promote local manufacturing through partnership with Integrated Polytechnic Regional Colleges (IPRCs) of Karongi and Musanze to help design prototype Tier 2 ICS to address local constraints.

3.2.4.2. Build business and marketing capacity of the cooperatives and youth led SMEs to act as the marketing channels of energy efficient cookstoves to CND community members based on wealth categories.

3.2.4.3 Create a specialized small business development fund, to help local women and youth groups build innovative business models for accelerated distribution of ICS.

3.2.4.4. Promote demand-through raising public awareness on the benefits of ICS by developing and disseminating messages on mass media and in community meetings.

Output 3.3 Financial services & private sector investment increased

Financing challenges and needs are particularly high in clean cooking, mainly for two reasons: (1) due to lack of income, the upfront purchase of clean cooking equipment presents an affordability challenge to most

households, which adds to the monthly fuel expenditures and (2) businesses operating in the sector are often unable to attract financing for investment in the value chain. Serving low-income households also generally yields low returns due to low customer base.

With a particular focus on marginalized communities, and vulnerable groups of youth and women in the CND, the Project will deploy a FSG model to capacitate the communities' productivity and financial resilience. The project will address barriers to access to financial services, specifically for women and youth by building the financial institutions to develop and financial products that meet the differentiated needs of vulnerable women and men in the CND. The project Gender Expert, Youth Focal Point along with the Community Mobilization Specialist and Community Engagement Specialist will provide support to address barriers to access, ensure women's financial skills are developed and there is community and financial institution awareness and support to ensure marginalized groups, including women and youth can access financial services including credit to invest in their businesses including purchase of the cookstoves. Besides this, to address the significant challenges facing MFIs and Participating Financial Institutes (PFI) such as limited understanding of forestry and agriculture sectors investment and the corresponding risks associated with climate change effects on credit repayment, this Project will coach the participating financial service providers to enhance their on assessing the impact of climate change on their clientele and loan portfolio, and ensure equitable access to financial assets by women, youth, and historically marginalized communities. Furthermore, the Project will facilitate local and international impact investors and PFIs to engage in investment and credit provision for farmer cooperatives and SMEs in the selected value chains.

In collaboration with TVET institutes/ICS youth producers and central (RFA, MININFRA) and local authorities (Districts), CND biomass and micro-finance experts will refine and test, for each type of ICS/fuel to be promoted, the category/profile of households to be targeted and the related subsidy/microcredit schemes to be implemented to facilitate stove adoption with minimum concessionality. These schemes will be designed to align with the procedures and rules of identified local partner financial institutions.

In addition, the financial products will be consistent with climate resilience and capacity-building for macadamia, avocado, vegetables, honey and ecotourism value chain actors under Outputs 3.2.3 and 3.2.4. Techniques will be developed to analyze and score forestry and agroforestry value chains that are climate-resilient.

Activities and sub activities under this output will be executed by the Rwanda Forestry Authority (RFA) with the technical support from WCS.

Activity 3.3.1 Facilitate access to finance & private sector investments

To de-risk the selected forestry and agroforestry-related value chains, the Project will organize sensitization sessions for value chain actors and private financial institutions to build a common understanding with regard to the selected value chains. This will also include introduction of monitoring and measurement systems to support the risk and climate impact assessment on the value chains to be undertaken by the financial service providers.

Communities involved in chosen forestry and agroforestry value chains in selected intervention sub-areas will be selected with a particular focus on landless people, historically marginalized communities, and vulnerable groups of youth and women in collaboration with local authorities (districts & sectors) and community representatives.

Furthermore, the interventions will capitalize on the fast outreach of informal saving groups. Saving groups are self-managed and self-capitalized entities that use members' savings to facilitate lending within the

group. They set up clear rules and regulations which are known and adhered to by all group members. Groups consist of a limited number (10-25) of people who have freely chosen to work together, and offer self-managed savings, insurance, and credit services in remote rural areas. Poorer segments of the community often do not have a regular source of income—however, within a savings group, people can save to purchase agricultural inputs, such as seeds, fertilizer or equipment. Savings groups may also provide mutual assistance for group members, such as medical insurance or loans to members in need.

Saving groups have transformed marginalized communities by mobilizing local savings, which provide members with a means to cope with emergencies, manage household cash-flow, build a capital base, and, crucially, re-build social networks, solidarity and trust. Savings group participants often do not have regular sources of income and commonly come from poorer segments of the community. Women are regularly involved in savings groups and have high participation rates. Saving group participants often come from the same socioeconomic background and know each other.

Activities of saving groups include:

- **SAVING:** The group determines a fixed amount for a given period that each group member will bring to the savings account. The allocation and management of these savings can vary according to the needs of the members or is used to support group activities.
- **LENDING:** The group can grant small internal credits to members. Loans are modest and repayable in the short term. Interest rate is fixed by members. At the request of the group, these credits are renewable almost automatically if there is no problem and if the amounts are within a ceiling set by the group.
- **SOCIAL:** The group not only improves the lives of its members financially but also socially as members are given the opportunity to socialize at each meeting and discuss other issues or concerns. The group further builds a small social fund to help group members in distress. Each group agrees on the amount to be saved, usually a smaller amount than the amount for regular savings, in every meeting for the social fund. This money is not loaned, but instead used to pay group member expenses for things such as funeral costs or medical treatment and are only used if agreed on by all group members.

In case of external loans from financial service providers, which most members can't afford, group members agree to mutually guarantee each other (co-guarantee) when taking a loan. If a member does not pay, the group ensures repayment for him/her. The group uses moral pressure to force repayment and therefore there is no requirement for physical guarantees with third parties such as financial institutions. The internal character assessment for group members is important.

The project will strengthen the capacity of saving groups and link them to more formalized financial institutions. The project will further enhance financial education and client protection principles.

This model is more convenient and more accepted by most women who cannot afford collateral loans. As such, this approach will attract more women and will facilitate their formal access to financing over time. Specific attention will also be paid to women's participation in other project financial products that will be developed, such as asset finance in activity 3.3.2.

The project will digitize saving groups, to reduce time and cost spent to access their savings, thereby facilitating transactions with formal financial institutions. Solutions such as “push and pull technology” (i.e., payments using cell phones, commonly referred to as “mobile money”) have proven to be useful and convenient for savings groups to send and receive money.

In activity 3.3.1, communities involved in chosen forestry and agroforestry value chains under this project in selected intervention sub-areas will be selected with a particular focus on landless people, historically marginalized communities, and vulnerable groups of youth and women in collaboration with local authorities (districts & sectors) and community representatives.

Sub-activities include:

3.3.1.1 Develop tailored credit assessment tool for selected value chains, while integrating climate impact data

3.3.1.2 Facilitate access to finance for selected value chains of avocado, macadamia, vegetables and ecotourism that are relevant to build climate resilience

3.3.1.3 Develop tailored financial products to meet the needs of women and youth

Activity 3.3.2 Set up and support savings & loan groups, enhance asset-building

To strengthen the capacity of local smallholder farmers in financial literacy and access to finance, farmers' cooperatives will be assisted and supported in setting up saving groups. Savings groups are a proven and effective approach to the financial inclusion of the most vulnerable and are increasingly recognized as one of the strongest forms of community organization available due to the high levels of social capital generated from its savings approach. Saving groups create a platform for linkage with financial institutions and other business development services. Further, savings groups are recognized for enabling high levels of female inclusion. As groups members co-guarantee each other for accessing a loan and groups are strong, it forms low lending risks for MFIs since members mutually reinforce each other's repayments.

Once grouped, small savings group members, especially women, grow from accessing internal group loans to accessing formal financial institution loans. As members' bankability and trustworthiness increases, members are able to start accessing individual loans, gradually building to larger loans and are able to open individual savings accounts at financial institutions.

Members of a saving group will be self-selected and the saving group will be managed by its members.

Self-selection will be based on the following:

- Knowledge of one another and similar economic background
- Close proximity to where the meetings will be held
- Reputation for honesty and reliability
- Ability to attend all meetings on time and follow rules
- Have a cooperative personality
- Be able to buy at least one share each week (or time set by the group)
- Be able to repay loans on time

Most saving groups are informal but transparent, democratic, and structured groups that have constitutions and clear rules. Members do not enter into contractual arrangements for their membership. Record-keeping is based on simple passbooks in which each member's savings and loans are recorded; all transactions are done in front of the members during group meetings.

Farmer savings groups (FSGs) will be created within cooperatives and farmers will decide which group to join. Groups will set up governing rules and conditions (savings amount, meetings, etc.), and criteria for new members who want to join. Particular attention will be given to social inclusion (including vulnerable communities and landless people).

Youth and women-owned businesses selected for activity 3.2.2 will be linked to targeted FSG beneficiaries for supplying inputs (modern beehives, seedlings, etc.) and energy-efficient cook-stoves. Furthermore, intensive capacity-building and technical support will be delivered to all groups with emphasis on financial literacy, entrepreneurship, and gender mainstreaming. The activity will also help link savings groups to PFIs.

For Activity 3.3.2, in collaboration with local authorities, the project will target existing or potential youth and women-owned businesses that can intervene at any level of selected value chains in activity 3.3.1. The project will ensure the integration of vulnerable and landless people.

Key sub-activity:

3.3.2.1 Establish/form saving groups for access to finance

3.3.2.2 Provide organizational, technical, and financial capacity-building to women, youth, and CSOs (such as cooperatives or VSLA) with focus on financial literacy and provide long-term coaching to beneficiaries

3.3.2.3 Support women, youth, and other marginalized groups to strengthen savings and loan groups and register for MFI accounts

Activity 3.3.3 Build the capacity of financial institutions to serve targeted value chains and communities

Many financial institutions demonstrate little knowledge about the science of agriculture, forestry, climate change and associated risks. A study by the International Food Policy and Research Institute (IFPRI)¹¹², noted that many agricultural activities are located in remote, widely dispersed regions and that financial institutions find it impossible to provide cost-effective and affordable services in this context. Hence for this specific activity, PFIs will be supported to access and adopt a digitalized forestry and agri-loan assessment tool, and to understand climate impacts on their portfolio to actively integrate and promote climate relevant financial products in their product offer.

The Project will work with these institutions by developing and refining appropriate financial products and financing operations that are better understood and user-friendly for smallholder farmers and vulnerable groups. The Project will also collaborate with selected PFIs to build their awareness of prospective farmer and low-income group clients and their exposure to climate impacts.

Key sub-activity:

3.3.3.1 Capacity-building of PFIs

Activity 3.3.4 Facilitate learning & knowledge sharing

To deliver an adaptive program that cooperates with market actors on inclusive and sustainable business models that have a positive impact on smallholder farmers, while generating and sharing lessons on which business models work and which do not, the project will adopt the program-embedded reflection and learning (PERL) model for facilitating learning and knowledge sharing to popularize the CND unique approaches to restoration and environment management.

By **proving**, we seek to know and show what works and what does not work by demonstrating results at all levels, generating deeper insights into farmer benefits and the effects on the wider ecosystem, and building business model sustainability.

¹¹² Kloeppinger-Todd, R. and Sharma, M. eds., 2010. *Innovations in rural and agriculture finance* (Vol. 18). Intl Food Policy Res Inst.

By **improving**, we learn by doing to enhance project adaptation and improvement, build a learning culture through discussion and reflection, generate knowledge to support project effectiveness, make comparisons over time, and enhance data quality procedures.

Through **sharing**, we disseminate high-level results and learnings to relevant audiences, contribute to visibility and communication, showcase successful, sustainable and scalable business models, and generate best practices.

Under this activity we will go through the following actions to prove, improve and share:

Capture - We will establish a KML framework and support case studies and discussion by project staff; improve data collection; collect insights from field visits; write human interest stories; hold quarterly program reviews; measure client satisfaction; and collect information on business models, market uptake, and ecosystem changes.

Synthesize - We will organize discussions and quarterly program reviews with program staff and stakeholders for assessment, interpretation, reflection, and learning from results. Outputs will include reports, briefs, and articles.

Disseminate - We will contribute to and support the project, presenting its achievements in newsletters, conferences, webinars, blogs, publications, media events, partner workshops and sector network presentations.

Take action - We will have regular discussions and reviews in which we revisit the work plan and processes in light of results and insights (e.g., based on outreach data, field visits, case studies, partner feedback, quarterly reports, changes in the ecosystem, etc.), and adapt operational planning when needed (within and outside of the normal project planning and reporting cycle).

Refine agenda - We will help to align the project intervention strategy with a shared vision for success.

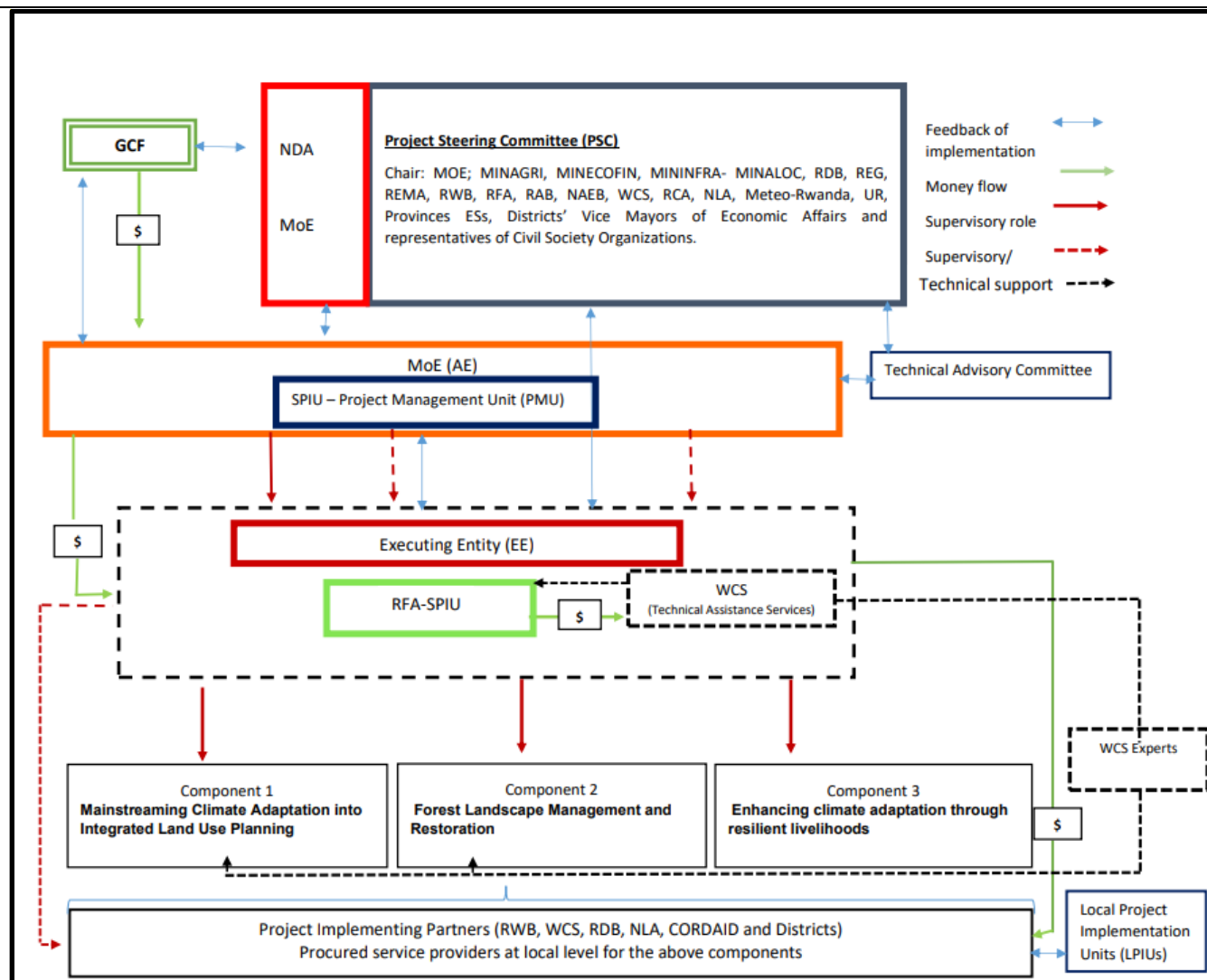
The table below summarizes the project activities to be implemented in Congo Nile Divide.

	Core PA Nodes	Stepping Stones	Landscape Linkages	Broader Farming Mosaic	Across the CND
	Volcanoes NP and Buffer	Gishwati NP and Buffer	Mukura NP to Gishwati NP Linkage	Nyungwe to Mukura Broader Farming Mosaic	
1.1.1 Synthesize & disseminate information on value of natural forests & ecosystem services					
1.1.2 Develop climate-resilient landscape land-use plan					
1.2.1 Create interagency taskforce institutionalizing integrated landscape planning & policy					
1.2.2 Build capacity for spatial planning in national agencies re climate change					
1.2.3 Develop an effective forest monitoring system to underpin forest management decisions					
2.1.1 Facilitate revision of PA management plans to address climate risks					
2.1.2 Establish long-term plans for CND financial sustainability post-GCF					
2.1.3 New fire management curriculum developed and operationalized in PAs and adjacent landscapes					
2.2.1 Secure key remaining natural areas outside PAs					
2.2.2 Restore natural forest cover in & outside PAs including riparian linkages					
2.2.3 Promote silvo-pastoralism with indigenous trees in Gishwati Pasture Stepping Stone areas					
3.1.1 Restore high slope areas (>55%) as protective forests					
3.1.2 Develop on-farm agroforestry for high caloric and indigenous tree species					
3.2.1 Develop agroforestry related value chain for markets access					
3.2.2 Facilitate & scale up capacity, value addition & marketing of select climate-resilient value chain products					
3.2.3 Facilitate access to input & output markets for vulnerable farmers					
3.2.4 Scale up marketing, production, sales, use of fuel-efficient cookstoves					
3.3.1 Facilitate access to finance & private sector investments					
3.3.2 Set up & support savings & loan groups, enhance asset-building					
3.3.3 Build capacity of financial institutions to serve targeted value chains & communities					
3.3.4 Facilitate learning & knowledge sharing					
4.1 - Performance monitoring plan developed					
4.2 - Project monitored and evaluated; lessons learnt integrated into adaptive management processes					
Primary Focus					
Secondary Focus					
Some activity					

Table 4: Summary of project activities (place based and non-place based) to be implemented across the Congo Nile Divide.

B.4. Implementation arrangements

The proposed project will be implemented in five years, from 2024–2029. The Ministry of Environment (MoE) is the Accredited Entity (AE) for the GCF and MoE will enter into a funded activity agreement with the GCF once the project is approved by the Board.



The governance arrangements for the project at the national level are designed to build upon the institutional structure of MoE (the accredited entity) and RFA (the Executing Entity).

Accredited Entity: The MoE on behalf of the Government of Rwanda is a Direct Access Entity of the GCF and will assume the oversight role on implementation, management, monitoring, evaluation and reporting on the CND Program elaborated in the FAA. The MoE will enter into a subsidiary agreement with the Executing Entity (EE) with clear roles and responsibilities for each party. Under the Ministry of Environment there is a Single Project Implementation Unit that has a number of staff charged with resource mobilization and providing oversight over different projects implemented through the Ministry of Environment. The MoE/SPIU has a track record of experience in project oversight, management, monitoring and evaluation of projects and programs including the GCF funded FP073 that is currently at the mid-term level of implementation with over USD 32.8 million, managed over USD 3.8 million for the GCF funded PPFs, overseen the implementation of USD 10 million from Adaptation Fund. The MoE is currently managing the World Bank Funded project preparation advance of USD 6 million and has been implementing since 2018 a program funded by the UNDP focused on Strengthening Capacities of Environment and Natural Resources Sector for a Green Economy Transformation of over USD 6 million. Additionally MoE assumes its oversight role through chairing of the Project Steering Committee for different projects implemented under the Environment and Natural Resources Sector as a whole namely TREPA, a GCF-funded project, Rwanda Urban Development Project Two funded by the World Bank and NDF, an Early Warning System project

(funded by LDCF), the Landscape Approach to Forest Restoration and Conservation (LAFREC) programme (funded by GEF/World Bank), Green Amayaga Project (funded by the GEF) and the “Building resilience of communities living in degraded forests, savannahs and wetlands of Rwanda through an ecosystem management approach” programme (funded by LDCF). A coordination platform between approved GCF projects will be established to identify, maximize linkages and ensure complementarity through learning and sharing best practices and failures arising from the implementation of existing GCF projects. Quarterly meetings will be organized under the supervision of the MoE between project directors and staff responsible for monitoring and evaluation to share lessons learned.

Project Financial Flow and AE Roles

Funds from GCF will be provided to the MoE GCF account (as the AE), under the terms of the FAA. MoE will be responsible for financial management and accountable for the use of GCF resources. Financial transactions will be subject to annual audits undertaken by internationally certified auditors. In addition, as the AE, MoE will: i) ensure that the project is executed in accordance with GCF standards; ii) supervise, oversee and manage the implementation of project interventions; iii) report on Project progress; and iv) ensure that Project activities are well coordinated and aligned with countries' national priorities. MoE will provide oversight to the project consisting of a) entering into subsidiary agreement with the EE; b) managing and disbursing GCF funds to EE on the approved work plans and budgets; c) reviewing financial expenditures and progress reports; d) overseeing Project implementation in accordance with the Project document and Annual Work Plans and Budgets, agreements with co-financiers and executing entity rules and procedures; e) providing technical guidance to ensure that the appropriate technical quality is applied to all Project activities; f) providing financial reports to the GCF for Project funds received; g) ensuring that the project complies with the terms agreed in the Project's respective FAA as well as the AMA signed between MoE and the GCF; and h) undertaking regular annual supervision missions according to the MoE's guidelines and convene Project Steering Committee (PSC) meeting twice a year aimed to review progress and approve work plans. The PMU hosted under the Single Project Implementation Unit of the Ministry of Environment will manage, monitor and report on day-to-day implementation of the Project activities by the EE. The EE will execute the tasks within the overall project management structure: a) implementing day-to-day activities as per the project work plan and budget, including the Environmental and Social Management Plan; b) undertaking procurement activities as per the agreement signed by GCF and AE as well as the sub-agreement signed by the AE and EE; c) managing contracts of suppliers and services providers; d) hiring and managing project staff relevant to the EE-managed project areas; e) implementing activities as per the project work plan; and f) carrying out financial and technical monitoring of activities, including the achievement of outputs and outcomes the EE is in charge of. EE will enter into Memorandum of Understanding with different implementing partner agencies that will provide technical know-how to the implementation of the project activities. The implementing partners including but not limited to RWB, WCS, NLA, RDB, CORDAID and Districts will provide technical and operation services articulated and agreed in respective MoUs. For the Implementing partners RFA will be paying service compensations and at District Level community approach will be used. Procured service providers will be contracted in accordance with MoE rules and regulations of procuring goods and services.

Project Steering Committee (PSC): An inter-ministerial Project Steering Committee (PSC) chaired by the Permanent Secretary will be established by MOE to serve as the project's coordination and decision-making body. The PSC will ensure the project delivers its outputs and achieves its outcomes. The Committee will meet on a semi-annual basis to review project progress and evaluations, facilitate implementation (ensuring the necessary resources and support are provided in a timely manner) and provide guidance to the PMU and EE/PIU. The Steering Committee will also facilitate effective coordination between the key stakeholders at the national and district levels and ensure the project aligns with Government strategies and programs.

The PSC will reflect the multi-sectoral nature of the project and will include senior-level representatives from GoR and other relevant Stakeholders. The PSC will be chaired by the Permanent Secretary, MOE or any Senior Manager from MoE delegated by the Permanent Secretary with representatives from: MINAGRI, MINECOFIN, MININFRA- MINALOC, RDB, REG, REMA, RWB, RFA, RAB, NAEB, WCS, RCA, NLA, UR, Provinces ESs, Districts' Vice Mayors of Economic Affairs and representatives of Civil Society Organizations.

Project Management Unit: It is under the MoE/SPIU that a dedicated Project Management Unit (PMU) of the CND program to supplement the existing MoE/SPIU structure for this complex program will be established for day-to-day management of the program interventions. The Project Management Unit (PMU) will be headed by the Project Coordinator who will serve as the secretary and the rapporteur of the PSC. The Project Coordinator and other PMU staff will be recruited as per the MoE/GoR recruitment practices. The Project Coordinator is responsible for the day-to-day operations of the PMU within approved project funded activity agreement and any other guidance provided by the AE and PSC including the tasks of managing and monitoring project risks initially identified and submitting new risks to the AE and PSC for consideration and decision-making on follow-up actions and updating the status of risks in the project risk log. The PMU Project Coordinator will report to the overall MoE/SPIU- Coordinator as the immediate supervisor and to the Permanent Secretary as the secondary supervisor.

The PMU will provide to the MoE/SPIU coordination and to the Chair of PSC with quarterly progress reports and close its operations when the final project evaluation report and other documentation required by the GCF have been completed and submitted to MoE. The PMU will be staffed with required professional and technically qualified personnel selected and recruited following the MoE/GoR procedures. In all, the PMU will strive to maintain a lean management structure. It will comprise a Project Coordinator, Monitoring & Evaluation (M&E) Specialist, E&S Specialist, Procurement Specialist, Gender Specialist, Financial Management Specialist, Communication and Outreach Specialist, Office Manager, and Driver as a dedicated team to manage the EE for the whole program.

Executing Entity: RFA will serve as the EE for the Project through a subsidiary agreement (SA) signed with MoE. The SA will establish clear roles and responsibilities for both parties for the delivery of the project activities, the schedule and conditions for installments, the determination of the prevailing fiduciary standards and the terms and conditions for arbitrations and termination of contract. The SA will include specific obligations for the EE on project execution, financial management, personnel administration and reporting, as well as on arbitration and liability terms.

Rwanda Forest Authority: RFA was established by the Government of Rwanda by the law No 72/2019 of 29/01/2020 with the mission of ensuring the growth of forest resources, their management and protection for sustainable development purposes. RFA is an affiliated agency under the MoE with a separate legal personality and a budget entity of the Government of Rwanda. RFA is currently implementing TREPA; a GCF funded project as one of the executing entities, COMBIO Project funded by the Swedish government, executed Forest Management and Biomass Energy project (FMBE) implemented in the Northern province; Sustainable forestry, agroforestry and biomass energy management for climate resilience in Gatsibo and Gicumbi districts; and Improving the Efficiency and Sustainability of Charcoal and Wood Fuel Value Chains in the North-western part of Rwanda supported by the Nordic Development Fund. RFA has three main divisions namely Forest Research Division, Forest Management, Administration and Finance and an established Single Project Implementation Unit (SPIU), composed of staff with the required financial management skills where the CND Staff will be hosted. The Director General of RFA as the Chief Budget Manager, the RFA SPIU Coordinator together with Program Manager and a Financial Management Specialist will be responsible for the day-to-day financial management of the Project. RFA SPIU will open a separate account at the National Bank of Rwanda (BNR), from which payments of the Project activities will

be made from. RFA follows the same financial and procurement policy as the one of the Ministry of Environment.

WCS Technical Assistance (TA) Services: As one of the project implementation partners, WCS will provide technical assistance to the EE (RFA) through an MOU signed with RFA. The MOU will detail all the activities that WCS will be directly supporting during the implementation of this project. The TA services will include advising RFA/SPIU, procuring and supervising international experts and consultants, and procuring materials to support the execution activities. Relevant technical experts from WCS will be embedded within RFA to advise SPIU on day-to-day project management to support the implementation of activities under outputs 1.2, 2.1, and 2.2. These experts will support RFA's technical teams and strengthen their capacities in different aspects of the project implementation. Furthermore, WCS technical assistance services will include supporting the development of ToRs and reviewing works of international consultants to ensure high quality deliverables.

CORDAID Technical Assistance Services: CORDAID will provide technical assistance to RFA mainly on output 3.2 and 3.3 through an MOU signed with RFA.

Other Government of Rwanda implementing partners will provide implementation support and RFA will enter into specific MOUs with these partners based on the type of support they will provide.

Table 5: Summary of project outputs. The EE will be responsible for all activities and sub-activities of the project, receiving support from technical implementation partners.

Outputs	Executing Entity per output	Technical Implementation Partners
Output 1.1. Landscape-wide land-use plan developed for climate-resilient livelihoods & forest ecosystems, integrating district development strategies (DDS)	RFA	WCS and NLA
Output 1.2. Local & national institutional capacities strengthened to integrate biodiversity & climate risks into land use planning & management	RFA	WCS and NLA
Output 2.1. Protected areas management effectiveness improved to address climate risks and adaptation	RFA	WCS and RDB
Output 2.2. Natural forest cover restored; biodiversity connections established	RFA	WCS
Output 3.1. Farming methods enhance productivity, reduce erosion & flooding risks, contribute to ecosystem services, and support connectivity	RFA	RWB

Output 3.2. Rural livelihoods generate alternative incomes and reduce pressure on forests	RFA	CORDAID
Output 3.3. Financial services & private sector investment engagement	RFA	CORDAID
Monitoring and Evaluation	PMU	

Technical Advisory Committee (TAC) will be responsible for ensuring technical guidance to both the project execution team and the PSC. The TAC will have technical representatives from all project stakeholders including MINAGRI, MINECOFIN, MININFRA, MINALOC, RDB, REG, REMA, RWB, RFA, RAB, NAEB, WCS, RCA, NLA, UR, Districts representatives and representatives of Civil Society Organizations. The TAC will be engaged in providing technical support and quality assurance of the project implemented activities as well as facilitating effective implementation and mainstreaming of project interventions beyond the life of the project. The institutional stakeholders will nominate relevant technical experts to the TAC, thus bringing expertise from their respective institutions to make technical contributions to the project implementation based on the integrated nature of the project.

The project will use the GoR fund management modalities based on an annual work plan developed by the PMU together with EE and subsequently approved by the PSC. The Annual Work Plan (AWP) indicates which activities should be covered at PMU and EE levels as per the allocated respective budgets. The EE, together with support from implementing partners, will establish Local Project Implementation Committees (LPICs) specific to the zones of intervention in the Western and Southern Provinces of Rwanda. Two LPICs will be established; one for the Southern province and another one for the Western province for the coordination of activities in Western (GMNP) and southern (NNP) Rwanda to ensure synergies and complementarity between partners. These teams will work closely with the Joint Action Development Forums (DJAF) based in each district located in these respective zones. LPIC will be composed of representatives of districts in charge of natural resources management, RDB, CSOs, and conservation organizations, women and youth councils and JADF. This arrangement will ensure that a) project delivery can proceed securely and efficiently despite the complex governance framework; and c) government partners play a leading role in project delivery and capacity development. PSC will provide policy direction and guidance to improve the coherence and efficiency of this innovative approach to be developed as an up-scalable model. At the local level, the EE and its service providers will coordinate closely with districts to ensure smooth local level coordination in project implementation, provide ownership and ensure sustainability. The District Executive Directors (DEDs) are expected to play an active role in project implementation, facilitation and monitoring and play a key role in the grievance redress mechanisms, as described in the Environmental and Social Management Plan of the project.

B.5. Justification for GCF funding request

Rwanda has achieved remarkable social, political, and economic progress over the past two decades. The Rwandan economy has grown well above average compared to its peers and the rest of the world, achieving growth rates of more than 8% regularly, with a 9.4% growth rate in 2019¹¹³. However, the economy which is

dependent on nature-based tourism and agriculture is extremely vulnerable to external shocks, including climate change.

Climate change is likely to increase variability in crop yields and agricultural production, cause severe flood damage to physical infrastructure, reduce labor productivity, and impact demand for tourism¹¹⁴. The recent Country Climate and Development Report (CCDR) estimates that if these risks materialize, Rwanda's GDP levels can drop by 5–7% below baseline in multiple years by 2050¹¹⁵, with negative impacts on private consumption, exports, and government revenues. By strengthening resilience to climate change, Rwanda can better prepare to manage the adverse impacts of climate change. Implementing the commitments articulated in the NDC is estimated to require new investments of US\$11 billion, of which close to US\$7 billion is conditional on new financing.

Rwanda is classified by the U.N. as a Least Developed Country (LDC)¹¹⁶ and lacks sufficient public financial resources to implement its climate change and resilient green development objectives. The implications of climate impact for Rwanda demands concerted action and support both locally and internationally to enable the mainstreaming of climate resilience into public, private and community sectors at the pace and scale required. This Project will allow Rwanda to implement its NDC commitments and pursue its aspirations under its National Forest Policy and Climate Resilient Green Economy plans for reversing trends of high deforestation, allowing for demonstration of sustainable reforestation models and reductions in demand for wood fuel. Rwanda has made an impressive commitment to using forests and forest protection as part of its strategy to mitigate the impacts from climate change, and to provide benefits of biodiversity conservation and livelihood improvements.

Yet, vulnerable citizens' needs, coupled with fragile landscape and forest restoration, will not be met solely with national resources in the rapid timeframe needed to arrest loss of economic assets for households and prevent further degradation of landscapes. Specific investments in awareness and forest governance, training at scale required to increase the resilience of CND communities to the impacts of climate change, and alignment and coordination of support on undertaking climate resilient restoration activities are not currently being developed by the GoR. The GoR does not have the financial means to implement all project activities identified to improve climate resilience and requires technical assistance and support. While project partners are contributing co-finance for this project implementation, this represents only a fraction of the resources required for implementation. The GCF grant investment can help initiate, expedite, and catalyze these critical interventions and assist with alignment of future climate resilience efforts. MoE is requesting grant funding for this Project due to Rwanda's status as an LDC, insufficient alternative finance options, and because the proposed activities will provide a public good on which the livelihoods of many of Rwanda's rural poor depend. The high levels of poverty among project beneficiaries in the CND as identified in the Feasibility Report necessitate the use of grant funds rather than a reimbursable financial option. While the GoR will contribute resources for the implementation and sustaining climate resilience efforts beyond the 5-year Project, this represents only a partial share of implementation resources. The GCF funding will crowd in additional private sector capital, such as efficient cookstoves and forest plantations, and will allow the government to cost-efficiently implement the adaptation capacity-building training and practices to ensure that CND landscape is restored, and climate resilient landscapes maintained.

¹¹⁴ "World Bank Group. 2019. *Rwanda Systematic Country Diagnostic*. World Bank, Washington, DC. World Bank. <https://openknowledge.worldbank.org/handle/10986/32113> License: CC BY 3.0 IGO."

¹¹⁵ World Bank Group. 2022. *Rwanda Country Climate and Development Report*. CCDR Series;. World Bank, Washington, DC. World Bank Group. <https://openknowledge.worldbank.org/handle/10986/38067> License: CC BY-NC-ND 3.0 IGO.

¹¹⁶ United Nations Department of Economic and Social Affairs, 2021. Least Developed Country Category: Rwanda Profile.

The lack of private capital structure, prevalence of subsistence farming, and inadequate market for agriculture and forested products, is the primary cause of the request for grants, not loans as little surplus products are produced and traded, and thus will be inadequate to secure or service any loans. The prevailing market conditions necessitate GCF grant finance, rather than loans, to raise the capacity and bridge the financial and technical gaps until the restoration and value chain models are brought to financial viability that can be sustained by private sector. Furthermore, our Economic and Financial Analysis (EFA) shows that the direct returns on investment in landscape restoration activities are too low to make them financially attractive or feasible for farmers, even if they had access to credit. Without GCF support, lack of private financing, large payback period, uncertainty and transaction cost of getting the loan, and upfront cost and benefits realized over a longer period of time reduces the financial attractiveness of climate resilience investments. It is unlikely that there will be a financial return on investment within the Project time frame to repay loans. The issue of societal benefit investment is accentuated by the fact that there cannot be direct uptake of benefits to communities from restoration within National Parks.

Serving low-income households yields low returns and the community lacks financial education and capacity to link them to formal private institutions. Most formal financial institutions view these as high risk and high cost, as the transactions are often small and the clients usually have neither physical collateral nor the education needed to organize their finances. Without the value chain development for potential revenue generating streams such as honey, tea plantation tourism, and efficient cookstoves, the private sector investments are not occurring. Henceforth, these potential revenue generating streams proposed in the project are not sustainable on their own and the GCF grant will not only provide financing support but will be critical in creating enabling opportunities for market development beyond the project to catalyze private investments. The full concessionality funded by GCF grants is requested so that an enabling environment can be created, and the project can overcome affordability challenges through developing a system of community co-guarantee. This will also go beyond the neoclassical economics model and focus on shifts in social dynamics that move the community as a whole to escape the poverty trap and increase financial resilience. The full concessionality for potential revenue generating streams will be passed on to end beneficiaries in the CND and the project will strengthen the capacity of community organizations on quality improvement and aggregation capacity, create market linkage with off-takers or buyers, and attract finance for development, marketing, and distribution.

Planting indigenous species, well-suited to the climatic conditions of the CND, is likely to increase the health and resilience of forest plantations and thus the erosion control benefits that these forests provide. However, when establishing protective forests, most existing projects in the CND use exotic species because they are fast growing, widely available, and low cost¹¹⁷. GCF grant resources are required to support restoration to help build natural capital (restored forest) which can be used sustainably and provide larger ecosystem benefits in the future.

The financial Net Present Value (NPV) of \$34.48 million for agroforestry, \$2.98 million for protective forests, and \$0.31 million for silvopasture intervention and economic NPV of \$58.93 million for agroforestry, \$30.22 million for protective forests, \$0.97 million for silvopasture, and \$10.59 million for energy efficient cookstoves interventions for 20-year analysis period when GCF support for the Project is considered¹¹⁸. These results are indicative of the investment in time and resources required to increase climate resilience, associated benefits, and restoration of landscape and forest productivity. Furthermore,

¹¹⁷ Mukuralinda, A., et al, 2016. *Taking to Scale Tree-Based Systems in Rwanda to Enhance Food Security, Restore Degraded Land, Improve Resilience to Climate Change and Sequester Carbon*. PROFOR, Washington D.C.

¹¹⁸ The provision of energy efficient cookstoves is a non-income generating intervention and is expected to have no financial cash inflows and no income generating opportunities, then no financial analyses is needed. The cookstoves analyses was pursued only at the economic analysis level.

agroforestry, restoration, and forest management require up-front investments that take 10-30 years to mature fully. However, these future benefits are depressed by the use of a high discount rate that downplays the importance of long-term investments. The discount rate considered in our economic analysis is 13%, which is Rwanda Central Bank's yield for a 20-year treasury bond, as issued in August 2021. This is the rate at which the government can borrow to fund equivalent investments in the absence of grant financing. For financial analysis, we used the discount rate of 15.44%, which is Rwanda Central Bank's lending rate for July 2021. Consequently, the GoR seeks maximum concessionality from the GCF for the urgent adaptation actions proposed under this Project. Without GCF funding, under a business as usual (BAU) scenario, inadequate technical and financial capacity at the national and local level will continue to hinder the efforts of rural communities in the CND to adapt their livelihoods. Consequently, the vicious negative cycle of unsustainable forest use and ecosystem degradation will persist, reducing the supply of goods and services which buffer impacts of climate change on rural communities, compounding their vulnerability. GCF funding is therefore vital to initiate progress towards a complete paradigm shift in GoR's approach to adapting the livelihoods of rural Rwandans to climate change.

The Project cost per direct beneficiary of \$31 (\$39.86 with co-funding) and 20-year economic rate of return of project interventions for agroforestry (49%), protective forests (72%), and silvopasture (51%) highlights its viability. The Economic rate of return for energy efficient cookstoves could not be estimated as there is no change in the sign in the net cash or resource flows from positive to negative over the analytical period; and the economic NPV of \$10,585.675 for this intervention highlights its feasibility. The cost of restoring natural forest for providing both adaptation and mitigation benefits is significantly greater than the cost of simply planting trees for mitigation. Restoration of natural forest in Rwanda is essential for sustaining the critical ecosystem services on which vulnerable rural communities and the national economy depend. The capital market in the country is weak and provides limited opportunities. It has low levels of direct investment which further limits the opportunity for financing. The country's capital markets currently are not able to serve the needs of rural and resource limited beneficiaries. The project addresses market failures and minimizes the risk of disrupting or unduly distorting markets or crowding out private finance, including new entrants, as well as other public investments. Taking the above-mentioned points into account the concessionality that the GCF provides is justified.

B.6. Exit strategy

The Project will establish a new inter-agency and local-to-national-scale spatial planning framework and capacity for an integrated approach to multi-sector analysis and planning. This will enable the analysis of land-use trade-offs that explicitly consider the value of ecosystem services, climate risks and risks of ecosystem degradation. This represents a radical shift away from business-as-usual and will have long-term, far-reaching implications for policy and financial planning in Rwanda. It will provide a spatially explicit framework for mainstreaming climate risk considerations across all sectors while also providing a mechanism for quantifying harmful financial flows to biodiversity from multiple sectors.

In addition, the multi-scale spatial planning system will be able to strengthen and link directly to the system under development by Rwanda's Ministry of Finance and Economic Planning (MINECOFIN) for policy coordination to integrate climate risks into fiscal planning, with support from the IMF Resilience and Sustainability Trust. Further, it will support the ability of both national and local governments to plan and implement the new disaster risk reduction strategy in a spatially explicit way, which previously has not been possible.

Mainstreaming climate change adaptation metrics. The Project will integrate climate resilience metrics into land use planning, district development strategies and annual performance contracts and harmonize cross-

sectoral monitoring and reporting mechanisms. This will lead to a systematic consideration of climate change risks and adaptation in policy planning that will be sustained beyond the Project. An MoU between key government agencies will operationalize the cross-sectoral planning task force, which will be supported by a ministerial order, giving it the mandate and financial support to operate beyond the life of the Project.

Rwanda has adopted a private-public partnership (PPP) business model to make its national parks operations financially viable. However, in Rwanda, as in the rest of the world, national parks and the biodiversity they are intended to protect are threatened by climate change and by the interaction of climate change with other direct anthropogenic threats. By “climate proofing” its national parks—i.e., by establishing a technical unit within RDB, by explicitly incorporating climate change considerations into the management plans of national parks, including climate-wise functional linkages, such as the one this project will create in the CND, and by training rangers and other park personnel to proactively manage climate-change-related threats—Rwanda will ensure that the ecosystem service benefits of the country’s biodiversity, including its national parks, will be sustained beyond the life of the Project. Specific finance instruments—such as payment for ecosystem services (PES) and structured impact conservation bonds to ensure ongoing revenues and benefit-sharing with surrounding communities—will be identified and implemented before the end of the project, in a manner consistent with the Biodiversity Finance Plan that the GoR has recently developed with support from BIOFIN. The Project through RFA will hire a Technical Advisor, to work with RDB, MoE, REMA, FONERWA and MINECOFIN to ensure that a portfolio of sustainable finance mechanisms is designed and implemented before the end of the project period to ensure the long-term sustainability of the Project outputs.

Strategies for Long-term Financial Sustainability. Two specific examples of the Government of Rwanda’s commitment and approaches to financial sustainability that will be key parts of the Exit Strategy are the Tourism Revenue Sharing Program (TRSP) of RDB and the Community Adaptation Fund of FONERWA, the Rwanda Green Fund.

Tourism Revenue Sharing Program. The long-term success of the three protected areas in the CND depends, to a substantial extent, on the support of communities living around protected areas. In recognition of this, in 2005 Rwanda established a Tourism Revenue Sharing Program, allocating 5% of gross tourism revenue to local communities. Currently Rwanda and Uganda are the only two African countries with formal tourism benefit sharing programs. RDB recently completed a 15-year review of its Tourism Revenue Sharing Program (Snyman et al. 2023) and it has increased the revenue sharing allocation from 5% to 10%.

The primary challenges identified in the 15-year review are increasing the effectiveness of community engagement, reaching the most vulnerable community members, and ensuring that the TRSP meets community needs. The GoR is interested in adapting the TRS Policy to more effectively engage the communities living around the PAs in the TRSP and to allocate more funding to directly enhance the lives of Rwandan citizens living around PAs. In this regard the Project will work with RDB and district governments, to increase transparency in the Program, ensure that the most vulnerable community members benefit from the TRSP and to link the cooperatives and youth-led SMEs to the Program. This will include working with RDB to use the insights from this project to revise the TRSP revenue allocation model and eligibility to create greater resilience among communities. The 15-year review recommended that the following percentages be used for the revenue allocation going forward: 70% for livelihood projects; 25% for infrastructure support; and 5% in an emergency fund. However, there is a need for input from this Project to ensure that the specific types of projects to be supported most effectively increase community resilience. This will ensure long-term benefits to communities.

The Community Adaptation Facility (CAF) within FONERWA was developed as part of Rwanda’s NDC Deep Dive: Advancing Financial Innovation to Scale Up Climate Action. The aim of the CAF is to provide grant

funding for community projects that scale up climate resilience. In this regard, the Project will support the cooperatives and youth-led SMEs engaged in Component 3 to prepare business plans and link them to innovative financing for community projects with the CAF. In line with the CAF priorities, the business plans will focus on the investment priorities of the Rwanda Green Fund (FONERWA) targeting conservation and sustainable management of natural resources especially ecosystem rehabilitation, sustainable land management, sustainable forest management and promotion and protection of biodiversity. The Project Technical Assistance will work with FONERWA to design specific long-term funding opportunities for communities in the CND to the CAF.

Additionally, the Project will enhance the environmental, social and economic resilience of communities in the CND by enhancing financial and accounting literacy, marketing and entrepreneurship, and vocational skills of community members (e.g., in native tree propagation, forest restoration, and fuel-efficient stove production). This will mean the Project will invest in businesses that restore or indirectly halt the degradation and loss of native forests by reducing or eliminating local stressors. These businesses and projects are expected to live beyond the lifetime of the project. The value chains, once organized profitably, can continue to grow without further support, while the financial service providers will continue to service savings groups and farmers and SMEs in the value chain with sustainable financial services also after the project ends. By raising the productivity from selected crops, diversifying agriculture towards high-value agriproducts, and strengthening their value chains through trainings in marketing, processing and business, the project will attract investments from the private sector in sustainable agriculture and agroforestry. All economic and environmental benefits from the project will be rigorously monitored, captured, compiled and disseminated as knowledge products, packaged for various stakeholders including investors and MFI, with which the project will engage to secure microcredits to beneficiary farmers.

(c) FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding	Total amount			Currency		
	39,056,421			USD (\$)		
GCF financial instrument	Amount	Tenor		Grace period	Pricing	
Grants	39,056,421	-		-	-	
(b) Co-financing information	Total amount			Currency		
	10,943,579			USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Rwanda Development Board	<u>Grant</u>	<u>5,000,000</u>	<u>USD (\$)</u>	-	-	-
Rwanda Forestry Authority	<u>Grant</u>	<u>831,307</u>	<u>USD (\$)</u>	-	-	-
Rwanda Water Resources Board	<u>Grant</u>	<u>5,000,000</u>	<u>USD (\$)</u>	-	-	-
Ministry of Environment	<u>In-kind</u>	<u>112,272</u>	<u>USD (\$)</u>	-	-	-
(c) Total financing (c) = (a)+(b)	Amount			Currency		
	<u>50,000,000</u>			<u>USD (\$)</u>		
(d) Other financing arrangements and contributions	The Government of Rwanda is providing grant financing via the Rwanda Forest Authority (RFA), Rwanda Water Board (RWB), and Rwanda Development Board (RDB) to support activities in component 3. In addition to their contributions, AE will cover office costs in the form of an in-kind contribution equivalent to \$112,271 for the Project Management Unit which will be hosted within AE premises. The co-financiers have been identified based on their existing mandates and activities they implement in the CND region hence this was the only assessment done and they have been able to commit their involvement in the project implementation. The provided co-financing commitments will serve as a basis of MoE engagement during the project implementation. During implementation of the project, annual work plans will be adjusted together with the executing entity and all the co-financiers to align with everyone’s activities. MoE/AE under its oversight role will ensure that all these stakeholders commit during planning on their co-financing and report on it during Annual Performance Reports submitted to GCF.					

C.2. Financing by component

Component	Output	Indicative cost USD (\$)	GCF financing		Co-financing		
			Amount USD (\$)	Financial Instrument	Amount USD (\$)	Financial Instrument	Name of Institutions
Component 1: Mainstreaming Climate adaptation into integrated land use planning	1.1 Landscape-wide land-use plan developed for climate-resilient livelihoods & forest ecosystems, integrating district strategies	1,729,172	1,729,172	Grant			
	1.2 Local & national institutional capacities strengthened to integrate biodiversity & climate risks into land use planning & management	1,195,652	1,195,652	Grant			
Component	Output	Indicative cost USD (\$)	GCF financing		Co-financing		
			Amount USD (\$)	Financial Instrument	Amount USD (\$)	Financial Instrument	Name of Institutions
Component 2: Forest and landscape management and restoration	2.1 PA management effectiveness improved re climate risks & adaptation	2,051,778	2,051,778	Grant			
	2.2 Natural forest cover restored, biodiversity connections established	16,917,145	14,271,796	Grant	2,645,349		Rwanda Water Board
Component 3: Enhancing climate adaptation through promotion of resilient livelihoods and climate-smart farming methods	3.1 Farming methods enhance productivity, reduce erosion & flooding risks, contribute to ecosystem services, and support connectivity	13,451,014	10,856,387	Grant	478,071	Grant	Rwanda Forestry Authority
	3.2 Rural livelihoods generate alternative incomes & reduce pressure on forests	8,612,188	3,850,283	Grant	4,761,905	Grant	Rwanda Development Board
	3.3 Financial services & private sector investment engage	559,890	559,890	Grant			

Monitoring & Evaluation	4.1 - Performance monitoring plan developed	555,475	555,475	Grant			
	4.2 - Project monitored and evaluated; lessons learnt integrated into adaptive management processes	645,390	645,390	Grant			
Project Management Costs	5.1 Project managed efficiently to achieve its goal to build resilience to climate change in CND	1,901,344	1,480,769	Grant	106,925	In-Kind	Ministry of Environment (AE)
					313,650	Grant	Rwanda Forestry Authority
<u>5% Contingency</u>		2,380,952	1,859,830	Grant	39,586	Grant	Rwanda Forestry Authority
					238,095	Grant	Rwanda Water Board
					238,095	Grant	Rwanda development Board
					5,346	In-Kind	Accredited Entity
Indicative total cost (USD)	50,000,000	39,056,421	10,943,579				

C.3 Capacity building and technology development/transfer

C.3.1 Does GCF funding finance capacity building activities? Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer? Yes ☒ No ☐

This project will build knowledge among land-use planning actors on the value of forests for climate resilience, and institutionalizing cross-sectoral meetings to encourage synergies and avoid overlapping

mandates and redundancy in different climate resilience interventions to ensure climate adaptation and climate resilience are explicitly incorporated into land-use planning processes in the CND, such that trade-offs between different sectors can be resolved while balancing the climate resilience of nature and people. Local land-use planning experts and community members will be trained and engaged to review and interpret the National Land Use and Development Master Plan, to ensure the plan accounts for current and future climate risks, and that these risks are also considered in district level planning schemes. The knowledge and insights gained from these activities will be synthesized and used to inform the design of an Integrated Land-use Plan for Resilient Livelihoods and Ecosystems, which will reconcile the cross-sectoral (e.g., tourism, environment, forestry, food production, cash crops for export) trade-offs necessary to sustain forest ecosystem functions, deliver critical ecosystem services and improve community livelihoods. An innovative scientifically credible system for monitoring trends in natural forest cover and forest types will also be established, and a comprehensive capacity building programme will improve use of remote sensing, ecosystem service modelling and spatial planning tools to inform decision making. **Budget: \$2,000,754**

To enhance forest resilience across the CND, through a combination of increased forest cover, improved forest condition, and improved management to reduce degradation and pressure on forests, the project will increase the extent and condition of natural forest within the CND's national parks, by facilitating regeneration of large areas previously degraded by fires, securing key remaining natural areas outside PAs and developing a financial sustainability plan for the CND. The project will invest in the capacity of government agencies, districts, and communities to manage forest fires. Community members will be trained in and provided technical assistance and equipment for restoration techniques, establishment, and management of indigenous tree nurseries. Management plans for the CND's national parks will also be updated and park staff trained to explicitly consider climate adaptation and gender issues and manage the parks as part of the larger CND landscape. The project will also develop the capacity of key Government ministries and agencies on sustainable conservation finance. **Budget: 5,420,837**

Additionally, this project will strengthen the livelihoods of landless and or the most vulnerable households, who are often most at risk of unsustainable use of their farmlands, protected areas and resources within protective forests. Erosion control measures and on-farm tree cover will be increased across the CND, helping reduce the risk of crop failure from extreme events like landslides, reduce land and water degradation caused by soil erosion, and increase provision of valuable products like wood fuel. To decrease pressure on forests for resource extraction, existing successful programs distributing improved cookstoves will be dramatically scaled up. Community members will be trained in and provided technical assistance for SLM practices, establishment, and management protective forests. Both cooperatives, individuals, participating financial institutions (PFIs), and value chain actors/small and mid-size enterprises (SMEs) will benefit from capacity-building to improve their business skills and access to finance, and to strengthen value chains (including beekeeping, ecotourism, agri value chains, and cookstoves), including the construction of processing facilities and marketing materials. Vocational training schools and local youth cooperatives will be supported to produce, test, and distribute Tier 2 cookstoves. **Budget: 13,893,987**

(d) EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

D.1. Impact potential

Adaptation Impact – This Project is designed to restore, sustain and expand critical forest ecosystem functions in the CND. More specifically, the project will protect, restore and expand natural forest in and around the core national parks; consolidate and expand some smaller but critical "stepping stones" of remaining natural habitat between these national parks (mainly small forest patches and wetlands/ riparian areas); improve the landscape linkage function of key high altitude between the protected areas, while at the same time securing ecosystem services through afforestation and improved native species mix for plantation forests on steep slopes; silvopastoral interventions on cultivated pasture; and agroforestry interventions in regenerative agricultural areas. These interventions will reduce erosion, decrease loss of soil organic matter, and thus increase the water infiltration rate, which increases water recharge which will enhance the resilience of forest ecosystems and smallholder farmers to climate change. The establishment of ecological connectivity between fragmented forests will promote landscape heterogeneity and support species migration to more favorable conditions within this human modified landscape and as a result help build the resilience of biodiversity to climate impacts. Increasing the capacity for species movement across these islands of forest habitat, will improve the adaptive capacity of multiple species that serve as important seed dispersers and insure the long-term viability of forests.

Project interventions will include 10,000 ha of land (including both forest and silvopastoral sites) under improved, climate-resilient management; and improve agroforestry practices on 3,346 ha of on-farm plantations, directly benefiting 1,254,242 people (654,404 women) with more climate resilient livelihoods from reduced exposure to landslides, floods and soil erosion and more knowledge on climate risks, value of forests and forests ecosystems, climate adaptation options, access to indigenous and agroforestry quality materials and improved ecosystem services.

The Project will provide direct benefits to increase the climate resilience of more than 1.5 million of the most vulnerable residents of the CND, including increased adaptive capacity through reduced exposure to flash floods, landslides, and soil erosion which will also increase their crop productivity and livelihoods (direct beneficiaries 52.2% female). Indirectly, 2,464,097 beneficiaries in the CND will receive co-benefits from improved and more sustained ecosystem services, including decreased sedimentation leading to improved water quality downstream and increased hydropower generation. This helps Rwanda produce cleaner hydropower energy at a lower cost, making it preferable to other more detrimental energy sources. The project will also indirectly benefit 9,260,745 people and create approximately 24,212 job opportunities in forest-dependent communities.

The Project will contribute altogether to the restoration of 5,000 ha of intact forest, 2,500 ha of protective forests and 3,346 ha of smallholder farmland, 1000 ha of pastures and 1,500 ha of riparian linkages. Overall, 275,617 ha of the landscape could potentially be positively impacted by the project as land use planning, improved natural forest management, improved buffer management and a stronger agroforestry sector would all contribute to improved outcomes beyond the specific areas directly targeted for restoration or improved management by the project. In particular, stronger capacity for land use planning across the CND and hence improved protection of key areas (e.g. steep slopes, riparian areas and protected area buffers) has strong potential to deliver significant mitigation benefits at a landscape rather than a site level. Although the project is focused on the priority landscapes covering 275,617 ha, it is likely that benefits due to increased adaptive capacity (achieved through improved land use planning and management of key features through

reduced exposure to flash floods, landslides, and soil erosion) could feasibly extend across the full 700,000 ha CND.

Additional adaptation benefits will include the training of 52,589 community members in forest management and restoration for climate resilience, which will be linked with ongoing restoration employment opportunities through other GoR and private sector projects; the strengthening of institutional and regulatory systems to promote climate resilience; and targeted capacity building for certain government institutions and ministries in forest monitoring, remote sensing, spatial planning, and analysis. These beneficiaries will include MoE, REMA, RDB, MINAGRI, and the University of Rwanda.

Complementary activities will address key drivers of deforestation while creating alternative livelihood pathways for youth and women through ecotourism and income-generating, climate-resilient crops, as well as strengthening supportive financial services.

Mitigation Impact – The Project will generate mitigation benefits by delivering sequestration of approximately 224,871 t CO₂-eq during the 5-year Project period, which is about 0.8% of the 28.2M t CO₂ reduction targeted by Rwanda's vision 2050.

- 5,000 ha of critical intact natural forests will be under improved management for climate resilience, accounting for an estimated total of 596,116t CO₂-eq over 20 years.
- 1,000 ha of pasture lands with increased coverage of native trees to secure reasonable landscape connectivity for forest species, accounting for an estimated total of 4,543t CO₂-eq over 20 years.
- 1,500 ha of riparian lands will be afforested which will contribute significantly to overall landscape connectivity, sequestering 110,138 t CO₂-eq over 20 years.
- 2,500 ha of protective forests and plantations will be established on slopes >55%, focusing on areas not currently planted in crops, accounting for an estimated total of 193,472 t CO₂-eq over 20 years.
- Erosion control and agroforestry practices will be implemented over 3,346 ha of smallholder farmland, sequestering 129,006t CO₂-eq over 20 years.
- Improved energy efficient stoves will be adopted by 8,500 households, securing 51,017t CO₂-eq of avoided emissions over 20 years.

Overall, there is potential for a significant contribution to GCF1 Core Indicator 4, hectares of natural resource areas brought under low emission or climate resilient practices. Conservatively, an area of 275,617 ha of the landscape could potentially be positively impacted by the project as land use planning, improved natural forest management, improved buffer management and a stronger agroforestry sector would all contribute to improved outcomes beyond the specific areas directly targeted for restoration or improved management by the project.

D.2. Paradigm shift potential

As described in Section B1, the Project provides a profound opportunity to implement new mechanisms of land-use planning and management that are based on consideration of the CND as a landscape, consisting of an inter-dependent mosaic of forests and farms, the management of each determining the health of the whole. Focusing on this larger scale, with coordination across both geographic and sectoral boundaries, while mainstreaming climate risks into management planning represents a fundamental paradigm shift for land-use planning and management in Rwanda. This approach to land-use planning and management is entirely new for Rwanda and is intended to maintain and recover the ecosystem services that are essential to the resilience of smallholder farming communities within the region and to the globally significant and

unique biodiversity in this part of Rwanda. This represents a reversal of the previous negative feedback loop of ecosystem degradation to a more positive, climate-resilient system.

Potential for scaling-up spatial planning and replication. Spatial planning is recognized by the IPCC, the EU and others as playing a key role in climate change adaptation and climate resilient development, but to date development planning in Rwanda has largely been based on tabular data, rather than on spatial planning. Adoption of spatial planning across ministries and on national regional and local levels, represents a significant paradigm shift for Rwanda. Piecemeal decisions and an ad hoc assemblage of isolated sector-by-sector projects will be replaced with a holistic, climate-smart, evidence-based approach to planning and development – enabling systematic, ecosystem-based, multi-sector analysis of land use tradeoffs. The Project will therefore establish a new inter-agency and local-national model and capacities for holistic planning and implementation – one that can be scaled up for use elsewhere in Rwanda and beyond. This will enable better-informed planning decisions, the identification and valuation of nature-based solutions to climate adaptation and mitigation, and overall will foster climate resilient development pathways in the CND and nationwide. Communication and collaboration with complementary GCF and other projects in Rwanda will accelerate this process.

Potential for knowledge and learning. Component 1 will invest in new technologies and climate-smart knowledge to improve national and district planning, scenario analysis and monitoring capacities. In doing so, new mechanisms of coordination and collaboration will be established, and shared across agencies. Component 2 will invest in best practices in forest management and restoration – some of which will require learning beyond the existing science of indigenous tree propagation and restoration (particularly in re-establishing rain forest habitat, establishing protective forests on slopes of >55%, and enriching farms with indigenous tree species). Component 3 will focus on processes to identify and pilot value-chain products (on-farm and off-farm) not yet in practice. All approaches will include the participation of local farmers, women, and youth to contribute to collective learning processes involving national and local government and vulnerable communities. Knowledge generated – and the processes by which it is generated – will be shared across the region. The project is based on collective learning, knowledge generation and dissemination at community, landscape and national levels. A focus on knowledge management through demonstrations, monitoring, and evaluation across the project activities will inform the implementation of current and future adaptation efforts.

Contribution to the creation of an enabling environment. Strengthening the Task Force in Component 1 will ensure a single, first point of contact for all sectors' planning development and investments in the CND and will ensure that planning for climate resilience is addressed. By establishing a mechanism for coordination and harmonization, it will radically improve current conditions for public and private interventions and will establish partnerships and collaborations that live beyond this Project. Also importantly, promotion of sustainable financing for each component will help to assure that initiatives will either be self-sustaining or have solid plans for continued financial support.

Contribution to the regulatory framework and policies. The Project will finance the development of the *Integrated Land-use Plan* which will be adopted by all ten CND districts as a framework for seeking synergies between rural livelihoods and forest and biodiversity protections, reconciling land use conflicts that arise, and guiding development decisions and climate change adaptation. It will update the National Land Use Master Plan, ensuring it incorporates climate resiliency, and is accurate and relevant to local contexts. As development of an Integrated Land-use Planning approach helps better reconcile cross-sectoral tradeoffs and improve climate adaptation planning in the CND, this approach will prove a successful model for application to other regions of Rwanda and to other forest/farming landscapes beyond. In addition,

revising 2 NP management plans to include climate concerns will serve as a model for the remaining parks of Rwanda, as well as for the East African region.

D.3. Sustainable development (max. 500 words, approximately 1 page)

The proposed Project is aligned with the United Nations 2030 Agenda for Sustainable Development, including SDGs 5, 13 and 15 and will contribute to the achievement of the SDGs 1, 2, 3, 5 and 10. The Project will promote an ecosystem-based approach to forest protection, restoration, and rehabilitation that includes:

- Reduction of erosion, sedimentation, and landslide risk as a result of restoration of degraded lands;
- Increased number of native trees on farms with improved agroforestry systems, improving biological connectivity;
- Increased carbon sequestration potential of plants and soils. While the primary objective of the Project is adaptation, the Project will also produce mitigation benefits of 204,079t CO₂-eq during the 5-year implementation period, and 1,033,275t CO₂-eq over the 20-year Project lifetime.
- Reduced GHG emissions through more fuel efficient cookstove technologies, yielding cumulative savings of approximately 51,017t CO₂-eq during the 20-year Project lifetime.

The Project will enable the transition to more climate resilient and productive livelihood models and empower the rural poor, significantly reducing impacts from extreme weather events and increasing their potential for improved wellbeing. Social benefits will include:

- Raised awareness about climate change effects and adaptation in 1,253,252 people living in the CND (at least 52.2% representing women).
- Capacity building of ~52,589 people (at least 50% women), in a wide range of topics aimed at increasing ecosystem and social resilience.
- Create significant social capital through co-designing and co-managing a range of adaptation strategies, indirectly promoting social cohesiveness among CND villages through joint saving and lending systems.
- Reduced inequality and poverty through increased financial inclusion and improved economic opportunities — especially for women, youth and historically marginalized groups.
- Forest and landscape restoration in the CND will have several economic co-benefits for vulnerable communities, including:
- Soil erosion control measures, protective forests on public lands and agroforestry on private lands, will increase agricultural productivity and associated income.
- The Project will create direct and indirect employment opportunities, benefiting approximately 24,212 households and stimulate the local economy.
- Leveraging private capital to invest in the value chains including cookstoves in the CND. The project will create important social and environmental benefits. However, the project will also have removed a number of risks (notably carbon risk) that worry more traditional (social-impact or sustainability) investors. As a result, the venture becomes an attractive investment for private capital interested in a “triple bottom line”, allowing the business to develop sustainably and scale to more CND areas. During project implementation we will therefore produce a bankable business plan that details the activities and resources required for expanding further.

The Project will result in positive outcomes around access to resources, improved livelihoods, and income generation opportunity and capacity for women. With support from a dedicated Gender Specialist, the Project will engage women in project planning, investment and decision making. With opportunities to generate additional income, women, who traditionally are already further vested in investing in their families, will be more able to respond to incentives that address their families' basic needs, such as better health and nutrition, linking to agriculture and food security improvements. The project will also get support from NGOs that engage men in programming and discussions to increase their participation in household and parental responsibilities and gain greater awareness of the importance of women's fair access and control of resources, including income. Women will benefit from training and educational activities, leadership, entrepreneurship and decision making. Annex 8, which will be further elaborated on at project onset, provides more information on gender-sensitive impacts.

D.4. Needs of recipient

Despite Rwanda's solid economic progress since 2000, poverty remains widespread and pervasive. Overall, 38.2% of the population lives in poverty and 16% in extreme poverty¹¹⁹. Women are more affected by poverty than men, with 47% of female-headed households living in poverty compared with 44.9% of all households. Rural households are more than twice as likely to be in poverty/extreme poverty than urban households. Districts of the Western and Southern Provinces in the CND region have the highest rates of poverty and extreme poverty in the country.

Most rural households are smallholder subsistence farmers and, because of high population density, high relief terrain and dependence on rain fed agriculture, the residents of the CND are highly vulnerable to climate change. These risks include exposure to flash floods¹²⁰, landslides¹²¹ and soil erosion¹²². During stakeholder consultations, communities indicated several socio-economic changes stemming from changing climate conditions. Notably, the longer dry season and unpredictable rains¹²³ have reduced agricultural yield significantly, leading to more widespread malnutrition and extreme poverty, and farmers can barely meet their own families' needs¹²⁴. In particular, women, youth, and historically marginalized people are feeling the impacts. As arable land is increasingly constrained, youth and historically marginalized people have few opportunities and are more focused on survival and short-term gains at the expense of education and sustainable livelihood development (see Annexes 6 and 8 for more detail).

As a result, the GoR has invested significantly in developing an enabling policy and institutional framework for a more climate-resilient development pathway. Yet, while adaptation is a top priority, Rwanda is classified by the U.N. as a LDC and lacks sufficient public financial resources to implement its climate change and resilient green development objectives. Currently there are no alternative funding options of the scale needed to secure Rwanda's priority forests to reduce vulnerability of rural communities to climate risks. The GoR also recognises that a piecemeal approach to decision making has exacerbated competition over scarce land resources and reduced the adaptive capacity of both ecosystems and rural populations. However, many

¹¹⁹ World Bank, 2020. *Poverty and Equity Brief: Rwanda*.

¹²⁰ Karamage, F., et al., 2017. *Modeling Rainfall-Runoff Response to Land Use and Land Cover Change in Rwanda (1990–2016)*. Water 2017, 9(2), 14.

¹²¹ Uwihirwe, J., et al., 2020. *Landslide precipitation thresholds in Rwanda*. Landslides.

¹²² World Bank, 2019. *Rwanda Systematic Country Diagnostic*.

¹²³ World Bank; CIAT. 2015. *Climate-Smart Agriculture in Rwanda*. CSA Country Profiles for Africa, Asia, and Latin America and the Caribbean Series. Washington D.C.: The World Bank Group.

¹²⁴ Clay N., King B., 2019. *Smallholders' uneven capacities to adapt to climate change amid Africa's 'green revolution': Case study of Rwanda's crop intensification program*. World Dev.

agencies operate within siloed environments, focusing on limited scopes framed by individual performance plans, and close coordination is often cited as a key recommendation toward strengthening climate change adaptation measures¹²⁵.

Additionally, district government staff and community members have limited technical understanding or awareness of the complex interactions between forest structure and functions, climate change, and the provision of ecosystem services. Accurate measurement of forest cover and forest type is notoriously difficult in high relief terrain, such as in the CND¹²⁶. Lack of access to and capacity to apply new methods for classifying and monitoring forest change has limited forest monitoring in Rwanda¹²⁷. Consequently, government forest data has historically been limited in scope and infrequently updated, and regional/global maps are too inaccurate to be of use. Capacity building around the technologies and processes to make landscape-scale planning decisions is crucial to the success of Rwanda's climate-resilient development pathway.

D.5. Country ownership

The proposed Project is closely aligned with the NAP's adaptation priorities: 1) promotion of non-agricultural income generating activities, 2) introduction of species resistant to environmental conditions, and 3) development of firewood alternative sources of energy. It is also aligned with several priorities identified in the 2020 NDC: promotion of afforestation/reforestation, improvement of forest management, integrated approach to planning and monitoring for sustainable land management, development of a harmonized and integrated spatial data management system for sustainable land use management. The Project will support implementation of the Green Growth and Climate Resilient Strategy and the Rwanda National Strategy for Transformation–NST1 (2017-2022) and is aligned with the goals of the Rwanda National Forestry Policy 2018, the Forest Sector Strategic Plan 2017-2022, the Strategic Plan for the Environment and Natural Resources Sector (2018-2024), and National Forest Management Plan 2017-2026, reflecting the government's intentions to address climate change impacts and mitigation targets by improving forest management. It is also aligned with the national Biomass Energy Strategy (BEST) 2020 which aims to ensure a more sustainable biomass supply and combustion technologies for homes and small enterprises. This Project aligns with the forthcoming GCF country program which prioritizes forest management through restoration and protection of natural forests, afforestation and agroforestry, as well as distribution of high-quality forestation inputs and capacity development.

While this Project proposal to GCF focuses on a different geographic area - with specific adaptation needs through land restoration, development of markets and finance mechanism, as well as support to mainstreaming climate change into land use planning - the early and ongoing lessons from the experiences from existing GCF-funded projects, "Strengthening Climate Resilience of Rural Communities in Northern Rwanda"¹²⁸ and "Transforming Eastern Province through adaptation" (TREPA)¹²⁹ will be applied during the

¹²⁵ Bagstad, K.J., 2019. *Towards ecosystem accounts for Rwanda: Tracking 25 years of change in flows and potential supply of ecosystem services*. British Ecological Society.

¹²⁶ Arakwiye, B., et al, 2021. *Thirty years of forest-cover change in Western Rwanda during periods of wars and environmental policy shifts*. Regional Environmental Change 21(2).

¹²⁷ Herold, M., Schiller, F., 2009. *An assessment of national forest monitoring capabilities in tropical non-Annex I countries: Recommendations for capacity building*. Global Observation of Forest and Land Cover Dynamics Land Cover Project Office.

¹²⁸ Rwanda Ministry of Environment, 2018. *FP073: Strengthening Climate Resilience of Rural Communities in Northern Rwanda*. Green Climate Fund.

¹²⁹ International Union for Conservation of Nature, 2021. *FP167: Transforming Eastern Province through Adaptation*. Green Climate Fund.

implementation of this Project in the Congo Nile Divide Landscape.

MoE is responsible for ensuring sustainable management of natural resources and for the development of policies, strategies, regulations, and mobilizing resources for the development of the sector. The MoE managed USD 10 million from the Adaptation Fund to implement the Reducing Vulnerability to Climate Change in North West Rwanda through Community Based Adaptation Programme¹³⁰ and has overseen a number of other climate projects including: an early warning system project (funded by LDCF), the Landscape Approach to Forest Restoration and Conservation (LAFREC), and the “Building resilience of communities living in degraded forests, savannahs and wetlands of Rwanda through an ecosystem management approach” programme (funded by LDCF)¹³¹. The OECD rated Rwanda as one of the countries that uses aid most effectively¹³², and Rwanda is on course to achieve the SDGs. The GoR has demonstrated comprehensive country ownership of, and capacity to implement, a funded project. The national designated authority (NDA) is the core interface between a country and the GCF and is mandated to align GCF-supported activities with strategic national objectives and priorities. The NDA coordinates with AEs on proposal development and completes a no-objection letter after it is determined that the proposal is aligned with the national priorities. This Project was initiated by the MoE and supported by the NDA, Rwanda Environment Management Authority, due to the high climate sensitivity of the CND.

Government agencies such as Rwanda Forestry Authority (RFA), Rwanda Development Board (RDB), Rwanda Environment Management Authority (REMA) and National Land Use Authority (NLA) will maintain strong ownership and support activities under component 1 and 2 as aligned with their mandates and the aforementioned plans and strategies.

To better understand the interests of stakeholders and potential impacts of the Project, and as preparation to design future engagement strategies, a comprehensive, gender-responsive, culturally sensitive, non-discriminatory, and inclusive stakeholder analysis was undertaken and is incorporated into the Stakeholder Engagement Plan (Annex 7).

D.6. Efficiency and effectiveness

Financing and concessional: The financial instrument chosen for this Project is grant financing of USD 39.06 million which constitutes roughly 78% of the total Project’s estimated cost. The relatively high level of concessionality reflects the public good character of the investments to be made by the Project, Rwanda’s status as an LDC¹³³ and its high vulnerability to climate change¹³⁴. Furthermore, it must be highlighted that most project expenditures relate to creating public goods through forest and landscape restoration or capacity-building of public institutions. These interventions are primarily non-income-generating; hence very hard to attract private sector investment. Attracting the private sector for public goods investments remains challenging in the context of many countries worldwide, including Rwanda.

Addressing low private sector investment in the forestry sector. One of the major reasons for lack of private-sector investment in Rwanda’s Forestry Sector is that most stands are too small to sustain an

¹³⁰ Adaptation Fund, 2014. Reducing Vulnerability to Climate Change in North West Rwanda through Community Based Adaptation. [Link](#).

¹³¹ UNEP, 2016. *Building resilience of communities living in degraded forests, savannahs and wetlands of Rwanda through an ecosystem management approach*.

¹³² Newfarmer, R., et al., 2013. *Managing Aid for Trade and Development Results: Rwanda Case Study*. OECD.

¹³³ United Nations Department of Economic and Social Affairs, 2021. Least Developed Country Category: Rwanda Profile.

¹³⁴ World Bank, 2021. *Rwanda*. Climate Change Knowledge Portal.

economically viable operation (see the Forest Sector Strategic Plan). By restoring smallholder protective forest plantations and organizing them into Private Forest Management Units (PFMUs), this Project will assist with creation of units that are economically attractive to private investors, helping to facilitate future private investment if landowners desire it. However, no private investors will be subsidized by the Project, and ownership of PFMUs will remain with the state/district governments and smallholders' cooperatives who own forest plantations. RFA in collaboration with RCA and the district government directly implement organization of the cooperatives and conflicts in the cooperatives are managed in accordance with internal rules and regulation governing each cooperative overseen by the Rwanda Cooperative Agency in case of escalated conflicts but specifically in the cooperatives that will be benefiting under this project, RFA and District government will also intervene in conflict settlement.

No excess subsidy will be provided to the private sector, and no private sector investors will be crowded out by these investments. Wherever these are required for institutional and capacity support for market development and crowding in for private investments these are undertaken with concessionality to community in mind. For example, the subsidy/microcredit schemes to facilitate the adoption of efficient cookstoves are undertaken with minimum concessionality. The value chain development for other potential revenue generating streams such as honey undertaken under output 3.2 requires institutional and capacity support funded by GCF grant. In the case of ecotourism for tea plantations, no project concessionality is planned and complete crowding in from the private sector is envisaged. The value chain development of beekeeping is 100% funded by GCF grants so that an enabling environment can be created, and the project can overcome affordability challenges through developing a system such as community co-guarantee, overcome lack of private capital structure and insufficient markets. Because of above mentioned reasons no private sector investors are expected to be crowded out by project investments.

GCF grant as a suitable financing instrument. The GCF grant can help create the enabling environment for the future engagement of the private sector in similar projects. The showcasing of potential revenue-generating ways, thanks to the concessional GCF financing, may help invigorate the potential for future investments in this sector, which in turn might result in a multiplier effect leading to a higher degree of development impact.

Results of the Economic and Financial Analysis (EFA). Following the GCF guidelines for developing the Annex 3 package, the EFA was prepared to assess the economic rationale for the proposed interventions. Consequently, four representative interventions were appraised in the EFA: (i). agroforestry, (ii). protective forests (iii). fuel-efficient cookstoves, (iv). silvo-pastoralism. Best practices in appraisal for public sector projects have been followed. The financial analysis for each intervention was calculated from the private perspective using a discount rate of 15.44% which was the lending rate of Rwanda Central Bank. For the three modeled interventions (see table below) the financial NPV is positive and financial IRR is higher than discount rate signifying that project activities are profitable. The GCF support will result in a positive financial return for farmers or farmers and private actors over a 20-year period and will help overcome high up-front investment costs during the project duration.

Financial analysis (US\$)	Agroforestry	Protective Forest	Silvopasture
FNPV	\$34,483,676	\$2,980,381	\$310,901
FIRR	42%	25%	35%

Aggregate financial analysis of with project (WP) interventions

The obtained ex-ante aggregate economic net present values (ENPVs) for these interventions over the 20 years' timeframe and at a 13% economic discount rate range between USD 0.97million and USD 78.93 million, with economic internal rates of return (EIRRs) between 49% and 72% (see table below).

Economic analysis (US\$)	Agroforestry	Protective Forest	Energy efficient cookstoves	Silvopasture
ENPV	\$58,934,340	\$30,218,468	\$10,585,674.50	\$966,331
EIRR	49 %	72%	N/A	51 %

Aggregate financial analysis of with project (WP) interventions¹³⁵

The sensitivity analysis that was carried out suggests that obtained results are relatively insensitive to increases in costs or decreases in benefits offering the robustness of obtained results. Therefore, the economic sustainability of the Project is likely, and the Project shows the economic rationale for implementation as benefits are likely to outweigh the costs.

Best Practices *Forest Restoration/Tree Planting*: The Project will use Rwanda's best practice guidelines for tree nursery establishment, tree planting and tree management¹³⁶. For assisted natural regeneration, methods developed and proven through multi-year research in Nyungwe and Gishwati-Mukura National Parks will be used¹³⁷. *Forest Monitoring*: The Project will utilize multiple contemporary methods for monitoring the success of project interventions and forest health. Forest plantations will be mapped and monitored using the Rwandan government's newly developed Forest Monitoring and Evaluation Software (FMES)¹³⁸, specifically developed for this purpose. The Project will develop a modern indigenous forest monitoring system, utilize the most high-resolution datasets currently available (<3m) and take advantage of modern remote-sensing advances including cloud computing and machine learning. Cost effective replication for forest restoration and climate resilience throughout Africa and beyond can be anticipated.

¹³⁵ No financial analyses for cookstoves intervention was warranted as energy efficient cookstoves are non-income generating intervention with expectation of no financial cash inflows. Only economic analyses was conducted for cookstoves.

¹³⁶ Unique Ltd. & RNRA. 2015. Tree plantation establishment and management manual for Rwanda. Kigali, Rwanda

¹³⁷ Masozera, A.B., 2004. Regeneration of Burned Forested Areas With Periodic Removal of Pteridium aquilinum. Wildlife Conservation Society Rwanda.

¹³⁸ Rwanda Forest Authority, Enabel, 2021. Technical guidelines on the use of the Forest Monitoring & Evaluation System - FMES. Kigali.

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

- ☐ Reduced emissions (mitigation)
☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Rwanda's natural forests are extremely depleted and degraded. While overall forest cover has increased since the 1980s, this is primarily made up of exotic monocultures, which have very low potential to provide ecosystem services and adaptation benefits. There are small-scale efforts to establish/restore native forests and increase overall forest resilience, but scaling these efforts is hampered by a limited supply of indigenous seedlings.	<u>Low</u>	Paradigm shift would involve a move away from widespread planting of exotic species for fuelwood, as well as improved erosion control measures on farmland and use of improved cookstoves, both of which will reduce pressure on Rwanda's forests. As the benefits of planting indigenous trees become more widely known, behavior changes in non-targeted communities are likely to be catalyzed.	The project is expected to bring 144,058 ha of land under climate-resilient management. This would represent a substantial step towards paradigm shift on increased resilience of forests and vulnerable communities, even before any replication effects, which will likely be catalyzed by project activities focusing on training, capacity building, and peer-to-peer knowledge exchange. The project will also deliver direct carbon reductions of 1,204,448 t CO ₂ -eq over 20 years, and potentially more through increased institutional capacity to mainstream climate adaptation and improve land-use planning.
Replicability	There are a small number of interventions occurring across Rwanda focused on improving resilience of vulnerable communities through forest restoration, agroforestry, livelihood diversification and clean cooking technologies. However, many interventions focus on exotic species which have very low potential to provide ecosystem services and adaptation benefits.	<u>Medium</u>	If increasing the extent and integrity of natural forests and improving capacity for climate-responsive land management can increase climate resilience of vulnerable communities in the CND, then the solution could be replicated throughout Rwanda and across East Africa. Further, if incorporating indigenous species into forest restoration and agroforestry can provide a similar/greater range of adaptation benefits compared to exotic species, and also avoid the drawbacks of exotic monocultures, then the solution could be expanded nationally. If development of an Integrated land use planning approach helps better reconcile cross-sectoral tradeoffs and improve climate adaptation planning	The project will invest in strengthening institutional capacity to promote climate resilience, directly increase the resilience of ecosystems and community livelihoods, and identify options for use of indigenous species in forest restoration and agroforestry, in order to better contribute to climate-resilient livelihoods. These topics have received little attention to date in Rwanda, and lessons learned will inform efforts to improve climate resilience across Rwanda and East Africa.

			in the CND, this approach could be applied to other regions of Rwanda.	
Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Sustainability	The GoR has invested in developing policies and institutional frameworks for a more climate-resilient development pathway. However, domestic funding to implement these policies remains lacking, and as yet they have not translated to wide-scale improvements to climate resilience. Changes in behavioral norms have been limited, and overall uptake of climate adaptation solutions is poor.	<u>Low</u>	Paradigm shift would see Rwanda demonstrate that pursuing development goals need not come at the cost of extensive environmental degradation. Climate-responsive Forest and landscape management will be standard, and government programmes will focus on increasing the resilience of forest ecosystems and enhancing their ecosystem services. Behavior change across the CND would support this, as communities recognise, value, and protect the benefits provided by forests.	The project will work closely with government and communities to build capacity to mainstream climate adaptation into forest and land-use planning and management. The project will facilitate and demonstrate the benefits of climate-resilient agricultural practices, forest management, and alternative livelihoods, raising awareness within and outside target communities.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Mitigation Results Area 4: Forests and land use	Core Indicator 1: GHG emissions reduced, avoided or removed/ sequestered	Greenhouse gas emission assessment	0 tonnes of CO ₂	107,686 tCO ₂ -eq	224,871tCO ₂ -eq (5-year implementation period) 1,084,291tCO ₂ -eq (20-year lifespan period)	Government priorities remain supportive of long-term forest restoration and afforestation plans. Absence of major natural disasters including forest fires and landslides in target areas The economic, social and political context in the country and project areas remain stable.

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Adaptation Results Area 1: Most vulnerable people and communities	Core Indicator 2: Direct and indirect beneficiaries reached	Annual Agricultural Survey ¹³⁹ National census Project Baseline and Completion Reports, including documentation of workers contracted for reforestation/afforestation work, farmers adopting agroforestry activities, good silvo-pastoral practices, restoration of riparian linkages, signed MoUs for Private Forest Management Units, and households receiving improved cookstoves	Direct beneficiaries: 0 Indirect beneficiaries: 0	Direct beneficiaries: 95,873 (50,046 females) Indirect beneficiaries: 2,809,381 (1,466,497 females)	Direct beneficiaries: 108,277 (56,521 females) Indirect beneficiaries: 2,809,381 (1,466,497 females)	Government priorities remain focused on long term forest restoration and afforestation plans Absence of major natural disasters including forest fires and landslides in target areas Rural landowners agree to adoption of agroforestry and reforestation activities
Adaptation Results Area 1: Most vulnerable people and communities	Supplementary Indicator 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options	Project Baseline and Completion Reports, including documentation of beekeepers using modern hives, farmers at different levels of value chains, people employed in ecotourism, have access to	Direct beneficiaries: 0 Indirect beneficiaries: 0	Direct beneficiaries: 26,160 (13,656 females) Indirect beneficiaries:	Direct beneficiaries: 52,319 (27,311 females) Indirect beneficiaries:	

¹³⁹ The National Institute of Statistics of Rwanda conducts the Agricultural Survey yearly. Available at: <https://www.statistics.gov.rw/statistical-publications/subject/agriculture-and-environment>

		finance for other climate-resilient livelihood options		2,809,381 (1,466,497 females)	2,809,381 (1,466,497 females)	
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Adaptation Results Area 4: Ecosystems and ecosystem services	Core Indicator 4: Hectares of natural resource areas brought under improved low emission and/or climate-resilient management practices	Updated National Park Management Plans complete (ha per park) Forest cover statistics from Forest Monitoring and Evaluation System (FMES) GIS Data/Remote Sensing Ranger Based Monitoring data from PAs Vigilance Committees monitoring data	No increase in climate-resilient management practices (o)	42,558 ha of land brought under climate-resilient management	130,104 ¹⁴⁰ ha of land brought under climate-resilient management	Absence of major natural disasters including forest fires in target areas
Adaptation Results Area 2: Health, well-being, food and water security	Supplementary Indicator 2.2: Beneficiaries (female/male) with improved food security	Annual Agricultural Survey Gender responsive Comprehensive Food Security and Vulnerability and Nutrition Analysis Survey (CFSVA) ¹⁴¹	TBD	Direct beneficiaries: 0 Indirect beneficiaries: 0	Direct beneficiaries: 1,598,536 (52.2% female) Indirect beneficiaries: 2,464,097 (52.2% female)	Absence of extreme natural disasters and economic shocks affecting yields and household economy; migration patterns do not significantly affect the number and status of households.

¹⁴⁰ 130,104 ha refer to 121,758 ha of all core PAs and 8,346 ha restored areas outside PAs

¹⁴¹ The National Institute of Statistics of Rwanda conducts the CFSVA every 3 years.

The level of beneficiary of this project are individual households, - who can be in a cooperative or a business community group - who will participate directly in the project, and thus benefit from its implementation. Beneficiaries include all people who own the land where activities will take place and/or those who will be employed by the project (except EE staff, consultants and implementation and contractual organizations). Dependents of household that directly benefit from the project are also considered as direct beneficiaries and were calculated based on 4.2 average household size¹⁴² and 0.37¹⁴³ average land size in the CNDL.

Adaptation Results Area 1, Core Indicator 2 direct beneficiaries are beneficiaries of activities 2.2.2, 2.2.3, 3.1.1, 1.1.2.

Adaptation Results Area 1, Supplementary Indicator 2.1 direct beneficiaries of activity 3.2.1, 3.2.2, 3.2.4 (excluding 8, 500 beneficiaries of cookstoves) and 3.3.2

Adaptation Results Area 2 Supplementary Indicator 2.2 direct beneficiaries are the total population in landscape implementation sectors of this project.

Indirect beneficiaries broadly refer to all other human populations living within the CNDL districts. More details on calculations are provided in annex 23.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)					
Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low-emission and climate-	The GoR is committed to developing an enabling policy and institutional framework for a more climate-resilient development pathway, as evidenced by the large number of government policies and strategies focused on improving climate mitigation and adaptation. However, climate change is not being comprehensively considered by all sectors, cross-sectoral planning is rare, and new institutions are needed to	<u>low</u>	An integrated planning approach is used to reconcile cross-sectoral tradeoffs while explicitly considering climate resilience actions. New institutions have permanent capacity in place to ensure that spatial data on climate, forests and ecosystem services is incorporated into land planning and management.	The project will facilitate development of an integrated land use plan for the CND which will reconcile the cross-sectoral (e.g., tourism, environment, forestry, food production, cash crops for export) trade-offs necessary to sustain forest ecosystem functions, deliver critical ecosystem services and improve community livelihoods.	National level (one country)

¹⁴² National Institute of Statistics of Rwanda (NISR); The Fifth Rwanda Population and Housing Census, Main Indicators Report, February 2023

¹⁴³ National Institute of Statistics of Rwanda (NISR), Agricultural Household Survey 2020 report, December 2021

<i>resilient development pathways in a country-driven manner</i>	provide relevant spatial data to inform planning			The project will facilitate the development and adoption of an integrated landscape level planning taskforce, in line with the JADF to bring the relevant stakeholders (ministries and their agencies, districts, private sector and civil society organizations) on board, to strengthen inter-sectoral collaboration on the planning and implementation of the FLR plans. The project will also support capacity development of relevant technical staff working on FLR and land use planning (from central government agencies, district government, and NGOs) and will assist the GoR in developing a dedicated SPU within an appropriate ministry to provide the technical expertise needed to process and integrate spatial planning into decision-making within and across sectors.	
Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core Indicator 6: <i>Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer, and innovation</i>	Climate-resilient development in Rwanda is hampered by a lack of modern technological advances, both in terms of generating information required for land-use management and decision making, and in deploying climate-resilient practices and technology. Rwanda has only recently developed a system to monitor plantation forests and lacks a robust system to monitor indigenous forests. Similarly, agricultural and forest management techniques are generally rudimentary, and most rural	<u>low</u>	The GoR has an innovative, scientifically credible system for monitoring indigenous forest cover and condition, as well as the technical capacity for incorporating spatial data into land-use management. Improved agricultural practices (e.g. terracing, modern beehives) are widespread, and locally produced improved cookstoves are in use by most households.	The project will design and finance the establishment of a new indigenous forest monitoring system for Rwanda. The project will also establish improved agricultural practices across 3,346 ha, as well as increasing use of modern beehives to reduce forest fire risk. Finally, the project will upskill technical colleges to enable local production and distribution of improved cookstoves to 8,500 households.	National level (one country)

	communities in the CND do not use modern cooking technologies.				
Core Indicator 8: <i>Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards.</i>	While GoR recognizes the need for developing an enabling policy and institutional framework for a more climate-resilient development pathway, the value of forests and ecosystem services for climate adaptation is poorly understood, both in government and by communities. There is no system to regularly produce data on indigenous forest cover and condition, and no specialized government unit dedicated to producing spatial data on climate change, forests or ecosystem services. Agricultural and forest management practices are generally rudimentary.	<u>low</u>	Rwanda has an innovative, scientifically credible system for generating and sharing information on natural forest cover and condition, and a government unit dedicated to ensuring spatial planning knowledge is incorporated into land-use management. Improved agricultural practices (e.g. terracing, modern beehives) are widespread, and degraded forest plantations are restored and managed according to government best-practices.	The project will design and finance the establishment of a new indigenous forest monitoring system for Rwanda, as well as establish a spatial planning unit within government that has the capacity to generate and integrate spatial planning knowledge into decision making. Project activities will also establish best-practice agricultural and forest management practices across 144,058 ha of land, and create community structures (e.g. farmer cooperatives) which facilitate knowledge and information sharing among community members.	National level (one country)

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 1: Landscape planning, policies & management effective & coordinated across sectors and scales to address climate risks & adaptation benefits						
Output 1.1. Landscape-wide land-use plan developed for climate-resilient livelihoods and forest ecosystems, integrating district strategies.	Operationalization of an Integrated Land-use Plan for Resilient Livelihoods and Ecosystems in the CND Climate change and biodiversity reflected in District Development Strategies (DDS)	Interim evaluation and final evaluation report Performance evaluations of the districts	0 integrated plans for the CND Climate change and biodiversity Indicators and targets not included in evaluations	1 draft integrated land use plan developed Climate change and biodiversity Indicators and targets fully integrated in evaluations	1 final integrated land use plan developed, approved and used Climate change and biodiversity Indicators and targets fully integrated in evaluations.	District land-use plans are updated within project timeframe Key stakeholders are supportive of cross sector collaboration. Sectoral ministries incorporate biodiversity and climate change into sectoral plans and programmes.
Output 1.2. Local and national institutional capacities strengthened to integrate biodiversity and climate risks into land use planning and management	District planners, staff within REMA, RDB, NLA, RDB with the skills and capacities to use spatial planning tools, generate data necessary for mainstreaming climate change, understand gender barriers and how to mainstream gender equality principles into planning, including supporting women, youth	Interim evaluation and final evaluation report Pre/post training assessments of unit members	0 units with capacity 0 staff reporting improved awareness and capacity.	Draft structure & function of spatial planning unit developed. Training materials in development	Spatial planning unit formally established. 36 Staff reporting improved awareness and capacity (18 Female, 18 Male)	Lack of leadership of the hosting institution/ government agency could delay the establishment of the unit. RFA is committed to lead and invest in the capacity of its staff to implement the monitoring system.

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	and marginalized groups participation in meetings					Capacity building programs can overcome the effects of high staff turn-over and frequent re-arrangement of the institutions of natural resources management (transfer of departments, merging and creation of new Ministries)
	UR GIS center with the capacity and skills to train students and staff from government agencies on spatial planning and climate modeling	Pre/post capacity assessment of the UR GIS Center	Staff with knowledge in spatial analysis and climate modeling but without quantitative skills to use new technologies/software to generate data for decision making	Training modules developed and staff to be trained identified	10 lectures and assistant lectures reporting improved skills, 20 students trained each year from year 3.	
	A system for monitoring forest trends is established and in use by the GOR	Interim evaluation and final evaluation report	0 systems in place	Operational structure & function of forest monitoring system developed	Forest monitoring system in operation	

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	Rural people, specifically women and youth, are trained and participate in land use planning processes at district level	Interim evaluation and final evaluation report	0	3 women per district 3 youth per district	30 youth 30 women	Rural women and youth groups in targeted districts are motivated to invest time and efforts in the activities.
Outcome 2: Natural forests protected, connected, more resilient to climate impacts & risks						
Output 2.1. Protected Area management effectiveness improved re climate risks and adaptation	Climate adaptation reflected in the new general park management plans	Progress report, mid-term evaluation and final evaluation	0 plans currently contain climate adaptation plans	3 General management plans updated and approved by key stakeholders	Updated general management plans guiding RDB management decisions	General Park management plans are in the process of revision during the project intervention

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	RDB has adopted a landscape approach to conservation. New iterations of park management plans reference integrated CND land use plan A framework for conservation planning at landscape level is operational, and RDB planning and monitoring dept at each park is actively engaging stakeholders	Progress report, mid-term and final evaluations	0 park management plans reference CND land use plan	1 new park management plan referencing CND land use plan RDB proactively engaging stakeholders in the land use planning process of the CND	3 new park management plans referencing CND land use plan	Limited engagement of RDB beyond parks' boundaries could delay the operationalization of the planning framework

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	Number of RDB, RFA, districts staff and local communities trained in fire forecasting, detection and management	Post-training assessments of change in capacity of trained park staff in designated positions on 1) fire detection, early warning methods and fire danger rating system and 2) management of climate information and its relevance for park management Post-training assessments of change in capacity of trained individuals in designated positions	0 staff reporting improved awareness and capacity 0 staff reporting improved capacity for fire management	15 Staff reporting improved awareness and capacity (8 Female, 7 Male) 15 staff reporting improved capacity for fire management	15 Staff reporting improved awareness and capacity (8 Female, 7 Male) 30 staff reporting improved capacity for fire management	Capacity building programs can overcome the effects of high staff turn-over and frequent re-arrangement of the institutions of natural resources management (transfer of departments, merging and creation of new Ministries) All stakeholders embrace the concept of information sharing and learning willingly and enthusiastically.
	Financial instruments to generate revenues post GCF identified and piloted	Interim and final evaluation reports Technical reports	0 financial instruments	1 financial instrument identified	3 financial instruments identified and 1 instrument piloted	GoR commitment to enactment of policies conducive to attracting investment in emerging financing solutions

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Output 2.2. Natural forest cover restored, biodiversity connections established	Area (# ha) of degraded natural forest restored in national parks	GIS mapping of cleared areas using field GPS data	0 ha restored	4,500 ha initially cleared	4,500 ha cleared at least four times, ensuring natural regeneration can take place in Nyungwe NP	Major fire outbreaks during project implementation may prevent reaching the targets
			0 ha restored	500 ha under restoration in GMNP	500 ha restored in GMNP	The park is well managed and there are no major illegal activities
	Area (#ha) of riparian lands restored with indigenous species in the CND	GIS mapping of restored areas using field GPS data	0 ha restored	500 ha	1500 ha restored	District officials are committed to protect and create 10 m buffer around streams/rivers as per land use policy
	Area of remnant natural forest loss avoided	Interim and final evaluation	TBA	Draft strategy and action for the protection and conservation of remnant forests	Strategy and action plan developed and implemented for the conservation of remnant natural forests	RDB, REMA and District officials are committed to protect remnant natural forests

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	Area (ha) of pastoral lands brought under climate resilient silvopastoral packages in GMNP Average tree density/ha in restored silvopastoral lands	GIS mapping Bi-annual technical report Interim Evaluation and Final Evaluation Report Annual forest inventory	TBA TBA TBA	500 ha Density: 50 tree/ha	1000 ha Density: 50 tree/ha	Farmers are willing to integrate trees on their pasture lands
Outcome 3: Vulnerable rural livelihoods more climate-resilient, diverse, economically sustainable & nature-positive						
Output 3.1 Farming methods enhance productivity, reduce erosion and flooding risks, contribute to ecosystem services, and support connectivity.	Area (# ha) of forest restored on slopes >55%	Interim Evaluation and Final Evaluation Report Annual forest cover report ¹⁴⁴ Catalogue of geo-referenced area under restoration	0 hectares	2,500 ha under restoration on slopes >55% planted with seedlings.	2,500 public and private hectares restored and maintained. density: 1,200 trees/ha	Farmers, women, and youth groups in targeted districts/sectors are motivated to invest time and efforts in the activities.

¹⁴⁴ The Annual Forest Report is prepared by Rwanda Forest Authority. The Monitoring system for the preparation of these reports has been developed.

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	% of tree species diversity in restored areas on slopes >50%	Interim Evaluation and Final Evaluation Report Annual forest inventory	0	30% indigenous species and 70% exotic species on private land 70% indigenous species and 30% exotic species on public land	30% indigenous species and 70% exotic species on private land 70% indigenous species and 30% exotic species on public land	
	Number of target beneficiaries (households) in target sites who have adopted agroforestry measures (and report improvements in soil and water management)	Interim Evaluation and Final Evaluation Report Household surveys	0	5,405 households (of which 50% are women headed households) adopted agroforestry measures	5,405 households beneficiaries (of which 50% are female).	
	Average tree density/ha in restored agroforestry lands	Interim Evaluation and Final Evaluation Report Household surveys Catalogue of geo-referenced area under restoration Annual forest inventory	0 TBA	 100 trees/ha	3,346 ha of agroforestry established. 100 trees/ha	

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Output 3.2. Rural livelihoods generate alternative incomes & reduce pressure on forests	Number of value chain products that are viable and marketed (Honey, Avocado, macadamia, horticulture)	Interim evaluation and final evaluation report Annual sales reports	0	3 products developed	5 products developed	Stakeholders in the value chains are collaborative
	Number of businesses/cooperatives with viable business plans established and operational for climate resilient value chains	Bi-annual technical report Interim Evaluation and Final Evaluation Report	37 beekeeping cooperatives with 1,965 members (32% women, 16% youth) 0 seed enterprise, nurseries farmer's cooperatives 0 cooperative of tourism guides	15 new cooperatives (40% women, 50% youth) 5 seed and nursery enterprises owned by youth and women. 2 cooperatives	20 new cooperatives (40% women, 50% youth) 10 seed and nursery enterprises owned by youth and women. 3 cooperatives	Willingness of farmers' representatives of and Rwanda Cooperative Agency (RCA) to participate in the project
	Increase in household income within community associations/cooperatives engaged in the project (disaggregated by gender and social category)	Household surveys	TBA in year 1	20% additional income in participating households in new income generating activities. Targets for health benefits, and time and firewood savings TBD	An additional 10%	Successful rolled out of capacity building and linkage to outside (input-output) market

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	# of target beneficiary households adopting cookstoves	Interim Evaluation and Final Evaluation Report Household surveys	0 target households using Tier 2 cookstoves	4000 target households of which 50% are women-headed households	8,500 target households using Tier 2 cookstoves	
	Number of employments generated from the selected value chains	Interim and final evaluations Annual reports on employees hired per value chain	TBA in year 1	# women:600 # young:400	# women:900 # young:600	No outbreak of a new pandemic Farmers, women and youth groups in targeted districts/sectors are motivated to invest time and efforts in the activities.
Output 3.3. Financial services & private sector investment engage	Agriculture Credit assessment tool upgraded to include forestry	Loan approval reports generated by the tool Interim and final evaluation	0	3 MFIs and PFIs are using the tool to provide loans to smallholder farmers 70% of the smallholder farmers and gradually grow	5 MFIs and PFIs are using the tool to provide loans to smallholder farmers	MFIs and PFIs are willing to participate in this project
	Number of financial products refined, contextualized , and rolled out to improve access to finance for agriproducts for 1) vulnerable groups/ smallholders and 2) horticulture, bee products,	Bi-annual technical report Interim Evaluation and Final Evaluation Report	TBA	4 financial products refined and rolled out . At least 1 for each value chain (4 total)	6 financial products already deployed by PFIs are refined, contextualized and at least 5 rolled out by PFIs for financing	MFIs are willing to contribute in terms of staff and resources

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	energy efficient cookstoves and ecotourism value chains				selected value chains. 10 approved loans' applications	
	Number of people including women and youth with organizational, technical and financial improved capacity	Interim and Final Evaluations	0	7,800 (40% youth, 50% women)	12,445 (40% youth, 50% women)	Farmers, women and youth groups in targeted districts/sectors are motivated to invest time and efforts in the activities.
	Number of men, women and youth who have access to thriving financial services	Interim and Final Evaluations	0	3,000 (40% youth, 50% women)	8,500 (40% youth, 50% women)	Men, women and youth are open to access financial services

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 1.1.1 Synthesize & disseminate information on value of natural forests & ecosystem services	Ensure that the value of natural forests and ecosystem services for supporting climate change resilient livelihoods is fully appreciated by all sectors involved in land use decision making	1.1.1.1: Map the sectors involved in land-use planning in the CND and review how forests, ecosystem services and climate resilience is incorporated into each sector's planning process. 1.1.1.2: Conduct comprehensive literature review & stakeholder consultation to collect & synthesize information on climate risks for various	A map of the sectors involved in the land-use planning in the CND Information package on climate risks, value of forests for increasing resilience of local communities, adaptation options developed and disseminated to ministries, local government and local NGOs

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		sectors (e.g. agriculture, forestry), the value of forests for increasing resilience of local communities, and highlight adaptation solutions with cross-sectoral implications 1.1.1.3: Host workshops & presentations with key ministries, government organizations, NGOs and community organizations to disseminate high-level knowledge on value of forests for increasing resilience of local communities 1.1.1.4: Develop guidelines for integrating climate risk into land use planning and cross sectoral planning 1.1.1.5: Develop outreach materials on climate risk in the CND and the value of forest ecosystems for increasing resilience 1.1.1.6: Implement outreach program tailored to different stakeholders (local government, civil society, communities) to enhance capacities for land-use planning, funding mobilization, and delivery of climate adaptation actions 1.1.1.7 Conduct climate literacy courses for local government and civil society organizations, aimed at increasing women and youth participation 1.1.1.8 Provide financial and logistical support to trained organizations in grassroots mobilization to increase	Guidelines for integrating climate risk into land use planning and cross sectoral plan developed 104 episodes of radio drama to educate communities on climate change and gender issues, climate risks, value of forests and forests ecosystems, climate adaptation options developed Broadcast of 104 radio program episodes on a popular national radio station

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		women and youth participation in climate adaptation planning 1.1.1.9 Introduce social safeguards at a high level at each meeting, including the GRM, FPIC, and Access Restrictions Mitigations as safeguards to be in place for work with local communities	
Activity 1.1.2 Develop climate-resilient landscape land-use plan	Ensure that climate change adaptation requirements are fully integrated into planning processes at local, district and national scales.	1.1.2.1: Engage district officials, JADF, and community members in the CND (especially women) in reviewing and interpreting the NLUDMP, to ensure the plan accounts for current and future climate risks, while building local support for climate sensitive planning. 1.1.2.2: Conduct participatory land-use planning process in communities from village to district level to support integrated climate resilient land use planning 1.1.2.3: Develop Integrated Land-use Plan that supports Resilient Livelihoods and Ecosystems in the CND, ensuring alignment with National Land Use and Development Master Plan and ensuring the CND plan guides the district plan 1.1.2.4: Develop and rollout a series of trainings (virtual sessions and online modules) on gender sensitivity and mainstreaming women and youth into planning	A summary of the current NLUDMP and its recommendations relating to forestry, agriculture and climate adaptation. Reviewed/updated district level planning schemes to ensure they account for current and future climate risks (e.g. DFMPs, District Development Strategies) A community natural resources mapping manual to outline main steps in community land-use planning process A comprehensive review of national and district development plans/strategies, guidelines and projects to understand planned future land-use & climate adaptation projects in the CND and identify synergies and conflicts between land-use & climate adaptation plans from different sectors Integrated Land Use Plan developed

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		1.1.2.5: Assess specific climate impacts on historically marginalized and Category c,d,e populations through a participatory NR process to ensure the component projects address their needs for adaptation	
Activity 1.2.1: Create interagency taskforce institutionalizing integrated landscape planning & policy	Collaborative integrated landscape planning supported, which secures climate resilience undertaken with appropriate social safeguards. This activity aims to strengthen collaborative efforts, in particular between institutions (from local to national level) in charge of natural resources management to encourage synergies and avoid overlapping mandates and redundancy in different climate resilience interventions.	1.2.1.1: Review & strengthen operationalization of the current cross-sectoral task force 1.2.1.2: Hold quarterly sectoral planning meetings with both national and district administrations in CND 1.2.1.3: Facilitate discussions and provide technical support (e.g., policy briefs) in decision-making for cross-sectoral actions around climate adaptation and forest resilience in the CND 1.2.1.4: Continue to introduce social safeguards at each meeting including the GRM, FPIC, and Access Restriction Mitigations as safeguards to be in place for work with local communities. For those at the district or community level, provide options for feedback on the process and best ways of communication with local partners	A cross-sectoral planning task force operational as recognized by a ministerial order
Activity 1.2.2: Build capacity for spatial planning in national agencies re climate change	Capacity for integrated land use planning which supports climate change resilience and delivery of ecosystem services is improved. This activity aims to build the technical capacity of staff in the Ministry	1.2.2.1: Recruit spatial planning expert to support spatial planning unit	Spatial Planning Unit Operational Capacity needs assessment completed

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
	of Environment and University of Rwanda to generate and use the data on forest cover and ecosystem service values necessary for mainstreaming climate adaptation and forest resilience into land-use planning..	1.2.2.2: Assess and identify the institutional home and operationalize the spatial planning unit 1.2.2.3: Conduct a capacity needs assessment and identify the appropriate tools for spatial planning to fit the Rwandan context and conduct familiarization 1.2.2.4: Led by the spatial planning unit, develop capacity within the University of Rwanda and the MoE to use remote sensing, climate change modeling and spatial planning tools (e.g., Marxan, InVEST, RIOS, SWAT) to incorporate climate risks into land use planning processes 1.2.2.5: Deliver training workshops on utilization of earth system models for land use planning purposes, utilization of remote sensing resources (lightning, satellite radiances) for hazards detection and climate monitoring and on regional climate monitoring and applications	Capacity for forest mapping, climate change modeling and spatial analysis built within UoR and MoE
Activity 1.2.3: Develop an effective forest monitoring system to underpin forest management decisions	An effective forest, climate change and land use monitoring system is established to support climate resilient forest management decisions.	1.2.3.1: Recruit Forest ecologist or remote sensing expert to support the design and implementation of indigenous forest monitoring system 1.2.3.2: Review existing forest mapping data and monitoring software, and assess utility for generating quantitative	A report on the status of existing forest mapping data and monitoring software A design of the structure and operation of forest monitoring Two weather stations installed and operational Forest monitoring system operationalized

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		assessments of indigenous forest cover from local to national scales 1.2.3.3: In collaboration with REMA, MINAGRI, RAB, RDB, RISA, MoE and affiliated agencies and University of Rwanda, design structure and operation of forest monitoring system, including required inputs (data, computing, personnel), desired outputs, and operational structure (e.g. location, reporting structure, funding) 1.2.3.4: Purchase and install 2 new weather automatic stations in the CND 1.2.3.5: Establish & operationalize forest monitoring system 1.2.3.6: Generate updated maps & statistics for indigenous forest cover, forest type etc. using forest monitoring system 1.2.3.7: Develop capacity within hosting institution for continued operation of forest monitoring system	Updated maps and statistics for indigenous/natural forest cover and type generated Trainings delivered to develop the capacity of staff within hosting institution in forest monitoring
Activity 2.1.1: Facilitate revision of PA management plans to address climate risks	Park planning process expanded to ensure full incorporation of buffer zone and landscape linkage requirements, and ability to engage with the larger CND and district landscape management planning processes.	2.1.1.1 Review and update existing national park management plans to ensure climate and related landscape changes risks, impacts and required management responses are integrated 2.1.1.2 Provide technical and financial support to planning, research and monitoring as well as community based conservation units within RDB in	General management plans for NNP, GMNP and VNP updated

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		managing the parks as part of larger CND landscapes 2.1.1.3 Train PA staff to integrate gender and social inclusion into programming 2.1.1.4 Train PA staff on integrating needs of women, youth, historically marginalized and Ubudehe Categories c,d,e populations into climate adaptation risks and responses 2.1.1.5 Train PA staff on SEAH, GRM, FPIC, and Access Restriction Mitigation Processes to ensure NP climate change responses are undertaken in a way which supports social inclusion and equity	
Activity 2.1.2 Establish long-term plans for CND financial sustainability post-GCF	Identify financial instruments that could be employed for revenue generation post GCF as part of a broad and comprehensive RDB business planning effort for the sustainable management of the CND landscape	2.1.2.1 RFA to Recruit a Technical Advisor to guide RDB, MoE, REMA, FONERWA and MINECOFIN on sustainable conservation finance.	Technical Advisor recruited CND long term financial sustainability developed
Activity 2.1.3 New fire management curriculum developed and operationalized in PAs	Capacity increased for improved fire management in core PAs and adjacent landscapes. A collaborative effort in fire management planning and implementation in CND districts will be carried out and facilitated.	2.1.3.1 Develop a curriculum tailored to needs and capacities of different stakeholders 2.1.3.2 Build the capacity of RDB to manage fire in National Parks 2.1.3.3 Build the capacity of RFA, local authorities and communities to collaboratively manage fire in forests outside PAs	A fire training program that addresses local needs and skills gaps developed Train the trainer process completed for 25 RDB staff allowing them to instruct future classes without assistance Fire detection, early warning, and fire danger ranger system is established and operationalized

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		2.1.3.4 Implement a “fire wise” outreach and awareness program for communities and local government (district, sector, cell), as well as Rwanda Forest Authority, surrounding natural forests to reduce the incidence of human-caused fires	200 community facilitators (youth) trained to raise awareness on fires in communities surrounding natural forests and forests plantations 10,000 community members reached through social marketing events
Activity 2.2.1 Secure key remaining natural areas outside PAs	Remaining priority natural forests and other ecosystems required for climate change resilience secured through appropriate land use planning processes	2.2.1.1 Raise awareness on remaining protected natural forests in CNDL to secure their protection 2.2.1.2 In collaboration with RFA, REMA, districts identify and implement actions that support conservation and management of remnant protected natural forests in CNDL	Action plan for the conservation of remnant natural forests developed and implemented
Activity 2.2.2 Restore natural forest cover in & outside PAs including riparian linkages	Support natural forest restoration processes in PAs through improved management of alien and invasive species; and actively restore nature forest in priority areas next to core PAs and protected forests to expand the natural forest estate	2.2.2.1 Review mapping of degraded natural forest areas in core NPs, stepping stones and unprotected riparian lands using updated imagery and ground truthing 2.2.2.2 Identify parcels for restoration in core PAs, stepping stones and riparian lands using desktop and field-based assessment 2.2.2.3 Establish indigenous tree seed nurseries in the CND to serve core PAs, stepping stones, riparian land restoration and promotion of indigenous trees on farms and in protective forests 2.2.2.4 Recruit, train in forest restoration methods and equip community workers to	Indigenous tree seed nurseries established and operational 4,500 ha of indigenous forest in NNP restored 500 ha restored in Gishwati Mukura NP 1,500 ha of riparian lands restored

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		be involved in core PAs restoration and riparian linkages 2.2.2.5 Restore 4500 ha of indigenous forest in NNP 2.2.2.6 Restore 500 ha Gishwati Mukura National Park 2.2.2.7 Restore 1,500 ha of riparian lands 2.2.2.8 Using permanent plots sampling, collect field monitoring data on tree species recruitment and growth for estimation of biomass, species richness etc. in restored parcels in Nyungwe and Gishwati-Mukura National Parks in restored parcels in Nyungwe and Gishwati-Mukura National Parks as well as on riparian lands to assess success of initiatives, then replicate or adapt as needed 2.2.2.9 Perform Environmental and Social Screening on all positions being created by the project 2.2.2.10 Establish a grievance process for laborers 2.2.2.11 Update emergency and preparedness plan including risk mitigation guidance to local conditions at restoration sites	
		2.2.2.12 Train all workers on their rights and how to access the GRM	

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		2.2.2.13 Provide code of conduct and emergency preparedness and safety training for all laborers 2.2.2.14 Hire and train labor and safety leads at each site to provide guidance to staff, be available for grievance issues, and monitor health and safety conditions for workers 2.2.2.15 Develop and implement strategy (including social marketing) to ensure that women and youth participate and benefit from forest restoration projects	
Activity 2.2.3 Promote silvopastoralism with indigenous trees in Gishwati Pasture areas	Existing pasture lands will be characterized and livestock farmers clustered according to the size of their grazing lands. Tree species that exist on their grazing land will be identified and grouped according to the level of their resilience to climate change. A participatory approach will be used to assess livestock owners' preferences for indigenous species on their farms. Sites will be identified for indigenous tree nursery establishment and nurseries will be established in project sites and managed by the livestock communities.	2.2.3.1 Assess the status of pasture lands in Gishwati area and identify key areas for restoration and the potential to introduce indigenous species 2.2.3.2 In collaboration with landowners, identify suitable indigenous species for fodder trees, shrubs, grasses, and herbaceous legumes that have potential to improve pasturelands and increase their climate adaptive capacity 2.2.3.3 Produce and disseminate fodder trees, shrubs, grasses, and herbaceous legumes to project beneficiaries 2.2.3.4 Train beneficiaries on improved livestock and pasture management	1,000 ha of pasture land in Gishwati grazing area restored with indigenous tree species

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 3.1.1 Restore high slope areas (>55%) as protective forests	Promotion of protective forest and agroforestry practices to improve landscape connectivity and ability to deliver ecosystem services	3.1.1.1 Introduce and raise awareness of indigenous species to target stakeholders in CND 3.1.1.2 In consultation with RFA, National Land Authority, districts and communities, determine fragile areas (steep slopes > 55%) to be allocated for protective forests and their ownership 3.1.1.3 Assess the current status of the indigenous tree species in selected areas for protective forests and select indigenous tree species appropriate to CNDL 3.1.1.4 Develop restoration plan for protective forests 3.1.1.5 Build capacity of local stakeholders (men and women) on PFMU approach and methods 3.1.1.6 Design and approve SFMPs of private FMUs 3.1.1.7 Ensure consent of smallholders prior to planting trees (SS) 3.1.1.8 In collaboration with smallholders reforest/restore 2,500 ha of public or private land with slopes >55% and ensure sustainable management under private FMUs according to approved SFMPs 3.1.1.9 Support monitoring and evaluation of restored private FMUs	2,500 ha of protective forests on high slope areas restored

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		3.1.1.10 Assess impacts of exotics on neighboring lands and mitigate their negative impacts (SS) 3.1.1.11 Develop and implement strategy (including social marketing) to ensure that women and youth participate and benefit from forest restoration projects (G) 3.1.1.12 Assess the benefits and costs of each proposed tree and plant species and how those benefits affect different population segments (SS)	

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 3.1.2 Develop on-farm agroforestry for high caloric and indigenous tree species	Implement agroforestry/sustainable land management (SLM) over 3,346 ha of smallholder farmland	3.1.2.1 Identify sub-areas of intervention for agroforestry dissemination in the CND 3.1.2.2 Introduce and raise awareness of agroforestry in target communities 3.1.2.3 Develop the capacity of extension agents at district/sector level and NGOs to support adoption of agroforestry technologies 3.1.2.4 Establish agroforestry/fruit tree nurseries to facilitate access to quality planting material 3.1.2.5 Promote sustainable land management practices by stabilizing existing terraces, and plant high calorific agroforestry species to provide a sustainable source of fuelwood for energy efficient cookstoves to protect sloping land against severe soil erosion 3.1.2.6 Train RFA, extension agents, Project participants and community members on specific techniques for identification of and management of invasive pests and pathogens (SS)	SLM/agroforestry established on 3,346 ha of smallholder land

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 3.2.1 Develop forestry and agroforestry-related value chains for market access	Vulnerable groups of youth and women identified and their capacity for production (quality and quantity) enhanced in sync with particular niche markets demands	3.2.1.1 Identification and mapping of vulnerable groups of youth and women within the CND 3.2.1.2 Capacity building for farmers' cooperatives/unions on good agricultural practices (GAP) for market standards in the selected agroforestry-related tree fruits and horticulture 3.2.1.3 Development of Global GAP standards modules and market access linkage for the selected value chains 3.2.1.4 CND Honey value chain capacity production, market, and financial development 3.2.1.5. Tourism value chain development	Vulnerable groups of youth and women within the CND identified and grouped into cooperatives 5 Value chains identified and beneficiaries selected GAP training for 900 people from targeted cooperatives and unions completed Train of trainers completed for 30 cooperative members New tourism trails and base camps outside of Gishwati-Mukura NP developed and operational Community freelance tour guides trained and operational
Activity 3.2.2 Facilitate and scale up climate-resilient value chain products	This activity aims to build the capacity of smallholder farmers in producing, processing and marketing quality products that meet national and international standards to be competitive on the market	3.2.2.1 Value chain contextualization in focus areas in the CND 3.2.2.2 Facilitating market accessibility	25 youth and women cooperatives/enterprises with the capacity to deliver quality products and services
Activity 3.2.3 Facilitate access to markets for vulnerable farmers	This activity will focus on enhancing productivity (quantity, quality, and sustainability at the farm level) and exploring the market linkages model between farmers, suppliers, and buyers.	3.2.3.1 Development of farmer-inclusive business models	900 cooperative members are linked to the market from farming inputs suppliers and production aggregators through inclusive farming-as-a-business.

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 3.2.4 Scale up production, use and sales of fuel-efficient cook-stoves	Evaluate and promote financing models that sustainably foster energy efficient cook-stove access to historically marginalized communities and other vulnerable groups of the CND	3.2.4.1. Identification and production of Energy Efficient Cooking-Stoves 3.2.4.2. Energy efficient cook-stoves distributed to CND community members and categorization based on UBUDEHE Categories	8500 households have access to EES
Activity 3.3.1 Facilitate access to finance & private sector investments	Build common understanding between value chain actors and private financial institutions towards selected value chains	3.3.1.1 Tailoring credit assessment tool for selected value chains, while integrating climate impact data 3.3.1.2 Facilitate access to finance for selected value chains	Agriculture Credit assessment tool upgraded and adopted by PFIs At least 15 projects of the 5 selected value chains funded by PFIs
Activity 3.3.2 Set up & support savings & loan groups, enhance asset-building	Strengthen the capacity of local smallholders' farmers in financial literacy and access to finance	3.3.2.1 Establish/form saving groups for access to finance	Cooperative members are linked to PFIs and at least 70% have accessed to financial service and loan
Activity 3.3.3 Build capacity of financial institutions to serve targeted value chains and communities	Undertake a climate screening of financial service providers using the green index tool and support PFIs to access and adopt a digitalized forestry and agri-loan assessment tool	3.3.3.1 Capacity-building of PFIs	PFIs have developed an environmental policy and strategy for their organization including staff awareness training and integration of climate and livelihood impact measurement systems
Activity 3.3.4 Facilitate learning & knowledge sharing	Produce and share lessons learned during the implementation of this project	3.3.4.1 Programme-embedded learning and reflection	ESMP

E.7. Monitoring, reporting and evaluation arrangements

The Project's logical framework outlines the expected results, indicators, means of verification, baseline values and target values at mid-term and end-term. The Project Coordinator will be responsible for the overall monitoring of Project progress and reporting to the PSC. A national M&E Specialist hosted under the PMU, supported by EE-based Specialists, will be employed to develop practical guidelines for — and operationalize — a performance monitoring framework to track the Project's progress towards achieving its targets. The national M&E Specialist will be responsible for: i) establishing a performance monitoring framework to define biannual targets; ii) measuring the indicators at least 1–2 times per year to evaluate the progress of the Project; iii) reporting to the Project Coordinator and PSC; and iv) participating in the production of reports. RFA M&E Specialist will support the National M&E Specialist by: i) following guidelines developed by the national M&E specialist to conduct performance monitoring; ii) measuring the program indicators at least 1–2 times per year; iii) ensuring all activities comply with the Project's ESMF process and procedures; iv) supporting environmental screening and helping to follow the Environmental Management Plans; and v) participating in production of reports. The M&E Specialists will be supported by a Gender Specialist to oversee and monitor the application of gender disaggregated indicators. The M&E Specialist and Gender Specialist will perform regular visits (at least once/year), randomly selecting villages to monitor implementation. They will use these visits to cross-check the information provided by EE and beneficiaries in monitoring reports. During each visit, the team will collect qualitative data on implementation outcomes and difficulties through focus group discussions with beneficiaries and interviews with key stakeholders. The national M&E Specialist, supported by the EE M&E Specialist will then compile quarterly Project reports outlining progress in each site, allowing for feedback and support from the PMU and PSC in case of delays or challenges. MoE, as the AE, will perform supervision missions from year two onwards, selecting sites to visit each year and assessing the status of implementation. In addition, the MoE procurement office — which oversees the destination and use of funds disbursed to the EE — will monitor the Project's compliance with policies and procedures, as well as alignment between implementation and objectives. The annual performance reports will include a narrative report (with supporting data) on implementation progress based on the logical framework in the Project, including a report on ESMS as well as gender. The report will be aligned with the modalities set out in the GCF results management framework and its performance measurement frameworks.

As per the AMA, an independent consultant will be contracted to conduct an Interim Evaluation and Final Evaluation. Both will comply with GCF evaluation policy and international standards and make use of qualitative and quantitative data analyses, including interviews with relevant stakeholders, data provided by the Project, EE M&E Specialists and Gender Specialist, and participative workshops and focus groups with beneficiaries. The MTE will analyze whether the Project is on track, any problems, and challenges, and required corrective actions. The PSC and PMU will develop a management response to the recommendations along with an implementation plan, and the PMU will monitor implementation of the recommendations. The Final Evaluation will be an independent assessment of Project performance against its contributions to a paradigm shift and enabling environment.

• (f) RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures

Selected Risk Factor 1 - *Lack of coordination, institutional resistance, and regulatory barriers to mainstreaming climate change into land use planning*

Category	Probability	Impact
<u>Macroeconomic/Political</u>	<u>Low</u>	<u>High</u>

Description

Despite its progressive policies, Rwanda has historically had limited capacity for mainstreaming climate change into government/institutional practices and given the pervasive ways in which the impacts of climate change impact across numerous sectors, ministries and government agencies will need to work across any perceived or real institutional barriers to align and mainstream climate change in land use planning. For example, the Rwanda Development Board has historically not worked beyond Park boundaries, and this will be integral for institutionalizing the climate change adaptation processes across the Congo Nile Divide as a large interconnected landscape. Additionally, insufficient coordination, collaboration and information-sharing between relevant government departments, and research institutions may hinder or limit participation of relevant institutions and agencies, reducing the timely delivery and effectiveness of the Project, especially the aspects related to mainstreaming climate change in land use planning and other components of the Project as well. Furthermore, the perception of an uneven playing field, with larger or more well-connected institutions and companies able to evade the payment of environmental fees and fines under existing legislation, may undermine the uptake of the other options for financial sustainability of improved PA and forest ecosystem management.

Mitigation Measure(s)

The Project is part of a larger suite of efforts supported by the multi-agency Project Steering Committee (PSC) to create a spirit of cooperation among the various agencies and institutions charged with addressing climate-related issues. The PSC demonstrates Rwanda's commitment to fulfilling mainstreaming of climate adaptation into regular government processes and procedures. As such numerous government agencies have already had the opportunity to work together on a variety of climate-related issues, align institutional policies and reduce any potential regulatory barriers to ensure successful implementation of projects related to climate change. The PSC will directly oversee the Project Management Unit, and with this larger guidance structure, the EE will communicate to the PMU and the PSC on a regular basis to prompt collaboration and address any unforeseen institutional or regulatory barriers encountered during the life of the Project for immediate attention and remediation. PSC will address the risk of evading the payment of environmental fees by facilitating buy-in from stakeholders such as RDB and MINECOFIN into compliance with environmental laws through greater emphasis on the links to climate resilience and NDC objectives. These measures will reduce the impact to LOW.

Selected Risk Factor 2 - *Extreme weather/climatic events hinder the ability to deliver Project objectives*

Category	Probability	Impact
<u>Other</u>	<u>Low</u>	<u>Medium</u>

Description

The activities that will deliver overall Project success involve dependencies on weather conditions for successful implementation of the projects that rely on accessing steep slopes, planting trees, and establishing beekeeping operations. Extreme events such as heavy rainfall or prolonged drought could severely limit the success of these endeavors making sites inaccessible, creating unfavorable conditions for seedling survival, and decreasing the potential success of new beekeeping operations. This region, given its topography and the expected climatic variability, is likely to experience these types of disruptions. Severe disruptions may impede implementation, including potential implementation delays and fall under GCF's policy on restructuring and cancellation. That said, it is unlikely these extreme conditions will present themselves in the next 5 years for the duration of the project.

Mitigation Measure(s)

As weather is unpredictable, any mitigation measures will not change the probability nor address the impact in the short term. As the entire project is envisioned to improve the resiliency of this landscape to climate variability and promote adaptation, the specific activities will be able to mitigate, and in some cases prevent, longer-term impacts. As our implementation arrangements involve bolstering the climate adaptation capacity of the government and the communities in the CND, the entire project itself will serve as mitigation to this unlikely outcome.

Selected Risk Factor 3 - Project staffing and capacity

Category	Probability	Impact
<u>Operational</u>	<u>Low</u>	<u>Medium</u>

Description

Implementation of the Project hinges on staffing of the Project Management Unit, hiring additional staff within the EE and recruiting thousands of additional laborers in the form of community engagement specialists, contractors for building terraces, individuals to remove ferns, contracts to construct and deliver stoves, consultants to contribute expertise on behavior change, invasive species/pest management, and gender sensitization. Many factors could intervene to slow or thwart hiring progress including the high turnover of staff in the implementing institutions as frequent changes in government bodies could result in a disruption and/or delays in implementation.

Mitigation Measure(s)

Rwanda has a successful track record of implementing large scale projects through government ministries. Competent staff will be available for project implementation, and their abilities will be enhanced through the commitment to training individuals involved in the program. This risk, if realized, may impede implementation including potential implementation delays and fall under GCF's policy on restructuring and cancellation. Therefore, the Project will invest heavily in strengthening institutional capacity to promote climate resilience and apply actions that directly increase the resilience of ecosystems and community livelihoods. The Project will also document decisions, best practices and lessons learned to support the institutional memory to sustain Project activities in case of high staff turnover. Labor and safety as well as community security plans to be established as part of the ESAP (Annex 6) will make the Project a desirable place to work. Mitigation measures reduce impact to LOW.

Selected Risk Factor 4 - Inadequate incorporation of gender, and environmental and social safeguard (ESS) considerations into the implementation of the proposed Project activities as well as its management structures and systems		
Category	Probability	Impact
<u>Technical and Operational</u>	<u>Low</u>	<u>Medium</u>
Description		
Should ESS be inadequately considered in the Project or in the management and governance structures, there could be social and environmental implications. To start there may be a disproportionate distribution of benefits to certain segments of society. Although the Project's primary beneficiaries include women and other most vulnerable people, there is a risk that the Project implementation will experience preferential treatment at sites that favor certain groups over others. For example, despite tremendous policy and legal efforts, a patriarchal culture and persistent disparities in education, employment, land ownership and other factors characterize gender relations in Rwanda and could cloud implementation efforts. Likewise, some vulnerable groups may be unable to benefit from the efforts, such as landless people unable to benefit from trees or having a safe place for a new cookstove. One serious risk afflicting major projects such as that proposed, if not properly addressed, includes sexual exploitation, sexual abuse, and sexual harassment (SEAH) which may be prevalent if preferential treatment occurs (e.g., interfamily SEAH in retaliation for benefits received by women rather than men) or there may be power dynamics with hired staff who work with communities and withhold Project benefits within a SEAH conflicts (e.g. requesting sex for an opportunity to participate in the project, harassment if potential beneficiary refuses). On the environmental side, several environmental risks detailed in Section F and Annex 6 could undermine the success of the Project including topsoil loss and possible collapse of the terraces, unsustainable practices for sourcing clay for the cookstoves, and inadvertent introduction of invasive species, pests or pathogens from the nurseries to the field sites.		
Mitigation Measure(s)		
The Project anticipates these challenges and has envisioned multiple participatory aspects to the Project that are guided by the gender action plan and the ESAP. To mitigate social risks, local community engagement will be facilitated by the EE in cooperation with local community organizations through the integrated landscape delivery teams (ILDT) and the Joint Action Development Forums (DJAF). Safeguards in place to support Labor and Community Health & Safety will require training/sensitization to SEAH and the repercussions for those who engage in such activities (i.e., termination of employment). In the hiring process, staff will be screened for prior infractions. Hired staff will be trained in survivor-centered and gender-responsive approaches to support any victims of SEAH in association with the Project activities. To minimize environmental risk, the PMU and the EE will adhere to the environmental action plan and oversee the hiring of contractors to perform the activities. Multiple consultants or specialists, if not part of subcontracts, will be recruited to train project participants on mitigating measures to reduce these risks. If the proposed ESAP is followed properly the impacts will be LOW.		
Selected Risk Factor 5 - Lack of adoption among target populations for livelihoods activities		
Category	Probability	Impact
<u>Technical and Operational</u>	<u>Medium</u>	<u>High</u>

Description		
The success of the Project depends upon the integration of activities such as beekeeping, improved cookstoves, planting of fruit trees and terrace construction on private lands by the project beneficiaries. If planned and executed incorrectly a myriad of issues could prevent beneficiaries from adopting the activities beyond the life of the Project. This includes but is not limited to overlapping or misalignment between the proposed Project activities and those being implemented by other government departments or non-government organizations in the Project area.		
Mitigation Measure(s)		
Numerous international development projects have experienced mixed results upon completion regarding the uptake of promoted activities into daily lives. This Project has a strong behavior change component based on current best practices in identifying the barriers to behavior change, removing those barriers, and changing norms thus ensuring, through adequate participation and training, that the activities will be adopted and maintained over time. The work plan involves ample collaboration with women and youth groups through the ILDTs to build long-term community ownership for the Project. Further, if there are activities being promoted by the government or other NGOs that either overlap with the proposed activities or contravene with the Project activities, adoption by intended beneficiaries may be disrupted. As such the NSC will serve as the overarching body to ensure such projects within similar geographic regions remain aligned. These mitigation measures reduce the impact to MEDIUM.		
Selected Risk Factor 6 - Unforeseen disruptions due to another COVID variant		
Category	Probability	Impact
<u>Other</u>	<u>Low</u>	<u>Medium</u>
Description		
As demonstrated during the recent years of the COVID pandemic, variants may cause unforeseen disruptions to travel and Project implementation.		
Mitigation Measure(s)		
The Project will likely start well after COVID has run its course; however, in the event of new variants the PMU and EE will be ready to implement necessary COVID-safety practices such as masking, social distancing and reducing travel to areas with low health care options. Mitigation will reduce impact to LOW.		
Selected Risk Factor 7 - Prohibited practices including money laundering and terrorist financing		
Category	Probability	Impact
<u>Prohibited Activities</u>	<u>Low</u>	<u>Medium</u>
Description		
The Project involves a large number of actors for successful implementation including ministries, government agencies, NGOs, sub-grantees, vendors, service providers, community members and beneficiaries. Specifically, there is an intention to distribute or disperse to beneficiaries both directly and indirectly commodities and other items of value such as the distribution of seedlings for trees that will produce fruits or contribute to enhancing biodiversity of farming systems, access to improved cook stoves as well as the		

possibility for distribution of bees to start honey operations. As with any project infusing large sums of money directly or indirectly with multiple actors, corruption could occur.

Mitigation Measure(s)

Since the 1990s Rwanda has set a social foundation for anti-corruption efforts by establishing policies and procedures to prevent corruption/reduce vulnerability to corruption and sanctioning misbehavior through prosecutions. Specifically, a reduction in administrative corruption is of note for a large project such as the one proposed. The systems in place will reduce the potential for prohibited activities. Additionally, commitment to campaigns for individuals to report corruption has created a societal norm for intolerance. As a result, Rwanda moved from a state of high corruption 20 years ago to a level on par with middle income countries. The International Monetary Fund published an assessment of Anti Money Laundering and Combating Financing of Terrorism in 2015 indicating that while the risk of AML/CFT is low, at that time more could be done to bolster the systems in place to address the issue. At present, the GoR has robust measures to fight money laundering and terrorist financing. This was done first by enacting the law N° 001/FIC/2022 of 16/02/2022 on Regulations relating to anti-money laundering, combating the financing of terrorism. The Rwanda government has also put in place a special agency for this purpose named Financial Intelligence Center (FIC) with a clear mission to promote financial system integrity by conducting effective financial intelligence to counter money laundering, the financing of terrorism and related financial crimes in governmental institutions, projects, and parastatal agencies under the supervision of the Ministry of Finance and economic planning. MoE and the GCF Project under its supervision will also comply with FIC regulations. The MoE also has an Internal Audit function as well as regular audit conducted by the Office of the Auditor General to carry out the efficiency and effectiveness of internal controls.

Regarding higher level security threats, at present, there are no UN Security Council Sanctions against Rwanda, though it is noted that there are sanctions in Democratic Republic of Congo, close to the Congo Nile Divide. The UN Security Council Consolidated Sanctions list indicates approximately 20 individuals with ties to Rwanda, a small percentage of whom may live in the country. This list has been circulated to the AEs and will be cross checked when hiring for the project and once project sites are selected to ensure these individuals will not be involved in any manner with the Project or its activities, either as a counterparty, implementer, or beneficiary.

To report and remedy any risks that may occur, the MoE will maintain a Project level grievance redress mechanism to which anyone can report on the occurrence of prohibited activities. All complaints will be investigated following the process identified in Annex 6. Any complaints believed to contribute to prohibited activities will be referred to the appropriate authorities in Rwanda.

Assuming all staff will be made aware of the prohibited activities, all subcontractors/counterparties will be screened for engagement of activities and made aware of their obligations to alert the PMU and/or submit grievances related to such activities occurring as part of the Project, the overall impact of the money laundering and financing of terrorism risk on the Project will be LOW.

Selected Risk Factor 8 - *Political Risk*

Category	Probability	Impact
Other	Low	Low
Description		

Political risk is in part driven by long-standing frictions with neighboring countries and the potential for domestic instability in these countries to spill over into Rwanda. These geopolitical challenges constrain Rwanda's longer-term development as a services hub. Although the government of Rwanda has been effective in its pursuit of social stability through targeted transfers and support and improving accountability at all levels from Villages and to national levels through village councils, JADF and Umushyikirano (National Dialogue Council - an annual forum that brings together leaders and citizens to reflect on the country's progress and share ideas to solve ongoing challenges, this political risk may impede implementation including potential implementation delays and fall under GCF's policy on restructuring and cancellation.

Mitigation Measure(s)

Much of the work of the project will be taking place within a Ministry and through work at the community level. Oftentimes, regardless of political instabilities, communities with devolved decision-making structures and local capacity will have a greater chance to withstand that instability and continue to operate. For this project, much of the work will be turned over to the local community leaders and the laborers hired locally who will be trained and will then own the capacity to do this work in the long-term, regardless of what national leadership is in place. The structures resulting from this workplace citizens participation in decision making which will constitute a buffer against potential domestic instability. Mitigation will reduce impact to LOW.

Selected Risk Factor 9 - Expansion of tea plantations

Category	Probability	Impact
Macroeconomic/Political	Low	High

Description

Projected climate changes predict significant elevation rises in CND agro ecological zones, including the potential for tea cultivation at much higher elevations than at present. This is unlikely to threaten Rwanda's existing montane forest protected areas, as foreign revenues from tourism in these reserves far surpass those from tea. However, tea remains a significant source of foreign revenue and a government target for expansion. Any increased tea production in the CND would therefore most likely affect lands outside of parks: smallholder, cooperative, and district holdings, many of which might also be targets for reforestation, agroforestry, and biodiversity stepping stone initiatives under this Project.

Mitigation Measure(s)

The integrated land use planning included in this Project design, will bring together tea industry, agriculture, forestry, biodiversity, and tourism interests to identify and reduce such potential conflicts over use of the CND's limited land resource base. Through this integrated land use planning we aim to establish a formal public-private sector forum at the landscape level to help improve communication across sectors, appreciation of environmental issues, and increase synergies between interventions to avoid conflicting targets and indicators between sectors and advocate for mainstreaming biodiversity and climate change in different economic sectors.

Selected Risk Factor 10 - Timely provision of Co-financing

Category	Probability	Impact
Other	Low	Low

Description
The Project co-financiers may delay providing their co-financing commitment which can jeopardize implementation and hence project intended objectives could not be met.
Mitigation Measure(s)
The Accredited Entity has already requested for commitments of co-financing from all the project co-financiers which are already Government of Rwanda institutions, and all the committed co-financing are already part of the Institutions Day to day mandates hence there is anticipated low impact on the project implementation

(e) GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment

As proposed, the environmental and social risk category for the Project is **Category B or ‘moderate risk’ according to GCF classification. The project was** partially screened for potential social and environmental risks, following GCF requirements (Annex 6). All preliminary potential negative impacts for activities and sub-activities were determined to be minimal, reversible and readily mitigated through generally accepted mitigation measures and standard international practices, thus in the low to moderate categories. While the impacts will occur within the Congo Nile Divide Project area boundary and implementation zone, the specific locations and communities affected by Project activities will be determined in Project implementation thus an Environmental and Social Management Framework has been developed for a more thorough screening of all activities by the AE/EE.

Summary of Risks	Mitigation Factors / Measure and Management Strategies

Environmental and social risks: For all Project components, disproportionate benefit distribution to some groups and not others including women, youth, historically marginalized people and Rwanda's poverty categories (Ubudehe) C, D, and E is a risk. Such power differentials could exacerbate exposure to sexual abuse, exploitation and harassment, a risk that may affect all Project components if unmitigated. The Project may cause social risks including loss of access to lands due to new land classifications or stricter implementation of land use and park laws, displacement of informal land occupants due to afforestation/reforestation, and loss of access to land for food production. The potential for invasive species spread in the reforestation/afforestation components, water usage during increased seedling production at nurseries, terrace stabilization for planting, loss of topsoil to erosion during planting, disturbance to wildlife during afforestation or reforestation, modification of water flow and topsoil that could increase exposure to disease, increased human-wildlife interactions with increased tree cover and fruit trees, air quality impacts from cookstove production at scale, and supply chain impacts for livelihoods interventions all have associated environmental and social risks to be thoroughly vetted as specific sites and beneficiaries are identified.	The PMU will hire an Environmental and Social Safeguards (ESS) Specialist to oversee the application of the ESMF for a more thorough assessment of impacts by specific activities during Project implementation. The ESS Specialist will also oversee stakeholder engagement, sensitize all relevant staff on social safeguards and community engagement including SEAH training, and ensure Project activities target vulnerable groups to build resilient livelihoods. Consent processes will be established with communities. The Project will have a Project-level grievance redress process available to address complaints that aligns and complements the grievance redress used by the MoE. The PMU will also hire an Environmental Management Specialist (EMS) who will work with the ESS on environmental assessment and implementing proper mitigation processes as required.
Summary of Risks	Mitigation Factors / Measure and Management Strategies
Labor and working conditions: The dangers involve the potential for sexual abuse, exploitation and harassment (SEAH) in the hiring process or while working, forced labor, child labor, and exposure to occupational health and safety risks. These risks include safety of laborers removing ferns and planting trees on steep slopes, exposure to potential disease vectors in standing water or soil and large scale fabrication of stoves, risks using the machinery in the production of stoves, exposure to bees in honey production as well as other similar risks.	The ESS will coordinate with AE/EE human resources staff to conduct a Labor and Working Conditions Risk Assessment that will inform the development of a Labor and Working Conditions Risk Management Plan. The ESS and HR staff will appoint field safety leads within each sector. These individuals will receive training on National Laws governing labor and working conditions, SEAH, and other risks relevant to the activities that involve the hiring of laborers during the course of the Project.

Resource efficiency and pollution prevention: The stabilization of terraces risks topsoil loss and terrace collapse due to deteriorating conditions and improper long-term maintenance and extreme weather events. Pollution/waste may be created across the supply chains for stove production and distribution. Pesticides and fertilizers may be used for seedling production. Water usage in nurseries could divert resources for others. The release of CO2 emissions in the sourcing and distribution of materials in reforestation/afforestation as well as the livelihood enhancement activities could also pose a risk.	The ESS, EMS, and a Procurement Specialist will conduct activity-specific assessment for resource efficiency and pollution control. During implementation they will oversee subcontracts for reforestation/afforestation and all livelihoods enhancement projects. For stabilization of terraces, the subcontractor will have an engineer oversee the technical components of stabilization to ensure best practices and reduction of risks. For livelihoods enhancement projects the ESS and Procurement Specialist will ensure environmentally sourced materials and establish measures to minimize production waste and maximize distribution efficiency of stoves. The EMS will work with nurseries on plans for mitigating pesticides, fertilizers and water use.
Community safety and security: Increased risk of forest fires due to increasing tree coverages, increased exposure to vector-borne diseases, human wildlife conflicts associated with reforestation/afforestation and the influx of international funding that shifts power dynamics and creates environments prone to increased incidents of SEAH. With expected changes in land classifications, interactions among law enforcement and informal land owners or communities who use the resources could become more prevalent and increase risks of harm if law enforcement lacks proper training for working with communities. Work on steep slippery slopes, working with bees, use of new cook stoves and work in remote locations (e.g., fern removal) will require safety considerations for participating community members. Extreme weather or global events (e.g. pandemic) may disrupt safety and security.	The ESS will screen each activity for community, safety and security risks. The EE will have at a minimum, an Emergency Response and Preparedness Plan to address specific local hazards and response at each site; all staff will be trained on the plan as well as on disease prevention and hygiene, SEAH prevention, and addressing human-wildlife conflicts. The PMU will need to address at a high level across the GoR the potential for escalating conflicts among landowners and law enforcement as land classifications change in the course of the Project.
Summary of Risks	Mitigation Factors / Measure and Management Strategies
Resource access/restrictions: Land classification processes could result in changes that affect resource access due to existing land use, occupancy, and land tenure in the CND especially among farmers, settlers, and traditional uses of the land and forest. At a smaller scale with less risk, as tree planting and transitions to native or value-added species (e.g. fruit trees) commence, access to tree resources may be diminished until trees grow larger and become usable.	The ESS will screen all Project activities for the potential to cause displacement, namely access restrictions to resources. The Project is committed to stakeholder engagement and will mitigate access restrictions risks through a process framework and mitigation hierarchy. The AE grievance redress process will manage access complaints.

Biodiversity conservation: The inadvertent introduction of non-native/invasive pest species/diseases during forestry/agroforestry related activities could negatively impact production and forest health. Wildlife may be disturbed in the course of fern removal and/or tree planting as well as other potential livelihood activities that will be implemented. Human-wildlife interactions may increase as a result of increased forest cover and fruit trees being available in the CND. This could cause landowners to retaliate against wildlife.	The ESS will screen all Project activities for biodiversity and conservation risks/impacts. Mitigation measures will be put in place to ensure tree planting, nursery management, and seedling distribution incorporate training by consultants on invasive species recognition and management and that options for managing human-wildlife conflicts are able should issues arise.
Historically marginalized peoples: Rwanda is a nation with a single/common culture, tribe, language and with a national constitution that recognizes all Rwandans are born and remain equal in rights and freedom (article 16 of Rwandan Constitution, 2015). That said, there are historically marginalized groups as well as categories c,d, e, groups in the areas of the project activities. The risks are that the impacts of project activities on these groups are not adequately considered and analyzed and further that the intended benefits for these groups do not materialize given a lack of attention to their particular needs and vulnerabilities. Should this occur, the overall intent of the project could be in jeopardy given the focus on making such communities more resilient in light of climate change uncertainties.	As the project focuses on providing benefits, in particular increased resilience to climate change for vulnerable communities, the ESS will pay particular attention to the screening for activity level impacts on historically marginalized and categories c,d,e. Thus far, 24 focus groups with 4 exclusively historically marginalized, 5 exclusively women and 3 exclusively youth participants informed the project design and provided insights on impacts and improved ways of benefiting individuals in these groups (see Annex 6, section 4). A detailed plan for engaging with women, youth, those below the poverty line and landless groups will be prepared by project staff in collaboration with district, sector, and local ngo leaders in the project implementation phase as specific projects and locations are determined. In addition, all interactions with these groups will include processes for obtaining consent as well as clarity regarding access to the grievance redress process should complaints arise during the project.

Risk details and screening assessment to be conducted in Project implementation are described in the Partial Environmental and Social Impact Assessment and Environmental and Social Management Framework (Annex 6). Positive project impacts include a focus on improving resilience of the most vulnerable communities, communities stewarding/ harnessing the carbon cycle and reducing greenhouse gasses by sequestering carbon (i.e., through conserving indigenous trees, increasing soil organic carbon, and practicing regenerative agriculture); and utilizing efficient cook stoves. The PMU/ESS will be responsible for conducting screenings, including screenings for project-level impacts on historically marginalized groups of each Project activity as specific locations and beneficiaries are determined and with implementing mitigation measures aligned with Rwanda National Laws and GCF policies and practices. These measures include a commitment to training in all relevant EE staff on stakeholder and community engagement; consent with local communities; access restrictions mitigations; SEAH and adhering to the labor and safety laws that meet international best practices. The full Stakeholder Engagement Plan can be found in Annex 7. The EE and the AE will develop a compliance monitoring, supervision and reporting schedule that meets requirements in the GCF Information Disclosure Policy and Environmental and Social Policy. The AE will establish a

Program-level grievance redress mechanism that includes a process of public notification, documentation, investigation and resolution. This GRM will complement the MoE GRM.

G.2. Gender assessment and action plan

Women, youth and marginalized groups in the CND are key stakeholders in the Project and its primary beneficiaries. The Project will be implemented across various sectors along the CND, where natural forests are particularly vulnerable due to high population, increasing dependence upon fuelwood, and illegal use of forests to improve income streams. The Gender Assessment uses a range of primary (24 focus groups and 18 key informant interviews) and secondary (published reports, policies and academic literature) sources to triangulate our understanding of gender in climate resilience and Project design. The Gender Analysis identified and analyzed gender risks and impacts of the program and climate change through four steps: scoping; legislative and policy review; stakeholder consultation and analysis; and baseline analysis. The Action Plan operationalizes the findings from the Assessment, the ESIA, and youth-based information collected during the feasibility studies and stakeholder consultation. Despite tremendous policy and legal efforts, a patriarchal culture and persistent disparities characterize gender relations in Rwanda. Disparities persist in post-primary education; in access to and control of property, including land and economic resources; in employment opportunities and entrepreneurship; in decision-making at household and community levels; in family responsibilities and unpaid care work; and in the experience of violence, harassment, conflict, and insecurity. Sexual and gender-based violence persists at high levels. While women have made impressive political progress, especially at the national level, their local political representation and engagement remains unequal. Economic opportunities remain markedly gender-differentiated. In Rwanda and the CND, structural barriers to economic and political equity and female empowerment have significantly impeded measures to adapt to climate change impacts. While these structural inequalities are lower compared to other countries because of leading gender-considerate policies across sectors, the point remains that women's ability to access, use and control natural resources, infrastructure and services is different and low compared to men. This means the degradation of natural resources and new infrastructure will affect genders differently and result in greater vulnerability for women. However, women are vital change agents and can be powerful leaders from community to global levels in mitigating and adapting to climate change. Three key themes in the Gender and Social Inclusion Action Plan include i) Change negative gender and social norms, ii) Provide access to skills building, leadership and programmatic inclusion, and iii) Facilitate women's financial autonomy. Youth account for over 50% of Rwanda's population. An analysis was done to determine how GCF funding could support activities to align with and strengthen the sustainability of the overall program by supporting youth through the following activity themes: i) Non land-based opportunities, ii) Access to skills building, and iii) Youth leadership and Programmatic Inclusion. Social norms are difficult to change, and deeply entrenched biases will require active engagement by men, women, and youth, in decision-making roles as well as grassroots stakeholders. By working through the themes outlined above, social and technical barriers to participating in decision-making, planning, and activities will be addressed. Specific targets will be developed after a brief inception phase. Further details are in Annex 8.

G.3. Financial management and procurement

Funds from GCF will be provided to the MOE's GCF account (as the AE), under the terms of the FAA. MoE will be responsible for financial management and accountable for the use of GCF resources. Financial transactions will be subject to annual audits undertaken by internationally certified auditors. In addition, as the AE, MoE will: i) ensure that the project is executed in accordance with GCF standards; ii) supervise,

oversee and manage the implementation of project interventions; iii) report on Project progress; and iv) ensure that Project activities are well coordinated and aligned with countries' national priorities. MoE will provide oversight to the project consisting of a) entering into a subsidiary agreement with the EE (RFA); b) managing and disbursing GCF funds to the EE, after providing no objection to work plans and budgets; c) reviewing financial expenditures and progress reports; d) overseeing Project implementation in accordance with the Project document and Annual Work Plans and Budgets, agreements with co-financiers and each executing entity rules and procedures; e) providing technical guidance to ensure that the appropriate technical quality is applied to all Project activities; f) providing financial reports to the GCF for Project funds received; g) ensuring that the project complies with the terms agreed in the Project's respective FAA as well as the AMA signed between MoE and the GCF; and h) undertaking regular annual supervision missions according to the MoE's guidelines and convene Project Steering Committee (PSC) meeting twice a year aimed to review progress and approve work plans. The PMU will manage, monitor and report on day-to-day implementation of the EE. The EE will execute the tasks within the overall project management structure: a) implementing day-to-day activities as per the project work plan and budget, including the Environmental and Social Management Plan; b) undertaking procurement activities as per the agreement signed by GCF and AE as well as the sub-agreement signed by the AE and EE; c) managing contracts of suppliers and services providers; d) hiring and managing project staff relevant to the EE-managed project areas; e) implementing activities as per the project work plan; and f) carrying out financial and technical monitoring of activities, including the achievement of outputs and outcomes the EE are in charge of. The EE will enter into a Memorandum of Understanding with different partner agencies that will provide technical know-how to the implementation of the project activities. Procured parties will be contracted in accordance with MoE rules and regulations of procuring goods and services. The financial management structure of the project is built on the Permanent Secretary of MoE, the Project Coordinator/ Chief of Party and the Finance Management Specialist at the PMU level who will oversee and monitor the project finances with project directors and finance managers from the EE (RFA).

The project will be executed by Rwanda Forestry Authority (RFA), a Rwanda government entity affiliated to the Ministry of Environment with a semi-autonomous legal entity. RFA has transparent and accountable financial management systems in place, which are essential elements for the effective, accountable management of the project. RFA will follow the Government of Rwanda procedures for planning, budgeting, and accounting, which are outlined in the law n° 12/2013/ol of 12/09/2013 on state finances and property, and ministerial order n°001/16/10/tc of 26/01/2016 relating to financials; and supporting operational manuals issued by the Ministry of Finance.

The MoE shall ascertain that the EE has in place efficient internal control systems, review financial reports and proceed funds disbursements and will also periodically inspect the project for compliance as well as coordinate audits exercises for the EE. RFA will enter into agreements with MoE that outline general modalities of the funds management in line with GCF grant requirements and RFA's policies. GCF funds will be transferred to an RFA designated bank account. Funds will be disbursed to the EE based on agreed annual work plans and budgets, and its reporting will be on a quarterly basis. The project financial processes will rely on the EE's charts of accounts, internal control systems, disbursement plans, budgeting, expenditure management and accounting, reporting, and assets management as assessed by MoE. Procurement regulations embed the principles of fairness, transparency, accountability and the promotion of competitive economy and efficiency. The Project will be audited annually and commissioned by the Accredited Entity (MoE) to produce certified financial statements and reports to ascertain that GCF funds are well managed. The project will maintain a regularly updated record of all assets and documents that will be retained for at least 5 years after the end of the project.

G.4. Disclosure of funding proposal

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

(f) ANNEXES		
H.1. Mandatory annexes		
☒	Annex 1	NDA no-objection letter(s)
☒	Annex 2	Feasibility study - and a market study, if applicable
☒	Annex 3	Economic and/or financial analyses in spreadsheet format
☒	Annex 4	Detailed budget plan
☒	Annex 5	Implementation timetable including key project/programme milestones
☒	Annex 6	E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3): ☒ Environmental and Social Impact Assessment (ESIA) or ☒ Environmental and Social Management Plan (ESMP) or ☐ Environmental and Social Management System (ESMS) ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
☒	Annex 7	Summary of consultations and stakeholder engagement plan
☒	Annex 8	Gender assessment and project/programme-level action plan
☒	Annex 9	Legal due diligence (regulation, taxation and insurance)
☒	Annex 10	Procurement plan
☒	Annex 11	Monitoring and evaluation plan
☒	Annex 12	AE fee request
☒	Annex 13	Co-financing commitment letter, if applicable
☒	Annex 14	Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule
H.2. Other annexes as applicable		
☒	Annex 15	Evidence of internal approval
☐	Annex 16	Map(s) indicating the location of proposed interventions
☐	Annex 17	Multi-country project/programme information
☐	Annex 18	Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
☒	Annex 19	Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
☐	Annex 20	First level AML/CFT (KYC) assessment
☒	Annex 21	Operations manual (Operations and maintenance)
☒	Annex 22	Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)



Republic of Rwanda

**RWANDA ENVIRONMENT
MANAGEMENT AUTHORITY
(REMA)**



Kigali, on **22 FEB 2022**
N° **0487**/NDA/2022

Deputy Executive Director
Green Climate Fund ("GCF")
Songdo Business District
175 Art Center-Daero
Yeonnsu-gu Incheon 22004
Republic of Korea

Re: Funding proposal for the GCF by the Ministry of Environment regarding "Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration Project"

Dear Sir,

We refer to the project titled "Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide Through Forest and Landscape Restoration Project" in Rwanda as included in the funding proposal submitted by the Ministry of Environment to us on 18 February 2022.

The undersigned is the duly authorized representative of Rwanda Environment Management Authority, the National Designated Authority of Rwanda.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Rwanda has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Rwanda;
- (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards, **FAUSTIN YAZIKWIYE** Faustin

For

Juliet KABERA
Director General/National GCF Focal Point



Cc:

-Permanent Secretary/ Ministry of Environment

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	Ministry of Environment (MoE) of the Republic of Rwanda
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Three Provinces of Rwanda (Western, Southern, and Northern) in 10 Districts (Karongi, Musanze, Ngororero, Nyabihu, Nyamagabe, Nyamasheke, Nyaruguru, Rubavu, Rusizi, and Rutsiro).
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Friday, September 22, 2023
Language(s) of disclosure	English and Kinyarwanda
Explanation on language	Kinyarwanda is the national language spoken by all Rwandans.
Link to disclosure	English: https://www.environment.gov.rw/index.php?eID=dumpFile&t=f&f=76268&token=0d001d8f3dc586127656b6b8280da983d72dda02 Kinyarwanda*: https://www.environment.gov.rw/index.php?eID=dumpFile&t=f&f=76248&token=a3f7cf9afd24f1eacdd08dc21dd9cd9299ad1e25
Other link(s)	N/A
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the "Environmental and Social Management Framework (ESMF)"
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Friday, September 22, 2023
Language(s) of disclosure	English and Kinyarwanda
Explanation on language	Kinyarwanda is the national language spoken by all Rwandans.
Link to disclosure	English: https://www.environment.gov.rw/index.php?eID=dumpFile&t=f&f=76268&token=0d001d8f3dc586127656b6b8280da983d72dda02 Kinyarwanda*: https://www.environment.gov.rw/index.php?eID=dumpFile&t=f&f=76248&token=a3f7cf9afd24f1eacdd08dc21dd9cd9299ad1e25

Other link(s)	N/A
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the “Environmental and Social Management Framework (ESMF)”
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity’s website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, Environmental and Social Management Framework (ESMF)	
Description of report/disclosure on accredited entity’s website	Indigenous Peoples Planning Framework
Language(s) of disclosure	English and Kinyarwanda
Explanation on language	Kinyarwanda is the national language spoken by all Rwandans.
Link to disclosure	English: https://www.environment.gov.rw/index.php?eID=dumpFile&t=f&f=76268&token=0d001d8f3dc586127656b6b8280da983d72dda02 Kinyarwanda*: https://www.environment.gov.rw/index.php?eID=dumpFile&t=f&f=76248&token=a3f7cf9afd24f1eacdd08dc21dd9cd9299ad1e25
Other link(s)	N/A
Remarks	A Historically Marginalized Peoples Plan, which serves as an Indigenous Peoples Planning Framework, is contained as an Annex to the ESMF.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, September 22, 2023
Place	To accommodate for those who may not have access to the Internet, a computer, or a reliable power source, the Ministry of Environment will be sharing the ESMF documents to the following districts: • Karongi district (2.0526° S, 29.4128° E), P.O Box: 23 Kibuye, village: Kamuvunyi, Cell: Gacaca, district: KARongi • Musanze district (1.5020° S, 29.6332° E), P.O Box: 03 Musanze, village: Byimana, Cell: Ruhengeri, District: Musanze • Ngororero district (1.8537° S, 29.6347° E), P.O Box: 103 Muhanga, village: Ngororero, Cell: Kazabe, district: Ngororero • Nyabihu district (1.5911° S, 29.5220° E), P.O Box: 125 Musanze village: Mukamira Cell: Rubaya District: Nyabihu

	• Nyamagabe district (2.4671° S, 29.5736° E), P.O BOX: 36 Gikongoro, village: Nyarusange, Cell: Nyamugari, district: Nyamagabe • Nyamasheke district (-2.342014193676258, 29.08901503907659), P.O Box: 72 Nyamasheke, village:Gikuyu, Cell: Ninzi, district:Nyamasheke • Nyaruguru district (2.6987° S, 29.5703° E), P.O Box: 215 Butare village: Nyarusovu Cell: Mubuga, district: Nyaruguru • Rubavu district (1.6736° S, 29.2886° E), P.O Box: 173 Gisenyi; village: Nyaburanga, Cell:Nengo, district: Rubavu • Rusizi district (2.4830° S, 28.9054° E), P.O Box: 253 Rusizi, village: Ntwali, Cell:Cyangugu, district: Rubavu • Rutsiro district (1.9617° S, 29.3857° E) P.O Box: 49 Kibuye, village: Nduba, Cell: Congo-Nil district: Rutsiro and request them to have physical copies of ESS reports in their local government offices, publish an Executive Summary of the ESMF on their notice board in both Kinyarwanda and English and publish it on their website to ensure it is accessed by all affected people in their location as well as to those who may have access to the Internet, a computer.
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, October 23, 2023

Note: This form was prepared by the accredited entity stated above.

** Subsequent to the disclosure of the ESS report to the Board and active observers on 22 September 2023, the following update has been made: the links to the Environmental & Social Management Framework (ESMF), in which the ESMF document -had been made available, have been updated for both English and Kinyarwanda. The footnote 86 has been removed.*

Secretariat's assessment of FP217

Proposal name:	Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration
Accredited entity:	Ministry of Environment of Rwanda
Country/(ies):	Rwanda
Project/programme size:	Small

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
This Congo Nile Divide project will safeguard Rwanda's biodiversity, enhance climate resilience, and support vulnerable communities.	The project should ensure linkage to other GCF funded projects in the country. In addition, we encourage MOE Rwanda to establish financial mechanisms to upscale across the wider landscape.
The project is located in the Albertine Rift, which is one of the most biologically rich regions in Africa, containing more than half of Africa's birds and 40% of its mammals.	The project should ensure monitoring and evaluation of the project interventions is undertaken.
With more frequent extreme rainfall events in recent years, this agricultural landscape is recognized as the region of Rwanda most vulnerable to floods and landslides.	
Project interventions will include improved management on 250,000 hectares (ha) (including promotion of natural regeneration/planting on 10,000 ha) and improved agroforestry practices on 2,000 ha of on-farm plantations.	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum XXI titled "List of proposed conditions and recommendations", respectively.

II. Summary of the Secretariat's assessment

2.1 Project background

3. The proposed project's goal is to improve the resilience of vulnerable communities to climate variability and shocks by increasing the extent and integrity of natural forests and by strengthening the capacity for climate-responsive forest and landscape management. The project will:

- (a) Strengthen institutional and regulatory frameworks for low-emission climate-resilient development pathways by mainstreaming climate change into integrated land-use planning.
 - (b) Increase the climate resilience of vulnerable rural communities through forest and landscape restoration and reducing demand for fuelwood.
4. The proposed interventions will target districts along the Congo Nile Divide (CND) region. Rwanda's CND, an area of 444,600 hectares (ha), is a high-elevation landscape of remnant natural forest reserves embedded in a mosaic of intensively managed smallholder farms and commercial plantations. The CND separates the drainage basins of the Congo and Nile Rivers. It is part of the Albertine Rift, which is one of the most biologically rich regions in Africa and "one of the most important regions for conservation" on the continent, containing more than half of Africa's birds and 40 per cent of its mammals.
5. Though covering barely 5 per cent of the country, this archipelago of protected forests is of critical importance for:
 - (a) Its globally recognized high biodiversity values;
 - (b) Its role in Rwanda's acclaimed ecotourism offerings, serving as the primary source of Rwanda's foreign revenue and as a major economic development driver;
 - (c) The ecosystem services provided for surrounding subsistence farmlands as well as commercial tea and coffee plantations offering employment opportunities for women while providing the second and third highest sources of foreign revenue;
 - (d) The regulation of water capture and release to growing urban centres on the CND periphery, the nation's extensive river network, and downstream hydroelectric operations; and
 - (e) The sequestration and storage of nearly 60 per cent of Rwanda's national carbon stocks.
6. These values and services are already compromised by a combination of anthropogenic and climatic factors; they will be much more severely threatened by the potential negative feedback loops associated with projected climate change impacts. The project will promote an ecosystem-based approach to forest protection and restoration and will increase the extent and resilience of forest ecosystems and species, including by functionally linking currently isolated forest fragments. Through these approaches, the project will protect, restore and rehabilitate forest ecosystems, thus enhancing biodiversity and reducing key threats while increasing the provision of critical ecosystem services to strengthen the adaptation of vulnerable communities.
7. Project interventions will directly benefit 1,254,242 people (654,404 women). The project will also indirectly benefit 9,260,745 people and create approximately 24,212 job opportunities in forest-dependent communities. The total mitigation impact is estimated to be 1,204,448 tonnes of carbon dioxide equivalent (t CO₂ eq) by 2043.
8. The total project financing is USD 50,000,000 with a request for GCF grants of USD 39,056,422. The Rwanda Forestry Authority (RFA), Rwanda Water Board (RWB), Rwanda Development Board (RDB) and the Ministry of Environment of Rwanda (MOE Rwanda) will provide co-financing of USD 10,831,308 in grants. MOE Rwanda will also provide an in-kind contribution of USD 112,271. The project is submitted under environmental and social safeguards (ESS) category B.

2.2 Component-by-component analysis

Component 1: Mainstreaming climate adaptation into integrated land-use planning (total cost: USD 2.9 million; GCF cost: USD 2.9 million)

9. This component aims to integrate climate change adaptation considerations into land-use planning at the national and subnational levels. The goal is to achieve climate resilience through collaboration among government agencies, civil society, the private sector and diverse partners. The component has two main outputs: The first is to develop a climate-resilient land-use plan for the CND by integrating climate adaptation information into existing land-use planning efforts. The second aims to strengthen institutional capacities in Rwanda to integrate biodiversity and climate risks into land-use planning.

Component 2: Forest and landscape management and restoration (total cost: USD 18.9 million; GCF cost: USD 16.3 million)

10. In this component, targeted interventions will integrate climate change awareness and adaptation into national forest park planning, policies and management, as well as restore degraded areas within those parks. These efforts will help sustain the forests' rich biodiversity, the lucrative tourism revenues that flow into local and national economies, and the ecosystem services needed for the climate resilience of vulnerable communities. This suite of activities will also identify priority areas outside of parks – in remnant natural stands, around wetlands and along streams – for protection and restoration.

Component 3: Enhancing climate adaptation through resilient livelihoods (total cost: USD 22.6 million; GCF cost: USD 15.3 million)

11. The activities under this component aim to combine environmental conservation with socioeconomic development, creating a multifaceted approach that resonates with the global commitment to sustainable growth. They include initiatives such as enhancing and restoring forest plantations, providing sustainable fuel sources, delivering carbon sequestration benefits totalling 522,606 t CO₂ eq over 20 years, and developing comprehensive land-use planning. The other activities focus on rural livelihoods, financial services and sustainable farming methods.

12. The component will support reforestation on steep slopes on both public and smallholder private lands. It will develop farm-based agroforestry for climate-resilient species that produce fuelwood and/or market products, while conserving and enriching soils. The component will complement land-based production with a scaling-up of alternative value-chain products for smallholders that are nature-positive, including, but not limited to, tourism-based income, beekeeping, and improved cooking stoves. Central to these activities will be the facilitation of access to market chains and financial mechanisms for establishing and maintaining businesses. Given that these activities are designed at the core to be profitable and promote financial sustainability, it is intended that they be replicable and sustained once GCF funding expires.

Project management (total cost: USD 3.1 million; GCF cost: USD 2.7 million)

13. The Project Management Unit (PMU) of the CND project will be established under MOE Rwanda/Single Project Implementation Unit (SPIU) to manage day-to-day operations of complex programme interventions. Headed by the Project Coordinator, who will also serve as the secretary and rapporteur of the Project Steering Committee, the PMU will be responsible for tasks such as managing and monitoring identified risks and reporting new risks to the accredited entity (AE) and the Project Steering Committee. The PMU will adhere to MOE Rwanda/Government of Rwanda recruitment practices and provide quarterly progress reports. The Project Coordinator will report to the MOE Rwanda/SPIU Coordinator as the immediate supervisor and to the Permanent Secretary as the secondary supervisor.

14. The PMU's structure will comprise various specialists, including for monitoring and evaluation, procurement, gender, financial management, communication and outreach, along with an office manager and driver. Its staff will be selected and recruited following MOE Rwanda/Government of Rwanda procedures, aiming for a lean management structure. Operations will conclude once the final project evaluation report and other required documents have been submitted to MOE Rwanda, signifying a comprehensive and organized approach to the management of the entire programme.
15. The GCF portion of this project management cost is less than 5 per cent of the total requested GCF funding and is compliant with the GCF Policy on Fees.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: N/A

16. The project has 1,254,24 direct beneficiaries, representing 30.9 per cent of the CND population, and 9,260,745 indirect beneficiaries, representing almost 70 per cent of Rwanda's population. Additional adaptation benefits will include the training of 45,992 community members in forest management and restoration for climate resilience. Altogether, the project will contribute to the restoration of 5,000 ha of intact forest, 2,500 ha of protective forests, 3,346 ha of smallholder farmland, 1,000 ha of pastures and 1,500 ha of riparian linkages. Overall, 275,617 ha of landscape could potentially be positively impacted by the project. Although the project is focused on priority landscapes covering 275,617 ha, it is likely that benefits due to increased adaptive capacity could feasibly extend across the full 700,000 ha CND. The project will generate mitigation benefits by sequestering approximately 250,841 t CO₂ eq during the five-year project period, which is about 0.9 per cent of the 28.2 million t CO₂ eq reduction target in Rwanda's Vision 2050. The total mitigation impact from reduced deforestation, the introduction of forest restoration and adoption of improved cookstoves is estimated to be 1,204,448 t CO₂ eq by 2043.
17. The project contributes to the implementation of Rwanda's updated nationally determined contribution (NDC) (2020) commitments. It is also aligned with the country's Green Growth and Climate Resilience Strategy and other national policies. The proposal objectives are also well aligned with the country's National Forest Policy and related plans for reversing trends of high deforestation.
18. With support from the NDC Partnership, Rwanda now has a single NDC implementation framework that captures both national and subnational climate actions and investments plans and includes a strategic programme for climate resilience, scaling up renewable energy programme, green climate fund strategic framework, public investment priorities, sector strategic plans and district development strategies. Therefore, the proposed project is well supported by this existing NDC implementation framework that provides an enabling environment that will ensure climate actions undertaken are integrated into national planning.
19. The proposed project further strengthens the country's national and subnational adaptation priorities, approaches and plans through promoting ecosystem-based adaptation while utilizing climate science, effective landscape planning, policies, cross-sectoral coordination and management in addressing climate risks and adaptation benefits.
20. The project in general seeks outcomes at a range of levels (from national through to catchment level) across the public sector, while scaling up lessons and initiatives from aligned programming. It will build the capacity of MOE Rwanda (AE) and other related agencies and stakeholders to coordinate and collaborate more effectively to help accelerate subnational and local climate action. It is envisaged that these funds will also crowd in additional private sector

capital (e.g. cookstoves and forest plantations) while enabling the government to implement the adaptation interventions. However, without a clear private sector framework or plan, this might be challenging to implement. Nonetheless, Rwanda has a GCF country programme and strategic framework and has managed GCF funded projects and programmes.

21. Furthermore, the project clearly articulates the alignment, additionality and/or complementarity with several ongoing or planned programmes funded by significant international and multilateral agencies (e.g. World Bank, Global Environment Facility and United Nations Development Programme, etc.), which reflects on the experience and track record of MOE Rwanda to deliver on this project interventions.

3.2 Paradigm shift potential

Scale: N/A

22. By utilizing integrated landscape and ecosystem-based adaptation approaches in addressing climate change impacts in Rwanda, the project offers myriad opportunities for upscaling and replication to similar ecological zones across the country. This approach to land-use planning and management is entirely new for Rwanda and will restore and sustain varied ecosystem services for both adjacent and nationwide populations. The project coordination and collaboration efforts across both geographic and sectoral boundaries, while mainstreaming climate risks in management planning, represents a fundamental paradigm shift for land-use planning and management in Rwanda.

23. The project has the potential to contribute to the regulatory frameworks and policies for driving climate investments in the forest and renewable energy sectors. The project will finance the development of the integrated land-use plan, which will be adopted by all ten CND districts as a framework for seeking synergies between rural livelihoods and forest and biodiversity protections, reconciling land-use conflicts that arise, and guiding development decisions and climate change adaptation. It will also update the National Land Use Master Plan to ensure it incorporates climate resiliency and is accurate and relevant to local contexts. The adoption of an integrated land-use planning approach helps to better reconcile cross-sectoral trade-offs and improve climate adaptation planning in the CND; this approach will prove a successful model for application in other regions of Rwanda and other forest/farming landscapes beyond. In addition, revising two national park management plans to include climate concerns will serve as a model for the remaining parks of Rwanda, as well as for the East Africa region.

24. Overall, the project's efforts in shifting policy, practice and collaborative learning processes symbolize a fundamental change towards climate-resilient development pathways in Rwanda, with implications for broader regional application.

3.3 Sustainable development potential

Scale: N/A

25. The project is closely aligned with several Sustainable Development Goals. Key environmental benefits include restoration of degraded lands to reduce erosion, sedimentation, and landslide risks; the planting of native trees to enhance biological connectivity; and mitigation of greenhouse gas emissions through both increased carbon sequestration and fuel-efficient cookstove technologies.

26. Social benefits extend to more than a million people, focusing on increased awareness of climate change, empowerment of the rural poor (with an emphasis on women's participation), and creation of social capital through collaborative adaptation strategies. The project aims to significantly reduce poverty and inequality. It will provide economic co-benefits such as increased agricultural productivity, employment opportunities for around 24,212 households, modern beekeeping practices yielding significant income, and a net present value (NPV) of USD 369 million over 20 years from forest restoration and other activities. Private capital

investment is also targeted, thus fostering a “triple bottom line” that balances social, environmental and economic benefits. Emphasis on gender-sensitive impacts ensures that women are actively engaged in planning and decision-making, leading to increased income, improved health and nutrition, and strengthened leadership and entrepreneurship skills.

27. Overall, the project offers a comprehensive and strategic approach to sustainable development, thereby promoting climate resilience, economic growth, social cohesion and gender equality within the CND region.

3.4 Needs of the recipient

Scale: N/A

28. The project proposal interventions were designed from key findings and recommendations of a climate change vulnerability assessment of the selected project geographical area (see annex 2). Climate exposure, sensitivity, adaptive capacity, impacts and barriers as well the greenhouse gas emissions profile were established and either quantified or supported with reference to scientific publications. Further, climate change impacts on the target sectors as well as size and economic and social development of the target population, including critical barriers to reduce their vulnerability, are elaborated herein. The proposed interventions are targeted to match the identified barriers. The vulnerability of the target sectors is quantified to the extent possible, however, gender disaggregation is covered in qualitative terms only.

29. The CND landscape districts support nearly one third of Rwanda’s population on barely one fifth of its land base. The total population is 1,598,536, of which women account for 52.2 per cent. Through the project interventions, targeted vulnerable groups will attain more climate-resilient livelihoods from reduced exposure to landslides, floods and soil erosion and more knowledge on climate risks, the value of forests and forest ecosystems, climate adaptation options, access to quality indigenous and agroforestry materials, and improved ecosystem services.

3.5 Country ownership

Scale: N/A

30. MOE Rwanda, which is the AE for this proposal, clearly played a pivotal role in defining and drafting the proposal stakeholder engagement report and plan (see annex 7) as it is aligned with Rwanda’s GCF country programme stakeholder engagement strategic framework. In its country programme, Rwanda articulated how it will conduct stakeholder engagement at the readiness stage and in delivering GCF concepts, projects and programmes. This process has provided Rwanda with an overarching framework that supports its stakeholder engagement processes, which is clearly seen in this proposal. This framework can support the strengthening of local climate action from national to local levels with clear feedback loops and avenues for incorporating learning.

31. The proposal indicates that a detailed stakeholder and gender analysis were conducted, augmented by extensive consultations of various actors from national, subnational and local levels. A range of stakeholders from the public and civil society were consulted, in addition to government agencies at all levels that will be engaged throughout the project cycle, cutting across all key sectors and scale. The general concern relates to local ownership and the transfer of knowledge that can sustain interventions after the project closure. It would be ideal if Indigenous knowledge and technology transfer is well factored into implementation.

3.6 Efficiency and effectiveness

Scale: N/A

32. Utilizing grant financing reflects the public good character of the investments, the project aims to create a model that overcomes barriers to private sector investment in Rwanda's forestry sector. With an economic internal rate of return of 43.32 per cent and a financial internal rate of return of 42 per cent, the project is estimated to generate an NPV of more than USD 120.48 million over 20 years, which translates to USD 64 per direct beneficiary. The mitigation cost is estimated to be USD 43.84 per t CO₂ eq. Implementing best practices for tree planting and utilizing cutting-edge monitoring technologies, the project is aligned with similar initiatives and sets the stage for potential replication in other regions. Its community-focused approach, including measures to develop economically viable units, contributes to a threefold increase in net economic benefits compared to a scenario without the project.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

33. Project background: The project aims to increase resilience of vulnerable rural communities in the Rwandan side of the CND by building capacity for integrated planning and management through forest and landscape restoration and reducing demand for fuelwood. The project will involve the formulation and implementation of plans (i.e. integrated district land-use plans and enhancements of general management plans of national parks), coordination with various agencies, capacity-building, and physical interventions on the ground. This includes afforestation/reforestation of protected forest and private landholdings, installation of soil conservation measures, ecological interventions such as fern removal in natural forests and restoration of degraded forests in Gishwati-Mukura National Park, production and distribution of fuel-efficient cookstoves, beekeeping activities, and support for the value chains of these enterprises. Among the environmental and social (E&S) co-benefits are the enhancement of the unique, endemic biodiversity of the area through decreased sedimentation, improved water quality and recharge, reduced erosion, decrease in loss of soil organic matter as a result of forest restoration and establishment of connectivity between the forests across the CND landscape. The project is also expected to contribute to the empowerment of the women and youth in the target areas.

34. E&S risk category and safeguards instrument: The overall project E&S risk category is Category B (moderate). The Secretariat confirms the proposed E&S risk category since the project is, in itself, an environmental and social enhancement project. The potential adverse E&S risks and impacts will mainly come from the implementation of ground activities, are expected to be mostly minimal and highly localized and can be avoided/mitigated through careful design and the application of generally accepted mitigation measures and standards. The Secretariat conforms to the assigned risk category and it is within the accreditation level of the AE. A partial environmental and social impact assessment and environmental and social management framework (ESMF) have been prepared for the project.

35. Compliance with the GCF ESS standards: The paragraphs below provide a brief description of the programme's compliance with ESS standards.

36. **ESS 1 (Assessment and management of environmental and social risks and impacts):** The ESMF discussed the baseline socioeconomic conditions of the project area. It has also provided a general assessment of the risks and impacts of the project. The assessment included an indication of the positive and negative components of the activities and subactivities, the consequence of the impact, the probability that the impact will occur, the duration for which the impacts are expected to occur, and the overall significance of the impact.

Nevertheless, given that much of the project activities cannot yet be fully specified in terms of location, ownership, design and approach and will only be identified and fully specified through consultations and coordination with various stakeholders during project implementation, the ESMF provided the basis for identifying the risks and impacts of the subprojects and defining the processes and procedures to be applied in assessing and managing the risks and impacts of individual project activities during implementation. The major anticipated negative impacts of the proposed activities and subactivities include, but are not limited to: disproportionate distribution of project benefits, potential for gender-based violence (GBV), risks related to labour and working conditions, community health and safety, potential for introduction of invasive species and pests on planted seedlings, water use at nurseries diverting from other uses, use of pesticides and fertilizers in seedling production, destabilization and erosion during construction/reconstruction of terraces, land classification that impact access and introduce restrictions, increased exposure to disease/vector-borne illnesses, potential for negative human-wildlife interactions, and environmental risks related to the supply chain. Among the mitigating measures include a commitment to stakeholder engagement, a process of consent, training on sexual exploitation, abuse and harassment (SEAH), establishing a grievance redress mechanism (GRM), adhering to the labour and safety laws and international good practices, providing training on invasive species management, providing oversight in the supply chain, and adoption of practices to reduce the use of chemicals and production of wastes.

37. **ESS 2 (Labour and working conditions):** The project will likely require the hiring of professionals, service personnel and assistants, and activities on the ground such as terracing, removal of ferns and reforestation/afforestation activities will require the hiring of field workers and labourers. Risks associated with these activities include, but are not limited to, the potential for SEAH in the hiring process or at work, forced labour, child labour, and exposure to occupational health and safety hazards. Workers may be exposed to hazardous working environments such as steep slopes, disease vectors in standing waters, safety hazards in the fabrication/production of stoves, and exposure to bee stings in honey production, among other things. The project will ensure proper labour and working conditions are provided to workers (e.g. in forest camps), and trainings on labour and occupational health and safety standards will be undertaken. A Code of Conduct as regards worker safety and security will also be reflected in contract documents. Moreover, the project will not support activities that include the use of forced labour, human trafficking and child labour. During implementation, risks and potential impacts related to labour and working conditions will be further assessed and managed through the development of a labour and working conditions risk management plan, including ensuring that workers have access to a GRM.

38. **ESS 3 (Resource efficiency and pollution control):** The nature of the proposed interventions does not entail the use of substantial amounts of fuel, water, electricity or raw materials, except for the operation of nurseries (for its water requirement) and the production and distribution of cookstoves (for raw materials and electricity). The project will however promote the increased use of energy-efficient cooking stoves to lessen the impact on forest resources for fuel. During implementation, further review on resource efficiency and pollution control risks will be undertaken to assess and manage issues as regards water use and water availability, use of pesticides and fertilizers, potential for soil loss, soil erosion and slope destabilization, and use of raw materials and waste generation in the cook stove fabrication/production and distribution. The project will also ensure raw materials are from sustainable sources and will establish measures to minimize production waste and maximize distribution efficiency of stoves. Appropriate management measures will be developed upon screening and assessment of these risks.

39. **ESS 4 (Community health, safety and security):** The project activities could have community health, safety and security issues, which include potential for increased exposure to vector-borne diseases should ecological balance be altered or stagnant water develop as a result of the reforestation/afforestation and fern removal activities in the area (which may increase

the population of certain disease vectors and insect pests). Some of the mitigating measures include proper drainage design of terraces. There could also be possibility for human–wildlife conflict in areas where fruit trees are planted, which may inadvertently increase physical interactions with wildlife that could result in crop loss or physical injury both to humans and wildlife. The risk related to the presence of migrant labour is considered low given that the project will hire residents from surrounding communities. Nevertheless, the activities will be assessed more closely with regard to such risks, with measures to be discussed and determined in consultation with the communities during implementation.

40. **ESS 5 (Land acquisition and involuntary resettlement):** The project does not include any activities that will require resettlement or involuntary taking of land. However, project activities could facilitate a potential loss of access to lands given that the afforestation/reforestation activities could result in stricter enforcement of land use and park laws in changes to rules of access or land classification. To address this, the project will not establish protective forests on lands used for food production. On-farm afforestation will not involve the acquisition of lands, land rights or land-use rights through expropriation or other compulsory procedures and will comply with the country's legal system. Communities will be engaged in an inclusive and participatory manner and will be part of the planning and afforestation processes, and agreements on land rights and responsibilities will be developed with them. An access restrictions mitigation process framework is developed and will be implemented as required.

41. **ESS 6 (Biodiversity conservation and sustainable management of living natural resources):** The project intends to enhance the region's ecosystem to reduce the vulnerability of communities to climate change impacts. For instance, the proposed interventions on protected forests will be restoration (e.g. restoration of the 500 ha of degraded forest in GMNP, and the fern removal covering 4,500 ha of Nyungwe National Park, while the proposed interventions in the Stepping Stones and Landscapes Linkages (located outside of the parks) will be composed of the protection of isolated forest patches and afforestation on steep slopes and riparian areas supplemented with agroforestry activities to enhance ecosystem services. Nevertheless, care should be exercised in implementing these activities to ensure that negative impacts such as potential for the introduction of alien or invasive species, pests and pathogens from poorly managed nurseries are adequately addressed. During project implementation, consultations with communities and stakeholders will aid in determining the selection of appropriate species suitable to the locality.

42. **GCF Indigenous Peoples Policy and ESS 7 (Indigenous Peoples):** The AE recognizes that while there are historically marginalized groups as well as "Categories C, D and E, groups" that may meet the GCF criteria for Indigenous Peoples that could potentially be included in the project area, the screening so far has indicated that Indigenous Peoples have not been found in the project area. Given that specific sites will be selected for project activities during project implementation, each subproject will be subject to more in-depth screening for impacts to historically marginalized peoples, with particular attention to the GCF criteria of Indigenous Peoples, during project implementation. The AE will conduct a screening for subprojects as outlined in the "Screening for Presence of and Risks to Historically Marginalized Groups" to evaluate the characteristics of historically marginalized and Categories C, D and E people. Should the screening process identify Indigenous Peoples, then the subactivity will require a historically marginalized peoples plan. The AE also recognizes that there are potential barriers to integrating historically marginalized peoples into the project. Nevertheless, there is interest in working with historically marginalized groups to identify how traditional livelihoods like beekeeping can be adapted and promoted and how negative impacts on these livelihoods will be mitigated. The AE has included in the project the hiring of an ESS specialist and community engagement specialist to minimize risks as well including a commitment to stakeholder engagement, free, prior and informed consent and the setting up of a project-level GRM. In line with its roles and functions, the GCF Indigenous Peoples Advisory Group (IPAG) is available to

provide advice to the AE, which is also encouraged to share emerging good practices and success stories with IPAG.

43. **ESS 8 (Cultural Heritage):** The physical interventions (i.e. afforestation, reforestation and terracing, etc.) will be undertaken in mountainous areas where there are few human settlements. The preliminary screening of target areas also indicated that it is not expected that the project activities will disturb any cultural heritage sites. Nevertheless, the project has developed a chance find procedure to cover chance encounters of artefacts during project implementation.

44. **Institutional implementation arrangements:** As the AE, MOE Rwanda will oversee and monitor the management and implementation of the project. The Wildlife Conservation Society (WCS) and RFA will serve as EEs for the project through a project cooperation agreement. All work will be coordinated by the PMU and supported by other relevant government agencies, including the National Land Authority and the RWB within MOE Rwanda. The PMU will have oversight of the day-to-day operations of the project, including monitoring and evaluation. It will engage specialists (e.g. Environmental and Social Safeguards Specialist, Invasive Species Specialist, Community Engagement Specialist, Behaviour Change Specialist, Gender Specialist, etc.) that will screen for the potential environmental and social risks and impacts and develop management measures for proposed activities. The PMU will also be responsible for coordination on trainings for all project EEs. As regards safety, a Labour and Safety Coordinator will also oversee working conditions in the field. The project will also engage Community Engagement Specialists and Labour and Safety Officers.

45. **Stakeholder engagement and GRM:** The project benefited substantively from an extensive stakeholder consultation process, which started during the project conceptualization. A Stakeholder Engagement Plan has also been prepared to engage the stakeholders throughout project implementation. The project will have a project-level GRM, which is described in the ESMF. The GRM is independent of governmental judicial procedures. The project also allows access to the GCF independent redress mechanism. In line with the GCF Indigenous Peoples Policy, the GCF Indigenous Peoples' focal point will be available for assistance at any stage, including before a claim has been made.

46. **Sexual exploitation, sexual abuse, and sexual harassment safeguarding:** The SEAH safeguarding approach proposed by the AE is considered by the Secretariat to be in compliance with the SEAH provisions pursuant to the revised GCF Environmental and Social Policy. The project identifies that all project components may exacerbate the existing power differentials and expose the project beneficiaries to potential SEAH risk. The AE will take a number of mitigation measures to prevent and manage SEAH, including ensuring that all staff are trained on survivor-centred approaches to address SEAH, and that the Human Resources and Administration Officer and the ESS Specialist of the project will work together to develop a Security and Risk Management Plan to prevent SEAH incidences. The project's Gender Specialist and Community Engagement Specialist will develop training on SEAH prevention, GBV programming and survivor-centred approach. The project will also safeguard against SEAH through a Code of Conduct that explicitly indicates zero tolerance for GBV. This Code of Conduct will be shared with all personnel and used for training all employees. In addition to the prevalence of GBV in the target country, the project also recognizes that the influx of funds and activities, combined with power dynamics, may disproportionately affect women, youth, historically marginalized people, and others in the poverty categories of Rwanda. As the specific locations and beneficiaries will be determined during full implementation, each subproject will be screened to avoid any subproject that may aggravate risks of GBV and SEAH. The project will respond to SEAH incidents through GRMs at the AE, EE, and project levels. Training will be organized for all project staff, implementing partners, key service providers and other partner organizations to identify and regularly assess SEAH risks along the project cycle as well as

ensure that both the AE and all the EEs respond to SEAH incidents with survivor-centred and gender responsive approaches.

4.2 Gender policy

47. The AE has provided a gender assessment and action plan therefore complies with the requirements of the GCF Gender Policy.

48. The gender assessment provided demonstrates the commitment of Rwanda to advancing gender equality and women's empowerment. The assessment shows that Rwanda has a strong political commitment to promoting gender equality, reflected in its policy and legal framework and various sectoral strategies. It is the first country to have a majority female Parliament, with 61 per cent of these positions being held by women, and has established through its constitution a minimum of 30 per cent female representation in public governance. The country has also made international commitments such as ratifying and implementing the Convention on the Elimination of all Forms of Discrimination against Women, the Beijing Platform for Action and the Maputo Protocol on the Rights of Women in Africa. An institutional framework for coordinating and ensuring oversight of gender equality and women's empowerment is in place, which mainly rests with the Ministry of Gender and Family Promotion. The Ministry is mandated to coordinate the implementation of the national gender policy and advocacy on gender issues, a gender monitoring office and the National Women's Council. All ministries and public institutions have gender focal points. However, there is divergence between the frameworks in place and implementation on the ground due to challenges such as lack of financial resources and technical capacity within the various ministries as well as existing norms and practices that run contrary to equality principles and practices.

49. The gender assessment was conducted through desk review and stakeholder consultation. Additional stakeholder consultation will be conducted at national, district and project site level in order to further adapt project activities to the local context. The assessment indicates that women and men have different roles and responsibilities within society and households. These roles rigidly set women as caregivers and homemakers, while men are considered breadwinners engaging in income-generating activities, wage employment and decision-making. Women own less land, have less access to extension services and financial support (loans), and are less educated and able to gain experience in handling money. Women rely more on informal sources of loans as they do not usually have collateral to put up to access loans from financial institutions. Women engage and represent 66 per cent of the agricultural workforce; however, they occupy 19.7 of the paid jobs. Women's roles in afforestation and agroforestry is also curtailed by their lack of ownership and or land rights. Lack of land rights hinders the financial benefits women can get from resources on land such as trees. Women mostly use trees that provide little commercial value; rather they are focused on fulfilling their household needs through fuelwood, fodder, bean poles and fruits. Discussion with women had indicated that women observed shortages of fuelwood, fodder and wood suitable to their bean poles, and also that there is declining soil fertility and crop yields. Fuelwood also is the main source for cooking, which has health hazards for women, and gaining access to cleaner, improved cook stoves is a challenge. Men usually decide not to spend money to purchase the improved cook stoves that can directly address the multiple challenges faced by women. Female-headed households are identified as much more vulnerable to climate change impacts compared to male-headed households owing to the prevalent gender inequality. Engagement in alternative source of incomes for women poses its own challenges to women, one of which is affordability of required inputs, and this is particularly so for the beekeeping sector. The interest of the government in this area is also high, and investments have taken place, but only 23 per cent of women engaged in the sector.

50. The action plan includes activities that will address the challenges identified through the gender assessment. Activities contain corresponding indicators, targets, gender expertise and budget while also including timelines for each of the actions. The project will direct investments to the areas of equal access to resources (e.g. land), services (finance), capacity development, decision-making and also the participation of women in management, as beneficiaries/partners and as critical stakeholders in processes. Project interventions will respond to and address the growing pressure on natural resources, thus ensuring that the resilience and adaptive capacities of communities, including that of women and disadvantaged people, are built. The gender action plan is focused on ensuring (1) climate adaptation programmes and land-use planning address the priorities of women and youth; (2) women, youth and marginalized households have increased capacity to effectively engage in and benefit from forest management and forest restoration; (3) women, youth and marginalized households have improved capacity to engage in green jobs and livelihood activities; and (4) monitoring and evaluation effectively measures how women, youth and marginalized people are benefiting to enable the adaptive management of the project. A capacity-building component that focuses on strengthening cooperatives and village savings and loans institutions through gender mainstreaming across decision-making, membership, and products and services is included to ensure women are able to benefit from and engage in all aspects of the project. Private landowners who will be willing to engage in land afforestation activities will create private forest management units, which will then be assisted in forming and registering cooperatives that are at least 50 per cent women. Further, the selection of tree species will take into consideration women's usage and priorities. Women will be engaged and supported through knowledge acquisition, trainings, and skills development processes so that they benefit from the project implementation, particularly focusing on critical challenges such as rights to land and access to decision-making positions. Further, the challenge of rights to land will be addressed through advocacy work with men and local judiciary systems. This will be complemented by activities that will advocate for women's land rights and access to redress mechanisms. The project will engage community leaders and women and men community influencers to further promote equal and fair control of resources to mitigate the risk of men taking control over incomes earned by women.

51. There are three recommendations to the AE. The first recommendation is that the AE needs to ensure that gender issues are mainstreamed across all components of the project; this is lacking for component 3 (output 3.1 of the gender action plan), which presents many important opportunities to address gender issues. Secondly, the AE needs to refine and clearly show what outcomes it intends to achieve as a result of the interventions. Activity 3.1.1 in the funding proposal states the following regarding the composition of administrative committees: "Note that gender consideration is mandatory (at least 30% women)". It is not clear if 30 per cent refers to "women in decision-making bodies" or "membership of women in cooperatives", or both. These results are different in nature and should be captured independently. The AE should provide clarification in this regard. Thirdly, as part of its expressed commitment to do so, the AE should consistently collect data and report in a sex-disaggregated manner on the gender action plan.

4.3 Risks

4.3.1. Overall programme assessment (low risk)

4.3.2. Accredited entity/executing entity capability to execute the current programme (low risk)

52. We acknowledge the important role of the Government of Rwanda via MOE Rwanda as the AE (direct access). The AE demonstrated its comparative advantage and capability in

relation to this project with references to its track record in executing projects of comparable size or above (including FP073 and FP167 with GCF) and existing affiliation with the EE, RFA.

53. The AE provided adequate assurance that RFA has the necessary capacity to undertake the role as EE.

4.3.3. Programme-specific execution risks (Medium risk)

54. Co-financing: We acknowledge the sizable amount of co-financing totalling USD 10,493,579 (grants and in-kind) for a co-financing ratio of 1:0.27. If co-financing does not materialize then this may impede project implementation. The AE identified this risk, and the covenants in the term sheet are adequate mitigation measures – noting that the term sheet for the proposed project has not yet been agreed.

55. Project viability (and concessionality): The use of grants (and 100 per cent concessionality) is justified given the country context, the beneficiaries in Rwanda's CND, and the fact that the proposed activities are public goods in nature, including creating enabling opportunities for market development beyond the project to catalyse private investments (see component 3: Enhancing climate adaptation through resilient livelihoods).

56. Capacity-building activities and technology development/transfer (i) represent 45 per cent of total GCF funding requested; (ii) may not result in realizing expected results in case issues arise during implementation; and (iii) are embedded in the project's design and thus crucial to achieving intended objectives and outcomes. On the other hand, we note that identification of risks and the mitigation measures proposed by the AE are adequate.

4.3.4. Compliance risk (medium risk)

57. MOE Rwanda, in its capacity as AE, confirmed that it has conducted a risk assessment of money laundering and terrorist financing (ML/TF) in relation to project implementation. The probability of occurrence was assessed as 'low', and the impact severity in the event of occurrence was assessed as 'medium'.

58. MOE Rwanda confirmed that co-financiers have been identified based on their existing mandates and activities implemented in the CND region. Co-financiers' commitments will serve as a basis of MOE Rwanda engagement during the project implementation. During implementation, annual work plans for the project will be adjusted in collaboration with the EE, and all co-financiers are to align with all activities. MOE Rwanda, per its oversight role, will ensure that all of these stakeholders commit during planning to their co-financing and report this in the annual performance reports submitted to GCF.

59. MOE Rwanda indicated that the Government of Rwanda currently has robust measures to fight ML/TF risks (e.g. law N° 001/FIC/2022 on regulations relating to anti-money laundering and countering the financing terrorism). The Government of Rwanda has also put in place a special agency for this purpose, the Financial Intelligence Center (FIC), which has a clear mission to promote financial system integrity by conducting effective financial intelligence to counter ML/TF and related financial crimes in governmental institutions, projects and parastatal agencies under the supervision of the Ministry of Finance and Economic Planning. Both MOE Rwanda and this project under its supervision will also comply with FIC regulations.

60. MOE Rwanda confirmed that it has an internal audit function as well as regular audits conducted by the Office of the Auditor General to ensure the efficiency and effectiveness of internal controls, which includes handling ML/TF risks. In addition, MOE Rwanda uses segregation of duties within financial management processes.

61. MOE Rwanda confirmed that it has conducted a risk assessment of prohibited practices in relation to project implementation. The probability of occurrence was assessed as Low, and the impact severity in the event of occurrence was assessed as Medium.
62. MOE Rwanda indicated that to ensure materials or technology procured under the project are used only for the purposes intended and not diverted or misused for unauthorized, improper or illicit purposes, a comprehensive set of controls will be used. This includes explicit terms and conditions in all procurement contracts or agreements that clearly (i) state the intended purpose of the materials or technology being provided and explicitly prohibit any unauthorized use or diversion; and (ii) define the individuals that are authorized to access and use the procured materials or technology. Furthermore, this will involve the implementation of strong authentication and access controls to ensure that only approved users have access. This refers to the record of any materials or technology procured under projects within an assets register. The records include assets identification, the person using the material and the location.
63. Monitoring and auditing the usage of the procured materials or technology will be undertaken to detect any anomalies or potential misuse, including periodic site visits. Training sessions will also be conducted to educate all stakeholders involved in the project about the importance of adhering to the intended use of the materials or technology and the consequences of misuse. For technologies that involve data transmission or storage, MOE Rwanda will ensure that appropriate encryption and data protection measures are in place to safeguard sensitive information.
64. MOE Rwanda confirmed that no projects will be undertaken in any jurisdiction or territory that is subject to or affected by sanctions or restrictive measures per the United Nations Security Council (UNSC) resolutions. Additionally, no individual or entity that is listed on any UNSC sanctions list, including the United Nations Consolidated Sanctions list, will be involved in any manner with the project or its activities, either as a counterparty, implementer or beneficiary.
65. MOE Rwanda acknowledged that there are United Nations sanctions in the neighbouring country of the Democratic Republic of the Congo, close to the CND. According to MOE Rwanda, the United Nations Consolidated Sanctions List indicates approximately 20 individuals with ties to Rwanda, a small percentage of whom may live in Rwanda. MOE Rwanda confirmed that the United Nations Consolidated Sanctions List has been circulated internally and will be cross checked when hiring for the project and once project sites are selected to ensure these individuals will not be involved in any manner with the project or its activities, either as a counterparty, implementer or beneficiary.
66. MOE Rwanda indicated that the project involves many actors for successful implementation, including ministries, government agencies, non-governmental organizations, subgrantees, vendors, service providers, community members and beneficiaries. MOE Rwanda confirmed that there is specifically an intention to distribute or disburse to beneficiaries, both directly and indirectly, commodities and other items of value such as (i) the distribution of seedlings for trees that will produce fruits or contribute to enhancing biodiversity of farming systems; (ii) access to improved cook stoves; as well as (iii) the option for the distribution of bees to start honey operations. MOE Rwanda acknowledged that, as with any project infusing large sums of money directly or indirectly with multiple actors, corruption could occur.
67. MOE Rwanda indicated that the distributions will be coordinated through procured contractors. The contractors are hired based on the Government of Rwanda/MOE Rwanda procurement policies, which are aligned with GCF policies and include due diligence in the hiring and contracting process. As for direct community contracting, the intended beneficiaries are screened and certified by local authorities in the area of project interventions, which mitigates the risk of prohibited practices.

68. MOE Rwanda confirmed that for reporting and remedying any risks that may occur, MOE Rwanda will maintain a project-level GRM through which anyone can report the occurrence of prohibited activities. All complaints will be investigated accordingly. Any complaints believed to contribute to prohibited activities will be referred to the appropriate authorities in Rwanda.

69. MOE Rwanda indicated that all staff will be made aware of the prohibited activities, and all subcontractors/counterparties will be screened for current and past engagement in such activities and made aware of their obligations to alert the PMU and/or submit grievances related to such activities occurring as part of the project.

70. **Compliance risk rating:** The Secretariat has conducted a review of the project in accordance with relevant GCF Board-approved policies and does not find any material issue or deviation with respect to compliance issues. Based on available information for this funding proposal, the Secretariat has determined a compliance risk rating of 'Medium' and has no objection to this proposal proceeding.

4.3.5. GCF portfolio concentration risk (low risk)

71. In the case of approval, the impact of this proposal on the GCF portfolio concentration in terms of result area and single proposal is not material.

4.3.6. Recommendation

72. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall programme	Low
Accredited entity/executing entity capability	Low
Project-specific execution	Medium
GCF portfolio concentration	Low
Compliance	Medium

4.4 Fiduciary

73. MOE Rwanda is the AE for the project and will provide oversight to the project consisting of, inter alia: (a) entering into a subsidiary agreement with the EE (RFA); (b) managing and disbursing GCF funds to the EE; (c) reviewing financial expenditures and progress reports; (d) overseeing project implementation following the project document and annual work plans and budgets, agreements with co-financiers and each EE's rules and procedures; and (e) providing technical guidance to ensure that the appropriate technical quality is applied to all project activities.

74. RFA will serve as the EE for the project through a subsidiary agreement signed with MOE Rwanda. RFA will execute the tasks within the overall project management structure as follows: (a) implementing day-to-day activities as per the project work plan and budget, including the environmental and social management plan; (b) undertaking procurement activities as per the signed agreements; and (c) managing contracts of suppliers and service providers. The EE will enter into a memorandum of understanding (MOU) with different partner agencies, including WCS and the Catholic Organisation for Relief and Development, which will provide technical assistance services to implement the project activities. Procured parties will be contracted following MOE Rwanda rules and regulations for procuring goods and services. The PMU will manage, monitor and report on the day-to-day implementation progress of the EE.

75. As one of the project implementation partners, WCS will provide technical assistance to RFA through an MOU signed with RFA. The MOU will detail all the activities that WCS will directly support during this project's implementation. The technical assistance services will include advising RFA/SPIU, procuring and supervising international experts and consultants, and procuring materials to support the execution activities. Relevant technical experts from WCS will be embedded within RFA to advise SPIU on day-to-day project management. At the same time, the Catholic Organisation for Relief and Development will provide technical assistance to RFA through an MOU.

76. The co-financiers (RDB, RFA, RWB and MOE Rwanda) will execute and manage the implementation of their activities.

77. MOE Rwanda shall ascertain that the EE has efficient internal control systems in place, review financial reports and proceed with funds disbursements. It will also periodically inspect the project for compliance and coordinate audit exercises for the EE. RFA will enter into agreements with MOE Rwanda that outline general modalities of the management of funds in line with GCF grant requirements and RFA's policies.

78. GCF funds will be transferred to an RFA-designated bank account. Based on agreed annual work plans and budgets, the funds will be disbursed to the EE, and its reporting will be quarterly. The project's financial processes will rely on the EE's charts of accounts, internal control systems, disbursement plans, budgeting, expenditure management and accounting, reporting, and assets management as assessed by MOE Rwanda.

79. The project will be audited annually and commissioned by MOE Rwanda to produce certified financial statements and reports to ascertain that GCF funds are well managed. The project will maintain a regularly updated record of all assets and documents that will be retained for at least five years after the end of the project.

4.5 Results monitoring and reporting.

80. The Secretariat has provided various rounds of feedback on results monitoring and reporting, which have been generally addressed by the AE. This includes a review of beneficiary numbers, revisions to the logical framework, mapping of project outcomes to GCF results areas, and timelines and resourcing for the mid-term and final evaluations. However, there are some issues related to beneficiary calculation that are pending as of the final review. These should be appropriately addressed as the project is mobilized as follows:

- (a) It would be helpful to provide gender disaggregation for beneficiary numbers wherever possible. This is currently done in section E (logical framework), but not in section A6 and annex 23;
- (b) In section E (logical framework), the AE should also provide a brief description of how beneficiary numbers against GCF results areas have been calculated. These details are currently missing, and while we have mid-term and final targets for various indicators related to beneficiary count, we do not have information to understand how these have been calculated, and how potential double-counting will be avoided. If space is limited in section E, the AE may consider including this information in a clearly marked section in annex 23 related to beneficiary count.

4.6 Legal assessment

81. The original Accreditation Master Agreement was signed with the Accredited Entity on 23 June 2016, and it became effective on 1 August 2017, which agreement was amended and restated pursuant to the first amendment and restatement dated as of May 14, 2019, which became effective on even date. The Accredited Entity's first term of accreditation lapsed on 31 July 2022. The Accredited Entity was recredited by the Board pursuant to Decision B.34/17. The second amendment and restatement agreement amending and restating the accreditation master agreement was signed on 22 November 2022, and the second amended and restated accreditation master agreement (the "AMA") became effective on 26 January 2023.

82. The Accredited Entity has provided a certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

83. The proposed project will be implemented in the Republic of Rwanda. The GCF has signed a bilateral agreement on privileges and immunities with the Government of the Republic of Rwanda. It was signed on 8 November 2022 and became effective on 25 May 2023.

4.7 List of proposed conditions (including legal)

84. To address matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and
- (b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP217

Proposal name:	Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration
Accredited entity:	Ministry of Environment of Rwanda
Country:	Rwanda
Project/programme size:	Small

I. Assessment of the independent Technical Advisory Panel

1. This funding proposal for a small-sized project in environmental and social safeguards (ESS) category B is submitted by the Government of Rwanda as a cross-cutting project contributing to both adaptation (92 per cent) and mitigation (8 per cent) results areas. The Ministry of Environment of Rwanda (MOE Rwanda) (a GCF direct access entity) is the accredited entity (AE) for the proposed project, and the executing entity (EE) is the Rwanda Forestry Agency, with several implementing partners, including other government agencies and the Wildlife Conservation Society (WCS) in Rwanda. The total cost of the funding proposal is USD 50 million, of which USD 39 million is GCF finance and USD 11 million is co-finance from the Government of Rwanda.

2. The project's goal is to apply a landscape approach to reduce the exposure of populations in Rwanda's Congo Nile Divide (CND) landscape to the impacts of unfavourable climatic trends and hazards intensified by climate change, including floods, landslides, soil erosion and wildfires. This is to be achieved through integrated land-use planning and management that allows for protecting, restoring and managing a matrix of interconnected ecosystems, including pristine forest in protected area nodes, important landscape linkages, pockets of forest in stepping stones, and agro-ecosystems in farmlands. Interventions in various spatially defined target areas include forest protection and fire management, assisted natural regeneration, afforestation of steep slopes, and support to silvopastoralism and agroforestry and its value chains.

3. The project has three planned outcomes (see the table below), with outputs under each outcome as well as the parties responsible for executing them (see table 5 in the funding proposal).

Project outcomes and outputs	Executing entity per output	Technical implementation partners
Outcome 1: Landscape planning, policies and management are effective and coordinated across sectors and scales to address climate risks and adaptation benefits		
Output 1.1. Landscape-wide land-use plan developed for climate-resilient livelihoods and forest ecosystems, integrating district development strategies	Rwanda Forest Agency (RFA)	Wildlife Conservation Society (WCS) and National Land Authority (NLA)
Output 1.2. Local and national institutional capacities strengthened to integrate biodiversity and climate risks into land use planning and management	RFA	WCS and NLA

Outcome 2: Natural forests are protected, connected, and more resilient to climate impacts and risks		
Output 2.1. Protected areas management effectiveness improved to address climate risks and adaptation	RFA	WCS and Rwanda Development Board
Output 2.2. Natural forest cover restored; biodiversity connections established	RFA	WCS
Outcome 3: Vulnerable rural livelihoods are more climate-resilient, diverse, economically sustainable and nature-positive		
Output 3.1. Farming methods enhance productivity, reduce erosion and flooding risks, contribute to ecosystem services, and support connectivity	RFA	Rwanda Water Board
Output 3.2. Rural livelihoods generate alternative incomes and reduce pressure on forests	RFA	Cordaid
Output 3.3. Financial services and private sector investment engagement	RFA	Cordaid
Monitoring and evaluation	Project Management Unit, hosted by the Ministry of Environment of Rwanda	

1.1 Impact potential

Scale: N/A

4. The cross-cutting impact potential of the proposed project and the relative weighting of the intended results can be seen in the attribution of GCF and co-financiers' contributions to the relevant GCF results areas as follows:

GCF results area	GCF contribution	Co-financiers' contribution
<u>Mitigation Result Area 4 (MR4)</u> Mitigation through forestry and land use	8%	0%
<u>Adaptation Results Area 1 (AR1)</u> Enhanced resilience of most vulnerable people and communities	13%	49%
<u>Adaptation Results Area 2 (AR2)</u> Enhanced resilience of health and well-being, and food and water security	2%	0%
<u>Adaptation Results Area 4 (AR4)</u> Enhanced resilience of ecosystems and ecosystem services	77%	51%

5. The emphasis on utilizing the GCF investment to enhance the resilience of ecosystems and ecosystem services is reflected in the project's focus on new mechanisms of land-use planning and management that consider the CND as a landscape consisting of an interdependent mosaic of forests and farms, the management of each determining the health of the whole. The independent Technical Advisory Panel (iTAP) believes that such an approach (i.e. targeting project interventions on the ground based on a systematic spatial planning process that emphasizes "ecological infrastructure" functioning as well as biodiversity) represents best practice for enhancing ecosystem resilience. Such an approach ensures that interventions are at sufficient scale to achieve a transformative effect: preserving landscape-wide ecosystem processes and functions that help buffer impacts of climate change. Maximizing connectivity is also important for maintaining the long-term health and stability of forests as carbon sinks.

6. **Adaptation impact potential:** The funding proposal outlines the major climate change-related hazards characterizing the landscape, in particular more frequent intense rainfall events increasing floods and landslides, and worsening soil erosion that is already widespread as a result of direct anthropogenic pressures. The project's main adaptation benefits will accrue to

vulnerable communities primarily engaged in agriculture on the CND's steep terrain, achieved through improved soil management, forestry and agroforestry to reduce catastrophic events affecting production systems, lives and livelihoods. This will be achieved through integrated landscape planning and management to enhance tree cover in crop and grazing lands and improve the connectivity of natural ecosystems and the key services they provide, including regulation of runoff and flows from increasingly extreme rainfall events; and through reduction of localized and downstream flooding, erosion and landslide incidence and risk.

7. Indicators for the main adaptation results are as follows:

- (a) 135,104 hectares (ha) of land to be brought under climate-resilient management through targeted interventions for protection, afforestation and agroforestry (AR4):
 - (i) Including 121,758 ha of core protected areas, linkages and stepping stones that will be brought under climate-risk informed planning and fire management to maintain landscape-scale ecosystem services; and
 - (ii) Including 13,346 ha for focused interventions (i.e. degraded forest in parks to be restored (5,000 ha), riparian lands restored (1,500 ha), pastures brought under silvopastoralism (1,000 ha), steep slopes afforested (2,500 ha), and farmland with new agroforestry practices (3,346 ha));
- (b) 1,254,242 people, resident in the landscape where targeted interventions will be carried out, will benefit directly from reduced exposure to climate-intensified disasters and/or enhanced adaptive capacity (AR 1 and 2); and
- (c) 9,260,745 people, approximately two thirds of the country's population, will benefit indirectly from a wide-scale education and outreach programme, including radio programming, on climate change and gender issues, climate risks, value of forests and forest ecosystems, and climate change adaptation options.

8. **Mitigation impact:** The project seeks to deliver greenhouse gas (GHG) emission reductions and removal benefits equivalent to 224,871 tonnes of carbon dioxide equivalent (t CO₂ eq) during the five-year project implementation period, reaching a total of 1,084,291 t CO₂ eq after 20 years. The estimation of reductions and removals is made based on the difference between the projected emissions with and without the project interventions – through improved cookstoves, and through on-the-ground interventions across a total of 13,346 ha of land. The land-based interventions will have cross-cutting benefits, since they contribute to core indicator 4 on hectares of land to be brought under climate-resilient management, as well as bringing mitigation benefits.

9. The following activities will contribute to achieving this mitigation potential: (i) assisted natural regeneration (ANR) in 4,500 ha of Nyungwe National Park through invasive fern removal; (ii) restoration of 500 ha of degraded forest through ANR in Gishwati-Mukura National Park; (iii) afforestation on 2,500 ha of public and private land with a slope >55 per cent; (iv) planting 1,500 ha of protective forest in riparian zones (public and private); (v) implementing erosion control and agroforestry practices on 3,346 ha of smallholder farmland; (vi) implementing silvopastoralism on 1,000 ha in the Gishwati area; and (vii) adoption of improved cookstoves by 8,500 households. Indicators for achievement of these targets are included in the project logical framework in section E.3 of the funding proposal.

10. The methodology for calculating GHG emission reductions and carbon sinks mirrors two approved consolidated methodologies for the A/R CDM, AR-ACM0003 and AR-AMS0007. This is considered a good practice because the applicability conditions of the methodologies are appropriate to the project conditions. The methodology is clear and conservative, considers the relevant guidance from the Intergovernmental Panel on Climate Change, and was used in an appropriate, clear and transparent manner when estimating the mitigation potential for each activity.

11. With regard to monitoring and reporting mitigation benefits, annex 22 proposes a clear and straightforward approach, but does not clarify the roles and responsibilities of the various actors that will be involved in the process over time, nor the frequency of the steps. Furthermore, the activities as presented in annex 22 are not reflected in annex 11 (monitoring and evaluation plan_rev August 2023). According to its updated nationally determined contribution (NDC) (2020, page 6)¹ Rwanda has already developed a national measurement, reporting and verification (MRV) framework. At the present stage it is not clear how this project will be aligned with the existing arrangements for the national MRV system, and/or how it will support or contribute to enhancing the functioning of this system across the country.
12. The iTAP assesses the impact potential of the project as high.

1.2 Paradigm shift potential

Scale: N/A

13. The proposed project is designed to bring about a paradigm shift away from the current situation of fragmented, unplanned decisions and a collection of isolated sector-by-sector projects that fail to take into account evolving climate risk and allow continued degradation. Instead, a new approach of integrated land use planning and management is promoted that preserves ecological connectivity across the landscape and helps maximize valuable ecosystem services for buffering the impacts of climate change. The new approach is based on international best practice and considers the CND as a landscape consisting of an interdependent mosaic of production and protected lands, the health of which is completely interconnected.
14. The idea behind the project is that a new holistic, climate-smart, evidence-based approach to planning and development will enable systematic, ecosystem-based, multi-sector analysis of land-use tradeoffs. Focusing on this larger scale, with coordination across both geographic and sectoral boundaries, while mainstreaming climate risks in management planning, represents a fundamental paradigm shift for land-use planning and management in Rwanda. Success in putting such a system into place could represent a crucial first step in reversing the previous negative feedback loop of ecosystem degradation to achieve a more positive, climate-resilient system.
15. Utilizing integrated landscape management and ecosystem-based adaptation approaches in addressing climate change impacts in Rwanda offers a myriad of opportunities for scaling up and replication to similar ecological zones across the country. The project also has the potential to contribute to the regulatory frameworks and policies for driving climate investments in the forest and renewable energy sectors. The project will finance the development of an integrated land-use plan, which will be adopted by all 10 CND districts as a framework for seeking synergies between rural livelihoods and forest and biodiversity protection, reconciling land-use conflicts that arise, and guiding development decisions and climate change adaptation. It will also update the National Land Use Master Plan, ensuring it incorporates climate resilience and local plans.
16. **Systematic conservation planning:** Although biodiversity conservation is considered a co-benefit of the project rather than a core focus, the project makes innovative use of systematic conservation planning to identify spatial areas in the landscape that are targeted for restoration, afforestation and agroforestry activities. This includes work to restore national parks as the core of an interconnected system, with landscape linkages and stepping stones identified for interventions to promote connectivity between remaining areas of natural habitat. Optimizing this kind of mosaic of land uses with management actions for protection, restoration and sustainable use helps to maximize the functionality of landscape-wide ecosystem services such

¹ https://unfccc.int/sites/default/files/NDC/2022-06/Rwanda_Updated_NDC_May_2020.pdf.

as nutrient cycling, soil fertility, slope stability, water infiltration, flood management and precipitation.

17. **Innovative financial mechanisms:** The project also includes an output to establish long-term plans for CND financial sustainability post-GCF by employing a technical advisor to support the establishment of innovative mechanisms such as those identified in the Biodiversity Finance Plan and also to refine existing financing mechanisms (for example, integrating climate resilience principles into the existing successful and growing Tourism Revenue Sharing Program). Support will be provided to beneficiaries of the project in applying to the existing Community Adaptation Facility within the Rwanda Green Fund (FONERWA). The Government of Rwanda, represented by the Rwanda Development Board, is also in advanced stages of negotiation with the World Bank for establishing a wildlife bond, potentially a gorilla conservation bond, modelled on the successfully established rhino conservation bond in South Africa, with interest shown by impact investors.

18. Overall, based on the above, the paradigm shift potential of the project is assessed as medium to high.

1.3 Sustainable development potential

Scale: N/A

19. The proposed project is aligned with the United Nations 2030 Agenda for Sustainable Development, including significant contributions towards Rwanda's efforts to meet Sustainable Development Goal (SDG) 5 (Gender Equality), SDG 13 (Climate Action) and SDG 15 (Life on Land), and will contribute to the achievement of SDG 1 (No Poverty), SDG 2 (Zero Hunger), SDG 3 (Good Health and Well-being) and SDG 10 (Reduced Inequalities).

20. **Environmental co-benefits:** Key environmental co-benefits (beyond the core benefits related to ecosystem-based adaptation) include restoration of soil fertility through silvopastoralism and regenerative mixed-use agroforestry, and the enhancement of biodiversity through planting 50 per cent native tree species. Promotion of improved cookstoves using significantly less fuel will contribute to relieving pressure on the remaining indigenous forest.

21. The iTAP was surprised to see the relatively low target of 50 per cent (30 per cent on private land and 70 per cent on public land) in the use of indigenous tree species in afforestation, reforestation and agroforestry interventions. The AE explained that exotic species are more suitable for afforestation of steep slopes, where they can easily establish themselves, "but this does not mean that exclusively exotic species will be used on such sites; there will be a mixture of both wherever applicable. In addition, while establishing blocks in the Participatory Forest Management Units, indigenous species will be used to separate one block from another. Indigenous species will be given due consideration in agroforestry to create positive interactions in the farm as well as promoting native species, avoiding monoculture and benefiting from their positive impacts in terms of resilience."

22. It is understood that there is often a preference for fast-growing, easily available and usually cheaper species like eucalyptus, pine and alnus for urgent needs such as slope stabilization and fuelwood, and also that a balance needs to be struck between promoting indigenous trees species and being responsive to farmers' wishes. At the same time, the project could potentially make more of the opportunity to rekindle traditional knowledge about indigenous species and their uses, and to build the nursery supply chain for quality native planting material given the vast superiority of indigenous tree species in terms of their multiple benefits, non-invasiveness, positive impact on soil and water quality, and resistance to pests and diseases that may worsen with a warming climate.

23. **Social and health co-benefits:** In enabling the transition to more climate-resilient and productive livelihood models, the project aims to increase household incomes and the nutritional status of families. It is expected that significant social capital will be created through

beneficiaries' participation in co-designing and co-managing a range of adaptation strategies, with an emphasis on women's participation. Social cohesiveness will also be enhanced through the strengthening of joint saving and lending systems. Social benefits will also accrue to indirect beneficiaries, for example those listening to radio programming that leads to increased awareness of climate change, or of opportunities for climate-adaptive agroforestry and agri-processing.

24. **Economic co-benefits:** Inequality and poverty will be addressed through increased financial inclusion and improved economic opportunities, especially for women, youth and historically marginalized groups. Forest and landscape restoration in the CND will bring economic benefits, since soil erosion control measures, protective forests on public lands, and agroforestry on private lands will all increase agricultural productivity and associated income. The project will create direct and indirect employment opportunities, benefiting an anticipated 24,212 households and stimulating the local economy. Private capital will be leveraged to invest in value chains, including those for improved cookstoves, irrigation pumps and beehives, as well as honey, woodlots, macadamia, avocado and vegetable production and processing.

25. **Gender benefits:** The project aims to achieve positive outcomes around access to resources, improved livelihoods, and income generation opportunities and capacities for women. With support from a dedicated gender specialist, the project will engage women in project planning, investment and decision-making. It is envisaged that women will respond to incentives that address their families' basic needs, such as better health and nutrition, linked to agriculture and food security improvements. The project will also get support from non-governmental organizations that engage men in programming and discussions to increase their participation in household and parental responsibilities and enhance their awareness of the importance of women's fair access to and control of resources, including income. Women will benefit from training and educational activities, leadership, entrepreneurship and decision-making, as shown in annex 8 (Gender Analysis/Assessment and Gender and Social Inclusion Action Plan).

26. The iTAP assesses sustainable development potential as high.

1.4 Needs of the recipient

Scale: N/A

27. Western Rwanda and its CND region are highly vulnerable to climate change, and are expected to experience a continuing upward trend in average annual temperatures, total annual rainfall (with a longer dry season), and more intense rainfall events and storms and increasing risk of flash floods, landslide, soil erosion and wildfires. Despite Rwanda's solid economic progress since 2000, poverty remains widespread and pervasive. Overall, 38.2 per cent of the country's population lives in poverty and 16 per cent in extreme poverty. Women are more affected by poverty than men, with 47 per cent of female-headed households living in poverty compared with 44.9 per cent of all households.

28. The CND landscape supports nearly one third of Rwanda's population on barely one fifth of its land base. Average population density is 736 people/square kilometre (km²), with some districts now exceeding 1,000 people/km². The highest rates of rural poverty in the country are reported from the CND, including its most vulnerable subsistence farmers. While high rainfall has historically permitted generally sufficient food production, the region's rugged topography requires farming on extremely steep slopes. With more frequent extreme rainfall events in recent years, this agricultural landscape is recognized as the region of Rwanda most vulnerable to flash floods and landslides. Between 2013–2016, heavy rains generated landslides that killed at least 174 people and destroyed the homes of more than 5,000. Landslides wipe out highly productive raised-bed cultivation systems in the CND's valleys and bottomlands, many hectares of crops are washed away, and an estimated 1.5 million tonnes of fertile soil are lost annually to erosion.

29. The funding proposal reports that during stakeholder consultations, communities indicated several socioeconomic changes stemming from changing climate conditions. Notably, the longer dry season and unpredictable rains have reduced agricultural yield significantly, leading to more widespread malnutrition and extreme poverty, and farmers can barely meet their own families' needs. In particular, women, youth, and historically marginalized people are feeling the impacts. As arable land is increasingly constrained, youth and historically marginalized people have few opportunities and are more focused on survival and short-term gains. This becomes a negative feedback loop, in which poverty leads to unsustainable practices such as overharvesting from forests and cultivation on extremely steep slopes, with degraded ecosystems then less able to provide services, worsening the cycle of poverty.

30. As a result, the Government of Rwanda has invested significantly in developing an enabling policy and institutional framework for a more climate-resilient development pathway. Yet, while adaptation is a top priority, as a least developed country, Rwanda lacks sufficient public financial resources to implement its climate change and resilient green development objectives. Currently there are no alternative funding options at the scale needed to secure Rwanda's priority forests to reduce the vulnerability of rural communities to climate risks. Rwanda's debt-to-gross domestic product ratio stands at 64 per cent, making further loans difficult, and the Government of Rwanda seeks maximum concessionality from the GCF for the urgent adaptation actions proposed under the project.

31. The Government of Rwanda recognizes that a piecemeal approach to decision-making has exacerbated competition over scarce land resources and reduced the adaptive capacity of both ecosystems and rural populations. Additionally, district government staff and community members have limited technical understanding of the complex interactions between forest structure and functions, climate change, and the provision of ecosystem services, and limited technical capacity for complex forest monitoring. In the face of worsening climate change impacts, there is an urgent need for a comprehensive approach that crosses siloes of government and achieves integrated land use planning and management informed by the evolving climate and socioeconomic conditions. The proposed interventions are well targeted to address the identified barriers and the great needs of these vulnerable communities.

32. Overall, based on the above, the needs of the recipients are rated as high.

1.5 Country ownership

Scale: N/A

33. **Policy and strategy alignment:** The project is closely aligned with Rwanda's national adaptation plan and three of its adaptation priorities: (i) promotion of non-agricultural income generating activities; (ii) introduction of species resistant to environmental conditions; and (iii) development of alternative sources of energy to firewood. It contributes to several priorities identified in the 2020 NDC: promotion of afforestation/reforestation; improvement of forest management; an integrated approach to planning and monitoring for sustainable land management; and development of a harmonized and integrated spatial data management system for sustainable land use management.

34. The project will support the implementation of Rwanda's Green Growth and Climate Resilient Strategy and the Rwanda National Strategy for Transformation (2017–2022) and is aligned with the goals of the Rwanda National Forestry Policy 2018, the Forest Sector Strategic Plan 2017–2022, the Strategic Plan for the Environment and Natural Resources Sector (2018–2024) and the National Forest Management Plan 2017–2026, thus reflecting the government's intention to address climate change impacts and mitigation targets by improving forest management. It is also aligned with the national Biomass Energy Strategy 2020, which aims to ensure a more sustainable biomass supply and combustion technologies for homes and small enterprises.

35. **GCF country programme:** According to the funding proposal (section D.5), the project aligns with the forthcoming GCF country programme, which “prioritizes forest management through restoration and protection of natural forests, afforestation and agroforestry, as well as distribution of high-quality reforestation inputs and capacity development”. While this project proposal to GCF focuses on a different geographic area, the early and ongoing lessons from the experiences from existing GCF-funded projects “Strengthening Climate Resilience of Rural Communities in Northern Rwanda”² and “Transforming Eastern Province through adaptation”³ will be applied during the implementation of this project in the CND landscape.

36. **Roles of key entities:** The proposed project was initiated by MOE Rwanda and supported by the national designated authority, Rwanda Environment Management Authority, due to the high vulnerability to climate change of the CND region of the country. MOE Rwanda is well placed to oversee the project as AE since the ministry is responsible for (i) ensuring the sustainable management of natural resources and for the development of policies, strategies and regulations; and (ii) mobilizing resources for the development of the sector. MOE Rwanda managed USD 10 million from the Adaptation Fund to implement the Reducing Vulnerability to Climate Change in North West Rwanda through Community Based Adaptation Programme⁴ and has overseen a number of other climate projects, including: an early warning system project (funded by the Least Developed Countries Fund (LDCF)), the Landscape Approach to Forest Restoration and Conservation and the “Building resilience of communities living in degraded forests, savannahs and wetlands of Rwanda through an ecosystem management approach” programme (funded by LDCF)⁵.

37. Given the centrality of forest protection, restoration and management to the project, the RFA is well placed to carry out the role of EE. It is noted that this would be RFA’s first GCF EE role, since MOE Rwanda’s other GCF project is executed by FONERWA. Strong support will be provided by a set of implementing partners – including the WCS and government agencies such as RDB, which are responsible for the national parks, RWB, Rwanda Environment Management Authority and the National Land Use Authority. The project activities overseen by each are shown in the table in the introduction above and are aligned with their mandates, as well as sectoral plans and strategies.

38. **Consultation and engagement with civil society:** To better understand the interests of stakeholders and potential impacts of the proposed project, and as a preparation for designing future engagement strategies, a comprehensive, gender-responsive, culturally sensitive, non-discriminatory, and inclusive stakeholder analysis was undertaken and is reflected in the Stakeholder Engagement Plan (annex 7). In terms of future engagement, the funding proposal indicates a number of specific areas where consultations will be ongoing to inform the rollout of activities, for example, consultations with communities and civil society organizations representing men, women, youth, and historically marginalized people in local communities, as well as district and national government agencies, on the following activities: Activity 1.1.1. Synthesize and disseminate information on value of natural forests and ecosystem services; Activity 1.1.2. Develop climate-resilient landscape land-use plans; Activity 2.2.3 Promote silvopastoralism with indigenous trees around Gishwati; Activity 3.1.1 Restore high slope areas (>55 per cent) as protective forests; and Activity 3.2.1 Develop an agroforestry-related value chain for market access.

²Rwanda Ministry of Environment, 2018. FP073: Strengthening Climate Resilience of Rural Communities in Northern Rwanda. Green Climate Fund.

³International Union for Conservation of Nature, 2021. FP167: Transforming Eastern Province through Adaptation. Green Climate Fund.

⁴Adaptation Fund, 2014. Reducing Vulnerability to Climate Change in North West Rwanda through Community Based Adaptation. [Link](#).

⁵United Nations Environment Programme, 2016. Building resilience of communities living in degraded forests, savannahs and wetlands of Rwanda through an ecosystem management approach.

39. Country ownership of the project is thus seen to be high.

1.6 Efficiency and effectiveness

Scale: N/A

40. Of the project's total proposed budget of USD 50,000,000, 78 per cent (USD 39,056,579) is proposed to be a GCF grant, and 22 per cent (USD 10,943,579) is made up of government co-financing to the project. The Government of Rwanda is providing grant financing via RFA, RWB, and RDB to support activities in component 3. In addition to these contributions totalling USD 10,831,307, the AE will cover office costs in the form of an in-kind contribution equivalent to USD 112,271 for the Project Management Unit, which will be hosted at MOE Rwanda.

41. **Government co-finance:** In response to a question from iTAP about the nature of the co-finance and the extent to which it represents new and additional funds, the AE explained that "[t]he co-financing commitments were provided based on the existing Government commitments provided every year in the project intervention area. These are earmarked funds that go into forest management, community revenue sharing around the parks, monitoring resources for catchment restoration interventions and also different in-kind contributions. All the details are indicated in the Annex 4 of the budget and activities to be covered by co-financing resources and Annex 13 indicates the commitment letters from the co-financiers." It is iTAP's understanding that these funds will not be spent in a business-as-usual manner, but will be newly directed towards climate risk management, through support to the project's adaptation and mitigation interventions and efforts to develop officials' capacity for climate risk-informed integrated land use planning and management.

42. **Role of financial institutions:** Participating financial institutions (PFIs) will be important in two ways – firstly as providers of credit to farmer cooperatives, savings groups and small enterprises in the targeted value chains, and secondly as recipients of training – to enhance their knowledge of good agricultural practices, equip them to assess the impact of climate change on their clientele and loan portfolio, and help ensure equitable access to financial assets by women, youth and historically marginalized communities.

43. In response to a question from iTAP about whether any consultation has happened with the potential PFIs, which are not mentioned in the extensive stakeholder consultations reported in annex 7, the AE stated that "At this stage [neither] Cordaid nor RFA have notified the participating financial institutions. As per the project implementation plan, a 6-month to 9-month inception phase will include in-depth analysis of existing financial institutions including microfinance institutions and saving and credit cooperative societies. They will be scored against a list of criteria to understand their leadership, financial position and financial products portfolio on a case-by-case basis."⁶

44. The iTAP believes this engagement will be crucial to the success of Output 3.3 on "Financial services and private sector investment engagement" in order to ensure sustainability of livelihood interventions, and urges the AE to start dialogue with potential partners as soon as possible since they will have experience and advice that can guide the way this output is approached for maximum effectiveness.

45. **Level of concessionality:** The project's proposed 100 per cent concessionality level is justified because the project is investing in multiple ecosystem goods and services that are public in nature, including landscape mosaics, at the core of which are public assets, in the form of national parks and other protected areas, as well as landscape linkages and stepping stones to promote connectivity and flood management, and steep slopes to be afforested to prevent landslides and erosion. Benefits will accrue also to privately owned small-scale family farms, but these are on a very small scale, and any profits made through processing and sale of

⁶ MOE Rwanda responses to iTAP, 4 September 2023.

agroforestry products will need to be retained at household level to buffer against shocks and stresses to increase and diversify incomes, and to improve families' nutritional status.

46. **Market transformation for cookstoves:** The project aims to avoid challenges which the feasibility study says were experienced with previous improved cookstoves (ICS) programmes in the region, which "have historically been characterized by poorly designed subsidy/incentive programs which in some instances have provided 100% subsidies to end-users" and have sometimes led to "unnecessary long-term dependence and poorly managed exits, which consequently created long-lasting ICS market distortions". Instead, the project adopts a market-oriented approach to strengthening the improved cookstove supply chain in the region, creating "strong linkages and business relationships among the key actors, including cookstove producers/youth small and medium-sized enterprises, farmers/savings groups, and microfinance institutions to ensure sustainable access and finance of cookstoves".

47. In collaboration with technical and vocational education and training institutes, youth producers, central and local authorities, and experts on biomass and micro-finance will refine and test, for each type of ICS/fuel to be promoted, the category/profile of households to be targeted and the related subsidy/microcredit schemes to be implemented to facilitate stove adoption with minimum concessionality. Subsidized cookstoves will be supplied to 4,000 poor households identified as being in a category of vulnerable groups, including youth, women, landless and disabled people. The stoves will be locally manufactured, with a competitive process leading to the award of a supply contract to local women- and youth-owned cooperatives or start-ups. A partnership with two integrated polytechnic regional colleges will enable the design of prototype affordable Tier 2 ICS customized to local conditions and customer preferences.

48. After year 4, the start-ups that will have graduated from the training and incubation programme (also conducted for the manufacture of beehives and irrigation pumps) will be fully empowered to maintain the supply of new equipment and repair of old equipment, as well as to market their products to producer organizations and households. Access to microfinance institutions will be facilitated to allow emerging cookstove business to grow. The members of village savings groups will be able to use income generated from the sale of fast-maturing crops (chili and French beans), coupled with savings accumulated from their groups, to buy new or replacement stoves over time. A specialized small business development fund will be established to support dissemination, accompanied by public awareness-raising of the benefits of ICS.

49. **Return on investment:** The project cost per direct beneficiary of USD 31 (USD 40 with co-finance) and 20-year economic rate of return of project interventions for agroforestry (49 per cent), protective forests (72 per cent), and silvopastoralism (51 per cent) all highlight the project's viability, as does the economic net present value of USD 10,586 for the improved cookstoves intervention. The mitigation cost is high, at an estimated USD 43.84 per t CO₂e, but comes with a multiplicity of adaptation benefits and socioeconomic and environmental co-benefits. The cost of restoring natural forest for providing both adaptation and mitigation benefits is significantly greater than the cost of simply planting trees for mitigation, since a complex multi-strata, multi-trophic level natural ecosystem (like a tropical rainforest) provides more varied and in many cases more robust ecosystem services than a simpler ecosystem (like a monocrop plantation). Restoration of natural forest in Rwanda is essential for sustaining the critical ecosystem services on which vulnerable rural communities and the national economy depend, and are now even more essential for buffering intensifying climate change impacts.

50. **Best practices:** The project will use Rwanda's best practice guidelines for tree nursery establishment, tree planting and tree management. For assisted natural regeneration, methods developed and proven through multi-year research in Nyungwe and Gishwati-Mukura National Parks will be used. The project will use multiple contemporary methods for monitoring the success of project interventions, as well as forest coverage and condition. Forest plantations will

be mapped and monitored using the Rwandan Government's newly developed Forest Monitoring and Evaluation Software specifically developed for this purpose. The project will develop a modern indigenous forest monitoring system, utilize the most high-resolution data sets currently available (<3 metres) and take advantage of recent remote-sensing advances, including cloud computing and machine learning. Although replication of effective and cost-effective project techniques through knowledge-sharing with other countries in Africa is not planned in the project design, this could be a by-product of the project.

51. The project's overall efficiency and effectiveness is rated as medium to high.

II. Overall remarks from the independent Technical Advisory Panel

52. The iTAP believes this cross-cutting public sector project from the Government of Rwanda is an important one that will make significant strides towards effective management and sustainable financing of forest-and-farm landscapes that play a vital role in climate change adaptation and mitigation in the Congo Nile Divide.

53. The iTAP recommends to the AE that they undertake the following actions to strengthen the measurement, reporting and verification (MRV) of the project's core mitigation benefits:

- (a) Clarify the roles and responsibilities of the various actors who will be involved in the steps of the MRV process over time, and the frequency of these steps;
- (b) Capture the MRV activities as presented in annex 22, also in annex 11, such that they become part of the project's overall monitoring and evaluation system; and
- (c) Clarify how this project will be aligned with the existing arrangements for the national MRV system and/or will support or contribute to enhancing the functioning of this system across the country.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP217)

Proposal name:	Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide through Forest and Landscape Restoration
Accredited entity:	Ministry of Environment of Rwanda
Country/(ies):	Rwanda
Project/programme size:	Small

Impact potential
The iTAP assessment of the impact potential of the project as high is well acknowledged and no further comments from the AE.
Paradigm shift potential
iTAP assessment as Medium to High is acknowledged by AE. The project is going to utilize integrated landscape management and ecosystem-based adaptation approaches in addressing climate change impacts in Rwanda which offers opportunities for scaling up and replication to similar ecological zones across the country.
Sustainable development potential
iTAP assessment as High is acknowledged by the AE.
Needs of the recipient
iTAP assessment as High is acknowledged by the AE
Country ownership
iTAP assessment as High is acknowledged by the AE
Efficiency and effectiveness
iTAP assessment as Medium to High is acknowledged by the AE. The AE has received the confirmation letter regarding the co-financing from the Rwanda Development Board, Rwanda Water Resources Board and Rwanda Forestry Authority to support complementary activities of this project. The AE will report on the co-financing in line with GCF Policy on co-financing annually in APRs.
Overall remarks from the independent Technical Advisory Panel:
The AE acknowledges iTAP overall assessment and recommendation for the Board approval.

**BUILDING RESILIENCE OF VULNERABLE COMMUNITIES TO
CLIMATE VARIABILITY IN RWANDA'S CONGO NILE DIVIDE
THROUGH FOREST AND LANDSCAPE RESTORATION**

**ANNEX 8 - GENDER ANALYSIS/ASSESSMENT AND GENDER &
SOCIAL INCLUSION ACTION PLAN**

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- **EXECUTIVE SUMMARY**

Since 1994, Rwanda has made significant progress in empowering women and countering the numerous inequities and discrimination to which women were historically subjected. Today Rwanda has one of the most progressive, gender equitable policy frameworks of any country in Africa, and has established and funded the Ministry of Gender and Family Promotion and the Gender Monitoring Office to oversee the implementation of these gender progressive policies.

There is no single definition nor understanding of gender equity and women's empowerment, and the meaning, and reality of these terms differs across communities and cultures and over time. That said, the Government of Rwanda holds the simple belief in the fundamental equality of all people. In a just society, men and women of all ages, at a minimum, should have the same rights to self-determination, should be equally represented at all levels of political leadership, and should not be subject to discrimination or gender-based violence.

Today rural women continue to play a vital role in ensuring the wellbeing and security of their families. But being the caretaker of the family is getting harder for women as farmland becomes increasingly scarce, as soil erosion and declining soil fertility lowers crop yields, and as fuelwood, fodder and clean water take ever longer to find and carry home. Changes in rainfall timing and intensity, driven by climate change, will further exacerbate the challenges women will face to meet the needs of their families.

Women, youth and marginalized groups are the primary targeted beneficiaries of the Green Climate Fund (GCF) proposal "Building Resilience of Vulnerable Communities to Climate Variability in Rwanda's Congo Nile Divide Through Forest and Landscape Restoration." Empowering women, youth and marginalized groups to play an equal role in land and resource use decisions will ensure that their interests and concerns are incorporated into plans designed to increase the wellbeing and security of families in the Congo Nile Divide (CND) by helping them to evolve their livelihood strategies and practices in ways that adapt to climate change.

Investing in women's lives and strengthening their capacity to overcome existing and looming challenges will do much to implement Rwanda's progressive gender policies and help reduce the continuing inequities and discrimination that rural women, in particular, still face in their daily lives. It will also reduce poverty, the incidence of child failure to thrive, and food and income insecurity within families across the CND. By ensuring a robust integration of women and youth in programming, the resilience of households in the Congo-Nile Divide will be strengthened.

The following summarizes the primary desired outcomes for women and youth that will be generated by implementing the four components of the GCF program in the CND.

Component 1: Mainstreaming Climate Adaptation into Integrated Land Use Planning

- Outcome 1: Climate adaptation programs in LUPs reflect and address the priorities of women and youth
- *Component 2: Forest Landscape Management and Restoration*
- Outcome 2: Women, youth and marginalized households have improved capacity to engage in forest management and restoration
- *Component 3: Enhancing climate adaptation through resilient livelihoods*
- Outcome 3: Women, youth and marginalized households have improved capacity to engage in green jobs

Component 4: Monitoring & Evaluation

- Outcome 4: Disaggregated and specialized monitoring generates data showing whether and how women, youth and marginalized people are benefiting from adaptive management systems

- **INTRODUCTION**

Decades of project design, implementation and impact evaluation have demonstrated conclusively that gender equality and women's empowerment are necessary conditions for effective investment and action to achieve environmental conservation, economic development and climate change resilience. This program seeks to ensure that women play equal roles in determining how to improve the wellbeing and security of their families and communities in ways that maintain the agro-ecological systems, biodiversity, and environmental services essential to the economies, nutrition and safety of their families in the Congo-Nile Divide (CND) now and in the future.

Our approach goes beyond trying merely to ensure that women are no longer ignored or marginalized but seeks to empower women so that they can decide how they can best engage in determining what support would best help them and their families to increase food and income security and resilience to climate shocks. To do this we need to complete a thorough gender analysis that identifies the differences in men and women's lives, including those that lead to sociocultural and economic inequalities, and applies this understanding to project development.

This gender analysis attempts to answer four questions:

1. What roles do men and women play in determining wellbeing objectives and the allocation of labor and capital, and how do these differ in male versus female headed households?
2. What are the inequalities, discriminations and rights denials faced by men and women in different social, economic and political contexts? How do these issues intersect with other factors such as age, ethnicity, disability, wealth class, etc.?
3. How might gender inequity impact on the effectiveness and sustainability of the project investments, actions and desired outcomes?
4. What are the risks that project efforts to reduce gender inequality by empowering women, actually exacerbates inequities and further undermines the status, and agency of women?

The purpose of this gender analysis is to ensure that the project proposal design and implementation will be informed by a thorough understanding of gender roles, power relations and dynamics. This assessment provides information to address the relevant environmental and climate change critical issues from a gender perspective. Information generated during the analysis will ensure that sufficient and appropriate attention is paid to gender issues within all CND project interventions.

Furthermore, these gender inclusion methods fall in line with the GCF's gender mainstreaming guidelines, designed to involve women, youth, and the historically marginalized in all climate adaptation, mitigation, and resilience strategies. The gender assessment will provide the foundation for a credible and practical gender action plan to be implemented by the GCF CND project.

1.1 Purpose of the Gender Analysis

Increased greenhouse gas emissions linked to human activity continue to threaten the livelihoods,

safety, and wellbeing of communities worldwide. However, climatic impacts will vary based on numerous factors, including geography, socioeconomic status, age, and gender. Women experience unequal access to resources, land, and power, reducing their ability to adapt to climate change as effectively as men. Women face greater domestic responsibility to maintain the home and raise children, reducing their time for education and careers. As a result, women are more tied to their land and more vulnerable to climatic impacts, yet also represent a potential group of leaders to spur mitigation and adaptation efforts.

Given the potential of women as environmental pioneers, climate resiliency efforts require a thorough understanding of gender roles in the region. MoE recognize the inextricable link between sustainable development of communities through gender planning and the conservation of wild places. Effective conservation strategy requires working to balance gender inequity and providing opportunities for women in the conservation space. Furthermore, the high dependence of women on agriculture as livelihood in Rwanda impacts both conservation and climate change mitigation efforts. Smart conservation programs therefore target agricultural interventions to aid marginalized groups such as women and youth. Such projects offer employment opportunities and reduce the threat of illegal or paramilitary activities that threaten conservation outcomes.

Shifting the imbalance of power and access for women is not only a conservation target, but a human rights issue. Sustainable development at all levels, local, national, and international, requires the empowerment of women. When women receive higher education, they have more flexibility in careers and are more likely to escape poverty. These women strengthen economies through diversified incomes, marry and have babies later, and have greater independence (NISR, 2018). However, climate change is likely to worsen financial stressors, particularly in rural communities, and increase the risk of gender based violence (GBV). As the frequency of extreme weather events increases, understanding how climate change impacts women is essential to combatting the human rights issue of gender equality.

There are numerous international human rights agreements that provide a framework for gender equity. Rwanda has signed on to several of these commitments, including the Convention on Elimination of all Forms of Violence against Women (CEDAW), the Maputo Protocol on the Rights of Women in Africa, and the Convention on Slavery and Repression of Human Trafficking (IUCN, 2021). Participating in these commitments requires periodic progress reporting and helps hold countries accountable.

In Rwanda, the government has made significant strides to address gender disparity following the Genocide. Thirty-four percent of households became run by widows post-Genocide, demonstrating an obvious need for the Rwandan government to address the newfound elevated standing of women (Vanhees, 2014). Several laws help ensure that on paper, Rwandan women have increased access and protection under the law. Such policies include the 1999 Inheritance and Marital Property Law (Law/n^o 22/99 of 12/11/1999), the 2008 Girls' Education Policy, and the 2010 National Gender Policy (IUCN, 2021). However, these policies are new and compete with a long history of patriarchal norms in Rwandan society, particularly outside of urban areas. Practice of these policies remains inconsistent, especially when women do not know their rights. Helping men and women understand gender policy is the first step to shifting cultural perceptions. But gender equality cannot be achieved by women alone, it also requires the active participation of men to accept the deconstruction of traditional gender roles and be aware of the intersectionality of women and other groups. For example, considering how to include youth and marginalized communities, especially the women in those groups, in equality strategies will

help lift up the most at-risk groups and reduce illegal activity.

1.2 Methodology

This Gender Assessment uses a range of primary (24 focus groups and 18 key informant interviews) and secondary (published reports, policies and academic literature) information sources to triangulate our understanding of the gender context for climate resilience and the project design.

Key Informant Interviews (KII) were conducted, using purposive sampling, with senior decision makers in governmental organizations at local, district and national levels. Results from these interviews were validated and enriched by 24 Focus Groups (FG) with 58 men and 91 women of different age groups within 21 communities bordering Nyungwe, Gishwati and Volcanoes national parks. Originally a large number of consultations was envisioned, but because of COVID constraints, including limitations on travel between districts, a smaller sample was completed. Additional stakeholder consultations will be conducted at national, district and project site level to inform the implementation of the project and to adapt activity implementation to the local context. These consultations will include the review of Environmental and Social Frameworks included in Annex 6. A desk study of secondary data sources gleaned from government reports and published policies, grey literature produced by civil society groups, and peer-reviewed academic papers has also fed into the report.

The following desired outcomes helped guide our interviews, the desk study and the analysis of all information garnered:

- Gender roles, needs, concerns and aspirations in the context of the proposed project are well understood and documented;
- Root causes of existing gender inequalities in the context of the proposed project are identified;
- Proven options for redressing traditional power imbalances between men and women are identified;
- Increased confidence that gender focused investments and actions proposed by the project will deliver the desired results.

The information generated from the gender analysis and assessment will be considered in all stages of the CND project cycle: design, work planning, implementation, and monitoring and evaluation. In each of these stages, managers will both be enabled and encouraged to view the project through a 'gender lens' so that project investments and actions:

- Reconcile gender inequalities that pre-date and might emerge from the project;
- Ensure the differential needs and concerns of women and men are addressed;
- Ensure women and men have equal access to resources, services, and human capital development;
- Ensure equal participation of women and men in management arrangements and as beneficiaries, partners and key stakeholders; and
- Ensure women's equal participation in decision-making processes.

• SITUATION ANALYSIS

1.3 Policy framework

The implementation of gender focused policy and legislation after 1994 has provided Rwanda with a strong legal framework for advancing women's status and rights. The high prevalence of female and orphan headed households, as a result of the Genocide, required the Rwandan government to reassess the country's property ownership laws. The 1999 Inheritance and Marital Property Law provided women with the right to own and inherit property, protecting widows and orphans from losing their land to male family members (Vanhees, 2014). However, this law only applies to formally registered, monogamous unions, and the children of such unions, denying inheritance rights to women engaged in consensual cohabitation or polygamous relationships and illegitimate children. Full equal rights to land titles and use of land for women were established by Law N° 43/2013 of 16/06/2013 Governing Land in Rwanda (IUCN, 2021). Furthermore, while the 1999 Inheritance and Marital Property Law established equal inheritance rights for male and female children, it was not mandated that all children should inherit equal amounts of property until Law N° 27/2016 of 08/07/2016 Governing Matrimonial Regimes, Donations and Successions (IUCN, 2021).

The Constitution of the Republic of Rwanda of 2003, revised in 2015, outlawed all forms of discrimination, provided legal equality within recognized marriages, and set a minimum of 30% female participation in the Senate and other public governance (East African Community, 2009). These high gender equality standards have helped propel Rwanda to both become the first country with a majority of female representation in Parliament and rank 7th out of 156 countries for the smallest gender gap (World Economic Forum, 2021). The 2010 National Gender Policy functioned to further advance gender mainstreaming and equality, confirming the Rwandan government's stance on gender as a cross-cutting issue that must be addressed for successful development (Republic of Rwanda, 2019).

This foundational principle of gender equality as a driving factor for growth is highlighted in Rwanda Vision 2020 and Vision 2050, which have been used as frameworks for Rwanda's development (Republic of Rwanda, 2000; Republic of Rwanda, 2020). While the primary objectives of these frameworks were to elevate Rwanda to a middle-income country by 2020 and a high-income country by 2050, these goals could not be accomplished without addressing gender inequality. Strategies for mitigating gender inequality included updating existing laws to account for gender aspects, increasing women's representation in decision-making positions, developing gender training programs, increasing women's involvement in associations and cooperatives, and increasing women's access to resources, such as education (IUCN, 2021). The 2008 Girls' Education Policy further aimed to increase girls' access to education by promoting elimination of gender disparities through the development of a 12 year basic education system, a school feeding program, separate girl's bathrooms, and increased infrastructure (Republic of Rwanda, 2019). Separate girl's bathrooms were introduced to limit gender-based violence (GBV) in schools, while the 2011 National Policy against Gender Based Violence aimed to progressively eliminate GBV through the development of a preventive, protective, supportive, and transformational environment (Republic of Rwanda, 2019).

To provide oversight and coordination of the gender policies, the Rwandan government established a number of gender machinery institutions. The Ministry of Gender and Family Promotion (MIGEPROF) implements national gender policy and ensures gender mainstreaming (IUCN, 2021). The main responsibilities of the Ministry are to implement and assess policies and provide resources for programs. The Gender Monitoring Office (GMO) implements gender audits in the public and private sector, and the

National Women's Council (NWC) advocates for women's rights and recruits female participation in development programs (IUCN, 2021). The National Gender Cluster is a forum of representatives from private sector, government, and development groups that meets to discuss the prioritization and coordination of gender equity efforts (IUCN, 2021). The Rwanda Women Parliamentarians oversee legal issues related to gender and help create new gender policy. Together, these groups work to ensure gender policy is not only implemented, but practiced in Rwanda.

At the international level, Rwanda has ratified several humanitarian agreements in support of gender equality. The Convention on Elimination of all Forms of Violence against Women (CEDAW) is an international bill of rights for women. All signatory countries are evaluated for national issues that may conflict with the agreement. In Rwanda, GBV and lower rates of education among women are areas for concern (UN Human Rights, 2017). Almost every country in Africa has committed to the African Charter on human and peoples' rights on the rights of women (ACHPR), which provides a continental framework for human rights. Most recently, Rwanda prioritized several of the United Nations Sustainable Development Goals, namely Quality Education (4), Gender Equality (5) and Reduced Inequalities (10) (United Nations Rwanda, 2021). These international agreements provide Rwanda with a level of accountability to implement and practice national gender policies.

While the Rwandan government has made significant progress to advance women's rights under the law, a gap remains for women not engaged in formal marriages. Approximately 30% of women in Rwanda participate in consensual partnerships but are not legally married, or are in polygamous marriages; neither of which are recognized under the law (Abbott & Malunda, 2016). This creates conflicts regarding land inheritance for women in these situations. Polygamous marriages do take place in Rwanda, and the status of wives is fluid; inheritance (without official marriage) can change based on seniority or who had children in the partnership (Vanhees, 2014). However, the 2008 Gender Based Violence Law does assert that if a person in a polygamous relationship wants to officially marry one of the partners, common belongings must be divided equally between the other spouses (Vanhees, 2014). This law does establish a precedent to address the rights of polygamous women, but remains insufficient. The Rwandan government sought to increase legal protections for women by hosting group weddings in rural areas, but further legislation to recognize women's rights as human rights, regardless of marital status, are necessary (IUCN, 2021).

1.4 National and international legal framework for gender equality

The Government of Rwanda, State of Gender Equity Report (2019) provides a comprehensive summary of constitutional provisions, and national laws that support the principle of gender equality and women's empowerment in Rwanda. Information for the legislative and policy review was drawn primarily from this report. Additional information was gleaned from Rwanda Vision 2020, the National Policy against Gender Based Violence (2011) and the National Gender Policy (2010).

Rwanda is internationally recognized as a world leader in promoting gender equality principles and women's empowerment. In the aftermath of the 1994 genocide, the Government undertook radical and far-reaching reforms to address the political, social, legal, and economic status of women. Legal reforms to give women property rights, and to enable them to inherit property, including land, were especially important. The Constitution, adopted in 2003 and amended in 2015, proactively promotes gender equality. It outlaws any form of gender discrimination, and enshrines the principle of equality within

marriage.

With 53% of the Rwanda population being female, the Vision 2020 national government strategy emphasizes that Gender equality will be one of the driving factors towards achieving rapid growth and sustainable development. Vision 2020 commits to continuously update and adapt laws on gender, strategies for an increased access to productive resources by women, representation in decision-making positions and apply positive discrimination in favour of women.

The National Gender Policy (NGP) of 2010 formulated to support government programs in various sectors that are directly aimed at addressing gender inequalities and women's rights. The NGP seeks to free Rwandan society from all forms of gender based discrimination and create an environment where both men and women equally contribute to and benefit from the national development goals.

The main goal of the NGP is to contribute to reducing gender inequalities in all sectors, as a key component of sustainable development. The NGP was informed by the Economic Development and Poverty Reduction Strategy (EDPRS I) of 2007 It helped to shape theThe Agriculture Gender Strategy (2010), the EDPRS II (2013), the National Strategy for Transformation (2017-2024), the Strategic Plan for the Transformation of Agriculture (PSTA3), the National Information and Communication Technology in Rural Agriculture (ICTRAg) Strategy (2016-2022), the National Policy against Gender- Based Violence (2011), the National Strategy on Climate Change and Low-Carbon Development (2011) and several others.

Constitutional provisions and gender

The Constitution of Rwanda of 4 June 2003, as amended on December 24, 2015, affirms the fundamental rights of all citizens of Rwanda, consistent with the United Nations Declaration of Human Rights and other human rights instruments. According to Article 11 of the Constitution: "All Rwandans are born and remain free and equal in rights and duties. Discrimination of whatever kind based on, inter alia, ethnic origin, tribe, clan, color, sex, region, social origin, religion or faith, opinion, economic status, culture, language, social status, physical or mental disability or any other form of discrimination is prohibited and punishable by law." Article 16 further enshrines the principle of gender equality, and the Government has committed to establishing equity and equality at all levels of society.

Key provisions within the constitution that relate to gender equality include the following:

- The preamble reaffirms Rwanda's adherence to human rights conventions, including the UN Convention on the Elimination of All Forms of Discrimination against Women (CEDAW 1979) and declares Rwanda's commitment to ensure equal rights between women and men;
- Articles 11 and 16 outlaw any form of gender discrimination;
- Articles 9, 76, and 82 mandate a minimum quota of 30% female representation in the Senate and other areas of public governance;
- Article 37 prohibits discrimination in employment; and
- Article 26 enshrines equality within monogamous marriage.

Since 1994, the Government of Rwanda has enacted a series of gender progressive policies. These include the following three key laws:

Gender equity in the right to inherit property

Enacted in 1999 Law n°22/99 (supplements book one of the civil code regarding succession) and revised

in 2016 (Law N°27/2016 of 08/07/2016 Governing Matrimonial Regimes, Donations and Successions) provides that both male and female children have the rights to inherit their parents property. This in terms of that law obviates customary practices where only sons inherit land and tangible assets.

Gender equity in the right to own land

Enacted in 2013, Law N° 43/2013 OF 16/06/2013 Governing Land in Rwanda, ensures that all women have equal rights to land titles registered in their name. This law enables women to secure loans from financial institutions using their land as collateral, which in turn removes a significant barrier from women engaging in commercial enterprise that require capital investment.

Gender-based violence

Enacted in 2008, Law n° 59/2008 of 10/09/2008 on Prevention and Punishment of Gender-Based Violence makes explicit that gender-based violence is a crime that is punishable under Rwanda law.

At the international level, Rwanda is party to key international instruments addressing women's and children's rights, including:

- The Convention on the Elimination of all forms of Discrimination against Women (CEDAW).
- The Convention on Slavery and Repression of Human Trafficking and its Additional Protocol repressing and punishing the sale and trafficking of children and women.
- The UN Convention on the Rights of the Child (UNCRC) and the Optional Protocol within the UNCRC on Child Trafficking, Child Prostitution and Child Pornography.
- The African Charter on Rights and Welfare of the Child (ACRWC)
- The African Charter on Human and Peoples' Rights and the Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa.
- ILO Convention 182 on the Worst Forms of Child Labor.

The government has also ratified and/or implemented numerous international conventions and instruments. Those include:

- The Convention on Elimination of all Forms of Violence against Women (CEDAW),
- The Beijing Platform for Action, the African Charter on human and people's rights on the rights of women (ACHPR),
- The International Convention on Social and Cultural Rights (ICCPR),
- The United Nations Security Council Resolution on Peace and Security (UNSCR 1325 and 1820) and
- The UN declaration on human rights (UDHR).

Lastly, the Government of Rwanda is committed to delivering on the 17 Sustainable Development Goals (SDGs) which includes gender equality. To fast track attainment of the SDGs, the Government of Rwanda argues that policies must be in place and effectively implemented to ensure that both women and men equally participate in and benefit from development investments and processes.

1.5 Institutional framework for implementing gender policies in Rwanda

Key efforts to implement the Government of Rwanda's gender progressive policies are included in the remit of the Agriculture, Justice, Health, Education, and Local Government ministries and their

executing agencies.

To help coordinate and oversee Rwanda's gender equality and women's empowerment efforts and ensure their compliance with national policies and regulations, Rwanda has established four key governmental and civil society entities. These include the Ministry of Gender and Family Promotion (MIGEPROF), the Gender Monitoring Office (GMO), the National Women's Council (NWC) and Rwanda Women Parliamentarians (RWPF/FFRP).

Ministry of Gender and Family Promotion (MIGEPROF)

The Ministry responsible for Gender has the mandate of coordinating the implementation of the national gender policy and advocating on gender issues at local to national levels. MIGEPROF formulates regulations and its staff support gender mainstreaming across the Government of Rwanda.

The Gender Monitoring Office (GMO)

The GMO documents and reports the extent to which a "gender lens" influences the investments and actions of public agencies and private organizations. The GMO undertakes monitoring and evaluation to assess the implementation and social impact of public and private sector implementation of activities that seek to comply with Government gender policies.

The National Women's Council (NWC)

The National Women's Council is a local to national organization of women that advocates for women's rights and the promotion of gender equality. Its members help mobilize women across the nation at all social, economic and political levels to ensure that their voices are heard and acted on in development and conservation programmes and activities. The National Women's Council provides a formal structure to give voice to women and through which women can raise ideas and concerns to inform policy. Lack of financial resources remains a constraint to the active engagement of village women in this process.

Rwanda Women Parliamentarians (RWPF/FFRP)

The RWPF/FFRP works to build the capacity of women Parliamentarians to navigate the halls of government successfully, to represent women's voices in the formulation of all policies not just those related to gender equality, and to hold the parliament and its members accountable for redressing historical inequities in the rights of women and the power that they wield in determining the future of all Rwandans.

The National Gender Cluster

This is a Government convened forum in which the Government of Rwanda, development partners, the Private Sector and Civil Society meet and discuss planning, coordination and prioritization of Gender Equality interventions.

1.6 Poverty and labor force participation

The poverty rate in Rwanda averages 39% (NISR, 2018). The Western (CND) region is the poorest in the nation with a poverty rate that averages 47.7% and ranges from 33.5% in Rusizi to 69.3% in Nyamasheke. Between 2013/14 and 2016/17 (NISR, 2018) 50.5% of female headed households had been poor during that time period, compared to 48.8% of male headed households. Poverty was more prevalent in households with more people and where the household head was older, had little education,

Labor force participation is lower in women (48.2%) than in men (65.6%) 16 years and older (NISR, 2020), and unemployment is higher in women (20.3%) than in men (15.9%). Young women and men are less likely to be in the labor force and more likely to be unemployed. In 2020, women were less likely to be employees and employers, and significantly more likely to be a contributing family worker than were men.

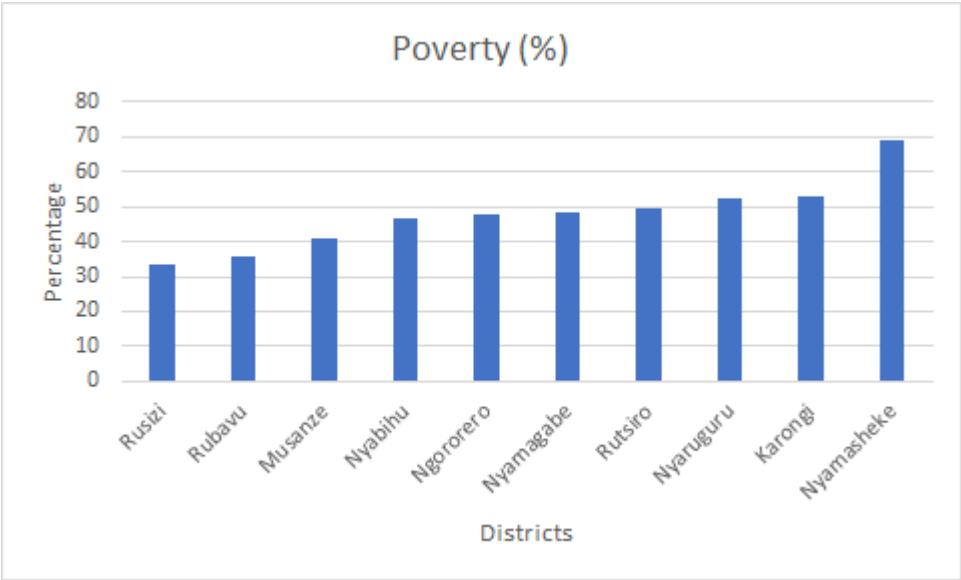


Figure 2: Poverty in districts within Congo Nile Divide (Source: EICV5 NISR, 2018)

12

Population 16 years old and over by labour force status, sex, age group, and urban/rural area, RLFS 2020								
	Total	Labour force status				Labour force participation rate (%)	Employment-population ratio (%)	Unemployment rate (%)
		Labour force	Employed	Unemployed	Outside labour force			
Total Population 16 yrs and over	7,472,601	4,212,972	3,460,860	752,112	3,259,630	56.4	46.3	17.9
16-24 yrs	2,309,913	1,080,767	828,459	252,308	1,229,146	46.8	35.9	23.3
25-34 yrs	1,874,035	1,356,005	1,098,949	257,056	518,080	72.4	58.6	19.0
35-54 yrs	2,195,647	1,427,063	1,230,818	196,245	768,584	65.0	56.1	13.8
55-64 yrs	617,306	271,530	233,426	38,104	345,775	44.0	37.8	14.0
65+ yrs	475,700	77,605	69,206	8,399	398,095	16.3	14.5	10.8
Male Pop. 16+ yrs	3,512,698	2,303,622	1,938,268	365,354	1,209,075	65.6	55.2	15.9
16-24 yrs	1,132,034	573,371	456,983	116,388	558,663	50.6	40.4	20.3
25-34 yrs	905,580	755,696	631,419	124,277	149,883	83.4	69.7	16.4
35-54 yrs	1,009,973	782,467	684,302	98,165	227,507	77.5	67.8	12.5
55-64 yrs	267,154	145,126	123,234	21,892	122,028	54.3	46.1	15.1
65+ yrs	197,957	46,963	42,331	4,632	150,994	23.7	21.4	9.9
Female Pop. 16+ yrs	3,959,903	1,909,349	1,522,592	386,757	2,050,554	48.2	38.5	20.3
16-24 yrs	1,177,879	507,396	371,476	135,920	670,483	43.1	31.5	26.8
25-34 yrs	968,455	600,309	467,530	132,779	368,146	62.0	48.3	22.1
35-54 yrs	1,185,674	644,597	546,517	98,080	541,077	54.4	46.1	15.2
55-64 yrs	350,152	126,405	110,193	16,212	223,747	36.1	31.5	12.8
65+ yrs	277,743	30,642	26,875	3,767	247,100	11.0	9.7	12.3
Urban Pop. 16+ yrs	1,581,477	1,060,077	867,565	192,512	521,399	67.0	54.9	18.2
16-24 yrs	501,826	248,767	191,104	57,663	253,060	49.6	38.1	23.2
25-34 yrs	493,474	399,957	321,737	78,220	93,516	81.0	65.2	19.6
35-54 yrs	445,849	357,316	310,082	47,234	88,532	80.1	69.5	13.2
55-64 yrs	78,623	42,609	34,597	8,012	36,014	54.2	44.0	18.8
65+ yrs	61,705	11,428	10,045	1,383	50,277	18.5	16.3	12.1
						0.0	0.0	0.0
Rural Pop. 16+ yrs	5,891,125	3,152,894	2,593,295	559,599	2,738,231	53.5	44.0	17.7
16-24 yrs	1,808,087	832,000	637,355	194,645	976,086	46.0	35.3	23.4
25-34 yrs	1,380,561	956,048	777,212	178,836	424,513	69.3	56.3	18.7
35-54 yrs	1,749,799	1,069,747	920,736	149,011	680,052	61.1	52.6	13.9
55-64 yrs	538,683	228,922	198,830	30,092	309,761	42.5	36.9	13.1
65+ yrs	413,995	66,177	59,162	7,015	347,818	16.0	14.3	10.6

Table 1: Population 16 years old and over by labour force status, sex, age group, and urban/rural area (Source: NISR Labor force survey report 2020)

Employed population by sex, status in employment, and urban/rural area, RLFS 2020			
	Total	Sex	
		Male	Female
Total employed population	3,460,860	56%	44%
Employee, Paid apprentice/intern	2,292,976	57%	43%
Employer	46,221	65%	35%
Own-account worker	968,858	57%	43%
Member of cooperative	5,792	64%	36%
Contributing family worker	147,013	22%	78%

Table 2: Employed population by sex, status in employment, and urban/rural area (Source: NISR Labor force survey report 2020)

1.7 Beliefs, perceptions and cultural norms

Pre-1994 Rwandan families and society in general were characterized by a patriarchal social structure that translated into male dominance and female subordination (Mukabera, 2016). In principle, in traditional Rwandan society, women's responsibilities included raising and educating children, welcoming visitors, managing the household, advising their husbands and maintaining traditions (Mukabera, 2016). In practice, though women were allowed substantial autonomy in their roles as child bearers and food producers men remained the ultimate arbiters of most family decisions.

Traditionally, the Rwandan woman was a symbol of fertility and weakness, while the man was a symbol of strength and protection who makes all major decisions. Rwandan women were valued as child bearers and more specifically the number of boys they gave birth to (Mukabera, 2016). Boys were taught to defend the interests of the family. It was their job to build or repair a house, to engage in activities related to cattle, cut firewood, go to battle, milk the cows, make decisions on farm management, and generate income for the family. Girls were raised to help their mothers with household chores, raise children, and most importantly to be modest, reserved, silent, obedient, and present a submissive attitude toward men (Mukabera, 2016).

These traditions effectively excluded women from accumulating material wealth and participating in economic and political decision making. The Family Code of 1992 precluded women from owning land. If a male head of household died, property passed to male heirs or to the man's brothers (Uwineza, 2009). In the case of divorce or the husband's death, a woman had no claim over the family property if she had not borne children; she would be forced to return to her birth family. Likewise, 'the Commercial Code', dating back to the colonial period, required the written consent of the husband whenever a wife wanted to engage in commercial activities, take up employment outside the household, or obtain credit, or take legal action (Mukabera, 2016).

Prior to 1994 men monopolized the public and political arena. Despite this, there have been some powerful women throughout Rwandan history, particularly the Queen Mothers who held substantial influence as advisers to the King in pre-colonial Rwandan society (Uwineza, 2009). In the customary system of justice, known as *gacaca*, women were not allowed to participate directly in public deliberations. Rather, they were expected to play an indirect role lobbying their husbands (Mukabera, 2016). Women could and did hold powerful positions as high priestesses within the religious realm, but this was undermined with colonization and the introduction of christianity.

Post-1994, the Rwandan government has taken large strides in improving gender equality and women's rights policies, however these policies have yet to reach their full potential in making substantial changes in women's lives. This may be due to gaps in policy and a dearth of resources applied to ensure proper functioning of the legislation and gender machinery institutions. There is a lack of capacity-building strategies to develop gender machinery institution staff's gender based knowledge and skills, and these staff include only a limited number of gender experts that rarely occupy decision-making positions (IUCN, 2021). Strategies to mainstream gender in capacity-building strategies of these institutions are not comprehensive, which contributes to functional issues such as lack of trained staff, staff retention measures, proper orientation for new hires, and clear handover procedures. This suggests that although the Rwandan government has passed strong gender-based legislation, implementation is limited by lack of concrete guidelines and limited resources.

While setbacks to establishing fully functional gender machinery institutions limit the effective execution

of gender equality laws, the current cultural norms and belief systems in place in Rwandan society profoundly diminish the positive effects that these laws have on women's daily lives. Discrimination against women continues to exist in multiple aspects of Rwandan society including economic opportunities, agriculture, politics, and division of labor. The long history of patriarchal dominance in Rwanda has led to the development of robust cultural norms, many of which define men's and women's roles in society, limiting women from taking advantage of the full range of opportunities within their legal rights. Other societal factors, such as the high prevalence of GBV, further limit women's ability to venture outside of their perceived household roles. In a recent study 52% of women surveyed in districts within the CND agreed that men were justified in beating them if they: burned the food, argued with him, went out without telling him, neglected the children, or refused to have sex with him (Government of Rwanda, 2016). Today, male and female policymakers are quick to acknowledge that many aspects of traditional Rwandan culture repressed women and abrogated their human rights (Uwineza, 2009).

1.8 Gender roles, responsibilities and labor allocation

Rwandan cultural norms divide household duties into activities that fall under either men's or women's responsibilities. Men are considered the breadwinners of the family, and engage in significantly more off-farm and wage employment than women (Republic of Rwanda, 2019). Men also hold domain over any valuable property or resources that the family owns or acquires, including cash crops, livestock, and valuable trees (IUCN, 2021; Kiyani et al., 2017). Men possess the majority of the decision-making power within the family, and are largely able to engage in any activities they desire.

Women's role in society is predominantly seen as being a mother and caretaker of both the family and the home. Primary duties include caring for children and elderly, cooking, growing food for the family, and providing labor on their husband's land. While the law allows for women to engage in other endeavors outside of their traditional duties, several factors often limit them from diverting from the cultural norm. The dominant barrier for women is lack of time, as the majority of their time is spent fulfilling their household responsibilities (IUCN, 2021). Women spend anywhere from 1 to 8 hours each day at their cookstove, not including the time it takes to collect the food, water, and fuel wood required for cooking (Campbell et al., 2012). Additionally, women are responsible for growing and tending to subsistence crops for the family, and are often required to take on other informal jobs on their husband's farms. These duties, on top of childcare, take a large toll on women's available time, preventing many from engaging in wage labor, training, and other forms of off-farm activities (Republic of Rwanda, 2019).

Innovations that reduce the time spent on these tasks, such as improved cookstoves and agricultural technologies, may increase women's ability to engage in activities outside of their traditional roles. Furthermore, male involvement in work traditionally delegated to women, such as childcare, would allow women to spend more time outside of the household. However, men are not expected, and are often unwilling to engage in what they perceive as women's work (Abbott & Malunda, 2016). As a result, some Rwandan women believe that government promotion of women's equality has increased their workload, as they are now expected to be bringing home money on top of their childcare, domestic, and farming duties (Abbott & Malunda, 2016). Furthermore, any additional income earned by women outside of traditional female tasks may be taken by the husband or result in increased domestic violence due to a perceived emasculation of the male role as breadwinner.

Violence is an issue for both men and women (30% vs 37%, respectively), but women are more prone to being victims, particularly sexual and gender-based violence (SGBV) (6% for men vs 23% for women,

DHS Survey, 2021). Although there are strong laws and policies against SGVB, trends of violence are not decreasing. Factors that impact higher rates of violence include lower education, earning money, household alcohol levels, and familial historical violence. In seeking support, both men and women turn to neighbors (32% vs 40%) and family (24% vs 33%), whilst some go to the local authorities (17%, 18%), police (17%, 11, ibid). Women also seek out their partners' relatives for support whilst men seek out friends. This demonstrates the importance of social networks much more than government agents.

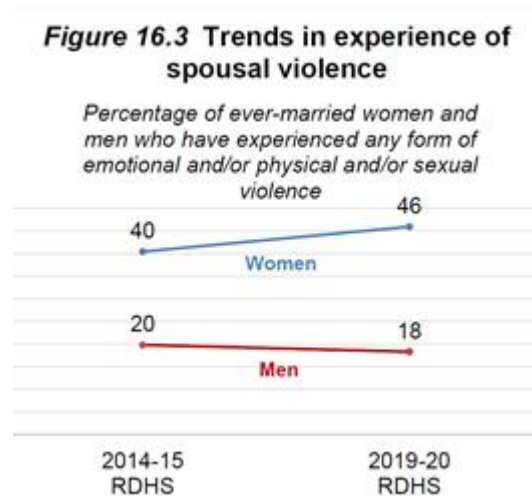


Figure 3, Trends in spousal violence over time. Source: Rwanda Demographic and Health Survey, 2021

1.9 Access to and control over capital and natural assets

The 1994 genocide left behind unprecedented distortions in social relations and exacerbated the already existing gender imbalances and inequalities. This is illustrated by the fact that today, 25% of households in Rwanda are headed by women. Women performing non traditional roles such as decision making, managing financial resources, constructing houses and many others have characterized the post genocide roles of women period in Rwanda.” (Gender and Community Development Analysis in Rwanda, EAC 2009).

Although collection and dissemination of credible data disaggregated by ethnicity is not available.¹ There has been some recognition of some ethnic minorities as Historically Marginalized People (HMPO, see Annex 6 for further details). While there is limited data based on ethnicity of HMPO, which critically impacts the baseline for monitoring activities potential impacts (positive and negative), we will refer to trends aligned with governmental policies supporting pro-poor programming, and rely on data from reports conducted by African Initiative for Mankind Progress Organization (AIMPO), Women’s Organization for Promoting Unity (WOPU) and Minority Rights Group International (MRG) leading a project titled, ‘Strengthening the Capacity of Civil Society Organizations (CSOs), Human Rights Defenders (HRDs) and Media Houses to Challenge Discrimination against HMPO and Promote their

¹ The project will be unable to disaggregate data based on traditionally marginalized people, however, all data collected will be disaggregated by sex and wealth category where possible

Human Rights in Rwanda’.

Women are likely to constitute the majority of genocide survivors which informs a need for intersectional, or the interconnected nature of social categorizations’ integration of history, gender norms, social disparity, power dynamics, rural/urban divide, and most importantly access to land rights. Despite policies and initiatives implemented by the government, other entities and projects, including this project to ensure HMPO are equitably included in project design, implementation and benefits, there are potential barriers which include: (1) Inadequate disaggregated data (2) Limited capacity and resources to analyze intersectionality (3) Limited access to decision-making (women) (4) Social inequality from increased long-term poverty of HMPO (5) Lack of promotion and protection of HMPO rights, including land tenures.

In addition, climate change is adversely impacting CND’s landscape with increasing temperatures and changing rainfall patterns contributing to ecosystem degradation and reduced livelihoods for the most vulnerable populations. The agricultural sector in Rwanda is primarily focused on subsistence farming, and most cultivated land is rain-fed, exposing farmers to significant climate-related food security risk. In addition, Rwanda’s efforts to reduce poverty are affected by high dependence on agriculture, especially when combined with climate change, declining plot sizes, and limited investment in intensification and erosion prevention. Thus, the impacts of climate change is a risk multiplier as it has exacerbated the social, cultural, and economic conditions of HMPO communities in Rwanda. In the face of the climate crisis, culturally-based climate resilient approaches should be adopted in policies along with ways to address the potential barriers listed above.

Compared to men, women are significantly less likely to own land (productive assets), they are less often engaged in income-earning activities, have less access to higher education (Govt of Rwanda, 2018), financial services (though this is rapidly becoming more equitable), health services, employment, direct payments, and non- school training opportunities. But that is changing according to the 2019 State of Gender Equity in Rwanda report. Women-headed households are more likely to use firewood (85% vs 78%), less likely to use charcoal than male-headed households (13% vs 19%), suggesting increased labor burdens for searching for fuelwood. However, women and male-headed households have the same access to water sources (EICV 5, 2018).

There are no significant differences by household’s poverty status [between male and female headed households], education or ownership of land (Abbott 2018). However, as men receive 75% of agricultural loans, and women only 25% this strongly suggests that land owned by married couples is in reality “owned” by the husband and not jointly, and the man in a married household is still culturally considered the head of the household. For married couples it is unclear whether women, on the death of their husbands, inherit their assets or are these, according to tradition, bequeathed to the sons. In addition, should women grow crops or trees, or raise livestock on marriage land do these assets belong to her or him or both. The power of women in illegal but still common polygamous marriages is even less than those in monogamous marriages.

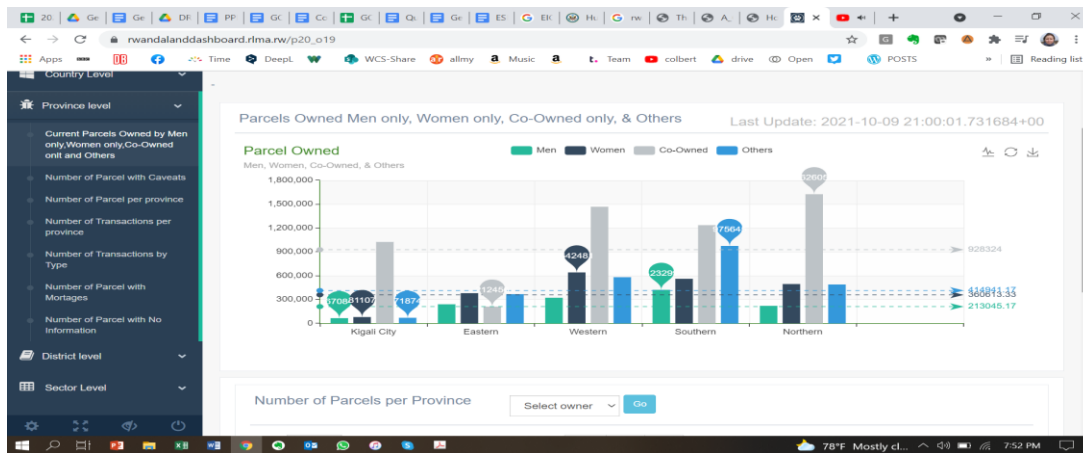


Figure 4. Land ownership by men, women, and co-owned parcels in 2021

(Source: Rwanda Land Dashboard https://rwandalanddashboard.rlma.rw/p20_o19, accessed October 10, 2021)

Though women are not the primary land or livestock owners in Rwanda they still play a key role in food production primarily to feed their family. In fact few Rwandans and almost exclusively men are involved in cash crops as land holdings are typically small and often barely enough to produce food sufficient to meet the nutritional security of the land owners' families.

Access to microloans and credit for women contributes to economic development of Rwandan society as a whole and helps women escape poverty (Mpakaniye, 2016). However, microloans remain difficult for the poor to attain for a number of reasons. Transaction fees reduce the value of already small accounts, while complicated account registration procedures, minimum balance requirements, and illiteracy all hinder access to loans, especially for women, who are more likely to be illiterate or have less experience handling money (Mpakaniye, 2016). Women typically rely on more informal loans from family because they lack collateral to access credit from a financial institution (IUCN, 2021). In 2018, of rural Rwandan women who accessed loans from a financial institution, 44.2% used saving and credits Cooperatives (SACCOs), compared to 39% of rural men (IUCN, 2021). 22% of rural men used a commercial bank, as opposed to only 16% of rural women (IUCN, 2021).

Helping women access loans from formal financial institutions will allow greater independence for women and more economic security for families. Several banks in Rwanda have already signed on to the African Guarantee Fund, aimed to reduce poverty and increase female access to loans (Mpakaniye, 2016). The fund helps those who lack collateral still receive credit (Mpakaniye, 2016). Annually, the number of microloans accessed by women in Rwanda is increasing, but additional support to enhance female financial literacy will help maintain the upward trend (Mpakaniye, 2016).

Agricultural extension services include training to innovate farming practices, and improve profitability and efficiency. The 2010 National Gender Policy (MIGEPROF), the 2018 Strategic Plan for the Transformation of Agriculture (PSTA3) and the 2010 Agriculture Gender Strategy (MINAGRI) have committed to increase extension staff in Rwanda, particularly women (IUCN, 2021). At present, women are significantly underrepresented as extension personnel, which furthers a cycle of reduced female access to extension services; 82% of extension workers are male in Rwanda (IUCN, 2021). Male extension workers are more likely to approach male farmers regarding men's crops, perpetuating the exclusion of women.

Similarly, farm promoters, village leaders who promulgate knowledge from extension services, are also dominated by men (IUCN, 2021). Women tend to be less educated than men and have greater domestic obligations that inhibit them from working with the community as promoters. Promoters are perceived as having elevated social status, leading men to more quickly jump on the opportunity than women (IUCN, 2021). By enabling more women to hold leadership positions as extension workers and farm promoters, more women can subsequently benefit from extension services, training and innovative farming practices.

This sector has vast potential for greater female involvement, but legal frameworks may be necessary to ensure more equal representation. Women represent 66% of the agricultural workforce, but only 19.7% of women in the sector occupy paid jobs (IUCN, 2021). Therefore, enabling women to become agricultural leaders through extension services and farm promoters may offer greater independence, financially and socially. Female-only extension service training may motivate women to participate and bring more attention to female crops, and establishing hiring quotas may enable more female leadership.

Women are equally likely as men to have access to a mobile phone and use mobile financial services. But are less likely to be mobile money account holders, or have access to bank services such as an ATM bank card. This is particularly true for women in rural areas, and ownership of capital goods is much less for women-headed households (see Table 3), demonstrating their marginalization across communities.

Head of household sex	Living room set	% owning at least one					Total surveyed
		Radio	TV	Computer	Mobile phone	Bicycle	
Male	19.6	78.6	11.9	3.6	71.1	16.2	2,031
Female	12.8	59.3	5.7	2.3	54.3	4.9	677

Table 3: Comparison of assets held by women and male-headed households (Data from EICV 5, 2018)

1.10 Women's health and access to health services

In 2010 life expectancy at birth was 71.13 years for women, and only 66.82 years for men (<https://www.statista.com/statistics/971106/life-expectancy-at-birth-in-rwanda-by-gender/>).

Based on the 2019/2020 Demographic and Health Survey report (NISR, 2020), the maternal mortality rate or ratio (MMR), which measures the obstetric risk associated with each live birth and is the age-standardized maternal mortality rate divided by the age-standardized fertility rate, declined from 10.7 per thousand live births in 2000, to 2.03 per 1000 in 2019-2020.

The mortality rate of children under 5 years of age continues to decline from 50 deaths per 1,000 live births in 2014 to 45 per 1,000 in 2020.

Women's health, proxied using Body Mass Index shows that on average 76% of women within CND districts are within the normal BMI range, 19% are considered obese and only 5% fall below 18.5 which is considered nutritionally compromised (Republic of Rwanda, 2016). The averages for Rwanda as a

whole are 73% normal, 21% obese, and 7% too thin. Similarly, only 18% of women exhibit signs of anemia, compared to 19% for the nation. Prevalence of malaria in children and women is low (0.5% and 0.4% respectively) compared to 2.2% and 0.6% for the country as a whole (Republic of Rwanda, 2016).

In 2017 74% of women had health insurance and 94% of women had access to community health services. The percentage of health insured households is lowest in Rubavu district (54%) and highest in Ngororero district (82%) according to the 2014-15 demographic and health survey (Government of Rwanda, 2016).

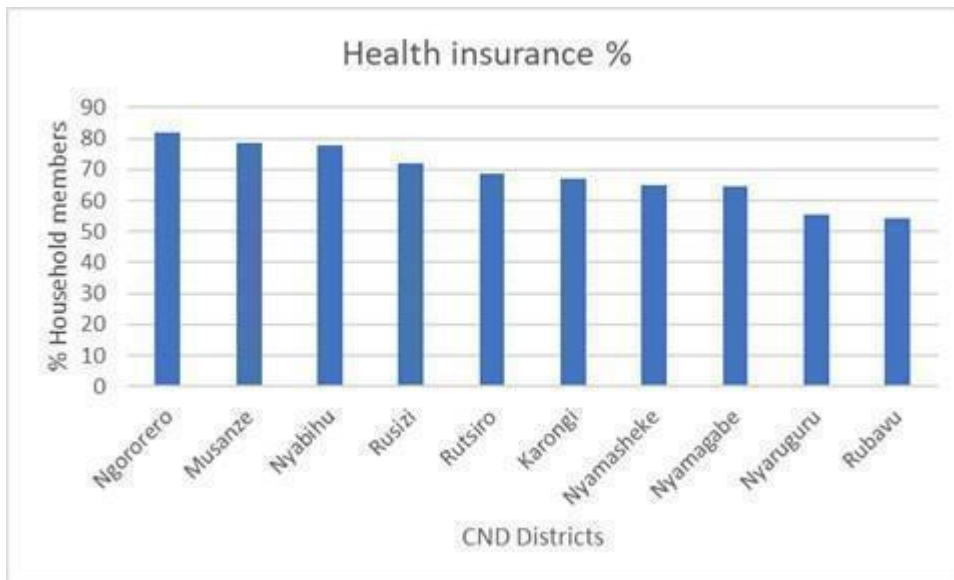


Figure 5: Percent population with health insurance by district (Source: Government of Rwanda, 2016)

Overall access to sanitation facilities is high among Rwandans (IUCN, 2021). However, due to unequal income distribution between men and women, households led by women are less likely to have sanitation access than male led households; 80.6% of female headed households have improved sanitation facilities compared to 88.0% of male headed households (IUCN, 2021). Traditional pit latrines are seen as more cost effective than modern sanitation systems, but pose risks of disease, overflow during the rainy season, groundwater contamination, mosquito breeding, and an increased risk of GBV for women due to distance from the house and night use (Tsinda et al., 2013). Lack of sanitation services poses both a serious social and environmental threat in Rwanda, particularly in informal settlements.

1.11 Women's educational attainment and access to education services

Education can lead to better health and greater access to resources (Allan et al., 2018:26). In Rwanda, the rate of people who have ever attended school is high, at 87%, but rates are higher in urban areas than rural areas (IUCN, 2021). The rates of primary education are approximately equal between boys and girls in Rwanda, and in secondary education women actually attend in higher numbers than men, 53% women to 46% men (Education Statistical Yearbooks, 2018). However, the disparity between men and women shifts with regard to technical training and tertiary education, in which men are more dominant.

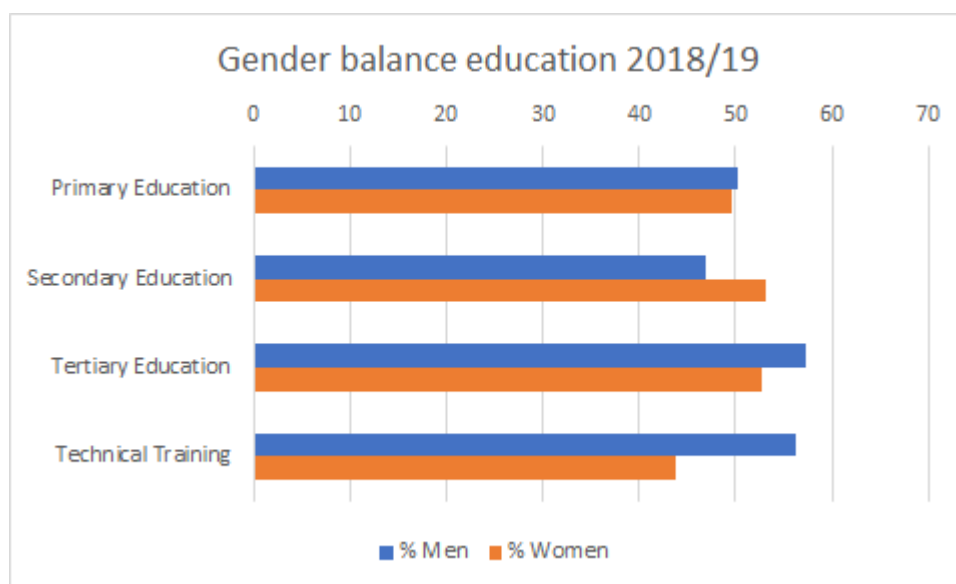


Figure 6: Gender balance in education in Rwanda (Source: State of Gender Equality in Rwanda, 2019)

In 2018, 92% of children (1.64 million boys and 1.68 million girls) were enrolled in pre-nursery to secondary education (Ministry of Education, 2018). Literacy rates continue to climb from 38.2% in 1978 to 73.2% in 2018. By this time youth literacy was at 85%. Girls and women lag behind boys and men. Only 56% of girls, 41% of women, and 28% of elderly women are literate.

Levels	Schools	Males	Females	Total	Percent
Pre-Nursery	96	3,199	3,292	6,491	0.2%
Nursery	3,210	112,044	114,662	226,706	6.3%
Primary	2,909	1,259,344	1,244,361	2,503,705	69.0%
Secondary	1,416	264,782	314,115	578,897	16.0%
TVET	360	57,643	44,842	102,485	2.8%
Tertiary	30	41,458	34,255	75,713	2.1%
Adult literacy	4,991	51,220	81,145	132,365	3.7%
Total	13,012	1,789,690	1,836,672	3,626,362	

Table 4: Levels of schooling in Rwanda (Source: Ministry of Education, 2018 Education Statistics)

The equal rates of primary education for boys and girls in Rwanda suggest a cultural value of education for all children, not only prioritization for boys. A lack of tuition fees for childhood education enables poor families to send boys and girls to school, and increasing rates of education contributes to falling illiteracy rates in Rwanda (IUCN, 2021). Higher levels of girls in secondary education may be attributed to Rwandan policies such as the 2008 Girls' Education Policy and the implementation of a school meal program (IUCN, 2021). Despite lower rates of women enrolled in tertiary education, the percentage is increasing due to flexible programming, such as online learning or night school, which enables women with domestic responsibilities to participate. According to the IUCN, the higher level of education men and women receive, the less likely they are to work in the agriculture sector (IUCN, 2021). In comparison, 92% of women with no education work in agriculture (IUCN, 2021).

Strong gender stereotypes in Rwanda still push girls into "traditional" soft trades including hairdressing, secretarial work and nursing (IUCN, 2021). 87% of women with a university degree work in the service sector (IUCN, 2021). Though women are more likely to be primary school teachers than men, they are rarely secondary, tertiary or technical/vocational teachers. In agriculture, women predominantly occupy informal roles, such as unpaid farming for the family, whereas the men typically hold higher paying positions or manage higher value products such as timber (IUCN, 2021). As a result, the majority of agroforesters are male in Rwanda (Anastase & Nahayo, 2020).

One of the greatest challenges to achieving economic stability lies in challenges in achieving quality educational attainment by female youth which leads to their lower literacy, especially in rural areas. Overall, social factors including traditional gender roles such as domestic chores and family care, historic marginalization of girls and women, patriarchal social traditions, and gender-based violence entrench girls' underperformance throughout their schooling which has a cyclical effect, as low performance in the primary leaving exam results in girls being admitted to lower quality secondary schools, and ultimately into higher education institutions (HIEs) in lower numbers (Huggins & Randell, 2007).

This results in limited access to paid labor and employment in sectors such as services or public government. Similarly, lower literacy has put them at a greater disadvantage in agriculture, as commercialization of farming has made it more difficult to feed families. While there are national laws and policies to protect employment, there is little protection for the vast majority of women who work in informal sectors which significantly affects women's rights within the country (Abbott et. al., 2008). Women in rural areas are primarily care-providers for children and the elderly, which is unpaid labor and invisibilized due to gender norms (Abbot, 2015). Thus, understanding these barriers and challenges and how to overcome them will inform interventions at project-level.

1.12 Patterns of power and decision-making

Rwanda is internationally acclaimed for being the first nation to have a majority female Parliament, with 61% of these positions held by women from 2018-2022 (Republic of Rwanda, 2019). The Constitution of the Republic of Rwanda established a minimum of 30% female representation in public governance, and this high level of female participation in upper levels of governance far exceeds this minimum (East African Community, 2009). However, female participation drastically decreases in many communities'

level governing bodies.

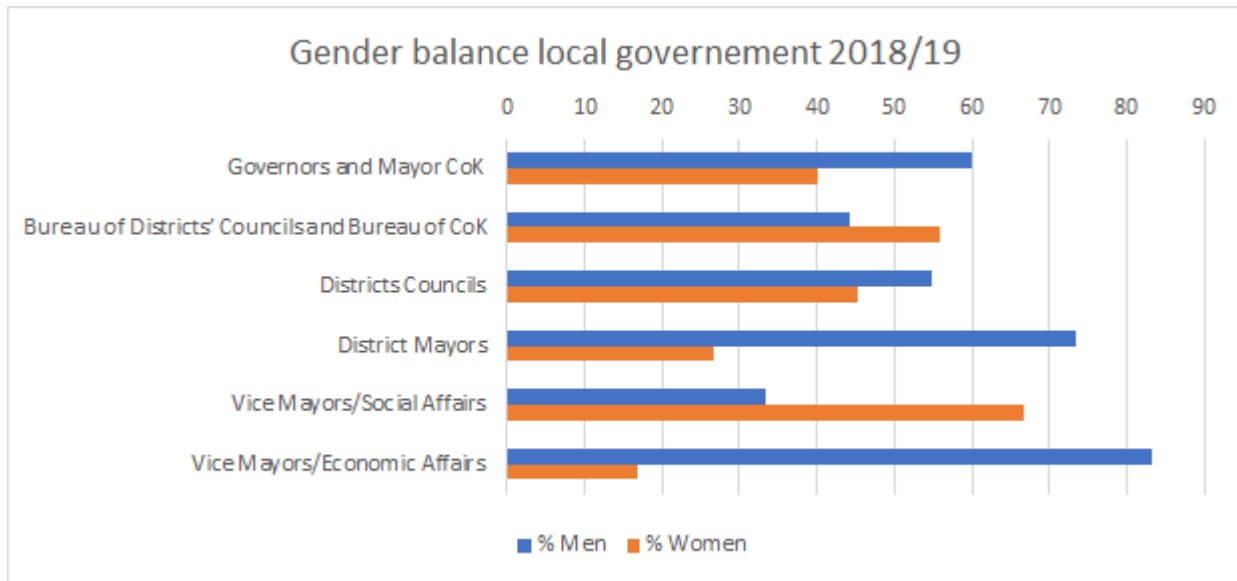


Figure 7: Female and male representation in different levels of local government in Rwanda in 2018-2019 (Republic of Rwanda, 2019)

Although overall female representation in the Rwandan government is increasing, higher levels of female representation in upper levels of governance mirror the disparity between gender equality policy and practice in the country. In higher levels of governance, these policies are implemented well and appear effective, but at community levels, larger gender gaps in prominent roles are clearly evident. Comparable gender gaps exist in central government roles that undergo recruitment processes, with many of these positions not reaching the 30% female representation minimum (Republic of Rwanda, 2019). Furthermore, female participation in the police force is extremely low, although representation is high in most other areas of the justice sector.

Similarly, in the private sector, women hold very few high level positions, particularly at the provincial level. Nationally, 30% of executive committee presidents are women, but there are 0 female presidents at the provincial level (Republic of Rwanda, 2019). Women have high levels of representation in second vice presidency in the private sector, with 70% of national and 80% of provincial second vice presidential roles occupied by women, but have low levels of representation in more powerful executive positions (Republic of Rwanda, 2019). In more local organizations, such as cooperatives, men hold the majority of the decision-making positions, while women tend to hold stereotypically female roles, such as secretary, vice president, and treasurer (IUCN, 2021).

At the household level, women have very little decision-making power. Although under the 1999 *Inheritance and Marital Property Law* women are legally allowed to own land, cultural barriers still inhibit their ability to control how this land is used. These barriers are predominantly based in the belief that men are the head of the household, and thus are the primary owners of all of the family's land and assets (IUCN, 2021). Women's lack of power within their households limits their ability to make decisions about the planting of crops and the use of land as collateral for credit, which would allow increased female engagement in other economic activities. Based on female reports, the majority of households (70%) make decisions jointly, while in 25% of households the husband makes the majority of the decisions

(IUCN, 2021). However, even in households that make joint decisions, women's decisions tend to revolve around day-to-day issues, while men's decisions are primarily focused on productive assets and income.

Women's ability to make decisions within both their own households and broader Rwandan society is further hampered by GBV. The effects of a longstanding patriarchal regime in the country has created and perpetuated the notion that men must be dominant and women must be submissive. Women break these gender norms by asserting their right to be involved in decision-making, and this can be seen as diminishing their husband's masculinity, which can result in violent retribution. In 2010, around 57% of Rwandan women reported experiencing GBV from their partner, and women who bring in more money tend to experience higher levels of GBV (RWAMREC, 2010). This suggests that increasing women's opportunities to make decisions and earn their own income will have a very limited effect on female empowerment until GBV is addressed. Engaging men and boys in gender equality advocacy and action highlights that GBV is not only a women's issue, and is believed to be successful in reducing the prevalence of GBV (UNDP, 2018).

1.13 Social service and support systems that contribute to gender empowerment, wellbeing, and security

Government mandated and running social support programs have reduced the percentage of extremely poor households from 23% and 26% in 2010 to 15% and 18% in 2017 for MHHs and FHHs respectively. Three state managed social support systems have contributed significantly to these observed reductions in poverty and increasing wellbeing and security.

The Girinka Program (One Cow per Family Program) of the Ministry of Agriculture (MINAGRI) started in 2006. By 2017 over 290,000 cows have been provided to vulnerable male and female headed households (approximately 6% of all Rwandan families) as a way to reduce child malnutrition, increase the productivity of agricultural labor, generate a new stream of income through surplus milk sales and strengthen social cohesion amongst Rwandan families and communities as cow owners pass on surplus calves to other needy families. Male headed households were, however, more likely than female headed households to receive a cow through this government program.

The largest social support program of the Government of Rwanda is the Vision 2020 Umurenge Programme (VUP) which was started in 2008 and is managed by the Local Administrative Entities Development Agency (LODA) within the Ministry of Local Government (MINALOC). VUP has four components: wage labor through public works, direct cash transfers, financial services and awareness raising. The majority of public works support agricultural production, land conservation and the maintenance of physical infrastructure. A review by FAO showed that VUP wages paid to women were often captured by men. Retaining control over VUP income was easier for women who owned an individual savings account (e.g., within a Savings and Credit Cooperative - SACCO), though most women only had accounts in Village Savings and Loan Group or informal tontines. Rural women were more disempowered than men through cultural constraints to speaking in public meetings, workloads that limited participation in community meetings and self-help associations, and limited access to credit and savings accounts. The FAO study also found that the majority of both MHHs and FHHs were unaware of the existence of a VUP GRM, or were skeptical about the effectiveness of this complaint process. The most important recommendations were to a) enable women to open their own formal bank accounts to increase their control over their incomes and strengthen their decision-making capacity, and

b) redirect households with high dependency ratios and/or low labor capacity away from public works wage labor and toward VUP direct cash transfers.

MINAGRI further supports poor rural families by offering subsidies for agricultural inputs, improved seeds and irrigation as well as offering credit through the Agriculture Guaranty Fund which provides specific opportunities for women.

● **CONCLUSIONS - ONGOING CHALLENGES TO GENDER EQUITY IN THE CONTEXT OF THE GCF CND PROJECT**

Despite tremendous efforts from the policy and legal perspectives, a patriarchal culture and persistent disparities continue to characterize gender relations in Rwanda. Disparities persist in post-primary education; in access to and control of property including land and economic resources; in employment opportunities and entrepreneurship; in decision-making at household and community levels; in family responsibilities and unpaid care work; and in the experience of violence, harassment, conflict, and insecurity. Sexual and gender-based violence persists at high levels in Rwanda. While women have made impressive political progress, especially at the national level, their progress in terms of local political representation and engagement remains unequal. Economic opportunities remain markedly gender-differentiated.

In Rwanda and in the Congo-Nile Divide, structural barriers to economic and political equity and female empowerment have significantly impeded measures to adapt to climate change impacts. In Rwanda, these structural inequalities are at a lower level compared to other countries in the world, because of leading gender-considerate policies across sectors. However, the point remains that women's ability to access, use and control natural resources, infrastructure and services differently is still low compared to men. This means the degradation of natural resources and new infrastructure will affect women and men differently, and will generally result in greater vulnerability of women. Women are vital agents of change and can be powerful leaders from the community to global level in mitigating and adapting to climate change.

1.14 Afforestation, agroforestry and tree tenure

Kiptot et al (2012) noted that an ICRAF survey conducted across countries in eastern Africa concluded that women rarely have tenure over trees unless they exclusively have tenure over the land. Moreover, they stated that "Women's rights to tree products are usually limited to products that are considered to have little or no commercial value. These products are mainly indigenous fruits....fodder and mulch." They also noted that: a) the survivorship of trees and fodder shrubs planted by men and women was effectively the same, b) low cost/input agroforestry techniques for replenishing soil fertility were attractive to women farmers given that they are both labor and capital limited. Fuelwood, fodder, green manure, and coppicing shrub production substantially increases benefits to women that cannot be captured by men. Women appear to grow trees/shrubs for domestic consumption (fuelwood, fodder, bean poles, fruit) and sell the surplus in local markets. It is unclear whether women who sell wood products from land that is owned by their husbands are able to capture that revenue stream.

Agroforestry offers multiple benefits to Rwandan farmers, in particular, women. Planting trees within farms represents a low effort pathway to high reward. To enhance climate resilience of communities, agroforestry practices sequester carbon, prevent erosion, and enhance soil quality for sustainable

farming. For women, trees provide additional income to elevate socioeconomic status and reduce the risk of GBV by eliminating the need to walk far distances for fuel wood (Anastase & Nahayo, 2020). Despite the majority of farms under female management in Rwanda, women are often excluded from agroforestry, as trees traditionally fall under male domain, women lack the necessary training and have less access to the retail market (Kiptot & Franzel, 2012). In order to better include women in agroforestry, training on forestry and marketing should be targeted to women. Tools and practices that reduce labor requirements for women, such as shorter trees (which decrease the need for male labor) or more modern processing techniques, will save women time and make the field more accessible (Kiptot & Franzel, 2012).

Afforestation and reforestation on public lands should include measures where women and youth formally have the right to capture a proportion of the value of the trees at harvest time. Assuming a harvest cycle of 15-20 years this will provide women with a long-term asset that could be used as retirement income, and/or given as gifts or bequeathed to children. Young men could use the trees as savings to be used to build or buy a house when they marry.

Investments in agroforestry on private farmlands should reflect the different interests and concerns of women and men. During focus groups women reported shortages of fuelwood, fodder and wood suitable for bean poles. They were also concerned about soil erosion on their steeply sloped farmland, and declining soil fertility and crop yields. On-farm tree planting, to address the concerns of women, should focus on fast growing, nitrogen fixing, coppicing species that can be planted in contours between rows of crops. The nutrient rich leaves of leguminous species can be used as fodder for livestock, and as green manure. Cutting the stems to ground level before planting ensures that the trees do not compete with the crops for sunlight and soil nutrients, but the tree roots that remain intact help stabilize the soil and prevent erosion caused by heavy rains. The cut stems can be dried for fuelwood or used as bean poles. Surplus fodder, fuelwood and bean poles can be sold to neighbors or at local markets. Coppicing species readily re-sprout from the roots allowing an annual harvest.

1.15 Improved cook stoves

In rural areas of Rwanda, 98% of households rely on wood as their main cooking fuel. The GoR completed a comprehensive report on Gender in the Energy Sector. The report noted that almost 60% of FHHs spent 1-3 hours per day collecting firewood compared to 36% of MHHs. The latter purchased fuelwood twice as often as FHHs who primarily rely on gathering fuelwood. Over 50% of both Rwandan FHHs and MHHs still cook using the three-stone method, and only 13.5% use efficient cookstoves. Over 50% of FHHs are unaware of the health benefits of improved cookstoves. The report argued for supporting women to transition from fuelwood to Liquid Petroleum Gas (LPG) primarily to reduce smoke inhalation and respiratory disease. The report did not address the fact that in higher altitude areas cooking fires also provide a heating source for the home. The use of clean cookstoves has major positive impacts on gender equality, by reducing household work for women and providing opportunities for women to engage in paid work.

Poor rural families, particularly FHH, are unlikely to be able to afford LPG stoves. Moreover, LPGs do not remain warm after the gas is turned off and thus do not provide a heat source for the home during the night or during cold spells. Tier 1 Canarumwe clay stoves are relatively inexpensive, substantially reduce woodfuel use, and provide a thermal mass that keeps radiating heat long after cooking is

completed. Though imperfect they are an improvement in efficiency compared to the 3 stones cookfire method. As men do not gather firewood nor do they cook they are often unwilling to spend “their” money to purchase an improved stove even though it substantially reduces fuelwood use and the time women must spend gathering fuelwood. Efforts to introduce improved cook stoves into MHHs in the CND must address this barrier.

1.16 Honey production and marketing

The Rwandan government has identified beekeeping as a potential source of additional revenue for farmers (Mushonga et al., 2019). The European demand for honey far outpaces the production in Rwanda, indicating a market incentive to raise bees beyond subsistence level (Mushonga et al., 2019). Furthermore, global decline of bee populations reflects a high biological need for more apiarists in Rwanda. Despite the economic and environmental benefits, along with low startup costs, beekeeping is perceived as a vocation for poor people. Out of 83,000 beekeepers in Rwanda, 86% are adult males and 71% lack primary education (Mushonga et al., 2019). To overcome the negative stigma around beekeeping, materials and training for modern hives need to be provided to women and youth. Modern hives produce a higher yield of honey, have lower risk of causing fires, and do not need to be placed as high up in trees as traditional hives, which deters some women due to cultural restrictions barring women from climbing trees (WCS, 2018). In a WCS study, 83% of beekeepers used traditional hives in Rwanda, demonstrating a need for more modern tools (WCS, 2018). Additionally, marketing training may help women to better sell the honey, as women still have relatively low market and distribution access in Rwanda. Despite the stigma, beekeeping remains a strong potential source of income, which could reduce reliance on more harmful forest livelihoods, such as deforestation and poaching (Fakhry et al., 2010).

A study by Mushonga (2019) suggests that people consider beekeeping for the poor and illiterate are largely the domain of older men no longer able to labor in agricultural production. Women do appear to be engaged in the production and marketing of honey beer. ADF provided \$343,000 in 2017 to strengthen the honey production sector in Rwanda. Production remains relatively low because of dependence on traditional hives rather than modern hives such as Kenya Top Bar or Langstroth (CODIT, 2009). The CODIT report funded by the SNV noted that in 2009, of the estimated 30,293 beekeepers in Rwanda only 23% were women. The report recommended that the GoR do away with subsidies to beekeepers and their co-operatives, focus on addressing the root causes of the prevailing problems not the effects, encouraging beekeepers to actively participate in their own development and advocating for the replication of knowledge and skills from common to individual apiaries.

During focus groups both women and marginalized groups noted that the cost of modern hives and other beekeeping equipment such as smokers was a barrier to entry as was lack of access to credit. A big concern was farmers’ use of pesticides that kill bees, making it impossible to locate apiaries near farmland.

1.17 Coronavirus (COVID-19) implications in advancing gender equity

The global coronavirus (COVID-19) pandemic left no country untouched from its impacts. While research is still emerging, the severity of the pandemic is clear; struggling economies, vast unemployment, and threats to health and wellbeing continue to affect the global community. Women in particular are already

vulnerable to socioeconomic inequity that is currently exacerbated by COVID-19. Therefore, the impact of the pandemic must be considered in gender planning into the future.

Rafaeli and Hutchinson (2020) compared the secondary impacts of COVID-19 to the Ebola crisis on women and girls in Africa, finding that women and girls are more likely to experience higher poverty and school dropout rates, an increase in unintended pregnancies, reduced income and financial empowerment, greater domestic responsibility (increased childcare due to school closures), and food insecurity due to the pandemic. In times of crisis, school closures and financial stress also lead to increased rates of gender based violence (Rafaeli & Hutchinson, 2020). Women face greater exposure to COVID through their roles as caretakers in families and in healthcare, and have less decision-making power, rendering their needs unmet during COVID-19 (Parry & Gordon, 2020). Yet few countries have enacted policies to address the gendered impacts of the pandemic.

In Rwanda, the Ministry of Gender and Family Promotion (MIGEPROF), in partnership with UN WOMEN and UNFPA, conducted a 2020 study on the gendered impacts of COVID-19. The report indicates that 44.2% of female respondents completely stopped their economic activities due to the pandemic, compared to only 17% of men (Bayisenge, Lo & Schreiner, 2020). In rural areas, the percentage was higher for women; 50% stopped their work and thus sources of income, compared to 17.7% of rural men (Bayisenge, Lo & Schreiner, 2020). As women occupy more informal and unsecured jobs in Rwanda, they are at greater risk of income loss than men.

The use of health care and maternal services in Rwanda considerably declined during the COVID-19 outbreak (Wanyana, Wong & Hakizimana, 2020). Declines in child health care services, particularly vaccinations, could be detrimental to avoiding future outbreaks of vaccine-preventable diseases (Wanyana, Wong & Hakizimana, 2020). Many women feared contracting the virus while attending regular checkups, which contributed to the decline of health services in Rwanda (Wanyana, Wong & Hakizimana, 2020).

Recommendations to mitigate the impacts of COVID-19 on women in Rwanda include (Bayisenge, Lo & Schreiner, 2020):

- Implement programs that offer employment or financial security in spite of COVID-19.
- Provide online or socially-distant courses to prevent school dropouts.
- Establish women-specific financial packages in the national economic recovery plan.
- Strengthen the mobilization of men and boys in childcare and domestic responsibilities to offset increased labor caused by the pandemic in parallel with on-going “Men engage” national programmes.
- Accelerate government interventions to address elevated levels of GBV.

1.18 Promoting gender equity within the CND GCF project

Gender equality and women’s empowerment should be considered in all stages of the CND project cycle: design, formulation, implementation, and monitoring and evaluation. In each of these stages, project managers should keep a ‘gender lens’ in mind, looking at ways the project can:

- Address gender inequalities that emerge from the project;
- Ensure the differential needs of women and men are addressed;
- Ensure women and men have equal access to resources, services, and capacity development;
- Ensure equal participation of women and men in management arrangements and as

beneficiaries, partners and key stakeholders; and

- Ensure women's equal participation in decision-making processes.

Based on key expected outcomes, this gender assessment provides the foundation for a realistic gender action plan to be implemented in the CND project.

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● GENDER, YOUTH AND SOCIAL INCLUSION ACTION PLAN

The goal of the programme is to increase climate resilience of vulnerable rural communities in the CND, reduce CO₂ emissions, build capacity for integrated spatial planning and increase the extent and integrity of forest ecosystems in the CND. The programme has three components directly targeting vulnerabilities within climate impact chains. Component (1) *Mainstreaming Climate Adaptation into Integrated Land Use Planning and Management* will lead to enhanced adaptive capacity of both ecosystems and rural populations, by strengthening coordination among sectors and integrating climate resilience considerations into planning and development. Component (2) *Landscape Management and Restoration* will implement high priority actions to enhance the resilience of vulnerable communities by increasing the extent and integrity of forest ecosystems and landscapes and by reducing demand for wood fuel. In addition, component (2) is designed to reduce the time women spend collecting and preparing fuelwood, and simultaneously to empower women to provide input to local-level decision making (see also Component (1)). As a co-benefit, greenhouse gas emissions will be reduced through restoration of natural forests, agroforestry, reduced fuelwood demand and fuel efficient cookstoves under components (1 and 2) are expected. Component (3) will focus on *Enhancing climate adaptation through resilient livelihoods*.

Women, youth and marginalized groups, including women-headed households in the CND are key stakeholders in the success of this project, and are the primary target beneficiaries of the project. The project will be implemented across various sectors along the CND, particularly focusing on communities around Nyungwe National Park and Gishwati-Mukura National Park--critical for Rwanda yet highly vulnerable forests due to high population and increasing demography who depend upon fuelwood, and actions taken by poor households that illegally use forests to improve income streams.

1.19 Methods

This Action Plan operationalizes the findings from the Gender Assessment, the ESIA, and youth-based information collected during the feasibility studies and stakeholder engagement during the Project Preparation Facility (PPF) phase. The Assessment used five standard lenses (USAID 2017^[4]) to conduct the analysis:

1. Institutions, Laws and Policies that influence the context in which men and women act and make decisions.
2. Beliefs, Perceptions, and Cultural Norms that shape how men and women, youth and elders are able (or expected) to practice activities, particularly those to be developed by this GCF project.
3. Gender Roles, Responsibilities, and Time Use in reproductive, productive, and community work.
4. Access to and Control over Assets and Resources are different between young and old, men and women.
5. Patterns of Power and Decision-making focused on how individuals are able to be represented and be in decision-making positions.

Based upon this framework of five lenses, the Gender Analysis identified and analyzed the gender risks and impacts of the program and climate change through four main steps:

Scoping: The GYA included a scoping process to review relevant literature and data for the project area, including local reports and data, consult with government and other partners, and conduct field visits to understand the project, identify and characterize the project area, and identify stakeholders,

including program-affected parties and other interested parties, who may have different concerns and priorities about project and climate change impacts, mitigation mechanisms and benefits.

Legislative and policy review: The GYA conducted a review of (1) Rwanda's applicable policy framework, national laws and regulations and institutional capabilities relating to gender and youth, (2) variations in country conditions and program context across the affected area, (3) relevant gender and youth studies, (4) Rwanda's obligations and commitments under relevant international agreements and treaties, (5) applicable requirements under GCF standards and (6) internationally accepted best practices.

Stakeholder consultation and analysis: Based on the identification of stakeholders identified in the scoping exercise, the GYA conducted initial consultations with stakeholders to gather information on concerns and priorities about the project and climate change, analyze the outcomes of the consultations and develop a strategy for continuing stakeholder engagement.

Baseline analysis: Drawing on the scoping exercise, the GYA assembled and analyzed existing information. The information was analyzed in light of the stakeholder consultation and analysis and legislative and policy review, to identify information gaps and ways in which these may be filled.

This information was then used to guide the Gender Youth Social Inclusion Action Plan. The details of the stakeholders consulted, including key informant interviews (18), focus group discussions (24), and district workshops (4) can be found in Annex 7. A literature review is available in the Gender Assessment (above) and ESIA (Annex 6).

1.20 Gender Issues

There are three key gender-based themes that influenced the design of the Gender Action Plan:

a. **Change negative gender and social norms.** The most profound barrier for this project is rural norms about the roles of men and women. Whilst the Government of Rwanda has made many advances and has implemented many policies (see Section 3 of the Gender Assessment) to achieve gender equality, particularly in national-level representation and inclusion, patriarchal privilege and decision-making remains embedded in rural society (IUCN, 2021). The consequences of this for this project is far-reaching. Men are generally seen as the head of household with the decision-making powers over all assets, resulting in male appropriation of household and farm-based assets (IUCN, 2021; Kiyani et al., 2017). Though female headed households face the same issues and challenges as male headed households, they are disproportionately impacted because of prevalent gender inequality issues (GMO 2017). Men in the CND are positioned to be significantly more involved in programs that result in monetary benefits, such as fern clearing, casual labor, and plantation projects. Although under Rwanda law, sons and daughters receive equal inheritance rights (Law/n° 22/99 of 12/11/1999) the PPF focus groups highlighted that due to cultural practices, parents favor sons in land inheritance, leaving young women in a position where they depend on marrying a man to gain access to farmland; without a formal marriage, these women have no land tenure. Broadly, these gender and social norms leave most women with little decision-making power and no real control over assets. The threat of sexual and gender based violence (SGBV) at the household level where women try to assert independence and achievement is high. For this project, the implications are that trees planted by women become men's property; money gained in employment (tree planting, nursery maintenance) or through farmland production (fruit trees, honey, etc) are ultimately controlled by men; and women shouldn't participate in activities without a

husband's permission. Although this seems bleak, this situation is not universal. The 24 CND community focus groups held in the PPF highlighted that many households decide together on how money should be spent; land tenure is broadly seen as joint ownership; and men realize the impact of dwindling fuelwood resources on women and girls. Gender equity is not yet universal and rural culture has a long trajectory to achieving it.

As a result, this program will focus heavily on social and behavior change communications (SBCC) as a means to positively change perceptions towards gender equity. Community members, including community leaders and women and men community influencers specifically will be targeted for sensitization on gender equity in order to shift their attitudes and to build a conducive environment for women and HMPO to be able to participate in project activities. Women have been engaged in the development of this project to ensure time constraints, work burden and reproductive responsibilities do not impede their ability to engage in the various activities designed by the project. Women will be regularly consulted throughout project implementation to ensure they are able to actively participate in project activities and activities take place at times which are convenient for them and at locations that are safe and contextually appropriate. At project start, once communities are selected for implementation, barriers faced by women and HMPO to participate in project activities will be reviewed and solutions to overcome barriers incorporated into programming through an additional screening process outlined in Annex 6. The project ESS Coordinator and Gender Officer will provide support to ensure all project staff understand barriers faced by women and HMPO and programming works to address barriers to ensure their active participation in activities. The project will also get support from NGOs that help men on masculinity and addressing SGVB which will include targeted outreach to mobilize their participation in household and parental responsibilities to ease women's responsibilities and sensitization on the importance of fair access and control of resources, including income. The engagement of community leaders and women and men community influencers to further promote equal and fair control of resources between men and women will further mitigate the risk of men taking over control of incomes earned and targeted for women. Lastly, the project will clearly explain participant selection, intended benefits and how participants can help monitor benefits and risks and reduce risks if necessary, including reducing the risk of men taking over control of incomes targeted for women. The program will have a safe, accessible Grievance Redress Mechanism (GRM) for all project stakeholders, particularly women, to access to pursue redress as needed. The project will also collaborate with existing Isange One Stop Centers in the project area in GBV prevention and immediate assistance to victims. All project staff will be trained and project beneficiaries informed on the purpose and use of the GRM, including reporting gender-based violence. The project will ensure confidentiality of the information and identity of the complainant. The grievance process will include options for victims to remain anonymous which will add a layer of protection for individuals who may be subject to gender-based violence. Throughout project implementation, project staff will disseminate information to beneficiaries on how they can use the GRM and solicit feedback on how the mechanism can be improved if needed.

b. **Provide access to skills building, leadership and programmatic inclusion.** Because of existing gender norms, there is much to do to build technical skills of rural women to be able to participate in the project. In addition to that, finding women leaders and building their leadership skills to be at the forefront of rural gender equity is crucial for addressing the root causes of women's exclusion in land use planning and decision-making. This project however will not just focus on rural women. Ensuring

that women play a strong role at all levels of climate programming in the country is important. Thus developing professional networks and internships to provide opportunities to strengthen leadership and technical skills will be incorporated into the program.

c. **Facilitate women's financial autonomy.** In order to support rural women to capture financial benefits of this project, this project will incorporate three activities: financial skills building, support in joining cooperatives; and facilitating access to entrepreneurship through financial/entrepreneurship training, access to village savings/loans and small grants, and mentoring. The project will ensure all trainings are gender-responsive and inclusive. Gender considerations will be integrated into training curriculum, content and delivery methods. Flexible training schedules and convenient venues will be selected to accommodate women's needs. Staff in charge of gender aspects will ensure gender mainstreaming and ensure the project addresses barriers to women's participation. Training will also use participatory approaches to create an inclusive learning environment and incorporate components on women's empowerment and leadership development, promoting gender equality and active participation. By facilitating independent access to financial services, women's income can be managed independently until shifting norms and household situations reach gender equity for households in adapting to climate change. Within the program, microenterprise, access to finance, beekeeping, and agroforestry/cookstove value chain support were chosen as key activities because they are feasible livelihoods within the target rural households and help strengthen resiliency of the rural economy. Further information on women in the value chains targeted by the project, including gender characteristics, composition of workforce and ownership etc. is detailed in the project Feasibility Study where possible and will be refined during project implementation.

1.21 Youth Analysis and Key Issues

Rwanda is a young country; 82% of the population is under the age of 40 and 51.5% are 19 or under (NISR, EICV5 2018, see Table 5). The sex ratio for youth, classified as people from 14-35 years old, is 92 males for every 100 females. The literacy rate for youth is quite high; averaging 91% in urban areas and 82% in rural areas (ibid). Numeracy rates are linked to wealth, with poorer families having lower capacity (72.4% vs 92.7%). Larger urban-rural discrepancies are found in computer literacy (28.5% vs 7%) and mobile phone ownership (70% vs 41%).

Table 5: Youth population in Rwanda by district. (Source: NISR EICV5 2018)

Youth Population in Rwanda (Source: NISR EICV5)			
District	Youth	Non-Youth	Total Population (000s)
Kigali City	36.5	63.5	1,631
South	24.3	75.7	2,739
West	25.1	74.9	2,685
North	26.8	73.2	1,841
East	24.6	75.4	2,998
All Rwanda	26.6	73.4	11,893

Table 6. Youth Usual Economic Activity by Sex and Province. Source: (NISR EICV5 2018).

Activity Type	Youth Usual Economic Activity by Sex and Province (Data source: EICV5)												
	Kigali City		Southern Province		Western Province		Northern Province		Eastern Province		Total		
	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	All
Worker	70.4	66.5	76	78	78.1	81.8	78.2	81.2	80.5	79.8	76.8	77.6	77.2
Seeking work	5.5	11.7	1	2.3	1.5	1.5	0.9	1.9	0.7	0.9	1.9	3.5	2.7
Student	21.7	15.9	18	14.7	16.8	12.1	17.5	14.6	15.2	14.9	17.7	14.4	16
Domestic	1.8	4.1	2.6	3.5	1.5	2.4	1.2	1.4	1.9	3	1.8	2.9	2.4
Disability	0.3	1	1.4	0.8	0.9	1.3	1.1	0.6	1.2	1.1	1	1	1
Other	0.4	0.7	1	0.7	1.2	0.8	1.2	0.3	0.5	0.3	0.8	0.6	0.7
Total	100	100	100	100	100	100	100	100	100	100	100	100	100
Usually	75.9	78.2	77.1	80.3	79.6	83.3	79.1	83	81.2	80.6	78.6	81.1	79.9
Inactive	24.1	21.8	22.9	19.7	20.4	16.6	20.9	17	18.8	19.4	21.4	18.9	20.1
Total	100	100	100	100	100	100	100	100	100	100	100	100	100

Employment figures for rural youth indicate that 54.7% are engaged in agriculture, and 32.9% in elementary activities. The rest are in services (7.8%), and very few in managerial, professional etc jobs. According to a recent analysis, youth “envision self-sufficiency achieved through economic independence and the ability to provide for and support their families, and they see this path as most available to them through self-employment (USAID 2019)”. The Youth Assessment highlighted key priority areas that youth identified for achieving their goals:

- Livelihoods and skills development for productive self-employment activities
- Education and skills for entering into the formal work sector
- Reproductive autonomy and health to avoid early pregnancy which is a key hurdle towards economic self-sufficiency

Poverty amongst youth has remained stable between EICV4 and EICV5, with about 30% of all youth below the poverty level, and 11.3% in extreme poverty. There are substantial gender differences, with young women increasingly vulnerable as they age (see Figures 7 and 8 with data from NISR’s EICV 5, 2018). Thus, programming ensuring attention on both young men and young women is vital for rural climate resiliency.

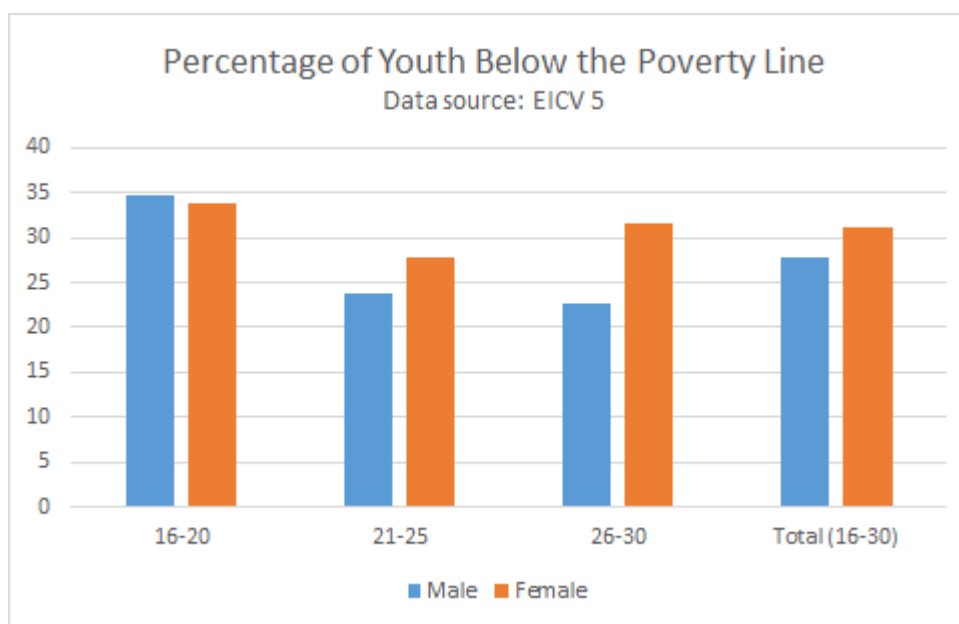


Figure 8: Percentage of youth below the poverty line by age (Source: NISR EICV5 2018)

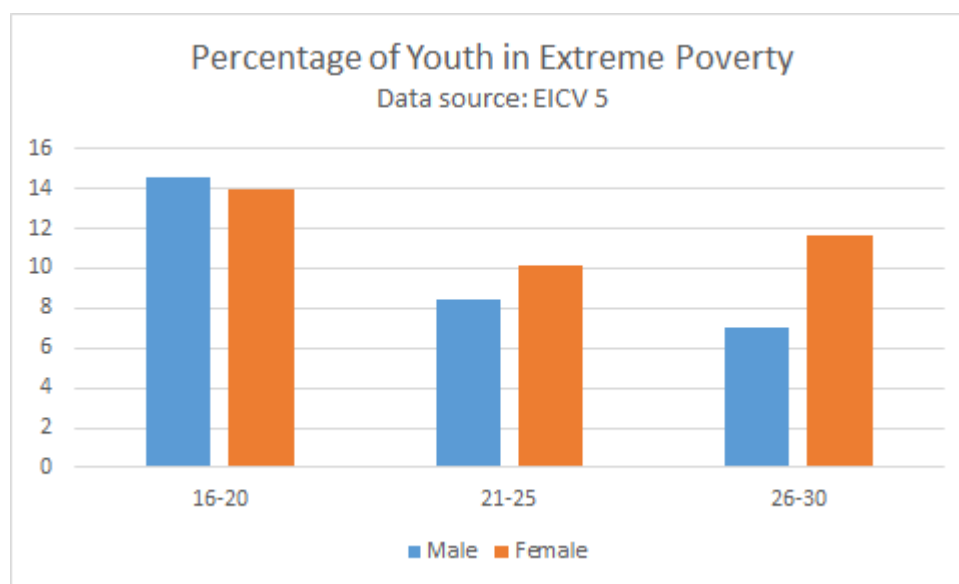


Figure 9: Percentage of youth in extreme poverty line in Rwanda by age. (Source: NISR EICV5 2018)

Translating this information into a climate adaptation program that supports young men and women in the CND, three key topics influenced the action plan:

- a. **Diversified opportunities.** With over 51.9% of the Rwandan population under the age of
- b. 15 and 18% between the ages of 15-24, youth make up a significant percentage of the population. There is already a scarcity of arable land, with many households owning less than what is needed for their subsistence. With the Rwandan population expected to increase from 12 million in 2019 to 20 million in 2042. The future for the upcoming generation of rural youth is worrisome--agricultural

lands will continue to grow scarcer and demand for resources and land increase. The youth interviewed are very cognizant of these constraints, and many are keen to diversify their livelihoods yet they are most interested in short-term gains. This GCF program intends to seek out vulnerable youth in critical areas of the CND to support entrepreneurship and diversified opportunities to reduce pressure on highly marginal lands.

c. **Access to skills building.** Many youth attend schools, but training in practical agroforestry, financial management, and forestry is lacking. This program intends to target youth for diversifying their skill sets and strengthening their resilience to adapt to climate change.

d. **Youth leadership and Programmatic Inclusion.** Many youths voiced their concerns over not being included in decision-making processes and programming, resulting in key stakeholders not being engaged in civil society. In order to engage the new generation of rural farmers and consumers, this project will identify and work with youth leaders in the CND to use SBCC to engage youth to benefit and support long-term tree initiatives. With youth involved in decision-making processes, innovative ideas to involve youth with best practice on government lands and to get youth to appropriate long-term thinking through tree planting as future investments.

1.22 Internal program considerations

In addition to programmatic activities with women and youth, this program will also have an internal focus to ensure best practices. These include:

a. **Avoid perpetuation of gender inequalities** and women's exclusion in the agro/forestry sector. SBCC will be used heavily to shift norms towards inclusivity and skills building to facilitate women and youth to join the sectors. In addition, women need to be engaged in the decision-making bodies of the project itself, including a Steering Committee, which will be at least 50% women, and the Project Management Unit with targets to achieve 50% women by the end of the program as mentoring and training increase capacity. The Gender Monitoring Office will be a part of the Steering Committee and along with the The National Women's Council and The Ministry of Gender and Family Promotion were consulted during proposal development and will continue to be involved at project implementation stage at the national and local level. There will be comprehensive consultation and feedback processes with these stakeholders through meetings and workshops to ensure perpetuating gender inequalities is avoided.

b. **Build partnerships.** This GCF project provides a unique opportunity to engage civil society at national and local levels. The team will work with men's, women's, and youth organizations to build their capacity and facilitate their engagement with project beneficiaries and local government actors.

c. **Gender and Diversity training.** Training will be an important component of a large, 5-year program. Building online training modules that can be viewed across the CND, and using new platforms like WhatsApp to communicate with urban and rural project staff will ensure that static, one-off training is avoided.

d. **Staffing.** Diversity and inclusion are important for the project, and hiring practices at all levels of the program will be reflected in hiring policies. The program management unit will hire a full-time dedicated gender expert and a full-time dedicated youth focal point (both AE staff) and associated costs are included in the project budget. These two staff will participate in all management meetings to support inclusion goals across the project, and will lead and oversee the implementation of the Gender, Youth, and Social Inclusion Action Plan. Another key role of these two positions will be to engage civil society

partners, particularly women's, men's, and youth groups to build capacity and increase the sustainability of achievements during this five-year project. Along with these positions, RFA staff will include a Community Mobilization Specialist and a Community Engagement Specialist who will be in charge of and support project gender mainstreaming. Their responsibilities include reviewing site specific barriers faced by women to participate in project activities, ensuring these barriers are addressed in order to support full participation of women, ensuring gender equity in all project activities and where needed help women access the Grievance Redress Mechanisms to pursue redress and connect women with Isange One Stop Centers in the project area to prevent gender-based violence and provide immediate assistance to victims.

e. **M&E.** Many stakeholder meetings highlighted the lack of community engagement as a key reason for project failures in the past. This project will incorporate suggestions from our stakeholder engagement meetings to ensure best practice is followed. This includes not only deeper local government engagement but also using a rights-based approach to ensure that communities are at the heart of planning our interventions. Targets for youth and gender are incorporated into the Gender and Youth Action Plan below and are set to ensure women and youth are equally targeted where possible and encouraged to actively participate. Exceptions include situations where group composition, such as government agents or community leaders, are outside the project's control or the activity solely targets men, such as masculinity sessions held with men to improve gender equity in program participation and benefits or assisted natural regeneration activities inside the park far from villages that will take male participants several weeks away from their homes. Overall targets for key activities (i.e., G4. 4.1.1.5, G4. 4.1.1.6) have not been set as the target will depend on the number of villages/sites where project activities take place, however disaggregated targets per village/site have been set. A project baseline will be conducted and baselines for all targets will be determined at project onset.

1.23 Gender and Youth Action Plan

Component 1: Mainstreaming Climate Adaptation into Integrated Land Use Planning						
Impact 1: Climate adaptation programming is successfully implemented because it is fundamentally informed by rural women and youth stakeholders <i>Indicator: Equal participation by gender groups in contributing to expected outcomes</i>						
Gender/Youth Outcome 1: Climate adaptation programs in LUPs reflect and address the priorities of women and youth <i>Indicator: % of men, women and youth participating in LUP processes who feel LUPs incorporate their needs as they relate to climate programming</i>						
Activity	Gender/Youth Outputs	Indicators	Targets	Timeline	Responsible	Costs
G1. 1.1.1.7 Conduct climate literacy seminars for local government and civil society organizations, aimed at increasing women and youth participation	Local institutions are trained to build capacity of rural people, specifically women and youth, to participate in climate adaptation processes	# of CSOs and government agencies trained	60 people (50% male, 50% female) from CSOs/Local institutions trained in Y1 and Y3 to increase participation of women/ youth/ marginalized in stakeholder consultations & activities	Y1,3	PMU/RFA	\$10,187
G1. 1.1.1.8 Provide financial and logistical support to trained organizations in grassroots mobilization to increase women and youth participation in climate adaptation planning	Trained institutions hold meetings to train women, youth and marginalized people on forests and mobilize them to participate in planning processes	# of meetings held by trained organizations with local communities on climate adaptation programming	At least 10 grassroots meetings per district are held in Y1 and Y3 in the CND	Y1,3	PMU/RFA/ Local CSOs (TBD)	\$12,262

G1. 1.1.2.4 Develop and rollout a series of trainings (virtual sessions and online modules) on gender sensitivity and mainstreaming women and youth into planning	Program Staff and government officials understand gender barriers, understand how to mainstream gender equality principles into planning and are strong facilitators to support women, youth and marginalized participation in meetings	# of people trained (disaggregated by gender) # project staff trained in social inclusion	- 65 government agents trained per year. - 100% project staff trained	Y1-5	PMU + Local Women's NGO (TBD)	\$40,523
Component 2: Forest Landscape Management and Restoration						
Gender/Youth Impact 2: Sustainable gains in restoration and landscape management are made because women and youth are enabled and motivated to adopt best practice <i>Impact Indicator: Targeted women and youth report an improved enabling environment for their participation in forest management and restoration</i>						
Gender/Youth Outcome 2: Women, youth and marginalized households have improved capacity to engage in forest management and restoration <i>Outcome Indicator: Equal participation by gender groups in forest landscape management and restoration</i>						
G2. 2.2.2.15 & 3.1.1.11 Develop and implement strategy (including social marketing) to ensure that women and youth benefit from forest restoration projects	Women and youth have the capacity, motivation, and opportunity to participate and benefit equitably in restoration programs. Programs will actively seek out direct participation of and benefits for women and youth through paid labor and direct involvement.	% Women beneficiaries % Youth beneficiaries	50% of beneficiaries are women 50% of beneficiaries are youth ²	Y1 Strategy Y2-5 Implementation	PMU, Women/Youth NGOs (TBD)	\$37,735
Component 3: Enhancing climate adaptation through resilient livelihoods						

² The total number of beneficiaries will be determined at project onset

Gender/Youth Impact 3: Women, marginalized and youth are empowered to actively participate in green jobs
Impact indicator: Target groups perception of economic empowerment

Gender/Youth Outcome 3: Women, youth and marginalized households have improved capacity to engage in green jobs
Outcome Indicator: Number of new women, youth and marginalized households that have engaged in green jobs

G3. 3.2.1.2 Develop and implement strategies to ensure that women,youth, and other marginalized groups benefit from rural livelihoods and financial services	Women and youth have the capacity, motivation, and opportunity to participate and benefit equitably in rural livelihoods and financial services	% Women beneficiaries % Youth beneficiaries	50% of beneficiaries are women 50% of beneficiaries are youth ³	Y1 & Y2	RFA & CORDAID	\$53,053
G3. 3.2.3.2 Organize Microenterprise training and develop and implement green grants programs targeting landless youth and poor, as well as women who are unable to access direct benefits from land-based project activities.	Landless poor, youth, and women have enterprise skills and grants to engage in green value chains	# of enterprises formed, beneficiaries disaggregate d by gender and social class	5,600 beneficiaries of which at least 50% are women 500 off-farm jobs created in production quality control	Y2,3 & 4	RFA & CORDAID	\$66,398
G3. 3.3.2.2 Provide organizational/technical/financial capacity building to women, youth, and CSOs (such as cooperatives or VSLA) with focus on financial literacy and provide long-term coaching to beneficiaries	Women and youth groups have the organizational, technical and financial skills to receive support from project	# CSOs/VSLAs trained # beneficiaries receiving coaching	300 CSOs/VSLAs 6,900 of which 40% youth 50% women	Y2,3 & 4	RFA & CORDAID	\$30,040

³ The total number of beneficiaries will be determined at project onset

G3.3.3.2.3. Support women, youth, other marginalized groups strengthen savings and loan groups and register for MFI accounts	Men, women and youth have access to thriving financial services	# people with new bank/MFI accounts	50% women	Y2,3,4	RFA & CORDAID	\$13,713
Component 4: Monitoring and Evaluation						
Gender/Youth Impact 4: Women and youth benefit from an adaptively managed program <i>Impact indicator: Reported attitudes, confidence, and a sense of empowerment towards engaging in climate programming</i>						
Gender/Youth Outcome 4: Disaggregated and specialized monitoring generates data showing whether and how women, youth and marginalized people are benefiting from adaptive management systems <i>Outcome indicator: Gender and youth disaggregated participation data</i>						
G4. 4.1.1.4: Develop an overarching gender and youth program unit within the GCF Project to ensure the project uses adaptive management to ensure strong inclusion of women, youth, and marginalized households such that equitable benefits are shared across the communities.	Gender/Youth program unit provides oversight and support to ensure gender/ youth targets are met	N/A	N/A	Y1-5	PMU	\$3,380
G4. 4.1.1.5: With the local government, project partners, and Youth CSO leaders, identify key male and female youth leaders in each project village to be project champions	Key youth leaders are identified and engaged	Youth leaders identified	2 Youth leaders (50% male, 50% female) per targeted village. ⁴	Y1	PMU, Youth NGO (TBD)	\$88,683

⁴ The number of targeted villages will be determined at project onset and will determine the total number of youth leaders to be reached. 50% male and 50% female youth leaders will be targeted per village.

G4. 4.1.1.6: With local government, project partners and women's civil society, identify key women leaders amongst project beneficiaries to be project champions	Key women leaders are identified and engaged	Women leaders identified	2 women leaders per village ⁵	Y1	PMU, Women's NGO (TBD)	\$87,903
G4. 4.1.1.7: Engage Youth & Women's NGOs to build leadership and coaching skills of identified project champions and facilitate their project involvement in order to advance project goals of greater participation amongst marginalized groups amid greater community-wide support for their participation	Leaders are equipped with skills and financial resources to build empowerment and facilitate women and youth engagement in the project	# of women and youth trained in coaching and leadership skills	200 (50% women, 25% male youth leaders, 25% female youth leaders) women and youth leaders trained as trainers	Y1-2	PMU	\$6,660
G.4. 4.1.1.8: Overarching social behavior change communications strategy and materials are developed to overcome barriers and enhance motivations for women, youth and marginalized people to participate in program activities	Gender sensitive and responsive communications programming supports new norms for social inclusion in project activities	# programs/ materials delivered in communities	At least 10 materials developed and distributed annually	Y1-5	PMU	\$97,552

⁵ The number of targeted villages will be determined at project onset and will determine the total number of women leaders to be reached. 100% of women leaders will be female

G4. 4.1.1.9: Informational campaigns on equity (land rights, access to resources, and decision-making) and programmatic opportunities to showcase the benefits of everyone (women, young men/women, marginalized) participating in programming to improve lives	Gender sensitive and responsive campaigns and informational material on inclusion of different social categories is developed and disseminated	# of people reached by campaigns	At least 200,000 (50% female, 50% male) people reached by campaigns over 5 years	Y1-5	PMU	\$17,030
G4. 4.1.1.10 Engage men's NGO to roll out programming for men as allies in women's economic empowerment and sexual/gender based violence reduction programming	Men in target areas participate in masculinity sessions to improve gender equity in program participation and benefits	# men participating in training sessions	At least 200 men participate in masculinity sessions each year from Y2-5	Y2-5	PMU Men's NGO	\$115,574
G4. 4.1.2.2 Identify and train men and women (adults and youth) to implement key M/E tools in their communities	Women and youth are active participants in M/E at central, sectoral, and community level	# Women and Youth trained on M/E tools	At least 5 women, 5 men and 5 youth (15 people total) per district are trained ⁶	Y1-5	PMU	\$3,380
G4. 4.1.3.2 Engage youth leaders in designing and monitoring targets for youth engagement, participation, and outcomes	Youth targets are identified and incorporated into M/E system	N/A	N/A	Y1-5	PMU	\$ 16,533
G4. 4.1.3.3 Engage women leaders in designing and monitoring targets for women engagement, participation, and outcomes	Women targets are identified and incorporated into M/E system	N/A	N/A	Y1-5	PMU	\$ 16,533

⁶ The number of targeted districts will be determined at project onset and will determine the total number of women, men and youth to be trained. The target for the number of individuals trained overall is 50% female, 50% male.

G4 .4.1.3.4: Monitor program using Gender Checklist to ensure gender is integrated in programming, HR, policy, and activity performance	Project has a strong gender monitoring program	- Project gender checklist developed - Annual checklist scores	Project scores on gender checklist increases yearly	Y1-5	PMU	\$ 16,533
G4. 4.2.2.1 Project partners (government and NGOs) are trained on gender-sensitive approaches	Project staff have skills to monitor inclusion of youth and women into programming	# staff trained	100% of M/E focal points from participating organizations/entities are trained	Y1-5	PMU	\$25,780
Project Gender and Youth staff time is incorporated into each activity cost						
Total Estimated Cost for Implementation of GAP: \$759,452						